Petal Realms 现行手册

V5.21 / S4 + S5 / 2026-10-27

本卷保留历史354 页，并增加现行玩家操作和工程补编。先阅读文末的中文玩家手册，再按需要查阅历史数学推导和当前工程规则。旧封面的“候选”“阻断”和生产版本是当时记录，不代表今天的上线状态。

内容	本文件物理页码
当前版本与阅读说明	1
历史规则正文，原编号保留	2-355
现行玩家手册	356-361
S4 / S5 机制与工程补编	362-365

当前补充重点：存档损坏停止静默重建、有效副本与备份恢复；单合/ 批量/ 此组全合；保存后短演出；风暴眩晕中断冲锋；版本与存档时间一致；隔离预览和发布验收。

软件入口：https://petal-realms.pages.dev/ 。源代码：https://github.com/liuzhide666/Petal-Realms 。帮助面板提供玩家版与完整手册下载。实际发布以GitHub、Cloudflare 部署和健康接口的相互核对为准。

运行时文件集指纹：

8312ef62908bd0edc01f62541c739e7aedfe3faf80a2812923087f5bb22e09a8

历史354 页不重新冒充当前源码的逐行计数。当前机制目录仍有143 个稳定标识；已实现、设计提案和待验证内容的身份按目录及补编说明区分。物理手机长时间运行、全面多人战斗和后期经济平衡不在此文档中虚报为完成。

Probability-Weighted Mechanics and Computation

Edition 3.2 with normative Supplements S1, S2 and S3

V5.19 candidate / Combat consistency / 23 October 2026

This manual preserves 347 historical pages and adds a normative correction for terminal combat, input ownership, damage acceptance and feedback, shared body geometry, bounded progression, save retries and encounter presentation. The original 138 mechanism identifiers remain registered.

Reading order and source authority

The next 347 pages retain their original text and printed numbering. S3 then supersedes only the behaviors it explicitly updates. Earlier source ledgers are historical and are not a V5.19 inventory. The following source pin identifies a local candidate, not a merged or deployed production release.

Frozen local runtime source: e0ece6621fd394858a4d5b8ce37b6da09c58db67

Publication boundary

Cloudflare production was observed at main commit 26aa9bf4aafc0c30841470921e57619686a5815b (V5.17). GitHub publication is blocked in this session. Complete Florr combat parity and physical iOS Safari remain unverified.

Probability-Weighted Mechanics and Computation

Edition 3.2 with normative Supplements S1 and S2

V5.18 / Original orbital field / 22 October 2026

This consolidated manual preserves the preceding 340 pages and appends the V5.18 presentation specification. The 138 registered mechanisms retain their status. Original outline artwork, bounded display settings and HUD changes do not introduce a new combat or reward probability law.

Reference boundary

The official Florr guest entry and animated background were observed in the browser. A playable session could not be established after one recovery attempt. This release therefore makes no claim of complete playtesting, one-to-one gameplay parity or verified Florr server formulas.

Reading order

The next 340 pages retain their original text and pagination, including earlier covers and the frozen source census. S2 follows them with separate page numbers. Earlier source inventories are not a V5.18 syntax census.

Frozen V5.18 runtime source: 08771687513e480a2cacd84c52ef1a6327b154fa

Probability-Weighted Mechanics and Computation

Edition 3.2 with normative Supplement S1

V5.17 / Interface continuity / 21 October 2026

This consolidated manual contains the original 333-page V5.16 standard and a source-pinned supplement for V5.17. The original 138 mechanism contracts remain registered. The supplement updates input, focus, scrolling, equipment access and session presentation rules. It adds no new combat or reward distribution.

Reading order

The next 333 pages retain their original printed page numbers. The supplement follows them, with separate S1 page numbers. The original source census remains explicitly tied to V5.16; it must not be interpreted as a V5.17 syntax inventory.

Deterministic interface transitions are probability-one cases of the state kernel. Preference weights, event probabilities and utility remain distinct.

Frozen V5.17 runtime source: 363a1faa631b10f88b9dfc5cd4539df4e5257fa5

Probability-Weighted Mechanics and Computation

Standard Handbook

PR-STD-003 | Edition 3.2 | 21 October 2026

A source-traceable, English specification for orbital combat, stochastic crafting, progression, ecology, interface calculations and persistent state.

Governing principle

```
Choose feasible policies by E[u(Z)]
Z = sum_i alpha_i * normalized_feature_i
Default u(z)=z; probability weights and utility weights remain separate.
```

Every deterministic transition is a probability-one case of the same state model. Random mechanics declare the distribution and conditioning state; tactics compare complete outcome distributions under a stated horizon. Reward, survival, time and resource costs use explicit units and normalization.

Coverage measure	Frozen source inventory
function nodes	626
operator nodes	7418
expression envelopes	3795
calls	4060
numeric calls	569
numeric literals	3824
random candidates	93
css computations	97

138 stable mechanism specifications; 51 numbered standard clauses; source fidelity and proposed runtime conformance are explicitly separated. The expression ledger records code syntax rather than pretending that every operator is a different gameplay mechanic.

Reader map and evidence boundary

This edition replaces the earlier research monograph as the normative project handbook. The document is an original engineering specification, not a peer-reviewed paper, an official Florr rulebook or a proof that all target mechanisms are already deployed. The scope is the frozen V5.16 source; future mechanisms require a new registered contract.

Read this part	Purpose
Chapters 1-19	Normative probability model, clocks, combat, ecology, resets and conformance
Part II: M001-M138	Stable formula and mechanism contracts
Part III	Numeric parameter registry, source references and fresh validation
Appendices A-H	Function ownership, complete operator envelopes, calls, literals, RNG, CSS and SQL

Source identity

Frozen runtime source commit: 5b61308db653691d68190b259274e08282058533

Frozen source baseline: V5.16. Runtime hashes below bind this standard to the implemented source. Publication of the matching game deployment is separately verified.

`site/index.html | SHA-256 dca2670e67380a168737355638b2e6a5af56160fbe414bc32d09c1d960904591`

`functions/api/[[path]].js | SHA-256 cfe7624c96017fdde76481b335eff2b05524b7a8348cbe4c682f1ae4ddc21a85`

`database/cloudflare/0001_accounts.sql | SHA-256 a4d9784904dc69c79c87bbd67b8844c2a94ba6866ddcbaa7a482da449f49fce4`

`database/migrations/20261017053742_create_petal_realms_saves_v1.sql | SHA-256 6cf6efe95a45b5e71ae818fd40bd2ef8c8ff676f58a80db4f4d65a09d02f7f37`

`site/_headers | SHA-256 a8f55e9a182245637225b859b192c8e051d701dd9a2f959fdda4aa3e06ea67b4`

`wrangler.jsonc | SHA-256 8c1ecd2fb0e760a66abdb01c8e72145ab259c0146f6e5e8842a64fc411ea7d5f`

`netlify.toml | SHA-256 b5d1bd23d9172f63c4a61b0f0bf6191ca30542b71532f4842e5f6080f23abc6e`

`site/rules.js | SHA-256 47aa2ab528f3f0c7301acced6f5d535bfc76fffb34c2e78a78771e7259649cea`

`site/art.js | SHA-256 05a991b371528a4c7dbe7ad2a8ea8c5ac5d85c647972ee496320a4c55051cd6e`

`site/workshop.js | SHA-256 ecb9bf79ef5590e02a52c3a2d8adaac33e164a2245dbf9b4277d17e8e9f89004`

`site/interface.js | SHA-256 72f47d051d2645be738e7f65fb122a8028be6429b1ccb984c6008938cba140b8`

`site/release.json | SHA-256 64309a419cc8b81a516c307d5e8e33b2fd5602ab8dfb79d16f1e7836f53b56bf`

`docs/standard/ruleset.json | SHA-256 2032dcd11a9227144ec13cc24e8131c35cedd60367bbf1f8c3b5261d7d115df1`

`docs/standard/mechanisms.json | SHA-256 aee1175e8c82de627d69ee2a2bef4ec7783b9731e32cf3f1dfe06c4de3e76e30`

Line locations are 1-based within the named source file. Function IDs F, operator IDs O, call IDs C and literal IDs N are stable only within this edition. Maximal expression envelopes preserve operand ordering and map every nested operator through standard_inventory.json. System tags are a navigation aid, not a proof of semantic data flow.

Contents

Standard Handbook 1

Reader map and evidence boundary 2

Contents 3

1. Scope, conformance and requirement language 5

2. The universal probability and expected-value model 6

3. Weight profiles, constraints and tactical decisions 7

4. Ordered composition and numerical semantics 8

5. Source-authoritative simulation sequence 9

6. Complete contact-effect and special-attack contracts 10

7. AI, spawn acceptance and world-event probabilities 11

8. Reset matrix and persistence boundaries 12

9. Loot, crafting and weighted reward valuation 13

10. Expected utility, risk and mathematical counterexamples 14

11. Numerical and implementation verification standard 15

12. Conformance roadmap and change control 16

13. Executable standard and conformance matrix 17

14. Implemented probability mechanics and exact expectations 18

15. Player-facing probability and tactical information 19

16. Visual engineering, content and migration compatibility 20

17. Exact player-facing tactical distributions 21

18. Motion, attack forecasts and material grammar 22

19. Constrained build planning and interface state 23

Part II. Standard mechanism contracts 25

Contracts M001-M010 26

Contracts M011-M020 31

Contracts M021-M030 36

Contracts M031-M040 41

Contracts M041-M050 46

Contracts M051-M060 51

Contracts M061-M070 56

Contracts M071-M080 61

Contracts M081-M090 66

Contracts M091-M100 71

Contracts M101-M110 76

Contracts M111-M120 81

Contracts M121-M130 86

Contracts M131-M138 91

Part III. Numeric parameter registry 95

Probability-weighted worked examples 98

Tactical utility: a complete small example 99

Validation and worked calculation evidence 100

References and evidence sources 101

Appendix A. Function and ownership registry 102

Appendix B. Complete operator expression ledger 113

Appendix C. Numerical call ledger 203

Appendix D. Complete call-site index 213

Appendix E. Complete numeric literal registry 281

Appendix F. Randomness and entropy register 327

Appendix G. CSS and database computation register 330

Appendix H. Coverage, limitations and change control 333

1. Scope, conformance and requirement language

Document identity: PR-STD-003, edition 3.2,21 October 2026. This is the project design, calculation and implementation-conformance standard for a probability-weighted expected-value game. The frozen implementation is Petal Realms V5.16 at commit 5b61308db653691d68190b259274e08282058533. The manual combines implementation-preserving requirements, derived mathematical laws and explicitly conditional future requirements. This edition describes the matching V5.16 source implementation. Deployment is an independently verified release step; source conformance is not server-authoritative reward validation.

STD-001: requirement words and status

MUST and MUST NOT identify mandatory requirements; SHOULD identifies a strong recommendation with a documented exception; MAY permits an option. Uppercase requirement words follow the BCP14 convention [R9,R10]. This project handbook is not an IETF standard or an official Florr specification. Each catalog record is tagged I, D or P: implemented behavior, derived analysis, or proposed extension. A P requirement applies only when that extension is enabled.

Conformance layer	Meaning	Current evidence
C0: source fidelity	Implementation statements match the frozen source and its evaluation order	Source pin, AST inventory, parameter tables, targeted code comparisons
C1: normative model	Every registered mechanic has inputs, units, transitions, distribution role and verification	This handbook and its reference calculations
C2: probability-governed runtime	A live ruleset uses versioned probabilities, utilities and scenario weights with validated policy outcomes	Partial: versioned frozen profiles, domain-separated streams and exact craft/reward analysis implemented; full policy optimization and replay remain targets
C3: authoritative economy	Reward eligibility and inventory transitions are server-authoritative and idempotent	Future architecture; cloud CAS alone does not establish C3

STD-002: completeness is version-bounded

The handbook MUST inventory every function node, operator node, numerical call, numerical literal, identified random call site, SQL operation and CSS computation within the declared runtime source files. This is an auditable source-coverage claim. It is not proof that every possible execution path, device, player policy or future feature has been balanced. New runtime files or generated code MUST expand the inventory before conformance is claimed. Vendor internals and browser cryptographic algorithms remain external platform contracts.

Supersession: edition 2 corrected the mean-variance/expected-utility distinction, selective reset behavior, dynamic spawn limits and additional combat paths. The original 120 stable mechanism IDs remain usable; eighteen implemented contracts M121-M138 extend the register; the expression and function appendices extend coverage beyond that conceptual catalog. Localized interface strings may be represented by stable LOCALIZED hash tokens in the English expression ledger; English mechanic selectors, numeric values and template interpolations are preserved.

2. The universal probability and expected-value model

STD-010: one state-transition model

```
s_next = F_v(s, a, z, h)
P_v(s_next | s,a) = distribution induced by z
E[g | s,a] = sum_j p_j(s,a)*g(s_next_j), sum_j p_j=1
```

State s includes positions, health, durability, inventory, cooldowns, enemy state, event phase, account revision and all other variables that affect future transitions. Action a contains player intent and explicit interface decisions. z is the ordered random input vector and v is the ruleset version. A deterministic mechanic is represented by a point-mass distribution: one next state has probability one. Therefore deterministic collision, movement and rendering can be analyzed within the same probabilistic framework without adding meaningless random rolls.

Every result MUST state the conditioning state, policy, horizon and stopping rule. The phrase expected damage is incomplete without saying whether it means requested damage, armor-mitigated damage, effective HP loss, damage during contact or damage per unit of expedition time. Render-only pixels are excluded from strategic reward unless a specifically declared perception or device-cost model includes them.

STD-011: probability weights and preference weights are different types

```
p_j = w_j / sum_k w_k, w_j>=0 and sum_k w_k>0
Z(outcome)=sum_i alpha_i*f_i(outcome), alpha_i>=0, sum_i alpha_i=1
Q(policy)=E[u(Z)] = sum_outcomes P(outcome|policy)*u(Z(outcome))
```

Probability weights describe how often an outcome occurs. Preference weights describe how much the design values an outcome feature. They MUST use separate field names, schemas and validation. Feature f_i MUST have declared units and a fixed normalization into[0,1], with larger values oriented as more desirable. Ratios MUST be divided by positive reference quantities before combining unlike units. The weights alpha are design choices, not facts inferred from rarity colors or the scale1.72^r.

STD-012: risk-neutral default and optional utility

```
Default: u(z)=z
Optional: u_eta(z)=(1-exp(-eta*z))/eta, eta>0
u_eta derivative=exp(-eta*z)>0; limit_eta_to_0 u_eta(z)=z
```

The default governing objective is weighted expected value, eta=0. An optional increasing concave utility may represent risk aversion, but its parameter and reason MUST be published with the profile. Evaluate utility on each outcome and then average; u(E[Z]) is generally different from E[u(Z)]. The exponential utility here is an original project option, not an existing runtime buff.

3. Weight profiles, constraints and tactical decisions

STD-020: profile schema and defaults

Field	Normative meaning
profile_id / version	Stable identity; every changed coefficient creates a new version
feature definitions	Unit, direction, normalization reference, clamp policy and horizon
utility_weights	Nonnegative coefficients with positive total, normalized explicitly
scenario_probabilities	Exposure-weighted encounter probabilities; separately normalized
eta	Zero for default linear expected value; optional nonnegative aversion
constraints	Survival, frame-time, inventory and account-integrity limits
provenance	Design choice, measured estimate or derived constant; never mix these labels

A usable illustrative profile assigns progress .40, survival .30, effective damage .20 and mobility .10, after each feature is converted to a stated[0,1] scale. These coefficients are a declared design example, not measured player preferences and not deployed balance parameters. A release profile SHOULD be calibrated using encounter traces and player objectives. The standard MUST NOT imply that this example is uniquely optimal.

STD-021: choose actions by expected outcome under constraints

```
a_star = argmax_a sum_j P(s_j | s,a)*u(Z(s_j))
subject to P(death before horizon | s,a)<=epsilon
and inventory>=0; cooldown and geometry feasibility hold
```

The candidate action set MUST contain only actions possible in the actual control and cooldown state. Keeping a safe escape, delaying a deck switch and accepting a lower immediate damage opportunity can be rational because they improve later expected outcomes. A policy comparison MUST include all scheduled outcomes, including deaths, timeouts, failed spawns and zero-reward episodes. Dropping failed episodes creates survivorship bias.

STD-022: scenario weighting and uncertainty

```
Q = sum_b pi_b*E[u(Z)|scenario_b]
Var(Z)=sum_b pi_b*Var(Z|b)+sum_b pi_b*(E[Z|b]-E[Z])^2
```

Scenario probabilities pi MUST describe the evaluated encounter policy. Equal map area, equal species count and equal observed exposure are different distributions. If a player deliberately farms one band, the weights change. The second variance term captures differences between scenario means; averaging conditional variances alone underestimates total uncertainty. Reports MUST separate sampling uncertainty from intentionally chosen utility coefficients.

4. Ordered composition and numerical semantics

STD-030: typed operator pipeline

```
state -> eligibility -> raw value -> modifiers -> mitigation
-> effective state change -> dependent effects -> expiry -> persistence
```

This is the documentation pipeline. The frozen source order is authoritative for C0, including exceptions. Each modifier MUST declare target, unit, operation, stacking group, cap, rounding stage, ownership, clock and expiry. Additions in a commutative group may be reordered; general state mutations may not. Overwrite poison, shared root charges, independent heal timers and exclusive stances MUST NOT be represented by one generic stack count.

Operation	Required semantics
Clamp	clamp(x,a,b)=max(a,min(b,x)); require a<=b
Rounding	JavaScript Math.round ties toward positive infinity; Python round is not an equivalent oracle
Binary operators	Keep parentheses and source operand order; + may concatenate strings
Logical operators	&&, || and ?? short-circuit and return operands; || treats zero as false
Bitwise PRNG operations	Signed32-bit coercion and unsigned right shift must match ECMAScript
Clock conversion	Simulation seconds, wall-clock milliseconds and server epoch seconds are distinct units
Damage accounting	Requested, mitigated and effective HP loss are separate variables

STD-031: fixed-step boundary rules

Simulation h is1/60 second. Timers are decremented and compared at discrete ticks, so expiry behavior can differ by one step from a continuous-time approximation. The current dash advances a full step when its timer was positive at the start; poison and venom similarly use tick ordering. A numeric reference MUST either reproduce this source order or label itself a continuous approximation. Tolerances MUST be stated for tests; rounding MUST NOT be moved across products to make a test pass.

All probabilities and weights MUST reject NaN and infinities. Zero-total weights are invalid. Arrays, counts and inventory ranks MUST remain within their declared domains. A theoretical formula with p=0 may have infinite expected waiting time; the interface MUST represent this explicitly rather than dividing by zero or displaying a finite budget.

5. Source-authoritative simulation sequence

The update function is a stateful ordered algorithm. The following sequence is source-confirmed [S1]. Each step can change later eligibility in the same tick. A refactor MUST preserve this order unless a versioned behavior change explicitly replaces it.

Stage	Current transition
1	Guard active/pause/dead at function entry; advance game.time; expire combo; derive stance; update events and presence
2	Derive body radius; advance dash/deck/boon clocks; activate nearby shrines
3	Resolve movement; derive velocity and orbital angle; apply venom HP drain; recompute zoom
4	Generate or clear root layers; update enemies, specials, motion, facing and poison
5	Resolve player/enemy and enemy/enemy body contacts, then two additional relaxation passes
6	Move orbital petals; reload/rock wear; melee exchange and dependent effects; emit petal projectiles
7	Apply nature synergy, defensive durability repair and individual healing
8	Advance hostile projectiles/interception and status; advance friendly shots/bounces
9	Attract and pick up drops, then remove expired ones; update cosmetic effects
10	Remove dead enemies, replenish population, update portals/camera/HUD, schedule local/cloud saves

STD-040: mid-tick death is not an immediate return

The entry guard stops an already-dead player on the next update call. If die() runs during the current update, it sets flags but the enclosing function continues its remaining statements. Thus later same-tick healing, attacks or cleanup can still execute. A simulator MUST preserve this behavior when reproducing C0. A future immediate-abort rule requires a deliberate transition change, not an undocumented assumption.

STD-041: direct status damage

```
venom HP drain per tick = h*(2.6+biomeDrop), while venomUntil>time
venom movement factor=.79; hostile venom hit sets expiry=time+3.5
```

Venom drain directly mutates HP and bypasses damagePlayer defense, dash immunity and global damage gating. Projectile venom status is assigned after damagePlayer, even if health loss was rejected by that call. Monster poison similarly subtracts HP directly before invoking the death pipeline. These status paths MUST be included in survival expectations rather than treated as ordinary mitigated hits.

6. Complete contact-effect and special-attack contracts

STD-050: effects following a qualifying orbital batch

Effect	Exact current rule	Required distinction
Ordinary durability	Subtract max(.5,mobDamage) per completed contact strike	Durability ratio changes before the next strike
Rock wear	13*h durability per overlapping retrieved rock, except glass	Multiple rock contacts can accumulate in one tick
Fang heal	min(Hmax,HP+max(1,.07*requestedBatchDamage))	Uses requested batch damage, including overkill/armor differences
Knockback	Bubble37 world units; other qualifying contact petals7	Displacement happens after the batch and changes later contact
Poison / iris	Set rate5*S_r and remaining duration2.4	Last writer replaces rate; not an independent sum
Lightning	Call damageMob with .20*requestedBatchDamage on each other living enemy within125	Armor applies separately to each secondary call
Puff	Calmed timer3.8 for non-guardian targets	A behavioral timer, not damage immunity
Rose/nectar	healWait decreases only after the petal reload guard; resetEquip initializes1.8	The heal clock pauses while reloading
Petal ranged shots	Missile/grape use power and rarity, but omit melee durability ratio, assault and overdrive	Do not copy all melee factors into their expected damage

STD-051: enemy specials

```
cooldown=max(guardian?4.2:5.2,10.3-.45*rank-(guardian?1.8:0))*(enraged?.75:1)
```

Trait	Current parameters
Activation	Start when distance<590 and time>=specialAt; windup .95 charger,1.1 otherwise
Sniper / venom shot	Speed=(venom?270:340)+min(120,1.7*level); damage=mobDamage*(venom?.48:.60); life2.3; radius10 venom or8 sniper
Sniper prediction	Aim offset=playerVelocity*clamp(distance/370,0,.65); venom has zero prediction
Rank spread	Rank>=3 emits side shots at-.18 and+.18 radians in addition to center
Charge	Duration .52; movement direction multiplier3.0 before enrage speed factor
Summon	Species selected by current region; HP=ceil(baseHP*(1+.045*summonerLevel)); damage analogous; level=max(1,summonerLevel-3); rank=min(6,summonerRank)
Nova	If player center distance<210, call damagePlayer(.65*mobDamage)
Shield	Set shieldUntil=time+4; damageMob adds8 armor during that interval

7. AI, spawn acceptance and world-event probabilities

STD-060: ordinary AI table

Mode	Exact current behavior
Activation	Ordinary engagement at distance<520 unless temporarily calmed; specials update separately
Ranged	Approach beyond280, retreat below175, strafe at .42 direction magnitude between; ordinary shot requires distance<440
Ranged shot timing	Reset interval uniform[1.65,2.85), multiplied by .60 when enraged; guardians add shots at+/-.32 radians
Rush	Chase; boost multiplier1.6 for .43 seconds when turn expires; next turn uniform[1.2,2.8)
Idle wander	Heading uniform[0,2*pi), turn interval uniform[.7,2.4), direction magnitude .46
Guardian enrage	At positive HP<=.35*maxHP; movement factor1.33; other damage/cooldown factors are path-specific
Neutral	Without retaliation, retreat only at d<180; otherwise neutralMotion uses heading magnitude .5. Retaliation uses aggroUntil=time+8.

STD-061: accepted spawn distribution

```
P(x | accepted,s)=q(x|s)*I(clearance(x,s))/Z(s)
P(no accepted proposal in60 attempts)=(1-Z)^60 only for iid fixed proposals
```

The actual proposal mixes near-player and world-wide sampling, with coordinate clamps that can create boundary point masses. Clearance depends on player, rocks, portals, existing bodies and guardian band. A report MUST NOT replace the accepted distribution with uniform area. Guardian selection is .025 only when the normal guardian is absent, before location acceptance; a forced attempt selects the guardian. Rank/level/elite draws occur after accepted location. Base-size clearance precedes elite size expansion.

STD-062: dynamic population loops

```
i=0; N=N0; while i<min(K,cap-N): attemptSpawn(); i=i+1
On successful attempt: N=N+1; on rejected attempt: N unchanged
```

Swarm opening uses cap130,K7. At N0=120 and all successful attempts it adds5, not7. Summon uses cap125,K3; at N0=122 all successful attempts add2, not3. The loop condition recomputes the remaining capacity each iteration. Expectations near caps require this state-dependent recurrence and acceptance probability; a one-time min(K,cap-N0) is wrong.

STD-063: event selection versus time exposure

```
P(event=e)=weight_e/100
Long-run active-time share_e=p_e*d_e/(sum_j p_j*d_j +69)
sum_j p_j*d_j=24.99 seconds; E[quiet gap]=48+42/2=69
```

This stationary continuous-time cycle approximation omits fixed-tick expiry/activation delays and assumes no map transition or initialization interruption. It differs from event selection probability and from probability conditional on some event being active. The current event durations are24,25,30,24,21 seconds with weights28,18,21,18,15. Transitions reschedule events using a different rule; those episodes MUST be modeled separately.

8. Reset matrix and persistence boundaries

State group	initGame	Region transition	Death / respawn
Player object	Recreated by newPlayer; position center	Same player; position center	Death flags; respawn full HP and center
Absolute game.time	Reset to0	Preserved	Preserved
nextSpawn	Reset to1.3 seconds	Preserved	Preserved
Skill cooldowns / buffs	Reset dictionaries	Cooldowns preserved; buffs cleared	Cooldowns preserved; buffs cleared
Venom/dash/root layers	New-player defaults	Venom/dash motion/iframes/root layers cleared	Same explicit clearing at respawn
Petal state	resetEquip full durability/timers	resetEquip full durability/timers	Respawn resets equipment
Loose drops/shots	Cleared	Cleared	Respawn clears; death alone is selective
Event/combo/boon	Reset	Cleared/rescheduled	Death clears event/combo/boon
Inventory/equipment	Load/sanitize saved arrays	Preserved	Preserved

STD-070: actual clocks and transient discrepancies

initGame now resets simulation time to0 and nextSpawn to1.3 seconds before assigning the first event deadline65+45U. This resolves the old same-page re-entry immediate-event defect. Region transition schedules current time+55+38U. Region changes and respawn clear venom, dash displacement/iframes, root layers, active skill buffs and boons, while preserving skill/dash cooldown deadlines to avoid free skill resets through portals. Quest leveling uses effectiveZoom consistently.

STD-071: cloud integrity

Native accounts use email-shaped identifier plus password without mailbox verification. Recovery uses a one-use256-bit secret, not an email OTP. The cloud service authenticates sessions, binds account identity and performs revision compare-and-swap. The browser holds one session Web Lock per account and refuses a competing tab. No lock fallback may silently enable unsafe multi-tab cloud entry. These controls MUST NOT be removed to simplify a probability model. They prevent access/lost-update faults, not forged client gameplay.

Local outboxes preserve pending snapshots and base revisions. A conflict MUST stop automatic overwrite. Snapshots intentionally omit transient world state, so loading a save is not resuming an event replay. Legacy Supabase authentication and position rooms remain isolated compatibility paths; their limits and semantics MUST NOT be confused with native Cloudflare account behavior.

9. Loot, crafting and weighted reward valuation

STD-080: a conditional joint reward model

```
E[V|m,state]=E[V(main)]+I_guardian*E[V(fixed bonus)]
+ .18*E[V(basic bonus)]
Var(total)=sum Var(channel)+2*sum Cov(channel_i,channel_j)
```

The ordinary main family is uniform over source list entries, not necessarily unique families. Main rank uses its rank-dependent distribution. Treasure shifts that sampled rank by one with probability .28, capped at7. Guardian and basic bonus ranks are derived from the same sampled main rank, so they are correlated. Expected values add without independence; total variance requires covariance. Expiry, travel, death and bag limits reduce collectible value relative to spawned value.

STD-081: three crafting quantities

```
E[consumption per attempt]=2.5+2.5p
E[cost to one output with replenishment]=5+2.5*(1-p)/p
Var(cost)=1.25*(1-p)/p+6.25*(1-p)/p^2
F_p(n)=0 for n<5; otherwise p+(1-p)*sum_j=1..4 F_p(n-j)/4
```

A successful attempt consumes5, while failure consumes a uniform integer1..4. A batch executes at most5 attempts and can stop early when stock falls below5. For expected batch output use the joint stock/output distribution, not5p when eligibility may be exhausted. The finite-inventory CDF answers completion chance; replenished mean answers long-run consumed cost. Neither is a guaranteed stock recommendation.

STD-082: transparent player-facing calculations

A probability-driven game SHOULD show probability, expected material demand, finite-stock completion chance and a chosen completion quantile as distinct values. A95-percent stock quantile is the smallest n with F_p(n)>=.95. Assumptions about independence, constant p, protection flags and stopping MUST accompany the model. Hidden changes to pity, salvage or rank probabilities are prohibited by the target standard and require a versioned economic-impact report.

Current numerical examples: at p=.64, mean6.40625 and95-percent budget12; at p=.32, mean10.3125 and95-percent budget24; at p=.08, mean33.75 and95-percent budget94. These are exact-model calculations validated against simulation, not production account telemetry.

10. Expected utility, risk and mathematical counterexamples

STD-090: increasing expected utility is the governing objective

For a normalized benefit outcome Z, an increasing utility preserves outcome-by-outcome improvement and first-order stochastic dominance under its usual integrability conditions. This is the appropriate core interpretation of the requested expected-value game. Mean-variance is retained only as a labeled diagnostic. Its penalty weights cannot be mistaken for probabilities or silently redefine the game objective.

```
A={1,10} equally likely; B=0
E[A]=5.5; Var(A)=20.25; E[A]-Var(A)=-14.75<0
For increasing u: E[u(A)]>u(0)=E[u(B)]
```

This counterexample disproves a universal dominance guarantee for mean-minus-variance scores. A report MUST distinguish preference over objective vectors from stochastic dominance of reward outcomes. Nonlinear transformations of expected reward are also not interchangeable with outcome-wise expected utility.

STD-091: rate and stopping rules

```
Long-run regenerative reward rate=E[cycle reward]/E[cycle duration]
In general: E[R/T] != E[R]/E[T]
V(s)=max(stopValue(s), E[V(next)|continue]-timeCost)
```

The regenerative ratio requires a consistent stationary cycle with positive finite expected duration. A fixed-horizon expedition has a different estimand. If death does not remove inventory in the current game, a model MUST NOT invent an item-loss penalty and call it current behavior. An optional extraction economy is a future rule and needs its own state and migration contract.

STD-092: practical tails

Completion probability, survival probability, expected progress and tail loss MUST remain separately visible. A high mean can coexist with a very large95-percent budget. Chance constraints and CVaR may be added as declared risk constraints, but their confidence levels, units and sampling uncertainty MUST be specified. Tiny failure probabilities cannot be certified from a small number of simulations with no observed failures.

11. Numerical and implementation verification standard

STD-100: independent reference calculations

The contact helper models one already-eligible batch. The caller must enforce .24-second gates only for ordinary single-exchange mode and 1-second gates for glass. Ordinary multi-exchange mode bypasses the pair throttle. This helper is not a temporal contact simulator. Source-oracle checks cover rank/event selectors and rounding, while combat, crafting and spawn tests validate the reference or declared hand-reviewed examples.

The accompanying probability_model.py implements validated weight normalization, bounded categorical sampling, outcome-wise utility, conditional drop distributions, correlated reward valuation, finite-inventory crafting, bounded batch distributions and a contact-exchange reference. It is a normative reference calculation package, not a deployed gameplay engine. Each helper states its supported domain and stopping assumption.

Test layer	Required evidence
Probability	Nonnegative finite masses, sum-to-one, threshold endpoint tests
Expected value	Exact small distributions and independent Monte Carlo consistency
Composition	Order-sensitive armor/crit, overwrite poison, depletion and death truncation
Inventory	Finite-stock DP versus full bounded batch enumeration; no negative stock
Utility	Dominance counterexample for mean-variance; increasing utility monotonicity
Source coverage	Every operator mapped to an expression envelope; calls/literals/functions all owned
Presentation	Searchable English text, complete IDs, table bounds, full rendered-page inspection
Release	Source commit/hash match; any runtime changes separately tested/deployed

STD-101: evidence limits

An AST traversal proves that source nodes were enumerated, not that inferred aliases are a complete semantic dataflow analysis. Function system tags are navigation labels. Source call indexes include asynchronous, rendering, validation and compatibility work. Native numerical functions and cryptographic primitives are external dependencies. Every coverage claim MUST identify this boundary. Future changes to strings, templates, modules, dynamic evaluation or generated scripts require parser/scope review.

The current scope includes every nonempty inline script in the production HTML, the complete native account API, both database schema files, the probability/art modules, canonical rule profile, hosting headers, Wrangler binding configuration and Netlify retirement configuration. Historical releases, tests and third-party packages are not current runtime mechanics. They are not duplicated into this handbook as if they were active gameplay.

12. Conformance roadmap and change control

Priority	Concrete deliverable	Acceptance criterion
P0: standardization	This handbook, source/operator/call/literal registries, exact parameter tables	All inventory records resolve to source hash and published manual section
P1: deterministic model	Separate RNG implemented; ordered input/event log remains pending	Matched-seed draw isolation passes; full replay needs a further event contract
P2: probability-driven UI	Craft CDF/batch/quantile and enemy reward lab implemented; effective-damage UI remains pending	Exact model/independent tree checks; explicit assumptions visible
P3: weighted balancing	Versioned feature normalizations, utility weights, scenario exposures and risk constraints	Policy comparisons with uncertainty and held-out scenarios
P4: behavior consistency	Clock/transient/zoom corrected; dynamic spawn caps and requested-damage effects retained explicitly	New source/runtime checks distinguish changed and preserved contracts
P5: authoritative economy	Server reward ledger and idempotent inventory transitions	Duplicate grants rejected; invalid encounter/material histories rejected

Changing a probability or utility coefficient MUST create a ruleset/profile revision and regenerate expected-value, variance, quantile and economy-flow reports. Save compatibility and player resource preservation MUST be reviewed before migration. Cosmetic changes MUST preserve collision and probability behavior unless explicitly versioned otherwise. A new mechanic MUST declare its distribution or deterministic point mass, source/sink effects, parameter units, composition order and verification before being admitted to the registry.

Current status: V5.16 implements the core probability module, versioned rule generation, exact finite-budget and bounded-batch analyses, covariance-aware reward valuation, separated random streams, prism critical hits, frost slow, explicit reset corrections and regional visual layers. The matched source is pinned in this edition. Full deterministic replay, global tactical-policy optimization and authoritative economy remain separately scoped work; partial implementation is not advertised as complete C2/C3 conformance.

13. Executable standard and conformance matrix

PR-STD-003 binds the frozen V5.16 source to the English mechanism register and its independent mathematical references. The first120 identifiers remain stable; M121-M124 and the six tactical-atlas contracts M125-M130 extend the register; M131-M138 add constrained build planning and interface state contracts. An implemented label describes the matching source path, not a claim of authoritative multiplayer or mathematically optimal automatic play.

Artifact	Authority / role
docs/standard/ruleset.json	Canonical versioned coefficients for critical chance/cap, frost and event weights; illustrative utility coefficients
scripts/build-rules.cjs	Generates the frozen runtime config literal and event weights; --check rejects source/profile divergence
site/rules.js	Pure, DOM-free probability, random stream, crafting, reward, utility and elemental computations
site/index.html	Actual ordered game transitions; controls, combat, inventory, rendering integration and cloud-save UI
site/workshop.js	DOM-free additive build allocation and release-profile validation
site/interface.js	Native dialogs, gate lifecycle, published-feed refresh and keyboard isolation
site/release.json	Versioned published announcement with a checked bundled fallback
site/art.js	Procedural materials, decorative anatomy, obstacle finish, biome light and telegraph presentation
functions/api/[[path]].js	Account/session authorization and29-family save validation; schema5 and revision guards preserved
docs/standard/mechanisms.json	Source-linked contract catalog matching this edition; I/D/P status retained

STD-110: a ruleset change is a behavior revision

The canonical profile MUST pass finite-domain and coefficient checks. Its version MUST match package.json. Generated bytes MUST match the source before release. Changing a probability or slow coefficient requires regenerating source, recalculating expectations and rerunning affected tests; changing a color alone does not justify consuming a gameplay random draw. The profile currently covers the first extracted numerical mechanisms; base creature/petal tables remain authoritative source-bound data and are inventoried separately.

STD-111: seed and stream ownership

```
seed = crypto.getRandomValues(Uint32Array(1))[0]
world / loot / craft / combat / visual use separate PRNG states
state_i = uint32(seed XOR imul(i+1,0x9e3779b9))
U_i = uint32(PRNG_mix(next_state_i)) / 2^32
```

World owns encounters, event timing and AI variability. Loot owns family/rank/bonus reward choices. Craft owns synthesis success/loss. Combat owns prism rolls. Visual owns decorative particle draws and loot orientation/placement jitter. Cached terrain and art use independent map-local seeds. The snapshot includes the session seed, rule version and per-domain counters. It is diagnostic context; a replay additionally needs initial state, complete input/event order and compatible numerical execution. The seed is not saved as a durable character identity.

14. Implemented probability mechanics and exact expectations

STD-120: prism contact critical hits

```
c = count(ready intact prism petals)
p(c)=min(.60,1-.85^c)
D_requested= D_precrit*(1+.60*I[U_combat<p(c)])
D_applied=max(1,D_requested-armor)
```

The critical factor applies once per qualifying orbital melee strike. It does not apply to body contact, poison or missile/grape projectile damage. A zero chance consumes no combat draw. The existing ten-strike batch may therefore contain several independent critical outcomes; a petal breaking or a target dying still truncates the batch. The visual batch feedback marks critical counts with a star. Rarity changes base damage/durability but does not change the per-copy chance.

```
E[applied|D,A,p]=(1-p)*max(1,D-A)+p*max(1,1.6D-A)
Example D=10,A=12,p=.15: E[applied]=1.45
max(1,E[raw]-A)=1, so averaging before mitigation is wrong.
```

STD-121: frost refresh and guardian reduction

```
c=max(1,count(ready surviving frost petals)) at a qualifying frost batch
slow=min(.35,.12*c)
move_factor=1-slow*(guardian?.5:1)
expiry=simulation_time+1.2 seconds
```

A frost batch refreshes the target timer; repeated applications do not sum durations. A qualifying source still contributes one copy when it breaks in that same batch. The factor multiplies enemy ordinary/charge displacement, while attack clocks and windup durations remain unchanged. Guardian maximum slow is17.5percent. The body-aligned frost outline is decorative. A browser scenario compares actual update displacement with and without a .88 factor; this is separate from a pure helper assertion.

15. Player-facing probability and tactical information

STD-130: distinguish every stopping rule

The craft panel displays per-attempt success, probability of at least one success within at most five eligible attempts, expected batch output, finite-stock first-success chance, replenished mean cost and a95-percent first-success material budget. A protected material group cannot execute; the analysis is explicitly conditional on removing protection. Quantile computation is bounded to99,999 copies; no threshold found is displayed as beyond the computation range, not as a guarantee.

```
Batch state=(stock,outputs); ineligible stock<5 is absorbing
E[outputs]=sum_state probability(state)*outputs(state)
P(batch success)=sum_outputs>=1 probability(state)
```

At stock12,p=.64, the bounded analysis shows an expected output near1.532. A single-attempt chance of64percent, a five-attempt success near96.46percent and a95-percent starting budget of12 are distinct quantities. They happen to be numerically related in this example because stock depletion limits feasible attempts; they must never be presented as the same formula.

STD-131: local enemy reward lab

The More menu opens the probability/tactics lab. It compares up to six living nearby enemies under current treasure-event state. Reward value is the declared rarity index1.72^r; it is neither currency nor an optimal equipment ranking. Main, guardian and basic-bonus channels share sampled rank. Exact joint enumeration includes covariance and reports mean and standard deviation. Travel time, expired loot, deaths and pickup failure remain outside this conditional spawned-reward analysis.

STD-132: utility is explicit, not a hidden automatic balance oracle

The pure module exports weightedValue and evaluatePolicy for normalized features and outcome-wise increasing utility. Feature entries must be finite within[0,1]; weights are normalized separately from probabilities. The illustrative progress/survival/damage/mobility profile is usable for offline policy studies and does not silently drive the enemies or choose the player loadout. An automatic policy requires an independently validated state-transition model and horizon.

16. Visual engineering, content and migration compatibility

Layer	Implemented contract
Creature body	Original collision ellipse remains authoritative; scaled limbs, segment marks, shell veins and eye highlights remain decorative
Guardians	Garden foliage, ocean appendages, factory shoulders/core and other regional motifs distinguish opponent roles
Map material	Each of eight biomes has a deterministic cached512pixel tile plus regional landmarks and obstacle finish
Light	Low-opacity directional tint, selected biome shafts and bounded shrine halos; low quality bypasses expensive fine layers
Telegraph	Directional warnings supplement existing source attack timing; no new invisible damage areas
UI	Botanical contrast, numeric cards and two-column probability grids adapt to portrait/landscape; scrollable panels preserve readable text
Content	29 petal families: original27 plus prism and frost; opponents43, regions8 and rarities10 remain
Cloud compatibility	Save schema5, identity binding and CAS preserved. Native API accepts the expanded family set and up to29*10 inventory keys

STD-140: reset correction and cooldown integrity

Character entry resets simulation time and spawn timer. Region/respawn clearing removes venom, active dash motion/iframes, root layers, skill buffs and boons; absolute skill cooldowns remain, preventing portal farming for free activations. Old source-preserved spawn-loop and requested-damage healing behavior remain documented rather than silently changed. The current PR does not add multiplayer authority, trading, hidden pity or an element interaction matrix.

STD-141: required acceptance evidence

The release MUST pass ruleset generation conformance, pure numeric/domain checks, independent bounded outcome enumeration, existing save/account regressions, real Worker compilation and browser operation/layout tests. Procedural drawing MUST preserve collision proxies and all nonvisual stream counters. Actual CJK fonts are used for Chinese UI screenshots; font metadata alone is insufficient. Screenshots and fixture measurements do not certify native iOS Safari or hardware frame-time budgets. Deployment SHA, Functions presence and database binding are verified independently from local tests.

17. Exact player-facing tactical distributions

STD-150: single-hit effective damage is outcome-wise

```
p=min(.60,1-.85^copies)
y0=min(H,max(1,X-A)); y1=min(H,max(1,1.6X-A))
mu=(1-p)y0+p*y1; variance=p(1-p)(y1-y0)^2
P(single-hit kill)=(1-p)I[y0>=H]+p*I[y1>=H]
```

X is the current raw orbital-melee hit, including durability (except glass), power talent, assault and overdrive. A is the exact current armor, including an active shield. H is remaining health. armorOf and rawPetalDamage serve actual combat and analysis. The health cap measures effective lost HP, even though the internal post-kill value may be negative. The panel evaluates one independently eligible hit per row; it does not combine future exchanges or promise contact uptime, DPS or an optimal loadout.

Counterexample	Correct expectation	Invalid shortcut
X=5,A=6,p=.15,H=100	.85*1+.15*2=1.15; variance=.1275	max(1,5*1.09-6)=1
X=10,A=0,p=.15,H=12	.85*10+.15*12=10.3; kill chance=.15	Uncapped raw mean10.9 overstates lost HP

STD-151: exact target acquisition uses a union, not a sum

```
a(r)=P(primary family matches)*I[r=targetRank]
g(r)=guardian forced channel matches exact family/rank
b(r)=.18*I[Basic bonus matches exact family/rank]
P(any target)=sum_r P(R=r)*(g(r)?1:a(r)+(1-a(r))*b(r))
```

Condition on the shared drop rank first. g is a Boolean guarantee, while primary family choice and the Basic Bernoulli are conditionally independent. Duplicate family entries retain multiplicity. Treasure mixing and the rank-7 exception are already included by dropRanks. The tracked goal requests exact family and exact quality, rather than at-least-quality. Ranks8 and9 honestly return zero under current drop rules. Defeat, collection and survival probabilities remain distinct; no travel or death adjustment is hidden inside this calculation.

STD-152: finite-batch output histogram

```
h[j]=sum_stock P(stock,j) after at most5 attempts
sum_j h[j]=1; E[outputs]=sum_j j*h[j]
P(any output)=1-h[0]
```

This marginal uses the existing exact stock/output DP: success consumes5 and adds one output; failure consumes a uniform1..4; fewer than5 stock is absorbing. All six bins0..5 remain visible, including zeros. In the live64% example with12 materials, P(any)=.964559584 and E[outputs]=1.531568864 (display rounding is separate). The first-success budget, capped batch and replenished expected cost remain different stopping rules. Analysis performs no random draw.

18. Motion, attack forecasts and material grammar

STD-153: time-composable smoothing with legacy gain

```
alpha(g,dt)=1-(1-g)^(60*dt)
alpha(g,1/60)=g
1-alpha(t1+t2)=(1-alpha(t1))*(1-alpha(t2))
x_next=x+(target-x)*alpha
```

Camera uses gain8/60; petal pursuit uses.25. Evaluate using -expm1(60*dt*log1p(-g)) for small-step stability. dt=0 returns0; gain0 returns0; gain1 returns1 for positive dt. The semigroup identity assumes a fixed target. It does not erase different collision, input or moving-target histories. Normal60Hz behavior retains its prior gain.

STD-154: shared hostile behavior and preview coefficients

Quantity	Shared rule
Windup	Charger .95s; other specials1.1s
Charge	Duration .52s; speed multiplier3; rage multiplier1.33; current frost factor
Nova	Strict player-center distance<210 world units
Projectile	Sniper340 or venom270 + min(120,1.7*level), world units/s; life2.3s
Spread	Rank>=3 emits0,-.18,+.18 radians in original spawn order
Sniper lead	Current player velocity times min(distance/370,.65) seconds

The renderer receives the same profile as runtime windup, projectiles and nova. Unobstructed projectile travel equals speed*life, plus its radius-sized spawn offset in the displayed ray. Charge reach uses a snapshot of speed/rage/frost. A later status expiry or rage transition can change realized travel, as can obstacles and fixed-step expiry boundaries. The capsule is a conservative visual body envelope, not a new collider. Windup rays track current aim and release recomputes it; sniper lead is an approximation, not an exact intercept solution. A progress arc communicates remaining windup. The nova ring replaces the old195-unit decorative mismatch with210.

STD-155: scale-aware anatomy and clear information hierarchy

Bee classification matches Bee exactly, avoiding the Beetle substring bug. Rat has two leg pairs, ears and a tail. Shell material traces are clipped to the authoritative ellipse, appear only above17 projected pixels outside low quality, and scale from body dimensions. Frost, poison and shield have distinct broken-ring radii and pip counts in addition to color. Regional vignette, shell traces and status geometry consume no gameplay or visual stream draws. Low quality omits shell and screen-light detail while retaining gameplay telegraphs.

The atlas separates reward moments, tracked-goal chance and collapsible contact forecasts. Craft output bars retain numeric percentages for every bin. On desktop, an active boss HUD hides the overlapping nearby-target card. Browser evidence must use real CJK fonts and wait for entry animations before judging text contrast. Seven new grouped tests include768 independent target-drop enumerations; real HTTP browser checks compare actual damage/projectile paths, nova drawing and RNG isolation. Native Safari hardware and anti-cheat authority remain unverified.

19. Constrained build planning and interface state

STD-156: preference weights are not probabilities

```
A_k = base_damage_k * rarity_scale_k
R_k = base_durability_k * rarity_scale_k / max(0.1, reload_seconds_k)
U_k = rarity_scale_k * I(support_family_k)
z_kj = f_kj / max_available(f_j), or0 if maximum=0
s_k = w_A*z_kA + w_R*z_kR + w_U*z_kU
```

Support-family tags are leaf, rose, nectar, starpetal, root, shell, soil, magnet, antenna, wing and frost. The tag is a preference feature, not a conversion of their different effects into healing units. Maxima use keys present in the current deck or backpack. The auxiliary score is dimensionless, the renewal feature has durability/second units before normalization, and the attack feature has health units before normalization. These quantities do not become a damage-per-second estimator merely because they are added after normalization.

Preference	Attack	Renewal	Support
Attack	.80	.15	.05
Sustain	.15	.35	.50
Balanced	.40	.30	.30

The weights sum to 1 as a normalization convention. They express player preference, whereas drop/craft weights describe uncertain outcomes. A deterministic preference proposal is a probability-one state transition. Existing expected utility and conditional damage models remain the basis for uncertain combat decisions.

STD-157: feasible allocation and exchange proof

```
pool_k = bag_k + count_current(k)
maximize sum_k x_k*s_k
subject to integer0<=x_k<=pool_k, sum_k x_k=N, 1<=N<=10
```

Sort keys by descending score and ascending ASCII key for ties, then take min(pool_k, remaining_slots) copies. The backup deck stays outside the pool. For an exchange proof, suppose a feasible solution selects a copy with score b while leaving available a copy with score a>b. Exchanging those copies preserves counts and capacity and increases the objective by a-b. Repeating exchanges yields the sorted solution. Ties can be exchanged without changing objective. Complexity is O(K log K + N), with at most 290 valid type/rank keys and 10 slots in the current content model.

This proof covers the additive preference objective only. Nonlinear critical stacking, shared cooldowns, health penalties, armor, enemy geometry and set synergies are deliberately outside that objective. A larger score is not evidence of improved survival or global tactical optimality. The explicit preview, item labels and editable manual deck let the player judge that boundary.

STD-158: per-key conservation and atomic application

```
bag_new[k] = pool[k] - x[k]
bag_new[k]+x[k] = bag_old[k]+count_old[k]
backup_new = backup_old
```

Construct the proposal from copied objects. Verify integer counts, registered keys and slot bounds. Returning current equipment may temporarily enlarge the pool, but the final bag must remain at most 99,999 for every key; reject a proposal that would overflow rather than silently clamp. Planning neither writes game state nor consumes random draws. After confirmation compare the complete owner/deck/slot/bag/backup signature; a changed signature rejects the old preview. Commit both the bag and current deck in one synchronous client task, then recompute equipment and save. This does not replace server identity binding or revision compare-and-swap.

STD-159: reload, health and random-stream boundary

```
petal_HP_i=0; reload_i=2.5 simulation_seconds
player_HP_new=min(player_HP_old,maxHP(new_deck))
```

Changing the deck recomputes synergies and zoom. A greater maximum health does not heal the player, and all newly applied petals must reload after the panel closes. Ordinary resetEquip initializes shot timers using the world stream; application shares that established behavior. Preview alone is RNG-free. Existing tests separately verify count conservation, unchanged backup, reload and preview isolation. The current save schema remains 5 and no new persistence field is necessary.

STD-160: one announcement per gate entry

The initial gate opens a native top-layer welcome dialog. Its dismissal promise gates optional automatic account restoration. Authentication and mode controls remain disabled until restoration settles, so a post-dismissal mode change cannot race the delayed session request. Actual transitions back to the gate open it again, including logout; switching an authentication tab does not. Manual reopen uses the same view. Duplicate calls while open reuse the existing promise. Hiding the gate closes a pending welcome so asynchronous account entry cannot leave it covering the world. No permanent hide flag suppresses future entries. The introduction describes the project without inventing a personal biography or unsupported credentials.

STD-161: current published updates and honest fallback

Fetch release.json from the same origin using no-store and an AbortController with a 5-second deadline. Validate schema 1, semantic numeric version, ISO UTC timestamp, bounded text fields and 1..20 highlight strings. Reject markup and decoded payloads longer than 16,384 characters. Only textContent renders feed values. At build time assert that the bundled JSON and checked-in feed are identical. Refresh when requested, on reopen, on return to a visible announcement and every 60 seconds while it remains visible; no polling is needed behind gameplay. Display the publisher timestamp in Asia/Shanghai and distinguish it from the fetch time. A failure preserves the last valid release or the bundled fallback and explicitly names the failed refresh. The feed does not claim live developer presence or automatically completed roadmap work.

STD-162: asynchronous confirmation guards

```
apply = accepted AND same(owner,account,active,view,player_object,conflict_state)
```

Native confirmation offers Cancel as its initial focus. Escape rejects. Concurrent pending requests reject without replacing the active operation. Because an await can outlive the originating view, askGame captures owner/account identity, player object, active flag, modal type and conflict state; mismatches reject. A conflict that was already present may be explicitly resolved, whereas a changed conflict invalidates the stale operation. The guard applies to build application, bounded bulk craft, talent respec, import/restore, migration and conflict resolution. The mutation still executes its ordinary payload and server ownership checks. No confirmation weakens authentication or allows a password reset without the existing recovery credential.

STD-163: component grammar, focus and device evidence

All game panels share forest surfaces, warm gold accents, clear section headings, consistent controls and quality labels. Inventory has a selected-item state, current/backup deck controls, comparison and collection routes; skill cards describe actual source formulas and set thresholds. The central playfield stays free of expanded secondary tools by default. Opening a game panel pauses simulation, clears held input, makes the HUD inert and focuses Close. Tab loops through visible panel controls. Closing clears input, removes inert and returns focus to the canvas. Native top-layer dialogs retain their own focus; recovery-code fields remain selectable. Decorative title rings have pointer-events:none, so they cannot block an announcement button. Reduced-motion media removes arrival animations. Five Chromium sizes and 14 panel types are reviewed with CJK fonts. Synthetic responsive checks do not certify physical iOS/Safari or prove subjective aesthetic perfection.

Part II. Standard mechanism contracts

M001-M138 are distinct conceptual mechanisms. Status I records implemented behavior, D a mathematical derivation under its assumptions, and P an extension that becomes mandatory only if enabled. Each record specifies its domain, transition, equation, composition, boundary, acceptance check and source/navigation anchor. The inventory appendices expand coverage to every registered expression and function, including interface, service and legacy paths outside these 138 concepts.

Symbol / input class	Domain and units
s / S; a / u[k]	Complete state; feasible action or sampled input, not utility u(z)
h, dt, t, T, cooldown	Nonnegative simulation seconds; h=1/60 s. Server timestamps use explicit epoch seconds; browser wall time uses ms.
x,y,R,radius,v	Finite world units; velocity world units/second. R used as reward in statistical sections is a separate scoped symbol.
H, HP, D, damage, armor	Nonnegative health or durability units. Requested, mitigated and effective damage are separate.
L, r, n, k, slot, rank	Integer level, rarity or count in the entry-specific bounds; r=0..9 for rarity.
p,q,w,pi,alpha	Probabilities in [0,1]; q=1-p. Raw weights finite and nonnegative with positive sum. Utility coefficients are a distinct type.
E[X], Var(X), SD, quantile	Expectation in X units; variance in squared units. Quantile belongs to a specified distribution and stopping condition.
lambda, mu, nu, eta	Explicitly chosen design coefficients; unit conversion and risk preference must be declared.
seed, revision, identity	Seed integer/byte stream, monotone integer snapshot revision, validated account identity. Cryptographic entropy is not a gameplay weight.

Contracts M001-M010

M001 | Fixed simulation clock

STATUS I | implemented baseline

Inputs / units: Clock/input state, finite frame interval and h>0; action intent bounded to the device control domain.

```
h = 1/60 s; S[k+1] = F(S[k], u[k], h)
```

Transition / meaning: The simulation advances in fixed increments; rendering follows the display clock. State S contains bodies, timers, health, inventory and encounters. A repeatable step size makes cooldown and contact semantics inspectable, but it does not make an unseeded random simulation reproducible.

Output / distribution role: Simulation state and input vector; deterministic conditional on input and random stream.

Trigger / clock: Per display callback or simulation tick; pause/overload policy applies.

Composition requirement: Input is sampled before movement; collision, attacks, healing, projectiles, loot and persistence follow in source order. Changing that order changes outcomes.

Boundary / exception: Floating-point arithmetic, iteration order and Math.random remain sources of divergence. Fixed step is not a claim of cross-platform lockstep.

Acceptance check: Compare position and cooldown after equal simulated time at different display schedules; hold the random stream fixed for a strict comparison.

Trace: loop; update [S1]; [R1]

Function registry: F0222, F0272, F0291

M002 | Accumulator overload policy

STATUS I | implemented baseline

Inputs / units: Clock/input state, finite frame interval and h>0; action intent bounded to the device control domain.

```
e = min((now-last)/1000, 0.08); steps <= 4
```

Transition / meaning: Elapsed wall time enters an accumulator. At most four fixed updates execute per display callback; excess accumulation is discarded under the existing guard. This deliberately prioritizes responsiveness over catching up after a stall.

Output / distribution role: Simulation state and input vector; deterministic conditional on input and random stream.

Trigger / clock: Per display callback or simulation tick; pause/overload policy applies.

Composition requirement: All simulation timers share the resulting game clock, so overload slows gameplay relative to wall time rather than selectively skipping enemy cooldowns.

Boundary / exception: The elapsed cap is 80 milliseconds; four steps cover about 66.7 milliseconds. The policy can lose time and is not real-time conservation.

Acceptance check: Inject 16, 40 and 200 millisecond display intervals; verify the step ceiling and document lost simulated time rather than reporting identical speed.

Trace: loop [S1]

Function registry: F0272

M003 | Pause and input neutralization

STATUS I | implemented baseline

Inputs / units: Clock/input state, finite frame interval and h>0; action intent bounded to the device control domain.

```
u = 0 when paused; accumulator = 0
```

Transition / meaning: Opening a modal pauses the simulation and resets held keys, mouse buttons and touch references. Blur, pointer cancellation and background transitions also clear input. This prevents a release event lost outside the page from leaving a permanent attack or movement command.

Output / distribution role: Simulation state and input vector; deterministic conditional on input and random stream.

Trigger / clock: Per display callback or simulation tick; pause/overload policy applies.

Composition requirement: Pause gates the update loop; resetInputs separately clears input state. Both layers matter because clearing one timer does not neutralize a captured pointer.

Boundary / exception: Rendered frames may continue during pause. Account dialogs and game modals have separate lifetimes and must preserve their own pause ownership.

Acceptance check: Hold attack, open a modal or dispatch blur, then resume without a new press; movement and attack must remain neutral.

Trace: resetInputs; showModal; loop [S1]

Function registry: F0272, F0273, F0414

M004 | Normalized keyboard movement

STATUS I | implemented baseline

Inputs / units: Clock/input state, finite frame interval and h>0; action intent bounded to the device control domain.

```
v = (dx,dy)/max(1, sqrt(dx^2+dy^2))
```

Transition / meaning: Physical key codes map WASD and arrows to a direction. Normalization prevents diagonal travel from exceeding axial speed by a factor of square root two. Keyboard movement takes priority over mouse following, while keyboard-only mode ignores pointer position.

Output / distribution role: Simulation state and input vector; deterministic conditional on input and random stream.

Trigger / clock: Per display callback or simulation tick; pause/overload policy applies.

Composition requirement: This unit vector is multiplied by stance, soil, talent, synergy and boon speed factors before collision-aware displacement.

Boundary / exception: Editable fields and native keyboard button activation retain their own semantics. A shortcut handler must not turn typing into movement.

Acceptance check: Check axial and diagonal displacement magnitudes; verify non-Latin keyboard layout codes and text fields do not change the intended mapping.

Trace: keyForEvent; moveInput [S1]

Function registry: F0208, F0412

M005 | Analog joystick saturation

STATUS I | implemented baseline

Inputs / units: Clock/input state, finite frame interval and h>0; action intent bounded to the device control domain.

```
j = delta/max(L, |delta|); |j| <= 1
```

Transition / meaning: A captured touch controls a circular joystick with maximum excursion L equal to roughly 31 percent of its control width. Below saturation, displacement retains analog magnitude; beyond the ring, only direction changes.

Output / distribution role: Simulation state and input vector; deterministic conditional on input and random stream.

Trigger / clock: Per display callback or simulation tick; pause/overload policy applies.

Composition requirement: The joystick direction enters the same movement pipeline as keyboard input. It does not have a separate faster movement constant.

Boundary / exception: Cancellation releases ownership and recenters the knob. One finger owns the joystick, while independent references may hold attack or defense.

Acceptance check: Drag to half radius and beyond full radius; compare relative speeds and ensure pointer cancellation cannot retain a nonzero vector.

Trace: joyMove; joyEnd; moveInput [S1]

Function registry: F0208, F0419, F0422

M006 | Multiplicative movement stack

STATUS I | implemented baseline

Inputs / units: Clock/input state, finite frame interval and h>0; action intent bounded to the device control domain.

```
speed = 252 * f_stance * f_soil * f_talent * f_scout * f_boon * f_venom
```

Transition / meaning: The current speed calculation multiplies independent categories. Defense uses 0.87 and attack 0.94; active venom uses 0.79. Talent stride contributes 1+0.04t. Scout and swift effects belong to additional categories, not an additive percentage pool.

Output / distribution role: Simulation state and input vector; deterministic conditional on input and random stream.

Trigger / clock: Per display callback or simulation tick; pause/overload policy applies.

Composition requirement: Compute each factor once from a snapshot, multiply, then pass the displacement through slideCircle. Do not round each intermediate product.

Boundary / exception: Correlated effects can produce surprising products. A nominal plus 20 percent and minus 20 percent compose to 0.96, not identity.

Acceptance check: Build a factor table for selected loadouts and compare numerical displacement in unobstructed terrain against the product.

Trace: update; TALENTS; bodySpeedFactor [S1]

Function registry: F0136, F0222, F0291

M007 | Camera smoothing and bounds

STATUS I | implemented baseline

Inputs / units: Clock/input state, finite frame interval and h>0; action intent bounded to the device control domain.

```
camera' = lerp(camera, target, 1-(1-8/60)^(60*dt))
```

Transition / meaning: Camera tracks the bounded player-centered target using frameAlpha with gain8/60. This preserves the old 60Hz gain and composes exactly under subdivided time for a fixed target.

Output / distribution role: Simulation state and input vector; deterministic conditional on input and random stream.

Trigger / clock: Per display callback or simulation tick; pause/overload policy applies.

Composition requirement: Zoom modifies the world extent visible through the viewport before the camera target and bounds are evaluated.

Boundary / exception: The fixed-target semigroup identity does not imply identical results for moving targets, varying input or collision histories. Camera remains clamped to world bounds.

Acceptance check: Move near all four world edges at minimum zoom; ensure no camera coordinate exceeds its legal interval.

Trace: site/index.html:update; site/rules.js:frameAlpha; M128

Function registry: F0222, F0291, F0611

M008 | View range and antenna composition

STATUS I | implemented baseline

Inputs / units: Clock/input state, finite frame interval and h>0; action intent bounded to the device control domain.

```
z = clamp(viewScale(level)/A, 0.38, 1); A = 1 + sum(0.12+0.018r)
```

Transition / meaning: Level progression broadens the field of view through a capped denominator. Equipped antenna contributions add within A, then divide the level-based zoom. Lower zoom reveals more world units without changing world-space movement or damage radii.

Output / distribution role: Simulation state and input vector; deterministic conditional on input and random stream.

Trigger / clock: Per display callback or simulation tick; pause/overload policy applies.

Composition requirement: Antennas stack additively with one another; their joint result composes multiplicatively with the level view scale through division.

Boundary / exception: The clamp creates diminishing or zero marginal benefit at the lower bound. Wider visibility can increase rendering cost and tactical information simultaneously.

Acceptance check: Enumerate zero, one and ten antennas across levels; confirm zoom stays bounded and does not alter collision results.

Trace: viewScale; antennaMultiplier; effectiveZoom [S1]

Function registry: F0123, F0124, F0127

M009 | Domain-separated seeded session random streams

STATUS I | implemented baseline

Inputs / units: Clock/input state, finite frame interval and h>0; action intent bounded to the device control domain.

```
seed <- crypto Uint32; domain_state_i=seed XOR imul(i+1,0x9e3779b9)
U_i=Mulberry32(domain_state_i); U_i in [0,1)
```

Transition / meaning: A fresh session seed creates world, loot, craft, combat and visual streams. Each domain has its own state and draw counter. All active gameplay draws are routed to their declared domain; terrain and procedural art use separate locally seeded generators.

Output / distribution role: Simulation state and input vector; deterministic conditional on input and random stream.

Trigger / clock: Per display callback or simulation tick; pause/overload policy applies.

Composition requirement: Cosmetic draw counts do not advance reward or combat streams. The profile version and seed/counters identify diagnostic context; no random seed is an authentication credential.

Boundary / exception: Session streams reset on character entry and are not restored in saves. A seed alone is not a replay: player input, iteration order and engine state must also be reproduced. Before active session setup, isolated compatibility paths may use Math.random.

Acceptance check: Generate matching stream instances; advance visual100 times and prove next world/loot/craft draws match the untouched instance. Browser rendering must preserve all nonvisual

counters.

Trace: streams; initGame; pulse; craftBatch; damageMob; chooseDropRank [S1]

Function registry: F0173, F0199, F0201, F0205, F0298, F0575

M010 | Versioned event replay

STATUS P | conditional proposed extension

Inputs / units: Clock/input state, finite frame interval and h>0; action intent bounded to the device control domain.

```
S_n = fold(F_v, S_0, events[0:n])
```

Transition / meaning: A future replay system records ordered input and reward events against a rules version v. Replaying an old event log with new balancing constants is not a valid reconstruction. Snapshots and event logs serve different recovery and audit purposes.

Output / distribution role: Simulation state and input vector; deterministic conditional on input and random stream.

Trigger / clock: Per display callback or simulation tick; pause/overload policy applies.

Composition requirement: Assign a monotone event sequence and deterministic tie-breaker; include random-stream positions from M009 and state checksums at checkpoints.

Boundary / exception: Client event logs are not trustworthy evidence of fair play without authoritative validation. Storage size and privacy retention must be bounded.

Acceptance check: Replay after restart and compare checkpoint hashes; reject missing versions, duplicated sequence numbers and reordered reward events.

Trace: Proposed event store; depends on M009

Contracts M011-M020

M011 | World-to-screen transform

STATUS I | implemented baseline

Inputs / units: Finite body positions, positive ellipse axes/radii and viewport dimensions; same world coordinate frame.

```
screen = zoom * (world - camera)
```

Transition / meaning: World positions use a 6800 by 5000 coordinate domain. The Canvas transform scales by zoom and translates by the camera. UI coordinates remain in CSS pixels, while rendering resolution separately incorporates device pixel ratio.

Output / distribution role: Contact/visibility decision, separating vector, position or pixel projection.

Trigger / clock: At movement/contact query or render call; rendering does not consume gameplay rewards.

Composition requirement: Apply camera subtraction before zoom. Inverse picking divides screen coordinates by zoom and restores the camera offset.

Boundary / exception: Confusing backing pixels with CSS pixels creates incorrect picking or apparent hitbox offsets on high-density devices.

Acceptance check: Transform known points and invert them at several zoom and DPR values; compare against a one-pixel display tolerance.

Trace: drawWorld; proj; resize [S1]

Function registry: F0234, F0238, F0271

M012 | Elliptical creature proxy

STATUS I | implemented baseline

Inputs / units: Finite body positions, positive ellipse axes/radii and viewport dimensions; same world coordinate frame.

```
(x_local/a)^2 + (y_local/b)^2 <= 1
```

Transition / meaning: Creature families have different axis ratios in mobCollider. Flying bodies are flatter, ant-like bodies longer, and armored or round bodies use other ratios. Facing rotates this authoritative ellipse; wings and thin legs are decorative appendages.

Output / distribution role: Contact/visibility decision, separating vector, position or pixel projection.

Trigger / clock: At movement/contact query or render call; rendering does not consume gameplay rewards.

Composition requirement: Use the same a, b and facing for geometric tests and body rendering. Elite scaling changes r before the family ratios are applied.

Boundary / exception: A decorative appendage is not an additional damaging hit region. Enlarging artwork without documenting this distinction misleads players.

Acceptance check: Render proxy overlays for each family at small, normal and elite radii; verify the opaque torso aligns with the ellipse.

Trace: mobCollider; drawCreatureBody [S1]

Function registry: F0210, F0247

M013 | Directional ellipse support

STATUS I | implemented baseline

Inputs / units: Finite body positions, positive ellipse axes/radii and viewport dimensions; same world coordinate frame.

```
s(n) = sqrt((a*n_x_local)^2 + (b*n_y_local)^2)
```

Transition / meaning: The support function gives the farthest projection of an ellipse onto a unit direction. It is used in separation estimates. It differs from the radius of the ellipse along a ray; substituting the latter underestimates diagonal support.

Output / distribution role: Contact/visibility decision, separating vector, position or pixel projection.

Trigger / clock: At movement/contact query or render call; rendering does not consume gameplay rewards.

Composition requirement: Rotate the world normal into the ellipse frame, normalize it, evaluate support, and combine two supports for a separation target.

Boundary / exception: This support-based center-direction response is not a complete exact ellipse-ellipse intersection solver or rigid-body impulse calculation.

Acceptance check: Test axis-aligned and 45-degree normals; compare support with maximized sampled boundary projections.

Trace: mobSupportRadius [S1]

Function registry: F0211

M014 | Exact ellipse-circle contact

STATUS I | implemented baseline

Inputs / units: Finite body positions, positive ellipse axes/radii and viewport dimensions; same world coordinate frame.

```
sum((axis_i*p_i)/(axis_i^2+t))^2 = 1
```

Transition / meaning: For an exterior point p in ellipse coordinates, a nonnegative Lagrange multiplier t determines the closest ellipse point. Eighteen bisection iterations approximate the root. Contact is accepted when the resulting distance is smaller than the circle radius.

Output / distribution role: Contact/visibility decision, separating vector, position or pixel projection.

Trigger / clock: At movement/contact query or render call; rendering does not consume gameplay rewards.

Composition requirement: An enclosing-radius rejection runs first, followed by an interior test, then root solving only for surviving candidates.

Boundary / exception: Tangency uses strict comparisons and finite precision. Very thin or degenerate axes would require additional guards outside current creature definitions.

Acceptance check: Compare against dense boundary sampling for grazing, interior, rotated and diagonal cases, including points just outside contact.

Trace: mobCircleOverlap [S1]

Function registry: F0212

M015 | Conservative broad-phase rejection

STATUS I | implemented baseline

Inputs / units: Finite body positions, positive ellipse axes/radii and viewport dimensions; same world coordinate frame.

```
|delta|^2 >= (R_a+R_b+margin)^2 => reject
```

Transition / meaning: A cheap squared-distance bound eliminates distant pairs before more expensive directional geometry. Conservativeness is the contract: false positives cost time; false negatives lose real contacts and change combat.

Output / distribution role: Contact/visibility decision, separating vector, position or pixel projection.

Trigger / clock: At movement/contact query or render call; rendering does not consume gameplay rewards.

Composition requirement: The enclosing radii must dominate all authoritative body axes. This filter precedes support calculations and separation passes.

Boundary / exception: This optimization does not remove the quadratic enumeration of all monster pairs in the present simulation.

Acceptance check: For every narrow-phase hit in randomized fixtures, require the broad phase to retain the pair; count candidates independently.

Trace: update collision loops; [R3]

Function registry: F0222, F0291

M016 | Static obstacle hash grid

STATUS I | implemented baseline

Inputs / units: Finite body positions, positive ellipse axes/radii and viewport dimensions; same world coordinate frame.

```
cell(x,y) = (floor(x/g), floor(y/g))
```

Transition / meaning: Rocks are assigned to a static spatial lookup during map construction. Movement queries inspect nearby cells rather than every rock. Grid size g is a spatial indexing parameter, not a gameplay unit conversion.

Output / distribution role: Contact/visibility decision, separating vector, position or pixel projection.

Trigger / clock: At movement/contact query or render call; rendering does not consume gameplay rewards.

Composition requirement: The current lookup scans a fixed 3 by 3 neighborhood at GRID_SIZE=150 and then runs obstacle resolution on retrieved candidates. It does not expand according to body radius.

Boundary / exception: Completeness requires interaction reach to fit the scanned neighborhood. Radius-adaptive queries are a future safeguard for unusually large bodies; map changes require index maintenance.

Acceptance check: Compare query results with a full rock scan at cell corners, for soil-enlarged players and maximum-sized creatures.

Trace: ROCK_GRID; nearbyRocks; buildMap [S1]

Function registry: F0177, F0193

M017 | Collision-aware sliding

STATUS I | implemented baseline

Inputs / units: Finite body positions, positive ellipse axes/radii and viewport dimensions; same world coordinate frame.

```
x_new = resolve(x_old + v*h, obstacles)
```

Transition / meaning: slideCircle advances circular movement proxies while resolving penetration against rocks and world limits. The practical result is sliding around obstacles rather than passing through them. Enemy rock proxies conservatively use the maximum ellipse axis.

Output / distribution role: Contact/visibility decision, separating vector, position or pixel projection.

Trigger / clock: At movement/contact query or render call; rendering does not consume gameplay rewards.

Composition requirement: Movement, dash and collision relaxation all share this resolver. Their order can alter the final separation around crowded obstacles.

Boundary / exception: This is not continuous collision detection; a very large displacement can tunnel through geometry without substepping or a swept test.

Acceptance check: Sweep representative speeds into corners and narrow gaps; verify no persistent overlap and measure maximum safe displacement.

Trace: slideCircle; mobRockRadius [S1]

Function registry: F0194, F0213

M018 | Pairwise crowd relaxation

STATUS I | implemented baseline

Inputs / units: Finite body positions, positive ellipse axes/radii and viewport dimensions; same world coordinate frame.

```
delta_a = -n*penetration/2; delta_b = +n*penetration/2
```

Transition / meaning: Monsters are treated as equally weighted solid bodies for symmetric positional correction. Two additional relaxation passes reduce residual overlap after chained contacts. This is an iterative position constraint, not momentum-conserving physical dynamics.

Output / distribution role: Contact/visibility decision, separating vector, position or pixel projection.

Trigger / clock: At movement/contact query or render call; rendering does not consume gameplay rewards.

Composition requirement: Rock resolution and player separation occur around crowd passes, so constraints can reintroduce errors into one another.

Boundary / exception: Finite iterations leave residual penetration in dense crowds; zero-distance pairs use a fixed fallback normal that can introduce directional bias.

Acceptance check: Test coincident bodies and packed rings, report residual penetration after each pass, and verify finite coordinates.

Trace: update crowd loops [S1]

Function registry: F0222, F0291

M019 | Dynamic monster spatial partition

STATUS P | conditional proposed extension

Inputs / units: Finite body positions, positive ellipse axes/radii and viewport dimensions; same world coordinate frame.

```
candidate work ~ O(N + K), worst case O(N^2)
```

Transition / meaning: A dynamic grid can replace all-pair enumeration when measured crowd cost warrants it. Insert enclosing monster bounds into covered cells and deduplicate candidate pairs. K is the number of candidate contacts, not the number of true collisions.

Output / distribution role: Contact/visibility decision, separating vector, position or pixel projection.

Trigger / clock: At movement/contact query or render call; rendering does not consume gameplay rewards.

Composition requirement: Use the same conservative bounds and narrow phase as M015. Rebuild after movement or maintain a tested incremental index.

Boundary / exception: Uniformly dense or huge objects can still yield quadratic candidates. Do not claim linear worst-case complexity or apply GPU benchmark speedups to JavaScript.

Acceptance check: Compare pair sets against brute force and benchmark uniform, clustered and boundary-heavy maps before enabling the new path.

Trace: Proposed dynamic grid; [R3]

M020 | Projected size and detail threshold

STATUS I | implemented baseline

Inputs / units: Finite body positions, positive ellipse axes/radii and viewport dimensions; same world coordinate frame.

```
q = r*zoom; detail = (quality != low) and (q >= 17)
```

Transition / meaning: The V5.14 baseline renderer enables fine anatomical marks only above a projected body size threshold. This avoids spending draw calls on details smaller than a useful display footprint. The body proxy remains unchanged at every level of detail.

Output / distribution role: Contact/visibility decision, separating vector, position or pixel projection.

Trigger / clock: At movement/contact query or render call; rendering does not consume gameplay rewards.

Composition requirement: Elite and guardian sizes affect projected size automatically; DPR increases sampling resolution but does not change CSS-size eligibility.

Boundary / exception: A hard threshold may cause visible detail switching near the boundary. Hysteresis is a possible future refinement, not present behavior.

Acceptance check: Inspect the same species at several zoom levels and sizes; assert identical collision geometry with detail enabled and disabled.

Trace: creatureVisualProfile [S1]

Function registry: F0252

Contracts M021-M030

M021 | Maximum health composition

STATUS I | implemented baseline

Inputs / units: Nonnegative current/max health and durability, target armor, level/rank and eligible contact state.

```
H_max = round((100+7*(L-1)+sum(24*S_r*cactus+42*S_r*soil))*(1+0.06t))
```

Transition / meaning: Base health grows linearly with level. Cactus and soil contribute rarity-scaled additions inside the vitality multiplier. Inventory items do not contribute unless equipped; the inactive backup deck is excluded.

Output / distribution role: Requested, mitigated and effective HP changes, durability/reload transition and derived damage rate.

Trigger / clock: Per eligible contact or continuous tick; exact batching and gates apply.

Composition requirement: Add item health first, multiply vitality once, then round the final total. Spending vitality also adjusts current health by the capacity increase.

Boundary / exception: Changing decks can reduce capacity and clamp current health. This is capacity management, not free proportional healing.

Acceptance check: Compare empty, cactus-heavy and mixed decks at known levels; verify inventory and backup copies contribute zero.

Trace: maxHP; spendTalent; equipInto [S1]

Function registry: F0019, F0131, F0276

M022 | Base outgoing damage stack

STATUS I | implemented baseline

Inputs / units: Nonnegative current/max health and durability, target armor, level/rank and eligible contact state.

```
D_raw = D_type*S_r*power*durability_ratio*assault*overdrive
```

Transition / meaning: Rarity scales base damage exponentially with S_r=1.72^r. Power is a talent multiplier. Ordinary petals lose damage as durability falls; glass omits that ratio. Assault and overdrive multiply after these factors.

Output / distribution role: Requested, mitigated and effective HP changes, durability/reload transition and derived damage rate.

Trigger / clock: Per eligible contact or continuous tick; exact batching and gates apply.

Composition requirement: The monster armor stage receives this raw value. Damage feedback totals must distinguish raw requested damage from effective health reduction.

Boundary / exception: Floating feedback currently accumulates requested damage in some paths, so it can exceed effective post-armor damage or overkill-adjusted loss.

Acceptance check: Use a known-health target with and without armor; compare requested, mitigated and effective health loss separately.

Trace: update petal exchange; damageMob [S1]

Function registry: F0201, F0222, F0291

M023 | Flat monster armor and damage floor

STATUS I | implemented baseline

Inputs / units: Nonnegative current/max health and durability, target armor, level/rank and eligible contact state.

```
D_applied = max(1, D_raw - armor)
```

Transition / meaning: Armor combines an active shield amount with a capped rank/tier contribution. The one-point floor prevents complete immunity from flat mitigation. Multiple small hits therefore interact with armor differently from one large hit of the same total raw value.

Output / distribution role: Requested, mitigated and effective HP changes, durability/reload transition and derived damage rate.

Trigger / clock: Per eligible contact or continuous tick; exact batching and gates apply.

Composition requirement: Apply armor per exchange strike and per chained damage call, not once to the sum of a ten-strike batch.

Boundary / exception: A damage floor creates discontinuities in marginal scaling. Requested zero damage can still become one point in this function.

Acceptance check: Check values below, at and above armor; verify batched small-hit damage is not silently replaced by a single aggregate hit.

Trace: damageMob [S1]

Function registry: F0201

M024 | Ten-strike durability exchange

STATUS I | implemented baseline

Inputs / units: Nonnegative current/max health and durability, target armor, level/rank and eligible contact state.

```
strikes = 10 if ordinary and H_petal > D_mob; else 1
```

Transition / meaning: An overlapping ordinary petal may perform up to ten exchange iterations in a single fixed update. The loop stops when the petal breaks or the monster dies. This rule is evaluated every tick; it is not a ten-hits-per-second cap.

Output / distribution role: Requested, mitigated and effective HP changes, durability/reload transition and derived damage rate.

Trigger / clock: Per eligible contact or continuous tick; exact batching and gates apply.

Composition requirement: Durability declines between strikes, so later raw damage in the same batch is weaker. Armor is also evaluated separately for each strike.

Boundary / exception: The theoretical loop ceiling is 600 strikes per second per sustained pair before durability and death constraints. Actual contact uptime is much smaller and state-dependent.

Acceptance check: Construct high-durability targets and log strikes per tick, durability decline and death truncation. Benchmark before changing this balance-sensitive rule.

Trace: update petal exchange [S1]

Function registry: F0222, F0291

M025 | Glass contact throttling

STATUS I | implemented baseline

Inputs / units: Nonnegative current/max health and durability, target armor, level/rank and eligible contact state.

```
N_hits(T) <= 1 + floor(T/1 s) for one sustained pair
```

Transition / meaning: Glass does not lose collision durability in ordinary exchanges and uses a per-target hit timestamp. A one-second interval restricts repeated damage to the same target. It trades burst exchange behavior for predictable sustained contact.

Output / distribution role: Requested, mitigated and effective HP changes, durability/reload transition and derived damage rate.

Trigger / clock: Per eligible contact or continuous tick; exact batching and gates apply.

Composition requirement: Each petal-target pair has separate timing. Multiple glass petals can hit the same monster independently; one petal can hit different monsters.

Boundary / exception: The bound depends on interval endpoint conventions. Glass immunity is path-specific and must not be generalized to every hypothetical future damage source.

Acceptance check: Hold one glass petal on one target for several seconds; compare timestamps, then repeat with two petals and two targets.

Trace: update; lastHits [S1]

Function registry: F0222, F0291

M026 | Ordinary single-exchange throttle

STATUS I | implemented baseline

Inputs / units: Nonnegative current/max health and durability, target armor, level/rank and eligible contact state.

```
tau_pair = 0.24 s when multi-exchange is unavailable
```

Transition / meaning: When a normal petal cannot use the ten-strike condition, the single-exchange path consults a per-target interval of 0.24 seconds. This produces a discontinuous behavior change as durability crosses the enemy-damage threshold.

Output / distribution role: Requested, mitigated and effective HP changes, durability/reload transition and derived damage rate.

Trigger / clock: Per eligible contact or continuous tick; exact batching and gates apply.

Composition requirement: This timer is distinct from global body damage immunity, petal reload time and healing cooldown. They must not be merged into one cooldown variable.

Boundary / exception: The multi-exchange path does not apply this interval. A DPS estimate based only on 1/0.24 is therefore wrong for durable petals.

Acceptance check: Test durability just above and just below monster damage and observe the branch change at a fixed contact geometry.

Trace: update petal exchange [S1]

Function registry: F0222, F0291

M027 | Body-contact damage clocks

STATUS I | implemented baseline

Inputs / units: Nonnegative current/max health and durability, target armor, level/rank and eligible contact state.

```
tau_player > 0.39 s; tau_mob > 0.56 s
```

Transition / meaning: Player contact damage uses a player-wide lastBody stamp, while body damage dealt to enemies uses a stamp on each enemy. The asymmetry affects swarming: many bodies do not necessarily cause all of their collision damage simultaneously.

Output / distribution role: Requested, mitigated and effective HP changes, durability/reload transition and derived damage rate.

Trigger / clock: Per eligible contact or continuous tick; exact batching and gates apply.

Composition requirement: Body overlap first triggers positional separation, then clock checks. Enemy contact damage can additionally include enrage and bloodmoon multipliers.

Boundary / exception: These clocks are not invulnerability guarantees for every damage path. Projectiles and specials must be inspected separately.

Acceptance check: Test two simultaneous enemy contacts and a projectile at the same tick; record which clocks gate each event.

Trace: update body collision; damagePlayer [S1]

Function registry: F0197, F0222, F0291

M028 | Reload downtime

STATUS I | implemented baseline

Inputs / units: Nonnegative current/max health and durability, target armor, level/rank and eligible contact state.

```
tau_reload = cd_type*(1-min(0.35, 0.05*t_reload))
```

Transition / meaning: A broken petal loses participation until its reload timer expires, then returns at full rarity-scaled durability. The talent cap currently limits rank to five, so the formula-level 35 percent clamp is not reached through normal talent allocation.

Output / distribution role: Requested, mitigated and effective HP changes, durability/reload transition and derived damage rate.

Trigger / clock: Per eligible contact or continuous tick; exact batching and gates apply.

Composition requirement: Cooldown advancement occurs at fixed h. Re-equipping and deck switching may impose separate reload delays and should not be confused with this break timer.

Boundary / exception: A nominal damage increase can be offset by increased break frequency. Uptime, not tooltip damage alone, determines sustained contribution.

Acceptance check: Break a known petal at each legal reload talent rank and compare observed simulation-time restoration.

Trace: petalReloadTime; resetEquip; update [S1]

Function registry: F0018, F0167, F0222, F0291

M029 | Defensive durability recovery

STATUS I | implemented baseline

Inputs / units: Nonnegative current/max health and durability, target armor, level/rank and eligible contact state.

```
dH_e/dt = 0.06*H_e_max after both 1.2 s quiet gates
```

Transition / meaning: An unbroken equipped petal repairs in defense only after the player and that petal have each avoided recent damage for 1.2 seconds. Repair is proportional to maximum durability, not missing durability.

Output / distribution role: Requested, mitigated and effective HP changes, durability/reload transition and derived damage rate.

Trigger / clock: Per eligible contact or continuous tick; exact batching and gates apply.

Composition requirement: Player health healing can occur concurrently, but a new hit independently resets the relevant quiet condition. Broken petals still reload rather than repair.

Boundary / exception: The rule is linear until clamped at full durability. It is not exponential recovery and cannot revive a broken petal early.

Acceptance check: Apply player-only and petal-only damage around the threshold; compare repair start times and full restoration duration.

Trace: recoverPetalDurability [S1]

Function registry: F0145

M030 | Renewal-reward sustained DPS

STATUS D | derived analysis

Inputs / units: Nonnegative current/max health and durability, target armor, level/rank and eligible contact state.

```
DPS = E[damage per cycle] / E[active time + reload time]
```

Transition / meaning: Treat one petal life and reload as a renewal cycle only when the encounter statistics are sufficiently stationary. This model explains why high burst damage with frequent breakage can lose to a weaker but durable loadout.

Output / distribution role: Requested, mitigated and effective HP changes, durability/reload transition and derived damage rate.

Trigger / clock: Per eligible contact or continuous tick; exact batching and gates apply.

Composition requirement: Include contact probability, per-strike mitigation, healing, durability loss and death truncation inside cycle reward rather than multiplying unrelated tooltip averages.

Boundary / exception: Nonstationary bosses, player adaptation and shared target deaths violate independent-cycle assumptions; use event simulation there.

Acceptance check: Compare analytical cycle estimates with long-run event logs under a controlled repeating encounter and report confidence intervals.

Trace: Derived from M022-M029

Contracts M031-M040

M031 | Orbit geometry and reversal

STATUS I | implemented baseline

Inputs / units: Orbital phase, equipped petal instances, skill/stance, timers, target geometry and projectile state.

```
theta_i = theta + 2*pi*i/n; sign(omega) in {-1,+1}
```

Transition / meaning: Equipped petals occupy evenly spaced angular phases around the player. Reversing orbit changes angular velocity sign while preserving the current angle, allowing a surviving petal to approach a target from the other direction.

Output / distribution role: Petal positions, healing, status/charge state, projectile or interception outcomes.

Trigger / clock: Per simulation tick or effect trigger; ownership and expiry are entry-specific.

Composition requirement: Stance changes radius, reach talents scale it and wing petals add a periodic offset. Those effects do not change equal base phase spacing.

Boundary / exception: Interpolation of petal positions causes transient lag after radius or direction changes. Reversal is not an instantaneous teleport to a new contact point.

Acceptance check: Reverse during sustained motion and compare phase continuity, angular sign and collision timing without changing inventory.

Trace: orbitRadius; reverseOrbit; update [S1]

Function registry: F0144, F0146, F0222, F0291

M032 | Stance radius and wing excursion

STATUS I | implemented baseline

Inputs / units: Orbital phase, equipped petal instances, skill/stance, timers, target geometry and projectile state.

```
R = (R_stance + I_wing*(42+35*sin(5t+0.7i+pi/6)))*reach
```

Transition / meaning: The base radii are 135 for attack, 51 for defense and 92 for normal stance. Wing petals oscillate around an additional positive offset. Reach multiplies the complete radius, including that oscillation.

Output / distribution role: Petal positions, healing, status/charge state, projectile or interception outcomes.

Trigger / clock: Per simulation tick or effect trigger; ownership and expiry are entry-specific.

Composition requirement: Radius affects exposure, target access and projectile interception. It composes with player movement but is not itself a movement-speed bonus.

Boundary / exception: The larger attack ring may miss a body already very close to the player. Greater reach is not universally greater damage.

Acceptance check: Plot minimum and maximum wing radii for each stance and legal reach rank; check actual contact against target size.

Trace: orbitRadius [S1]

Function registry: F0146

M033 | Petal position response

STATUS I | implemented baseline

Inputs / units: Orbital phase, equipped petal instances, skill/stance, timers, target geometry and projectile state.

```
x_next = lerp(x, orbit_target, 1-.75^(60*dt))
```

Transition / meaning: Petal centers pursue orbit coordinates using a time-composable alpha whose 60Hz gain is.25. Repeated steady-target updates decay the positional error geometrically.

Output / distribution role: Petal positions, healing, status/charge state, projectile or interception outcomes.

Trigger / clock: Per simulation tick or effect trigger; ownership and expiry are entry-specific.

Composition requirement: Collision uses the resulting petal position, so this smoothing is gameplay-relevant, unlike cosmetic wing motion in the renderer.

Boundary / exception: Moving orbit targets and contact/rock interactions keep their simulation ordering; only constant-target smoothing has exact subdivision equivalence.

Acceptance check: Measure lag after a radius step and compare the geometric decay of error over fixed updates.

Trace: site/index.html:update; site/rules.js:frameAlpha; M128

Function registry: F0222, F0291, F0611

M034 | Continuous leaf and threshold star healing

STATUS I | implemented baseline

Inputs / units: Orbital phase, equipped petal instances, skill/stance, timers, target geometry and projectile state.

```
leaf: 1.9*S_r HP/s; star: 3.1*S_r HP/s below 0.75*H_max
```

Transition / meaning: Each active leaf contributes continuous healing. Star petals contribute only while player health is below three quarters of capacity. Contributions are processed per petal and clamped to maximum health.

Output / distribution role: Petal positions, healing, status/charge state, projectile or interception outcomes.

Trigger / clock: Per simulation tick or effect trigger; ownership and expiry are entry-specific.

Composition requirement: Separate equipped copies add their rates. Threshold checks occur sequentially, so one healer can switch off another threshold-based contribution in the same tick.

Boundary / exception: Healing estimates must include petal break downtime. Full-health or above-threshold time yields no useful health gain.

Acceptance check: Integrate health over known durations below and above the threshold, including multiple healers and one broken copy.

Trace: update healing loop [S1]

Function registry: F0222, F0272, F0291

M035 | Rose and nectar trigger healing

STATUS I | implemented baseline

Inputs / units: Orbital phase, equipped petal instances, skill/stance, timers, target geometry and projectile state.

```
rose: round(19*S_r), 6 s; nectar: round(35*S_r), 11 s
```

Transition / meaning: Rose triggers below 97 percent health, nectar below 50 percent, when their independent healing timers are ready. A proc heals the smaller of its nominal value and missing health. Timers are separate from durability reload timers.

Output / distribution role: Petal positions, healing, status/charge state, projectile or interception outcomes.

Trigger / clock: Per simulation tick or effect trigger; ownership and expiry are entry-specific.

Composition requirement: Multiple copies maintain independent timers. Sequential processing and capacity clamping can waste later procs when the first one fills the player.

Boundary / exception: Tooltip nominal healing divided by cooldown overstates effective healing when thresholds are rarely crossed or procs overheal.

Acceptance check: Schedule controlled damage around both thresholds and measure useful healing, overheal and timer reset events.

Trace: update healing loop [S1]

Function registry: F0222, F0272, F0291

M036 | Root armor charge inventory

STATUS I | implemented baseline

Inputs / units: Orbital phase, equipped petal instances, skill/stance, timers, target geometry and projectile state.

```
A_layers <= 10; one layer added per 2 s while root active
```

Transition / meaning: Active root petals enable a shared armor-layer accumulator. The layer value derives from the sum of active root rarity scales. Taking player damage consumes a layer through the existing mitigation path; losing all active roots clears stored layers.

Output / distribution role: Petal positions, healing, status/charge state, projectile or interception outcomes.

Trigger / clock: Per simulation tick or effect trigger; ownership and expiry are entry-specific.

Composition requirement: Root copies increase mitigation strength through rootedArmor, not the number of simultaneous independent ten-layer pools.

Boundary / exception: Burst damage timing and charge availability matter. Averaging armor as a constant reduction ignores consumption and empty-pool periods.

Acceptance check: Charge with one and multiple roots, apply repeated hits, then break the last root and inspect layer count and mitigation.

Trace: rootActive; rootedArmor; damagePlayer [S1]

Function registry: F0137, F0140, F0197

M037 | Poison refresh rather than independent stacks

STATUS I | implemented baseline

Inputs / units: Orbital phase, equipped petal instances, skill/stance, timers, target geometry and projectile state.

```
poison_rate = 5*S_r; poison_time = 2.4 s on a qualifying hit
```

Transition / meaning: Poison and iris assign a damage rate and refresh duration on a monster. They do not create a list of independently ticking poison instances. Later writes can replace a stronger rate with a weaker one, making update order relevant.

Output / distribution role: Petal positions, healing, status/charge state, projectile or interception outcomes.

Trigger / clock: Per simulation tick or effect trigger; ownership and expiry are entry-specific.

Composition requirement: Direct strike damage still occurs. Poison ticks reduce health separately; completion invokes the ordinary death pipeline when health reaches zero.

Boundary / exception: The last writer is not necessarily the strongest source. Proposed max-potency or additive poison semantics would be a balance change, not a documentation correction.

Acceptance check: Alternate low- and high-rarity poison hits and inspect stored rate, duration and total damage.

Trace: update exchange; poison tick [S1]

Function registry: F0222, F0291

M038 | Chain lightning and crowd dependence

STATUS I | implemented baseline

Inputs / units: Orbital phase, equipped petal instances, skill/stance, timers, target geometry and projectile state.

```
D_chain = 0.20*requested_batch_damage per nearby secondary target
```

Transition / meaning: A lightning contact iterates over other living enemies within 125 world units and calls their damage path. Dense encounters can multiply total contribution. The quantity passed is derived from requested exchange damage, not necessarily the primary target health actually lost.

Output / distribution role: Petal positions, healing, status/charge state, projectile or interception outcomes.

Trigger / clock: Per simulation tick or effect trigger; ownership and expiry are entry-specific.

Composition requirement: Armor and floors are applied again on each secondary target. Multiple lightning petals repeat the process independently.

Boundary / exception: Without a target cap, total chain output grows with local crowd count. Overkill and requested-versus-effective damage must be distinguished in balance metrics.

Acceptance check: Place targets at 124 and 126 units and vary armor and primary remaining health; log each applied secondary hit.

Trace: update lightning branch [S1]

Function registry: F0222, F0291

M039 | Homing choice and grape bounce budget

STATUS I | implemented baseline

Inputs / units: Orbital phase, equipped petal instances, skill/stance, timers, target geometry and projectile state.

```
missile range 570; grape range 480; grape extra bounces <= 2
```

Transition / meaning: Ranged petals choose a nearest eligible enemy at emission. Projectile travel range is finite; grapes can retarget previously unhit enemies after contact while an explicit bounce budget remains. A hit-ID list prevents revisiting the same enemy.

Output / distribution role: Petal positions, healing, status/charge state, projectile or interception outcomes.

Trigger / clock: Per simulation tick or effect trigger; ownership and expiry are entry-specific.

Composition requirement: Power and rarity affect projectile damage; artillery can shorten shot interval. Obstacles and range expiration can terminate shots before a hit.

Boundary / exception: The current friendly-shot contact uses a circle-distance proxy rather than the exact ellipse-circle path used by melee petals.

Acceptance check: Check three-target chains, duplicate IDs, obstacle interruption, range expiry and equidistant target tie behavior.

Trace: update friendlyShots and shotWait [S1]

Function registry: F0222, F0291

M040 | Interception coverage as a geometry process

STATUS D | derived analysis

Inputs / units: Orbital phase, equipped petal instances, skill/stance, timers, target geometry and projectile state.

```
P(block) = measure(covered approach phases)/(2*pi)
```

Transition / meaning: Petals can intercept hostile projectiles before body contact. Under uniformly distributed approach angle and a static ideal ring, angular coverage estimates block probability. Real moving petals and nonuniform enemy aim invalidate a simple independent-angle assumption.

Output / distribution role: Petal positions, healing, status/charge state, projectile or interception outcomes.

Trigger / clock: Per simulation tick or effect trigger; ownership and expiry are entry-specific.

Composition requirement: Defense radius, number of intact petals, projectile radius and broken-slot uptime jointly change coverage. Gaps correlate over time.

Boundary / exception: Adding individual arc probabilities can exceed one; use the union of covered arcs, not an unbounded sum.

Acceptance check: Compute arc unions for a static fixture, then compare with simulated uniformly sampled approach angles and moving-ring experiments.

Trace: Derived from projectile and orbit geometry

Contracts M041-M050

M041 | Experience required per level

STATUS I | implemented baseline

Inputs / units: Integer progression state, owned equipment, talent allocation, stance/skill readiness and boon state.

```
XP_need(L) = 35 + floor(18L + 0.9L^2)
```

Transition / meaning: The next-level threshold grows approximately quadratically in level. Reward processing repeatedly subtracts thresholds, allowing several levels from one large reward. The integer floor is applied to the combined polynomial term.

Output / distribution role: Level/capacity/modifier, deck/action transition or marginal build utility.

Trigger / clock: On progression, equipment/action decision or timer update.

Composition requirement: Kill reward, combo bonus and event bonus determine experience inflow; the threshold determines how that inflow translates into progression.

Boundary / exception: A quadratic next-level requirement implies roughly cubic cumulative experience at high levels, not exponential progression. Imported levels are separately bounded.

Acceptance check: Sum thresholds and compare multi-level reward processing with sequential level-ups; test equality at exact thresholds.

Trace: xpNeed; damageMob [S1]

Function registry: F0201, F0203

M042 | Cumulative progression cost

STATUS D | derived analysis

Inputs / units: Integer progression state, owned equipment, talent allocation, stance/skill readiness and boon state.

```
XP_total(1->L) = sum(j=1..L-1) XP_need(j)
```

Transition / meaning: Cumulative cost is a discrete sum, not the threshold at the destination level. Ignoring floors gives a useful cubic approximation using standard sums of integers and squares. Exact progression calculators should retain the floors.

Output / distribution role: Level/capacity/modifier, deck/action transition or marginal build utility.

Trigger / clock: On progression, equipment/action decision or timer update.

Composition requirement: Divide this cost by a policy-specific expected experience rate only under a stable farming regime; route changes and deaths alter that rate.

Boundary / exception: The approximation error from each floor is less than one, so cumulative error is bounded by the number of summed thresholds.

Acceptance check: Compare the closed polynomial approximation with exact summation at levels 5, 12, 20, 30 and 100.

Trace: Derived from M041

M043 | Discrete equipment capacity

STATUS I | implemented baseline

Inputs / units: Integer progression state, owned equipment, talent allocation, stance/skill readiness and boon state.

```
n(L) = 6,7,8,9,10 at L >= 1,5,12,20,30
```

Transition / meaning: Five milestones unlock equipment slots for both active and backup decks. Capacity is a step function; a small experience gain near a milestone can matter more tactically than the same gain elsewhere.

Output / distribution role: Level/capacity/modifier, deck/action transition or marginal build utility.

Trigger / clock: On progression, equipment/action decision or timer update.

Composition requirement: New slots initialize with basic common petals. More slots change phase spacing, possible synergies, health contributions and interception coverage together.

Boundary / exception: Slot count is capped at ten; a linear extrapolation beyond level thirty would fabricate equipment that does not exist.

Acceptance check: Check just below and at every milestone, including both decks and imported saves with insufficient slot arrays.

Trace: SLOT_MILESTONES; ensureSlots [S1]

Function registry: F0169

M044 | Talent budget and allocation cap

STATUS I | implemented baseline

Inputs / units: Integer progression state, owned equipment, talent allocation, stance/skill readiness and boon state.

```
B(L) = floor((L-1)/2); 0 <= t_i <= 5; sum(t_i) <= B
```

Transition / meaning: Six talent categories share a level-derived budget. Sanitization floors, clamps and allocates available budget in a fixed category order, preventing imported values from exceeding capacity. A reset returns points without creating new ones.

Output / distribution role: Level/capacity/modifier, deck/action transition or marginal build utility.

Trigger / clock: On progression, equipment/action decision or timer update.

Composition requirement: Within one talent category, ranks add linearly to that category multiplier; separate categories enter their own downstream formulas.

Boundary / exception: When an invalid save overspends, sanitization order affects which categories retain points. This is a recovery policy, not an optimization algorithm.

Acceptance check: Test negative, fractional, enormous and overspent allocations; verify budget conservation and deterministic repaired output.

Trace: talentEarned; sanitizeTalents [S1]

Function registry: F0010, F0011

M045 | Set-family threshold synergies

STATUS I | implemented baseline

Inputs / units: Integer progression state, owned equipment, talent allocation, stance/skill readiness and boon state.

```
s_family = 0 if count<2; 1 if 2<=count<4; 2 if count>=4
```

Transition / meaning: Set families count equipped copies belonging to named type lists. Their two threshold tiers are discontinuous: the second or fourth qualifying copy can unlock a larger marginal gain than the third.

Output / distribution role: Level/capacity/modifier, deck/action transition or marginal build utility.

Trigger / clock: On progression, equipment/action decision or timer update.

Composition requirement: An active deck can trigger several families if its type composition supports them. Copies count; distinct species are not required by the current function.

Boundary / exception: The family count does not create an unbounded linear bonus after four copies. A fifth qualifying petal may add only its own direct contribution.

Acceptance check: Enumerate counts zero through ten per family and check tier boundaries, backup exclusion and mixed-family decks.

Trace: SET_FAMILIES; computeSynergies [S1]

Function registry: F0149

M046 | Deck exchange opportunity cost

STATUS I | implemented baseline

Inputs / units: Integer progression state, owned equipment, talent allocation, stance/skill readiness and boon state.

```
tau_switch = 2.6 s; initial reload_i = max(2.5 s, reload_i)
```

Transition / meaning: Switching swaps active and backup arrays, flips the deck label and imposes cooldown plus reload downtime. Current health is clamped to the new maximum. The switch is therefore a tactical commitment rather than a free instantaneous counter.

Output / distribution role: Level/capacity/modifier, deck/action transition or marginal build utility.

Trigger / clock: On progression, equipment/action decision or timer update.

Composition requirement: Slot effects, health capacity, zoom, synergies and petal state all update from the incoming deck; the outgoing inventory is not duplicated.

Boundary / exception: Immediate repeated switching is blocked. Returning to a deck does not preserve old in-fight petal durability as a hidden second health pool.

Acceptance check: Check inventory conservation across repeated swaps and verify all incoming petals remain inactive until their reload expires.

Trace: swapLoadout; resetEquip [S1]

Function registry: F0167, F0196

M047 | Dash displacement and immunity window

STATUS I | implemented baseline

Inputs / units: Integer progression state, owned equipment, talent allocation, stance/skill readiness and boon state.

```
distance_nominal = 920*0.20 = 184; cooldown = 5.2 s; iFrames = 0.27 s
```

Transition / meaning: Dash normalizes the current movement direction, or falls back to the orbit angle when nearly stationary. Displacement is collision-aware. The immunity timer slightly exceeds nominal dash movement duration.

Output / distribution role: Level/capacity/modifier, deck/action transition or marginal build utility.

Trigger / clock: On progression, equipment/action decision or timer update.

Composition requirement: Ordinary movement factors do not multiply the hardcoded dash speed. Obstacle resolution can reduce actual distance below the nominal value.

Boundary / exception: Immunity applies through damagePlayer and should not be assumed to cover future direct health mutations. The last frame can overshoot duration by a fixed-step fraction.

Acceptance check: Measure open-space and blocked distance, then apply damage immediately before and after the immunity timer expires.

Trace: performDash; update; damagePlayer [S1]

Function registry: F0195, F0197, F0222, F0291

M048 | Temporary shrine boon gates

STATUS I | implemented baseline

Inputs / units: Integer progression state, owned equipment, talent allocation, stance/skill readiness and boon state.

```
ready_at = t + 100 s; swift and ward duration = 18 s
```

Transition / meaning: Shrines grant location-based temporary benefits with independent cooldowns. Swift modifies speed and ward modifies incoming damage; bloom uses an immediate restoration effect rather than a duration-based multiplier.

Output / distribution role: Level/capacity/modifier, deck/action transition or marginal build utility.

Trigger / clock: On progression, equipment/action decision or timer update.

Composition requirement: The active boon is one game-level state, so acquiring another boon is not an automatically stacking list of every visited shrine.

Boundary / exception: Cooldowns use simulation time and therefore pause with the game. Map reconstruction and session reload do not constitute persistent shrine progression.

Acceptance check: Visit multiple shrine types before expiration, verify replacement semantics, and compare pause time with simulation cooldown time.

Trace: SHRINE_TYPES; updateShrines; activeBoon [S1]

Function registry: F0023, F0024

M049 | Soil health-size-speed tradeoff

STATUS I | implemented baseline

Inputs / units: Integer progression state, owned equipment, talent allocation, stance/skill readiness and boon state.

```
radius = 24+7n_soil; speed_factor = max(0.55, 0.89^n_soil)
```

Transition / meaning: Soil adds rarity-scaled health but enlarges the player body and slows ordinary movement. Body radius and speed penalty depend on copy count, not rarity. This creates a real survival tradeoff because a larger body is easier to hit and harder to route through gaps.

Output / distribution role: Level/capacity/modifier, deck/action transition or marginal build utility.

Trigger / clock: On progression, equipment/action decision or timer update.

Composition requirement: Health additions enter M021, speed enters M006, and body radius enters contact and obstacle geometry. These effects must be evaluated together.

Boundary / exception: Increasing rarity improves the health side without increasing count-based penalties. A health-only score hides exposure and maneuverability costs.

Acceptance check: Compare paths, projectile hit probability and survival for equal-health builds achieved with different soil counts.

Trace: maxHP; bodyRadius; bodySpeedFactor [S1]

Function registry: F0131, F0135, F0136

M050 | Marginal build contribution

STATUS D | derived analysis

Inputs / units: Integer progression state, owned equipment, talent allocation, stance/skill readiness and boon state.

```
Delta_i = J(build + item_i) - J(build)
```

Transition / meaning: A petal has no universal scalar value independent of its build. Marginal contribution can include a synergy threshold, extra interception, health, range or a healing trigger

interaction. Evaluate changes against a defined tactical objective J.

Output / distribution role: Level/capacity/modifier, deck/action transition or marginal build utility.

Trigger / clock: On progression, equipment/action decision or timer update.

Composition requirement: Compute the full build before and after the replacement, returning the displaced item to inventory. Include capacity and reload opportunity costs.

Boundary / exception: Marginals need not add: two items may be complements or substitutes. Optimizing independent item scores can miss an entire family breakpoint.

Acceptance check: Compare greedy item ranking with exhaustive search on a small inventory where the optimal combination is known.

Trace: Derived from equipment and synergy rules

Contracts M051-M060

M051 | Radial danger bands

STATUS I | implemented baseline

Inputs / units: Region/biome, danger band, level/rank, spawn geometry, population and event clocks.

```
rho = hypot((x-cx)/cx,(y-cy)/cy)/sqrt(2)
```

Transition / meaning: Six normalized thresholds at 0.15, 0.27, 0.39, 0.52, 0.65 and 0.78 partition the rectangular world into seven danger bands. The normalization measures elliptical distance relative to half-extents, not Euclidean distance in screen pixels.

Output / distribution role: Conditional encounter distribution, AI intent, accepted spawn or world-event state.

Trigger / clock: On spawn/event attempt or AI tick; failed spawns are explicit zero-output outcomes.

Composition requirement: Band controls minimum monster level, rank and elite probability at spawn; camera zoom does not change band membership.

Boundary / exception: Boundary membership uses strict comparisons. A spawn just across a threshold can have discontinuously different reward and risk.

Acceptance check: Check center, corners and points immediately on either side of each band boundary.

Trace: regionBand; BAND_THRESHOLDS [S1]

Function registry: F0128

M052 | Biome and band level synthesis

STATUS I | implemented baseline

Inputs / units: Region/biome, danger band, level/rank, spawn geometry, population and event clocks.

```
L_min = 1 + 4*biome_drop + offset_band; L = L_min + U{0,1,2,3}
```

Transition / meaning: Biome difficulty and radial band combine additively before a uniform four-level offset. Band offsets are 0, 4, 10, 18, 30, 46 and 66. Exceptional super spawns add a further eighteen levels.

Output / distribution role: Conditional encounter distribution, AI intent, accepted spawn or world-event state.

Trigger / clock: On spawn/event attempt or AI tick; failed spawns are explicit zero-output outcomes.

Composition requirement: Level enters health and damage scaling, while band separately affects rank and elite chance. These are related but distinct variables.

Boundary / exception: The display range describes normal spawns; super variants and special event boosts can exceed it.

Acceptance check: For each biome-band pair, enumerate the normal level support and compare a seeded sample after controlling the random stream.

Trace: minZoneLevel; spawnMob [S1]

Function registry: F0129, F0185

M053 | Elite probability and stat multipliers

STATUS I | implemented baseline

Inputs / units: Region/biome, danger band, level/rank, spawn geometry, population and event clocks.

```
p_elite = 0.05 + 0.017*band; size *=1.16; HP *=1.45; damage *=1.20
```

Transition / meaning: Non-guardian monsters independently test the current elite chance after location and rank decisions. Elite health, damage and physical size all increase, while the underlying species behavior remains recognizable.

Output / distribution role: Conditional encounter distribution, AI intent, accepted spawn or world-event state.

Trigger / clock: On spawn/event attempt or AI tick; failed spawns are explicit zero-output outcomes.

Composition requirement: Elite factors multiply level scaling and possible super factors. Size also changes collision footprint and visual-detail eligibility.

Boundary / exception: Independence applies only to the draw conditional on preceding decisions, not to observed encounters after spawn rejection or player selection.

Acceptance check: Sample conditional elite outcomes per band and check size, health and damage ratios allowing final integer rounding.

Trace: spawnMob [S1]

Function registry: F0185

M054 | Rare outer-band super variant

STATUS I | implemented baseline

Inputs / units: Region/biome, danger band, level/rank, spawn geometry, population and event clocks.

```
P(super | outer band, ordinary species) = 0.003
```

Transition / meaning: An outermost-band non-guardian can become rank seven, gain eighteen levels and receive further health and damage factors. The rarity of this conditional branch is not the probability of meeting a super monster per minute.

Output / distribution role: Conditional encounter distribution, AI intent, accepted spawn or world-event state.

Trigger / clock: On spawn/event attempt or AI tick; failed spawns are explicit zero-output outcomes.

Composition requirement: The encounter rate additionally depends on accepted spawn locations, species choice, map population, travel policy and time spent in the outer band.

Boundary / exception: An unconditional 0.3 percent claim about all monsters would be false. Location conditioning must remain visible in reports.

Acceptance check: Force outer and inner band fixtures, compare branch frequencies, and verify guardians are excluded.

Trace: spawnMob [S1]

Function registry: F0185

M055 | Rejection-sampled safe spawning

STATUS I | implemented baseline

Inputs / units: Region/biome, danger band, level/rank, spawn geometry, population and event clocks.

```
accept x only if every clearance constraint holds; attempts <= 60
```

Transition / meaning: Candidate positions are rejected near the player, rocks, portals and existing monsters. Guardians must also satisfy a minimum danger band. After sixty failed attempts, spawning can fail rather than violating clearance.

Output / distribution role: Conditional encounter distribution, AI intent, accepted spawn or world-event state.

Trigger / clock: On spawn/event attempt or AI tick; failed spawns are explicit zero-output outcomes.

Composition requirement: The resulting position distribution is conditioned on all clearances. A near-player proposal branch biases later accepted spawns toward the active player.

Boundary / exception: Rejection sampling does not guarantee a spawn in crowded maps. Treat a failed attempt as no new entity, not a hidden zero-health monster. Clearance uses base species size before elite enlargement; post-scaling separation is not guaranteed.

Acceptance check: Construct crowded fixtures and verify finite termination, no forbidden overlap, and correct accounting when no candidate is accepted.

Trace: spawnMob; nearbyRocks [S1]

Function registry: F0185, F0193

M056 | Ranged kiting policy

STATUS I | implemented baseline

Inputs / units: Region/biome, danger band, level/rank, spawn geometry, population and event clocks.

```
approach if d>280; retreat if d<175; strafe otherwise
```

Transition / meaning: Ranged monsters use distance thresholds to preserve space while aiming projectiles. The thresholds create a tactical annulus rather than a simple chase. Facing responds to desired motion and is also used by the body proxy.

Output / distribution role: Conditional encounter distribution, AI intent, accepted spawn or world-event state.

Trigger / clock: On spawn/event attempt or AI tick; failed spawns are explicit zero-output outcomes.

Composition requirement: Ordinary shots, special attacks, enrage and calm/stun conditions overlay the base movement decision in source order.

Boundary / exception: Obstacle resolution can prevent the desired escape and cause corner traps. The annulus is a desired policy, not a guaranteed reachable orbit.

Acceptance check: Place the player at each threshold, beside a wall and in a corner; record movement direction and shot timing.

Trace: update monster AI; updateMobFacing [S1]

Function registry: F0214, F0222, F0291

M057 | Telegraphed special attack state

STATUS I | implemented baseline

Inputs / units: Region/biome, danger band, level/rank, spawn geometry, population and event clocks.

```
ready -> windup -> execute -> cooldown
```

Transition / meaning: Special traits include venom, sniper, charge, summon, nova and shield. Windup records warning coordinates; execution may aim using current position or prediction according to the trait. A warning is a state transition with finite timing, not decorative randomness.

Output / distribution role: Conditional encounter distribution, AI intent, accepted spawn or world-event state.

Trigger / clock: On spawn/event attempt or AI tick; failed spawns are explicit zero-output outcomes.

Composition requirement: Stuns, death and ordinary movement must be evaluated consistently with windup processing. Visual warning toggles do not cancel the underlying attack.

Boundary / exception: Warnings should not promise a fixed impact point if execution retargets. Future design should distinguish locked aim from tracking aim explicitly.

Acceptance check: Trigger every trait with a moving player; compare the displayed warning geometry, execution point and actual damage area.

Trace: updateMobSpecial; performMobSpecial [S1]

Function registry: F0164, F0166

M058 | Neutral retaliation and calming

STATUS I | implemented baseline

Inputs / units: Region/biome, danger band, level/rank, spawn geometry, population and event clocks.

```
aggro_until = t+8 s on damage; calm duration = 3.8 s for non-guardians
```

Transition / meaning: Wandering neutral bodies retreat when close unless their retaliation timer is active. Puff can temporarily calm an ordinary enemy, while guardians are excluded from that calming path. These are timed behavior gates rather than persistent faction reputation.

Output / distribution role: Conditional encounter distribution, AI intent, accepted spawn or world-event state.

Trigger / clock: On spawn/event attempt or AI tick; failed spawns are explicit zero-output outcomes.

Composition requirement: A new damaging hit refreshes retaliation. Calm checks can override nearby pursuit, and special attack processing has its own placement in the update order.

Boundary / exception: Poison timing and special windup can make perceived calm behavior differ from a promise of complete harmlessness; no such promise is made.

Acceptance check: Test damage, expiry, puff contact, guardian exclusion and a special already winding up.

Trace: neutralMotion; damageMob; update [S1]

Function registry: F0201, F0215, F0222, F0291

M059 | Guardian enrage threshold

STATUS I | implemented baseline

Inputs / units: Region/biome, danger band, level/rank, spawn geometry, population and event clocks.

```
enraged = guardian and 0 < HP <= 0.35*HP_max
```

Transition / meaning: Guardians enter an enraged state below 35 percent health. Existing paths increase movement speed, some damage and some firing frequency. The state depends on current health each update, not on a separate irreversible phase transition.

Output / distribution role: Conditional encounter distribution, AI intent, accepted spawn or world-event state.

Trigger / clock: On spawn/event attempt or AI tick; failed spawns are explicit zero-output outcomes.

Composition requirement: Enrage combines with rank, event multipliers and special cooldown logic. Each path uses its own explicit multiplier.

Boundary / exception: Health restoration above the threshold would remove enrage under this implementation. Do not document an irreversible boss phase unless that rule changes.

Acceptance check: Place health above, at and below threshold and inspect speed, ordinary shot behavior, special cooldown and indicator.

Trace: update; shoot; specialCooldown [S1]

Function registry: F0162, F0222, F0231, F0291

M060 | Weighted world event scheduling

STATUS I | implemented baseline

Inputs / units: Region/biome, danger band, level/rank, spawn geometry, population and event clocks.

```
P(event=e) = weight_e/100; weights = 28,18,21,18,15
```

Transition / meaning: The five events are swarm, treasure, bloodmoon, renewal and meteor. Their weights sum to one hundred, while durations vary from twenty-one to thirty seconds. A subsequent quiet interval is added when an event starts.

Output / distribution role: Conditional encounter distribution, AI intent, accepted spawn or world-event state.

Trigger / clock: On spawn/event attempt or AI tick; failed spawns are explicit zero-output outcomes.

Composition requirement: Event effects compose with existing damage, experience, spawning, healing or drop paths. Transitioning maps clears the current event and reschedules.

Boundary / exception: Event share in wall-clock time is not identical to event selection probability because durations and pauses differ.

Acceptance check: Enumerate cumulative sampling intervals, verify normalization and simulate long-run time fractions with the actual duration schedule.

Trace: WORLD_EVENTS; weightedEvent; updateWorldEvents [S1]

Function registry: F0154, F0159

Contracts M061-M070

M061 | Base drop-family choice

STATUS I | implemented baseline

Inputs / units: Eligible enemy/drop family, rank, active event; craft stock integer>=0 and p in [0,1].

```
P(type=t | species=m) = 1/|drop_list_m| for listed t
```

Transition / meaning: An ordinary kill chooses one petal type uniformly from that species drop list. Lists differ across species and include later-added families. This conditional type distribution should be extracted from code rather than inferred from color or biome names.

Output / distribution role: Reward joint distribution, inventory transition, consumed cost or finite-budget completion probability.

Trigger / clock: On eligible death/pickup or craft attempt; no attempt with fewer than five copies.

Composition requirement: Rank is sampled separately in the ordinary death path. Additional basic or guardian drops add further rewards and must be counted as separate channels.

Boundary / exception: Uniform list selection assumes no duplicate type entries. A duplicated entry would intentionally or accidentally change its weight.

Acceptance check: Check list uniqueness and estimate conditional frequencies for several species, excluding extra reward channels.

Trace: mobs.drop; damageMob [S1]

Function registry: F0201

M062 | Rank-dependent drop distribution

STATUS I | implemented baseline

Inputs / units: Eligible enemy/drop family, rank, active event; craft stock integer>=0 and p in [0,1].

```
for k>=3: P(R=k,k-1,k-2,k-3) = (0.12,0.61,0.21,0.06)
```

Transition / meaning: The ordinary rank sampler subtracts up to three ranks from the monster rank, flooring at zero. At low ranks, several branches collapse into common rewards, so the four displayed probabilities must be aggregated by the resulting rank.

Output / distribution role: Reward joint distribution, inventory transition, consumed cost or finite-budget completion probability.

Trigger / clock: On eligible death/pickup or craft attempt; no attempt with fewer than five copies.

Composition requirement: Treasure may increase the sampled rank afterward, capped at seven. Type choice and rank transformation occur in distinct stages.

Boundary / exception: Rank-seven super monsters use a different distribution and cannot be evaluated with this four-way rule.

Acceptance check: Enumerate exact probabilities for ranks zero through six, including merged zero outcomes, and check all masses sum to one.

Trace: chooseDropRank [S1]

Function registry: F0199

M063 | Super drop exception and event boost

STATUS I | implemented baseline

Inputs / units: Eligible enemy/drop family, rank, active event; craft stock integer>=0 and p in [0,1].

```
rank 7: P(R=6,5,7) = (0.68,0.28,0.04); treasure boost p=0.28
```

Transition / meaning: Super rank-seven monsters preferentially drop rank six, then five, with a small chance of seven. Treasure can increase an already sampled drop by one rank, with a cap of seven. The two stages produce a mixture distribution.

Output / distribution role: Reward joint distribution, inventory transition, consumed cost or finite-budget completion probability.

Trigger / clock: On eligible death/pickup or craft attempt; no attempt with fewer than five copies.

Composition requirement: For a baseline probability mass vector p, the boosted vector is a capped one-step shift mixed with p using weights 0.28 and 0.72.

Boundary / exception: The event does not produce ranks eight or nine through this path. Those ranks rely on crafting under current rules.

Acceptance check: Calculate the mixture exactly and compare event-on and event-off samples, especially mass accumulated at the cap.

Trace: chooseDropRank; damageMob [S1]

Function registry: F0199, F0201

M064 | Additional reward channels

STATUS I | implemented baseline

Inputs / units: Eligible enemy/drop family, rank, active event; craft stock integer>=0 and p in [0,1].

```
E[count per kill] = 1 + 0.18 + I_guardian
```

Transition / meaning: Every kill creates a main drop; an independent basic-petal branch has probability 0.18. Guardians also create a second specified-family drop with its own rank transformation. Expected count is not the same as expected collectible value.

Output / distribution role: Reward joint distribution, inventory transition, consumed cost or finite-budget completion probability.

Trigger / clock: On eligible death/pickup or craft attempt; no attempt with fewer than five copies.

Composition requirement: Sum channel expectations without requiring independence; use covariance terms when computing total value variance. The guardian second drop shares the main sampled rank.

Boundary / exception: The count formula assumes all drops can be collected. Expiry, death, route choice and inventory limits reduce realized collection.

Acceptance check: Count spawned drops separately from collected drops and compare conditional rank correlation within guardian reward pairs.

Trace: damageMob; drop pickup [S1]

Function registry: F0201

M065 | Rarity scaling and crafting probability

STATUS I | implemented baseline

Inputs / units: Eligible enemy/drop family, rank, active event; craft stock integer>=0 and p in [0,1].

```
S_r = 1.72^r; p_r = 0.64/2^r except p_7=1
```

Transition / meaning: Rarity scales many petal stats exponentially. Craft success becomes less likely by rank, except for a guaranteed rank-seven-to-eight conversion. Rank nine is terminal and cannot be crafted upward.

Output / distribution role: Reward joint distribution, inventory transition, consumed cost or finite-budget completion probability.

Trigger / clock: On eligible death/pickup or craft attempt; no attempt with fewer than five copies.

Composition requirement: Damage, durability and some utility effects use the same rarity scale, but copy-count penalties and certain timers do not.

Boundary / exception: The guaranteed rank-seven step and difficult rank-eight step create a nonmonotone success-probability profile; do not fit one smooth curve through the exception.

Acceptance check: Export the complete ten-row parameter table from source and test terminal and exceptional branches explicitly.

Trace: rarity; craftBatch [S1]

Function registry: F0298

M066 | Random material loss per craft

STATUS I | implemented baseline

Inputs / units: Eligible enemy/drop family, rank, active event; craft stock integer>=0 and p in [0,1].

```
C = 5 on success; C = U{1,2,3,4} on failure
```

Transition / meaning: A craft requires at least five copies to start. Success removes five and creates one next-rank petal; failure removes a uniformly sampled one to four copies and creates no output. Protection blocks both single and batch operations.

Output / distribution role: Reward joint distribution, inventory transition, consumed cost or finite-budget completion probability.

Trigger / clock: On eligible death/pickup or craft attempt; no attempt with fewer than five copies.

Composition requirement: Each attempt changes inventory before the next batch attempt checks eligibility. The batch limit is five attempts, not five successes.

Boundary / exception: One-click displayed success probability does not describe the probability of any success in a batch with limited inventory.

Acceptance check: Run deterministic success and failure fixtures, check inventory conservation and verify protected materials remain unchanged.

Trace: craftBatch; takeBag; addBag [S1]

Function registry: F0120, F0121, F0298

M067 | Expected input consumed per attempt

STATUS D | derived analysis

Inputs / units: Eligible enemy/drop family, rank, active event; craft stock integer>=0 and p in [0,1].

```
E[C] = 5p + 2.5(1-p) = 2.5+2.5p
```

Transition / meaning: The conditional failure mean is 2.5. Averaging over success and failure gives expected consumption for an eligible attempt. This value alone does not tell the cost of obtaining a successful output.

Output / distribution role: Reward joint distribution, inventory transition, consumed cost or finite-budget completion probability.

Trigger / clock: On eligible death/pickup or craft attempt; no attempt with fewer than five copies.

Composition requirement: To evaluate a budget, combine consumption with success count, eligibility and stopping. Do not subtract next-rank output as if it were an interchangeable input copy.

Boundary / exception: Protection or insufficient inventory means no attempt occurs; the expectation is conditional on a craft actually being submitted.

Acceptance check: Enumerate the five outcome branches analytically and compare a large fixed-p sample mean.

Trace: Derived from M066

M068 | Expected replenished cost to one output

STATUS D | derived analysis

Inputs / units: Eligible enemy/drop family, rank, active event; craft stock integer>=0 and p in [0,1].

```
E[copies until success] = 5 + 2.5*(1-p)/p
```

Transition / meaning: The number of failures before success has mean (1-p)/p. Each failure loses an average 2.5 input copies, and the terminal success costs five. This is a replenished-inventory model: the player can obtain more material whenever eligibility would otherwise fail.

Output / distribution role: Reward joint distribution, inventory transition, consumed cost or finite-budget completion probability.

Trigger / clock: On eligible death/pickup or craft attempt; no attempt with fewer than five copies.

Composition requirement: Recursive base-rank-equivalent cost multiplies these stage costs only under independent replenishment and no alternative sources of higher-rank inputs.

Boundary / exception: For p=0.64 the cost is 6.40625 copies; for p=0.0025 it is 1002.5. These are expectations, not guaranteed budgets.

Acceptance check: Validate with exact recurrence and Monte Carlo; contrast with finite-inventory stopping to expose the changed question.

Trace: Derived from M065-M067

M069 | Finite-budget craft success recursion

STATUS D | derived analysis

Inputs / units: Eligible enemy/drop family, rank, active event; craft stock integer>=0 and p in [0,1].

```
F(n)=0 for n<5; F(n)=p+(1-p)*sum(j=1..4)F(n-j)/4
```

Transition / meaning: F(n) is the chance of at least one successful next-rank output before input inventory falls below five. A success stops immediately. On failure, the remaining inventory is n-j. This recursion accounts for randomized losses and eligibility.

Output / distribution role: Reward joint distribution, inventory transition, consumed cost or finite-budget completion probability.

Trigger / clock: On eligible death/pickup or craft attempt; no attempt with fewer than five copies.

Composition requirement: Evaluate n in ascending order using dynamic programming; the dependency graph is acyclic because each failure removes at least one copy.

Boundary / exception: This is not the same as 1-(1-p)^floor(n/5), which ignores retained material after failures.

Acceptance check: Compare exact values for small n with full outcome-tree enumeration and simulation confidence intervals.

Trace: Derived finite-inventory model

Function registry: F0565

M070 | Craft cost variance and tail risk

STATUS D | derived analysis

Inputs / units: Eligible enemy/drop family, rank, active event; craft stock integer>=0 and p in [0,1].

```
Var(total cost)=1.25*q/p + 6.25*q/p^2; q=1-p
```

Transition / meaning: Conditional failure loss has variance 1.25. Summing a geometrically distributed number of independent failure losses and adding the fixed success cost yields the displayed variance. Standard deviation can be large even when expected cost seems acceptable.

Output / distribution role: Reward joint distribution, inventory transition, consumed cost or finite-budget completion probability.

Trigger / clock: On eligible death/pickup or craft attempt; no attempt with fewer than five copies.

Composition requirement: Quantiles, not just means, should guide inventory recommendations. Multi-stage rarity progression compounds uncertainty and may create heavy practical grind tails.

Boundary / exception: The formula assumes constant p, replenishment and independent failure-loss draws. Pity or rate changes require a different model.

Acceptance check: Check p=1 gives zero variance and compare simulated moments and upper quantiles at several intermediate probabilities.

Trace: Derived random-sum model

Function registry: F0618

Contracts M071-M080

M071 | Binomial encounter yield

STATUS D | derived analysis

Inputs / units: Declared independent or conditional outcome samples, stopping rule, policy and confidence level.

```
P(K=k)=choose(n,k)*p^k*(1-p)^(n-k)
```

Transition / meaning: K counts a particular reward across n eligible independent trials at fixed probability p. The mean is np and variance is np(1-p). A kill with several correlated reward channels is not automatically one Bernoulli trial.

Output / distribution role: Yield/waiting distribution, interval, diagnostic or proposed hazard transition.

Trigger / clock: Per observation or experiment; the independence assumption is not automatic.

Composition requirement: Condition first on species, rank and event state, then aggregate. Mixing changing p values produces a Poisson-binomial distribution rather than this binomial model.

Boundary / exception: Inventory limits, guaranteed guardian bonuses and adaptive rates violate the identical-trial assumption.

Acceptance check: Check total probability equals one; compare analytical first and second moments with enumerated distributions.

Trace: Derived from M061-M064

M072 | Geometric waiting-time quantile

STATUS D | derived analysis

Inputs / units: Declared independent or conditional outcome samples, stopping rule, policy and confidence level.

```
Q_alpha = ceil(log(1-alpha)/log(1-p)); 0<p<1
```

Transition / meaning: The number of attempts through the first success has survival probability (1-p)^n. Q_alpha is the smallest attempt count whose success probability reaches alpha. This describes attempts, not consumed materials or elapsed play time.

Output / distribution role: Yield/waiting distribution, interval, diagnostic or proposed hazard transition.

Trigger / clock: Per observation or experiment; the independence assumption is not automatic.

Composition requirement: Combine with an independently modeled encounter time only when the independence assumption is justified. Craft failures consume variable quantities, so use M069 for finite inventory.

Boundary / exception: For p=1 the answer is one. For p=0 a finite quantile does not exist. Means conceal a substantial upper tail.

Acceptance check: For p=0.64 confirm the 95 percent attempt quantile is three; test the defining inequalities at Q and Q-1.

Trace: Derived survival calculation

M073 | Multiple-output waiting distribution

STATUS D | derived analysis

Inputs / units: Declared independent or conditional outcome samples, stopping rule, policy and confidence level.

```
P(T=n)=choose(n-1,k-1)*p^k*(1-p)^(n-k); n>=k
```

Transition / meaning: T is the attempts needed for k successes under independent constant-rate Bernoulli trials. E[T]=k/p and Var(T)=k(1-p)/p^2. This is useful for a target build requiring repeated copies, before accounting for rank-to-rank input production.

Output / distribution role: Yield/waiting distribution, interval, diagnostic or proposed hazard transition.

Trigger / clock: Per observation or experiment; the independence assumption is not automatic.

Composition requirement: A full rarity pipeline is a branching resource process: recursively multiply expected input requirements only under replenishment assumptions and separately propagate variance.

Boundary / exception: Shared pity counters or duplicate protection create dependence. Their state must be included explicitly.

Acceptance check: Check k=1 reduces to M072 and sum probabilities over an adequate tail-controlled range.

Trace: Derived negative-binomial model

M074 | Conditional reward-rate thinning

STATUS D | derived analysis

Inputs / units: Declared independent or conditional outcome samples, stopping rule, policy and confidence level.

```
lambda_target=lambda_kill*P(type)*P(rank|type,state)*E[channel count]
```

Transition / meaning: Under a stationary kill stream and conditionally independent target selection, this gives expected target copies per unit time. The factors are conditional probabilities, not arbitrary global averages.

Output / distribution role: Yield/waiting distribution, interval, diagnostic or proposed hazard transition.

Trigger / clock: Per observation or experiment; the independence assumption is not automatic.

Composition requirement: Partition time into event and non-event strata and sum exposure-weighted rates. For nonidentical bonus channels compute each channel separately and add expectations.

Boundary / exception: Poisson waiting-time claims need stronger assumptions than this mean-rate identity. Travel, death and inventory management change effective kill rate.

Acceptance check: Compare a stratified synthetic log with direct target-count divided by elapsed time.

Trace: Derived from M061-M064

M075 | Absorbing budget-state chain

STATUS D | derived analysis

Inputs / units: Declared independent or conditional outcome samples, stopping rule, policy and confidence level.

```
v = Q*v + b; v=(I-Q)^(-1)*b
```

Transition / meaning: Q contains transition probabilities among unfinished resource states and b contains one-step success probabilities. The solution gives eventual completion chances before reaching an absorbing failure or success state.

Output / distribution role: Yield/waiting distribution, interval, diagnostic or proposed hazard transition.

Trigger / clock: Per observation or experiment; the independence assumption is not automatic.

Composition requirement: Add pity, inventory and craft rank to state only when they change future probabilities. M069 is the acyclic one-resource specialization and is cheaper to evaluate.

Boundary / exception: The inverse expression requires transient Q with spectral radius below one. Large state spaces need sparse solves or simulation.

Acceptance check: Check each full transition row sums to one; compare the one-rank solution with M069 to machine precision.

Trace: Derived absorbing-chain formulation

M076 | Wilson interval for observed success

STATUS P | conditional proposed extension

Inputs / units: Declared independent or conditional outcome samples, stopping rule, policy and confidence level.

```
center=(phat+z^2/(2n))/(1+z^2/n); half=z*sqrt(phat*(1-phat)/n+z^2/(4n^2))/(1+z^2/n)
```

Transition / meaning: This proposed monitoring interval expresses finite-sample uncertainty for a fixed Bernoulli probability. Use z=1.96 for an approximate 95 percent interval and label n, successes and sampling method.

Output / distribution role: Yield/waiting distribution, interval, diagnostic or proposed hazard transition.

Trigger / clock: Per observation or experiment; the independence assumption is not automatic.

Composition requirement: Calculate separate strata for species, rank, event and client version. Do not infer a changed drop rate from a small mixed sample.

Boundary / exception: Optional stopping, dependence and multiple comparisons compromise nominal coverage. No interval is produced at n=0.

Acceptance check: Test zero successes, all successes and n=100 with 50 successes; ensure bounds remain in [0,1].

Trace: Statistical monitoring proposal; [R8]

M077 | Paired common-random-number experiment

STATUS P | conditional proposed extension

Inputs / units: Declared independent or conditional outcome samples, stopping rule, policy and confidence level.

```
Delta_i=J(A,R_i)-J(B,R_i); SE=sd(Delta)/sqrt(n)
```

Transition / meaning: Compare candidate builds or balance versions using matched encounter seeds. Pairing can reduce noise when outcomes are positively correlated, making small tactical differences easier to measure.

Output / distribution role: Yield/waiting distribution, interval, diagnostic or proposed hazard transition.

Trigger / clock: Per observation or experiment; the independence assumption is not automatic.

Composition requirement: Domain-separated random streams prevent one extra visual or loot draw from shifting every subsequent encounter. Record scenario and seed with each pair.

Boundary / exception: Pairing does not guarantee variance reduction. An unseeded current client cannot claim these reproducible experiments yet.

Acceptance check: Compare paired and unpaired variance estimates; retain raw outcomes and report confidence intervals for the difference.

Trace: Proposed experiment harness; M009

M078 | Local balance elasticity

STATUS D | derived analysis

Inputs / units: Declared independent or conditional outcome samples, stopping rule, policy and confidence level.

```
E_x=(x/J)*dJ/dx; dJ/dx approx (J(x+e)-J(x-e))/(2e)
```

Transition / meaning: Elasticity measures proportional response of a chosen outcome J to a small parameter change. It can identify whether damage, downtime or travel speed dominates a build under a stated scenario.

Output / distribution role: Yield/waiting distribution, interval, diagnostic or proposed hazard transition.

Trigger / clock: Per observation or experiment; the independence assumption is not automatic.

Composition requirement: Use paired seeds for finite differences. Thresholds such as a new equipment slot require discrete differences rather than pretending the outcome is smooth.

Boundary / exception: Undefined at J=0 and sensitive to step size near discontinuities. One scenario is not a universal balance verdict.

Acceptance check: Repeat at e and e/2; require stable sign away from thresholds and show uncertainty in simulated J.

Trace: Derived sensitivity method

M079 | Pity hazard with capped waiting

STATUS P | conditional proposed extension

Inputs / units: Declared independent or conditional outcome samples, stopping rule, policy and confidence level.

```
h(k)=min(1,p0+delta*max(0,k-k0)); S(n)=product(k=0..n-1)(1-h(k))
```

Transition / meaning: h(k) is success probability after k consecutive failures. The survival product defines the complete waiting distribution; a guaranteed terminal hazard bounds the longest unlucky streak.

Output / distribution role: Yield/waiting distribution, interval, diagnostic or proposed hazard transition.

Trigger / clock: Per observation or experiment; the independence assumption is not automatic.

Composition requirement: Pity state belongs to a precisely defined account, reward family and season. Reset on qualifying success and specify whether bonus channels advance the counter.

Boundary / exception: This is not shipped. It changes economy supply and expected cost, so matching only the base probability is insufficient.

Acceptance check: Compute E[T]=sum(n>=0)S(n), verify monotone survival, and compare economy inflation against the current constant-rate model.

Trace: Proposed fairness mechanism

M080 | Probability calibration score

STATUS P | conditional proposed extension

Inputs / units: Declared independent or conditional outcome samples, stopping rule, policy and confidence level.

```
Brier=(1/n)*sum(i)(p_i-y_i)^2
```

Transition / meaning: If an interface forecasts an encounter outcome, score the stated probability p_i against binary result y_i. Calibration diagrams compare predicted and observed frequencies within disclosed bins.

Output / distribution role: Yield/waiting distribution, interval, diagnostic or proposed hazard transition.

Trigger / clock: Per observation or experiment; the independence assumption is not automatic.

Composition requirement: Log predictions before outcomes are known. Stratify by model version and report both calibration and sample count.

Boundary / exception: A low aggregate score may hide poor rare-event calibration. This is an analytics proposal, not a current personalized prediction feature.

Acceptance check: Use synthetic calibrated and deliberately biased forecasts; verify the biased model worsens under the same sample.

Trace: Proposed forecasting validation

Contracts M081-M090

M081 | Tactical Markov decision model

STATUS P | conditional proposed extension

Inputs / units: Complete tactical state, feasible action set, transition probabilities, utility profile and horizon.

```
V(s)=max_a {r(s,a)+gamma*sum(s2)P(s2|s,a)*V(s2)}; 0<=gamma<1
```

Transition / meaning: State includes player and enemy health, cooldowns, positions, equipment, encounter timers and relevant persistent counters. Actions include movement, stance, dash, retreat and deck choice. A discounted objective makes tactical planning explicit.

Output / distribution role: Expected utility, risk constraint, route/policy or Pareto set under stated assumptions.

Trigger / clock: At decision/analysis time; proposed planners are not claimed live.

Composition requirement: Use a small discretized arena first. Frame-by-frame continuous control and long-term inventory decisions need different time scales.

Boundary / exception: This is a planning model, not an implemented bot. A state omitting cooldowns or poison is generally not Markov.

Acceptance check: Solve a tiny hand-enumerable arena and verify the chosen policy against exhaustive finite-horizon search.

Trace: Proposed tactical foundation

M082 | Mean-variance diagnostic risk score

STATUS P | conditional proposed extension

Inputs / units: Complete tactical state, feasible action set, transition probabilities, utility profile and horizon.

```
J_MV=E[R]-lambda*Var(R)-mu*E[loss]-nu*E[time]
```

Transition / meaning: This proposed diagnostic score subtracts dispersion, loss and time from expected reward. It is not generally an expected utility of the outcome and need not preserve first-order stochastic dominance. It MUST NOT be substituted for the normative E[u(X)] objective without an explicit design exception.

Output / distribution role: Expected utility, risk constraint, route/policy or Pareto set under stated assumptions.

Trigger / clock: At decision/analysis time; proposed planners are not claimed live.

Composition requirement: Coefficients convert each term into a common declared score unit. Report the components separately. The normative expected-utility model evaluates utility on each outcome before averaging.

Boundary / exception: For rewards A={1,10} with equal probability and B=0, A dominates B outcome by outcome, yet E[A]-Var(A)=-14.75<0. Nonnegative penalty weights alone do not ensure monotonicity.

Acceptance check: Verify this counterexample and separately test increasing utility E[u(A)]>E[u(B)]. A mean-variance score is labeled diagnostic, not the universal game objective.

Trace: Proposed objective family

M083 | Expedition ruin constraint

STATUS P | conditional proposed extension

Inputs / units: Complete tactical state, feasible action set, transition probabilities, utility profile and horizon.

```
maximize E[reward] subject to P(HP reaches 0 before extraction)<=epsilon
```

Transition / meaning: This separates a survival requirement from the reward objective. A cautious route may satisfy the constraint while a high-mean farming route fails it.

Output / distribution role: Expected utility, risk constraint, route/policy or Pareto set under stated assumptions.

Trigger / clock: At decision/analysis time; proposed planners are not claimed live.

Composition requirement: Estimate ruin under the same enemy, latency and input model used for route evaluation. Include healing cooldown and inventory opportunity cost.

Boundary / exception: A finite simulation sample cannot certify a very small failure probability without a sufficiently tight upper confidence bound.

Acceptance check: Use exact small-state ruin probabilities as a benchmark and report an upper confidence bound for simulated routes.

Trace: Proposed constrained planning

M084 | Conditional value at risk for loss

STATUS P | conditional proposed extension

Inputs / units: Complete tactical state, feasible action set, transition probabilities, utility profile and horizon.

```
CVaR_alpha(L)=min_eta {eta+E[max(L-eta,0)]/(1-alpha)}
```

Transition / meaning: L is expedition loss, including consumed resources and declared death penalties. CVaR focuses on the average of the worst tail rather than just its threshold.

Output / distribution role: Expected utility, risk constraint, route/policy or Pareto set under stated assumptions.

Trigger / clock: At decision/analysis time; proposed planners are not claimed live.

Composition requirement: Combine with expected reward in a constrained or weighted objective. Preserve the same unit of account across both terms.

Boundary / exception: For finite discrete outcomes this variational definition avoids ambiguous treatment of mass exactly at the quantile. Alpha must be below one.

Acceptance check: Calculate a small discrete loss distribution by enumeration and match the optimization result against its tail allocation.

Trace: Proposed tail-risk criterion

M085 | Danger-aware route search

STATUS P | conditional proposed extension

Inputs / units: Complete tactical state, feasible action set, transition probabilities, utility profile and horizon.

```
edge_cost=distance/speed + a*expected_damage + b*exposure; a,b>=0; speed>0
```

Transition / meaning: Dijkstra search over a navigation graph yields a minimum cost route when edge costs are nonnegative. The danger layer can encode ranged firing lanes, guardian zones and obstacle bottlenecks.

Output / distribution role: Expected utility, risk constraint, route/policy or Pareto set under stated assumptions.

Trigger / clock: At decision/analysis time; proposed planners are not claimed live.

Composition requirement: Recompute locally when an event or crowd changes. A lower-bound travel-time heuristic supports A-star if it never overestimates the final edge cost.

Boundary / exception: This differs from the existing unweighted region-hop helper. Dynamic enemies make stale route costs unreliable.

Acceptance check: Compare against exhaustive paths on a small graph, then verify collision-free traversal in the actual movement resolver.

Trace: Proposed route planner

M086 | Breadth-first region routing

STATUS I | implemented baseline

Inputs / units: Complete tactical state, feasible action set, transition probabilities, utility profile and horizon.

```
distance[next]=distance[current]+1 on first visit
```

Transition / meaning: The current nextHop helper explores the region connection graph to find the next portal destination. With equal edge costs, breadth-first traversal minimizes region hops, not combat danger or continuous travel length.

Output / distribution role: Expected utility, risk constraint, route/policy or Pareto set under stated assumptions.

Trigger / clock: At decision/analysis time; proposed planners are not claimed live.

Composition requirement: Show a portal direction for the selected region hop. Actual movement still follows player input and collision resolution.

Boundary / exception: Region-hop count ignores where the player and portals are located. This is not obstacle-aware automatic navigation.

Acceptance check: Check a direct neighbor, a multi-region destination and an unreachable graph node; verify the next-hop result.

Trace: nextHop; zone links [S1]

Function registry: F0257

M087 | Optimal stopping for extraction

STATUS P | conditional proposed extension

Inputs / units: Complete tactical state, feasible action set, transition probabilities, utility profile and horizon.

```
V(s)=max{cash_out(s), E[V(next)|continue]-time_cost}
```

Transition / meaning: At a safe extraction opportunity, compare secured inventory value with the continuation value of another encounter. This gives a principled retreat rule under a stated loss model.

Output / distribution role: Expected utility, risk constraint, route/policy or Pareto set under stated assumptions.

Trigger / clock: At decision/analysis time; proposed planners are not claimed live.

Composition requirement: Continuation depends on healing readiness, remaining dash cooldown and the next region. It should consume the same reward model used by route planning.

Boundary / exception: Extraction with a new loss rule is not shipped. Current local persistence must not be reinterpreted as an invented death-loss economy.

Acceptance check: Solve a two-encounter example by backward induction and verify that greater loss exposure cannot improve the stopping value.

Trace: Proposed expedition mode

Function registry: F0065

M088 | Orbit contact duty cycle

STATUS D | derived analysis

Inputs / units: Complete tactical state, feasible action set, transition probabilities, utility profile and horizon.

```
DPS_contact = DPS_while_touching * fraction_of_time_in_contact
```

Transition / meaning: Attack stance increases orbital reach, but useful damage depends on how often a petal intersects a target and remains operational. Orbit phase, enemy body width and relative movement determine contact time.

Output / distribution role: Expected utility, risk constraint, route/policy or Pareto set under stated assumptions.

Trigger / clock: At decision/analysis time; proposed planners are not claimed live.

Composition requirement: Estimate duty cycle from fixed-step collision logs; apply the actual armor and exchange rule inside each contact rather than multiplying an already averaged damage value twice.

Boundary / exception: This factorization is descriptive when damage and contact are dependent. A single universal duty cycle would hide reload and durability effects.

Acceptance check: Compare a stationary target with radial chase and tangential kiting traces at matched equipment.

Trace: Derived from M024, M030-M033

M089 | Boss telegraph reaction budget

STATUS P | conditional proposed extension

Inputs / units: Complete tactical state, feasible action set, transition probabilities, utility profile and horizon.

```
slack=windup-(input_delay+decision_time+escape_time)
```

Transition / meaning: Positive slack means the chosen escape can finish before the predicted strike under the stated model. Escape time includes collision-aware travel and the player hit radius.

Output / distribution role: Expected utility, risk constraint, route/policy or Pareto set under stated assumptions.

Trigger / clock: At decision/analysis time; proposed planners are not claimed live.

Composition requirement: Dash immunity and geometry offer separate escape paths; select the feasible action with the least future cooldown cost.

Boundary / exception: The current game does not measure individual reaction time. Input delay distributions need real device measurements before difficulty is calibrated to them.

Acceptance check: Test early, boundary and late responses against the exact attack time and collision check, including paused and resumed states.

Trace: Proposed telegraph audit

Function registry: F0626

M090 | Pareto equipment search

STATUS P | conditional proposed extension

Inputs / units: Complete tactical state, feasible action set, transition probabilities, utility profile and horizon.

```
A dominates B iff all objectives(A)>=objectives(B) and at least one is greater
```

Transition / meaning: Search a bounded inventory for builds trading damage, survival, reach and mobility. Keep nondominated candidates instead of forcing every player into a single scalar ranking.

Output / distribution role: Expected utility, risk constraint, route/policy or Pareto set under stated assumptions.

Trigger / clock: At decision/analysis time; proposed planners are not claimed live.

Composition requirement: Prune only with valid bounds. Family thresholds and armor breakpoints make individual item scores insufficient for safe greedy pruning.

Boundary / exception: The combinatorial search grows rapidly with slots and rarity variants. Proposed first scope is a small owned-inventory subset.

Acceptance check: Compare the pruned frontier with exhaustive enumeration for a small inventory and show how scenario changes alter the frontier.

Trace: Proposed build adviser

Contracts M091-M100

M091 | Typed modifier operator order

STATUS P | conditional proposed extension

Inputs / units: Typed modifier/economy/cooperation/migration inputs with nonnegative stock and version identity.

```
result=clamp(round((base+sum(flat))*product(mult)),lo,hi)
```

Transition / meaning: Every future modifier should declare its target, unit, stacking group, operation and expiry. This proposed canonical pipeline prevents two descriptions of the same bonus from silently applying at different stages.

Output / distribution role: Ordered state transition, reward allocation, conservation ledger or versioned snapshot.

Trigger / clock: On enabled extension trigger; explicit P contract is required before deployment.

Composition requirement: Apply overrides in a separately documented precedence stage. M021-M039 remain authoritative for current formulas until a migration intentionally changes them.

Boundary / exception: Addition and multiplication do not commute. Rounding after each term is different from rounding once at the end.

Acceptance check: Use order-sensitive examples and verify a serialized modifier list produces the same result regardless of insertion order within a commutative group.

Trace: Proposed modifier contract

M092 | Stacked prism critical hits before armor

STATUS I | implemented baseline

Inputs / units: Typed modifier/economy/cooperation/migration inputs with nonnegative stock and version identity.

```
p_crit=min(.60,1-.85^c)
raw=D*(1+.60*Bernoulli(p_crit))
E[applied]=(1-p)*max(1,D-A)+p*max(1,1.6D-A)
```

Transition / meaning: Each ready intact prism contributes an independent15-percent chance to the combined at-least-one critical probability, capped at60percent. Every qualifying orbital melee strike samples the combat stream when p>0; a success multiplies requested damage by1.6.

Output / distribution role: Ordered state transition, reward allocation, conservation ledger or versioned snapshot.

Trigger / clock: On enabled extension trigger; explicit P contract is required before deployment.

Composition requirement: Count ready equipped copies at the strike. Crit follows durability, talent, assault and overdrive multiplication, then the target armor and minimum-one floor apply. Projectiles, poison and body contact do not use this critical path.

Boundary / exception: Zero prisms consume no critical draw. A prism broken by the current strike may contribute to that strike but not later ones. Critical chance does not scale with rarity. Expected raw multiplier is not the multiplier of armor-mitigated damage.

Acceptance check: Check0,1,2 and many-copy caps; forced success/failure; and D10,A12 produces E[applied]=1.45 at p.15, while applying armor to mean raw damage incorrectly gives1.

Trace: criticalChance; criticalFactor; criticalMultiplier; update; damageMob [S1]

Function registry: F0201, F0220, F0222, F0291, F0589, F0590

M093 | Element interaction matrix

STATUS P | conditional proposed extension

Inputs / units: Typed modifier/economy/cooperation/migration inputs with nonnegative stock and version identity.

```
D_element=D*E[attack_element,defense_element]
```

Transition / meaning: E is an explicitly versioned nonnegative matrix. It creates encounter-specific strengths without hard-coding scattered conditional multipliers.

Output / distribution role: Ordered state transition, reward allocation, conservation ledger or versioned snapshot.

Trigger / clock: On enabled extension trigger; explicit P contract is required before deployment.

Composition requirement: Apply element transformation at one declared stage before or after armor. Multi-element attacks require a normalized mixture or an explicit precedence rule.

Boundary / exception: No new elemental advantage is shipped. Unbounded multiplicative cycles can create dominant builds if combined with other bonuses.

Acceptance check: Check neutral diagonal behavior, nonnegative entries, bounded row maxima and scenario coverage for every element.

Trace: Proposed elemental extension

M094 | Diminishing resistance curve

STATUS P | conditional proposed extension

Inputs / units: Typed modifier/economy/cooperation/migration inputs with nonnegative stock and version identity.

```
mitigation=R/(K+R); incoming=D*K/(K+R), R>=0, K>0
```

Transition / meaning: This smooth resistance proposal yields diminishing mitigation per added resistance point. K sets the half-damage resistance level and must have the same unit as R.

Output / distribution role: Ordered state transition, reward allocation, conservation ledger or versioned snapshot.

Trigger / clock: On enabled extension trigger; explicit P contract is required before deployment.

Composition requirement: Keep it separate from current flat armor and root layers. Replacing either requires a deliberate balance migration and damage-floor review.

Boundary / exception: Negative resistance needs a separate bounded rule. Adding this on top of existing defenses can make sustain effectively unlimited.

Acceptance check: Check R=0 gives D, R=K gives D/2 and marginal reduction decreases as R increases.

Trace: Proposed defensive curve

M095 | Potency-duration conserving poison merge

STATUS P | conditional proposed extension

Inputs / units: Typed modifier/economy/cooperation/migration inputs with nonnegative stock and version identity.

```
new_duration=(old_rate*old_time+new_rate*new_time)/max(old_rate,new_rate)
```

Transition / meaning: For nonnegative rates with at least one positive rate, choose the stronger rate and preserve total remaining damage budget. It avoids a weak application reducing a strong existing poison.

Output / distribution role: Ordered state transition, reward allocation, conservation ledger or versioned snapshot.

Trigger / clock: On enabled extension trigger; explicit P contract is required before deployment.

Composition requirement: Only merge effects in the same poison family. Define whether armor and resistance apply at application or at each tick.

Boundary / exception: The current implementation is last-writer refresh, M037. This proposal changes damage totals and is not active.

Acceptance check: For two effects verify new_rate*new_duration equals the sum of remaining potency-duration; test zero-duration input.

Trace: Proposed alternative to M037

M096 | Craft salvage conservation

STATUS P | conditional proposed extension

Inputs / units: Typed modifier/economy/cooperation/migration inputs with nonnegative stock and version identity.

```
net_sink=consumed_inputs-salvage_value_in_input_units
```

Transition / meaning: A failed craft could return a bounded salvage currency. Expected net sink must remain positive to prevent unlimited craft loops that generate value.

Output / distribution role: Ordered state transition, reward allocation, conservation ledger or versioned snapshot.

Trigger / clock: On enabled extension trigger; explicit P contract is required before deployment.

Composition requirement: Convert salvage redemption into the same input-copy unit and include every bonus or discount channel before testing a cycle.

Boundary / exception: No salvage currency is shipped. A positive local sink does not prove every multi-rank trade cycle is safe.

Acceptance check: Enumerate redemption cycles across ranks and verify every closed cycle has expected nonpositive net value.

Trace: Proposed economy extension

M097 | Stock-flow economy accounting

STATUS P | conditional proposed extension

Inputs / units: Typed modifier/economy/cooperation/migration inputs with nonnegative stock and version identity.

```
stock[t+1]=stock[t]+sources[t]-sinks[t]+net_transfers[t]
```

Transition / meaning: Account separately for loot creation, craft destruction, migration, administrative grants and transfers. A conserved ledger makes unexpected inflation diagnosable.

Output / distribution role: Ordered state transition, reward allocation, conservation ledger or versioned snapshot.

Trigger / clock: On enabled extension trigger; explicit P contract is required before deployment.

Composition requirement: Aggregate by rarity and family, then use an explicit conversion valuation only for cross-rank summaries. Keep raw item counts available.

Boundary / exception: The current client-authoritative inventory is not an auditable competitive economy. Server event authority is required before valuable trading.

Acceptance check: Reconcile a synthetic session event ledger against the final inventory exactly, including failed crafts and reloads.

Trace: Proposed economy ledger

M098 | Arbitrage cycle detection

STATUS P | conditional proposed extension

Inputs / units: Typed modifier/economy/cooperation/migration inputs with nonnegative stock and version identity.

```
w(i,j)=-log(exchange_rate(i,j)), rate>0; negative cycle => multiplicative gain
```

Transition / meaning: If trading or exchange shops are introduced, represent conversion edges with log costs. A negative cycle signals a deterministic gross gain before transaction constraints.

Output / distribution role: Ordered state transition, reward allocation, conservation ledger or versioned snapshot.

Trigger / clock: On enabled extension trigger; explicit P contract is required before deployment.

Composition requirement: Include fees and capacity limits. Randomized exchanges need risk and expected log-growth analysis beyond deterministic cycle detection.

Boundary / exception: No trading market is shipped. Preventing graph arbitrage does not prevent client forgery or duplication.

Acceptance check: Construct a three-currency profitable loop and a fee-adjusted nonprofitable loop; verify cycle detection changes correctly.

Trace: Proposed exchange audit

M099 | Contribution-normalized cooperative reward

STATUS P | conditional proposed extension

Inputs / units: Typed modifier/economy/cooperation/migration inputs with nonnegative stock and version identity.

```
share_i=eligible_i*(epsilon+contribution_i)/(sum_j eligible_j*(epsilon+contribution_j))
```

Transition / meaning: An eligible team member receives a fraction of a fixed shared reward pool. Contributions can include damage, protection and healing, with caps to avoid farming trivial actions.

Output / distribution role: Ordered state transition, reward allocation, conservation ledger or versioned snapshot.

Trigger / clock: On enabled extension trigger; explicit P contract is required before deployment.

Composition requirement: Calculate eligibility server-side and bound repeated support credit. A positive epsilon prevents complete exclusion of low-output participants.

Boundary / exception: This is a multiplayer proposal. Current local enemies and client simulation must not be described as authoritative shared combat.

Acceptance check: Check shares sum to one and test zero contributions, disconnects and intentionally inflated healing loops.

Trace: Proposed cooperative allocation

M100 | Versioned seasonal state migration

STATUS P | conditional proposed extension

Inputs / units: Typed modifier/economy/cooperation/migration inputs with nonnegative stock and version identity.

```
S_new = migrate(version_old, version_new, S_old)
```

Transition / meaning: A migration must state which resources persist, which convert and which expire. It should produce an audit summary before a seasonal ruleset becomes active.

Output / distribution role: Ordered state transition, reward allocation, conservation ledger or versioned snapshot.

Trigger / clock: On enabled extension trigger; explicit P contract is required before deployment.

Composition requirement: Validate first, migrate a copy, then commit atomically with a revision guard. Keep a recoverable pre-migration snapshot.

Boundary / exception: No season reset is proposed for current production accounts. This entry defines a safe future contract rather than authorizing destructive changes.

Acceptance check: Check idempotence for already migrated data and round-trip preservation of fields designated permanent.

Trace: Proposed migration contract

Contracts M101-M110

M101 | Revision compare-and-swap persistence

STATUS I | implemented baseline

Inputs / units: Validated account/snapshot identity, integer revision, bounded payload and explicit wall-clock timers.

```
UPDATE saves SET revision=revision+1 WHERE player_id=id AND revision=expected
```

Transition / meaning: The production API writes only when the stored revision matches the client base revision. An insert has a separate no-existing-row condition. A conflict preserves the local outbox instead of silently overwriting another device.

Output / distribution role: Persistence result, retry/lock state, authentication verdict or proposed remote interpolation.

Trigger / clock: On account/save/network event; stochastic gameplay weights must not weaken security guards.

Composition requirement: Authenticate first, bind the account identity, validate the payload, then execute the guarded statement and inspect whether a row was returned.

Boundary / exception: CAS prevents lost updates, not cheating in a client-generated payload. It does not merge two divergent inventories.

Acceptance check: Issue two writes from one revision and verify exactly one succeeds while the other receives a conflict.

Trace: putSave in functions/api/[[path]].js [S1]; [R7]

Function registry: F0524

M102 | Response identity binding

STATUS I | implemented baseline

Inputs / units: Validated account/snapshot identity, integer revision, bounded payload and explicit wall-clock timers.

```
request identity header == authenticated account; response epoch == capturedEpoch
```

Transition / meaning: The server checks the requested account identity against the authenticated session. The frontend separately checks the captured account and session epoch before applying an asynchronous save response.

Output / distribution role: Persistence result, retry/lock state, authentication verdict or proposed remote interpolation.

Trigger / clock: On account/save/network event; stochastic gameplay weights must not weaken security guards.

Composition requirement: Use this guard with revision checks and account-specific outbox keys. Neither substitutes for server authentication.

Boundary / exception: A stale response may arrive successfully after a switch; ignoring it is required even if its HTTP status is 200.

Acceptance check: Delay an account-A response, switch to B, then verify no A data or revision becomes active in B.

Trace: cfAPI account header; cfFlush epoch/account guard [S1]

Function registry: F0074, F0079

M103 | Cross-tab save exclusion

STATUS I | implemented baseline

Inputs / units: Validated account/snapshot identity, integer revision, bounded payload and explicit wall-clock timers.

```
at most one held save lock per origin and account
```

Transition / meaning: Web Locks hold one same-origin account session lock. A competing tab is rejected immediately using ifAvailable, rather than queued. Server CAS remains the final cross-device arbiter.

Output / distribution role: Persistence result, retry/lock state, authentication verdict or proposed remote interpolation.

Trigger / clock: On account/save/network event; stochastic gameplay weights must not weaken security guards.

Composition requirement: Account-specific lock names isolate accounts. If Web Locks are unavailable, cloud entry fails with guidance to upgrade the browser or use a local character.

Boundary / exception: This lock does not span devices or browser profiles and cannot replace CAS. A suspended tab continues to hold its account lock.

Acceptance check: Run competing same-account tabs and verify rejection of the second tab; test cloud-entry refusal without the API and server conflict protection.

Trace: cfTakeLock; cfReleaseLock [S1]

Function registry: F0069, F0070

M104 | Durable local outbox

STATUS I | implemented baseline

Inputs / units: Validated account/snapshot identity, integer revision, bounded payload and explicit wall-clock timers.

```
pending={owner,baseRevision,payload}; persist pending before network write
```

Transition / meaning: The browser retains the account-bound pending snapshot when the cloud is unavailable. A later retry uses its original base revision so divergence is detected rather than overwritten.

Output / distribution role: Persistence result, retry/lock state, authentication verdict or proposed remote interpolation.

Trigger / clock: On account/save/network event; stochastic gameplay weights must not weaken security guards.

Composition requirement: After successful acknowledgement, clear only the matching pending state. A newer local snapshot created during the request must remain queued.

Boundary / exception: Browser storage can be cleared or fail. The outbox is a recovery aid, not an unlimited backup guarantee.

Acceptance check: Interrupt a save, reload offline, then reconnect; verify progress survives and conflicting remote changes remain protected.

Trace: cfOutboxKey; cfQueue; cfFlush [S1]

Function registry: F0068, F0078, F0079

M105 | Bounded exponential retry schedule

STATUS I | implemented baseline

Inputs / units: Validated account/snapshot identity, integer revision, bounded payload and explicit wall-clock timers.

```
delay=max(RetryAfter, min(30 s, 2 s*2^retry)); automatic retries bounded
```

Transition / meaning: The native cloud client backs off temporary failures and respects server Retry-After. Authentication failures and revision conflicts stop the automatic retry path.

Output / distribution role: Persistence result, retry/lock state, authentication verdict or proposed remote interpolation.

Trigger / clock: On account/save/network event; stochastic gameplay weights must not weaken security guards.

Composition requirement: Preserve the outbox across exhaustion and expose an understandable retry action. A retry must retain identity and revision guards.

Boundary / exception: Current deterministic delays can synchronize many clients. Random jitter is a future improvement, not claimed as shipped.

Acceptance check: Use fake time to check the cap, server delay precedence, retry exhaustion, conflict stop and 401 stop.

Trace: cfFlush; cfSchedule [S1]

Function registry: F0076, F0079

M106 | Bounded save validation

STATUS I | implemented baseline

Inputs / units: Validated account/snapshot identity, integer revision, bounded payload and explicit wall-clock timers.

```
accept iff byte_size<=limit and schema, ranges, lengths and inventory bounds pass
```

Transition / meaning: The API rejects oversized or malformed snapshots and constrains known numeric and collection fields. This protects storage and prevents accidental invalid state from reaching the database.

Output / distribution role: Persistence result, retry/lock state, authentication verdict or proposed remote interpolation.

Trigger / clock: On account/save/network event; stochastic gameplay weights must not weaken security guards.

Composition requirement: Validate before serialization and guarded persistence. Client sanitization remains useful for user experience but is not the security boundary.

Boundary / exception: Validation is not proof that claimed rewards were legitimately earned; the gameplay simulation is still client-authoritative.

Acceptance check: Test malformed JSON, boundary values, excessive collections, unexpected types and valid maximum-size snapshots.

Trace: validateSave; bodyOf [S1]

Function registry: F0515, F0521

M107 | Memory-hard password derivation

STATUS I | implemented baseline

Inputs / units: Validated account/snapshot identity, integer revision, bounded payload and explicit wall-clock timers.

```
key=scrypt(password,salt,N=16384,r=8,p=5,dkLen=32)
```

Transition / meaning: The production account service stores a versioned salted password derivation rather than a plaintext password. Verification uses constant-time comparison after deriving the candidate key.

Output / distribution role: Persistence result, retry/lock state, authentication verdict or proposed remote interpolation.

Trigger / clock: On account/save/network event; stochastic gameplay weights must not weaken security guards.

Composition requirement: Authenticate through HTTPS cookies and retain request limits. Recovery uses a separately generated one-use secret; it is not an email ownership claim.

Boundary / exception: These are actual implementation parameters, not a universal security recommendation. Browser gameplay randomness must never generate credentials.

Acceptance check: Run the real Workers-compatible runtime health self-test plus valid and invalid password verification, without logging secrets.

Trace: makePassword; checkPassword; verifyPasswordRuntime [S1]

Function registry: F0509, F0511, F0512

M108 | Fixed-window authentication rate guard

STATUS I | implemented baseline

Inputs / units: Validated account/snapshot identity, integer revision, bounded payload and explicit wall-clock timers.

```
attempts = 1 after reset; else attempts+1; reject above action-specific limit
```

Transition / meaning: The API maintains bounded windows for sensitive account actions and returns a Retry-After delay when a window is exhausted. Registration uses email-shaped address plus password without verification mail.

Output / distribution role: Persistence result, retry/lock state, authentication verdict or proposed remote interpolation.

Trigger / clock: On account/save/network event; stochastic gameplay weights must not weaken security guards.

Composition requirement: Rate guards constrain abuse independently of authentication semantics. Reaching a limit must not turn an unauthenticated request into a successful login.

Boundary / exception: No fallback may bypass password or recovery proof. A fixed window permits a boundary burst; tighter sliding windows are a possible future revision.

Acceptance check: Advance fake time across a reset and verify limits, independent buckets and no credential-bypass behavior.

Trace: limit; auth routes [S1]

Function registry: F0516

M109 | Server-authoritative reward event

STATUS P | conditional proposed extension

Inputs / units: Validated account/snapshot identity, integer revision, bounded payload and explicit wall-clock timers.

```
inventory_next = apply(validated_server_event, inventory_current)
```

Transition / meaning: For competitive progression, the server must determine reward eligibility and resulting inventory transitions. A client may submit input intent, not an arbitrary successful craft or drop result.

Output / distribution role: Persistence result, retry/lock state, authentication verdict or proposed remote interpolation.

Trigger / clock: On account/save/network event; stochastic gameplay weights must not weaken security guards.

Composition requirement: Assign an idempotency key and monotonically ordered event sequence; atomically update event ledger and inventory with a revision guard.

Boundary / exception: This is not current architecture. Moving a formula into a Worker alone does not make position, damage or encounter history trustworthy.

Acceptance check: Replay a duplicate reward request and require exactly one grant; reject impossible encounter transitions and stale event sequences.

Trace: Proposed authority upgrade

M110 | Remote snapshot interpolation

STATUS P | conditional proposed extension

Inputs / units: Validated account/snapshot identity, integer revision, bounded payload and explicit wall-clock timers.

```
render_state(t)=lerp(snapshot_a,snapshot_b,(t-ta)/(tb-ta))
```

Transition / meaning: For future shared encounters, render remote bodies behind the latest authoritative time to absorb ordinary packet jitter. Local input prediction is a separate system with reconciliation.

Output / distribution role: Persistence result, retry/lock state, authentication verdict or proposed remote interpolation.

Trigger / clock: On account/save/network event; stochastic gameplay weights must not weaken security guards.

Composition requirement: Interpolate only continuous compatible fields; health events, deaths and equipment changes require discrete versioned transitions.

Boundary / exception: No networked world simulation is shipped. Excessive extrapolation can show impossible positions and conceal real collision disagreement.

Acceptance check: Inject latency, packet loss and out-of-order delivery; check bounded visual error and correct discrete event order.

Trace: Proposed multiplayer rendering

Contracts M111-M120

M111 | Seeded cached terrain material

STATUS I | implemented baseline

Inputs / units: Finite geometry, view size, device scale, visual preferences, validated state and release evidence.

```
tile=paint(seed+813); repeat tile in world space
```

Transition / meaning: The V5.13 candidate builds a small biome material once and reuses it as a world-aligned canvas pattern. Subtle mottles and curved strokes add texture without a large downloaded bitmap.

Output / distribution role: Pixels, layout, animation budget, verification result or release decision.

Trigger / clock: Per presentation/event/release; decorative quality does not alter authoritative hit areas.

Composition requirement: The terrain generator uses a local seeded generator and never consumes combat or loot randomness. Low quality bypasses the decorative layer.

Boundary / exception: The cache is per biome and resolution, not an infinitely growing per-position atlas. This is procedural 2D art, not a 3D texture asset claim.

Acceptance check: Override Math.random to throw during rendering; verify material generation still succeeds and patterns remain anchored during camera movement.

Trace: paintTerrain; terrainMaterials [S1]; [R4]

Function registry: F0251

M112 | Radial material light model

STATUS I | implemented baseline

Inputs / units: Finite geometry, view size, device scale, visual preferences, validated state and release evidence.

```
C(r)=interpolate(highlight,base,shadow; normalized radial distance)
```

Transition / meaning: Creature bodies use a restrained radial gradient with a stable light direction, darker rim and selected internal anatomy. Low quality substitutes the base fill to reduce drawing work.

Output / distribution role: Pixels, layout, animation budget, verification result or release decision.

Trigger / clock: Per presentation/event/release; decorative quality does not alter authoritative hit areas.

Composition requirement: Apply body material before clipped shell detail, eyes and hurt feedback. Ground shadow is decorative and must not influence contact.

Boundary / exception: This is an artistic canvas gradient, not physically based rendering or measured light transport. Highly reflective effects must not obscure rank and health cues.

Acceptance check: Inspect all species at three radii and compare low/high quality; verify no clipping or invisible outlines against each biome.

Trace: materialFill; drawCreatureModel [S1]

Function registry: F0250, F0253

M113 | Collision-aligned anatomy scaling

STATUS I | implemented baseline

Inputs / units: Finite geometry, view size, device scale, visual preferences, validated state and release evidence.

```
body_axes = mobCollider(m); decorative_length = coefficient*radius
```

Transition / meaning: The body silhouette derives from the same authoritative ellipse proxy as contact geometry. Legs, wings, petals and claws scale with body radius but remain decorative.

Output / distribution role: Pixels, layout, animation budget, verification result or release decision.

Trigger / clock: Per presentation/event/release; decorative quality does not alter authoritative hit areas.

Composition requirement: Use bounded stroke widths and projected-size detail gating so giant creatures gain structure while small creatures remain legible.

Boundary / exception: Decorative appendages can extend beyond the hit body. The gallery explicitly shows dashed body proxies rather than suggesting every wing is collidable.

Acceptance check: Compare collider JSON before and after drawing for every species and size; inspect the English gallery with overlaid proxies.

Trace: creatureVisualProfile; drawCreatureModel [S1]

Function registry: F0252, F0253

M114 | Speed-linked articulated animation

STATUS I | implemented baseline

Inputs / units: Finite geometry, view size, device scale, visual preferences, validated state and release evidence.

```
phase=sin(t*(3+0.025*speed)+0.73*id)
```

Transition / meaning: Leg and wing articulation uses a bounded sinusoid keyed to creature identity and nominal speed. It adds a distinct living pose without moving the gameplay body.

Output / distribution role: Pixels, layout, animation budget, verification result or release decision.

Trigger / clock: Per presentation/event/release; decorative quality does not alter authoritative hit areas.

Composition requirement: Reduced-motion preference sets this decorative phase to zero. Simulation time makes the pose pause with gameplay.

Boundary / exception: This candidate is speed-linked, not measured foot-plant animation. A stationary enemy may still have an idle cycle.

Acceptance check: Check finite transforms at every species size and phase=0 under prefers-reduced-motion. Drawing must not mutate world coordinates.

Trace: creatureVisualProfile [S1]

Function registry: F0252

M115 | Contextual enemy information density

STATUS I | implemented baseline

Inputs / units: Finite geometry, view size, device scale, visual preferences, validated state and release evidence.

```
show_HP = near or injured or guardian; show_name = near or guardian or debug
```

Transition / meaning: Health and labels appear when tactically relevant rather than occupying every distant body. Label font size is bounded to keep very large enemies from dominating the view.

Output / distribution role: Pixels, layout, animation budget, verification result or release decision.

Trigger / clock: Per presentation/event/release; decorative quality does not alter authoritative hit areas.

Composition requirement: Preserve guardian identification and damage feedback. Information rules use gameplay distance, not random visual sampling.

Boundary / exception: This reduces clutter but does not replace a full accessibility mode. Color alone should not carry rank or danger information.

Acceptance check: Inspect crowded small-screen scenes and distant guardian scenes; confirm injured enemies retain health feedback.

Trace: drawMob [S1]

Function registry: F0254

M116 | Progressive disclosure of secondary tools

STATUS I | implemented baseline

Inputs / units: Finite geometry, view size, device scale, visual preferences, validated state and release evidence.

```
visible_secondary = tools_open; primary controls remain visible
```

Transition / meaning: The candidate places crafting, codex, quests, statistics and account tools behind a More control. Movement, combat stance, map, inventory and pause remain directly reachable.

Output / distribution role: Pixels, layout, animation budget, verification result or release decision.

Trigger / clock: Per presentation/event/release; decorative quality does not alter authoritative hit areas.

Composition requirement: Closing a panel collapses the tools group and resets aria-expanded. Keyboard shortcuts still operate through the existing guarded input path.

Boundary / exception: Controls must remain discoverable and usable on touch. A hidden button is not a removed feature.

Acceptance check: Test collapsed and expanded states, keyboard focus, small landscape layout and panel close behavior.

Trace: toolsMenuBtn; closeModal; sideBtns [S1]

Function registry: F0274

M117 | Safe-area viewport and touch ownership

STATUS I | implemented baseline

Inputs / units: Finite geometry, view size, device scale, visual preferences, validated state and release evidence.

```
usable_width=viewport-safe_left-safe_right; usable_height=viewport-safe_top-safe_bottom
```

Transition / meaning: The existing responsive shell accounts for device safe areas, uses appropriately sized touch targets and prevents text selection on game controls while allowing intentional text entry.

Output / distribution role: Pixels, layout, animation budget, verification result or release decision.

Trigger / clock: Per presentation/event/release; decorative quality does not alter authoritative hit areas.

Composition requirement: Pointer capture and distinct movement/action ownership prevent one thumb from taking over the other control. Desktop mouse and keyboard retain their own input path.

Boundary / exception: Browser emulation is not a native iOS Safari certification. Input fields must remain selectable for passwords, recovery codes and backups. Chromium testing uses actual Noto CJK fonts; native iOS Safari and hardware timing remain separate unverified surfaces.

Acceptance check: Test portrait and landscape, resizing, two simultaneous contacts, input focus and absence of page overflow.

Trace: viewport CSS; pointer handlers [S1]

M118 | Canvas backing-store cost model

STATUS D | derived analysis

Inputs / units: Finite geometry, view size, device scale, visual preferences, validated state and release evidence.

```
pixels=W_css*H_css*dpr^2; raw_RGBA_bytes=4*pixels
```

Transition / meaning: Doubling device pixel ratio quadruples backing-store pixels and raw color-buffer storage. Multiple layers and browser internal buffers multiply actual memory beyond this lower-level estimate.

Output / distribution role: Pixels, layout, animation budget, verification result or release decision.

Trigger / clock: Per presentation/event/release; decorative quality does not alter authoritative hit areas.

Composition requirement: Cap DPR according to the existing quality mode and keep CSS geometry separate from backing-store dimensions. Texture caching trades one-time work and memory for fewer repeated primitives.

Boundary / exception: Pixel count is not a frame-time benchmark. Device GPU, overdraw, gradients, browser and scene complexity also matter.

Acceptance check: For 390 by 844 CSS pixels compare DPR 1 and 2 pixel counts; measure real frame times separately on representative hardware.

Trace: Derived canvas cost model; [R4]

M119 | Reduced-motion presentation policy

STATUS I | implemented baseline

Inputs / units: Finite geometry, view size, device scale, visual preferences, validated state and release evidence.

```
decorative_animation = 0 when reduced_motion; essential simulation unchanged
```

Transition / meaning: The V5.13 V5.14 baseline disables new panel arrival and meter transitions and freezes new anatomical oscillation when reduced motion is requested. The game remains mechanically playable.

Output / distribution role: Pixels, layout, animation budget, verification result or release decision.

Trigger / clock: Per presentation/event/release; decorative quality does not alter authoritative hit areas.

Composition requirement: Keep collision and cooldown clocks identical across preferences. Future review should inventory pre-existing particles, portals and camera effects separately.

Boundary / exception: This scoped change is not a claim of full WCAG conformance or complete elimination of all legacy motion. W3C SC 2.3.3 is Level AAA guidance.

Acceptance check: Emulate both media preferences and compare animation styles and anatomy phase; verify gameplay numerical invariants remain unchanged.

Trace: reducedMotion; CSS media rule [S1]; [R5,R6]

M120 | Layered verification and release budget

STATUS P | conditional proposed extension

Inputs / units: Finite geometry, view size, device scale, visual preferences, validated state and release evidence.

```
accept = syntax and numerical and account and visual and deployment checks
```

Transition / meaning: The release gate combines source and ruleset generation checks, numerical contracts, account/save isolation, worker compilation, browser input/layout regression, visual RNG and collider invariants, and handbook source coverage. Each evidence type has its own scope.

Output / distribution role: Pixels, layout, animation budget, verification result or release decision.

Trigger / clock: Per presentation/event/release; decorative quality does not alter authoritative hit areas.

Composition requirement: Pin the source revision, record numerical seeds, preserve failure artifacts and compare the deployed health version with the reviewed commit.

Boundary / exception: Passing synthetic tests cannot prove every device or all future interactions. Declare gaps and add evidence when a new subsystem changes.

Acceptance check: Require at least 120 uniquely numbered catalog entries, complete fields, English-only PDF text, 30 or more pages and full rendered-page inspection for this monograph.

Trace: Proposed ongoing engineering contract

Contracts M121-M130

M121 | Bounded batch stock-output dynamic programming

STATUS I | implemented baseline

Inputs / units: Finite probability/profile inputs, stock/output state, elemental copies and presentation parameters in the declared domains.

```
P_next(n-5,k+1)+=p*P(n,k)
P_next(n-j,k)+=(1-p)/4*P(n,k), j=1..4
If n<5, retain state; run at most5 attempts
```

Transition / meaning: The exact joint distribution tracks remaining materials and produced outputs. The displayed batch success probability is sum of mass with output>=1; expected output and remaining stock are separate moments.

Output / distribution role: Versioned profile/analysis result, bounded slow or decorative material output.

Trigger / clock: Per eligible batch, profile generation, simulation expiry or render call as specified.

Composition requirement: The same per-attempt chance, five-input success cost, uniform failure loss and stock eligibility used by craftBatch feed this pure analyzer. Protection blocks execution; protected analyses are explicitly hypothetical.

Boundary / exception: Finite-stock first success, bounded batch success, expected consumed input and replenished mean have different stopping rules. The95-percent stock budget uses first-success DP, not a five-attempt guarantee.

Acceptance check: Compare forward DP with independent recursive enumeration for stocks0,4,5,8,12,25; all mass sums to one and stock is nonnegative.

Trace: craftAnalysis; craftCDF; renderCraftAnalysis; craftBatch [S1]

Function registry: F0298, F0323, F0585, F0586

M122 | Frost refresh, bounded slow and guardian resistance

STATUS I | implemented baseline

Inputs / units: Finite probability/profile inputs, stock/output state, elemental copies and presentation parameters in the declared domains.

```
s=min(.35,.12*c); factor=1-s*(guardian?.5:1)
expiry=simulation_time+1.2 seconds
```

Transition / meaning: A successful frost melee batch refreshes one target slow timer. Slow affects ordinary movement and charging displacement but does not change attack cooldowns or telegraph durations.

Output / distribution role: Versioned profile/analysis result, bounded slow or decorative material output.

Trigger / clock: Per eligible batch, profile generation, simulation expiry or render call as specified.

Composition requirement: Count surviving ready frost petals after the contact batch, with at least one qualifying source for the batch even if it just broke. Refresh overwrites rather than adding durations; guardian reduction is multiplicative on slow magnitude.

Boundary / exception: Slow is not stun and not an element-resistance matrix. Max ordinary slow35percent, guardian17.5percent. Expired slow has factor1. Region/respawn state is transient, not a stored permanent debuff.

Acceptance check: Check0/1/3/10 copies, guardian half-strength, exact expiry boundary and a live update displacement scenario.

Trace: slowFactor; activeCopies; update; anatomy [S1]

Function registry: F0217, F0222, F0291, F0591, F0620

M123 | Ruleset generation and source conformance

STATUS I | implemented baseline

Inputs / units: Finite probability/profile inputs, stock/output state, elemental copies and presentation parameters in the declared domains.

```
canonical JSON -> generated config literal -> frozen runtime object
check(config.version == package.version)
```

Transition / meaning: docs/standard/ruleset.json owns deployed critical, frost, event and illustrative utility profile values. The build script emits the runtime literal; --check rejects divergence.

Output / distribution role: Versioned profile/analysis result, bounded slow or decorative material output.

Trigger / clock: Per eligible batch, profile generation, simulation expiry or render call as specified.

Composition requirement: Every probability/profile coefficient revision changes the ruleset version, then regeneration and source/model checks run. Utility weights are examples and do not automatically control the player.

Boundary / exception: Deep freeze prevents accidental mutation, not malicious client modification. The remaining legacy base stats are source-bound tables; this first standard module does not claim full server-authoritative economy.

Acceptance check: Run build-rules --check; deliberately change one config coefficient in a disposable copy and require mismatch detection.

Trace: build-rules; config; streams [S1]

Function registry: F0575

M124 | Regional material and light compositing

STATUS I | implemented baseline

Inputs / units: Finite probability/profile inputs, stock/output state, elemental copies and presentation parameters in the declared domains.

```
C_out=alpha*C_light+(1-alpha)*C_scene
phase=0 if reduced_motion else f(time,identity)
```

Transition / meaning: Cached512pixel regional textures, landmarks, obstacle overlays and directional light layers improve material identity without changing collision proxies. Species and guardian anatomy use bounded strokes and body-aligned decoration.

Output / distribution role: Versioned profile/analysis result, bounded slow or decorative material output.

Trigger / clock: Per eligible batch, profile generation, simulation expiry or render call as specified.

Composition requirement: World geometry remains authoritative; appendages, light pools and decorative ponds add no blockers. Material generation has a local deterministic stream; low quality bypasses terrain/lighting and fine rock detail.

Boundary / exception: Gradients are an artistic appearance model, not physical radiometry. Reduced motion fixes decorative phase, but legacy portal/particle effects retain separately declared scope.

Acceptance check: Render43 species at three sizes, all8 biomes, low/high and reduced motion; compare collider objects and nonvisual RNG counters before/after drawing.

Trace: terrain; landmark; anatomy; lighting; rock; creatureVisualProfile; drawWorld [S1]

Function registry: F0238, F0252, F0617, F0619, F0620, F0621, F0622

M125 | Outcome-wise post-armor critical moments

STATUS I | implemented baseline

Inputs / units: Raw damage X >=0; armor A>=0; remaining HP H>=0; intact Prism copies c. Damage/HP units.

```
Y = min(H, max(1, X*C-A)); C in {1,1.6}; E[Y] = (1-p)y0+p*y1; Var(Y)=p(1-p)(y1-y0)^2
```

Transition / meaning: Single already-eligible melee hit; raw damage uses current durability, talents, assault and overdrive. Armor and overkill are evaluated before averaging.

Output / distribution role: Mean/variance, attainable range and single-hit kill probability.

Trigger / clock: Frozen analysis state; simulation seconds for attack/status timing; presentation-only pixels for rendering.

Composition requirement: Freeze current state; branch critical; subtract armor; floor1; cap lost HP; weight. No time, trajectory or subsequent exchange approximation.

Boundary / exception: Zero HP loses no more HP; raw 0 still floors1 if HP>0; no nonfinite inputs. Never substitute max(1,E[XC]-A).

Acceptance check: Independent armor-floor and HP-cap examples; compare against actual damageMob branches.

Trace: site/rules.js:damageMoments; site/index.html:rawPetalDamage, armorOf, renderDamageForecast

Function registry: F0200, F0219, F0325, F0600

M126 | Exact target-drop union probability

STATUS I | implemented baseline

Inputs / units: Nonempty family list, enemy rank0..7, guardian/treasure flags; exact family and target rank0..9.

```
Pr(target at exact rank k) = sum_r Pr(R=r) * [g(r) ? 1 : a(r)+(1-a(r))*b(r)]
```

Transition / meaning: Primary, fixed guardian and optional Basic channels share rank. Duplicate drop-family entries retain their selection multiplicity.

Output / distribution role: Probability of at least one target item, conditional on defeat.

Trigger / clock: Frozen analysis state; simulation seconds for attack/status timing; presentation-only pixels for rendering.

Composition requirement: Enumerate rank; force guardian target if present; otherwise union independent primary choice and 0.18 Basic channel; sum rank mass.

Boundary / exception: Ranks8..9 return0 under existing channels; no travel, survival, collection, inventory or craft chance included.

Acceptance check: Independent enumeration of main family, shared rank and extra-drop Bernoulli across 768 rank/event/target combinations.

Trace: site/rules.js:targetDropChance; site/index.html:renderProbabilityLab

Function registry: F0327, F0608

M127 | Bounded batch production histogram

STATUS I | implemented baseline

Inputs / units: Independent attempt success p in[0,1], integer stock0..99999, attempt cap0..5.

```
h[j] = sum_s Pr(stock=s, outputs=j); E[outputs]=sum_j j*h[j]; Pr(any)=1-h[0]
```

Transition / meaning: Marginalize the existing exact stock/output DP after at most5 attempts. Stopped states retain their mass.

Output / distribution role: Output probabilities0..attemptCap, mean, at-least-one chance.

Trigger / clock: Frozen analysis state; simulation seconds for attack/status timing; presentation-only pixels for rendering.

Composition requirement: Success consumes5 and adds1 output; failure consumes uniform1..4; stock<5 is absorbing.

Boundary / exception: p=0, p=1, stock<5 and zero attempts remain normalized. A probability budget is not a guarantee.

Acceptance check: All bin masses sum1; first moment equals existing outputs; p1/stock25 gives exactly5.

Trace: site/rules.js:craftAnalysis; site/index.html:renderCraftDistribution

Function registry: F0321, F0586

M128 | Time-composable presentation smoothing

STATUS I | implemented baseline

Inputs / units: Gain g in[0,1], dt>=0 seconds; camera gain8/60 and orbital gain0.25.

```
alpha(g,dt) = 1-(1-g)^(60*dt); x_next = x+(target-x)*alpha
```

Transition / meaning: Retain the legacy 60Hz gain while giving exact constant-target composition over subdivided time.

Output / distribution role: Stable interpolation alpha in[0,1].

Trigger / clock: Frozen analysis state; simulation seconds for attack/status timing; presentation-only pixels for rendering.

Composition requirement: Evaluate with -expm1(60*dt*log1p(-g)); apply to camera and petal orbit targets.

Boundary / exception: dt0 alpha0; g0 alpha0; g1 alpha1 for dt>0. Moving targets and collision feedback do not imply frame-independent gameplay.

Acceptance check: Compare two half-steps to one step for fixed target; exact old gain at dt1/60.

Trace: site/rules.js:frameAlpha; site/index.html:update

Function registry: F0222, F0291, F0611

M129 | Shared hostile special and telegraph profiles

STATUS I | implemented baseline

Inputs / units: Special type; enemy level,rank,speed; current rage and frost state; current player position and velocity.

```
nova radius210; charge reach=v*3*0.52*enrage*frost; projectile reach=(base+min(120,1.7*level))*2.3
```

Transition / meaning: Shared coefficients drive runtime windup/projectiles/nova and the preview. Sniper uses a capped linear lead approximation, not an exact intercept solver.

Output / distribution role: Windup, speed, duration, spread, predicted path and radius.

Trigger / clock: Frozen analysis state; simulation seconds for attack/status timing; presentation-only pixels for rendering.

Composition requirement: Compute profile; render progress and ray/corridor; at release recompute current aim. Obstacles can shorten paths.

Boundary / exception: Prediction may move during windup; charge uses current speed snapshot; future status expiry can change reach. Nova tests player center distance; render does not add collision.

Acceptance check: Actual spawned projectile speed/lifetime matches profile; canvas nova ring has actual210 radius.

Trace: site/rules.js:specialProfile,specialAim; site/index.html:performMobSpecial; site/art.js:telegraph

Function registry: F0164, F0612, F0613, F0626

M130 | Scale-aware material and status presentation

STATUS I | implemented baseline

Inputs / units: Body rx,ry,r; zone palette; status deadlines; quality and reduced-motion state.

```
body-detail enabled iff quality != low and projectedRadius>=17px; overlay geometry scales with
authoritative body dimensions
```

Transition / meaning: Decorative shell traces are clipped to body ellipse. Frost, poison and shield retain distinct ring geometry. Regional vignette uses deterministic gradients.

Output / distribution role: Canvas pixels only; no reward, seed, state or collider writes.

Trigger / clock: Frozen analysis state; simulation seconds for attack/status timing; presentation-only pixels for rendering.

Composition requirement: Base silhouette; clipped material; anatomy; eyes; world status rings; shared telegraph; screen light.

Boundary / exception: Low/small sprites omit shell details. Color is supplemented by broken-ring structure and status pip counts. Native device color profiles remain unverified.

Acceptance check: 43 species at3 sizes; four viewports; collider/state immutability and RNG isolation; screenshots.

Trace: site/art.js:surface,status,lighting; site/index.html:drawCreatureModel,drawMob

Function registry: F0253, F0254, F0621, F0624, F0625

Contracts M131-M138

M131 | Normalized build preferences

STATUS I | implemented baseline

Inputs / units: Owned item keys, finite nonnegative features and one registered preference.

```
z[k,j] = f[k,j]/max_available(f[:,j]); zero maximum gives z=0; s[k] = sum_j w[j]*z[k,j]
```

Transition / meaning: The three features are base damage times rarity scale, base durability times scale divided by max(0.1,reload seconds), and rarity scale for a declared support-family tag, otherwise zero. Weights are attack=(.8,.15,.05), sustain=(.15,.35,.5), balanced=(.4,.3,.3). Normalize over the union of backpack and current deck keys.

Output / distribution role: Per-item dimensionless score and normalized weight tuple.

Trigger / clock: User decision and browser presentation clock; gameplay cooldowns use simulation seconds.

Composition requirement: Compute the same feature vector for preview and application; preference coefficients are not random probabilities.

Boundary / exception: This additive preference ignores hit frequency, synergies, health penalties, crits, slows and enemy armor. It is not DPS or a global combat optimum.

Acceptance check: Check all-zero feature domains, each policy and stable repeated previews.

Trace: PetalWorkshop.plan; buildWorkshopPlan [S15]

Function registry: F0333, F0528

M132 | Inventory-constrained greedy build

STATUS I | implemented baseline

Inputs / units: Bag integer counts 0..99999, current deck and feature registry.

```
n[k]=bag[k]+count_current(k); maximize sum_k x[k]*s[k] subject to 0<=x[k]<=n[k], sum_k x[k]=N
```

Transition / meaning: Sort by descending per-copy additive score, then ascending ASCII item key; take the available copies until N slots are filled. The exchange argument proves optimality for this additive model: replacing any selected lower score by an available higher score cannot decrease the objective.

Output / distribution role: Full N-slot proposal, remaining bag and before/after scores.

Trigger / clock: User decision and browser presentation clock; gameplay cooldowns use simulation seconds.

Composition requirement: The backup deck is excluded from the pool. Exactly N current slots are filled, N in 1..10. No RNG is consumed by planning.

Boundary / exception: An optimum of this preference model need not win a battle. Equal-score selection is deterministic and may rearrange the current deck.

Acceptance check: Enumerate small bounded candidate pools and compare with exhaustive feasible allocations; assert preview purity.

Trace: PetalWorkshop.plan [S15]

Function registry: F0528

M133 | Atomic equipment conservation

STATUS I | implemented baseline

Inputs / units: Proposed bag/deck and current owner/view signature.

```
bag_new[k] + count_new(k) = bag_old[k] + count_old(k)
```

Transition / meaning: Plan from a copy of inventory, preserve the backup deck, and check an owner/deck/bag/backup signature after confirmation. Commit the proposal only when the signature still matches. Reject any resulting stack above99999 rather than clamp away equipment.

Output / distribution role: Either unchanged state or one conserved replacement.

Trigger / clock: User decision and browser presentation clock; gameplay cooldowns use simulation seconds.

Composition requirement: All current slots are exchanged together; no intermediate takeBag/addBag sequence is exposed.

Boundary / exception: This is a synchronous client transaction during a paused view. It is not a server-authoritative anti-cheat transaction or a cross-device lock.

Acceptance check: Check each item key, cancellation, stale identity and unchanged backup; test a nearly-full returning stack.

Trace: workshopIdentity; renderBuildWorkshop; PetalWorkshop.plan [S15]

Function registry: F0336, F0337, F0528

M134 | Build application reload and health clamp

STATUS I | implemented baseline

Inputs / units: Confirmed full deck and current nonnegative player HP.

```
cd_i=2.5 s; hp_i=0; player_HP_new=min(player_HP_old,maxHP(new_deck))
```

Transition / meaning: Applying a build uses the established equip reload path, recomputes synergies, view scale and HUD, then saves. It does not grant healing or immediate ready petals.

Output / distribution role: Reloading equipment and clamped current health.

Trigger / clock: User decision and browser presentation clock; gameplay cooldowns use simulation seconds.

Composition requirement: New cooldowns advance only after leaving the paused panel. HP cannot rise merely because the new deck has more maximum health.

Boundary / exception: resetEquip also initializes shotWait using the world stream, as ordinary equip does. Only preview is RNG-free.

Acceptance check: Assert all reloads equal2.5 at commit, HP nonincrease and persisted current/backup compatibility.

Trace: resetEquip; renderBuildWorkshop; maxHP [S03]

Function registry: F0131, F0167, F0337

M135 | Gate-entry announcement lifecycle

STATUS I | implemented baseline

Inputs / units: Gate visibility and user dismissal events.

```
gate_visible_entry -> welcome_open -> dismiss -> optional_session_restore
```

Transition / meaning: Every initial load and transition back to the gate opens the developer/update announcement. Manual reopen is available. First session restoration waits for dismissal; a hidden gate closes a pending welcome. Repeated calls while open share the same close promise.

Output / distribution role: One current welcome view and a resolved dismissal promise.

Trigger / clock: User decision and browser presentation clock; gameplay cooldowns use simulation seconds.

Composition requirement: Gate visibility owns welcome presentation. Changing authentication tabs does not create a new gate entry. Authentication and mode controls are reserved until initial restoration settles, preventing a post-dismissal mode race.

Boundary / exception: The developer text describes the project without inventing personal biography. No forever-hide preference suppresses subsequent gate entries.

Acceptance check: Check first load, manual reopen, Escape, logout, focus and restored-session ordering.

Trace: PetalUI.install; PetalUI.welcome; logout [S16]

Function registry: F0403, F0549, F0553

M136 | Validated announcement freshness

STATUS I | implemented baseline

Inputs / units: Untrusted same-origin JSON response and trusted bundled fallback.

```
valid(feed) -> render_text(feed); fetch_failure -> retain(last_good or bundled)
```

Transition / meaning: Fetch same-origin release.json without cache, with a5-second abort, maximum16384 decoded characters, schema1, numeric semantic version, ISO UTC timestamp and1..20 bounded highlights. Render text using textContent, never feed HTML.

Output / distribution role: Sanitized release object, fetch timestamp and honest status.

Trigger / clock: User decision and browser presentation clock; gameplay cooldowns use simulation seconds.

Composition requirement: Manual reopen/refresh and returning visibility request current published content. The bundled object must exactly match the checked-in release JSON at build time.

Boundary / exception: This is current published release information, not developer presence or an unimplemented roadmap feed. A failed fetch visibly names cached/bundled fallback.

Acceptance check: Reject malformed schemas, markup, oversized fields and invalid dates; exercise HTTP failure and successful refresh.

Trace: PetalWorkshop.release; PetalUI.refresh; PetalUI.paintNews [S16]

Function registry: F0540, F0545, F0546

M137 | Guarded asynchronous confirmation

STATUS I | implemented baseline

Inputs / units: Operation description, decision and captured owner/view state.

```
apply iff accepted AND owner,account,active,view,player,conflict identities remain equal
```

Transition / meaning: A native dialog offers cancel as the initial focus. Escape rejects. The suspended operation resumes only if the account, player object and view still match the captured state. A second pending request rejects instead of replacing the active decision.

Output / distribution role: Boolean acceptance bound to a stable operation context.

Trigger / clock: User decision and browser presentation clock; gameplay cooldowns use simulation seconds.

Composition requirement: Apply this boundary to bulk crafting, talent reset, imports, restore, migration, conflict resolution and build application. A pre-existing conflict may be explicitly resolved; a newly changed conflict invalidates a stale action.

Boundary / exception: Confirmation does not replace server ownership, revision checks or payload validation.

Acceptance check: Test cancellation without mutation, initial cancel focus, stale owner and accepted guarded commit.

Trace: askGame; PetalUI.ask [S16]

Function registry: F0332, F0551

M138 | Panel focus and presentation isolation

STATUS I | implemented baseline

Inputs / units: Viewport, focus, input event and panel state.

```
panel_open -> pause AND resetInputs AND inert(background); close -> resetInputs AND restore(canvas_focus)
```

Transition / meaning: Shared panel header, navigation, materials, spacing and device breakpoints apply to inventory, equipment, crafting, skills, talents, stats, journal, map, codex, probability, preferences and help. Tab cycles visible panel controls; native top-layer dialogs own their own focus.

Output / distribution role: Paused, navigable panel or unobstructed gameplay view.

Trigger / clock: User decision and browser presentation clock; gameplay cooldowns use simulation seconds.

Composition requirement: Reduced-motion media disables entry animation. Native input fields remain selectable. Decorative brand rings cannot intercept pointer input.

Boundary / exception: These are presentation and interaction contracts. They do not change colliders, reward weights or authentication requirements. Hardware Safari remains outside synthetic Chromium coverage.

Acceptance check: Inspect all14 panel types at five viewports including320x568 and568x320, with actual CJK fonts; verify body overflow and keyboard restoration.

Trace: showModal; closeModal; renderModal; PetalUI.install [S08]

Function registry: F0273, F0274, F0346, F0553

Part III. Numeric parameter registry

The following parameter objects are freshly extracted from the frozen source, not inferred from colors or displayed names. Pets use base hp/dmg (health units), cd (seconds) and size (world units). Mobs use base hp/dmg, speed (world units/second), size (world units), ai policy selector and drop family list. Runtime modifiers in the source ledger apply after these base parameters. Regions use integer identifiers/bands/ports and base numeric factors in their named source fields. Rarity rows use zero-based rank, multiplicative scale and craft probability; terminal rank is not a craft input.

Pets

Key	Exact numeric / policy fields
basic	{"hp":34,"dmg":13,"cd":2.1,"size":16}
rose	{"hp":30,"dmg":6,"cd":3.4,"size":17}
leaf	{"hp":40,"dmg":7,"cd":2,"size":18}
stinger	{"hp":11,"dmg":31,"cd":4.3,"size":13}
cactus	{"hp":60,"dmg":11,"cd":3.3,"size":20}
bubble	{"hp":27,"dmg":9,"cd":1.9,"size":19}
sand	{"hp":38,"dmg":16,"cd":2.2,"size":16}
claw	{"hp":49,"dmg":19,"cd":2.7,"size":17}
iris	{"hp":34,"dmg":18,"cd":2.7,"size":17}
lightning	{"hp":26,"dmg":24,"cd":3.1,"size":17}
gear	{"hp":80,"dmg":21,"cd":3.5,"size":18}
fang	{"hp":35,"dmg":19,"cd":2.4,"size":17}
magma	{"hp":58,"dmg":29,"cd":3.7,"size":20}
shell	{"hp":73,"dmg":12,"cd":3,"size":19}
antenna	{"hp":41,"dmg":21,"cd":2.5,"size":16}
poison	{"hp":32,"dmg":14,"cd":2.7,"size":17}
starpetal	{"hp":44,"dmg":9,"cd":3,"size":19}
nectar	{"hp":24,"dmg":4,"cd":2.3,"size":16}
missile	{"hp":26,"dmg":23,"cd":2.9,"size":16}
grape	{"hp":37,"dmg":15,"cd":2.6,"size":19}
puff	{"hp":32,"dmg":6,"cd":2.2,"size":18}
bloom	{"hp":48,"dmg":16,"cd":2.4,"size":19}
glass	{"hp":34,"dmg":22,"cd":2.5,"size":17}
root	{"hp":44,"dmg":7,"cd":3,"size":18}
soil	{"hp":54,"dmg":10,"cd":2.8,"size":20}
wing	{"hp":22,"dmg":19,"cd":3,"size":17}
magnet	{"hp":44,"dmg":7,"cd":2.6,"size":18}
prism	{"hp":30,"dmg":16,"cd":2.8,"size":17}
frost	{"hp":39,"dmg":10,"cd":2.6,"size":18}

Mobs

Key	Exact numeric / policy fields
ladybug	{"hp":63,"dmg":8,"speed":66,"size":22,"ai":"wander","drop":["basic","rose","nectar"]}
bee	{"hp":48,"dmg":13,"speed":107,"size":19,"ai":"rush","drop":["stinger","rose"]}
ant	{"hp":45,"dmg":8,"speed":103,"size":16,"ai":"rush","drop":["basic","antenna"]}
spider	{"hp":82,"dmg":15,"speed":100,"size":23,"ai":"chase","drop":["poison","iris"]}
centipede	{"hp":96,"dmg":13,"speed":86,"size":23,"ai":"chase","drop":["basic","cactus","soil"]}
crab	{"hp":115,"dmg":18,"speed":96,"size":28,"ai":"rush","drop":["claw","shell"]}
jelly	{"hp":73,"dmg":12,"speed":67,"size":26,"ai":"ranged","drop":["lightning","bubble"]}
starfish	{"hp":95,"dmg":13,"speed":58,"size":24,"ai":"chase","drop":["bubble","shell","starpetal","grape"]}
bubble	{"hp":45,"dmg":5,"speed":69,"size":22,"ai":"wander","drop":["bubble","leaf","starpetal"]}
urchin	{"hp":118,"dmg":18,"speed":51,"size":24,"ai":"chase","drop":["shell","stinger"]}
scorpion	{"hp":95,"dmg":17,"speed":98,"size":25,"ai":"chase","drop":["stinger","sand"]}
beetle	{"hp":105,"dmg":14,"speed":85,"size":24,"ai":"rush","drop":["cactus","sand"]}
scarab	{"hp":140,"dmg":17,"speed":70,"size":26,"ai":"chase","drop":["sand","shell","glass"]}
fireAnt	{"hp":69,"dmg":14,"speed":126,"size":18,"ai":"rush","drop":["antenna","magma"]}
mantis	{"hp":106,"dmg":21,"speed":103,"size":26,"ai":"chase","drop":["leaf","claw","root"]}
dragonfly	{"hp":67,"dmg":14,"speed":145,"size":19,"ai":"ranged","drop":["iris","leaf","grape","wing"]}
wasp	{"hp":85,"dmg":21,"speed":122,"size":20,"ai":"ranged","drop":["stinger","poison","missile"]}
helBeetle	{"hp":180,"dmg":27,"speed":122,"size":27,"ai":"rush","drop":["magma","fang"]}
helSpider	{"hp":166,"dmg":26,"speed":113,"size":25,"ai":"chase","drop":["poison","magma"]}
helWasp	{"hp":138,"dmg":25,"speed":122,"size":24,"ai":"ranged","drop":["magma","stinger","missile"]}
ember	{"hp":110,"dmg":21,"speed":100,"size":21,"ai":"ranged","drop":["magma","iris"]}
soldier	{"hp":115,"dmg":18,"speed":124,"size":22,"ai":"chase","drop":["antenna","fang","glass"]}
queenAnt	{"hp":280,"dmg":22,"speed":55,"size":33,"ai":"ranged","drop":["antenna","cactus","soil"]}
termite	{"hp":75,"dmg":11,"speed":101,"size":19,"ai":"rush","drop":["antenna","leaf"]}
rat	{"hp":110,"dmg":16,"speed":119,"size":23,"ai":"rush","drop":["fang","poison"]}
slime	{"hp":130,"dmg":12,"speed":56,"size":26,"ai":"chase","drop":["bubble","poison"]}
mosquito	{"hp":53,"dmg":17,"speed":148,"size":16,"ai":"ranged","drop":["stinger","fang","wing"]}
drone	{"hp":105,"dmg":19,"speed":117,"size":23,"ai":"ranged","drop":["gear","lightning","magnet"]}
mechaSpider	{"hp":180,"dmg":21,"speed":102,"size":26,"ai":"chase","drop":["gear","poison"]}
mechaWasp	{"hp":120,"dmg":23,"speed":134,"size":22,"ai":"ranged","drop":["gear","stinger","missile"]}
barrel	{"hp":180,"dmg":13,"speed":31,"size":24,"ai":"wander","drop":["gear","shell"]}
mechaFlower	{"hp":225,"dmg":26,"speed":70,"size":28,"ai":"ranged","drop":["gear","lightning","magnet"]}
guardianGarden	{"hp":480,"dmg":28,"speed":84,"size":45,"ai":"rush","guardian":true,"drop":["leaf","rose","cactus"]}
guardianOcean	{"hp":620,"dmg":31,"speed":86,"size":47,"ai":"chase","guardian":true,"drop":["starpetal","claw","shell"]}
guardianDesert	{"hp":700,"dmg":34,"speed":91,"size":45,"ai":"rush","guardian":true,"drop":["sand","stinger","cactus"]}
guardianJungle	{"hp":730,"dmg":39,"speed":112,"size":44,"ai":"chase","guardian":true,"drop":["iris","poison","leaf"]}
guardianHel	{"hp":1100,"dmg":48,"speed":83,"size":52,"ai":"ranged","guardian":true,"drop":["magma","fang","lightning"]}

Key	Exact numeric / policy fields
guardianAnt	{"hp":830,"dmg":37,"speed":67,"size":51,"ai":"ranged","guardian":true,"drop":["antenna","cactus","fang"]}
guardianSewer	{"hp":770,"dmg":33,"speed":88,"size":46,"ai":"rush","guardian":true,"drop":["poison","fang","bubble"]}
guardianFactory	{"hp":1080,"dmg":43,"speed":77,"size":53,"ai":"ranged","guardian":true,"drop":["gear","lightning","shell"]}
wildFlower	{"hp":110,"dmg":12,"speed":49,"size":24,"ai":"wander","drop":["bloom","leaf","puff"]}
healerBloom	{"hp":170,"dmg":15,"speed":72,"size":27,"ai":"ranged","drop":["rose","nectar","bloom"]}
sporeBloom	{"hp":210,"dmg":23,"speed":68,"size":28,"ai":"ranged","drop":["poison","puff","bloom","root"]}

Zones

Key	Exact numeric / policy fields
garden	{"drop":0,"ports":[["ocean",2230,700],["desert",610,1480],["jungle",2090,1430],["ant",1260,1510],["sewer",1240,460],["factory",2190,350]],"seed":122}
ocean	{"drop":1,"ports":[["garden",340,690],["jungle",2040,1400]],"seed":213}
desert	{"drop":1,"ports":[["garden",1940,380],["ant",580,1420],["hel",2130,1400]],"seed":314}
jungle	{"drop":2,"ports":[["garden",310,420],["ocean",2100,420],["hel",1240,1550]],"seed":415}
hel	{"drop":3,"ports":[["desert",360,530],["jungle",1220,260],["factory",2100,1280]],"seed":516}
ant	{"drop":1,"ports":[["garden",1260,280],["desert",2030,1330],["sewer",360,1300]],"seed":617}
sewer	{"drop":1,"ports":[["garden",1250,280],["factory",2120,900],["ant",380,1350]],"seed":718}
factory	{"drop":2,"ports":[["garden",380,500],["sewer",380,1290],["hel",2120,950]],"seed":819}

Rarity

Key	Exact numeric / policy fields
0	{"en":"common","scale":1,"chance":0.64}
1	{"en":"unusual","scale":1.72,"chance":0.32}
2	{"en":"real","scale":2.9583999999999997,"chance":0.16}
3	{"en":"epic","scale":5.088448,"chance":0.08}
4	{"en":"legend","scale":8.75213056,"chance":0.04}
5	{"en":"mythic","scale":15.053664563199998,"chance":0.02}
6	{"en":"ultra","scale":25.892303048703997,"chance":0.01}
7	{"en":"super","scale":44.53476124377087,"chance":1}
8	{"en":"unique","scale":76.5997893392859,"chance":0.0025}
9	{"en":"eternal","scale":131.75163766357176,"chance":0.00125}

Parameter completeness beyond base objects

Every numeric literal occurrence is registered in Appendix E with function owner, value, count and source line. Every derived or conditional parameter is preserved in Appendix B expression envelopes and Appendix C numerical calls. Chapters 5-8 supply the combat, special AI, event and reset interpretation needed to avoid treating a literal value as an unconditional buff. Numeric SQL guards and CSS computations have separate ledgers.

Probability-weighted worked examples

A finite budget must be evaluated under the actual eligibility rule: fewer than five inputs cannot make another attempt. The following curves use the exact first-success dynamic program, not a normal approximation or an expected-cost budget.

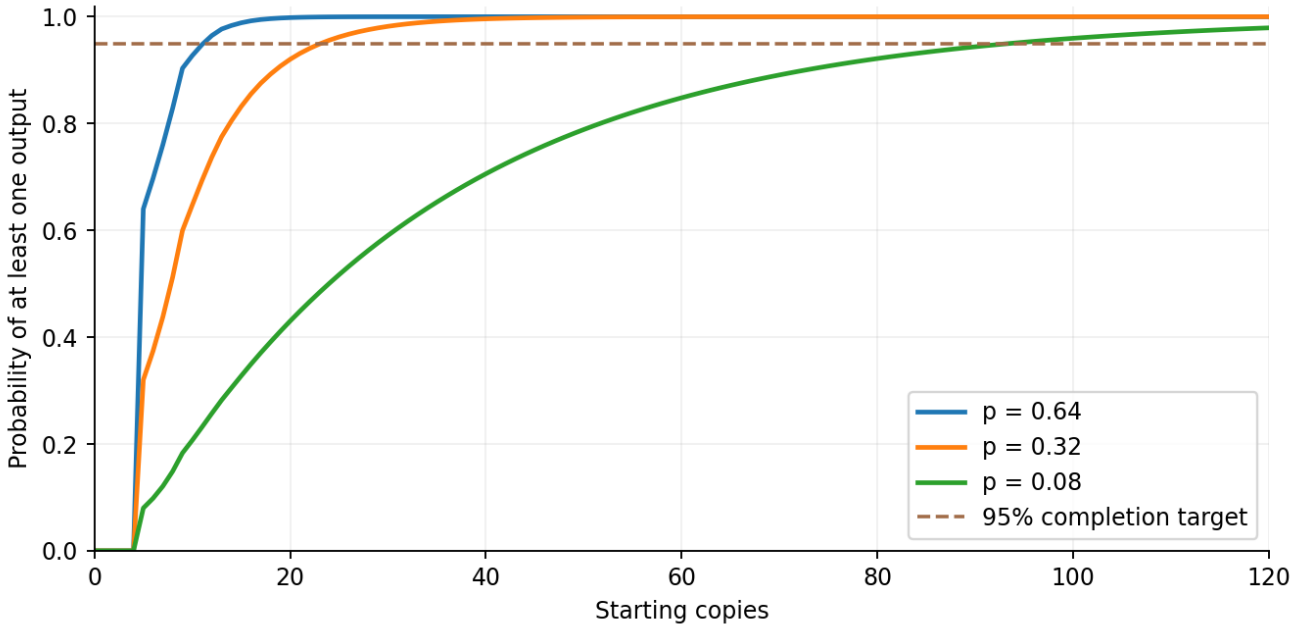


Figure 1. Exact completion chance from F_p(n). The 95% target is a choice of service level, not a guarantee. All curves assume independent, constant success probability and uniform failure loss.

Scenario probabilities and time exposure

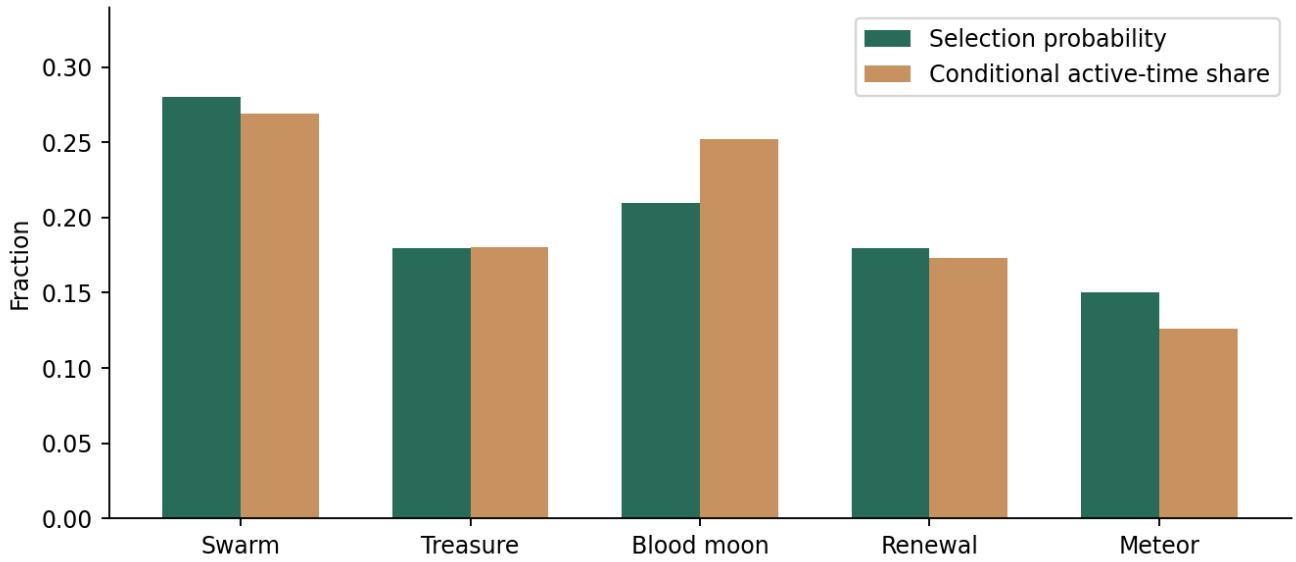


Figure 2. Event selection and conditional active-time composition differ because durations differ. This stationary continuous-time illustration excludes quiet gaps from the conditional active shares; unconditional active fractions include the mean 69-second quiet gap and fixed ticks add small timing discrepancies.

Tactical utility: a complete small example

Use an illustrative normalized feature profile alpha=(.40,.30,.20,.10) for progress, survival, effective damage and mobility. Policy A has a .75 safe outcome with features (.4,1,.6,.5) and a .25 death outcome with features (.8,0,.9,.2). Policy B has one sure outcome (.45,1,.5,.6). These numbers are an original explanatory scenario, not live balance coefficients.

Outcome	Probability	Weighted benefit Z
A safe	.75	.40*.4 + .30*1 + .20*.6 + .10*.5 = .63
A death	.25	.40*.8 + .30*0 + .20*.9 + .10*.2 = .52
B sure	1	.40*.45 + .30*1 + .20*.5 + .10*.6 = .64

```
E[Z_A]=.75*.63+.25*.52=.6025
E[Z_B]=.64
Under default linear utility, B has greater expected benefit.
If the death constraint is epsilon=.10, A is also infeasible.
```

The probabilities multiply full outcome utilities; utility coefficients combine normalized features inside each outcome. They are different levels of weighting. A future runtime planner must derive outcome probabilities from its tactical state model, not assume this tiny example applies to all enemies. Positive progress in a death outcome does not imply a current game death penalty: the example feature values are declared scenario assumptions.

Validation and worked calculation evidence

243 fresh independent reference-model assertions passed, plus 66 existing JavaScript assertions and 7 new grouped tactical-atlas regressions (including 768 independently enumerated drop cases). This is model and source-contract evidence. The build/feed suite adds 218 assertions, 48 exhaustive allocations, an overflow boundary and 6 asynchronous guard cases; 245 browser assertions cover five viewports and 14 panel types. These counts are not a new live deployment or a full browser/production certification. Test cases and source-oracle results are included in the source archive.

Check family	Evidence / interpretation
Finite crafting	Normalized distributions, exact finite-stock recursion, forward stock/output DP and independent Monte Carlo
Combat exchange	Requested/mitigated/effective separation; durability truncation and special glass path
Reward covariance	Joint shared-rank reward enumerator; guardian channels are not independent marginals
Utility	Outcome-wise increasing utility, Jensen inequality and mean-variance dominance counterexample
Source oracle	JavaScript extracted drop/event functions and rounding boundaries; 66 additional JS module assertions and actual browser elemental/reset checks
Coverage	All operator nodes belong to a maximal envelope; all calls have registered owners; all identified RNG aliases resolved


```
Contact example: {"applied": 26.647058823529413, "effective": 26.647058823529413, "mob_hp":
73.3529411764706, "petal_hp": 0, "requested": 34.411764705882355, "strikes": 5}

Guardian illustrative reward value moments: {"mean": 28.480977647446455, "variance": 181.84729377564545}
```


The guardian reward moments use the reference valuation and declared example configuration in validate_model.py. They are not measured production currency income. Monte Carlo confidence bounds test the stochastic implementation; exact enumeration remains the reference where available.

Reproducibility and falsification

Run validate_model.py to regenerate the recorded assertions, then build_handbook.py from the archived inventory. The optional source oracle requires the external frozen checkout at the declared commit and Node; extract.cjs additionally uses the Acorn parser bundled with Node v24.19.0. No repository checkout, live credentials or player save data are included in the archive. To audit a changed commit, regenerate the AST inventory and review every inventory difference rather than reusing this edition IDs blindly.

References and evidence sources

S1

Petal Realms frozen repository main commit 5b61308db653691d68190b259274e08282058533; files and hashes listed in Reader map. https://github.com/liuzhide666/Petal-Realms (repository identity must be verified against the checkout remote).

R1

Glenn Fiedler. Fix Your Timestep! https://gafferongames.com/post/fix_your_timestep/

R2

PCG project. Random number generator concepts. https://www.pcg-random.org/

R3

NVIDIA. GPU Gems 3, Chapter 32: Broad-Phase Collision Detection with CUDA. https://developer.nvidia.com/gpugems/gpugems3/part-v-physics-simulation/chapter-32-broad-phase-collision-detection-cuda

R4

MDN. Optimizing canvas. https://developer.mozilla.org/en-US/docs/Web/API/Canvas_API/Tutorial/Optimizing_canvas

R5

web.dev. Rendering performance. https://web.dev/articles/rendering-performance

R6

W3C. Media Queries Level 5: prefers-reduced-motion. https://www.w3.org/TR/mediaqueries-5/#prefers-reduced-motion

R7

Cloudflare. D1 documentation. https://developers.cloudflare.com/d1/

R8

NIST/SEMATECH e-Handbook of Statistical Methods. https://www.itl.nist.gov/div898/handbook/

R9

Leiba. RFC 8174 (2017), Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words. https://www.rfc-editor.org/rfc/rfc8174.html

R10

Bradner. RFC 2119 (1997), Key words for use in RFCs to Indicate Requirement Levels. https://www.rfc-editor.org/rfc/rfc2119.html

R11

Ecma International / TC39. ECMAScript Math.random and Math.round semantics, living specification. https://tc39.es/ecma262/multipage/numbers-and-dates.html

Primary normative external sources R9-R11 were consulted for this revision on 20 October 2026 (Asia/Shanghai and UTC). R1-R8 are retained research/context references, not evidence that proposed game rules are deployed. Equations, numeric game derivations and reference algorithms are original project analysis unless attributed. No proprietary Florr formulas or artwork are asserted.

Appendix A. Function and ownership registry

All 626 function nodes, including callbacks. Parent ownership prevents arithmetic in an anonymous callback from being attributed to its outer function. L means source line; api means functions/api/[[path]].js, html means site/index.html; rules/art mean their site JavaScript modules. System tags S01-S14 classify navigation, and counts are lexical nodes, not execution frequencies.

S01 Runtime and input

S02 Geometry and movement

S03 Player, equipment and progression

S04 Combat, skills and effects

S05 Enemies and ecology

S06 World, events and transitions

S07 Inventory, crafting and goals

S08 Presentation and interface

S09 Native account and persistence

S10 Local and legacy compatibility

S11 Legacy presence protocol

S12 Account service and database

S15 Constrained build workshop

S16 Announcement and confirmation

S13 Executable probability standard

S14 Procedural art and lighting

```
F0001 anonymous callback() | html:L238-1546 | parent client1:root | S08 | ops 52 calls 124
F0002 $(id) | html:L239-239 | parent F0001 | S08 | ops 0 calls 1
F0003 clamp(v, a, b) | html:L241-241 | parent F0001 | S02 | ops 0 calls 2
F0004 lerp(a, b, t) | html:L241-241 | parent F0001 | S02 | ops 3 calls 0
F0005 dist(a, b, c, d) | html:L241-241 | parent F0001 | S02 | ops 2 calls 1
F0006 rnd(a, b) | html:L241-241 | parent F0001 | S08 | ops 4 calls 1
F0007 SEED(n) | html:L242-242 | parent F0001 | S06 | ops 0 calls 0
F0008 anonymous callback() | html:L242-242 | parent F0007 | S06 | ops 15 calls 2
F0009 anonymous callback([en,name,hex], i) | html:L243-243 | parent F0001 | S08 | ops 3 calls 2
F0010 talentEarned(level) | html:L305-305 | parent F0001 | S03 | ops 2 calls 2
F0011 sanitizeTalents(raw, level) | html:L306-306 | parent F0001 | S03 | ops 5 calls 12
F0012 anonymous callback(k) | html:L306-306 | parent F0011 | S03 | ops 0 calls 0
F0013 talentRank(key) | html:L307-307 | parent F0001 | S03 | ops 1 calls 1
F0014 talentPoints() | html:L308-308 | parent F0001 | S03 | ops 3 calls 4
F0015 anonymous callback(sum, n) | html:L308-308 | parent F0014 | S03 | ops 1 calls 0
F0016 talentModifier(key) | html:L309-309 | parent F0001 | S03 | ops 2 calls 1
F0017 petalDamageMultiplier() | html:L310-310 | parent F0001 | S03 | ops 0 calls 1
F0018 petalReloadTime(t) | html:L311-311 | parent F0001 | S03 | ops 3 calls 2
F0019 spendTalent(key) | html:L312-312 | parent F0001 | S03 | ops 16 calls 11
F0020 resetTalents() | html:L313-313 | parent F0001 | S03 | ops 3 calls 7
F0021 shrineCooldown(shrine) | html:L315-315 | parent F0001 | S06 | ops 1 calls 1
F0022 activateShrine(shrine) | html:L316-316 | parent F0001 | S06 | ops 14 calls 7
F0023 updateShrines(map) | html:L317-317 | parent F0001 | S06 | ops 4 calls 3
F0024 activeBoon(type) | html:L318-318 | parent F0001 | S06 | ops 3 calls 0
F0025 cloudForgetStoredSession() | html:L333-333 | parent F0001 | S10 | ops 0 calls 1
F0026 cloudStoreSession() | html:L334-334 | parent F0001 | S10 | ops 3 calls 2
F0027 cloudReadStoredSession() | html:L335-335 | parent F0001 | S10 | ops 8 calls 2
F0028 cloudOutboxKey(uid) | html:L337-337 | parent F0001 | S10 | ops 1 calls 0
F0029 cloudLegacyFlag(uid) | html:L338-338 | parent F0001 | S10 | ops 1 calls 0
F0030 cloudRev(value) | html:L339-339 | parent F0001 | S10 | ops 6 calls 3
F0031 cloudReadOutbox(uid) | html:L340-340 | parent F0001 | S10 | ops 9 calls 7
```

F0032 cloudWriteOutbox(uid, snapshot, baseRevision) | html:L341-341 | parent F0001 | S10 | ops 0 calls 7
F0033 cloudClearOutbox(uid) | html:L342-342 | parent F0001 | S10 | ops 0 calls 4
F0034 cloudReady() | html:L343-343 | parent F0001 | S10 | ops 1 calls 2
F0035 cloudAuthMessage(err, action='login') | html:L346-375 | parent F0001 | S10 | ops 22 calls 23
F0036 anonymous callback() | html:L379-379 | parent F0001 | S08 | ops 5 calls 6
F0037 mailWait(action) | html:L380-380 | parent F0001 | S10 | ops 3 calls 3
F0038 mailPause(action, seconds) | html:L381-381 | parent F0001 | S10 | ops 4 calls 7
F0039 isMailUnavailable(err, action='register') | html:L382-386 | parent F0001 | S10 | ops 7 calls 4
F0040 refreshMailControls() | html:L387-394 | parent F0001 | S10 | ops 27 calls 7
F0041 mailRequest(action, path, options) | html:L395-403 | parent F0001 | S10 | ops 9 calls 10
F0042 cloudRequest(path, {method='GET',body,token,prefer}={}) | html:L404-414 | parent F0001 | S10 | ops 26 calls 19
F0043 setCloudSession(a) | html:L415-415 | parent F0001 | S10 | ops 14 calls 3
F0044 cloudToken() | html:L416-418 | parent F0001 | S10 | ops 10 calls 6
F0045 cloudLoad() | html:L419-419 | parent F0001 | S10 | ops 5 calls 6
F0046 cloudSchedule(ms=1200) | html:L420-420 | parent F0001 | S10 | ops 5 calls 1
F0047 anonymous callback() | html:L420-420 | parent F0046 | S10 | ops 2 calls 1
F0048 cloudQueue(snapshot) | html:L421-423 | parent F0001 | S10 | ops 7 calls 5
F0049 cloudFlush() | html:L424-440 | parent F0001 | S10 | ops 50 calls 18
F0050 cloudEnter(a) | html:L441-454 | parent F0001 | S10 | ops 19 calls 16
F0051 cloudVerifiedSignupReady() | html:L456-464 | parent F0001 | S10 | ops 4 calls 4
F0052 cloudAuth(name, pass) | html:L465-479 | parent F0001 | S10 | ops 16 calls 20
F0053 cloudAutoResume() | html:L480-493 | parent F0001 | S10 | ops 16 calls 13
F0054 cloudSendRecovery() | html:L494-504 | parent F0001 | S10 | ops 11 calls 15
F0055 cloudRecoveryFromURL() | html:L505-528 | parent F0001 | S10 | ops 31 calls 28
F0056 cloudUpdatePassword(e) | html:L529-544 | parent F0001 | S10 | ops 27 calls 21
F0057 cloudChangePasswordFromSettings(event) | html:L546-566 | parent F0001 | S10 | ops 28 calls 17
F0058 cloudExportBackup() | html:L567-576 | parent F0001 | S10 | ops 17 calls 20
F0059 anonymous callback(k) | html:L568-568 | parent F0058 | S10 | ops 0 calls 1
F0060 anonymous callback() | html:L574-574 | parent F0058 | S10 | ops 0 calls 1
F0061 configureAccountMode(cloud) | html:L577-591 | parent F0001 | S10 | ops 37 calls 23
F0062 switchCloudMode() | html:L592-595 | parent F0001 | S10 | ops 3 calls 4
F0063 configured() | html:L596-596 | parent F0001 | S08 | ops 0 calls 1
F0064 isCloudLogin() | html:L596-596 | parent F0001 | S08 | ops 2 calls 0
F0065 mode() | html:L596-596 | parent F0001 | S08 | ops 0 calls 0
F0066 pending() | html:L596-596 | parent F0001 | S08 | ops 1 calls 0
F0067 conflict() | html:L596-596 | parent F0001 | S08 | ops 0 calls 0
F0068 cfOutboxKey(id) | html:L600-600 | parent F0001 | S09 | ops 1 calls 0
F0069 cfReleaseLock() | html:L603-603 | parent F0001 | S09 | ops 2 calls 1
F0070 cfTakeLock(id) | html:L604-613 | parent F0001 | S09 | ops 2 calls 2
F0071 anonymous callback(resolve, reject) | html:L607-612 | parent F0070 | S09 | ops 1 calls 2
F0072 anonymous callback(lock) | html:L608-611 | parent F0071 | S09 | ops 2 calls 3
F0073 anonymous callback(done) | html:L610-610 | parent F0072 | S09 | ops 1 calls 0
F0074 cfAPI(path, {method='GET',body,accountId=cfAccount?.id}={}) | html:L614-620 | parent F0001 | S09 | ops 11 calls 11
F0075 cfReadOutbox(id) | html:L621-625 | parent F0001 | S09 | ops 10 calls 7
F0076 cfSchedule(delay=1500) | html:L626-626 | parent F0001 | S09 | ops 4 calls 1
F0077 anonymous callback() | html:L626-626 | parent F0076 | S09 | ops 2 calls 1
F0078 cfQueue(payload) | html:L627-632 | parent F0001 | S09 | ops 13 calls 11
F0079 cfFlush() | html:L633-648 | parent F0001 | S09 | ops 40 calls 17
F0080 cfConfigure() | html:L649-654 | parent F0001 | S09 | ops 10 calls 11
F0081 cfShowRecovery(code, account) | html:L655-663 | parent F0001 | S09 | ops 8 calls 12
F0082 anonymous callback(e) | html:L660-660 | parent F0081 | S09 | ops 0 calls 1
F0083 $('cfDownloadRecovery').onclick() | html:L661-661 | parent F0081 | S09 | ops 0 calls 1
F0084 $('cfCloseRecovery').onclick() | html:L662-662 | parent F0081 | S09 | ops 4 calls 2
F0085 cfDownload(data, name) | html:L664-664 | parent F0001 | S09 | ops 2 calls 7
F0086 anonymous callback() | html:L664-664 | parent F0085 | S09 | ops 0 calls 1
F0087 cfEnter(account) | html:L665-680 | parent F0001 | S09 | ops 24 calls 17
F0088 cfAuthenticate(e) | html:L681-690 | parent F0001 | S09 | ops 16 calls 15
F0089 cfResume() | html:L691-695 | parent F0001 | S09 | ops 10 calls 5
F0090 cfLogout() | html:L696-705 | parent F0001 | S09 | ops 12 calls 8

F0091 anonymous callback(resolve) | html:L700-700 | parent F0090 | S09 | ops 0 calls 1
F0092 cfExportPending() | html:L706-709 | parent F0001 | S09 | ops 4 calls 6
F0093 cfUseCloud() | html:L710-716 | parent F0001 | S09 | ops 11 calls 15
F0094 cfUseLocal() | html:L717-729 | parent F0001 | S09 | ops 11 calls 17
F0095 cfPasswordForm(event) | html:L730-734 | parent F0001 | S09 | ops 7 calls 10
F0096 cfPanelControls() | html:L735-746 | parent F0001 | S09 | ops 21 calls 12
F0097 $('cfSaveNow').onclick() | html:L740-740 | parent F0096 | S09 | ops 1 calls 2
F0098 $('cfBackupPending').onclick() | html:L740-740 | parent F0096 | S09 | ops 1 calls 2
F0099 $('cfExportProgress').onclick() | html:L741-741 | parent F0096 | S09 | ops 1 calls 4
F0100 $('cfImportProgress').onchange(e) | html:L742-742 | parent F0096 | S09 | ops 1 calls 1
F0101 cfImportProgress(file) | html:L747-753 | parent F0001 | S09 | ops 15 calls 14
F0102 cfRestoreProgress() | html:L754-756 | parent F0001 | S09 | ops 5 calls 9
F0103 cfMigrate(event) | html:L757-774 | parent F0001 | S09 | ops 20 calls 29
F0104 cfRecoveryForm() | html:L775-782 | parent F0001 | S09 | ops 9 calls 6
F0105 $('cfRecoverCancel').onclick() | html:L779-779 | parent F0104 | S09 | ops 1 calls 1
F0106 form.onsubmit(e) | html:L780-781 | parent F0104 | S09 | ops 7 calls 13
F0107 validateLocalProgress(payload) | html:L784-788 | parent F0001 | S10 | ops 6 calls 7
F0108 anonymous callback(x) | html:L786-786 | parent F0107 | S10 | ops 3 calls 1
F0109 anonymous callback(k) | html:L786-786 | parent F0107 | S10 | ops 1 calls 1
F0110 sanitizeProtectedPetals(raw) | html:L790-794 | parent F0001 | S07 | ops 11 calls 3
F0111 isPetalProtected(key) | html:L795-795 | parent F0001 | S07 | ops 1 calls 0
F0112 togglePetalProtection(key) | html:L796-802 | parent F0001 | S07 | ops 12 calls 7
F0113 parseLocalBackup(raw) | html:L803-812 | parent F0001 | S10 | ops 35 calls 11
F0114 exportLocalProgress() | html:L813-819 | parent F0001 | S10 | ops 8 calls 14
F0115 anonymous callback() | html:L818-818 | parent F0114 | S10 | ops 0 calls 1
F0116 importLocalProgress(file) | html:L820-830 | parent F0001 | S10 | ops 16 calls 13
F0117 restoreBeforeImport() | html:L831-838 | parent F0001 | S10 | ops 8 calls 12
F0118 inventoryKey(t, r) | html:L839-839 | parent F0001 | S07 | ops 2 calls 0
F0119 itemOf(k) | html:L839-839 | parent F0001 | S07 | ops 9 calls 4
F0120 addBag(t, r, n=1) | html:L839-839 | parent F0001 | S07 | ops 6 calls 4
F0121 takeBag(t, r, n=1) | html:L840-840 | parent F0001 | S07 | ops 3 calls 2
F0122 slotCapacity(lvl) | html:L845-845 | parent F0001 | S03 | ops 2 calls 0
F0123 viewScale(lvl) | html:L846-846 | parent F0001 | S03 | ops 4 calls 2
F0124 antennaMultiplier() | html:L846-846 | parent F0001 | S03 | ops 1 calls 2
F0125 anonymous callback(s) | html:L846-846 | parent F0124 | S03 | ops 1 calls 0
F0126 anonymous callback(sum, s) | html:L846-846 | parent F0124 | S03 | ops 3 calls 0
F0127 effectiveZoom() | html:L846-846 | parent F0001 | S03 | ops 1 calls 3
F0128 regionBand(x, y) | html:L847-847 | parent F0001 | S03 | ops 8 calls 1
F0129 minZoneLevel(zone, band) | html:L848-848 | parent F0001 | S03 | ops 3 calls 0
F0130 bandLabel(zone, band) | html:L849-849 | parent F0001 | S03 | ops 5 calls 1
F0131 maxHP() | html:L850-850 | parent F0001 | S03 | ops 5 calls 4
F0132 anonymous callback(n, s) | html:L850-850 | parent F0131 | S03 | ops 7 calls 0
F0133 soilCount() | html:L851-851 | parent F0001 | S03 | ops 0 calls 1
F0134 anonymous callback(s) | html:L851-851 | parent F0133 | S03 | ops 1 calls 0
F0135 bodyRadius() | html:L852-852 | parent F0001 | S03 | ops 2 calls 1
F0136 bodySpeedFactor() | html:L853-853 | parent F0001 | S03 | ops 0 calls 3
F0137 rootedArmor() | html:L854-854 | parent F0001 | S03 | ops 0 calls 2
F0138 anonymous callback(e) | html:L854-854 | parent F0137 | S03 | ops 5 calls 0
F0139 anonymous callback(total, e) | html:L854-854 | parent F0137 | S03 | ops 2 calls 0
F0140 rootActive() | html:L855-855 | parent F0001 | S03 | ops 0 calls 1
F0141 anonymous callback(e) | html:L855-855 | parent F0140 | S03 | ops 5 calls 0
F0142 magnetRadius() | html:L856-856 | parent F0001 | S03 | ops 2 calls 2
F0143 anonymous callback(n, e) | html:L856-856 | parent F0142 | S03 | ops 7 calls 0
F0144 reverseOrbit() | html:L858-862 | parent F0001 | S03 | ops 14 calls 2
F0145 recoverPetalDurability(e, dt) | html:L863-867 | parent F0001 | S03 | ops 17 calls 1
F0146 orbitRadius(t, i, mode) | html:L868-868 | parent F0001 | S03 | ops 15 calls 2
F0147 newPlayer() | html:L870-870 | parent F0001 | S03 | ops 0 calls 0
F0148 traitOf(t) | html:L880-880 | parent F0001 | S05 | ops 0 calls 2
F0149 computeSynergies(slots) | html:L883-883 | parent F0001 | S08 | ops 4 calls 2
F0150 anonymous callback(s) | html:L883-883 | parent F0149 | S08 | ops 0 calls 1

F0151 syner(name) | html:L884-884 | parent F0001 | S08 | ops 1 calls 0
F0152 activeSkill(name) | html:L885-894 | parent F0001 | S04 | ops 59 calls 25
F0153 skillRemaining(name) | html:L895-895 | parent F0001 | S04 | ops 2 calls 1
F0154 weightedEvent(rng=game.random?.world||Math.random) | html:L897-897 | parent F0001 | S06 | ops 4 calls 4
F0155 anonymous callback(s, e) | html:L897-897 | parent F0154 | S06 | ops 1 calls 0
F0156 spawnNearPlayer(tierBoost=0) | html:L898-898 | parent F0001 | S05 | ops 17 calls 4
F0157 beginWorldEvent(type) | html:L899-902 | parent F0001 | S06 | ops 35 calls 19
F0158 anonymous callback(x) | html:L901-901 | parent F0157 | S06 | ops 0 calls 0
F0159 updateWorldEvents(dt) | html:L903-910 | parent F0001 | S06 | ops 38 calls 16
F0160 anonymous callback(x) | html:L909-909 | parent F0159 | S06 | ops 1 calls 0
F0161 specialProfileFor(m) | html:L911-911 | parent F0001 | S08 | ops 8 calls 2
F0162 specialCooldown(m) | html:L912-912 | parent F0001 | S05 | ops 8 calls 1
F0163 spawnSpecialProjectile(m, dx, dy, trait, offset=0) | html:L913-913 | parent F0001 | S04 | ops 16 calls 7
F0164 performMobSpecial(m, map) | html:L914-921 | parent F0001 | S05 | ops 46 calls 19
F0165 anonymous callback(o) | html:L918-918 | parent F0164 | S05 | ops 2 calls 1
F0166 updateMobSpecial(m, dt, d, map) | html:L922-926 | parent F0001 | S05 | ops 21 calls 5
F0167 resetEquip(delay=0) | html:L927-927 | parent F0001 | S03 | ops 2 calls 2
F0168 anonymous callback(s) | html:L927-927 | parent F0167 | S03 | ops 6 calls 1
F0169 ensureSlots(notify=false) | html:L928-928 | parent F0001 | S03 | ops 4 calls 7
F0170 baseSave() | html:L929-929 | parent F0001 | S10 | ops 0 calls 0
F0171 readSave() | html:L930-930 | parent F0001 | S10 | ops 2 calls 2
F0172 saveGame() | html:L931-931 | parent F0001 | S10 | ops 13 calls 10
F0173 initGame(saved) | html:L932-938 | parent F0001 | S10 | ops 103 calls 68
F0174 anonymous callback(_, i) | html:L934-934 | parent F0173 | S10 | ops 3 calls 0
F0175 anonymous callback(_, i) | html:L935-935 | parent F0173 | S10 | ops 3 calls 0
F0176 anonymous callback([k]) | html:L936-936 | parent F0173 | S10 | ops 1 calls 0
F0177 buildMap(id) | html:L939-943 | parent F0001 | S06 | ops 50 calls 38
F0178 anonymous callback([to,x,y]) | html:L939-939 | parent F0177 | S06 | ops 4 calls 0
F0179 anonymous callback(p) | html:L941-941 | parent F0177 | S06 | ops 1 calls 1
F0180 anonymous callback(o) | html:L941-941 | parent F0177 | S06 | ops 3 calls 1
F0181 anonymous callback(o) | html:L943-943 | parent F0177 | S06 | ops 2 calls 1
F0182 anonymous callback(o) | html:L943-943 | parent F0177 | S06 | ops 2 calls 1
F0183 anonymous callback(o) | html:L943-943 | parent F0177 | S06 | ops 1 calls 1
F0184 anonymous callback(m) | html:L943-943 | parent F0177 | S06 | ops 1 calls 0
F0185 spawnMob(id, initial=false, forced=false) | html:L944-950 | parent F0001 | S05 | ops 91 calls 32
F0186 anonymous callback(m) | html:L944-944 | parent F0185 | S05 | ops 1 calls 0
F0187 anonymous callback(o) | html:L948-948 | parent F0185 | S05 | ops 3 calls 1
F0188 anonymous callback(o) | html:L948-948 | parent F0185 | S05 | ops 3 calls 1
F0189 anonymous callback(o) | html:L948-948 | parent F0185 | S05 | ops 3 calls 1
F0190 createLandmarks(id, r, rocks, portals) | html:L952-956 | parent F0001 | S06 | ops 19 calls 7
F0191 anonymous callback(p) | html:L954-954 | parent F0190 | S06 | ops 1 calls 1
F0192 drawLandmark(l, zone) | html:L957-957 | parent F0001 | S08 | ops 1 calls 2
F0193 nearbyRocks(rocks, x, y) | html:L959-959 | parent F0001 | S02 | ops 13 calls 5
F0194 slideCircle(a, dx, dy, r, rocks) | html:L960-960 | parent F0001 | S02 | ops 37 calls 6
F0195 performDash() | html:L961-961 | parent F0001 | S04 | ops 16 calls 7
F0196 swapLoadout() | html:L962-962 | parent F0001 | S03 | ops 18 calls 8
F0197 damagePlayer(dmg) | html:L964-964 | parent F0001 | S04 | ops 29 calls 9
F0198 recordMob(m, killed=false) | html:L965-965 | parent F0001 | S07 | ops 6 calls 1
F0199 chooseDropRank(m) | html:L966-966 | parent F0001 | S07 | ops 15 calls 3
F0200 armorOf(m) | html:L966-966 | parent F0001 | S08 | ops 8 calls 1
F0201 damageMob(m, dmg, x, y, quiet=false) | html:L966-966 | parent F0001 | S04 | ops 89 calls 38
F0202 registerComboKill() | html:L967-967 | parent F0001 | S07 | ops 8 calls 3
F0203 xpNeed(lvl) | html:L968-968 | parent F0001 | S07 | ops 5 calls 1
F0204 floating(x, y, text, color) | html:L969-969 | parent F0001 | S08 | ops 1 calls 2
F0205 pulse(x, y, color, num=8) | html:L970-970 | parent F0001 | S08 | ops 17 calls 8
F0206 transition(to) | html:L971-971 | parent F0001 | S06 | ops 33 calls 7
F0207 die() | html:L972-972 | parent F0001 | S06 | ops 15 calls 4
F0208 moveInput() | html:L973-973 | parent F0001 | S01 | ops 36 calls 12
F0209 petalContactRadius(def) | html:L977-977 | parent F0001 | S02 | ops 2 calls 0
F0210 mobCollider(m) | html:L978-989 | parent F0001 | S02 | ops 17 calls 7

F0211 mobSupportRadius(m, dx, dy) | html:L990-998 | parent F0001 | S02 | ops 12 calls 6
F0212 mobCircleOverlap(m, x, y, r) | html:L999-1023 | parent F0001 | S02 | ops 60 calls 7
F0213 mobRockRadius(m) | html:L1024-1024 | parent F0001 | S02 | ops 0 calls 2
F0214 updateMobFacing(m, vx, vy, dt) | html:L1025-1031 | parent F0001 | S02 | ops 13 calls 6
F0215 neutralMotion(m, dx, dy, d) | html:L1032-1032 | parent F0001 | S05 | ops 9 calls 2
F0216 actMode() | html:L1033-1033 | parent F0001 | S01 | ops 5 calls 3
F0217 activeCopies(type) | html:L1034-1034 | parent F0001 | S08 | ops 0 calls 1
F0218 anonymous callback(e) | html:L1034-1034 | parent F0217 | S08 | ops 5 calls 0
F0219 rawPetalDamage(e) | html:L1035-1035 | parent F0001 | S08 | ops 13 calls 3
F0220 criticalMultiplier() | html:L1036-1036 | parent F0001 | S08 | ops 0 calls 2
F0221 clearTransientState() | html:L1037-1037 | parent F0001 | S08 | ops 9 calls 0
F0222 update(dt) | html:L1038-1063 | parent F0001 | S04 | ops 615 calls 172
F0223 anonymous callback(o) | html:L1058-1058 | parent F0222 | S04 | ops 2 calls 1
F0224 anonymous callback(b) | html:L1058-1058 | parent F0222 | S04 | ops 1 calls 0
F0225 anonymous callback(o) | html:L1059-1059 | parent F0222 | S04 | ops 2 calls 1
F0226 anonymous callback(s) | html:L1059-1059 | parent F0222 | S04 | ops 1 calls 0
F0227 anonymous callback(d) | html:L1060-1060 | parent F0222 | S04 | ops 1 calls 0
F0228 anonymous callback(f) | html:L1061-1061 | parent F0222 | S04 | ops 1 calls 0
F0229 anonymous callback(f) | html:L1061-1061 | parent F0222 | S04 | ops 1 calls 0
F0230 anonymous callback(m) | html:L1062-1062 | parent F0222 | S04 | ops 1 calls 0
F0231 shoot(m, dx, dy) | html:L1064-1064 | parent F0001 | S04 | ops 11 calls 1
F0232 screenFx(x, y, c) | html:L1065-1065 | parent F0001 | S08 | ops 0 calls 1
F0233 updatePortals(dt, map) | html:L1066-1066 | parent F0001 | S06 | ops 20 calls 13
F0234 proj(x, y) | html:L1067-1067 | parent F0001 | S08 | ops 4 calls 0
F0235 visible(x, y, r=60) | html:L1068-1068 | parent F0001 | S08 | ops 15 calls 0
F0236 circle(x, y, r, fill, stroke=null, lw=1) | html:L1069-1069 | parent F0001 | S08 | ops 3 calls 4
F0237 drawShrine(shrine) | html:L1070-1070 | parent F0001 | S08 | ops 22 calls 13
F0238 drawWorld() | html:L1071-1080 | parent F0001 | S08 | ops 76 calls 66
F0239 drawRegionBands(id) | html:L1081-1081 | parent F0001 | S08 | ops 28 calls 10
F0240 drawDecor(d, zone) | html:L1082-1082 | parent F0001 | S08 | ops 49 calls 31
F0241 drawRock(o, zone) | html:L1083-1083 | parent F0001 | S08 | ops 109 calls 35
F0242 drawPortal(p) | html:L1084-1084 | parent F0001 | S08 | ops 44 calls 16
F0243 drawLoot(l) | html:L1085-1085 | parent F0001 | S08 | ops 14 calls 11
F0244 drawFlower(x, y, r) | html:L1086-1086 | parent F0001 | S08 | ops 44 calls 25
F0245 drawPetal(e) | html:L1087-1087 | parent F0001 | S08 | ops 143 calls 102
F0246 drawMobHitbox(m) | html:L1088-1092 | parent F0001 | S08 | ops 5 calls 13
F0247 drawCreatureBody(m, fill, stroke, lineWidth=2) | html:L1093-1098 | parent F0001 | S08 | ops 3 calls 5
F0248 tint(hex, amount) | html:L1101-1101 | parent F0001 | S08 | ops 8 calls 6
F0249 anonymous callback(c) | html:L1101-1101 | parent F0248 | S08 | ops 3 calls 3
F0250 materialFill(color, r) | html:L1102-1102 | parent F0001 | S08 | ops 8 calls 6
F0251 paintTerrain() | html:L1103-1112 | parent F0001 | S08 | ops 31 calls 23
F0252 creatureVisualProfile(m) | html:L1113-1113 | parent F0001 | S08 | ops 12 calls 3
F0253 drawCreatureModel(m) | html:L1114-1140 | parent F0001 | S08 | ops 324 calls 135
F0254 drawMob(m) | html:L1141-1151 | parent F0001 | S08 | ops 65 calls 32
F0255 drawNavArrow() | html:L1153-1153 | parent F0001 | S08 | ops 27 calls 18
F0256 anonymous callback(p) | html:L1153-1153 | parent F0255 | S08 | ops 1 calls 0
F0257 nextHop(from, to) | html:L1154-1154 | parent F0001 | S06 | ops 3 calls 4
F0258 minimap() | html:L1155-1157 | parent F0001 | S08 | ops 70 calls 31
F0259 updateBossHud() | html:L1158-1158 | parent F0001 | S08 | ops 25 calls 10
F0260 updateSkillDock() | html:L1159-1159 | parent F0001 | S08 | ops 21 calls 13
F0261 anonymous callback([id]) | html:L1159-1159 | parent F0260 | S08 | ops 0 calls 1
F0262 anonymous callback([id,v]) | html:L1159-1159 | parent F0260 | S08 | ops 3 calls 1
F0263 syncHud(force=false) | html:L1160-1162 | parent F0001 | S08 | ops 172 calls 74
F0264 anonymous callback(s) | html:L1160-1160 | parent F0263 | S08 | ops 1 calls 0
F0265 fillSlots() | html:L1163-1163 | parent F0001 | S08 | ops 33 calls 8
F0266 anonymous callback() | html:L1163-1163 | parent F0265 | S08 | ops 0 calls 1
F0267 toast(msg) | html:L1164-1164 | parent F0001 | S08 | ops 2 calls 4
F0268 anonymous callback() | html:L1164-1164 | parent F0267 | S08 | ops 0 calls 1
F0269 render() | html:L1165-1165 | parent F0001 | S08 | ops 1 calls 2
F0270 drawGateBackground() | html:L1166-1166 | parent F0001 | S08 | ops 12 calls 10

F0271 resize() | html:L1167-1167 | parent F0001 | S08 | ops 28 calls 9
F0272 loop(now) | html:L1168-1168 | parent F0001 | S01 | ops 19 calls 4
F0273 showModal(type) | html:L1169-1169 | parent F0001 | S08 | ops 3 calls 7
F0274 closeModal() | html:L1170-1170 | parent F0001 | S08 | ops 4 calls 8
F0275 openPanel(type, index) | html:L1171-1171 | parent F0001 | S08 | ops 6 calls 1
F0276 equipInto(key, slot) | html:L1172-1172 | parent F0001 | S03 | ops 17 calls 13
F0277 bindBagDrag(root) | html:L1174-1174 | parent F0001 | S08 | ops 1 calls 8
F0278 resetHighlights() | html:L1174-1174 | parent F0277 | S08 | ops 0 calls 1
F0279 anonymous callback(s) | html:L1174-1174 | parent F0278 | S08 | ops 0 calls 1
F0280 anonymous callback(e) | html:L1174-1174 | parent F0277 | S08 | ops 0 calls 2
F0281 anonymous callback() | html:L1174-1174 | parent F0277 | S08 | ops 0 calls 1
F0282 anonymous callback(e) | html:L1174-1174 | parent F0277 | S08 | ops 1 calls 4
F0283 anonymous callback(e) | html:L1174-1174 | parent F0277 | S08 | ops 1 calls 1
F0284 anonymous callback(e) | html:L1174-1174 | parent F0277 | S08 | ops 1 calls 3
F0285 move(event) | html:L1174-1174 | parent F0284 | S08 | ops 15 calls 9
F0286 finish(event) | html:L1174-1174 | parent F0284 | S08 | ops 7 calls 9
F0287 inventoryFilterCards(mode, root) | html:L1178-1199 | parent F0001 | S08 | ops 32 calls 13
F0288 anonymous callback(a, b) | html:L1182-1187 | parent F0287 | S08 | ops 9 calls 6
F0289 anonymous callback(card, i) | html:L1188-1188 | parent F0287 | S08 | ops 1 calls 1
F0290 bindInventoryFilters(mode, root) | html:L1200-1210 | parent F0001 | S08 | ops 11 calls 5
F0291 update() | html:L1203-1203 | parent F0290 | S04 | ops 0 calls 1
F0292 query.oninput() | html:L1204-1204 | parent F0290 | S08 | ops 2 calls 1
F0293 tier.onchange() | html:L1205-1205 | parent F0290 | S08 | ops 2 calls 1
F0294 only.onchange() | html:L1207-1207 | parent F0290 | S08 | ops 1 calls 1
F0295 order.onchange() | html:L1208-1208 | parent F0290 | S08 | ops 1 calls 1
F0296 inventoryControls(mode) | html:L1211-1218 | parent F0001 | S08 | ops 7 calls 2
F0297 anonymous callback(r, i) | html:L1214-1214 | parent F0296 | S08 | ops 6 calls 0
F0298 craftBatch(key, requested=1) | html:L1221-1238 | parent F0001 | S07 | ops 29 calls 11
F0299 sanitizeCollectionGoal(value) | html:L1239-1242 | parent F0001 | S07 | ops 8 calls 3
F0300 collectionOwned() | html:L1243-1246 | parent F0001 | S07 | ops 3 calls 2
F0301 anonymous callback(s) | html:L1245-1245 | parent F0300 | S07 | ops 3 calls 0
F0302 creatureRegions(type) | html:L1247-1247 | parent F0001 | S07 | ops 0 calls 2
F0303 anonymous callback(id) | html:L1247-1247 | parent F0302 | S07 | ops 2 calls 1
F0304 collectionSources(type) | html:L1248-1250 | parent F0001 | S07 | ops 0 calls 2
F0305 anonymous callback(id) | html:L1249-1249 | parent F0304 | S07 | ops 0 calls 1
F0306 anonymous callback(t) | html:L1249-1249 | parent F0305 | S07 | ops 0 calls 1
F0307 setCollectionGoal(type, rank) | html:L1251-1255 | parent F0001 | S07 | ops 7 calls 6
F0308 updateCollectionGoal() | html:L1256-1262 | parent F0001 | S07 | ops 19 calls 8
F0309 anonymous callback(id) | html:L1260-1260 | parent F0308 | S07 | ops 0 calls 0
F0310 renderCollectionGuide() | html:L1263-1266 | parent F0001 | S08 | ops 9 calls 4
F0311 anonymous callback(id) | html:L1265-1265 | parent F0310 | S08 | ops 4 calls 0
F0312 bindCollectionGuide(root) | html:L1267-1270 | parent F0001 | S08 | ops 1 calls 4
F0313 anonymous callback(btn) | html:L1268-1268 | parent F0312 | S08 | ops 1 calls 0
F0314 btn.onclick() | html:L1268-1268 | parent F0313 | S08 | ops 3 calls 3
F0315 $('clearCollectionGoal').onclick() | html:L1269-1269 | parent F0312 | S08 | ops 2 calls 3
F0316 petalBaseStats(key) | html:L1271-1274 | parent F0001 | S07 | ops 3 calls 3
F0317 renderPetalWorkbench(key) | html:L1275-1282 | parent F0001 | S08 | ops 48 calls 12
F0318 delta(n) | html:L1279-1279 | parent F0317 | S08 | ops 5 calls 0
F0319 anonymous callback([,family]) | html:L1280-1280 | parent F0317 | S08 | ops 0 calls 1
F0320 anonymous callback([,family]) | html:L1280-1280 | parent F0317 | S08 | ops 0 calls 0
F0321 renderCraftDistribution(mass) | html:L1283-1283 | parent F0001 | S08 | ops 2 calls 2
F0322 anonymous callback(p, n) | html:L1283-1283 | parent F0321 | S08 | ops 10 calls 1
F0323 renderCraftAnalysis(rate, stock, locked) | html:L1284-1287 | parent F0001 | S08 | ops 20 calls 6
F0324 pct(n) | html:L1285-1285 | parent F0323 | S08 | ops 2 calls 1
F0325 renderDamageForecast(m) | html:L1288-1293 | parent F0001 | S08 | ops 15 calls 9
F0326 anonymous callback(e) | html:L1289-1289 | parent F0325 | S08 | ops 3 calls 0
F0327 renderProbabilityLab() | html:L1294-1305 | parent F0001 | S08 | ops 62 calls 20
F0328 anonymous callback(m) | html:L1295-1295 | parent F0327 | S08 | ops 1 calls 0
F0329 anonymous callback(a, b) | html:L1295-1295 | parent F0327 | S08 | ops 1 calls 2
F0330 anonymous callback(_, r) | html:L1297-1297 | parent F0327 | S08 | ops 0 calls 0

F0331 anonymous callback(t) | html:L1300-1300 | parent F0327 | S08 | ops 0 calls 0
F0332 askGame(message) | html:L1307-1307 | parent F0001 | S16 | ops 16 calls 1
F0333 buildWorkshopPlan() | html:L1309-1314 | parent F0001 | S15 | ops 5 calls 7
F0334 anonymous callback(s) | html:L1311-1311 | parent F0333 | S15 | ops 0 calls 1
F0335 anonymous callback(s) | html:L1313-1313 | parent F0333 | S15 | ops 0 calls 1
F0336 workshopIdentity() | html:L1315-1315 | parent F0001 | S15 | ops 0 calls 1
F0337 renderBuildWorkshop() | html:L1316-1331 | parent F0001 | S15 | ops 51 calls 17
F0338 anonymous callback(x) | html:L1325-1325 | parent F0337 | S15 | ops 1 calls 0
F0339 x.onclick() | html:L1325-1325 | parent F0338 | S15 | ops 1 calls 1
F0340 $('workshopBag').onclick() | html:L1325-1325 | parent F0337 | S15 | ops 0 calls 1
F0341 $('workshopSkills').onclick() | html:L1325-1325 | parent F0337 | S15 | ops 0 calls 1
F0342 $('applyWorkshop').onclick() | html:L1326-1330 | parent F0337 | S15 | ops 7 calls 14
F0343 anonymous callback(k) | html:L1329-1329 | parent F0342 | S15 | ops 0 calls 1
F0344 renderSkillAtlas() | html:L1332-1341 | parent F0001 | S08 | ops 59 calls 23
F0345 anonymous callback(t) | html:L1339-1339 | parent F0344 | S08 | ops 0 calls 0
F0346 renderModal() | html:L1343-1348 | parent F0001 | S08 | ops 8 calls 9
F0347 anonymous callback(b) | html:L1346-1346 | parent F0346 | S08 | ops 1 calls 2
F0348 renderModalContents() | html:L1349-1434 | parent F0001 | S08 | ops 645 calls 171
F0349 anonymous callback(btn) | html:L1349-1349 | parent F0348 | S08 | ops 1 calls 0
F0350 btn.onclick() | html:L1349-1349 | parent F0349 | S08 | ops 0 calls 1
F0351 $('respecTalents').onclick() | html:L1349-1349 | parent F0348 | S08 | ops 0 calls 2
F0352 $('openWorkshop').onclick() | html:L1360-1360 | parent F0348 | S08 | ops 0 calls 1
F0353 anonymous callback(x) | html:L1361-1361 | parent F0348 | S08 | ops 1 calls 0
F0354 x.onclick() | html:L1361-1361 | parent F0353 | S08 | ops 2 calls 1
F0355 anonymous callback(x) | html:L1362-1362 | parent F0348 | S08 | ops 1 calls 0
F0356 x.onclick() | html:L1362-1362 | parent F0355 | S08 | ops 2 calls 1
F0357 anonymous callback(x) | html:L1363-1363 | parent F0348 | S08 | ops 1 calls 0
F0358 x.onclick() | html:L1363-1363 | parent F0357 | S08 | ops 2 calls 2
F0359 $('equipPreview').onclick() | html:L1364-1364 | parent F0348 | S08 | ops 0 calls 1
F0360 $('protectPreview').onclick() | html:L1365-1365 | parent F0348 | S08 | ops 0 calls 1
F0361 $('trackPetal').onclick() | html:L1366-1366 | parent F0348 | S08 | ops 2 calls 3
F0362 $('craftPreview').onclick() | html:L1367-1367 | parent F0348 | S08 | ops 1 calls 1
F0363 anonymous callback(x) | html:L1386-1386 | parent F0348 | S08 | ops 1 calls 0
F0364 x.onclick() | html:L1386-1386 | parent F0363 | S08 | ops 1 calls 1
F0365 lockBtn.onclick() | html:L1388-1388 | parent F0348 | S08 | ops 0 calls 1
F0366 doBtn.onclick() | html:L1389-1396 | parent F0348 | S08 | ops 13 calls 5
F0367 multiBtn.onclick() | html:L1397-1407 | parent F0348 | S08 | ops 24 calls 8
F0368 anonymous callback(x) | html:L1409-1409 | parent F0348 | S08 | ops 1 calls 0
F0369 x.onclick() | html:L1409-1409 | parent F0368 | S08 | ops 4 calls 2
F0370 anonymous callback(s) | html:L1411-1411 | parent F0348 | S08 | ops 0 calls 0
F0371 anonymous callback(s, d) | html:L1411-1411 | parent F0348 | S08 | ops 1 calls 0
F0372 anonymous callback(id) | html:L1414-1414 | parent F0348 | S08 | ops 0 calls 0
F0373 anonymous callback(id) | html:L1415-1415 | parent F0348 | S08 | ops 0 calls 0
F0374 anonymous callback(t) | html:L1416-1416 | parent F0348 | S08 | ops 0 calls 0
F0375 anonymous callback(r, i) | html:L1416-1416 | parent F0348 | S08 | ops 4 calls 0
F0376 anonymous callback(t) | html:L1416-1416 | parent F0348 | S08 | ops 4 calls 0
F0377 anonymous callback(btn) | html:L1417-1417 | parent F0348 | S08 | ops 1 calls 0
F0378 btn.onclick() | html:L1417-1417 | parent F0377 | S08 | ops 0 calls 3
F0379 anonymous callback(x) | html:L1417-1417 | parent F0348 | S08 | ops 1 calls 0
F0380 x.onclick() | html:L1417-1417 | parent F0379 | S08 | ops 1 calls 1
F0381 anonymous callback(x) | html:L1417-1417 | parent F0348 | S08 | ops 1 calls 0
F0382 x.onclick() | html:L1417-1417 | parent F0381 | S08 | ops 1 calls 1
F0383 anonymous callback(s) | html:L1418-1418 | parent F0348 | S08 | ops 1 calls 0
F0384 anonymous callback(x) | html:L1421-1421 | parent F0348 | S08 | ops 1 calls 0
F0385 anonymous callback(x) | html:L1424-1424 | parent F0348 | S08 | ops 1 calls 0
F0386 x.onclick() | html:L1424-1424 | parent F0385 | S08 | ops 15 calls 12
F0387 $('movementSelect').onchange(e) | html:L1425-1425 | parent F0348 | S08 | ops 1 calls 2
F0388 $('qualitySelect').onchange(e) | html:L1425-1425 | parent F0348 | S08 | ops 1 calls 2
F0389 $('numberToggle').onclick() | html:L1425-1425 | parent F0348 | S08 | ops 2 calls 2
F0390 $('telegraphToggle').onclick() | html:L1425-1425 | parent F0348 | S08 | ops 2 calls 2

F0391 $('hitboxToggle').onclick() | html:L1425-1425 | parent F0348 | S08 | ops 2 calls 2
F0392 $('particleToggle').onclick() | html:L1425-1425 | parent F0348 | S08 | ops 2 calls 2
F0393 $('modalSwap').onclick() | html:L1425-1425 | parent F0348 | S08 | ops 0 calls 2
F0394 $('settingsFromPause').onclick() | html:L1427-1427 | parent F0348 | S08 | ops 0 calls 1
F0395 $('statsFromPause').onclick() | html:L1427-1427 | parent F0348 | S08 | ops 0 calls 1
F0396 $('helpFromPause').onclick() | html:L1427-1427 | parent F0348 | S08 | ops 0 calls 1
F0397 $('manualSave').onclick() | html:L1427-1427 | parent F0348 | S08 | ops 1 calls 2
F0398 $('importLocalSave').onchange(e) | html:L1431-1431 | parent F0348 | S08 | ops 1 calls 1
F0399 sortedBag() | html:L1435-1435 | parent F0001 | S07 | ops 0 calls 3
F0400 anonymous callback([k,n]) | html:L1435-1435 | parent F0399 | S07 | ops 2 calls 1
F0401 anonymous callback(a, b) | html:L1435-1435 | parent F0399 | S07 | ops 2 calls 3
F0402 stackedBag() | html:L1435-1435 | parent F0001 | S07 | ops 2 calls 7
F0403 logout() | html:L1436-1436 | parent F0001 | S10 | ops 11 calls 20
F0404 anonymous callback() | html:L1436-1436 | parent F0403 | S10 | ops 4 calls 0
F0405 accounts() | html:L1437-1437 | parent F0001 | S10 | ops 7 calls 3
F0406 toHex(arr) | html:L1438-1438 | parent F0001 | S10 | ops 0 calls 2
F0407 anonymous callback(b) | html:L1438-1438 | parent F0406 | S10 | ops 0 calls 2
F0408 hashPass(pass, salt) | html:L1439-1439 | parent F0001 | S10 | ops 1 calls 8
F0409 anonymous callback(x) | html:L1439-1439 | parent F0408 | S10 | ops 0 calls 1
F0410 setAuthMode(mode) | html:L1440-1455 | parent F0001 | S10 | ops 46 calls 26
F0411 onAuth(e) | html:L1456-1456 | parent F0001 | S10 | ops 39 calls 50
F0412 keyForEvent(e) | html:L1458-1463 | parent F0001 | S01 | ops 4 calls 7
F0413 isEditing(node) | html:L1464-1464 | parent F0001 | S01 | ops 2 calls 1
F0414 resetInputs() | html:L1465-1470 | parent F0001 | S01 | ops 9 calls 9
F0415 onCanvasMove(e) | html:L1471-1471 | parent F0001 | S01 | ops 7 calls 0
F0416 bindTouchButton(node, key) | html:L1472-1477 | parent F0001 | S01 | ops 0 calls 5
F0417 anonymous callback(e) | html:L1474-1474 | parent F0416 | S01 | ops 4 calls 4
F0418 end(e) | html:L1475-1475 | parent F0416 | S01 | ops 3 calls 2
F0419 joyMove(e) | html:L1479-1479 | parent F0001 | S01 | ops 24 calls 3
F0420 anonymous callback(e) | html:L1480-1480 | parent F0001 | S08 | ops 6 calls 3
F0421 anonymous callback(e) | html:L1481-1481 | parent F0001 | S08 | ops 1 calls 2
F0422 joyEnd(e) | html:L1482-1482 | parent F0001 | S01 | ops 5 calls 0
F0423 anonymous callback(e) | html:L1486-1486 | parent F0001 | S08 | ops 11 calls 3
F0424 anonymous callback(e) | html:L1487-1487 | parent F0001 | S08 | ops 5 calls 0
F0425 anonymous callback(e) | html:L1489-1489 | parent F0001 | S08 | ops 5 calls 2
F0426 anonymous callback() | html:L1490-1490 | parent F0001 | S08 | ops 1 calls 0
F0427 anonymous callback(e) | html:L1491-1491 | parent F0001 | S08 | ops 1 calls 2
F0428 anonymous callback(e) | html:L1492-1492 | parent F0001 | S08 | ops 1 calls 2
F0429 anonymous callback(e) | html:L1493-1493 | parent F0001 | S08 | ops 2 calls 3
F0430 anonymous callback(e) | html:L1494-1494 | parent F0001 | S08 | ops 2 calls 3
F0431 anonymous callback(e) | html:L1495-1511 | parent F0001 | S08 | ops 36 calls 33
F0432 anonymous callback(x) | html:L1496-1496 | parent F0431 | S08 | ops 0 calls 1
F0433 anonymous callback(id) | html:L1506-1506 | parent F0431 | S08 | ops 1 calls 0
F0434 anonymous callback(e) | html:L1512-1512 | parent F0001 | S08 | ops 0 calls 2
F0435 anonymous callback() | html:L1513-1513 | parent F0001 | S08 | ops 1 calls 2
F0436 anonymous callback() | html:L1514-1514 | parent F0001 | S08 | ops 0 calls 2
F0437 $('toolsMenuBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 4
F0438 $('skillMenuBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 4
F0439 $('talentBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0440 $('partyBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0441 anonymous callback(b) | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0442 anonymous callback() | html:L1515-1515 | parent F0441 | S08 | ops 0 calls 1
F0443 $('cfModeBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 4 calls 2
F0444 $('localModeBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 3 calls 1
F0445 $('cfUseCloud').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 1 calls 1
F0446 $('cfUseLocal').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 1 calls 1
F0447 $('localFallbackBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 5 calls 2
F0448 $('tabLogin').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0449 $('tabRegister').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0450 $('guestBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 2 calls 1

F0451 anonymous callback(b) | html:L1515-1515 | parent F0001 | S08 | ops 1 calls 0
F0452 b.onclick() | html:L1515-1515 | parent F0451 | S08 | ops 0 calls 1
F0453 $('gateNews').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 1 calls 1
F0454 anonymous callback(e) | html:L1515-1515 | parent F0001 | S08 | ops 1 calls 2
F0455 $('respawnBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 10 calls 7
F0456 $('mapBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0457 $('bagBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0458 $('craftBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0459 $('probabilityBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0460 $('codexBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0461 $('questBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0462 $('questStrip').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0463 $('collectionGoal').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0464 $('statsBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0465 $('helpBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0466 $('pauseBtn').onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0467 mini.onclick() | html:L1515-1515 | parent F0001 | S08 | ops 0 calls 1
F0468 updateHostingStatus() | html:L1517-1521 | parent F0001 | S08 | ops 11 calls 2
F0469 partyRoomName(raw) | html:L1527-1527 | parent F0001 | S11 | ops 1 calls 2
F0470 partySend(event, payload, topic) | html:L1528-1528 | parent F0001 | S11 | ops 8 calls 3
F0471 partySnapshot() | html:L1529-1529 | parent F0001 | S11 | ops 4 calls 4
F0472 partyPresence() | html:L1530-1530 | parent F0001 | S11 | ops 1 calls 2
F0473 leaveParty() | html:L1531-1531 | parent F0001 | S11 | ops 6 calls 4
F0474 joinParty(room) | html:L1532-1539 | parent F0001 | S11 | ops 18 calls 10
F0475 ws.onopen() | html:L1533-1533 | parent F0474 | S11 | ops 2 calls 2
F0476 anonymous callback() | html:L1533-1533 | parent F0475 | S11 | ops 2 calls 2
F0477 ws.onmessage(e) | html:L1534-1537 | parent F0474 | S11 | ops 44 calls 20
F0478 ws.onerror() | html:L1538-1538 | parent F0474 | S11 | ops 2 calls 0
F0479 ws.onclose() | html:L1538-1538 | parent F0474 | S11 | ops 4 calls 1
F0480 updateParty() | html:L1540-1540 | parent F0001 | S11 | ops 13 calls 8
F0481 drawRemotePlayers() | html:L1541-1541 | parent F0001 | S11 | ops 13 calls 9
F0482 renderPartyPanel() | html:L1542-1543 | parent F0001 | S11 | ops 21 calls 16
F0483 anonymous callback() | html:L1543-1543 | parent F0482 | S11 | ops 1 calls 4
F0484 anonymous callback() | html:L1543-1543 | parent F0482 | S11 | ops 0 calls 2
F0485 randomState() | html:L1544-1544 | parent F0001 | S08 | ops 1 calls 1
F0486 snapshot() | html:L1544-1544 | parent F0001 | S08 | ops 8 calls 17
F0487 anonymous callback(e) | html:L1544-1544 | parent F0486 | S08 | ops 0 calls 0
F0488 anonymous callback(x) | html:L1544-1544 | parent F0486 | S08 | ops 0 calls 0
F0489 anonymous callback(k) | html:L1544-1544 | parent F0486 | S08 | ops 2 calls 2
F0490 anonymous callback() | html:L1544-1544 | parent F0001 | S08 | ops 2 calls 2
F0491 anonymous callback() | html:L1544-1544 | parent F0001 | S08 | ops 3 calls 1
F0492 anonymous callback() | html:L1544-1544 | parent F0001 | S08 | ops 1 calls 2
F0493 anonymous callback() | html:L1544-1544 | parent F0001 | S08 | ops 4 calls 3
F0494 anonymous callback() | html:L1544-1544 | parent F0001 | S08 | ops 3 calls 1
F0495 accountId() | html:L1545-1545 | parent F0001 | S08 | ops 1 calls 0
F0496 anonymous callback(e) | html:L1545-1545 | parent F0001 | S08 | ops 1 calls 1
F0497 hash(value) | api:L12-12 | parent account-api:root | S12 | ops 0 calls 3
F0498 now() | api:L13-13 | parent account-api:root | S12 | ops 1 calls 2
F0499 anonymous callback(status, message, retry=0) | api:L15-15 | parent account-api:root | S12 | ops 2 calls 1
F0500 fail(status, message) | api:L17-17 | parent account-api:root | S12 | ops 0 calls 0
F0501 statement(db, sql, ...args) | api:L18-18 | parent account-api:root | S12 | ops 0 calls 2
F0502 reply(data, status=200, headers={}) | api:L19-21 | parent account-api:root | S12 | ops 0 calls 1
F0503 username(value, emailOnly=false) | api:L22-34 | parent account-api:root | S12 | ops 26 calls 19
F0504 anonymous callback(x) | api:L28-28 | parent F0503 | S12 | ops 0 calls 1
F0505 password(value) | api:L35-38 | parent account-api:root | S12 | ops 6 calls 1
F0506 derive(pass, salt) | api:L39-39 | parent account-api:root | S12 | ops 0 calls 0
F0507 anonymous callback(resolve, reject) | api:L39-39 | parent F0506 | S12 | ops 0 calls 1
F0508 anonymous callback(err, key) | api:L39-39 | parent F0507 | S12 | ops 1 calls 2
F0509 verifyPasswordRuntime() | api:L41-46 | parent account-api:root | S12 | ops 1 calls 1
F0510 anonymous callback() | api:L42-45 | parent F0509 | S12 | ops 1 calls 3

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F0511 makePassword(pass) | api:L47-50 | parent account-api:root | S12 | ops 0 calls 5
F0512 checkPassword(pass, record) | api:L51-63 | parent account-api:root | S12 | ops 25 calls 7
F0513 anonymous callback(resolve, reject) | api:L58-58 | parent F0512 | S12 | ops 0 calls 1
F0514 anonymous callback(err, key) | api:L58-58 | parent F0513 | S12 | ops 1 calls 2
F0515 bodyOf(request) | api:L64-72 | parent account-api:root | S12 | ops 12 calls 18
F0516 limit(db, bucket, max, seconds) | api:L73-79 | parent account-api:root | S12 | ops 3 calls 4
F0517 cookieToken(request) | api:L80-83 | parent account-api:root | S12 | ops 2 calls 2
F0518 accountFor(db, request) | api:L84-88 | parent account-api:root | S12 | ops 1 calls 5
F0519 publicAccount(a) | api:L89-89 | parent account-api:root | S12 | ops 0 calls 0
F0520 sessionCookie(db, account, remember) | api:L90-98 | parent account-api:root | S12 | ops 7 calls 13
F0521 validateSave(payload) | api:L99-113 | parent account-api:root | S12 | ops 60 calls 26
F0522 anonymous callback(k) | api:L102-102 | parent F0521 | S12 | ops 4 calls 1
F0523 item(x) | api:L103-103 | parent F0521 | S12 | ops 9 calls 2
F0524 putSave(db, id, expected, payload) | api:L114-122 | parent account-api:root | S12 | ops 10 calls 9
F0525 migrate(db, account, body) | api:L123-144 | parent account-api:root | S12 | ops 19 calls 27
F0526 onRequest({request,env}) | api:L146-214 | parent account-api:root | S12 | ops 61 calls 104
F0527 anonymous callback() | api:L2-24 | parent workshop:root | S15 | ops 1 calls 5
F0528 plan(bag, slots, defs, policy) | api:L4-17 | parent F0527 | S15 | ops 24 calls 29
F0529 anonymous callback(x) | api:L9-9 | parent F0528 | S15 | ops 3 calls 1
F0530 anonymous callback(x, i) | api:L9-9 | parent F0528 | S15 | ops 1 calls 1
F0531 score(k) | api:L10-10 | parent F0528 | S15 | ops 0 calls 1
F0532 anonymous callback(s, x, i) | api:L10-10 | parent F0531 | S15 | ops 4 calls 0
F0533 anonymous callback(k) | api:L11-11 | parent F0528 | S15 | ops 1 calls 0
F0534 anonymous callback(a, b) | api:L11-11 | parent F0528 | S15 | ops 7 calls 2
F0535 anonymous callback(n) | api:L15-15 | parent F0528 | S15 | ops 1 calls 0
F0536 anonymous callback([,n]) | api:L16-16 | parent F0528 | S15 | ops 1 calls 0
F0537 anonymous callback(s, k) | api:L16-16 | parent F0528 | S15 | ops 1 calls 1
F0538 anonymous callback(s, k) | api:L16-16 | parent F0528 | S15 | ops 1 calls 1
F0539 anonymous callback(k) | api:L16-16 | parent F0528 | S15 | ops 0 calls 1
F0540 release(x) | api:L18-22 | parent F0527 | S15 | ops 23 calls 9
F0541 text(s) | api:L19-19 | parent F0540 | S15 | ops 8 calls 1
F0542 anonymous callback() | api:L2-44 | parent interface:root | S16 | ops 1 calls 1
F0543 $(id) | api:L3-3 | parent F0542 | S16 | ops 0 calls 1
F0544 closeWelcome() | api:L5-5 | parent F0542 | S16 | ops 0 calls 3
F0545 paintNews(state) | api:L6-13 | parent F0542 | S16 | ops 15 calls 12
F0546 refresh() | api:L14-19 | parent F0542 | S16 | ops 2 calls 3
F0547 anonymous callback() | api:L16-16 | parent F0546 | S16 | ops 0 calls 1
F0548 anonymous callback() | api:L17-17 | parent F0546 | S16 | ops 9 calls 12
F0549 welcome() | api:L20-25 | parent F0542 | S16 | ops 5 calls 11
F0550 anonymous callback(r) | api:L23-23 | parent F0549 | S16 | ops 1 calls 0
F0551 ask(message, title="LOCALIZED_4673fa63") | api:L26-30 | parent F0542 | S16 | ops 0 calls 1
F0552 anonymous callback(resolve) | api:L29-29 | parent F0551 | S16 | ops 4 calls 7
F0553 install(feed) | api:L31-42 | parent F0542 | S16 | ops 7 calls 16
F0554 anonymous callback() | api:L34-34 | parent F0553 | S16 | ops 3 calls 3
F0555 $('confirmAccept').onclick() | api:L35-35 | parent F0553 | S16 | ops 0 calls 2
F0556 $('confirmCancel').onclick() | api:L35-35 | parent F0553 | S16 | ops 0 calls 2
F0557 anonymous callback() | api:L36-36 | parent F0553 | S16 | ops 3 calls 3
F0558 anonymous callback() | api:L37-37 | parent F0553 | S16 | ops 1 calls 4
F0559 anonymous callback() | api:L38-38 | parent F0553 | S16 | ops 3 calls 2
F0560 anonymous callback(e) | api:L39-39 | parent F0553 | S16 | ops 0 calls 2
F0561 anonymous callback() | api:L40-40 | parent F0553 | S16 | ops 3 calls 2
F0562 anonymous callback() | rules:L2-28 | parent rules:root | S13 | ops 1 calls 2
F0563 freeze(o) | rules:L4-4 | parent F0562 | S13 | ops 0 calls 3
F0564 anonymous callback(v) | rules:L4-4 | parent F0563 | S13 | ops 4 calls 1
F0565 finite(x) | rules:L6-6 | parent F0562 | S13 | ops 1 calls 1
F0566 probability(p) | rules:L7-7 | parent F0562 | S13 | ops 3 calls 1
F0567 integer(n, max=99999) | rules:L8-8 | parent F0562 | S13 | ops 5 calls 1
F0568 normalize(values) | rules:L9-9 | parent F0562 | S13 | ops 3 calls 5
F0569 anonymous callback(x) | rules:L9-9 | parent F0568 | S13 | ops 3 calls 1
F0570 anonymous callback(x) | rules:L9-9 | parent F0568 | S13 | ops 1 calls 0
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F0571 anonymous callback(s, x) | rules:L9-9 | parent F0568 | S13 | ops 1 calls 0

F0572 anonymous callback(x) | rules:L9-9 | parent F0568 | S13 | ops 1 calls 0

F0573 choose(weights, rng) | rules:L10-10 | parent F0562 | S13 | ops 5 calls 4

F0574 anonymous callback(x) | rules:L10-10 | parent F0573 | S13 | ops 1 calls 0

F0575 streams(seed) | rules:L11-11 | parent F0562 | S13 | ops 6 calls 3

F0576 out[key]() | rules:L11-11 | parent F0575 | S13 | ops 17 calls 2

F0577 out.snapshot() | rules:L11-11 | parent F0575 | S13 | ops 0 calls 0

F0578 dropRanks(rank, treasure=false) | rules:L12-12 | parent F0562 | S13 | ops 8 calls 11

F0579 add(r, p) | rules:L12-12 | parent F0578 | S13 | ops 2 calls 2

F0580 anonymous callback([rank,p]) | rules:L12-12 | parent F0578 | S13 | ops 0 calls 0

F0581 anonymous callback([rank,p]) | rules:L12-12 | parent F0578 | S13 | ops 0 calls 0

F0582 rewardStats(types, rank, guardian, treasure, value) | rules:L13-13 | parent F0562 | S13 | ops 13 calls 12

F0583 anonymous callback(s, x) | rules:L13-13 | parent F0582 | S13 | ops 2 calls 0

F0584 anonymous callback(s, x) | rules:L13-13 | parent F0582 | S13 | ops 4 calls 0

F0585 craftCDF(p, n) | rules:L14-14 | parent F0562 | S13 | ops 15 calls 2

F0586 craftAnalysis(p, n, attempts=5) | rules:L15-15 | parent F0562 | S13 | ops 41 calls 14

F0587 add(stock, wins, mass) | rules:L15-15 | parent F0586 | S13 | ops 4 calls 2

F0588 anonymous callback(x) | rules:L15-15 | parent F0586 | S13 | ops 1 calls 0

F0589 criticalChance(copies) | rules:L16-16 | parent F0562 | S13 | ops 3 calls 2

F0590 criticalFactor(copies, rng) | rules:L17-17 | parent F0562 | S13 | ops 4 calls 3

F0591 slowFactor(copies, guardian=false) | rules:L18-18 | parent F0562 | S13 | ops 4 calls 2

F0592 expectedUtility(outcomes, eta=0) | rules:L19-19 | parent F0562 | S13 | ops 3 calls 4

F0593 anonymous callback(s, x) | rules:L19-19 | parent F0592 | S13 | ops 1 calls 1

F0594 anonymous callback(s, x) | rules:L19-19 | parent F0592 | S13 | ops 8 calls 3

F0595 weightedValue(features, weights) | rules:L20-20 | parent F0562 | S13 | ops 2 calls 3

F0596 anonymous callback(x) | rules:L20-20 | parent F0595 | S13 | ops 5 calls 1

F0597 anonymous callback(s, w, i) | rules:L20-20 | parent F0595 | S13 | ops 2 calls 0

F0598 evaluatePolicy(outcomes, weights, eta=0) | rules:L21-21 | parent F0562 | S13 | ops 0 calls 2

F0599 anonymous callback(x) | rules:L21-21 | parent F0598 | S13 | ops 0 calls 1

F0600 damageMoments(raw, armor, copies, hp=null) | rules:L22-22 | parent F0562 | S13 | ops 17 calls 15

F0601 anonymous callback(s, x) | rules:L22-22 | parent F0600 | S13 | ops 2 calls 0

F0602 anonymous callback(s, x) | rules:L22-22 | parent F0600 | S13 | ops 4 calls 0

F0603 anonymous callback(x) | rules:L22-22 | parent F0600 | S13 | ops 1 calls 0

F0604 anonymous callback(s, x) | rules:L22-22 | parent F0600 | S13 | ops 0 calls 1

F0605 anonymous callback(x) | rules:L22-22 | parent F0600 | S13 | ops 1 calls 0

F0606 anonymous callback(s, x) | rules:L22-22 | parent F0600 | S13 | ops 0 calls 1

F0607 anonymous callback(s, x) | rules:L22-22 | parent F0600 | S13 | ops 3 calls 0

F0608 targetDropChance(types, rank, guardian, treasure, target, wanted) | rules:L23-23 | parent F0562 | S13 | ops 10 calls 5

F0609 anonymous callback(t) | rules:L23-23 | parent F0608 | S13 | ops 1 calls 0

F0610 anonymous callback(s, row) | rules:L23-23 | parent F0608 | S13 | ops 18 calls 2

F0611 frameAlpha(gain, dt) | rules:L24-24 | parent F0562 | S13 | ops 9 calls 4

F0612 specialProfile({type,rank=0,level=1,speed=0,enraged=false,frostFactor=1}) | rules:L25-25 | parent F0562 | S13 | ops 34 calls 6

F0613 specialAim(m, p) | rules:L26-26 | parent F0562 | S13 | ops 14 calls 9

F0614 anonymous callback() | art:L2-55 | parent art:root | S14 | ops 2 calls 1

F0615 ellipse(c, x, y, rx, ry, color, angle=0) | art:L5-5 | parent F0614 | S14 | ops 1 calls 3

F0616 halo(c, x, y, r, color, alpha=.2) | art:L6-6 | parent F0614 | S14 | ops 7 calls 6

F0617 terrain(c, zone, seed) | art:L7-15 | parent F0614 | S14 | ops 71 calls 45

F0618 random() | art:L7-7 | parent F0617 | S14 | ops 4 calls 1

F0619 landmark(c, l, zone, time, reduced) | art:L16-21 | parent F0614 | S14 | ops 102 calls 26

F0620 anatomy(c, m, body, detail, time, reduced) | art:L22-39 | parent F0614 | S14 | ops 303 calls 124

F0621 lighting(c, zone, width, height, time, reduced) | art:L40-40 | parent F0614 | S14 | ops 39 calls 21

F0622 rock(c, o, zone) | art:L41-41 | parent F0614 | S14 | ops 56 calls 29

F0623 traits(type) | art:L42-42 | parent F0614 | S14 | ops 2 calls 5

F0624 surface(c, m, body, zone, detail) | art:L43-47 | parent F0614 | S14 | ops 75 calls 28

F0625 status(c, m, time) | art:L48-48 | parent F0614 | S14 | ops 33 calls 13

F0626 telegraph(c, m, player, time, profile) | art:L49-53 | parent F0614 | S14 | ops 57 calls 45

Appendix B. Complete operator expression ledger

All 7,418 operator nodes occur inside these 3,795 maximal expression envelopes. Nested operators are preserved in their parent expression, with a one-to-envelope mapping and original expression hashes in standard_inventory.json. This ledger deliberately includes comparison, logical, assignment, update, string and bitwise operators: all affect program computation, but not all are arithmetic reward formulas. A localized literal is replaced by its LOCALIZED hash token; selectors and numerical interpolation remain. Operators inside a nested callback have that callback owner.

F0001 | anonymous callback

O00001 L240:10 Math.PI*2

O00002 L240:39 W/2

O00003 L240:46 H/2

F0004 | lerp

O00004 L241:61 a+(b-a)*t

F0005 | dist

O00007 L241:98 a-c

O00008 L241:102 b-d

F0006 | rnd

O00009 L241:118 a+(game.random?.world||Math.random)()*(b-a)

F0008 | anonymous callback

O00013 L242:21 n=(n+0x6d2b79f5)|0

O00016 L242:56 n^(n>>>15)

O00018 L242:67 1|n

O00019 L242:72 t^=t+Math.imul(t^(t>>>7),61|t)

O00024 L242:109 ((t^(t>>>14))>>>0)/4294967296

F0009 | anonymous callback

O00028 L243:338 i===7?1:.64/Math.pow(2,i)

F0001 | anonymous callback

O00031 L245:416 pets.antenna.desc="LOCALIZED_64c443f3"

F0010 | talentEarned

O00032 L305:48 (level-1)/2

F0011 | sanitizeTalents

O00034 L306:184 raw&&Object.prototype.hasOwnProperty.call(raw,k)?raw[k]:0

O00036 L306:251 Number.isFinite(n)?Math.max(0,Math.min(TALENTS[k].max,Math.floor(n))):0

O00037 L306:323 out[k]=Math.min(v,budget)

O00038 L306:349 budget-=out[k]

F0013 | talentRank

O00039 L307:32 Number(game.talents?.[key])||0

F0014 | talentPoints

O00040 L308:42 talentEarned(game.p?.level||1)-Object.values(game.talents||{}).reduce((sum,n)=>sum+n,0)

F0015 | anonymous callback

O00043 L308:121 sum+n

F0016 | talentModifier

O00044 L309:36 1+talentRank(key)*TALENTS[key].gain

F0018 | petalReloadTime

O00046 L311:35 pets[t].cd*(1-Math.min(.35,talentRank('reload')*.05))

F0019 | spendTalent

O00049 L312:29 !game.active||!TALENTS[key]||game.dead||talentPoints()<1||talentRank(key)>=TALENTS[key].max
O00057 L312:168 game.talents[key]++
O00058 L312:191 key==='vitality'
O00059 L312:208 game.p.hp=Math.min(maxHP(),old+(maxHP()-oldHP))
O00062 L312:301 TALENTS[key].name+"LOCALIZED_0c8ee481"+talentRank(key)+"LOCALIZED_42a30123"

F0020 | resetTalents

O00065 L313:27 !game.active
O00066 L313:53 game.talents=sanitizeTalents({},game.p.level)
O00067 L313:99 game.p.hp=Math.min(game.p.hp,maxHP())

F0021 | shrineCooldown

O00068 L315:50 shrine.readyAt-game.time

F0022 | activateShrine

O00069 L316:35 shrineCooldown(shrine)>0
O00070 L316:110 shrine.readyAt=game.time+data.cooldown
O00072 L316:152 shrine.type==='bloom'
O00073 L316:175 game.p.hp=Math.min(maxHP(),game.p.hp+maxHP()*.30)
O00076 L316:243 game.p.y-55
O00077 L316:284 game.boon={type:shrine.type,until:game.time+data.duration}
O00079 L316:388 "LOCALIZED_26167b41"+data.name+(shrine.type==='bloom'?"LOCALIZED_ace9cda0":"LOCALIZED_0a99e9a6")

F0023 | updateShrines

O00083 L317:48 map.shrines||[]
O00084 L317:67 shrineCooldown(shrine)===0&&dist(game.p.x,game.p.y,shrine.x,shrine.y)<82

F0024 | activeBoon

O00087 L318:33 game.boon?.type===type&&game.boon.until>game.time

F0001 | anonymous callback

O00090 L321:17 innerWidth/2
O00091 L321:32 innerHeight/2

F0026 | cloudStoreSession

O00092 L334:32 !cloudRemember||!cloudSession

F0027 | cloudReadStoredSession

O00095 L335:91 !raw
O00096 L335:139 a?.format===1&&typeof a.refresh_token==='string'&&a.refresh_token.length>10?a.refresh_token:null

F0028 | cloudOutboxKey

O00103 L337:26 'petal_realms_cloud_outbox_'+uid

F0029 | cloudLegacyFlag

O00104 L338:27 'petal_realms_cloud_pending_'+uid

F0030 | cloudRev

O00105 L339:28 value===null||value===undefined
O00108 L339:97 !Number.isSafeInteger(n)||n<1

F0031 | cloudReadOutbox

O00111 L340:90 !text

O00112 L340:139 data?.format!==1||!data.snapshot||typeof data.snapshot!=='object'||Array.isArray(data.snapshot)

O00119 L340:342 "LOCALIZED_37ec7aec"+err.message

F0034 | cloudReady

O00120 L343:29
/^https:\/\/[^/]+\.supabase\.co$/i.test(CLOUD_CONFIG.url)&&/^sb_publishable_/.test(CLOUD_CONFIG.publishableKey)

F0035 | cloudAuthMessage

O00121 L347:19 err?.code||''

O00122 L348:18 err?.message||''

O00123 L350:4 code==='invalid_credentials'||/invalid.login.credentials|invalid.grant|invalid.password|email.not.found|user.not.found/.test(message)

O00125 L352:4 ['email_address_not_authorized','email_not_authorized'].includes(code)||/email.address.not.authorized|email.recipient.not.authorized/.test(message)

O00126 L354:4 (/email|smtp|mail/.test(message)&&/sending|send|failed|error|provider|smtp/.test(message))||['email_provider_error','email_send_failed'].includes(code)

O00128 L356:4 code==='email_not_confirmed'||/email.not.confirmed|email.not.verified|confirm.your.email/.test(message)

O00130 L358:4 ['over_email_send_rate_limit','over_request_rate_limit','over_sms_send_rate_limit','rate_limit_exceeded'].includes(code)||/too.many.requests|rate.limit|email.rate.limit|email.send.rate/.test(message)

O00131 L360:4 code==='email_exists'||code==='user_already_exists'||/already.registered|already.exists/.test(message)

O00135 L362:4 ['signup_disabled','email_provider_disabled'].includes(code)||/signups?.(are.)?disabled/.test(message)

O00136 L364:4
['weak_password','password_too_short'].includes(code)||/password.should.be.at.least|weak.password/.test(message)

O00137 L366:4 code==='email_address_invalid'||/invalid.email|email.address.is.invalid/.test(message)

O00139 L370:4 code==='captcha_failed'||/captcha/.test(message)

O00141 L374:8 action==='register'?"LOCALIZED_26cbaeda":"LOCALIZED_a453e8f7"

F0036 | anonymous callback

O00143 L379:48 localStorage.getItem(MAIL_LIMIT_KEY)||'{}'

O00144 L379:139 Number.isFinite(raw[k])&&raw[k]>Date.now()

O00146 L379:182 out[k]=Math.min(raw[k],Date.now()+86400000)

F0037 | mailWait

O00148 L380:54 ((mailTimers[action]||0)-Date.now())/1000

F0038 | mailPause

O00151 L381:35 mailTimers[action]=Math.max(mailTimers[action]||0,Date.now()+Math.min(86400,Math.max(1,seconds))*1000)

F0039 | isMailUnavailable

O00155 L383:19 err?.code||''

O00156 L384:8 ['email_not_confirmed','email_address_not_authorized','email_not_authorized','email_provider_error','email_send_failed','over_email_send_rate_limit','mail_cooldown'].includes(code)||
(action!=='login'&&(err?.status===429||/SMTP|??????|Confirm Email|??/.test(String(err?.message||''))))

F0040 | refreshMailControls

O00162 L389:4 !send||!recover||!hint

O00167 L391:4 cloudMode&&authMode==='register'&&!cloudRestoring&&!authWorking

O00173 L391:68 send.disabled=signup>0

O00175 L392:1 recover.disabled=cloudMode&&(reset>0||mailSending.has('recovery'))

O00179 L393:1 hint.textContent=cloudMode&&((authMode==='register'&&signup)||reset)?"LOCALIZED_d6f1ead3"+Math.max(authMode==='register'?signup:0,reset)+"LOCALIZED_1d590e3a":''

F0041 | mailRequest

O00189 L396:32 wait||mailSending.has(action)

O00190 L396:112 e.code='mail_cooldown'

O00191 L399:15 err.status===429||err.code==='over_email_send_rate_limit'

O00194 L400:14 err.retryAfter||(err.code==='over_email_send_rate_limit'?3600:60)

O00197 L401:5 err.code==='over_email_send_rate_limit'

F0042 | cloudRequest

O00198 L405:4 !cloudReady()

O00199 L407:10 headers.Authorization='Bearer '+token

O00201 L408:11 headers.Prefer=prefer

O00202 L409:24 CLOUD_CONFIG.url+path

O00203 L409:67 body===undefined?undefined:JSON.stringify(body)

O00205 L411:5 data=raw?JSON.parse(raw):null

O00207 L411:53 "LOCALIZED_d500cfcf"+resp.status

O00208 L412:4 !resp.ok

O00209 L412:22 data?.msg||data?.message||data?.error_description||data?.error||('HTTP '+resp.status)

O00214 L412:137 e.code=String(data?.code||data?.error_code||'')

O00217 L412:185 e.status=resp.status

O00218 L412:267 retry?(Number.isFinite(Number(retry))?Number(retry):(Date.parse(retry)-Date.now())/1000):0

O00222 L412:358 e.retryAfter=Number.isFinite(seconds)?Math.min(86400,Math.max(0,Math.ceil(seconds))):0

F0043 | setCloudSession

O00224 L415:31 !a?.access_token||!a?.refresh_token||!a?.user?.id

O00229 L415:115 cloudSession={uid:a.user.id,access:a.access_token,refresh:a.refresh_token,expires:Date.now()+Math.max(30,a.expires_in||3600)*1000}

O00233 L415:246 cloudRevision=null

O00234 L415:265 cloudPending=null

O00235 L415:283 cloudConflict=false

O00236 L415:303 cloudRetryCount=0

O00237 L415:321 cloudLastUid=a.user.id

F0044 | cloudToken

O00238 L416:31 !cloudSession

O00239 L416:71 Date.now()<cloudSession.expires-60000

O00241 L418:1 cloudSession.access=a.access_token

O00242 L418:36 cloudSession.refresh=a.refresh_token||cloudSession.refresh

O00244 L418:95 cloudSession.expires=Date.now()+Math.max(30,a.expires_in||3600)*1000

F0045 | cloudLoad

O00248 L419:88 '/rest/v1/petal_realms_saves?user_id=eq.'+encodeURIComponent(cloudSession.uid)+'&select=revision,payload&limit=1'

O00250 L419:214 !Array.isArray(rows)

O00251 L419:260 cloudRevision=cloudRev(rows[0]?.revision)

O00252 L419:309 rows[0]?.payload??null

F0046 | cloudSchedule

O00253 L420:35 cloudSaveTimer!==null||cloudConflict||!cloudSession

O00257 L420:94 cloudSaveTimer=setTimeout(()=>{cloudSaveTimer=null;void cloudFlush()},ms)

F0047 | anonymous callback

O00258 L420:125 cloudSaveTimer=null

O00259 L420:145 void cloudFlush()

F0048 | cloudQueue

O00260 L421:33 !cloudSession||!game.active

O00263 L421:95 cloudPending=snapshot

O00264 L421:117 cloudGeneration++

O00265 L422:71 existing?existing.baseRevision:cloudRevision

O00266 L422:128 $('saveText').textContent="LOCALIZED_6e39d558"

F0049 | cloudFlush

O00267 L424:31 cloudBusy||cloudConflict||!cloudSession||cloudPending===null

O00272 L425:1 cloudBusy=true

O00273 L425:77 cloudPending=null

O00274 L428:5 expected===null

O00275 L428:22 rows=await cloudRequest('/rest/v1/petal_realms_saves?select=revision',{method:'POST',token,prefer:'retur
n=representation',body:{user_id:uid,revision:1,schema_version:5,payload:snapshot,updated_at:new Date().toISOString()}})

O00276 L429:7 rows=await cloudRequest('/rest/v1/petal_realms_saves?user_id=eq.'+encodeURIComponent(uid)+'&revision=eq.'
+encodeURIComponent(expected)+'&select=revision',{method:'PATCH',token,prefer:'return=representation',body:{revision:ex
pected+1,schema_version:5,payload:snapshot,updated_at:new Date().toISOString()}})

O00282 L430:5 !Array.isArray(rows)||rows.length!==1

O00285 L430:44 cloudConflict=true

O00286 L431:2 cloudRevision=cloudRev(rows[0].revision)

O00287 L431:43 success=true

O00288 L431:56 cloudRetryCount=0

O00289 L432:5 generation===cloudGeneration

O00290 L432:60 game.active&&game.user==='cloud-'+uid

O00293 L432:98 $('saveText').textContent="LOCALIZED_37ddd2c2"

O00294 L433:10 cloudPending!==null

O00295 L434:16 cloudPending===null

O00296 L434:36 cloudPending=snapshot

O00297 L435:5 game.active&&game.user==='cloud-'+uid

O00300 L435:43 $('saveText').textContent=cloudConflict?"LOCALIZED_4ebd83a3":"LOCALIZED_9cb5a870"

O00302 L437:10 cloudBusy=false

O00303 L438:5 success&&cloudPending!==null

O00305 L439:10 !success&&!cloudConflict&&cloudPending!==null&&cloudSession&&cloudRetryCount++<6

O00314 L439:120 2000*2**(cloudRetryCount-1)

F0050 | cloudEnter

O00317 L446:5 !user?.id||user.id!==cloudSession.uid

O00320 L448:5 !pending&&localStorage.getItem(cloudLegacyFlag(uid))==='1'

O00323 L449:17 pending.baseRevision!==cloudRevision

O00324 L449:120 resume=true

O00325 L450:2 game.user='cloud-'+uid

O00327 L450:25 game.guest=false

O00328 L450:51 resume?pending.snapshot:snapshot

O00329 L450:85 $('authPass').value=''

O00330 L450:108 $('authConfirm').value=''

O00331 L452:13 cloudPending=pending.snapshot

O00332 L452:43 cloudGeneration++

O00333 L453:13 cloudSession=null

O00334 L453:31 cloudRevision=null

O00335 L453:50 cloudPending=null

F0051 | cloudVerifiedSignupReady

O00336 L458:5 settings=await cloudRequest('/auth/v1/settings')

O00337 L461:4 settings?.autoconfirm!==false&&settings?.mailer_autoconfirm!==false

F0052 | cloudAuth

O00340 L466:4 !cloudReady()

O00341 L467:4 !/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(name)||name.length>254||pass.length<6

O00346 L468:1 cloudRemember=$('rememberMe').checked

O00347 L468:42 !cloudRemember

O00348 L469:4 authMode==='register'

O00349 L470:5 pass.length<8

O00350 L471:5 pass!==$('authConfirm').value

O00351 L473:37 '/auth/v1/signup?redirect_to='+encodeURIComponent(CLOUD_SITE_URL)

O00352 L474:5 !a?.access_token

O00353 L474:74 $('authError').textContent="LOCALIZED_fc2c56e0"

O00354 L474:172 $('authPass').value=''

O00355 L474:195 $('authConfirm').value=''

F0053 | cloudAutoResume

O00356 L481:4 !cloudMode||game.active||cloudRestoring||cloudRecovering

```
O00360 L482:43 !refresh
O00361 L483:1 cloudRestoring=true
O00362 L483:21 $('authError').textContent="LOCALIZED_8833cf87"
O00363 L483:63 $('authSubmit').disabled=true
O00364 L486:2 cloudRemember=true
O00365 L486:21 $('rememberMe').checked=true
O00366 L488:2 $('authError').textContent=''
O00367 L491:2 $('authError').textContent="LOCALIZED_9ee29132"+cloudAuthMessage(err,'login')+"LOCALIZED_34737059"
O00370 L492:10 cloudRestoring=false
O00371 L492:31 $('authSubmit').disabled=false
```

F0054 | cloudSendRecovery

```
O00372 L495:4 !cloudMode
O00373 L497:4 !/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email)
O00374 L497:48 $('authError').textContent="LOCALIZED_c341480c"
O00375 L498:34 btn.disabled=true
O00376 L498:52 $('authError').textContent="LOCALIZED_059c8425"
O00377 L500:31 '/auth/v1/recover?redirect_to='+encodeURIComponent(CLOUD_SITE_URL)
O00378 L501:2 $('authError').textContent="LOCALIZED_ed528f36"
O00379 L502:13 $('authError').textContent="LOCALIZED_ac2e2925"+cloudAuthMessage(err,'recovery')
O00381 L502:130 !isMailUnavailable(err,'recovery')
O00382 L503:9 btn.disabled=false
```

F0055 | cloudRecoveryFromURL

```
O00383 L506:4 !location.hash||!cloudReady()
O00386 L509:4 !access||!refresh||!['recovery','signup','email','magiclink'].includes(type)
O00391 L511:30 location.pathname+location.search
O00392 L512:1 cloudRestoring=true
O00393 L512:21 $('authSubmit').disabled=true
O00394 L515:5 !user?.id
O00395 L516:5 type==='recovery'
O00396 L517:4 cloudRemember=$('rememberMe').checked
O00397 L517:45 !cloudRemember
O00398 L518:79 Number(hash.get('expires_in'))||3600
O00399 L519:4 cloudRecovering=true
O00400 L519:25 $('authForm').style.display='none'
O00401 L519:60 $('forgotPasswordBtn').style.display='none'
O00402 L519:104 $('resetForm').style.display='block'
O00403 L520:4 $('authHeading').textContent="LOCALIZED_c7bdb2ce"
O00404 L520:41 $('authError').textContent="LOCALIZED_d733f950"
O00405 L522:4 cloudRemember=$('rememberMe').checked
O00406 L522:45 !cloudRemember
O00407 L523:80 Number(hash.get('expires_in'))||3600
O00408 L525:13 cloudSession=null
O00409 L525:31 cloudRecovering=false
O00410 L525:53 $('authError').textContent=err.message||"LOCALIZED_85ccb5e7"
O00412 L526:9 cloudRestoring=false
O00413 L526:30 $('authSubmit').disabled=false
```

F0056 | cloudUpdatePassword

```
O00414 L531:4 !cloudRecovering||!cloudSession
O00417 L533:4 pwd.length<8||pwd!==confirmed
O00420 L533:35 $('authError').textContent="LOCALIZED_62c9997b"
O00421 L534:28 btn.disabled=true
O00422 L537:5 !user?.id||user.id!==cloudSession.uid
O00425 L538:2 $('resetNewPass').value=''
O00426 L538:29 $('resetConfirmPass').value=''
O00427 L539:2 cloudRecovering=false
O00428 L539:24 $('resetForm').style.display='none'
```

```
O00429 L539:60 $('authForm').style.display='block'
O00430 L539:96 $('forgotPasswordBtn').style.display='block'
O00431 L540:112 (cloudSession.expires-Date.now())/1000
O00433 L542:13 $('authError').textContent=err.message||"LOCALIZED_75a3d64e"
O00435 L542:62 cloudRecovering=false
O00436 L542:84 $('resetForm').style.display='none'
O00437 L542:120 $('authForm').style.display='block'
O00438 L542:156 $('forgotPasswordBtn').style.display='block'
O00439 L542:201 cloudSession=null
O00440 L543:9 btn.disabled=false
```

F0057 | cloudChangePasswordFromSettings

```
O00441 L549:4 !form||!cloudSession||!game.active
O00446 L553:4 !old||fresh.length<8||fresh!==repeat||old===fresh
O00453 L553:55 msg.textContent="LOCALIZED_71fbd4d7"
O00454 L554:1 btn.disabled=true
O00455 L554:19 msg.textContent="LOCALIZED_04ea2526"
O00456 L558:5 user?.id!==uid||typeof user.email!=='string'
O00460 L560:5 !check?.access_token||check.user?.id!==uid
O00463 L562:5 updated?.id!==uid
O00464 L563:15 msg.textContent="LOCALIZED_ef61ddf1"
O00465 L564:13 msg.textContent="LOCALIZED_0536e8dd"+cloudAuthMessage(err,'login')
O00467 L565:9 form.querySelector('#currentGamePassword').value=''
O00468 L565:61 btn.disabled=false
```

F0058 | cloudExportBackup

```
O00469 L568:4 !uid
O00470 L568:103 all.length===1
O00471 L568:118 uid=all[0].slice('petal_realms_cloud_outbox_'.length)
O00472 L569:4 !uid
O00473 L570:63 'petal_realms_save_cloud-'+uid
O00474 L571:4 !pending&&!saved
O00477 L572:103 pending||null
O00478 L572:128 saved?JSON.parse(saved):null
O00479 L574:1 a.href=url
O00480 L574:12 a.download='petal-realms-cloud-backup-'+new Date().toISOString().slice(0,10)+'.json'
O00483 L575:1 $('authError').textContent="LOCALIZED_877e57c3"
O00484 L576:13 $('authError').textContent=err.message||"LOCALIZED_f263a786"
```

F0061 | configureAccountMode

```
O00486 L578:1 cfMode=false
O00487 L578:43 $('authForm').style.display='block'
O00488 L579:1 cloudMode=!!cloud
O00491 L579:19 cloudSession=null
O00492 L579:37 cloudRevision=null
O00493 L579:56 cloudPending=null
O00494 L579:74 cloudConflict=false
O00495 L580:1 $('authName').type=cloudMode?'email':'text'
O00497 L580:45 $('authName').name=cloudMode?'email':'username'
O00499 L580:93 $('authName').inputMode=cloudMode?'email':'text'
O00501 L580:142 $('authName').maxLength=cloudMode?254:18
O00503 L581:1 $('authNameLabel').textContent=cloudMode?"LOCALIZED_46541c3e":"LOCALIZED_67e9700c"
O00505 L582:1 $('authName').value=''
O00506 L582:24 $('authName').placeholder=cloudMode?'name@example.com':"LOCALIZED_fbd4a5df"
O00508 L583:1 $('authPass').value=''
O00509 L583:24 $('authConfirm').value=''
O00510 L584:1 $('cloudModeBtn').textContent=cloudMode?"LOCALIZED_a58f225b":"LOCALIZED_cb23b286"
O00512 L585:1 $('cloudBackupBtn').style.display=cloudMode?'block':'none'
O00514 L585:60 $('rememberWrap').style.display=cloudMode?'flex':'none'
```

O00516 L586:1 $('forgotPasswordBtn').style.display=cloudMode?'block':'none'

O00518 L586:63 $('guestBtn').style.display='none'

O00519 L588:1 $('authIntro').textContent=cloudMode?"LOCALIZED_58cfc4ed":"LOCALIZED_52f1994b"

O00521 L589:1 $('authDisclosure').textContent=cloudMode?"LOCALIZED_291cb7cf":"LOCALIZED_414c3645"

F0062 | switchCloudMode

O00523 L593:4 cloudRecovering||cloudRestoring||authWorking

O00525 L593:50 $('authError').textContent="LOCALIZED_8727e4fa"

F0001 | anonymous callback

O00526 L596:0 window.PetalCloudStatus=Object.freeze({configured:()=>cloudReady(),isCloudLogin:()=>!!cloudSession,mode:(
)=>cloudMode,pending:()=>cloudPending!==null,conflict:()=>cloudConflict})

F0064 | isCloudLogin

O00527 L596:84 !!cloudSession

F0066 | pending

O00529 L596:130 cloudPending!==null

F0068 | cfOutboxKey

O00530 L600:22 'petal_realms_cf_outbox_'+id

F0069 | cfReleaseLock

O00531 L603:43 cfLockRelease=null

O00532 L603:62 cfLockId=null

F0070 | cfTakeLock

O00533 L605:4 cfLockId===id

O00534 L606:4 !navigator.locks?.request

F0071 | anonymous callback

O00535 L608:26 'petal-realms-save-'+id

F0072 | anonymous callback

O00536 L609:6 !lock

O00537 L610:3 cfLockId=id

F0073 | anonymous callback

O00538 L610:45 cfLockRelease=done

F0074 | cfAPI

O00539 L616:5 response=await fetch('/api/'+path,{method,credentials:'same-origin',signal:AbortSignal.timeout(15000),hea
ders:{'Content-Type':'application/json',...(accountId?{'X-Petal-Account':accountId}:{})},body:body===undefined?undefine
d:JSON.stringify(body)})

O00544 L618:14 data=await response.json()

O00545 L619:4 !response.ok

O00546 L619:32 data?.error||"LOCALIZED_c38fb6f8"

O00547 L619:56 e.status=response.status

O00548 L619:81 e.retryAfter=Math.min(86400,Math.max(0,Number(response.headers.get('Retry-After'))||0))

F0075 | cfReadOutbox

O00550 L622:52 !raw

O00551 L623:32
p.format!==1||p.owner!==id||(p.baseRevision!==null&&(!Number.isSafeInteger(p.baseRevision)||p.baseRevision<1))

F0076 | cfSchedule

O00560 L626:35 cfTimer||cfConflict||!cfAccount

O00563 L626:74 cfTimer=setTimeout(()=>{cfTimer=null;void cfFlush()},delay)

F0077 | anonymous callback

O00564 L626:98 cfTimer=null
O00565 L626:111 void cfFlush()

F0078 | cfQueue

O00566 L628:4 !cfAccount||cfLoggingOut
O00568 L629:4 cfLockId!==cfAccount.id
O00569 L629:29 cfConflict=true
O00570 L629:45 $('saveText').textContent="LOCALIZED_d8a7685b"
O00571 L630:45 old&&old.baseRevision!==cfRevision
O00573 L630:81 cfConflict=true
O00574 L630:97 $('saveText').textContent="LOCALIZED_19055897"
O00575 L630:150 cfPending={format:1,owner:cfAccount.id,baseRevision:cfRevision,payload,time:new Date().toISOString()}
O00576 L630:326 $('saveText').textContent=cfConflict?"LOCALIZED_4ebd83a3":"LOCALIZED_6a5837e1"
O00578 L631:10 $('saveText').textContent="LOCALIZED_f3a5e0e0"

F0079 | cfFlush

O00579 L634:4 cfBusy||cfConflict||!cfAccount||!cfPending
O00584 L635:57 cfBusy=true
O00585 L638:5 epoch!==cfEpoch||cfAccount?.id!==account.id
O00588 L639:2 cfRevision=result.revision
O00589 L639:29 cfRetry=0
O00590 L640:5 cfPending===pending
O00591 L640:29 localStorage.getItem(cfOutboxKey(account.id))!==JSON.stringify(pending)
O00592 L640:102 cfConflict=true
O00593 L640:118 $('saveText').textContent="LOCALIZED_01d7c3bc"
O00594 L640:217 cfPending=null
O00595 L640:232 $('saveText').textContent="LOCALIZED_37ddd2c2"
O00596 L641:21 cfPending={...cfPending,baseRevision:cfRevision}
O00597 L643:5 epoch!==cfEpoch
O00598 L644:5 e.status===409
O00599 L644:21 cfConflict=true
O00600 L644:37 $('saveText').textContent="LOCALIZED_23f5ad95"
O00601 L645:7 $('saveText').textContent=e.status===401?"LOCALIZED_79bf215f":"LOCALIZED_f18ce847"
O00604 L646:5 !cfConflict&&e.status!==401&&cfRetry++<6
O00610 L646:66 e.retryAfter*1000
O00611 L646:99 2000*2**cfRetry
O00613 L647:13 epoch===cfEpoch
O00614 L647:30 cfBusy=false
O00615 L647:46 cfPending&&!cfConflict&&cfRetry===0

F0080 | cfConfigure

O00619 L650:29 cfMode=true
O00620 L650:41 $('authNameLabel').textContent="LOCALIZED_2e412e03"
O00621 L650:79 $('authName').maxLength=254
O00622 L650:107 $('authName').inputMode='email'
O00623 L650:139 $('authName').autocomplete='username'
O00624 L650:177 $('authName').placeholder='name@example.com'
O00625 L651:1 $('authIntro').textContent="LOCALIZED_86202d17"
O00626 L652:1 $('authDisclosure').textContent="LOCALIZED_68f530bc"
O00627 L653:1 $('cloudModeBtn').textContent="LOCALIZED_9266ff36"
O00628 L653:40 $('rememberWrap').style.display='flex'

F0081 | cfShowRecovery

O00629 L657:31 game.paused=true
O00630 L658:45 card.id='cfRecoveryCard'
O00631 L658:70 card.className='botanical-dialog'
O00632 L658:157 card.style.cssText='width:min(520px,calc(100vw - 28px))'

O00633 L659:1 card.innerHTML="LOCALIZED_a31a7a7a"

O00634 L660:28 $('cfRecoveryValue').value=code

O00635 L661:1 $('cfDownloadRecovery').onclick=()=>cfDownload({kind:'petal-realms-recovery-code',username:recoveryUserna
me,recoveryCode:code},'petal-realms-recovery.json')

O00636 L662:1 $('cfCloseRecovery').onclick=()=>{card.close();card.remove();game.paused=!!modal||game.dead}

F0084 | $('cfCloseRecovery').onclick

O00637 L662:62 game.paused=!!modal||game.dead

F0085 | cfDownload

O00641 L664:158 a.href=url

O00642 L664:169 a.download=name

F0087 | cfEnter

O00643 L666:4 !account?.id||!account.username

O00646 L669:58 data.accountId!==account.id

O00647 L672:4 pending&&pending.baseRevision!==data.revision

O00649 L673:15 pending?.payload||data.payload

O00650 L674:1 cfEpoch++

O00651 L674:11 cfAccount=account

O00652 L674:29 cfRevision=data.revision

O00653 L674:54 cfPending=pending

O00654 L674:72 cfConflict=false

O00655 L674:89 cfRetry=0

O00656 L674:99 cfBusy=false

O00657 L675:35 cfTimer=null

O00658 L676:1 cloudSession=null

O00659 L676:19 cloudPending=null

O00660 L676:37 game.user='cf-'+account.id

O00662 L676:64 game.guest=false

O00663 L677:19 $('authPass').value=''

O00664 L677:42 $('authConfirm').value=''

O00665 L677:68 $('authError').textContent=''

O00666 L678:62 game.paused=true

F0088 | cfAuthenticate

O00667 L682:42 authWorking=true

O00668 L682:59 $('authSubmit').disabled=true

O00669 L682:89 $('authError').textContent="LOCALIZED_859752bc"

O00670 L684:5 authMode==='register'&&$('authPass').value!==$('authConfirm').value

O00673 L685:25 'auth/'+(authMode==='register'?'register':'login')

O00676 L688:11
$('authError').textContent=e.message+(e.retryAfter?"LOCALIZED_447b6547"+e.retryAfter+"LOCALIZED_37a7d40c":'')

O00681 L689:9 authWorking=false

O00682 L689:27 $('authSubmit').disabled=false

F0089 | cfResume

O00683 L692:4 !cfMode||game.active||authWorking

O00686 L692:45 authWorking=true

O00687 L692:62 cloudRestoring=true

O00688 L692:82 $('authSubmit').disabled=true

O00689 L693:92 $('authError').textContent=e.message

O00690 L694:9 authWorking=false

O00691 L694:27 cloudRestoring=false

O00692 L694:48 $('authSubmit').disabled=false

F0090 | cfLogout

O00693 L697:4 cfLoggingOut||cfMigrating

O00694 L697:48 game.paused=true

O00695 L697:65 cfLoggingOut=true
O00696 L702:2 cfEpoch++
O00697 L702:12 cfAccount=null
O00698 L702:27 cfPending=null
O00699 L702:42 cfBusy=false
O00700 L702:105 cfTimer=null
O00701 L703:2 cfLoggingOut=false
O00702 L704:11 cfLoggingOut=false
O00703 L704:36 "LOCALIZED_4ba587b6"+e.message+"LOCALIZED_0828937d"

F0092 | cfExportPending

O00705 L707:14 cfAccount?.id||localStorage.getItem('petal_realms_cf_last_uid')
O00706 L707:82 id&&localStorage.getItem(cfOutboxKey(id))
O00707 L707:127 !raw
O00708 L708:10 $('authError').textContent=e.message

F0093 | cfUseCloud

O00709 L711:4 authWorking||game.active||!await askGame("LOCALIZED_5ff9d359")
O00712 L712:1 authWorking=true
O00713 L713:99 !account||account.id!==id
O00716 L713:182 !cfExportPending()
O00717 L714:22 'petal_realms_cf_before_cloud_'+id
O00718 L715:27 $('authError').textContent=e.message
O00719 L715:72 authWorking=false

F0094 | cfUseLocal

O00720 L718:4 authWorking||game.active||!await askGame("LOCALIZED_cf7afc66")
O00723 L719:1 authWorking=true
O00724 L722:5 !account||account.id!==id
O00727 L723:36 !pending
O00728 L725:5 !cfExportPending()
O00729 L728:27 $('authError').textContent=e.message
O00730 L728:72 authWorking=false

F0095 | cfPasswordForm

O00731 L732:4 $('cfNewPassword').value!==$('cfConfirmPassword').value
O00732 L732:61 msg.textContent="LOCALIZED_441bc454"
O00733 L732:96 btn.disabled=true
O00734 L733:162 msg.textContent="LOCALIZED_d03e27d0"
O00735 L733:208 msg.textContent=e.message
O00736 L733:242 btn.disabled=false
O00737 L733:261 $('cfOldPassword').value=''

F0096 | cfPanelControls

O00738 L736:27 !b||game.guest
O00740 L737:4 cfAccount&&['pause','settings'].includes(modal)
O00741 L739:2 $('cfAccountStatus').textContent="LOCALIZED_3f662b6d"+cfAccount.displayName+"LOCALIZED_588da410"+(cfConflict?"LOCALIZED_3c35aca5":cfPending?"LOCALIZED_dfdfa5d4":"LOCALIZED_ef6ae691"+(cfRevision??"LOCALIZED_53b650d6"))
O00749 L740:2 $('cfSaveNow').onclick=()=>{saveGame();void cfFlush()}

F0097 | $('cfSaveNow').onclick

O00750 L740:41 void cfFlush()

F0096 | cfPanelControls

O00751 L740:57 $('cfBackupPending').onclick=()=>{if(!cfExportPending())toast("LOCALIZED_c6e8b769")}

F0098 | $('cfBackupPending').onclick

O00752 L740:94 !cfExportPending()

F0096 | cfPanelControls

O00753 L741:2 $('cfExportProgress').onclick=()=>{saveGame();cfDownload({kind:'petal-realms-local-save',format:1,payload
:JSON.parse(localStorage.getItem('petal_realms_save_'+game.user))},'petal-realms-progress.json')}

F0099 | $('cfExportProgress').onclick

O00754 L741:140 'petal_realms_save_'+game.user

F0096 | cfPanelControls

O00755 L742:2 $('cfImportProgress').onchange=e=>{void cfImportProgress(e.target.files[0])}

F0100 | $('cfImportProgress').onchange

O00756 L742:37 void cfImportProgress(e.target.files[0])

F0096 | cfPanelControls

O00757 L742:79 $('cfRestoreProgress').onclick=cfRestoreProgress

O00758 L742:128 $('cfPasswordForm').onsubmit=cfPasswordForm

O00759 L743:10 modal==='pause'&&!cfAccount

O00762 L744:735 $('cfMigrateForm').onsubmit=cfMigrate

F0101 | cfImportProgress

O00763 L748:43 !file||!account||cfConflict

O00767 L749:8 file.size>1048576

O00768 L749:94 d.kind!=='petal-realms-local-save'||d.format!==1

O00771 L750:4 game.user!==user||cfAccount!==account

O00774 L750:73 !await askGame("LOCALIZED_3efa9864")

O00775 L751:33 'petal_realms_before_import_'+user

O00776 L751:89 'petal_realms_save_'+user

O00777 L752:17 "LOCALIZED_eceb664a"+e.message

F0102 | cfRestoreProgress

O00778 L755:4 !cfAccount||cfConflict

O00780 L755:69 'petal_realms_before_import_'+game.user

O00781 L755:113 !raw

O00782 L755:194 !await askGame("LOCALIZED_c749c64b")

F0103 | cfMigrate

O00783 L760:4 pass!==$('cfMigrateConfirm').value

O00784 L760:40 msg.textContent="LOCALIZED_749e2da1"

O00785 L761:4 !await askGame("LOCALIZED_56ddcff4")

O00786 L762:1 cfMigrating=true

O00787 L762:18 btn.disabled=true

O00788 L764:67 cloudPending||cloudConflict

O00789 L764:138 token=await cloudToken()

O00790 L765:49 'petal_realms_save_'+source

O00791 L766:15 data=await cfAPI('auth/register',{method:'POST',body:{username:name,password:pass,remember:true}})

O00792 L766:126 e.status!==409

O00793 L766:149 data=await cfAPI('auth/login',{method:'POST',body:{username:name,password:pass,remember:true}})

O00794 L768:5 game.user!==source

O00795 L772:27 msg.textContent="LOCALIZED_cdf16d90"+e.message

O00797 L773:9 cfMigrating=false

O00798 L773:27 btn.disabled=false

O00799 L773:46 $('cfMigratePassword')&&($('cfMigratePassword').value='')

O00801 L773:104 $('cfMigrateConfirm')&&($('cfMigrateConfirm').value='')

F0104 | cfRecoveryForm

O00803 L776:4 !cfMode||authWorking

O00805 L777:78 panel.style.display='block'

O00806 L778:1 panel.style.display='none'

O00807 L778:28 form=document.createElement('form')

O00808 L778:64 form.id='cfRecoverForm'

O00809 L778:88 form.innerHTML="LOCALIZED_968c645f"

O00810 L779:1 $('cfRecoverCancel').onclick=()=>{form.remove();panel.style.display='block'}

F0105 | $('cfRecoverCancel').onclick

O00811 L779:49 panel.style.display='block'

F0104 | cfRecoveryForm

O00812 L780:1 form.onsubmit=async e=>{e.preventDefault();if(authWorking)return;if($('cfRecoverPassword').value!==$('cfRecoverConfirm').value){$('cfRecoverError').textContent="LOCALIZED_749e2da1";return}authWorking=true; try{const data=await cfAPI('auth/recover',{method:'POST',body:{username:$('cfRecoverName').value,recoveryCode:$('cfRecoverCode').value,newPassword:$('cfRecoverPassword').value}});cfShowRecovery(data.recoveryCode,data.account);form.remove();panel.style.display='block';await cfEnter(data.account)}catch(err){const error=$('cfRecoverError')||$('authError');error.textContent=err.message}finally{authWorking=false}}

F0106 | form.onsubmit

O00813 L780:69 $('cfRecoverPassword').value!==$('cfRecoverConfirm').value

O00814 L780:129 $('cfRecoverError').textContent="LOCALIZED_749e2da1"

O00815 L780:179 authWorking=true

O00816 L781:241 panel.style.display='block'

O00817 L781:320 $('cfRecoverError')||$('authError')

O00818 L781:356 error.textContent=err.message

O00819 L781:394 authWorking=false

F0107 | validateLocalProgress

O00820 L786:4 !Object.hasOwn(zones,payload.zone)||payload.slots.length>10||!payload.slots.every(x=>x&&itemOf(x.t+'@'+x.r))||Object.keys(payload.bag).some(k=>!itemOf(k))

F0108 | anonymous callback

O00826 L786:89 x&&itemOf(x.t+'@'+x.r)

F0109 | anonymous callback

O00829 L786:147 !itemOf(k)

F0110 | sanitizeProtectedPetals

O00830 L791:18 !raw||typeof raw!=='object'||Array.isArray(raw)

O00835 L792:64 enabled===true&&itemOf(key)&&++count<=400

O00840 L792:106 safe[key]=true

F0111 | isPetalProtected

O00841 L795:38 game.protectedPetals?.[key]===true

F0112 | togglePetalProtection

O00842 L797:4 !game.active||!itemOf(key)||(game.bags[key]||0)<1

O00848 L798:4 !game.protectedPetals

O00849 L798:26 game.protectedPetals={}

O00850 L799:26 delete game.protectedPetals[key]

O00851 L799:64 game.protectedPetals[key]=true

O00852 L800:32 (isPetalProtected(key)?"LOCALIZED_211abcc5":"LOCALIZED_0d36c8c0")+pets[itemOf(key)[0]].name

F0113 | parseLocalBackup

O00854 L804:4 typeof raw!=='string'||raw.length>1048576

O00858 L806:4 data?.kind!=='petal-realms-local-save'||data.format!==1||!p||p.version!==5||!p.p||!Number.isInteger(p.p.level)||p.p.level<1||p.p.level>999|| !p.bag||typeof p.bag!=='object'||Array.isArray(p.bag)||!Array.isArray(p.slots)

O00881 L809:39 !Number.isFinite(p.p[k])||p.p[k]<0

O00884 L810:40 !Number.isInteger(n)||n<0||n>99999

F0114 | exportLocalProgress

O00889 L814:4 !game.active||game.guest||cloudSession
O00892 L815:40 !payload
O00893 L818:1 a.href=url
O00894 L818:12 a.download='petal-realms-progress-'+new Date().toISOString().slice(0,10)+'.json'

F0116 | importLocalProgress

O00897 L821:4 !file||!game.active||game.guest||cloudSession
O00902 L823:5 file.size>1048576
O00903 L824:77 !game.active||game.user!==owner||cloudSession
O00907 L825:5 !await askGame("LOCALIZED_d312624f")
O00908 L826:36 'petal_realms_save_'+game.user
O00909 L827:33 'petal_realms_before_import_'+game.user
O00910 L828:23 'petal_realms_save_'+game.user
O00911 L829:19 "LOCALIZED_eceb664a"+(err.message||"LOCALIZED_03854dd7")

F0117 | restoreBeforeImport

O00913 L832:4 !game.active||game.guest||cloudSession
O00916 L833:36 'petal_realms_before_import_'+game.user
O00917 L833:80 !raw
O00918 L835:5 !await askGame("LOCALIZED_31ed872e")
O00919 L836:23 'petal_realms_save_'+game.user
O00920 L837:19 "LOCALIZED_6e958e70"+err.message

F0118 | inventoryKey

O00921 L839:34 t+'@'+r

F0119 | itemOf

O00923 L839:64 typeof k!=='string'
O00925 L839:148 !match
O00926 L839:210 Object.hasOwn(pets,t)&&Number.isInteger(r)&&r>=0&&r<rarity.length?[t,r]:null

F0120 | addBag

O00932 L839:315 !itemOf(inventoryKey(t,r))
O00933 L839:375 game.bags[k]=clamp((game.bags[k]||0)+n,0,99999)
O00936 L839:426 !game.bags[k]
O00937 L839:440 delete game.bags[k]

F0121 | takeBag

O00938 L840:55 (game.bags[k]||0)<n
O00940 L840:99 -n

F0122 | slotCapacity

O00941 L845:69 lvl>=o.level
O00942 L845:82 v=o.slots

F0123 | viewScale

O00943 L846:31 1/Math.min(1.75,1+.05*Math.floor(lvl/10))

F0124 | antennaMultiplier

O00947 L846:109 1+game.slots.filter(s=>s.t==='antenna').reduce((sum,s)=>sum+.12+.018*s.r,0)

F0125 | anonymous callback

O00948 L846:132 s.t==='antenna'

F0126 | anonymous callback

O00949 L846:165 sum+.12+.018*s.r

F0127 | effectiveZoom

O00952 L846:223 viewScale(game.p.level)/antennaMultiplier()

F0128 | regionBand

O00953 L847:35 Math.hypot((x-cx)/cx,(y-cy)/cy)/Math.SQRT2

O00958 L847:90 i<BAND_THRESHOLDS.length

O00959 L847:115 i++

O00960 L847:122 rho<BAND_THRESHOLDS[i]

F0129 | minZoneLevel

O00961 L848:40 1+zones[zone].drop*4+ZONE_BANDS[band].offset

F0130 | bandLabel

O00964 L849:68 ZONE_BANDS[band].title+"LOCALIZED_e4dd17df"+lv+"LOCALIZED_d4f85d36"+(lv+3)

F0131 | maxHP

O00969 L850:35 (100+Math.floor((game.p.level-1)*7)+game.slots.reduce((n,s)=>n+(s.t==='cactus'?24*rarity[s.r].scale:s.t==='soil'?42*rarity[s.r].scale:0),0))*talentModifier('vitality')

F0132 | anonymous callback

O00974 L850:96 n+(s.t==='cactus'?24*rarity[s.r].scale:s.t==='soil'?42*rarity[s.r].scale:0)

F0134 | anonymous callback

O00981 L851:49 s.t==='soil'

F0135 | bodyRadius

O00982 L852:29 24+soilCount()*7

F0138 | anonymous callback

O00984 L854:62 e.t==='root'&&e.cd<=0&&e.hp>0

F0139 | anonymous callback

O00989 L854:127 total+2.5*rarity[e.r].scale

F0141 | anonymous callback

O00991 L855:51 e.t==='root'&&e.cd<=0&&e.hp>0

F0142 | magnetRadius

O00996 L856:31 (38+game.equipped.reduce((n,e)=>n+(e.t==='magnet'&&e.cd<=0?95+e.r*15:0),0))*talentModifier('harvest')

F0143 | anonymous callback

O00998 L856:63 n+(e.t==='magnet'&&e.cd<=0?95+e.r*15:0)

F0144 | reverseOrbit

O01005 L859:4 !game.active||game.paused||game.dead||modal

O01009 L860:1 game.p.orbitDirection=game.p.orbitDirection===-1?1:-1

O01014 L861:17 "LOCALIZED_9428536c"+(game.p.orbitDirection===-1?"LOCALIZED_c782be60":"LOCALIZED_8dc5be02")+"LOCALIZED_295405f5"

F0145 | recoverPetalDurability

O01019 L864:4 game.mode!=='defend'||e.cd>0||e.hp<=0||game.time-game.p.lastHit<1.2||game.time-(e.lastDamage??0)<1.2

O01031 L865:11 pets[e.t].hp*rarity[e.r].scale

O01032 L866:1 e.hp=clamp(e.hp+max*.06*dt,0,max)

F0146 | orbitRadius

O01036 L868:42 mode==='attack'?135:mode==='defend'?51:92

O01040 L868:91 (t==='wing'?base+(42+35*Math.sin(game.time*5+i*.7+Math.PI/6)):base)*talentModifier('reach')

F0149 | computeSynergies

O01051 L883:163 count>=def.min
O01052 L883:178 result[name]=count>=4?2:1

F0151 | syner

O01055 L884:28 game.synergies?.[name]||0

F0152 | activeSkill

O01056 L885:30 !game.active||game.paused||game.dead||!SKILL_CONFIG[name]
O01061 L886:33 (game.skills[name]||0)>game.time
O01063 L887:44 game.skills[name]=game.time+cfg.cd
O01065 L888:4 name==='storm'
O01066 L888:33 170+30*syner('assault')
O01068 L888:87 m.dead||dist(m.x,m.y,p.x,p.y)>radius+m.r
O01071 L888:149 (35+game.p.level*3)*petalDamageMultiplier()*(1+.18*syner('assault'))
O01077 L888:227 m.stunUntil=game.time+(mobs[m.t].guardian?.25:.75)
O01080 L889:4 name==='volley'
O01081 L889:31 (17+game.p.level*1.7)*petalDamageMultiplier()*(1+.25*syner('artillery'))
O01087 L889:116 i<10
O01088 L889:121 i++
O01089 L889:134 i*TAU/10+p.angle
O01092 L889:178 p.x+Math.cos(a)*38
O01094 L889:199 p.y+Math.sin(a)*38
O01096 L889:221 Math.cos(a)*500
O01097 L889:240 Math.sin(a)*500
O01098 L890:4 name==='barrier'
O01099 L890:22 game.skillBuffs.barrier=game.time+4.5+syner('fortress')
O01102 L891:4 name==='heal'
O01103 L891:42 maxHP()*(.13+.06*syner('nature'))
O01106 L891:77 p.hp=Math.min(maxHP(),p.hp+value)
O01108 L891:124 p.y-60
O01109 L891:131 '+'+value+' HP'
O01111 L892:4 name==='overdrive'
O01112 L892:24 game.skillBuffs.overdrive=game.time+7
O01114 L893:7 "LOCALIZED_776c414d"+cfg.name

F0153 | skillRemaining

O01115 L895:48 (game.skills[name]||0)-game.time

F0154 | weightedEvent

O01117 L897:27 game.random?.world||Math.random

F0155 | anonymous callback

O01118 L897:112 s+e.weight

F0154 | weightedEvent

O01119 L897:128 rng()*total
O01120 L897:192 (v-=e.weight)<0

F0156 | spawnNearPlayer

O01122 L898:75 map.living.length>=130
O01123 L898:187 map.living.length-1
O01124 L898:211 map.living.length>before&&tierBoost>0
O01127 L898:250 m.level+=Math.round(tierBoost*2)
O01129 L898:283 m.hp=Math.ceil(m.hp*(1+tierBoost*.15))
O01133 L898:322 m.max=m.hp
O01134 L898:333 m.dmg=Math.ceil(m.dmg*(1+tierBoost*.10))
O01138 L898:381 map.living.length>before

F0157 | beginWorldEvent

```
O01139 L899:34 !WORLD_EVENTS[type]
O01140 L899:67 game.worldEvent={type,until:game.time+WORLD_EVENTS[type].duration,nextPulse:game.time+3,meteors:[]}
O01143 L899:167 game.nextEventAt=game.time+WORLD_EVENTS[type].duration+48+(game.random?.world||Math.random)()*42
O01149 L900:4 type==='swarm'
O01150 L900:31 i<Math.min(7,130-game.mapData[game.zone].living.length)
O01152 L900:87 i++
O01153 L901:4 type==='treasure'
O01154 L901:106 i<7
O01155 L901:110 i++
O01156 L901:123 bag[Math.floor((game.random?.world||Math.random)()*bag.length)]||'basic'
O01159 L901:204 (game.random?.world||Math.random)()*TAU
O01161 L901:246 75+(game.random?.world||Math.random)()*210
O01164 L901:314 game.p.x+Math.cos(a)*d
O01166 L901:340 W-55
O01167 L901:354 game.p.y+Math.sin(a)*d
O01169 L901:380 H-55
O01170 L901:412 zones[game.zone].drop+Math.floor((game.random?.world||Math.random)()*3)
O01173 L902:7 "LOCALIZED_cb7f1000"+WORLD_EVENTS[type].name
```

F0159 | updateWorldEvents

```
O01174 L903:56 !e
O01175 L903:63 game.time>=game.nextEventAt
O01176 L903:134 game.time>=e.until
O01177 L903:154 game.worldEvent=null
O01178 L904:4 e.type==='renewal'
O01179 L904:23 game.p.hp=Math.min(maxHP(),game.p.hp+dt*2.2)
O01182 L905:4 game.time>=e.nextPulse
O01183 L905:28 e.nextPulse=game.time+(e.type==='meteor'?3.1:6.3)
O01187 L906:4 e.type==='swarm'
O01188 L906:33 i<2
O01189 L906:37 i++
O01190 L907:4 e.type==='meteor'
O01191 L907:31 (game.random?.world||Math.random)()*TAU
O01193 L907:73 50+(game.random?.world||Math.random)()*150
O01196 L907:140 game.p.x+Math.cos(a)*d
O01198 L907:166 W-60
O01199 L907:180 game.p.y+Math.sin(a)*d
O01201 L907:206 H-60
O01202 L907:215 game.time+1.1
O01203 L909:4 e.type==='meteor'
O01204 L909:57 impact.hit||game.time<impact.at
O01206 L909:98 impact.hit=true
O01207 L909:155 dist(impact.x,impact.y,game.p.x,game.p.y)<impact.r+game.p.r
O01209 L909:228 24+zones[game.zone].drop*6
O01211 L909:256 e.meteors=e.meteors.filter(x=>!x.hit)
```

F0160 | anonymous callback

```
O01212 L909:286 !x.hit
```

F0161 | specialProfileFor

```
O01213 L911:80 m.specialType||traitOf(m.t)
O01214 L911:113 m.rank||0
O01215 L911:129 m.level||1
O01216 L911:146 m.speed||0
O01217 L911:165 !!m.enraged
O01219 L911:189 m.frostUntil>game.time?m.frostFactor:1
```

F0162 | specialCooldown

```
001221 L912:35 Math.max(mobs[m.t].guardian?4.2:5.2,10.3-(m.rank||0)*.45-(mobs[m.t].guardian?1.8:0))*(m.enraged?.75:1)
```

F0163 | spawnSpecialProjectile

```
001229 L913:116 Math.atan2(dy,dx)+offset
001230 L913:178 m.x+Math.cos(a)*m.r
001232 L913:200 m.y+Math.sin(a)*m.r
001234 L913:223 Math.cos(a)*v
001235 L913:240 Math.sin(a)*v
001236 L913:256 trait==='venom'?10:8
001238 L913:281 m.dmg*(trait==='venom'?.48:.6)
001241 L913:340 trait==='venom'?'#9de893':'#ffe3a1'
001243 L913:383 trait==='venom'?'venom':null
```

F0164 | performMobSpecial

```
001245 L914:59 !t||m.dead
001247 L915:49 p.x-m.x
001248 L915:62 p.y-m.y
001249 L915:75 Math.hypot(dx,dy)||1
001250 L915:99 dx/len
001251 L915:109 dy/len
001252 L916:4 t==='sniper'||t==='venom'
001255 L917:4 t==='charger'
001256 L917:19 m.chargeFor=profile.duration
001257 L917:48 m.chargeVX=nx
001258 L917:62 m.chargeVY=ny
001259 L918:4 t==='summon'
001260 L918:27 game.zone==='ant'?'soldier':game.zone==='jungle'?'mantis':'ant'
001264 L918:103 i<Math.min(3,125-map.living.length)
001266 L918:139 i++
001267 L918:152 i*TAU/3
001269 L918:169 m.x+Math.cos(a)*95
001271 L918:191 W-70
001272 L918:206 m.y+Math.sin(a)*95
001274 L918:228 H-70
```

F0165 | anonymous callback

```
001275 L918:274 dist(xx,yy,o.x,o.y)<o.r+25
```

F0164 | performMobSpecial

```
001277 L918:343 def.hp*(1+.045*m.level)
001280 L918:388 ++mobUid
001281 L918:464 m.level-3
001282 L918:511 def.dmg*(1+.045*m.level)
001285 L918:636 game.time+6
001286 L919:4 t==='nova'
001287 L919:19 len<profile.radius
001288 L919:51 m.dmg*.65
001289 L920:4 t==='shield'
001290 L920:18 m.shieldUntil=game.time+4
001292 L920:57 m.y-44
```

F0166 | updateMobSpecial

```
001293 L922:41 m.specialAt===undefined
001294 L922:65 m.specialAt=game.time+5+(game.random?.world||Math.random)()*3
001299 L923:4 !m.specialType
001300 L923:19 m.specialType=traitOf(m.t)
001301 L923:49 !m.specialType
```

O01302 L924:4 m.windup>0

O01303 L924:16 m.windup-=dt

O01304 L924:32 m.windup<=0

O01305 L924:45 m.windup=0

O01306 L924:81 m.specialAt=game.time+specialCooldown(m)

O01308 L925:4 d<590&&game.time>=m.specialAt

O01311 L925:35 m.windup=specialProfileFor(m).windup

O01312 L925:72 m.warningX=game.p.x

O01313 L925:92 m.warningY=game.p.y

F0167 | resetEquip

O01314 L927:29 game.synergies=computeSynergies(game.slots)

O01315 L927:73 game.equipped=game.slots.map(s=>({t:s.t,r:s.r,hp:delay?0:pets[s.t].hp*rarity[s.r].scale,cd:delay,reloadTotal:delay,lastDamage:0,healWait:1.8,shotWait:.5+(game.random?.world||Math.random)(),lastHits:new Map(),x:game.p?.x??cx,y:game.p?.y??cy}))

F0168 | anonymous callback

O01316 L927:122 delay?0:pets[s.t].hp*rarity[s.r].scale

O01318 L927:223 .5+(game.random?.world||Math.random)()

O01320 L927:283 game.p?.x??cx

O01321 L927:299 game.p?.y??cy

F0169 | ensureSlots

O01322 L928:78 game.slots.length>=wanted

O01323 L928:117 game.slots.length<wanted

O01324 L928:231 "LOCALIZED_648a0902"+game.slots.length+"LOCALIZED_ebdb8d7d"

F0171 | readSave

O01326 L930:68 localStorage.getItem('petal_realms_save_'+game.user)||'null'

F0172 | saveGame

O01328 L931:23 !game.active||game.guest||!game.p

O01332 L931:89 'petal_realms_save_'+game.user

O01333 L931:466 $('saveText').textContent=cloudSession?"LOCALIZED_4cdc8b56":"LOCALIZED_1bd91a7d"

O01335 L931:576 'petal_realms_save_'+game.user

O01336 L931:613 cloudSession&&game.user==='cloud-'+cloudSession.uid

O01339 L931:708 'petal_realms_save_'+game.user

O01340 L931:748 $('saveText').textContent="LOCALIZED_8b921fba"

F0173 | initGame

O01341 L932:95 game.random=globalThis.PetalRules.streams(seed)

O01342 L932:143 game.time=0

O01343 L932:155 game.nextSpawn=1.3

O01344 L932:182 saved&&typeof saved==='object'?saved:baseSave()

O01348 L932:230 game.zone=zones[a.zone]?a.zone:'garden'

O01350 L932:270 game.mapData={}

O01351 L932:286 game.p=newPlayer()

O01352 L932:305 game.combo.count=0

O01353 L932:324 game.combo.until=0

O01354 L932:343 game.p.level=clamp(Math.floor(+a.p?.level||1),1,999)

O01357 L932:396 game.p.xp=clamp(Math.floor(+a.p?.xp||0),0,1e9)

O01360 L932:443 game.p.kills=clamp(Math.floor(+a.p?.kills||0),0,1e9)

O01363 L932:496 game.talents=sanitizeTalents(a.talents,game.p.level)

O01364 L932:549 game.boon=null

O01365 L932:564 game.skills={}

O01366 L932:579 game.skillBuffs={}

O01367 L932:598 game.worldEvent=null

O01368 L932:619 game.nextEventAt=65+(game.random?.world||Math.random)()*45

O01372 L932:678 game.bags={}

O01373 L932:725 a.bag||{}

O01374 L932:739 itemOf(k)&&Number.isFinite(v)&&v>0

O01377 L932:774 game.bags[k]=clamp(Math.floor(v),0,99999)

O01378 L933:0 game.protectedPetals=sanitizeProtectedPetals(a.protectedPetals)

O01379 L934:37 game.slots=Array.from({length:cap},(_,i)=>{const s=a.slots?.[i];return s&&pets[s.t]&&rarity[s.r]?{t:s.t,r:s.r}:{t:'basic',r:0}})

F0174 | anonymous callback

O01380 L934:108 s&&pets[s.t]&&rarity[s.r]?{t:s.t,r:s.r}:{t:'basic',r:0}

F0173 | initGame

O01383 L935:0 game.backupSlots=Array.from({length:cap},(_,i)=>{const q=a.backupSlots?.[i];return q&&pets[q.t]&&rarity[q.r]?{t:q.t,r:q.r}:{t:'basic',r:0}})

F0175 | anonymous callback

O01384 L935:83 q&&pets[q.t]&&rarity[q.r]?{t:q.t,r:q.r}:{t:'basic',r:0}

F0173 | initGame

O01387 L936:0 game.deck=a.deck===1?1:0

O01390 L936:25 game.deckCooldown=0

O01391 L936:45 game.journal={}

O01392 L936:121 game.journal[id]={kills:clamp(Math.floor(Number(j?.kills)||0),0,1e8),claimed:!!j?.claimed}

O01396 L936:212 game.guardians={}

O01397 L936:264 a.guardians||{}

O01398 L936:294 game.guardians[id]=clamp(Math.floor(Number(v)||0),0,1e6)

O01400 L936:351 game.settings={...DEFAULT_SETTINGS,...Object.fromEntries(Object.entries(a.settings||{}).filter(([k])=>k in DEFAULT_SETTINGS))}

F0176 | anonymous callback

O01402 L936:453 k in DEFAULT_SETTINGS

F0173 | initGame

O01403 L936:481 !['auto','low','high'].includes(game.settings.quality)

O01404 L936:536 game.settings.quality='auto'

O01405 L936:568 !['mouse','keyboard'].includes(game.settings.movement)

O01406 L936:623 game.settings.movement='mouse'

O01407 L936:654 game.goal=sanitizeCollectionGoal(a.goal)

O01408 L936:695 game.goalAnnounced=collectionOwned()>0

O01410 L937:0 game.codex={}

O01411 L937:48 a.codex||{}

O01412 L937:64 mobs[t]&&d&&typeof d==='object'

O01416 L937:96 game.codex[t]={seen:!!d.seen,kills:clamp(Math.floor(Number(d.kills)||0),0,100000000),maxLevel:clamp(Math.floor(Number(d.maxLevel)||0),0,999)}

O01421 L938:0 $('ui').inert=false

O01422 L938:33 game.p.hp=clamp(+a.p?.hp||100,1,maxHP())

O01425 L938:74 game.p.x=cx

O01426 L938:86 game.p.y=cy

O01427 L938:98 game.drops=[]

O01428 L938:112 game.bullets=[]

O01429 L938:128 game.friendlyShots=[]

O01430 L938:150 game.particles=[]

O01431 L938:168 game.floats=[]

O01432 L938:183 game.mode='normal'

O01433 L938:202 game.portalCool=2

O01434 L938:220 game.portalTimer=0

O01435 L938:239 game.portalTarget=null

O01436 L938:262 game.navTarget=null

O01437 L938:282 game.nearest=null

O01438 L938:300 game.zoom=effectiveZoom()

O01439 L938:326 game.p.dashCooldown=0

O01440 L938:348 game.active=true

O01441 L938:365 game.dead=false

O01442 L938:381 game.paused=false

O01443 L938:399 modal=null

O01444 L938:410 game.uiTime=0

O01445 L938:424 game.saveTime=0

O01446 L938:630 $('userBadge').textContent=cfAccount?cfAccount.username.split('@')[0].slice(0,18):game.user||"LOCALIZED_8214f290"

O01449 L938:728 $('saveText').textContent=game.guest?"LOCALIZED_738a182f":"LOCALIZED_1cb59e2f"

F0178 | anonymous callback

O01451 L939:166 x*(W/2500)

O01453 L939:179 y*(H/1760)

F0177 | buildMap

O01455 L940:12 i<950

O01456 L940:18 i++

O01457 L940:36 75+r()*(W-150)

O01460 L940:53 75+r()*(H-150)

O01463 L940:70 2+r()*10

O01465 L940:81 r()*TAU

O01466 L940:105 r()*5

O01467 L941:12 i<285

O01468 L941:18 i++

O01469 L941:31 115+r()*(W-230)

O01472 L941:49 115+r()*(H-230)

O01475 L941:69 19+r()*33

O01477 L941:82 dist(x,y,cx,cy)<260||portals.some(p=>dist(x,y,p.x,p.y)<135)||rocks.some(o=>dist(x,y,o.x,o.y)<rad+o.r+13)

F0179 | anonymous callback

O01480 L941:119 dist(x,y,p.x,p.y)<135

F0180 | anonymous callback

O01481 L941:157 dist(x,y,o.x,o.y)<rad+o.r+13

F0177 | buildMap

O01484 L941:234 r()*3

O01485 L941:243 r()*TAU

O01486 L942:65 o.x/GRID_SIZE

O01487 L942:94 o.y/GRID_SIZE

O01488 L942:113 kx+','+ky

O01490 L942:126 !lookup.has(key)

O01491 L943:162 n<75

O01492 L943:167 n++

O01493 L943:175 rocks.every(o=>dist(sx,sy,o.x,o.y)>o.r+115)&&portals.every(o=>dist(sx,sy,o.x,o.y)>o.r+145)&&dist(sx,sy,cx,cy)>300&&shrines.every(o=>dist(sx,sy,o.x,o.y)>240)

F0181 | anonymous callback

O01496 L943:190 dist(sx,sy,o.x,o.y)>o.r+115

F0182 | anonymous callback

O01498 L943:237 dist(sx,sy,o.x,o.y)>o.r+145

F0183 | anonymous callback

O01501 L943:307 dist(sx,sy,o.x,o.y)>240

F0177 | buildMap

O01502 L943:338 sx=290+r()*(W-580)

O01506 L943:357 sy=290+r()*(H-580)

O01510 L943:505 game.mapData[id]=result

O01511 L943:541 i<125

O01512 L943:547 i++

O01513 L943:572 !living.some(m=>m.t===GUARDIANS[id])

F0184 | anonymous callback

O01514 L943:588 m.t===GUARDIANS[id]

F0185 | spawnMob

O01515 L944:91 !map||map.living.length>=145

O01518 L944:135 game.random?.world||Math.random

O01519 L944:170
forced||(r()<.025&&!map.living.some(m=>m.t===GUARDIANS[id]))?GUARDIANS[id]:z.mob[Math.floor(r()*z.mob.length)]

F0186 | anonymous callback

O01524 L944:209 m.t===GUARDIANS[id]

F0185 | spawnMob

O01526 L945:12 i<60

O01527 L945:17 i++

O01528 L946:11 !initial&&id===game.zone&&r()<.66&&!data.guardian

O01535 L946:62 x=near?clamp(game.p.x+rnd(-980,980),100,W-100):100+r()*(W-200)

O01543 L946:125 y=near?clamp(game.p.y+rnd(-760,760),100,H-100):100+r()*(H-200)

O01551 L947:3 id===game.zone&&dist(x,y,game.p?.x??cx,game.p?.y??cy)<(initial?360:310)

O01557 L947:87 data.guardian&&regionBand(x,y)<2

O01559 L948:3 nearbyRocks(map.rocks,x,y).some(o=>dist(x,y,o.x,o.y)<o.r+data.size+10)||map.portals.some(o=>dist(x,y,o.x,
o.y)<o.r+data.size+42)||map.living.some(o=>dist(x,y,o.x,o.y)<o.r+data.size+9)

F0187 | anonymous callback

O01561 L948:38 dist(x,y,o.x,o.y)<o.r+data.size+10

F0188 | anonymous callback

O01564 L948:95 dist(x,y,o.x,o.y)<o.r+data.size+42

F0189 | anonymous callback

O01567 L948:151 dist(x,y,o.x,o.y)<o.r+data.size+9

F0185 | spawnMob

O01570 L948:195 good=true

O01571 L948:214 !good

O01572 L949:44 band===6&&!data.guardian&&r()<.003

O01577 L949:93 superSpawn?7:band

O01578 L949:123 minZoneLevel(id,band)+Math.floor(r()*4)+(superSpawn?18:0)

O01582 L949:193 !data.guardian&&r()<.05+band*.017

O01587 L949:237 (.82+.15*level)*(elite?1.45:1)*(superSpawn?2.4:1)

O01593 L950:20 ++mobUid

O01594 L950:37 data.size*(elite?1.16:1)

O01596 L950:100 data.hp*mul

O01597 L950:128 data.hp*mul

O01598 L950:156 data.dmg*(.84+level*.095)*(elite?1.20:1)*(superSpawn?1.7:1)

O01605 L950:223 data.speed*(1+z.drop*.04)

O01608 L950:261 r()*TAU

O01609 L950:274 r()*3

O01610 L950:286 r()*2

O01611 L950:334 elite?1:0

O01612 L950:354 game.time+3+r()*7

O01615 L950:426 r()*TAU

F0190 | createLandmarks

O01616 L954:13 i<n
O01617 L954:17 i++
O01618 L954:41 x=430+r()*(W-860)
O01622 L954:59 y=400+r()*(H-800)
O01626 L954:77 tries++
O01627 L954:91 tries<18&&(dist(x,y,cx,cy)<450||portals.some(p=>dist(x,y,p.x,p.y)<160))

F0191 | anonymous callback

O01631 L954:139 dist(x,y,p.x,p.y)<160

F0190 | createLandmarks

O01632 L954:180 75+r()*90
O01634 L954:205 i%2
O01635 L954:215 r()*99

F0192 | drawLandmark

O01636 L957:49 l.r+32

F0193 | nearbyRocks

O01637 L959:69 !lookup
O01638 L959:110 x/GRID_SIZE
O01639 L959:137 y/GRID_SIZE
O01640 L959:167 -1
O01641 L959:170 a<=1
O01642 L959:175 a++
O01643 L959:189 -1
O01644 L959:192 b<=1
O01645 L959:197 b++
O01646 L959:226 (gx+a)+','+(gy+b)

F0194 | slideCircle

O01650 L960:51 a.x+dx
O01651 L960:58 r+27
O01652 L960:63 W-r-27
O01654 L960:80 a.y+dy
O01655 L960:87 r+27
O01656 L960:92 H-r-27
O01658 L960:112 n<2
O01659 L960:116 n++
O01660 L960:169 nx-o.x
O01661 L960:179 ny-o.y
O01662 L960:210 r+o.r+1
O01664 L960:221 d<min
O01665 L960:231 d<.001
O01666 L960:239 vx=1
O01667 L960:244 vy=0
O01668 L960:249 d=1
O01669 L960:253 nx=o.x+vx/d*min
O01673 L960:269 ny=o.y+vy/d*min
O01677 L960:285 nx=clamp(nx,r+27,W-r-27)
O01681 L960:310 ny=clamp(ny,r+27,H-r-27)
O01685 L960:337 a.x=nx
O01686 L960:344 a.y=ny

F0195 | performDash

O01687 L961:26 !game.active||game.paused||game.dead||game.p.dashCooldown>0
O01692 L961:142 Math.hypot(dx,dy)<.12

```
O01693 L961:165 dx=Math.cos(game.p.angle)
O01694 L961:191 dy=Math.sin(game.p.angle)
O01695 L961:228 Math.hypot(dx,dy)||1
O01696 L961:249 game.p.dashVX=dx/norm
O01698 L961:271 game.p.dashVY=dy/norm
O01700 L961:293 game.p.dashTime=.20
O01701 L961:313 game.p.dashIFrames=.27
O01702 L961:336 game.p.dashCooldown=5.2
```

F0196 | swapLoadout

```
O01703 L962:26 !game.active||game.paused||game.dead||game.deckCooldown>0
O01708 L962:113 game.slots=game.backupSlots
O01709 L962:141 game.backupSlots=temp
O01710 L962:163 game.deck=1-game.deck
O01712 L962:185 game.deckCooldown=2.6
O01713 L962:207 game.p.hp=Math.min(game.p.hp,maxHP())
O01714 L962:288 e.hp=0
O01715 L962:295 e.cd=e.reloadTotal=Math.max(2.5,petalReloadTime(e.t))
O01717 L962:378 "LOCALIZED_f682cd08"+(game.deck===0?'A':'B')+"LOCALIZED_fa66ee6e"
```

F0197 | damagePlayer

```
O01721 L964:30 game.dead||game.p.dashIFrames>0||game.time-game.p.lastHit<.32
O01726 L964:110 (game.mode==='defend'?.69:1)*(activeBoon('ward')?.72:1)
O01730 L964:186 dmg*shield*(game.skillBuffs.barrier>game.time?.58:1)*(syner('fortress')?1-.09*syner('fortress'):1)
O01738 L964:288 game.p.armorLayers>0
O01739 L964:310 protectedDamage=Math.max(1,protectedDamage-rootedArmor())
O01741 L964:368 game.p.armorLayers--
O01742 L964:407 game.p.y-58
O01743 L964:437 game.p.hp-=protectedDamage
O01744 L964:464 game.p.lastHit=game.time
O01745 L964:489 game.p.glow=.5
O01746 L964:522 game.p.y-35
O01747 L964:534 "LOCALIZED_afba2418"+Math.ceil(protectedDamage)
O01748 L964:579 game.p.hp<=0
O01749 L964:593 game.p.hp=0
```

F0198 | recordMob

```
O01750 L965:43 game.codex[m.t]||(game.codex[m.t]={seen:true,kills:0,maxLevel:0})
O01752 L965:109 s.seen=true
O01753 L965:121 s.maxLevel=Math.max(s.maxLevel,m.level||1)
O01755 L965:174 s.kills++
```

F0199 | chooseDropRank

```
O01756 L966:38 m.rank??Math.min(6,m.band||0)
O01758 L966:77 game.random?.loot||Math.random
O01759 L966:114 rank===7
O01760 L966:130 r<.68?6:r<.96?5:7
O01764 L966:155 r<.12?rank:Math.max(0,rank-(r<.73?1:r<.94?2:3))
```

F0200 | armorOf

```
O01771 L966:230 (m.shieldUntil>game.time?8:0)+Math.min(14,(m.rank||0)*1.25+(m.tier||0)*2)
```

F0201 | damageMob

```
O01779 L966:386 dmg=Math.max(1,dmg-armor)
O01781 L966:412 m.hp-=dmg
O01782 L966:422 m.hit=.12
O01783 L966:435 dmg>0
O01784 L966:441 m.aggroUntil=game.time+8
O01786 L966:469 dmg>0&&!quiet
```

O01789 L966:496 "LOCALIZED_afba2418"+Math.round(dmg)
O01790 L966:530 m.hp<=0
O01791 L966:539 m.dead=true
O01792 L966:569 game.p.kills++
O01793 L966:657 mission&&!mission.claimed
O01795 L966:684 mission.kills++
O01796 L966:703 mission.kills===QUESTS[game.zone].goal
O01797 L966:793 game.guardians[game.zone]=(game.guardians[game.zone]||0)+1
O01800 L966:858 "LOCALIZED_3843c069"+mobs[m.t].name
O01801 L966:897 8+(m.level||1)*4+(m.tier?10:0)
O01806 L966:965 baseXP*(1+combo.bonus)*(game.worldEvent?.type==='bloodmoon'?1.28:1)
O01811 L966:1035 game.p.xp+=earnedXP
O01812 L966:1058 combo.count===5||combo.count===10||combo.count===20
O01817 L966:1116 "LOCALIZED_2c96f697"+combo.count+"LOCALIZED_ff914c01"+Math.round(combo.bonus*100)+'%'
O01822 L966:1181 combo.count>1&&combo.count%3===0
O01826 L966:1227 m.y-34
O01827 L966:1234 '+'+earnedXP+' XP'
O01829 L966:1270 game.p.xp>=xpNeed(game.p.level)
O01830 L966:1303 game.p.xp-=xpNeed(game.p.level)
O01831 L966:1335 game.p.level++
O01832 L966:1350 game.zoom=effectiveZoom()
O01833 L966:1376 game.p.hp=clamp(game.p.hp+22,0,maxHP())
O01835 L966:1440 "LOCALIZED_616cce48"+game.p.level+"LOCALIZED_8c953e5c"+Math.round(1/game.zoom*100)+'%'
O01841 L966:1584 (game.random?.loot||Math.random)()*mob.drop.length
O01843 L966:1663 game.worldEvent?.type==='treasure'&&(game.random?.loot||Math.random)()<.28
O01847 L966:1738 rank=Math.min(7,rank+1)
O01849 L966:1808 mob.guardian?80:42
O01850 L966:1831 (game.random?.visual||Math.random)()*TAU
O01852 L966:1909 m.x+32
O01853 L966:1918 m.y+10
O01854 L966:1952 rank-1
O01855 L966:1979 (game.random?.loot||Math.random)()<.18
O01857 L966:2037 m.x+((game.random?.visual||Math.random)()*34-17)
O01861 L966:2088 m.y+((game.random?.visual||Math.random)()*34-17)
O01865 L966:2160 rank-2
O01866 L966:2215 game.p.kills%5===0

F0202 | registerComboKill

O01868 L967:48 c.count=game.time<=c.until?Math.min(100,c.count+1):1
O01872 L967:101 c.until=game.time+7
O01874 L967:160 Math.max(0,c.count-1)*.05

F0203 | xpNeed

O01876 L968:28 35+Math.floor(lvl*18+lvl*lvl*.9)

F0204 | floating

O01881 L969:37 game.floats.length>55

F0205 | pulse

O01882 L970:44 i<num
O01883 L970:50 i++
O01884 L970:63 (game.random?.visual||Math.random)()*TAU
O01886 L970:106 30+(game.random?.visual||Math.random)()*130
O01889 L970:178 Math.cos(a)*s
O01890 L970:195 Math.sin(a)*s
O01891 L970:214 .3+(game.random?.visual||Math.random)()*.5
O01894 L970:272 2+(game.random?.visual||Math.random)()*3
O01897 L970:318 game.particles.length>220
O01898 L970:368 game.particles.length-220

F0206 | transition

```
O01899 L971:27 !zones[to]
O01900 L971:56 game.zone=to
O01901 L971:91 game.worldEvent=null
O01902 L971:112 game.nextEventAt=game.time+55+(game.random?.world||Math.random)()*38
O01907 L971:181 game.p.x=cx
O01908 L971:193 game.p.y=cy
O01909 L971:205 game.combo.count=0
O01910 L971:224 game.combo.until=0
O01911 L971:243 game.portalCool=2.7
O01912 L971:263 game.portalTimer=0
O01913 L971:282 game.portalTarget=null
O01914 L971:305 game.bullets=[]
O01915 L971:321 game.friendlyShots=[]
O01916 L971:343 game.drops=[]
O01917 L971:357 game.particles=[]
O01918 L971:375 game.boon=null
O01919 L971:432 p.x=game.p.x
O01920 L971:445 p.y=game.p.y
O01921 L971:458 game.camera.x=game.p.x-game.width/game.zoom/2
O01925 L971:504 game.camera.y=game.p.y-game.height/game.zoom/2
O01929 L971:557 "LOCALIZED_d7ce0bc1"+zones[to].name+"LOCALIZED_588da410"+zones[to].flavor
```

F0207 | die

```
O01932 L972:15 game.worldEvent=null
O01933 L972:36 game.dead=true
O01934 L972:51 game.paused=true
O01935 L972:68 game.boon=null
O01936 L972:83 game.combo.count=0
O01937 L972:102 game.combo.until=0
O01938 L972:121 pointer.left=false
O01939 L972:140 pointer.right=false
O01940 L972:160 touch.atk=false
O01941 L972:176 touch.def=false
O01942 L972:192 $('deathDesc').textContent="LOCALIZED_44927a08"+game.p.kills+"LOCALIZED_37497bfa"+zones[game.zone].name
+"LOCALIZED_219f12a0"
```

F0208 | moveInput

```
O01947 L973:38 keys.has('w')||keys.has('arrowup')
O01948 L973:73 dy-=1
O01949 L973:82 keys.has('s')||keys.has('arrowdown')
O01950 L973:119 dy+=1
O01951 L973:128 keys.has('a')||keys.has('arrowleft')
O01952 L973:165 dx-=1
O01953 L973:174 keys.has('d')||keys.has('arrowright')
O01954 L973:212 dx+=1
O01955 L973:221 joy.id!==null
O01956 L973:236 dx=joy.dx
O01957 L973:246 dy=joy.dy
O01958 L973:264 !dx&&!dy&&pointer.active&&game.settings.movement!=='keyboard'
O01964 L973:336 pointer.x-game.width/2
O01966 L973:362 pointer.y-game.height/2
O01968 L973:409 d>53
O01969 L973:415 dx=mx/d*clamp((d-53)/95,0,1)
O01974 L973:444 dy=my/d*clamp((d-53)/95,0,1)
O01979 L973:507 l>1?{dx:dx/l,dy:dy/l}:{dx,dy}
```

F0209 | petalContactRadius

```
O01983 L977:40 def.size*PETAL_WORLD_SCALE*.74
```

F0210 | mobCollider

```
O01985 L979:9 m.t||''
O01986 L979:19 m.r||1
O01987 L982:55 wx=.80
O01988 L982:62 wy=.61
O01989 L983:63 wx=.65
O01990 L983:70 wy=.93
O01991 L984:48 wx=.88
O01992 L984:55 wy=.71
O01993 L985:28 wx=.81
O01994 L985:35 wy=.80
O01995 L986:34 wx=.76
O01996 L986:41 wy=.77
O01997 L987:34 wx=.78
O01998 L987:41 wy=.78
O01999 L988:12 r*wx
O02000 L988:20 r*wy
O02001 L988:31 Number.isFinite(m.facing)?m.facing:0
```

F0211 | mobSupportRadius

```
O02002 L992:4 len<1e-8
O02003 L994:9 (dx*ca+dy*sa)/len
O02007 L994:29 (-dx*sa+dy*ca)/len
O02012 L997:19 o.rx*u
O02013 L997:26 o.ry*v
```

F0212 | mobCircleOverlap

```
O02014 L1003:10 x-m.x
O02015 L1003:19 y-m.y
O02016 L1003:31 dx*dx+dy*dy
O02019 L1004:13 Math.max(a,b)+r
O02020 L1005:4 dist2>=outer*outer
O02022 L1007:19 dx*cs+dy*sn
O02025 L1007:44 -dx*sn+dy*cs
O02029 L1008:4 (px/a)*(px/a)+(py/b)*(py/b)<=1
O02037 L1009:4 r<=0
O02038 L1012:10 a*a
O02039 L1012:17 b*b
O02040 L1013:13 2*Math.max(a*px,b*py)
O02043 L1014:13 i<18
O02044 L1014:18 i++
O02045 L1015:10 (lo+hi)*.5
O02047 L1016:10 a*px/(aa+t)
O02050 L1016:24 b*py/(bb+t)
O02053 L1017:5 u*u+v*v>1
O02057 L1017:15 lo=t
O02058 L1017:25 hi=t
O02059 L1019:9 (lo+hi)*.5
O02061 L1020:10 aa*px/(aa+t)
O02064 L1020:26 bb*py/(bb+t)
O02067 L1021:10 px-qx
O02068 L1021:19 py-qy
O02069 L1022:8 ex*ex+ey*ey<r*r
```

F0214 | updateMobFacing

```
O02074 L1026:4 vx*vx+vy*vy<.009
O02078 L1027:12 Math.atan2(vy,vx)+Math.PI/2
O02080 L1028:15 Number.isFinite(m.facing)?m.facing:goal
```

O02081 L1029:33 goal-current

O02082 L1029:56 goal-current

O02083 L1030:1 m.facing=current+delta*Math.min(1,dt*9)

F0215 | neutralMotion

O02087 L1032:37 m.aggroUntil>game.time

O02088 L1032:82 d<180?{dx:-dx*.65,dy:-dy*.65}:{dx:Math.cos(m.a)*.5,dy:Math.sin(m.a)*.5}

F0216 | actMode

O02096 L1033:22 touch.def||pointer.right||keys.has('shift')

O02098 L1033:84 touch.atk||pointer.left||keys.has(' ')||keys.has('space')

F0218 | anonymous callback

O02101 L1034:59 e.t===type&&e.hp>0&&e.cd<=0

F0219 | rawPetalDamage

O02106 L1035:77 e.t==='glass'?1:clamp(e.hp/(def.hp*scale),0,1)

O02110 L1035:131
def.dmg*scale*petalDamageMultiplier()*ratio*(1+.09*syner('assault'))*(game.skillBuffs.overdrive>game.time?1.17:1)

F0221 | clearTransientState

O02119 L1037:46 p.venomUntil=0

O02120 L1037:61 p.dashTime=0

O02121 L1037:74 p.dashIFrames=0

O02122 L1037:90 p.dashVX=0

O02123 L1037:101 p.dashVY=0

O02124 L1037:112 p.armorLayers=0

O02125 L1037:128 p.armorTimer=0

O02126 L1037:143 game.skillBuffs={}

O02127 L1037:162 game.boon=null

F0222 | update

O02128 L1038:23 !game.active||game.paused||game.dead

O02131 L1038:67 game.time+=dt

O02132 L1038:84 game.combo.count&&game.time>game.combo.until

O02134 L1038:130 game.combo.count=0

O02135 L1038:149 game.combo.until=0

O02136 L1038:168 game.portalCool=Math.max(0,game.portalCool-dt)

O02138 L1038:258 game.mode=actMode()

O02139 L1038:314 p.r=bodyRadius()

O02140 L1038:331 p.dashCooldown=Math.max(0,p.dashCooldown-dt)

O02142 L1038:376 p.dashIFrames=Math.max(0,p.dashIFrames-dt)

O02144 L1038:419 game.deckCooldown=Math.max(0,game.deckCooldown-dt)

O02146 L1038:473 game.boon&&game.boon.until<=game.time

O02148 L1038:511 game.boon=null

O02149 L1038:571 p.speed*(p.venomUntil>game.time?.79:1)*(syner('scout')?1+.07*syner('scout'):1)*bodySpeedFactor()*talen
tModifier('stride')*(activeBoon('swift')?1.22:1)*(game.mode===='defend'?.87:game.mode===='attack'?.94:1)

O02165 L1038:805 p.dashTime>0

O02166 L1038:819 p.dashTime-=dt

O02167 L1038:848 p.dashVX*920*dt

O02169 L1038:864 p.dashVY*920*dt

O02171 L1038:914 v.dx*speed*dt

O02173 L1038:928 v.dy*speed*dt

O02175 L1039:0 p.vx=(p.x-oldPX)/dt

O02178 L1039:20 p.vy=(p.y-oldPY)/dt

O02181 L1039:40 p.angle+=dt*(game.mode===='defend'?1.55:2.65)*(p.orbitDirection===-1?-1:1)

O02190 L1039:114 p.glow=Math.max(0,p.glow-dt)

O02192 L1039:146 p.venomUntil>game.time

O02193 L1039:169 p.hp=Math.max(0,p.hp-dt*(2.6+zones[game.zone].drop))

```
O02197 L1039:225 p.hp<=0&&!game.dead
O02200 L1039:251 game.zoom=effectiveZoom()
O02201 L1040:17 p.armorTimer+=dt
O02202 L1040:40 p.armorTimer>=2
O02203 L1040:57 p.armorTimer-=2
O02204 L1040:73 p.armorLayers=Math.min(10,p.armorLayers+1)
O02206 L1040:122 p.armorLayers=0
O02207 L1040:138 p.armorTimer=0
O02208 L1042:101 mob.enraged=!!(spec.guardian&&mob.hp>0&&mob.hp<=mob.max*.35)
O02216 L1042:165 d<Math.max(520,game.width/game.zoom*.6)
O02219 L1042:220 mob.hit=Math.max(0,mob.hit-dt)
O02221 L1042:251 mob.turn-=dt
O02222 L1042:264 mob.shoot-=dt
O02223 L1042:295 mob.calmed>0&&!spec.guardian
O02226 L1042:324 mob.calmed-=dt
O02227 L1042:373 mob.stunUntil>game.time
O02228 L1042:398 vx=0
O02229 L1042:403 vy=0
O02230 L1042:416 d<520&&!(mob.calmed>0&&!spec.guardian)
O02236 L1042:465 (p.x-mob.x)/Math.max(1,d)
O02238 L1042:494 (p.y-mob.y)/Math.max(1,d)
O02240 L1042:523 spec.ai==='ranged'
O02241 L1042:546 d>280
O02242 L1042:553 vx=dx
O02243 L1042:559 vy=dy
O02244 L1042:573 d<175
O02245 L1042:580 vx=-dx
O02247 L1042:587 vy=-dy
O02249 L1042:599 vx=-dy*.42
O02252 L1042:610 vy=dx*.42
O02254 L1042:623 mob.shoot<=0&&d<440
O02257 L1042:723 -.32
O02258 L1042:752 a+off
O02259 L1042:768 a+off
O02260 L1042:776 mob.shoot=rnd(1.65,2.85)*(mob.enraged?.60:1)
O02263 L1042:830 spec.ai==='wander'
O02264 L1042:885 vx=v.dx
O02265 L1042:893 vy=v.dy
O02266 L1042:910 spec.ai==='rush'
O02267 L1042:928 vx=dx
O02268 L1042:934 vy=dy
O02269 L1042:943 mob.turn<0
O02270 L1042:955 mob.turn=rnd(1.2,2.8)
O02271 L1042:977 mob.boost=.43
O02272 L1042:994 mob.boost>0
O02273 L1042:1007 mob.boost-=dt
O02274 L1042:1021 vx*=1.6
O02275 L1042:1029 vy*=1.6
O02276 L1042:1043 vx=dx
O02277 L1042:1049 vy=dy
O02278 L1042:1064 mob.turn<0
O02279 L1042:1076 mob.a=rnd(0,TAU)
O02280 L1042:1093 mob.turn=rnd(.7,2.4)
O02281 L1042:1114 vx=Math.cos(mob.a)*.46
O02283 L1042:1137 vy=Math.sin(mob.a)*.46
O02285 L1042:1163 mob.chargeFor>0
O02286 L1042:1180 mob.chargeFor=Math.max(0,mob.chargeFor-dt)
O02288 L1042:1223 vx=mob.chargeVX*globalThis.PetalRules.config.special.chargeMultiplier
O02290 L1042:1293 vy=mob.chargeVY*globalThis.PetalRules.config.special.chargeMultiplier
```

O02292 L1042:1409 vx*mob.speed*dt*(mob.enraged?globalThis.PetalRules.config.special.enrageSpeed:1)*(mob.frostUntil>game.time?mob.frostFactor:1)

O02299 L1042:1535 vy*mob.speed*dt*(mob.enraged?globalThis.PetalRules.config.special.enrageSpeed:1)*(mob.frostUntil>game.time?mob.frostFactor:1)

O02306 L1042:1694 mob.poisonTime>0

O02307 L1042:1712 mob.poisonTime-=dt

O02308 L1042:1731 mob.hp-=mob.poison*dt

O02310 L1042:1756 mob.hp<=0

O02311 L1044:12 i<map.living.length

O02312 L1044:32 i++

O02313 L1044:88 p.x-a.x

O02314 L1044:100 p.y-a.y

O02315 L1044:161 p.r+.5

O02316 L1044:180 mobSupportRadius(a,pdx,pdy)+p.r+.5

O02318 L1044:222 a.x-p.x

O02319 L1044:233 a.y-p.y

O02320 L1044:243 Math.hypot(dx,dy)||1

O02321 L1044:281 min-d

O02322 L1044:297 dx/d

O02323 L1044:305 dy/d

O02324 L1044:310 a.x+=nx*over*.65

O02327 L1044:327 a.y+=ny*over*.65

O02330 L1044:358 -nx*over*.35

O02333 L1044:371 -ny*over*.35

O02336 L1044:402 game.time-p.lastBody>.39

O02338 L1044:441 a.dmg*(a.enraged?1.20:1)*(game.worldEvent?.type==='bloodmoon'?1.18:1)

O02343 L1044:512 p.lastBody=game.time

O02344 L1044:536 game.time-(a.bodyStamp||0)>.56

O02347 L1044:568 a.bodyStamp=game.time

O02348 L1044:602 9+game.p.level*.4

O02350 L1044:640 i+1

O02351 L1044:644 j<map.living.length

O02352 L1044:664 j++

O02353 L1044:719 b.x-a.x

O02354 L1044:730 b.y-a.y

O02355 L1044:744 a.r+b.r+2

O02357 L1044:757 dx*dx+dy*dy>=upper*upper

O02362 L1044:820 mobSupportRadius(a,dx,dy)+mobSupportRadius(b,-dx,-dy)+1

O02366 L1044:879 d<mn

O02367 L1044:894 d<.01?1:dx/d

O02370 L1044:910 d<.01?0:dy/d

O02373 L1044:928 (mn-d)*.5

O02375 L1044:938 a.x-=nx*over

O02377 L1044:951 a.y-=ny*over

O02379 L1044:964 b.x+=nx*over

O02381 L1044:977 b.y+=ny*over

O02383 L1046:15 pass<2

O02384 L1046:22 pass++

O02385 L1046:42 i<map.living.length

O02386 L1046:62 i++

O02387 L1046:164 i+1

O02388 L1046:168 j<map.living.length

O02389 L1046:188 j++

O02390 L1046:243 b.x-a.x

O02391 L1046:254 b.y-a.y

O02392 L1046:268 a.r+b.r+2

O02394 L1046:281 dx*dx+dy*dy>=upper*upper

O02399 L1046:345 mobSupportRadius(a,dx,dy)+mobSupportRadius(b,-dx,-dy)+.5

O02403 L1046:405 d<min

O02404 L1046:421 d<.001?1:dx/d

```
O02407 L1046:438 d<.001?0:dy/d
O02410 L1046:454 (min-d)*.5
O02412 L1046:465 a.x-=nx*k
O02414 L1046:475 a.y-=ny*k
O02416 L1046:485 b.x+=nx*k
O02418 L1046:495 b.y+=ny*k
O02420 L1047:12 i<game.equipped.length
O02421 L1047:35 i++
O02422 L1047:101 p.angle+i*TAU/game.equipped.length
O02425 L1047:187 p.x+Math.cos(angle)*radius
O02427 L1047:217 p.y+Math.sin(angle)*radius
O02429 L1047:244 e.x=lerp(e.x,tx,globalThis.PetalRules.frameAlpha(.25,dt))
O02430 L1047:302 e.y=lerp(e.y,ty,globalThis.PetalRules.frameAlpha(.25,dt))
O02431 L1047:363 e.cd>0
O02432 L1047:371 e.cd-=dt
O02433 L1047:383 e.cd<=0
O02434 L1047:392 e.hp=def.hp*rar.scale
O02436 L1047:433 e.shotWait=.5
O02437 L1047:533 dist(e.x,e.y,o.x,o.y)<o.r+petalContactRadius(def)
O02439 L1047:587 e.t!=='glass'
O02440 L1047:602 e.hp-=13*dt
O02442 L1047:614 e.lastDamage=game.time
O02443 L1047:641 e.hp<=0
O02444 L1047:650 e.hp=0
O02445 L1047:657 e.cd=e.reloadTotal=petalReloadTime(e.t)
O02447 L1047:708 e.cd>0
O02448 L1049:30 m.dead||e.hp<=0
O02450 L1049:58 !mobCircleOverlap(m,e.x,e.y,petalContactRadius(def))
O02451 L1049:132 e.t==='glass'
O02452 L1049:158 !glass&&m.dmg<e.hp
O02455 L1049:191 multi?10:1
O02456 L1049:249 hit<strikes
O02457 L1049:261 hit++
O02458 L1049:271 e.hp<=0||m.dead
O02460 L1049:296 !multi&&game.time-(e.lastHits.get(m.id)||-10)<(glass?1:.24)
O02467 L1049:404 rawPetalDamage(e)*crit
O02468 L1049:457 actual++
O02469 L1049:469 crit>1
O02470 L1049:476 critCount++
O02471 L1049:488 dealt+=dmg
O02472 L1049:502 !glass
O02473 L1049:510 e.hp-=Math.max(.5,m.dmg)
O02474 L1049:535 e.lastDamage=game.time
O02475 L1049:562 !multi
O02476 L1049:603 e.hp<=0
O02477 L1049:612 e.hp=0
O02478 L1049:619 e.cd=e.reloadTotal=petalReloadTime(e.t)
O02480 L1049:669 !actual
O02481 L1049:703 "LOCALIZED_afba2418"+Math.round(dealt)+(actual>1?"LOCALIZED_4d4655d1"+actual:'')+(critCount?"LOCALIZED
_20d58620"+critCount:'')
O02489 L1049:781 critCount?'#efc3f6':'#fff1bb'
O02490 L1049:815 e.t==='fang'
O02491 L1049:828 p.hp=clamp(p.hp+Math.max(1,dealt*.07),0,maxHP())
O02494 L1049:880 e.t==='poison'||e.t==='iris'
O02497 L1049:910 m.poison=5*rar.scale
O02499 L1049:931 m.poisonTime=2.4
O02500 L1049:951 e.t==='lightning'
O02501 L1049:999 a!==m&&!a.dead&&dist(a.x,a.y,m.x,m.y)<125
O02506 L1049:1053 dealt*.2
O02507 L1049:1074 e.t==='frost'
```

```
O02508 L1049:1089 m.frostUntil=game.time+globalThis.PetalRules.config.frost.duration
O02510 L1049:1156
m.frostFactor=globalThis.PetalRules.slowFactor(Math.max(1,activeCopies('frost')),!!mobs[m.t].guardian)
O02513 L1049:1262 e.t==='puff'&&!mobs[m.t].guardian
O02516 L1049:1296 m.calmed=3.8
O02517 L1049:1317 Math.hypot(m.x-p.x,m.y-p.y)||1
O02520 L1049:1354 e.t==='bubble'?37:7
O02522 L1049:1374 m.x+=(m.x-p.x)/d*knock
O02526 L1049:1397 m.y+=(m.y-p.y)/d*knock
O02530 L1049:1444 actual>1?5:2
O02532 L1049:1461 e.cd>0
O02533 L1050:3 e.cd>0
O02534 L1050:19 e.shotWait-=dt
O02535 L1050:37 (e.t==='missile'||e.t==='grape')&&e.shotWait<=0
O02540 L1050:208 d<best&&d<(e.t==='missile'?570:480)
O02545 L1050:245 best=d
O02546 L1050:252 aim=m
O02547 L1050:286 aim.y-e.y
O02548 L1050:296 aim.x-e.x
O02549 L1050:347 Math.cos(a)*(e.t==='missile'?570:420)
O02552 L1050:388 Math.sin(a)*(e.t==='missile'?570:420)
O02555 L1050:432 e.t==='missile'?570:480
O02557 L1050:486 e.t==='missile'?10:12
O02559 L1050:512 def.dmg*rar.scale*petalDamageMultiplier()*(e.t==='missile'?2.4:1.4)
O02564 L1050:588 e.t==='grape'?2:0
O02566 L1050:618 e.shotWait=(e.t==='missile'?2.6:1.95)*(syner('artillery')?.84:1)
O02571 L1052:3 syner('nature')&&p.hp<maxHP()
O02573 L1052:33 p.hp=Math.min(maxHP(),p.hp+dt*.6*syner('nature'))
O02577 L1054:62 e.cd>0
O02578 L1055:3 e.t==='leaf'&&p.hp<maxHP()
O02581 L1055:30 p.hp=Math.min(maxHP(),p.hp+dt*1.9*rs)
O02585 L1056:3 e.t==='starpetal'&&p.hp<maxHP()*.75
O02589 L1056:39 p.hp=Math.min(maxHP(),p.hp+dt*3.1*rs)
O02593 L1057:3 e.t==='rose'||e.t==='nectar'
O02596 L1057:33 e.healWait=Math.max(0,e.healWait-dt)
O02598 L1057:84 e.t==='rose'?.97:.50
O02600 L1057:108 p.hp<maxHP()*trigger&&e.healWait<=0
O02604 L1057:167 (e.t==='rose'?19:35)*rs
O02607 L1057:212 maxHP()-p.hp
O02608 L1057:231 p.hp+=gain
O02609 L1057:242 e.healWait=e.t==='rose'?6:11
O02612 L1057:284 p.y-48
O02613 L1057:291 '+'+Math.round(gain)+' HP'
O02615 L1057:343 e.t==='rose'?'#ffa9bc':'#fce19a'
O02617 L1058:29 b.x+=b.vx*dt
O02619 L1058:42 b.y+=b.vy*dt
O02621 L1058:55 b.life-=dt
O02622 L1058:69 b.life<0||nearbyRocks(map.rocks,b.x,b.y).some(o=>dist(b.x,b.y,o.x,o.y)<o.r+b.r)
```

F0223 | anonymous callback

```
O02624 L1058:118 dist(b.x,b.y,o.x,o.y)<o.r+b.r
```

F0222 | update

```
O02626 L1058:150 b.dead=true
O02627 L1058:203 e.cd<=0&&dist(b.x,b.y,e.x,e.y)<b.r+pets[e.t].size*.65
O02632 L1058:261 e.t!=='glass'
O02633 L1058:276 e.hp-=b.dmg*.9
O02635 L1058:291 e.lastDamage=game.time
O02636 L1058:315 b.dead=true
O02637 L1058:358 e.hp<=0
```

O02638 L1058:367 e.hp=0
O02639 L1058:374 e.cd=e.reloadTotal=petalReloadTime(e.t)
O02641 L1058:423 !b.dead&&dist(b.x,b.y,p.x,p.y)<b.r+p.r
O02645 L1058:463 b.dead=true
O02646 L1058:498 b.status==='venom'
O02647 L1058:518 p.venomUntil=game.time+3.5
O02649 L1058:558 p.y-45
O02650 L1058:613 game.bullets=game.bullets.filter(b=>!b.dead)

F0224 | anonymous callback

O02651 L1058:649 !b.dead

F0222 | update

O02652 L1059:69 shot.vx*dt
O02653 L1059:83 shot.vy*dt
O02654 L1059:94 shot.x+=dx
O02655 L1059:105 shot.y+=dy
O02656 L1059:116 shot.traveled+=Math.hypot(dx,dy)
O02657 L1059:152
shot.traveled>shot.range||nearbyRocks(map.rocks,shot.x,shot.y).some(o=>dist(shot.x,shot.y,o.x,o.y)<o.r+shot.radius)

F0225 | anonymous callback

O02659 L1059:223 dist(shot.x,shot.y,o.x,o.y)<o.r+shot.radius

F0222 | update

O02661 L1059:269 shot.dead=true
O02662 L1059:323 m.dead||shot.hitIds.includes(m.id)||dist(shot.x,shot.y,m.x,m.y)>m.r+shot.radius
O02666 L1059:474 shot.bounces>0
O02667 L1059:490 shot.bounces--
O02668 L1059:558 n.dead||shot.hitIds.includes(n.id)
O02669 L1059:639 d<best
O02670 L1059:647 best=d
O02671 L1059:654 next=n
O02672 L1059:688 next.y-shot.y
O02673 L1059:702 next.x-shot.x
O02674 L1059:723 shot.vx=Math.cos(a)*v
O02676 L1059:745 shot.vy=Math.sin(a)*v
O02678 L1059:774 shot.dead=true
O02679 L1059:796 game.friendlyShots=game.friendlyShots.filter(s=>!s.dead)

F0226 | anonymous callback

O02680 L1059:844 !s.dead

F0222 | update

O02681 L1060:30 drop.life-=dt
O02682 L1060:44 drop.ang+=dt*2
O02684 L1060:120 pd<reach&&pd>36
O02687 L1060:151 (reach-pd)/Math.max(reach,1)
O02689 L1060:188 drop.x+=(p.x-drop.x)/pd*dt*(170+500*v)
O02696 L1060:227 drop.y+=(p.y-drop.y)/pd*dt*(170+500*v)
O02703 L1060:269 dist(drop.x,drop.y,p.x,p.y)<38
O02704 L1060:323 drop.life=-1
O02706 L1060:359 '+'+pets[drop.t].name
O02707 L1060:404 drop.r>=6
O02708 L1060:420 "LOCALIZED_7bbbdf12"+rarity[drop.r].en+' '+pets[drop.t].name
O02711 L1060:472 game.drops=game.drops.filter(d=>d.life>0)

F0227 | anonymous callback

O02712 L1060:504 d.life>0

F0222 | update

O02713 L1061:32 fx.x+=fx.vx*dt

O02715 L1061:47 fx.y+=fx.vy*dt

O02717 L1061:62 fx.vx*=.97

O02718 L1061:73 fx.vy*=.97

O02719 L1061:84 fx.life-=dt

O02720 L1061:96 game.particles=game.particles.filter(f=>f.life>0)

F0228 | anonymous callback

O02721 L1061:136 f.life>0

F0222 | update

O02722 L1061:174 f.y-=dt*37

O02724 L1061:185 f.life-=dt

O02725 L1061:196 game.floats=game.floats.filter(f=>f.life>0)

F0229 | anonymous callback

O02726 L1061:230 f.life>0

F0222 | update

O02727 L1062:0 map.living=map.living.filter(m=>!m.dead)

F0230 | anonymous callback

O02728 L1062:32 !m.dead

F0222 | update

O02729 L1062:41 game.nextSpawn-=dt

O02730 L1062:63 game.nextSpawn<=0

O02731 L1062:85 map.living.length<112

O02732 L1062:127 game.nextSpawn=1.3

O02733 L1063:0 game.camera.x=lerp(game.camera.x,clamp(p.x-game.width/game.zoom/2,0,Math.max(0,W-game.width/game.zoom)),globalThis.PetalRules.frameAlpha(8/60,dt))

O02740 L1063:147 game.camera.y=lerp(game.camera.y,clamp(p.y-game.height/game.zoom/2,0,Math.max(0,H-game.height/game.zoom)),globalThis.PetalRules.frameAlpha(8/60,dt))

O02747 L1063:296 game.uiTime-=dt

O02748 L1063:312 game.saveTime+=dt

O02749 L1063:333 game.uiTime<=0

O02750 L1063:359 game.uiTime=.13

O02751 L1063:378 game.saveTime>12

O02752 L1063:407 game.saveTime=0

F0231 | shoot

O02753 L1064:36 195+(zones[game.zone].drop*20)

O02755 L1064:88 m.x+dx*m.r

O02757 L1064:101 m.y+dy*m.r

O02759 L1064:115 dx*speed

O02760 L1064:127 dy*speed

O02761 L1064:144 m.dmg*.70*(m.enraged?1.20:1)

F0233 | updatePortals

O02764 L1066:34 game.portalCool>0

O02765 L1066:152 dist(portal.x,portal.y,game.p.x,game.p.y)<portal.r+game.p.r*.42

O02768 L1066:217 close=portal

O02769 L1066:239 !close

O02770 L1066:247 game.portalTimer=0

O02771 L1066:266 game.portalTarget=null

O02772 L1066:341 close!==game.portalTarget

O02773 L1066:368 game.portalTimer=0

O02774 L1066:387 game.portalTarget=close
O02775 L1066:411 game.portalTimer+=dt
O02776 L1066:477 $('portalMessage').textContent="LOCALIZED_98a08803"+zones[close.to].name+"LOCALIZED_77aaed7d"
O02779 L1066:546 $('portalFill').style.width=clamp(game.portalTimer/.9*100,0,100)+'%'
O02783 L1066:618 game.portalTimer>=.9

F0234 | proj

O02784 L1067:26 (x-game.camera.x)*game.zoom
O02786 L1067:54 (y-game.camera.y)*game.zoom

F0235 | visible

O02788 L1068:34
x+r>game.camera.x&&x-r<game.camera.x+game.width/game.zoom&&y+r>game.camera.y&&y-r<game.camera.y+game.height/game.zoom

F0236 | circle

O02803 L1069:91 ctx.fillStyle=fill
O02804 L1069:132 ctx.strokeStyle=stroke
O02805 L1069:155 ctx.lineWidth=lw

F0237 | drawShrine

O02806 L1070:31 !visible(shrine.x,shrine.y,65)
O02807 L1070:112 shrineCooldown(shrine)===0
O02808 L1070:186 ready&&game.settings.quality!=='low'
O02810 L1070:275 ctx.globalAlpha=ready?1:.35
O02812 L1070:313 reducedMotion.matches?0:Math.sin(game.time*2+shrine.x)*3
O02816 L1070:384 data.color+'35'
O02817 L1070:456 ctx.fillStyle=data.color
O02818 L1070:481 ctx.font='bold 26px system-ui'
O02819 L1070:512 ctx.textAlign='center'
O02820 L1070:548
shrine.type==='bloom'?"LOCALIZED_e6e1c70b":shrine.type==='swift'?"LOCALIZED_c07d0664":"LOCALIZED_3191b526"
O02824 L1070:606 bob+9
O02825 L1070:613 ctx.font='bold 12px system-ui'
O02826 L1070:657 ready?data.name:Math.ceil(shrineCooldown(shrine))+"LOCALIZED_00fe8dda"

F0238 | drawWorld

O02828 L1071:74 ctx.fillStyle=z.bg
O02829 L1071:190 -game.camera.x
O02830 L1071:205 -game.camera.y
O02831 L1071:263 ctx.strokeStyle=z.grid+'14'
O02833 L1071:291 ctx.lineWidth=1
O02834 L1071:333 Math.floor(game.camera.x/84)*84
O02836 L1071:365 x<Math.min(W,game.camera.x+game.width/game.zoom+84)
O02840 L1071:417 x+=84
O02841 L1071:488 game.camera.y+game.height/game.zoom
O02843 L1071:536 Math.floor(game.camera.y/84)*84
O02845 L1071:568 y<Math.min(H,game.camera.y+game.height/game.zoom+84)
O02849 L1071:621 y+=84
O02850 L1071:690 game.camera.x+game.width/game.zoom
O02852 L1071:742 ctx.lineWidth=36
O02853 L1071:759 ctx.strokeStyle=z.wall
O02854 L1071:801 W-16
O02855 L1071:806 H-16
O02856 L1071:812 ctx.strokeStyle='#e2ffe929'
O02857 L1071:840 ctx.lineWidth=3
O02858 L1071:877 W-70
O02859 L1071:882 H-70
O02860 L1072:15 map.landmarks||[]
O02861 L1072:79 map.shrines||[]

```
O02862 L1072:126 i<map.decor.length
O02863 L1072:145 i++
O02864 L1072:174 game.settings.quality==='low'&&i%3!==0
O02868 L1073:44 o.r+10
O02869 L1075:154 ctx.strokeStyle='#ff8c67'
O02870 L1075:180 ctx.lineWidth=3
O02871 L1076:45 m.r+35
O02872 L1076:115
game.settings.showTelegraph&&mobs[m.t].ai==='ranged'&&m.shoot>0&&m.shoot<.32&&dist(m.x,m.y,game.p.x,game.p.y)<470
O02880 L1076:241 ctx.strokeStyle='#fff1a299'
O02881 L1076:292 ctx.lineWidth=2
O02882 L1077:67 b.r+4
O02883 L1077:73 b.color+'45'
O02884 L1078:73 s.radius+4
O02885 L1078:84 s.t==='missile'?'#ffb77655':'#b79afa55'
O02887 L1078:149 s.t==='missile'?'#ffdb81':'#a18bda'
O02889 L1079:91 e.cd<=0
O02890 L1080:16 game.settings.particles?game.particles:[]
O02891 L1080:60 ctx.globalAlpha=clamp(f.life/f.max,0,1)
O02893 L1080:128 ctx.globalAlpha=1
O02894 L1080:174 ctx.globalAlpha=clamp(f.life/.85,0,1)
O02896 L1080:212 ctx.font='bold 12px system-ui'
O02897 L1080:243 ctx.textAlign='center'
O02898 L1080:266 ctx.fillStyle='#182624'
O02899 L1080:310 f.x+1
O02900 L1080:316 f.y+1
O02901 L1080:323 ctx.fillStyle=f.color
O02902 L1080:374 ctx.globalAlpha=1
O02903 L1080:409 game.settings.quality!=='low'
```

F0239 | drawRegionBands

```
O02904 L1081:52 i>=1
O02905 L1081:57 i--
O02906 L1081:70 BAND_THRESHOLDS[i-1]*Math.SQRT2
O02908 L1081:136 cx*k
O02909 L1081:141 cy*k
O02910 L1081:155 ctx.strokeStyle=rarity[i].hex+'48'
O02912 L1081:190 ctx.lineWidth=2.2
O02913 L1081:266 ctx.font='700 21px system-ui'
O02914 L1081:296 ctx.textAlign='center'
O02915 L1081:331 i<7
O02916 L1081:335 i++
O02917 L1081:350 (i===0?.07:(BAND_THRESHOLDS[i-1]+(BAND_THRESHOLDS[i]??.92))/2)*Math.SQRT2
O02924 L1081:438 cx+cx*rho
O02926 L1081:452 W-160
O02927 L1081:459 ctx.fillStyle='#122b2bc9'
O02928 L1081:498 x-102
O02929 L1081:504 cy-17
O02930 L1081:518 ctx.fillStyle=rarity[i].hex
O02931 L1081:581 cy+5
```

F0240 | drawDecor

```
O02932 L1082:89 ctx.lineWidth=1.8
O02933 L1082:110 zone===='garden'||zone==='jungle'
O02936 L1082:144 ctx.strokeStyle=zone==='garden'?'#d1e4a78a':'#9acb89a0'
O02939 L1082:229 s*.7
O02940 L1082:246 -s*.45
O02942 L1082:253 -s*.45
O02944 L1082:274 s*.7
O02945 L1082:291 s*.4
```

```
O02946 L1082:296 -s*.67
O02948 L1082:320 d.type===0
O02949 L1082:341 -s*.5
O02951 L1082:371 zone==='ocean'
O02952 L1082:387 ctx.strokeStyle='#bbe6e9a3'
O02953 L1082:443 s*.62
O02954 L1082:451 TAU*.8
O02955 L1082:475 d.type===2
O02956 L1082:520 zone==='desert'
O02957 L1082:537 ctx.strokeStyle='#fff0b56a'
O02958 L1082:595 Math.PI*.15
O02959 L1082:607 Math.PI*.85
O02960 L1082:641 zone==='hel'
O02961 L1082:655 ctx.strokeStyle='#ffb38a70'
O02962 L1082:710 -s
O02963 L1082:729 s*.2
O02964 L1082:748 -s*.8
O02966 L1082:776 zone==='factory'
O02967 L1082:794 ctx.fillStyle='#b7c5c459'
O02968 L1082:833 -s*.5
O02970 L1082:839 -s*.5
O02972 L1082:853 d.type===0
O02973 L1082:876 s*.3
O02974 L1082:898 ctx.strokeStyle=zone==='ant'?'#e6c99d90':'#a1bfb16f'
O02977 L1082:978 -s
O02978 L1082:1013 -s*.4
O02980 L1082:1033 s*.4
```

F0241 | drawRock

```
O02981 L1083:83 r+4
O02982 L1083:103 zone==='garden'||zone==='jungle'
O02985 L1083:140 o.kind===1
O02986 L1083:195 -r*.23
O02988 L1083:202 -r*.25
O02990 L1083:209 r*.63
O02991 L1083:215 zone==='garden'?'#8cb478':'#5ca979'
O02993 L1083:259 r*.26
O02994 L1083:265 r*.15
O02995 L1083:271 r*.47
O02996 L1083:293 ctx.fillStyle=zone==='jungle'?'#2e634c':'#477e59'
O02999 L1083:355 k<5
O03000 L1083:359 k++
O03001 L1083:372 k*TAU/5
O03003 L1083:387 Math.cos(a)*r*.45
O03005 L1083:405 Math.sin(a)*r*.45
O03007 L1083:423 r*.61
O03008 L1083:429 k%2?'#416e4e':'#7aa170'
O03010 L1083:465 r*.4
O03011 L1083:490 zone==='ocean'
O03012 L1083:554 k<6
O03013 L1083:558 k++
O03014 L1083:571 k*TAU/6
O03016 L1083:586 Math.cos(a)*r*.57
O03018 L1083:604 Math.sin(a)*r*.55
O03020 L1083:622 r*.24
O03021 L1083:628 k%2?'#7ccac4':'#e5b69a'
O03023 L1083:662 zone==='desert'
O03024 L1083:722 -r*.25
O03026 L1083:729 -r*.22
```

```
O03028 L1083:736 r*.52
O03029 L1083:761 zone===='hel'
O03030 L1083:811 ctx.strokeStyle='#f6a476'
O03031 L1083:837 ctx.lineWidth=3
O03032 L1083:880 -r*.4
O03034 L1083:886 -r*.5
O03036 L1083:920 r*.5
O03037 L1083:925 r*.65
O03038 L1083:953 zone===='factory'
O03039 L1083:971 ctx.fillStyle='#70828a'
O03040 L1083:995 ctx.strokeStyle='#b4c4c4'
O03041 L1083:1021 ctx.lineWidth=3
O03042 L1083:1050 -r*.76
O03044 L1083:1057 -r*.76
O03046 L1083:1064 1.52*r
O03047 L1083:1071 1.52*r
O03048 L1083:1094 -r*.76
O03050 L1083:1101 -r*.76
O03052 L1083:1108 1.52*r
O03053 L1083:1115 1.52*r
O03054 L1083:1137 -.47
O03055 L1083:1161 -.47
O03056 L1083:1178 x*r
O03057 L1083:1182 y*r
O03058 L1083:1207 zone===='sewer'
O03059 L1083:1223 ctx.fillStyle='#42675a'
O03060 L1083:1260 -r*.85
O03062 L1083:1267 -r*.7
O03064 L1083:1273 r*1.7
O03065 L1083:1279 r*1.4
O03066 L1083:1286 ctx.strokeStyle='#95bdad'
O03067 L1083:1312 ctx.lineWidth=3
O03068 L1083:1343 -r*.85
O03070 L1083:1350 -r*.7
O03072 L1083:1356 r*1.7
O03073 L1083:1362 r*1.4
O03074 L1083:1379 -1
O03075 L1083:1382 k<=1
O03076 L1083:1387 k++
O03077 L1083:1419 k*r*.45
O03079 L1083:1427 -r*.6
O03081 L1083:1445 k*r*.45
O03083 L1083:1453 r*.6
O03084 L1083:1521 -r*.2
O03086 L1083:1527 -r*.2
O03088 L1083:1533 r*.48
O03089 L1083:1567 game.settings.quality!=='low'
```

F0242 | drawPortal

```
O03090 L1084:45 p.r+Math.sin(game.time*3+p.x)*2
O03094 L1084:122 r+13
O03095 L1084:152 k<3
O03096 L1084:156 k++
O03097 L1084:189 i<=6
O03098 L1084:194 i++
O03099 L1084:205 i*TAU/6+game.time*(k%2?-.4:.5)+k*.4
O03108 L1084:245 r*(.64+k*.17)
O03111 L1084:265 Math.cos(a)*rad
O03112 L1084:283 Math.sin(a)*rad
```

```
O03113 L1084:302 i===0
O03114 L1084:361 ctx.strokeStyle=k===2?'#e6fbe2':z.accent
O03117 L1084:402 ctx.globalAlpha=.45+k*.14
O03120 L1084:428 ctx.lineWidth=3.4-k*.5
O03123 L1084:464 ctx.globalAlpha=1
O03124 L1084:493 r*.58
O03125 L1084:523 r*.35
O03126 L1084:529 z.accent+'73'
O03127 L1084:544 ctx.fillStyle='#f9fff0'
O03128 L1084:568 ctx.font='bold 13px system-ui'
O03129 L1084:599 ctx.textAlign='center'
O03130 L1084:644 ctx.font='bold 13px system-ui'
O03131 L1084:675 ctx.fillStyle='#fffaf0'
O03132 L1084:712 "LOCALIZED_a5393359"+z.name
O03133 L1084:726 r+30
```

F0243 | drawLoot

```
O03134 L1085:88 l.y+Math.sin(game.time*3+l.ang)*4
O03138 L1085:155 c+'36'
O03139 L1085:163 ctx.fillStyle='#162d2b'
O03140 L1085:187 ctx.strokeStyle=c
O03141 L1085:205 ctx.lineWidth=3
O03142 L1085:251 -14
O03143 L1085:255 -14
O03144 L1085:292 ctx.font='bold 19px system-ui'
O03145 L1085:323 ctx.fillStyle=pet.color
O03146 L1085:347 ctx.textAlign='center'
O03147 L1085:370 ctx.textBaseline='middle'
```

F0244 | drawFlower

```
O03148 L1086:75 game.time-p.lastHit<.25
O03150 L1086:99 ctx.globalAlpha=.68
O03151 L1086:122 p.armorLayers>0
O03152 L1086:139 ctx.strokeStyle='#a6efde'
O03153 L1086:165 ctx.lineWidth=2+Math.min(p.armorLayers,5)*.5
O03156 L1086:238 r+18
O03157 L1086:274 game.mode==='defend'?'#b0dafa44':game.mode==='attack'?'#ffcda94a':'#e2e9bd27'
O03161 L1086:363 r+12
O03162 L1086:386 a<6
O03163 L1086:390 a++
O03164 L1086:405 a*TAU/6
O03166 L1086:440 ctx.fillStyle='#f7e0a9'
O03167 L1086:464 ctx.strokeStyle='#a6b784'
O03168 L1086:490 ctx.lineWidth=2
O03169 L1086:536 -r*.82
O03171 L1086:543 r*.48
O03172 L1086:549 r*.67
O03173 L1086:645 -8
O03174 L1086:648 -5
O03175 L1086:675 -5
O03176 L1086:727 Math.PI-.15
O03177 L1086:740 ctx.strokeStyle='#8a6b53'
O03178 L1086:766 ctx.lineWidth=2.5
O03179 L1086:797 ctx.globalAlpha=1
O03180 L1086:831 p.hp/maxHP()
O03181 L1086:849 ctx.fillStyle='#0d282688'
O03182 L1086:888 -28
O03183 L1086:892 -r-22
O03185 L1086:904 ctx.fillStyle=pct>.38?'#8fe1a0':'#f1a08e'
```

```
O03188 L1086:959 -28
O03189 L1086:963 -r-22
O03191 L1086:969 56*pct
```

F0245 | drawPetal

```
O03192 L1087:45 d.size*PETAL_WORLD_SCALE
O03193 L1087:115 game.p.angle*.25
O03194 L1087:144 size*.84
O03195 L1087:182 ctx.fillStyle=c
O03196 L1087:198 ctx.strokeStyle=rarity[e.r].hex
O03197 L1087:230 ctx.lineWidth=2.6
O03198 L1087:251 e.t==='missile'
O03199 L1087:300 -2
O03200 L1087:315 -size*.7
O03202 L1087:324 -size*.6
O03204 L1087:345 -size*.45
O03206 L1087:369 -size*.7
O03208 L1087:378 size*.6
O03209 L1087:434 -size*.35
O03211 L1087:446 size*.16
O03212 L1087:474 e.t==='grape'
O03213 L1087:512 -.6
O03214 L1087:518 -.35
O03215 L1087:523 -.15
O03216 L1087:534 -.15
O03217 L1087:541 -.22
O03218 L1087:569 x*size
O03219 L1087:576 y*size
O03220 L1087:583 size*.3
O03221 L1087:622 e.t==='puff'
O03222 L1087:659 -.37
O03223 L1087:690 -.25
O03224 L1087:708 x*size
O03225 L1087:715 y*size
O03226 L1087:722 size*k
O03227 L1087:768 e.t==='prism'
O03228 L1087:812 -size
O03229 L1087:830 size*.7
O03230 L1087:871 -size*.7
O03232 L1087:899 ctx.fillStyle=materialFill(d.color,size)
O03233 L1087:993 -size
O03234 L1087:1030 -size*.7
O03236 L1087:1053 size*.7
O03237 L1087:1064 ctx.strokeStyle='#fff1f7'
O03238 L1087:1090 ctx.lineWidth=1
O03239 L1087:1127 e.t==='frost'
O03240 L1087:1142 ctx.strokeStyle=d.color
O03241 L1087:1166 ctx.lineWidth=3
O03242 L1087:1194 k<6
O03243 L1087:1198 k++
O03244 L1087:1225 k*TAU/6
O03246 L1087:1279 -size
O03247 L1087:1299 -size*.65
O03249 L1087:1321 -size*.22
O03251 L1087:1331 -size*.83
O03253 L1087:1355 -size*.65
O03255 L1087:1377 size*.22
O03256 L1087:1386 -size*.83
O03258 L1087:1433 e.t==='glass'
```

```
O03259 L1087:1477 -size*.9
O03261 L1087:1498 size*.86
O03262 L1087:1523 size*.9
O03263 L1087:1543 -size*.86
O03265 L1087:1572 ctx.fillStyle='#c7f7f5b0'
O03266 L1087:1630 e.t==='root'
O03267 L1087:1644 ctx.strokeStyle='#ddc6a2'
O03268 L1087:1670 ctx.lineWidth=4
O03269 L1087:1698 k<3
O03270 L1087:1702 k++
O03271 L1087:1736 -size*.5
O03273 L1087:1767 (k-1)*size*.62
O03276 L1087:1784 (k-1)*size*.7
O03279 L1087:1798 size*.7
O03280 L1087:1829 e.t==='wing'
O03281 L1087:1875 size*.98
O03282 L1087:1884 size*.5
O03283 L1087:1892 -.5
O03284 L1087:1935 e.t==='soil'
O03285 L1087:1960 size*.92
O03286 L1087:2005 -size*.28
O03288 L1087:2015 -size*.26
O03290 L1087:2025 size*.23
O03291 L1087:2053 e.t==='magnet'
O03292 L1087:2080 size*.86
O03293 L1087:2129 size*.43
O03294 L1087:2169 e.t==='antenna'
O03295 L1087:2197 size*.78
O03296 L1087:2238 size*.4
O03297 L1087:2349 size*.9
O03298 L1087:2357 size*.63
O03299 L1087:2366 Math.PI/4
O03300 L1087:2415 e.t==='stinger'||e.t==='fang'
O03303 L1087:2475 -size
O03304 L1087:2493 size*.72
O03305 L1087:2502 size*.75
O03306 L1087:2523 -size*.72
O03308 L1087:2533 size*.75
O03309 L1087:2591 e.t==='bubble'
O03310 L1087:2618 size*.74
O03311 L1087:2665 -5
O03312 L1087:2668 -5
O03313 L1087:2690 e.t==='gear'
O03314 L1087:2716 k<8
O03315 L1087:2720 k++
O03316 L1087:2747 k*TAU/8
O03318 L1087:2769 -4
O03319 L1087:2772 -size
O03320 L1087:2808 size*.72
O03321 L1087:2879 k<5
O03322 L1087:2883 k++
O03323 L1087:2910 k*TAU/5
O03325 L1087:2949 -size*.47
O03327 L1087:2959 size*.33
O03328 L1087:2968 size*.53
O03329 L1087:3035 size*.38
O03330 L1087:3075 ctx.strokeStyle=rarity[e.r].hex
O03331 L1087:3107 ctx.lineWidth=1.7
O03332 L1087:3153 size+4
O03333 L1087:3245 ctx.strokeStyle='#f5dc88dd'
```

```
O03334 L1087:3273 ctx.lineWidth=1.4
```

F0246 | drawMobHitbox

```
O03335 L1090:56 ctx.strokeStyle='#b8f1e4ad'
O03336 L1090:84 ctx.lineWidth=1.6
O03337 L1091:46 -hit.ry
O03338 L1091:55 ctx.strokeStyle='#ffe19e'
O03339 L1091:81 ctx.lineWidth=2
```

F0247 | drawCreatureBody

```
O03340 L1096:1 ctx.fillStyle=fill
O03341 L1097:12 ctx.strokeStyle=stroke
O03342 L1097:35 ctx.lineWidth=lineWidth
```

F0248 | tint

```
O03343 L1101:77 -1
O03344 L1101:90 a<0?0:255
O03346 L1101:121 '#'+[n>>16,(n>>8)&255,n&255].map(c=>Math.round(c+(target-c)*f).toString(16).padStart(2,'0')).join('')
```

F0249 | anonymous callback

```
O03351 L1101:168 c+(target-c)*f
```

F0250 | materialFill

```
O03354 L1102:34 game.settings.quality==='low'
O03355 L1102:110 -r*.33
O03357 L1102:117 -r*.4
O03359 L1102:123 r*.06
O03360 L1102:133 r*1.1
O03361 L1102:228 -.38
```

F0251 | paintTerrain

```
O03362 L1104:4 game.settings.quality==='low'
O03363 L1106:4 !pattern
O03364 L1106:58 tile.width=tile.height=512
O03366 L1106:127 zones[game.zone].seed+813
O03367 L1106:194 zones[game.zone].seed+813
O03368 L1107:14 i<95
O03369 L1107:19 i++
O03370 L1107:32 random()*256
O03371 L1107:47 random()*256
O03372 L1107:62 1+random()*12
O03374 L1107:76 c.fillStyle=i%3?'#e7f3cb0b':'#183e3512'
O03377 L1107:146 s*.43
O03378 L1107:152 random()*TAU
O03379 L1108:2 c.lineWidth=.65
O03380 L1108:18 c.strokeStyle='#f4edc20b'
O03381 L1108:56 i<18
O03382 L1108:61 i++
O03383 L1108:74 random()*256
O03384 L1108:89 random()*256
O03385 L1108:149 x+8
O03386 L1108:153 y-6
O03387 L1108:157 x+14
O03388 L1108:162 y+2
O03389 L1109:2 pattern=ctx.createPattern(tile,'repeat')
O03390 L1111:12 ctx.fillStyle=pattern
O03391 L1111:75 game.width/game.zoom
O03392 L1111:96 game.height/game.zoom
```

F0252 | creatureVisualProfile

```
O03393 L1113:100 game.settings.quality!=='low'&&m.r*game.zoom>=17
O03397 L1113:155 reducedMotion.matches?0:Math.sin(game.time*(3+m.speed*.025)+(m.id||0)*.73)
O03404 L1113:241 m.r*.055
```

F0253 | drawCreatureModel

```
O03405 L1117:1 ctx.strokeStyle=tint(d.color,-.50)
O03407 L1117:36 ctx.lineWidth=v.line
O03408 L1117:57 ctx.lineCap='round'
O03409 L1118:4 insect||t==='rat'
O03411 L1118:35 t.includes('spider')?4:t==='rat'?2:3
O03414 L1118:84 k<pairs
O03415 L1118:92 k++
O03416 L1118:115 -1
O03417 L1118:130 (k-(pairs-1)/2)*r*.37
O03422 L1118:157 phase*(k%2?1:-1)*r*.07
O03428 L1118:207 side*v.rx*.58
O03430 L1118:235 side*r*.96
O03432 L1118:246 y-r*.17+wave
O03435 L1118:271 side*r*1.16
O03437 L1118:283 y+r*.12+wave
O03440 L1118:329 side*r*.96
O03442 L1118:340 y-r*.17+wave
O03445 L1118:353 v.line*.7
O03446 L1119:31 -1
O03447 L1119:63 side*r*.40
O03449 L1119:74 -r*.18
O03451 L1119:93 side*(.56+phase*.13)
O03454 L1119:115 ctx.fillStyle=machine?'#afefed7a':'#e8f8ed99'
O03456 L1119:161 ctx.strokeStyle='#edfff2c2'
O03457 L1119:189 ctx.lineWidth=v.line*.6
O03459 L1119:241 side*r*.32
O03461 L1119:252 -r*.30
O03463 L1119:259 r*.44
O03464 L1119:265 r*.77
O03465 L1119:271 side*.12
O03466 L1119:367 side*r*.51
O03468 L1119:378 -r*.70
O03470 L1119:397 side*r*.15
O03472 L1119:408 -r*.25
O03474 L1119:427 side*r*.58
O03476 L1119:438 -r*.29
O03478 L1120:4 t==='crab'
O03479 L1120:35 -1
O03480 L1120:42 ctx.strokeStyle=tint(d.color,-.2)
O03482 L1120:76 ctx.lineWidth=r*.16
O03484 L1120:123 side*r*.6
O03486 L1120:133 -r*.1
O03488 L1120:161 side*r*1.1
O03490 L1120:172 -r*.35
O03492 L1120:179 side*r*1.07
O03494 L1120:191 -r*.7
O03496 L1120:218 side*r*1.07
O03498 L1120:230 -r*.7
O03500 L1120:236 r*.25
O03501 L1120:276 v.line*.6
O03502 L1120:287 ctx.strokeStyle=tint(d.color,-.4)
O03504 L1120:321 ctx.lineWidth=v.line
O03505 L1120:369 side*r*1.1
```

```
O003507 L1120:380 -r*.88
O003509 L1120:399 side*r*.98
O003511 L1120:410 -r*.66
O003513 L1121:4 t==='jelly'||t==='slime'
O003516 L1121:40 -2
O003517 L1121:43 k<=2
O003518 L1121:48 k++
O003519 L1121:53 ctx.strokeStyle=tint(d.color,.15)+'a8'
O003521 L1121:92 ctx.lineWidth=v.line*1.1
O003523 L1121:144 k*r*.25
O003525 L1121:152 r*.45
O003526 L1121:177 k*r*.36+phase*4
O003530 L1121:193 r*.77
O003531 L1121:199 k*r*.18-phase*3
O003535 L1121:215 r*.98
O003536 L1121:221 k*r*.27
O003538 L1121:229 r*1.16
O003539 L1122:4 t==='starfish'||t==='urchin'
O003542 L1122:46 k<(t==='urchin'?12:5)
O003545 L1122:68 k++
O003546 L1122:81 k*TAU/(t==='urchin'?12:5)-Math.PI/2
O003552 L1122:117 ctx.strokeStyle=tint(d.color,-.18)
O003554 L1122:152 ctx.lineWidth=t==='urchin'?v.line:r*.22
O003558 L1122:219 Math.cos(a)*r*.50
O003560 L1122:237 Math.sin(a)*r*.50
O003562 L1122:267 Math.cos(a)*r*1.12
O003564 L1122:286 Math.sin(a)*r*1.12
O003566 L1123:40 k<6
O003567 L1123:44 k++
O003568 L1123:71 k*TAU/6
O003570 L1123:80 ctx.fillStyle=materialFill(d.color,r)
O003571 L1123:118 ctx.strokeStyle=tint(d.color,.58)
O003572 L1123:152 ctx.lineWidth=v.line*.65
O003574 L1123:207 -r*.53
O003576 L1123:214 r*.31
O003577 L1123:220 r*.51
O003578 L1123:272 ctx.strokeStyle=tint(d.color,-.15)
O003580 L1123:336 -r*.28
O003582 L1123:357 -r*.83
O003584 L1124:33 t==='bubble'?'#90d8df':d.color
O003586 L1124:80 -.48
O003587 L1126:44 v.rx*.96
O003588 L1126:53 v.ry*.96
O003589 L1127:13 ctx.fillStyle=tint(d.color,-.25)
O003591 L1127:59 -r*.48
O003593 L1127:66 -r*.42
O003595 L1127:73 r*.96
O003596 L1127:79 r*.87
O003597 L1127:86 ctx.strokeStyle=tint(d.color,.6)
O003598 L1127:119 ctx.lineWidth=v.line*.65
O003600 L1127:159 -r*.44
O003602 L1127:166 -r*.38
O003604 L1127:173 r*.88
O003605 L1127:179 r*.79
O003606 L1127:195 r*.06
O003607 L1127:201 r*.24
O003608 L1127:227 v.line*.7
O003609 L1127:247 r*.06
O003610 L1127:253 r*.11
O003611 L1127:259 m.enraged?'#ffa76d':'#a5f0e0'
```

```
O03612 L1127:318 -1
O03613 L1127:340 -1
O03614 L1127:353 a*r*.34
O03616 L1127:361 b*r*.30
O03618 L1127:369 r*.045
O03619 L1128:9 t==='ladybug'
O03620 L1128:24 ctx.strokeStyle='#49353b'
O03621 L1128:50 ctx.lineWidth=v.line
O03622 L1128:100 -v.ry
O03623 L1128:159 -.38
O03624 L1128:164 -.30
O03625 L1128:175 -.30
O03626 L1128:182 -.39
O03627 L1128:211 x*r
O03628 L1128:215 y*r
O03629 L1128:219 r*.13
O03630 L1128:255 (x-.025)*r
O03632 L1128:266 (y-.03)*r
O03634 L1128:276 r*.035
O03635 L1129:17 ctx.strokeStyle=tint(d.color,-.55)
O03637 L1129:52 ctx.lineWidth=r*.16
O03639 L1129:88 -.15
O03640 L1129:130 -r*.7
O03642 L1129:136 r*y
O03643 L1129:152 r*.7
O03644 L1129:157 r*y
O03645 L1130:51 ctx.strokeStyle=tint(d.color,-.43)
O03647 L1130:86 ctx.lineWidth=v.line*.85
O03649 L1130:127 -.30
O03650 L1130:172 r*y
O03651 L1130:176 v.rx*.89
O03652 L1130:185 r*.13
O03653 L1131:40 r*.28
O03654 L1131:46 r*.33
O03655 L1131:65 -.20
O03656 L1131:85 ctx.strokeStyle=tint(d.color,.38)
O03657 L1131:119 ctx.lineWidth=v.line*.65
O03659 L1131:171 -r*.20
O03661 L1131:178 r*.19
O03662 L1131:198 r*.4
O03663 L1131:215 r*.2
O03664 L1131:220 r*.19
O03665 L1132:34 ctx.strokeStyle=tint(d.color,-.47)
O03667 L1132:69 ctx.lineWidth=v.line
O03668 L1132:119 -v.ry
O03669 L1132:156 -v.rx
O03670 L1132:162 -r*.3
O03672 L1132:192 -r*.05
O03674 L1132:204 -r*.3
O03676 L1133:14 ctx.fillStyle='#fff8de38'
O03677 L1133:68 -v.rx*.28
O03679 L1133:78 -v.ry*.36
O03681 L1133:88 v.rx*.30
O03682 L1133:97 v.ry*.15
O03683 L1133:106 -.6
O03684 L1133:128 ctx.strokeStyle='#fff7d83b'
O03685 L1133:156 ctx.lineWidth=v.line*.6
O03687 L1133:212 v.rx*.87
O03688 L1133:221 v.ry*.87
O03689 L1133:232 Math.PI*.9
```

O03690 L1133:243 Math.PI*1.6
O03691 L1137:12 -r*.35
O03693 L1137:30 r*.105
O03694 L1137:65 -1
O03695 L1137:79 side*r*.25
O03697 L1137:95 eyeR*1.15
O03698 L1137:123 side*r*.25
O03700 L1137:134 eyeY-eyeR*.08
O03702 L1137:148 eyeR*.66
O03703 L1137:187 side*r*.25-eyeR*.17
O03707 L1137:207 eyeY-eyeR*.3
O03709 L1137:220 eyeR*.21
O03710 L1138:16 ctx.strokeStyle='#f9d49c'
O03711 L1138:42 ctx.lineWidth=v.line
O03712 L1138:90 -r*.33
O03714 L1138:97 -r*.67
O03716 L1138:116 -r*.24
O03718 L1138:123 -r*.91
O03720 L1138:144 -r*.77
O03722 L1138:163 r*.24
O03723 L1138:169 -r*.91
O03725 L1138:188 r*.33
O03726 L1138:194 -r*.67
O03728 L1139:4 m.hit>0

F0254 | drawMob

O03729 L1142:59 ctx.fillStyle='#17392e30'
O03730 L1142:115 r*.24
O03731 L1142:121 r*1.05
O03732 L1142:128 r*.74
O03733 L1142:165 m.facing??0
O03734 L1143:25 r+6
O03735 L1144:164 m.windup>0?specialProfileFor(m):null
O03737 L1145:13 dist(m.x,m.y,game.p.x,game.p.y)<290
O03738 L1145:54 d.guardian||m.hp<m.max||close
O03741 L1146:4 !show
O03742 L1146:67 m.hp/m.max
O03743 L1146:91 r*.12
O03744 L1147:1 ctx.fillStyle='#183c36cf'
O03745 L1147:57 -r
O03746 L1147:60 -r-13
O03748 L1147:66 r*2
O03749 L1147:86 ctx.fillStyle=pct>.3?'#c5e7a3':'#efad90'
O03752 L1147:140 -r+1
O03754 L1147:145 -r-12
O03756 L1147:162 (r*2-2)*pct
O03759 L1147:186 h-2
O03760 L1148:18 r*.81
O03761 L1148:24 -r*.77
O03763 L1149:4 m.windup>0
O03764 L1149:24 r*.85
O03765 L1150:4 close||d.guardian||game.settings.showHitboxes
O03767 L1150:51 ctx.textAlign='center'
O03768 L1150:74 ctx.font='600 '+clamp(r*.32,10,14)+'px system-ui'
O03772 L1150:124 ctx.lineWidth=3
O03773 L1150:140 ctx.strokeStyle='#193c35ed'
O03774 L1150:168 ctx.fillStyle=rarity[m.rank||0].hex
O03776 L1150:216
(d.guardian?"LOCALIZED_6e61ccdb":'')+d.name+"LOCALIZED_588da410"+rarity[m.rank||0].name+"LOCALIZED_e4dd17df"+m.level
O03783 L1150:313 -r-22

O03785 L1150:341 -r-22
O03787 L1150:378 ctx.font='10px system-ui'
O03788 L1150:404 ctx.fillStyle='#f2dcc0'
O03789 L1150:438 'ATK '+Math.round(m.dmg*(m.enraged?1.20:1))
O03792 L1150:503 r+18
O03793 L1150:528 r+18

F0255 | drawNavArrow

O03794 L1153:27 !game.navTarget||game.navTarget===game.zone
O03797 L1153:124 !next

F0256 | anonymous callback

O03798 L1153:188 p.to===next

F0255 | drawNavArrow

O03799 L1153:204 !portal
O03800 L1153:269 x>pad&&y>pad&&x<game.width-pad&&y<game.height-pad
O03809 L1153:335 x-game.width/2
O03811 L1153:353 y-game.height/2
O03813 L1153:412 game.width-pad
O03814 L1153:443 game.height-pad
O03815 L1153:508 ctx.fillStyle='#fff2c8'
O03816 L1153:532 ctx.strokeStyle='#3c5440'
O03817 L1153:558 ctx.lineWidth=2
O03818 L1153:618 -12
O03819 L1153:622 -11
O03820 L1153:638 -7
O03821 L1153:655 -12

F0257 | nextHop

O03822 L1154:29 from===to
O03823 L1154:210 first||other
O03824 L1154:226 other===to

F0258 | minimap

O03825 L1155:22 !game.active
O03826 L1155:122 mctx.fillStyle=z.wall
O03827 L1155:169 mctx.fillStyle=z.bg
O03828 L1155:207 ww-8
O03829 L1155:212 hh-8
O03830 L1155:227 ww/W
O03831 L1155:235 hh/H
O03832 L1156:12 i>=0
O03833 L1156:17 i--
O03834 L1156:30 BAND_THRESHOLDS[i]*Math.SQRT2
O03835 L1156:90 cx*sx
O03836 L1156:96 cy*sy
O03837 L1156:102 cx*k*sx
O03839 L1156:110 cy*k*sy
O03841 L1156:127 mctx.strokeStyle=rarity[i+1].hex+'77'
O03844 L1156:165 mctx.lineWidth=2
O03845 L1157:24 mctx.fillStyle='#20383477'
O03846 L1157:65 o.x*sx-1.5
O03848 L1157:76 o.y*sy-1.5
O03850 L1157:117 mctx.fillStyle=a.level>=20?'#f69b7c':'#ffe0a8'
O03853 L1157:190 a.x*sx
O03854 L1157:197 a.y*sy
O03855 L1157:258 drop.r<3
O03856 L1157:288 mctx.globalAlpha=.75+.25*Math.sin(game.time*5)

O03860 L1157:335 mctx.fillStyle=rarity[drop.r].hex

O03861 L1157:395 drop.x*sx

O03862 L1157:405 drop.y*sy

O03863 L1157:415 drop.r>=6?4:2.7

O03865 L1157:485 m.shrines||[]

O03866 L1157:512 mctx.globalAlpha=shrineCooldown(shrine)===0?1:.35

O03869 L1157:562 mctx.fillStyle=SHRINE_TYPES[shrine.type].color

O03870 L1157:623 shrine.x*sx-3

O03872 L1157:637 shrine.y*sy-3

O03874 L1157:720 peer.zone!==game.zone

O03875 L1157:751 mctx.fillStyle='#a2f6ff'

O03876 L1157:790 peer.x*sx-3

O03878 L1157:802 peer.y*sy-3

O03880 L1157:850 mctx.fillStyle='#e3fff0'

O03881 L1157:901 portal.x*sx

O03882 L1157:913 portal.y*sy

O03883 L1157:954 game.p.x*sx

O03884 L1157:968 game.p.y*sy

O03885 L1157:980 mctx.fillStyle='#fce39e'

O03886 L1157:1005 mctx.strokeStyle='#283e2d'

O03887 L1157:1032 mctx.lineWidth=3

O03888 L1157:1114 mctx.strokeStyle='#ffffff91'

O03889 L1157:1159 game.camera.x*sx

O03890 L1157:1176 game.camera.y*sy

O03891 L1157:1193 game.width/game.zoom*sx

O03893 L1157:1217 game.height/game.zoom*sy

F0259 | updateBossHud

O03895 L1158:140 m.dead||!mobs[m.t].guardian

O03897 L1158:217 distance<nearest

O03898 L1158:235 nearest=distance

O03899 L1158:252 boss=m

O03900 L1158:290 !boss

O03901 L1158:327 !!boss?.enraged

O03903 L1158:347 !boss

O03904 L1158:360 $('bossTitle').textContent=mobs[boss.t].name+(boss.enraged?"LOCALIZED_23524901":"LOCALIZED_e4cbd6b6")

O03907 L1158:440 $('bossFill').style.width=clamp(boss.hp/boss.max*100,0,100)+'%'

O03911 L1158:504 $('bossStat').textContent='LV '+boss.level+"LOCALIZED_588da410"+Math.ceil(Math.max(0,boss.hp))+' /
'+boss.max+' HP'+(boss.enraged?"LOCALIZED_60d7c5c2":'')

F0260 | updateSkillDock

O03920 L1159:137 btn.disabled=r>.05

O03922 L1159:185 btn.innerHTML=label.key.toUpperCase()+'
'+label.name+'<small>'+(r>0?Math.ceil(r)+'s':"LOCALIZED_b073db9c")+'</small>'

F0262 | anonymous callback

O03931 L1159:368 v.name+' '+syner(id)+"LOCALIZED_6e2d7f7c"

F0260 | updateSkillDock

O03934 L1159:394
$('synergyLabel').textContent="LOCALIZED_1dd26112"+(names.join("LOCALIZED_588da410")||"LOCALIZED_0adf850e")

O03937 L1159:525 !e

O03938 L1159:534 $('eventRibbon').textContent="LOCALIZED_ec94013b"+WORLD_EVENTS[e.type].name+"LOCALIZED_588da410"+Math.
ceil(e.until-game.time)+'s'

F0263 | syncHud

O03944 L1160:33 !game.active

O03945 L1160:193 !cloudSession

O03946 L1160:208
$('orbitBtn').textContent="LOCALIZED_de90a45c"+(p.orbitDirection===-1?"LOCALIZED_ace4c580":"LOCALIZED_ad244ea4")+' H'

O03952 L1160:317 "LOCALIZED_5e563206"+(p.orbitDirection===-1?"LOCALIZED_c782be60":"LOCALIZED_8dc5be02")

O03956 L1160:368 $('hpFill').style.width=clamp(p.hp/maxHP()*100,0,100)+'%'

O03960 L1160:426 $('xpFill').style.width=clamp(p.xp/xpNeed(p.level)*100,0,100)+'%'

O03964 L1160:492 $('hpText').textContent=Math.ceil(p.hp)+' / '+Math.ceil(maxHP())+' HP'

O03968 L1160:563 $('xpText').textContent=Math.floor(p.xp)+' / '+xpNeed(p.level)+' XP'

O03972 L1160:632 $('levelBadge').textContent='LV '+p.level

O03974 L1160:674 $('kills').textContent="LOCALIZED_7a157a5b"+p.kills+(game.combo.count>=2?"LOCALIZED_333d6a7c"+game.combo.count:'')

O03980 L1160:762 $('bodyDamage').textContent="LOCALIZED_69b0f8d0"+Math.round(9+p.level*.4)

O03984 L1160:823 $('slotSummary').textContent="LOCALIZED_8ff79e5d"+game.slots.length

O03986 L1160:878 $('zoneLabel').textContent=z.name+' '+z.en

O03989 L1160:921 $('zoneFlavor').textContent=z.flavor

O03990 L1160:958 $('zoneTier').textContent=bandLabel(game.zone,band)

O03991 L1160:1010 $('modeTag').textContent=game.mode==='attack'?"LOCALIZED_549aed57":game.mode==='defend'?"LOCALIZED_bd5404bd":"LOCALIZED_e8a7b9fa"

O03996 L1160:1121 $('zoneLabel').style.color=z.accent

O03997 L1160:1157 $('progression').textContent='LV '+p.level+"LOCALIZED_588da410"+game.slots.length+"LOCALIZED_04449f87"+Math.round(100/game.zoom)+'%'+(game.slots.length<10?"LOCALIZED_fa7018e2"+SLOT_MILESTONES.find(s=>s.slots>game.slots.length).level:'')

F0264 | anonymous callback

O04009 L1160:1329 s.slots>game.slots.length

F0263 | syncHud

O04010 L1160:1366 $('talentBtn').textContent="LOCALIZED_32096086"+(talentPoints()?"LOCALIZED_588da410"+talentPoints()+"LOCALIZED_d1fe6435":'')

O04015 L1160:1501 !game.boon

O04016 L1160:1527 tag.textContent=SHRINE_TYPES[game.boon.type].name+"LOCALIZED_588da410"+Math.ceil(Math.max(0,game.boon.until-game.time))+"LOCALIZED_00fe8dda"

O04021 L1160:1637 $('deckLabel').textContent="LOCALIZED_21d2387c"+(game.deck?'B':'A')+"LOCALIZED_33d4c922"+(game.deckCooldown>0?"LOCALIZED_10656897"+game.deckCooldown.toFixed(1)+"LOCALIZED_00fe8dda":"LOCALIZED_b073db9c")

O04030 L1160:1829 j.kills>=q.goal

O04031 L1160:1900 qi.innerHTML=''+(j.claimed?"LOCALIZED_856989a3":"LOCALIZED_dbc8c508"+(done?"LOCALIZED_54cd6730":"LOCALIZED_dc9591e5"))+"LOCALIZED_ac1f6963"+Math.min(j.kills,q.goal)+' / '+q.goal+"LOCALIZED_a60c4957"+Math.min(100,j.kills/q.goal*100)+'%"></i> '

O04045 L1160:2170 $('dashTag').textContent="LOCALIZED_7c9490ec"+(p.dashCooldown>0?p.dashCooldown.toFixed(1)+'s':"LOCALIZED_b073db9c")

O04050 L1160:2299 !game.navTarget||game.navTarget===game.zone

O04053 L1160:2347 game.navTarget&&game.navTarget!==game.zone

O04055 L1160:2390 $('navGuide').textContent="LOCALIZED_fdca40f2"+zones[game.navTarget].name+"LOCALIZED_4221a25a"+zones[nextHop(game.zone,game.navTarget)]?.name

O04059 L1161:12 i<game.equipped.length

O04060 L1161:35 i++

O04061 L1161:52 'slot'+i

O04062 L1161:84 dom&&e

O04063 L1161:104 e.cd>0?e.cd.toFixed(1)+'s':Math.ceil(e.hp/(pets[e.t].hp*rarity[e.r].scale)*100)+'%'

O04070 L1161:188 dom.querySelector('.slot-state').textContent=state

O04071 L1161:272 e.cd>0

O04072 L1161:310 "LOCALIZED_cba64ad6"+(i+1)+' '+pets[e.t].name+' '+(e.cd>0?"LOCALIZED_080d0e10":"LOCALIZED_2b52906e")+state+"LOCALIZED_7e612fb8"

O04082 L1161:381 dom.querySelector('.cooldown').style.width=(e.cd>0?clamp(e.cd/(e.reloadTotal||petalReloadTime(e.t))*100,0,100):0)+'%'

O04089 L1161:499 dom.querySelector('.durability').style.width=clamp(e.hp/(pets[e.t].hp*rarity[e.r].scale)*100,0,100)+'%'

O04094 L1162:122 d<best

O04095 L1162:130 best=d

O04096 L1162:137 near=m

O04097 L1162:145 game.nearest=near

O04098 L1162:220 !near

O04099 L1162:260 node.innerHTML=''+data.name+(near.tier?"LOCALIZED_11657542":'')+"LOCALIZED_588da410"+rarity[near.rank||0].en+"LOCALIZED_e4dd17df"+near.level+"LOCALIZED_9886578b"+Math.round(near.dmg)+"LOCALIZED_9cbcd64f"+Math.ceil(near.hp)+' / '+near.max+''

F0265 | fillSlots

O04117 L1163:94 document.body.dataset.slots=game.slots.length

O04118 L1163:172 i<game.slots.length

O04119 L1163:192 i++

O04120 L1163:272 b.id='slot'+i

O04122 L1163:286 b.type='button'

O04123 L1163:302 b.className='slot'

O04124 L1163:321 b.title="LOCALIZED_cba64ad6"+(i+1)+"LOCALIZED_588da410"+p.name+"LOCALIZED_588da410"+rarity[s.r].name+"LOCALIZED_d387a937"+Math.round(p.dmg*rarity[s.r].scale)+"LOCALIZED_6ef447cd"+Math.round(p.hp*rarity[s.r].scale)+"LOCALIZED_734b05ee"

O04138 L1163:476 b.innerHTML=''+(i+1)+''+p.glyph+''+p.name+''

F0267 | toast

O04150 L1164:39 d.textContent=msg

O04151 L1164:106 toastTimer=setTimeout(()=>d.classList.remove('show'),2600)

F0269 | render

O04152 L1165:40 !game.active

F0270 | drawGateBackground

O04153 L1166:30 ctx.fillStyle='#264c3b'

O04154 L1166:125 i<140

O04155 L1166:131 i++

O04156 L1166:142 r()*game.width

O04157 L1166:159 r()*game.height

O04158 L1166:178 r()*14+2

O04160 L1166:214 ctx.globalAlpha=.25

O04161 L1166:246 x<game.width

O04162 L1166:259 x+=85

O04163 L1166:266 ctx.strokeStyle='#b6d49a'

O04164 L1166:363 ctx.globalAlpha=1

F0271 | resize

O04165 L1167:18 game.width=canvas.clientWidth||innerWidth

O04167 L1167:60 game.height=canvas.clientHeight||innerHeight

O04169 L1167:105 game.dpr=Math.min(devicePixelRatio||1,game.settings.quality==='low'?1:game.settings.quality==='high'?2:Math.min(2,game.width<850?1.5:2))

O04177 L1167:242 canvas.width=Math.round(game.width*game.dpr)

O04179 L1167:287 canvas.height=Math.round(game.height*game.dpr)

O04181 L1167:389 game.camera.x=clamp(game.p.x-game.width/game.zoom/2,0,Math.max(0,W-game.width/game.zoom))

O04187 L1167:479 game.camera.y=clamp(game.p.y-game.height/game.zoom/2,0,Math.max(0,H-game.height/game.zoom))

F0272 | loop

O04193 L1168:75 (now-last)/1000

O04195 L1168:96 last=now

O04196 L1168:108 game.active&&!game.paused&&!game.dead

O04200 L1168:147 acc+=elapsed

O04201 L1168:175 acc>=1/60&&it++<4

O04206 L1168:201 1/60

O04207 L1168:207 acc-=1/60

O04209 L1168:220 it>=4

O04210 L1168:226 acc=0

O04211 L1168:237 acc=0

F0273 | showModal

O04212 L1169:25 modal=type

O04213 L1169:36 game.paused=true

O04214 L1169:110 $('ui').inert=true

F0274 | closeModal

O04215 L1170:123 modal=null
O04216 L1170:174 $('ui').inert=false
O04217 L1170:211 !game.dead
O04218 L1170:223 game.paused=false

F0275 | openPanel

O04219 L1171:34 !game.active||game.dead
O04221 L1171:65 selectedSlot=index??0
O04223 L1171:87 selectedDeck=0
O04224 L1171:102 craftSelect=null

F0276 | equipInto

O04225 L1172:57 !parsed
O04226 L1172:99 selectedDeck===0?game.slots:game.backupSlots
O04228 L1172:147 !Number.isInteger(slot)||slot<0||slot>=editing.length
O04233 L1172:211 !takeBag(t,r)
O04234 L1172:296 editing[slot]={t,r}
O04235 L1172:316 selectedSlot=slot
O04236 L1172:337 selectedDeck===0
O04237 L1172:371 game.p.hp=clamp(game.p.hp,0,maxHP())
O04238 L1172:408 game.zoom=effectiveZoom()
O04239 L1172:477 "LOCALIZED_2e5d6902"+rarity[r].en+' '+pets[t].name

F0282 | anonymous callback

O04242 L1174:544 +slot.dataset.slot

F0277 | bindBagDrag

O04243 L1174:591 item.draggable=true

F0283 | anonymous callback

O04244 L1174:649 e.dataTransfer.effectAllowed='move'

F0284 | anonymous callback

O04245 L1174:835 e.pointerType==='mouse'

F0285 | move

O04246 L1174:979 event.pointerId!==e.pointerId
O04247 L1174:1037 event.clientX-origin.x
O04248 L1174:1060 event.clientY-origin.y
O04249 L1174:1087 gap>12&&!dragged
O04252 L1174:1105 dragged=true
O04253 L1174:1118 ghost=document.createElement('div')
O04254 L1174:1154 ghost.className='drag-ghost'
O04255 L1174:1183 ghost.textContent=pets[itemOf(key)[0]].name
O04256 L1174:1263 !dragged
O04257 L1174:1302 ghost.style.left=event.clientX+'px'
O04259 L1174:1338 ghost.style.top=event.clientY+'px'

F0286 | finish

O04261 L1174:1525 event.pointerId!==e.pointerId
O04262 L1174:1720
dragged&&event.type==='pointerup'?document.elementFromPoint(event.clientX,event.clientY)?.closest('[data-slot]'):null
O04265 L1174:1884 suppressItemClick=performance.now()+500
O04267 L1174:1948 +target.dataset.slot

F0287 | inventoryFilterCards

O04268 L1179:15 mode==='craft'

O04269 L1179:38 isCraft?craftSearch:bagSearch

O04270 L1180:20 isCraft?craftRarity:bagRarity

O04271 L1180:58 isCraft?'rarity':bagSort

O04272 L1181:39 isCraft?'[data-craft]':'[data-equip]'

F0288 | anonymous callback

O04273 L1184:5 sorting==='count'

O04274 L1184:30 bn-an||br-ar

O04277 L1185:5 sorting==='name'

O04278 L1185:29 a.dataset.name.localeCompare(b.dataset.name,'zh')||br-ar

O04280 L1186:9 br-ar||a.dataset.name.localeCompare(b.dataset.name,'zh')

F0289 | anonymous callback

O04282 L1188:27 card.style.order=String(i)

F0287 | inventoryFilterCards

O04283 L1192:13 !!(search&&!card.dataset.name.toLocaleLowerCase().includes(search))||
(rarityFilter!=='all'&&rank!==Number(rarityFilter))||
(isCraft&&craftReadyOnly&&(count<5||rank>=rarity.length-1||isPetalProtected(card.dataset.craft)))

O04299 L1195:2 card.hidden=hide

O04300 L1195:22 !hide

O04301 L1195:28 shown++

O04302 L1198:12 counter.textContent="LOCALIZED_2166772c"+shown+' /
'+total+"LOCALIZED_a5ec0e4f"+(isCraft&&craftReadyOnly?"LOCALIZED_68ef3772":'')

F0290 | bindInventoryFilters

O04310 L1201:15 mode==='craft'

O04311 L1202:1 query.value=isCraft?craftSearch:bagSearch

O04313 L1202:43 tier.value=isCraft?craftRarity:bagRarity

O04315 L1204:1 query.oninput=()=>{if(isCraft)craftSearch=query.value;else bagSearch=query.value;update()}

F0292 | query.oninput

O04316 L1204:31 craftSearch=query.value

O04317 L1204:60 bagSearch=query.value

F0290 | bindInventoryFilters

O04318 L1205:1 tier.onchange=()=>{if(isCraft)craftRarity=tier.value;else bagRarity=tier.value;update()}

F0293 | tier.onchange

O04319 L1205:31 craftRarity=tier.value

O04320 L1205:59 bagRarity=tier.value

F0290 | bindInventoryFilters

O04321 L1206:62 only.checked=craftReadyOnly

O04322 L1207:3 only.onchange=()=>{craftReadyOnly=only.checked;update()}

F0294 | only.onchange

O04323 L1207:22 craftReadyOnly=only.checked

F0290 | bindInventoryFilters

O04324 L1208:50 order.value=bagSort

O04325 L1208:70 order.onchange=()=>{bagSort=order.value;update()}

F0295 | order.onchange

O04326 L1208:90 bagSort=order.value

F0296 | inventoryControls

O04327 L1212:13 mode==='craft'

O04328 L1213:8 "LOCALIZED_2ddd80ea"+ "LOCALIZED_02bb7f30"+rarity.map((r,i)=>'<option value="'+i+'">'+r.en+"LOCALIZED_588da410"+r.name+'</option>').join('')+'</select></label>'+ (craft?"LOCALIZED_60160a1a": "LOCALIZED_972d067f")+ ' '

F0297 | anonymous callback

O04333 L1214:89 '<option value="'+i+'">'+r.en+"LOCALIZED_588da410"+r.name+'</option>'

F0298 | craftBatch

O04340 L1223:13 Number.isFinite(numeric)&&numeric>0?Math.min(5,Math.trunc(numeric)):0

O04343 L1225:4 !item||!limit||isPetalProtected(key)

O04347 L1227:4 rank>=rarity.length-1||!rarity[rank].chance

O04351 L1228:13 i<limit

O04352 L1228:21 i++

O04353 L1229:5 (game.bags[key]||0)<5

O04355 L1230:12 rank===7||(game.random?.craft||Math.random)()<rarity[rank].chance

O04359 L1231:17 win?5:1+Math.floor((game.random?.craft||Math.random)()*4)

O04363 L1232:5 !takeBag(type,rank,consumed)

O04364 L1233:2 result.attempts++

O04365 L1233:20 result.spent+=consumed

O04366 L1234:22 rank+1

O04367 L1234:30 result.success++

O04368 L1235:7 result.failed++

F0299 | sanitizeCollectionGoal

O04369 L1240:4 !value||typeof value.t!=='string'||!Number.isInteger(value.r)||!itemOf(inventoryKey(value.t,value.r))

F0300 | collectionOwned

O04377 L1244:22 !g

O04378 L1245:8 (game.bags[inventoryKey(g.t,g.r)]||0)+[...game.slots,...game.backupSlots].filter(s=>s.t===g.t&&s.r===g.r).length

F0301 | anonymous callback

O04380 L1245:92 s.t===g.t&&s.r===g.r

F0303 | anonymous callback

O04383 L1247:68 zones[id].mob.includes(type)||GUARDIANS[id]===type

F0307 | setCollectionGoal

O04385 L1252:55 !goal

O04386 L1253:1 game.goal=goal

O04387 L1253:16 game.goalAnnounced=collectionOwned()>0

O04389 L1254:7 "LOCALIZED_3ed73a82"+rarity[rank].name+' '+pets[type].name

F0308 | updateCollectionGoal

O04392 L1257:35 !node

O04393 L1258:32 !game.goal

O04394 L1258:47 !game.goal

O04395 L1260:1 node.textContent=(owned?"LOCALIZED_06cd9b2e":"LOCALIZED_780c60fb")+rarity[g.r].name+' '+pets[g.t].name+" LOCALIZED_588da410"+(owned?"LOCALIZED_5b5f87cd"+owned+"LOCALIZED_408a6f9d":sources.map(id=>zones[id].name).slice(0,2).join(' / '))

O04405 L1261:4 owned&&!game.goalAnnounced

O04407 L1261:32 game.goalAnnounced=true

O04408 L1261:62 "LOCALIZED_bd9475d7"+rarity[g.r].name+' '+pets[g.t].name

F0310 | renderCollectionGuide

O04411 L1264:22 !g

O04412 L1265:8 "LOCALIZED_aec18596"+rarity[g.r].name+'
'+pets[g.t].name+"LOCALIZED_54390a58"+collectionOwned()+"LOCALIZED_61f6e68b"+collectionSources(g.t).map(id=>'<button
type="button" data-goal-zone="'+id+'">'+zones[id].name+"LOCALIZED_5854f3d1").join('')+"LOCALIZED_c5cad4f1"

F0311 | anonymous callback

O04420 L1265:229 '<button type="button" data-goal-zone="'+id+'">'+zones[id].name+"LOCALIZED_5854f3d1"

F0313 | anonymous callback

O04424 L1268:56 btn.onclick=()=>{game.navTarget=btn.dataset.goalZone;closeModal();syncHud();toast(game.navTarget===game
.zone?"LOCALIZED_b47cf264":"LOCALIZED_b6d7f4f5")}

F0314 | btn.onclick

O04425 L1268:73 game.navTarget=btn.dataset.goalZone

O04426 L1268:138 game.navTarget===game.zone?"LOCALIZED_b47cf264":"LOCALIZED_b6d7f4f5"

F0312 | bindCollectionGuide

O04428 L1269:29
$('clearCollectionGoal').onclick=()=>{game.goal=null;game.goalAnnounced=false;saveGame();syncHud();renderModal()}

F0315 | $('clearCollectionGoal').onclick

O04429 L1269:67 game.goal=null

O04430 L1269:82 game.goalAnnounced=false

F0316 | petalBaseStats

O04431 L1272:27 !item

O04432 L1273:27 d.dmg*scale

O04433 L1273:54 d.hp*scale

F0317 | renderPetalWorkbench

O04434 L1276:27 !item

O04435 L1277:62 selectedDeck===0?game.slots:game.backupSlots

F0318 | delta

O04437 L1279:16 n===0?"LOCALIZED_a9105fc0":(n>0?'+':'')+n

F0317 | renderPetalWorkbench

O04442 L1280:11 Object.entries(SET_FAMILIES).filter(([,family])=>family.types.includes(t)).map(([,family])=>family.name
).join("LOCALIZED_c4a40ea8")||"LOCALIZED_84ef9ca4"

O04443 L1281:8 "LOCALIZED_89ddb28f"+rarity[r].hex+'">'+d.glyph+' '+rarity[r].en+' '+d.name+'</h3><p>'+d.desc+"LOCALIZED
_75edfb0d"+stats.damage+"LOCALIZED_da047c04"+delta(stats.damage-old.damage)+"LOCALIZED_56a2b791"+stats.hp+"LOCALIZED_da
047c04"+delta(stats.hp-old.hp)+"LOCALIZED_d62370fc"+stats.reload+"LOCALIZED_176b5645"+(selectedDeck===0?"LOCALIZED_cb62
ebd6":"LOCALIZED_fcdd0cb0")+"LOCALIZED_e10c4d6a"+(selectedSlot+1)+"LOCALIZED_f114611c"+pets[previous.t].name+"LOCALIZED
_588da410"+rarity[previous.r].en+"LOCALIZED_eaa88303"+set+"LOCALIZED_125a9cb8"+(game.bags[key]||0)+"LOCALIZED_588da410"
+(isPetalProtected(key)?"LOCALIZED_0cd6b666":"LOCALIZED_6a48ed44")+"LOCALIZED_6d142fe0"+(isPetalProtected(key)?"LOCALIZ
ED_f81a018d":"LOCALIZED_f3717448")+"LOCALIZED_43893882"

F0321 | renderCraftDistribution

O04487 L1283:46 "LOCALIZED_c9307341"+mass.map((p,n)=>' <div
aria-hidden="true"> <b>'+n+"LOCALIZED_cfd9b489"+(p*100).toFixed(2)+'%</small></
div>').join('')+' '

F0322 | anonymous callback

O04489 L1283:180 ' <i
style="height:'+(p*100)+'%"> <b>'+n+"LOCALIZED_cfd9b489"+(p*100).toFixed(2)+'%</small> '

F0324 | pct

O04499 L1285:66 (n*100).toFixed(2)+'%'

F0323 | renderCraftAnalysis

O04501 L1286:8 "LOCALIZED_5d8d934e"+globalThis.PetalRules.version+"LOCALIZED_640aee4d"+pct(a.success)+"LOCALIZED_595dd119"+a.outputs.toFixed(3)+"LOCALIZED_2f71f21d"+pct(a.completion)+"LOCALIZED_313166fd"+a.meanCost.toFixed(2)+"LOCALIZED_1ae90649"+(a.q95===null?"LOCALIZED_928b6b2d":a.q95+"LOCALIZED_e9e0fdc2")+"LOCALIZED_0e0aacfc"+(locked?"LOCALIZED_4c78f91d":'')+"LOCALIZED_84cc4426"+renderCraftDistribution(a.outputMass)+'</section>'

F0326 | anonymous callback

O04521 L1289:37 e.cd<=0&&e.hp>0

F0325 | renderDamageForecast

O04524 L1290:4 !ready.length

O04525 L1291:128 rows+='<tr><th scope="row">'+pets[e.t].name+'</th><td>'+a.mean.toFixed(2)+'</td><td>'+(a.killChance*100).toFixed(1)+'%</td></tr>'

O04533 L1292:8 "LOCALIZED_8c9e9b28"+ready.length+"LOCALIZED_24748830"+rows+"LOCALIZED_bc251bc6"+armor.toFixed(2)+"LOCALIZED_610d7a55"

F0328 | anonymous callback

O04539 L1295:61 !m.dead

F0329 | anonymous callback

O04540 L1295:82 dist(a.x,a.y,game.p.x,game.p.y)-dist(b.x,b.y,game.p.x,game.p.y)

F0327 | renderProbabilityLab

O04541 L1295:182 game.worldEvent?.type==='treasure'

O04542 L1296:10 "LOCALIZED_32704e5c"+zones[game.zone].name+"LOCALIZED_6117fe5d"+(goal?"LOCALIZED_4c942a35"+rarity[goal.r].name+' '+pets[goal.t].name+"LOCALIZED_4c78b2b5":"LOCALIZED_d3f07b47")+' '

O04551 L1297:93 m.rank??0

O04552 L1297:103 !!spec.guardian

O04554 L1297:159 goal?globalThis.PetalRules.targetDropChance(spec.drop,m.rank??0,!!spec.guardian,treasure,goal.t,goal.r):null

O04558 L1298:2 html+='<article class="probability-panel encounter-card"> '+(spec.guardian?"LOCALIZED_3191b526":"LOCALIZED_4dfa7ad5")+' <h3>'+spec.name+'</h3><small>'+rarity[m.rank||0].en+"LOCALIZED_e4dd17df"+m.level+"LOCALIZED_588da410"+Math.round(dist(m.x,m.y,game.p.x,game.p.y))+"LOCALIZED_e9d895e6"+stats.mean.toFixed(2)+"LOCALIZED_2b806e0d"+Math.sqrt(stats.variance).toFixed(2)+' '

O04575 L1299:10 html+='<div class="goal-chance" data-goal-chance="'+chance+"LOCALIZED_39cc5eec"+(chance*100).toFixed(2)+'% '

O04584 L1300:2 html+="LOCALIZED_36e37f23"+[...new Set(spec.drop)].map(t=>pets[t].name).join("LOCALIZED_c4a40ea8")+(spec.guardian?"LOCALIZED_85f0b07c":'')+'</p>'+renderDamageForecast(m)+'</article>'

O04591 L1302:4 !near.length

O04592 L1302:17 html+="LOCALIZED_c719ddd7"

O04593 L1303:1 html+="LOCALIZED_f10915d0"+(globalThis.PetalRules.criticalChance(activeCopies('prism'))*100).toFixed(2)+"LOCALIZED_45b9e355"+((1-globalThis.PetalRules.slowFactor(activeCopies('frost')))*100).toFixed(0)+"LOCALIZED_572e4272"+globalThis.PetalRules.version+"LOCALIZED_5039e4f0"

F0332 | askGame

O04603 L1307:195 yes&&owner===game.user&&account===cfAccount?.id&&active===game.active&&view===modal&&player===game.p&&conflict===cfConflict&&(!active||!game.dead)

F0333 | buildWorkshopPlan

O04619 L1312:77 defs[k]={attack:d.dmg*scale,renewal:d.hp*scale/Math.max(.1,d.cd),support:support.has(t)?scale:0}

F0337 | renderBuildWorkshop

O04624 L1317:1 $('panelTitle').textContent="LOCALIZED_eb052878"

O04625 L1318:5 plan=buildWorkshopPlan()

O04626 L1318:39 b.textContent="LOCALIZED_59e493a1"+e.message

O04628 L1321:47 html+='<button type="button" data-policy="'+k+'" aria-pressed="'+(workshopPolicy===k)+'">'+name+'</button>'

O04636 L1322:1 html+="LOCALIZED_bc587ae9"+(game.deck===0?'A':'B')+' '

O04641 L1323:72 html+='<div class="workshop-slot" data-proposed="'+k+"LOCALIZED_73942a65"+(i+1)+'</small>'+d.glyph+''+d.name+'<small

style="color:'+rarity[r].hex+'">'+rarity[r].name+'</small> '

O04657 L1324:1 html+="LOCALIZED_232af156"+plan.before.toFixed(3)+"LOCALIZED_668c95db"+plan.after.toFixed(3)+"LOCALIZED_3da7bad3"+Math.round(plan.weights[0]*100)+"LOCALIZED_b5dfcfa2"+Math.round(plan.weights[1]*100)+"LOCALIZED_accfe0e6"+Math.round(plan.weights[2]*100)+"LOCALIZED_8d94fcf7"

O04671 L1325:1 b.innerHTML=html

F0338 | anonymous callback

O04672 L1325:65 x.onclick=()=>{workshopPolicy=x.dataset.policy;renderModal()}

F0339 | x.onclick

O04673 L1325:80 workshopPolicy=x.dataset.policy

F0337 | renderBuildWorkshop

O04674 L1325:128 $('workshopBag').onclick=()=>showModal('bag')

O04675 L1325:174 $('workshopSkills').onclick=()=>showModal('skills')

O04676 L1326:35 $('applyWorkshop').onclick=async()=>{ if(!await askGame("LOCALIZED_1c021520"))return; if(identity!==workshopIdentity()){toast("LOCALIZED_45213383");renderModal();return} game.bags={...plan.bag};game.slots=plan.slots.map(k=>{const[t,r]=itemOf(k);return {t,r}});resetEquip(2.5);game.p.hp=Math.min(game.p.hp,maxHP());game.zoom=effectiveZoom();fillSlots();saveGame();syncHud(true);renderModal();toast("LOCALIZED_ef12e966"+policies[workshopPolicy]); }

F0342 | $('applyWorkshop').onclick

O04677 L1327:5 !await askGame("LOCALIZED_1c021520")

O04678 L1328:5 identity!==workshopIdentity()

O04679 L1329:2 game.bags={...plan.bag}

O04680 L1329:26 game.slots=plan.slots.map(k=>{const[t,r]=itemOf(k);return {t,r}})

O04681 L1329:108 game.p.hp=Math.min(game.p.hp,maxHP())

O04682 L1329:146 game.zoom=effectiveZoom()

O04683 L1329:229 "LOCALIZED_ef12e966"+policies[workshopPolicy]

F0344 | renderSkillAtlas

O04684 L1333:1 $('panelTitle').textContent="LOCALIZED_bd6f053b"

O04685 L1334:22 "LOCALIZED_825b79a0"+(170+30*syner('assault'))+"LOCALIZED_afcf8091"+((35+level*3)*power*(1+.18*syner('assault'))).toFixed(1)+"LOCALIZED_b9421e94"

O04697 L1334:155 "LOCALIZED_36eea6f7"+((17+level*1.7)*power*(1+.25*syner('artillery'))).toFixed(1)+"LOCALIZED_61f15d9c"

O04705 L1334:265 "LOCALIZED_caa0ad55"+(4.5+syner('fortress')).toFixed(1)+"LOCALIZED_7768bc60"

O04708 L1334:329 "LOCALIZED_13194d41"+Math.round(maxHP()*(.13+.06*syner('nature')))+"LOCALIZED_d6603e93"

O04713 L1336:50 html+='<article class="skill-reference"><kbd>'+cfg.key.toUpperCase()+'</kbd><h3>'+cfg.name+'</h3> '+effects[k]+"LOCALIZED_7df24e80"+cfg.cd+"LOCALIZED_54efdb03"+(skillRemaining(k)>0?"LOCALIZED_486d6e6d"+skillRemaining(k).toFixed(1)+"LOCALIZED_00fe8dda":"LOCALIZED_ab27f80d")+'</small></article>'

O04728 L1337:1 html+="LOCALIZED_e1de9187"

O04729 L1339:49 html+='<article class="inspect-box">'+d.name+"LOCALIZED_588da410"+(syner(id)?"LOCALIZED_8fa28a06"+syner(id)+"LOCALIZED_5262a9b2":"LOCALIZED_fdc1183b")+''+d.types.map(t=>pets[t].name).join("LOCALIZED_588da410")+' '+bonuses[id]+' </article>'

O04741 L1340:1 $('panelBody').innerHTML=html+' '

F0346 | renderModal

O04743 L1345:12 labels[modal]||['FIELD NOTEBOOK','']

O04744 L1345:49 $('panelEyebrow').textContent=meta[0]

O04745 L1345:87 $('panelCaption').textContent=meta[1]

O04746 L1345:125 $('panelLayer').dataset.panel=modal

F0347 | anonymous callback

O04747 L1346:58 b.dataset.panel===modal

F0346 | renderModal

O04748 L1347:45 focus?.id||null

O04749 L1347:86 identity&&$(identity)&&focus!==document.activeElement

F0348 | renderModalContents

O04752 L1349:34 modal==='workshop'

O04753 L1349:86 modal==='skills'

O04754 L1349:156 modal==='probability'

O04755 L1349:179 $('panelTitle').textContent="LOCALIZED_9085afd2"

O04756 L1349:215 b.innerHTML=renderProbabilityLab()

O04757 L1349:260 modal==='party'

O04758 L1349:306 modal==='talents'

O04759 L1349:325 $('panelTitle').textContent="LOCALIZED_b640bdd6"

O04760 L1349:376 "LOCALIZED_4f640805"+game.p.level+"LOCALIZED_2f07fafb"+talentEarned(game.p.level)+"LOCALIZED_f29984f0"
+talentPoints()+"LOCALIZED_6c120c8c"

O04766 L1349:677 html+=' '+t.name+"LOCALIZED_588da410"+rank+'/5<small>'+t.desc
+"LOCALIZED_546db4b6"+Math.round(rank*t.gain*100)+'%</small><button type="button" data-talent="'+k+'"
'+(rank>=5||!talentPoints()?'disabled':'')+"LOCALIZED_133cda56"

O04785 L1349:939 html+="LOCALIZED_1c4e468f"

O04786 L1349:1041 b.innerHTML=html

F0349 | anonymous callback

O04787 L1349:1107 btn.onclick=()=>spendTalent(btn.dataset.talent)

F0348 | renderModalContents

O04788 L1349:1156 $('respecTalents').onclick=async()=>{if(await askGame("LOCALIZED_86a2fed7"))resetTalents()}

O04789 L1350:8 modal==='bag'

O04790 L1351:1 $('panelTitle').textContent="LOCALIZED_eed9181b"

O04791 L1351:45 "LOCALIZED_42b65b39"+renderCollectionGuide()+"LOCALIZED_0841aa17"

O04793 L1352:1 html+="LOCALIZED_00779997"+(selectedDeck===0?'active':'')+'">'+(game.deck===0?'A':'B')+"LOCALIZED_5d6e82
dd"+(selectedDeck===1?'active':'')+'">'+(game.deck===0?'B':'A')+"LOCALIZED_b9a0caa4"+'</button> '

O04811 L1353:15 selectedDeck===0?game.slots:game.backupSlots

O04813 L1354:1 html+=' '

O04814 L1355:13 i<editing.length

O04815 L1355:30 i++

O04816 L1355:54 html+='<button data-slot="'+i+'" class="'+(i===selectedSlot?'active':'')+'"
aria-pressed="'+(i===selectedSlot)+'"><span
style="color:'+rarity[x.r].hex+"LOCALIZED_ab70f789"+(i+1)+"LOCALIZED_588da410"+pets[x.t].name+'</button>'

O04833 L1355:258 html+=' '

O04834 L1356:4 !itemOf(bagPreviewKey)||(game.bags[bagPreviewKey]||0)<1

O04838 L1356:60 bagPreviewKey=sortedBag()[0]?.[0]||null

O04840 L1356:100 html+=renderPetalWorkbench(bagPreviewKey)

O04841 L1357:1 html+="LOCALIZED_caf9e67e"+inventoryControls('bag')+' '

O04844 L1358:64 html+='<button type="button" class="item-card" aria-pressed="'+(bagPreviewKey===k)+'"
data-equip="'+k+'" data-name="'+d.name+' '+t+'" data-rank="'+r+'" data-count="'+n+"LOCALIZED_9ada4d64"+n+'<span
class="itemicon" style="color:'+d.color+'">'+d.glyph+''+d.name+'<small style="color:'+rarity[r]
.hex+'">'+rarity[r].en+"LOCALIZED_d387a937"+Math.round(d.dmg*rarity[r].scale)+"LOCALIZED_786249f9"+Math.round(d.hp*rari
ty[r].scale)+(isPetalProtected(k)?"LOCALIZED_7855f189":'')+'</small></button>'

O04878 L1359:1 html+=' '

O04879 L1359:19 !sortedBag().length

O04880 L1359:39 html+="LOCALIZED_56b9d779"

O04881 L1360:1 b.innerHTML=html

O04882 L1360:18 $('openWorkshop').onclick=()=>showModal('workshop')

F0353 | anonymous callback

O04883 L1361:46 x.onclick=()=>{selectedDeck=+x.dataset.deck;renderModal()}

F0354 | x.onclick

O04884 L1361:61 selectedDeck=+x.dataset.deck

F0355 | anonymous callback

O04886 L1362:46 x.onclick=()=>{selectedSlot=+x.dataset.slot;renderModal()}

F0356 | x.onclick

O04887 L1362:61 selectedSlot=+x.dataset.slot

F0357 | anonymous callback

O04889 L1363:47
x.onclick=()=>{if(performance.now()<suppressItemClick)return;bagPreviewKey=x.dataset.equip;renderModal()}

F0358 | x.onclick

O04890 L1363:65 performance.now()<suppressItemClick

O04891 L1363:108 bagPreviewKey=x.dataset.equip

F0348 | renderModalContents

O04892 L1364:22 $('equipPreview').onclick=()=>equipInto(bagPreviewKey,selectedSlot)

O04893 L1365:24 $('protectPreview').onclick=()=>togglePetalProtection(bagPreviewKey)

O04894 L1366:20 $('trackPetal').onclick=()=>{const
[t,r]=itemOf(bagPreviewKey);setCollectionGoal(t,Math.min(rarity.length-1,r+1))}

F0361 | $('trackPetal').onclick

O04895 L1366:112 rarity.length-1

O04896 L1366:128 r+1

F0348 | renderModalContents

O04897 L1367:22 $('craftPreview').onclick=()=>{craftSelect=bagPreviewKey;showModal('craft')}

F0362 | $('craftPreview').onclick

O04898 L1367:53 craftSelect=bagPreviewKey

F0348 | renderModalContents

O04899 L1370:8 modal==='craft'

O04900 L1371:1 $('panelTitle').textContent="LOCALIZED_ffa26a60"

O04901 L1372:14 craftSelect&&itemOf(craftSelect)?itemOf(craftSelect):null

O04903 L1373:15 selected?(game.bags[craftSelect]||0):0

O04905 L1374:10 selected&&selected[1]<rarity.length-1?rarity[selected[1]].chance:0

O04909 L1375:11 !!selected&&selected[1]===7

O04913 L1377:24 html+='<b style="color:'+rr.hex+"LOCALIZED_77e0ac79"+rr.en+''

O04918 L1378:1 html+=' '+(selected?rarity[selected[1]].en+' '+pet
s[selected[0]].name:"LOCALIZED_ae91a5cf")+''+(selected?"LOCALIZED_55df57d3"+available+"LOCALIZED_91df224
c"+(isPetalProtected(craftSelect)?"LOCALIZED_4b1409e3":selected[1]===rarity.length-1?"LOCALIZED_509f3399":(forge?"LOCAL
IZED_fca0dd10":"LOCALIZED_d17faf50"+(rate*100).toFixed(rate<.01?3:2)+'%')):"LOCALIZED_15e500c7")+''+(selected&&!
forge&&rate>0?"LOCALIZED_7ce02fbd":'')+' <button id="doCraft" type="button" class="small-primary" '+(!selected||av
ailable<5||!rate||isPetalProtected(craftSelect)?'disabled':'')+'>'+(forge?"LOCALIZED_1950b688":"LOCALIZED_5b10709d")+"L
OCALIZED_33617d88"+(!selected||available<10||!rate||isPetalProtected(craftSelect)?'disabled':'')+"LOCALIZED_7b65aee5"+(
!selected?'disabled':'')+'>'+(selected&&isPetalProtected(craftSelect)?"LOCALIZED_f81a018d":"LOCALIZED_9f68538d")+'</but
ton> '

O04976 L1379:4 selected&&rate>0

O04978 L1379:21 html+=renderCraftAnalysis(rate,available,isPetalProtected(craftSelect))

O04979 L1380:1 html+="LOCALIZED_4fdb77d4"+inventoryControls('craft')+' '

O04982 L1382:64 had=true

O04983 L1383:2 html+='<button type="button" class="item-card '+(craftSelect===k?'active':'')+'" data-craft="'+k+'"
data-name="'+d.name+' '+t+'" data-rank="'+r+'" data-count="'+n+'"
aria-pressed="'+(craftSelect===k)+"LOCALIZED_89ddf509"+n+'<span class="itemicon"
style="color:'+d.color+'">'+d.glyph+''+d.name+'<small style="color:'+rarity[r].hex+'">'+rarity[
r].en+(isPetalProtected(k)?"LOCALIZED_7855f189":n>=5&&r<rarity.length-1?"LOCALIZED_68ef3772":'')+'</small></button>'

O05020 L1384:4 !had

O05021 L1384:9 html+="LOCALIZED_1077520a"

O05022 L1385:1 html+=' '

O05023 L1385:16 b.innerHTML=html

F0363 | anonymous callback

O05024 L1386:47 x.onclick=()=>{craftSelect=x.dataset.craft;renderModal()}

F0364 | x.onclick

O05025 L1386:62 craftSelect=x.dataset.craft

F0348 | renderModalContents

O05026 L1388:47 lockBtn.onclick=()=>{if(craftSelect)togglePetalProtection(craftSelect)}

O05027 L1389:10 doBtn.onclick=()=>{ if(!craftSelect)return; const item=itemOf(craftSelect);if(!item)return; const result=craftBatch(craftSelect,1);if(!result.attempts)return; const [t,r]=item; toast(result.success?(r===7?"LOCALIZED_0682d305":"LOCALIZED_7456e9f6")+rarity[r+1].en+' '+pets[t].name:"LOCALIZED_f404d5f9"+result.spent+"LOCALIZED_88a6dc2f"+pets[t].name); saveGame();renderModal(); }

F0366 | doBtn.onclick

O05028 L1390:5 !craftSelect

O05029 L1391:36 !item

O05030 L1392:44 !result.attempts

O05031 L1394:8 result.success?(r===7?"LOCALIZED_0682d305":"LOCALIZED_7456e9f6")+rarity[r+1].en+' '+pets[t].name:"LOCALIZED_f404d5f9"+result.spent+"LOCALIZED_88a6dc2f"+pets[t].name

F0348 | renderModalContents

O05041 L1397:13 multiBtn.onclick=async()=>{ const key=craftSelect,item=key&&itemOf(key); if(!item||isPetalProtected(key)||Number(game.bags[key]||0)<10)return; const [t,r]=item; const msg="LOCALIZED_c20640e3"+pets[t].name+"LOCALIZED_0929240f"+game.bags[key]+"LOCALIZED_1869bc65"; if(!await askGame(msg))return; const result=craftBatch(key,5); if(!result.attempts)return; toast((r===7?"LOCALIZED_0448a153":"LOCALIZED_1f8f3c56")+' '+result.attempts+"LOCALIZED_31e39590"+result.success+"LOCALIZED_03ae4504"+result.failed+"LOCALIZED_b0906b23"+result.spent+"LOCALIZED_88a6dc2f"+pets[t].name); saveGame();renderModal(); }

F0367 | multiBtn.onclick

O05042 L1398:29 key&&itemOf(key)

O05043 L1399:5 !item||isPetalProtected(key)||Number(game.bags[key]||0)<10

O05048 L1401:12 "LOCALIZED_c20640e3"+pets[t].name+"LOCALIZED_0929240f"+game.bags[key]+"LOCALIZED_1869bc65"

O05052 L1402:5 !await askGame(msg)

O05053 L1404:5 !result.attempts

O05054 L1405:8 (r===7?"LOCALIZED_0448a153":"LOCALIZED_1f8f3c56")+' '+result.attempts+"LOCALIZED_31e39590"+result.success+"LOCALIZED_03ae4504"+result.failed+"LOCALIZED_b0906b23"+result.spent+"LOCALIZED_88a6dc2f"+pets[t].name

F0348 | renderModalContents

O05066 L1409:8 modal==='map'

O05067 L1409:23 $('panelTitle').textContent="LOCALIZED_68ee24fb"

O05068 L1409:553 svg+='<line x1="'+a[0]*700+'" y1="'+a[1]*360+'" x2="'+b[0]*700+'" y2="'+b[1]*360+'" stroke="#bdd6bc77" stroke-width="2" stroke-dasharray="5 6"/>'

O05081 L1409:767 a[0]*700

O05082 L1409:778 a[1]*360

O05083 L1409:794 id===game.zone

O05084 L1409:809 svg+='<circle cx="'+x+'" cy="'+y+'" r="'+(active?26:21)+'" fill="'+z.bg+'" stroke="'+(active?'#fff4b8':'#c7e0bf')+'" stroke-width="'+(active?4:2)+'"/><text x="'+x+'" y="'+(y+4)+'" text-anchor="middle" font-size="13" font-weight="800" fill="#fffbee">'+z.name+'</text>'

O05107 L1409:1077 svg+="LOCALIZED_f47f94ec"+zones[game.zone].name+"LOCALIZED_1b702fa5"

O05110 L1409:1340 svg+='<button data-nav="'+id+'" style="'+(game.navTarget===id?'border-color:#ffe2aa':'')+'">'+z.name+(id===game.zone?"LOCALIZED_4b2232f7":'')+'</button>'

O05122 L1409:1481 svg+="LOCALIZED_aa3acd10"

O05123 L1409:1612 svg+=' '+z.name+"LOCALIZED_782e143d"+bandLabel(id,0)+"LOCALIZED_09ec8776"+bandLabel(id,6)+' '

O05130 L1409:1727 svg+=' '

O05131 L1409:1741 b.innerHTML=svg

F0368 | anonymous callback

O05132 L1409:1801 x.onclick=()=>{game.navTarget=x.dataset.nav;closeModal();toast(game.navTarget===game.zone?"LOCALIZED_d4bb88b2":"LOCALIZED_def386f6"+zones[game.navTarget].name)}

F0369 | x.onclick

O05133 L1409:1816 game.navTarget=x.dataset.nav

O05134 L1409:1864 game.navTarget===game.zone?"LOCALIZED_d4bb88b2":"LOCALIZED_def386f6"+zones[game.navTarget].name

F0348 | renderModalContents

O05137 L1410:8 modal==='codex'

O05138 L1410:25 $('panelTitle').textContent="LOCALIZED_00c43b44"

F0371 | anonymous callback

O05139 L1411:108 s+d.kills

F0348 | renderModalContents

O05140 L1411:128 codexChoice&&mobs[codexChoice]?codexChoice:Object.keys(mobs)[0]

O05142 L1411:192 codexChoice=select

O05143 L1412:9 "LOCALIZED_3c5822b2"+seen+'/'+Object.keys(mobs).length+"LOCALIZED_5f2a8ce6"+kills+"LOCALIZED_5eee8c5f"

O05149 L1413:0 html+=' <button data-filter="all" class="'+(codexFilter==='all'?'active':'')+"LOCALIZED_75b3d566"+(codexFilter==='seen'?'active':'')+"LOCALIZED_069cd3fb"+(codexFilter==='missing'?'active':'')+"LOCALIZED_613dabfa"

O05162 L1414:72 !!info?.seen

O05164 L1414:88 codexFilter==='seen'&&!found||codexFilter==='missing'&&found

O05170 L1414:225 html+='<button data-creature="'+t+'" class="codex-card '+(codexChoice===t?'selected':'')+'">'+(found?"LOCALIZED_94631675":'?')+''+(found?m.name:"LOCALIZED_1c395656")+'<small>'+(found?"LOCALIZED_4f5cfdd9"+m.dmg+"LOCALIZED_bccd1689"+m.hp:"LOCALIZED_4a33df83")+'</small><small>'+(found?"LOCALIZED_7a157a5b"+(info.kills||0)+"LOCALIZED_588da410"+inZones:"LOCALIZED_9a7b90c8")+'</small></button>'

O05198 L1414:589 html+=' '

O05199 L1415:51 !!info?.seen

O05201 L1416:0 html+=' <h3>'+(found?d.name:"LOCALIZED_fd849220")+'</h3>'+(found?"LOCALIZED_8a05010e"+d.hp+"LOCALIZED_2b83d0a6"+d.dmg+"LOCALIZED_b7aa7672"+d.speed+"LOCALIZED_e0b9065e"+({'wander':"LOCALIZED_444ec2a6",'rush':"LOCALIZED_63f94102",'chase':"LOCALIZED_c0348304",'ranged':"LOCALIZED_7ba00927"}[d.ai]||d.ai)+"LOCALIZED_3799c21e"+(info?.maxLevel||1)+"LOCALIZED_4129b1b5"+(info?.kills||0)+"LOCALIZED_80a274d5"+areas+"LOCALIZED_457ae4be"+d.drop.map(t=>pets[t].name).join("LOCALIZED_c4a40ea8")+"LOCALIZED_84c42b4c"+rarity.map((r,i)=>'<option value="'+i+'">'+r.name+'</option>').join('')+'</select></label> '+d.drop.map(t=>'<button type="button" data-track-drop="'+t+"LOCALIZED_2f1529af"+pets[t].name+'</button>').join('')+"LOCALIZED_248e8583":"LOCALIZED_2da205ff")+' '

F0375 | anonymous callback

O05231 L1416:522 '<option value="'+i+'">'+r.name+'</option>'

F0376 | anonymous callback

O05235 L1416:634 '<button type="button" data-track-drop="'+t+"LOCALIZED_2f1529af"+pets[t].name+'</button>'

F0348 | renderModalContents

O05239 L1417:0 b.innerHTML=html

O05240 L1417:27 $('codexGoalRank').value=String(game.goal?.r??0)

F0377 | anonymous callback

O05242 L1417:129 btn.onclick=()=>setCollectionGoal(btn.dataset.trackDrop,Number($('codexGoalRank').value))

F0379 | anonymous callback

O05243 L1417:267 x.onclick=()=>{codexFilter=x.dataset.filter;renderModal()}

F0380 | x.onclick

O05244 L1417:282 codexFilter=x.dataset.filter

F0381 | anonymous callback

O05245 L1417:376 x.onclick=()=>{codexChoice=x.dataset.creature;renderModal()}

F0382 | x.onclick

O05246 L1417:391 codexChoice=x.dataset.creature

F0348 | renderModalContents

O05247 L1418:8 modal==='stats'

O05248 L1418:25 $('panelTitle').textContent="LOCALIZED_2ba98198"

F0383 | anonymous callback

O05249 L1418:130 s.t==='leaf'

F0348 | renderModalContents

O05250 L1420:0 html+="LOCALIZED_15daf4e7"+p.level+"LOCALIZED_8ae5dd2c"+Math.ceil(p.hp)+' / '+Math.ceil(maxHP())+"LOCALIZED_b9c40f00"+Math.round(9+p.level*.4)+"LOCALIZED_fe0b5ed5"+p.speed+"LOCALIZED_95d3aaf8"+p.r+"LOCALIZED_1cbbec72"+p.armorLayers+"LOCALIZED_f3c164b2"+Math.round(rootedArmor())+"LOCALIZED_ef9ea93f"+Math.round(bodySpeedFactor()*100)+"LOCALIZED_c7a6b5ce"+Math.round(magnetRadius())+"LOCALIZED_c51c5392"+p.kills+"LOCALIZED_21c5b38b"+Math.floor(p.xp)+' / '+xpNeed(p.level)+' '

O05280 L1421:0 html+="LOCALIZED_1dd07778"+game.slots.length+"LOCALIZED_537c9114"+Math.round(game.width/game.zoom)+"LOCALIZED_20dd86b3"+Math.round(game.height/game.zoom)+"LOCALIZED_ed3716ea"+Math.round(100/game.zoom)+"LOCALIZED_4cd34e97"+(SLOT_MILESTONES.find(x=>x.slots>game.slots.length)?.level||"LOCALIZED_fd7d711f")+"LOCALIZED_cd23c20d"+leaves+"LOCALIZED_6f4b39da"

F0384 | anonymous callback

O05297 L1421:257 x.slots>game.slots.length

F0348 | renderModalContents

O05298 L1422:0 html+="LOCALIZED_57cbfaab"+zones[game.zone].name+"LOCALIZED_f1bbaba2"+W+"LOCALIZED_20dd86b3"+H+"LOCALIZED_9fec507c"+Math.round(p.x)+', '+Math.round(p.y)+"LOCALIZED_d68bffa7"+bandLabel(game.zone,region)+"LOCALIZED_d7673656"+game.mapData[game.zone].living.length+"LOCALIZED_c1b9e500"+(game.nearest?mobs[game.nearest.t].name+' (LV '+game.nearest.level+')':"LOCALIZED_484d5561")+"LOCALIZED_2bb1b3d3"

O05319 L1423:12 i<game.slots.length

O05320 L1423:32 i++

O05321 L1423:125 s.t==='root'?"LOCALIZED_69b8ecf3":s.t==='soil'?"LOCALIZED_dbdc65ba"+Math.round(42*scale)+"LOCALIZED_ba050e9d":s.t==='glass'?"LOCALIZED_f8d10f5e":s.t==='magnet'?"LOCALIZED_cd129328":s.t==='wing'?"LOCALIZED_7e10d50d":s.t==='leaf'?"LOCALIZED_caa0ad55"+(1.9*scale).toFixed(1)+"LOCALIZED_609eb6a4":s.t==='starpetal'?"LOCALIZED_798b9985"+(3.1*scale).toFixed(1)+"LOCALIZED_609eb6a4":s.t==='rose'?"LOCALIZED_c0edac0d"+Math.round(19*scale):s.t==='nectar'?"LOCALIZED_2c37046a"+Math.round(35*scale):s.t==='fang'?"LOCALIZED_6a4ca29f":"LOCALIZED_484d5561"

O05354 L1423:486 html+=' '+(i+1)+"LOCALIZED_588da410"+rarity[s.r].name+' '+q.name+"LOCALIZED_7ad65f9d"+Math.round(q.dmg*scale)+"LOCALIZED_b009f353"+Math.ceil(e.hp)+' / '+Math.round(q.hp*scale)+"LOCALIZED_c75f612c"+q.cd+"LOCALIZED_618df70f"+heal+' '

O05374 L1423:709 html+=' '

O05375 L1423:724 b.innerHTML=html

O05376 L1424:8 modal==='quests'

O05377 L1424:26 $('panelTitle').textContent="LOCALIZED_a1611ea2"

O05378 L1424:271 j.kills>=q.goal

O05379 L1424:287 html+=' '+z.name+(id===game.zone?"LOCALIZED_4b2232f7":'')+"LOCALIZED_ac1f6963"+Math.min(j.kills,q.goal)+' / '+q.goal+"LOCALIZED_284f659b"+(game.guardians[id]||0)+"LOCALIZED_5e1a4909"+rarity[Math.min(2,z.drop+1)].name+' '+pets[q.petal].name+"LOCALIZED_05790e9c"+(60+z.drop*35)+'
<button class="small-primary" data-claim="'+id+'" '+(!ok||j.claimed?'disabled':'')+'>'+(j.claimed?"LOCALIZED_426e458a":ok?"LOCALIZED_ea96728e":"LOCALIZED_04281137")+'</button> '

O05412 L1424:693 html+=' '

O05413 L1424:708 b.innerHTML=html

F0385 | anonymous callback

O05414 L1424:771 x.onclick=()=>{const id=x.dataset.claim,j=game.journal[id],q=QUESTS[id];if(j.claimed||j.kills<q.goal)return;j.claimed=true;const rank=Math.min(2,zones[id].drop+1);addBag(q.petal,rank,2);game.p.xp+=60+zones[id].drop*35;while(game.p.xp>=xpNeed(game.p.level)){game.p.xp-=xpNeed(game.p.level);game.p.level++;game.p.hp=Math.min(maxHP(),game.p.hp+30);ensureSlots(true)}game.zoom=viewScale(game.p.level);saveGame();renderModal();syncHud(true);toast("LOCALIZED_a30d0e14"+pets[q.petal].name+"LOCALIZED_0a42eaba")}

F0386 | x.onclick

O05415 L1424:846 j.claimed||j.kills<q.goal

O05417 L1424:879 j.claimed=true

O05418 L1424:916 zones[id].drop+1

O05419 L1424:957 game.p.xp+=60+zones[id].drop*35

O05422 L1424:995 game.p.xp>=xpNeed(game.p.level)

O05423 L1424:1028 game.p.xp-=xpNeed(game.p.level)

O05424 L1424:1060 game.p.level++

O05425 L1424:1075 game.p.hp=Math.min(maxHP(),game.p.hp+30)

O05427 L1424:1134 game.zoom=viewScale(game.p.level)

O05428 L1424:1213 "LOCALIZED_a30d0e14"+pets[q.petal].name+"LOCALIZED_0a42eaba"

F0348 | renderModalContents

O05430 L1425:8 modal==='settings'

O05431 L1425:28 $('panelTitle').textContent="LOCALIZED_04726542"

O05432 L1425:96 "LOCALIZED_9702ab0d"+(st.showNumbers?"LOCALIZED_8da97ddd":"LOCALIZED_3fd47edc")+"LOCALIZED_e3097284"+(st.showTelegraph?"LOCALIZED_8da97ddd":"LOCALIZED_3fd47edc")+"LOCALIZED_0dff9a02"+(st.showHitboxes?"LOCALIZED_8da97ddd":"LOCALIZED_3fd47edc")+"LOCALIZED_1d0d4c47"+(st.particles?"LOCALIZED_8da97ddd":"LOCALIZED_3fd47edc")+"LOCALIZED_5897238a"+(game.deck===0?'B':'A')+"LOCALIZED_f723aabb"

O05448 L1425:1184 cloudSession&&game.user==='cloud-'+cloudSession.uid

O05451 L1425:1236 html+="LOCALIZED_2140e578"

O05452 L1425:1932 b.innerHTML=html

O05453 L1425:1976 $('changePasswordForm').onsubmit=cloudChangePasswordFromSettings

O05454 L1425:2041 $('movementSelect').value=st.movement

O05455 L1425:2079 $('movementSelect').onchange=e=>{st.movement=e.target.value;resetInputs();saveGame()}

F0387 | $('movementSelect').onchange

O05456 L1425:2112 st.movement=e.target.value

F0348 | renderModalContents

O05457 L1425:2165 $('qualitySelect').value=st.quality

O05458 L1425:2201 $('qualitySelect').onchange=e=>{st.quality=e.target.value;resize();saveGame()}

F0388 | $('qualitySelect').onchange

O05459 L1425:2233 st.quality=e.target.value

F0348 | renderModalContents

O05460 L1425:2280 $('numberToggle').onclick=()=>{st.showNumbers=!st.showNumbers;saveGame();renderModal()}

F0389 | $('numberToggle').onclick

O05461 L1425:2311 st.showNumbers=!st.showNumbers

F0348 | renderModalContents

O05463 L1425:2368 $('telegraphToggle').onclick=()=>{st.showTelegraph=!st.showTelegraph;saveGame();renderModal()}

F0390 | $('telegraphToggle').onclick

O05464 L1425:2402 st.showTelegraph=!st.showTelegraph

F0348 | renderModalContents

O05466 L1425:2463 $('hitboxToggle').onclick=()=>{st.showHitboxes=!st.showHitboxes;saveGame();renderModal()}

F0391 | $('hitboxToggle').onclick

O05467 L1425:2494 st.showHitboxes=!st.showHitboxes

F0348 | renderModalContents

O05469 L1425:2553 $('particleToggle').onclick=()=>{st.particles=!st.particles;saveGame();renderModal()}

F0392 | $('particleToggle').onclick

O05470 L1425:2586 st.particles=!st.particles

F0348 | renderModalContents

O05472 L1425:2639 $('modalSwap').onclick=()=>{closeModal();swapLoadout()}

O05473 L1426:8 modal==='help'

O05474 L1426:24 $('panelTitle').textContent="LOCALIZED_b665e558"

O05475 L1426:59 b.innerHTML="LOCALIZED_8ff16205"

O05476 L1427:8 modal==='pause'

O05477 L1427:25 $('panelTitle').textContent="LOCALIZED_8b4a2dc9"

O05478 L1427:61 b.innerHTML="LOCALIZED_c4f0dad2"

O05479 L1427:377 $('settingsFromPause').onclick=()=>showModal('settings')

O05480 L1427:434 $('statsFromPause').onclick=()=>showModal('stats')

O05481 L1427:485 $('helpFromPause').onclick=()=>showModal('help')

O05482 L1427:534 $('resumeBtn').onclick=closeModal

O05483 L1427:568 $('manualSave').onclick=()=>{saveGame();toast(cfAccount?"LOCALIZED_4b5a9e03":"LOCALIZED_eb2dd02d")}

F0397 | $('manualSave').onclick

O05484 L1427:614 cfAccount?"LOCALIZED_4b5a9e03":"LOCALIZED_eb2dd02d"

F0348 | renderModalContents

O05485 L1427:650 $('leaveBtn').onclick=logout

O05486 L1428:4 modal==='map'

O05487 L1429:4 modal==='pause'&&!game.guest&&!cloudSession&&!cfAccount

O05494 L1431:1 $('exportLocalSave').onclick=exportLocalProgress

O05495 L1431:50 $('restoreLocalSave').onclick=restoreBeforeImport

O05496 L1431:100 $('importLocalSave').onchange=e=>{void importLocalProgress(e.target.files[0])}

F0398 | $('importLocalSave').onchange

O05497 L1431:134 void importLocalProgress(e.target.files[0])

F0400 | anonymous callback

O05498 L1435:70 itemOf(k)&&n>0

F0401 | anonymous callback

O05500 L1435:151 br-ar||pets[at].name.localeCompare(pets[bt].name,'zh')

F0402 | stackedBag

O05502 L1435:309 !grouped.has(t)

O05503 L1435:362 grouped.get(t).hidden+=n

F0403 | logout

O05504 L1436:78 void cfLogout()

O05505 L1436:204 void cloudFlush().finally(()=>{if(cloudSession===departingCloud){cloudSession=null;cloudPending=null;cloudRevision=null}})

F0404 | anonymous callback

O05506 L1436:238 cloudSession===departingCloud

O05507 L1436:269 cloudSession=null

O05508 L1436:287 cloudPending=null

O05509 L1436:305 cloudRevision=null

F0403 | logout

O05510 L1436:327 $('ui').inert=false

O05511 L1436:347 game.active=false

O05512 L1436:365 game.paused=true

O05513 L1436:382 game.user=null

O05514 L1436:397 game.guest=false

O05515 L1436:414 modal=null

O05516 L1436:569 $('authPass').value=''

O05517 L1436:613 $('authError').textContent=''

O05518 L1436:643 pointer.active=false

F0405 | accounts

O05519 L1437:84 localStorage.getItem(acctKey)||'{}'

O05520 L1437:128 a&&typeof a==='object'&&!Array.isArray(a)?a:{}

F0408 | hashPass

O05526 L1439:38 !crypto?.subtle

F0410 | setAuthMode

O05527 L1442:1 authMode=mode

O05528 L1442:33 mode==='register'

O05529 L1443:4 !cloudRestoring

O05530 L1443:20 $('authSubmit').disabled=false

O05531 L1444:1 $('confirmWrap').style.display=registering?'block':'none'

O05533 L1444:59 $('authConfirm').required=registering

O05534 L1445:1 $('forgotPasswordBtn').style.display=cloudMode&&!registering?'block':'none'

O05538 L1446:41 !registering

O05539 L1447:1 $('authSubmit').textContent=registering?(cloudMode?"LOCALIZED_0e69aad6":"LOCALIZED_308d4be7"):"LOCALIZED_ffedc27e"

O05542 L1448:1 $('authHeading').textContent=cloudMode?(registering?"LOCALIZED_6a9ac309":"LOCALIZED_a6448932"):(registering?"LOCALIZED_e547a22e":"LOCALIZED_2a60f506")

O05546 L1449:1 $('authError').textContent=''

O05547 L1449:31 $('authPass').autocomplete=registering?'new-password':'current-password'

O05549 L1450:1 $('authPass').minLength=registering?8:cloudMode?6:4

O05552 L1450:53 $('authPass').placeholder=registering?"LOCALIZED_b93ecd96":"LOCALIZED_728a7b60"

O05554 L1451:1 $('authModeHint').textContent=cloudMode?(registering?"LOCALIZED_46875175":"LOCALIZED_4dc83e21"):(registering?"LOCALIZED_226fd84c":"LOCALIZED_0264c036")

O05558 L1452:12 $('authName').type=registering?'email':'text'

O05560 L1452:58 $('authName').maxLength=254

O05561 L1452:86 $('authNameLabel').textContent="LOCALIZED_2e412e03"

O05562 L1452:124 $('rememberWrap').style.display='flex'

O05563 L1452:163 $('forgotPasswordBtn').style.display='none'

O05564 L1452:207 $('authPass').minLength=8

O05565 L1452:233 $('authHeading').textContent=registering?"LOCALIZED_f319bf28":"LOCALIZED_e174a2ff"

O05567 L1452:292 $('authModeHint').textContent=registering?"LOCALIZED_fc03bef0":"LOCALIZED_b6f2dd52"

O05569 L1453:1 $('cfRecoverBtn').style.display=cfMode?'block':'none'

O05571 L1453:55 $('cfPendingControls').style.display=cfMode?'block':'none'

F0411 | onAuth

O05573 L1456:101 authWorking=true

O05574 L1456:158 btn.disabled=true

O05575 L1456:223 $('authError').textContent="LOCALIZED_d5a0356a"

O05576 L1456:344 $('authError').textContent=''

O05577 L1456:385 $('authError').textContent=cloudAuthMessage(err,authMode)

O05578 L1456:491 !isMailUnavailable(err,authMode)

O05579 L1456:533 authWorking=false

O05580 L1456:551 btn.disabled=false

O05581 L1456:728 !/^[\p{L}\p{N}_-]{2,18}$/u.test(name)||['__proto__','constructor','prototype'].includes(name)||pass.length<4

O05585 L1456:838 $('authError').textContent="LOCALIZED_af2ce028"

O05586 L1456:906 authWorking=false

O05587 L1456:931 btn.disabled=true

O05588 L1456:949 $('authError').textContent="LOCALIZED_2a9087b3"

O05589 L1456:1012 authMode==='register'

O05590 L1456:1038 pass.length<8||pass!==$('authConfirm').value

O05593 L1456:1352 a[name]={salt,hash}

O05594 L1456:1433 Object.prototype.hasOwnProperty.call(a,name)?a[name]:null

O05595 L1456:1494 !u||typeof u.salt!=='string'||typeof u.hash!=='string'

O05602 L1456:1622 hash!==u.hash

O05603 L1456:1657 game.user=name

O05604 L1456:1672 game.guest=false

O05605 L1456:1689 $('authPass').value=''

O05606 L1456:1712 $('authConfirm').value=''

O05607 L1456:1738 $('authError').textContent=''

O05608 L1456:1811 $('authError').textContent=err?.message||"LOCALIZED_4577e006"

O05610 L1456:1884 authWorking=false

O05611 L1456:1902 btn.disabled=false

F0412 | keyForEvent

O05612 L1459:22 e.code||''

O05613 L1460:24 e.code||''

O05614 L1461:4 e.code==='Space'

O05615 L1462:15 e.key||''

F0413 | isEditing

O05616 L1464:32 !!node?.closest?.('input,textarea,select,[contenteditable=""],[contenteditable="true"]')

F0414 | resetInputs

O05618 L1466:1 pointer.left=pointer.right=pointer.active=false
O05621 L1466:49 touch.atk=touch.def=false
O05623 L1466:88 joy.id=null
O05624 L1466:100 joy.dx=joy.dy=0
O05626 L1469:17 $('joyKnob').style.transform='translate(-50%,-50%)'

F0415 | onCanvasMove

O05627 L1471:28 e.pointerType!=='mouse'||!game.active||game.paused
O05631 L1471:86 pointer.x=e.clientX
O05632 L1471:106 pointer.y=e.clientY
O05633 L1471:126 pointer.active=true

F0417 | anonymous callback

O05634 L1474:44 !game.active||game.paused||game.dead
O05637 L1474:176 touch[key]=true

F0418 | end

O05638 L1475:40 touch[key]=refs.size>0
O05640 L1475:66 !touch[key]

F0419 | joyMove

O05641 L1479:60 box.left+box.width/2
O05643 L1479:84 box.top+box.height/2
O05645 L1479:108 e.clientX-mx
O05646 L1479:124 e.clientY-my
O05647 L1479:163 box.width*.31
O05648 L1479:202 d?1/d:0
O05650 L1479:210 joy.dx=dx*unit*n/limit
O05654 L1479:233 joy.dy=dy*unit*n/limit
O05658 L1479:256 kn.style.transform='translate(calc(-50% + '+(joy.dx*limit)+'px),calc(-50% + '+(joy.dy*limit)+'px))'

F0420 | anonymous callback

O05665 L1480:60 !game.active||game.paused||game.dead||joy.id!==null
O05670 L1480:119 joy.id=e.pointerId

F0421 | anonymous callback

O05671 L1481:41 e.pointerId===joy.id

F0422 | joyEnd

O05672 L1482:22 e.pointerId===joy.id
O05673 L1482:44 joy.id=null
O05674 L1482:56 joy.dx=joy.dy=0
O05676 L1482:72 kn.style.transform='translate(-50%,-50%)'

F0423 | anonymous callback

O05677 L1486:45 e.pointerType==='touch'||!game.active||game.paused
O05681 L1486:157 pointer.active=true
O05682 L1486:177 pointer.x=e.clientX
O05683 L1486:197 pointer.y=e.clientY
O05684 L1486:220 e.button===0
O05685 L1486:233 pointer.left=true
O05686 L1486:254 e.button===2
O05687 L1486:267 pointer.right=true

F0424 | anonymous callback

O05688 L1487:43 e.pointerType==='touch'

O05689 L1487:77 e.button===0

O05690 L1487:90 pointer.left=false

O05691 L1487:112 e.button===2

O05692 L1487:125 pointer.right=false

F0425 | anonymous callback

O05693 L1489:41 e.detail>0&&game.active&&!game.paused&&e.target.closest?.('#ui button')

F0426 | anonymous callback

O05698 L1490:44 pointer.active=false

F0427 | anonymous callback

O05699 L1491:47 !isEditing(e.target)

F0428 | anonymous callback

O05700 L1492:47 !isEditing(e.target)

F0429 | anonymous callback

O05701 L1493:44 e.target.closest('button,canvas')&&!isEditing(e.target)

F0430 | anonymous callback

O05703 L1494:89 e.target.closest?.('#game,#ui')&&!isEditing(e.target)

F0431 | anonymous callback

O05705 L1496:4 modal&&e.key==='Tab'

O05707 L1496:226 -1

O05708 L1496:233 e.shiftKey&&(document.activeElement===first||!$('panelLayer').contains(document.activeElement))

O05712 L1496:371 !e.shiftKey&&(document.activeElement===last||!$('panelLayer').contains(document.activeElement))

O05717 L1497:4 !game.active||e.isComposing||e.ctrlKey||e.metaKey||e.altKey

O05722 L1498:4 k==='escape'

O05723 L1500:4 document.activeElement?.tagName==='BUTTON'&&(k===' '||k==='enter')

O05728 L1502:4 modal||game.dead

O05729 L1503:82 pointer.active=false

O05730 L1504:4 k===' '||k==='shift'

F0433 | anonymous callback

O05733 L1506:48 SKILL_CONFIG[id].key===k

F0431 | anonymous callback

O05734 L1509:9 k==='q'

O05735 L1509:39 k==='r'

O05736 L1509:69 k==='h'

O05737 L1510:9 k==='p'&&cloudSession

O05739 L1510:92 k==='0'?9:Number(k)-1

F0435 | anonymous callback

O05742 L1513:57 !isEditing(document.activeElement)

F0001 | anonymous callback

O05743 L1515:0 $('toolsMenuBtn').onclick=()=>{const
open=document.body.classList.toggle('tools-open');$('toolsMenuBtn').setAttribute('aria-expanded',String(open))}

O05744 L1515:149 $('skillMenuBtn').onclick=()=>{const
open=document.body.classList.toggle('skills-open');$('skillMenuBtn').setAttribute('aria-expanded',String(open))}

O05745 L1515:299 $('talentBtn').onclick=()=>openPanel('talents')

O05746 L1515:347 $('partyBtn').onclick=()=>openPanel('party')

O05747 L1515:509 $('cfModeBtn').onclick=()=>{if(!authWorking&&!cloudRestoring){cfConfigure();void cfResume()}}

F0443 | $('cfModeBtn').onclick

O05748 L1515:540 !authWorking&&!cloudRestoring

O05751 L1515:585 void cfResume()

F0001 | anonymous callback

O05752 L1515:603 $('localModeBtn').onclick=()=>{if(!authWorking&&!cloudRestoring)configureAccountMode(false)}

F0444 | $('localModeBtn').onclick

O05753 L1515:637 !authWorking&&!cloudRestoring

F0001 | anonymous callback

O05756 L1515:696 $('cfRecoverBtn').onclick=cfRecoveryForm

O05757 L1515:737 $('cfExportPending').onclick=cfExportPending

O05758 L1515:782 $('cfUseCloud').onclick=()=>{void cfUseCloud()}

F0445 | $('cfUseCloud').onclick

O05759 L1515:811 void cfUseCloud()

F0001 | anonymous callback

O05760 L1515:830 $('cfUseLocal').onclick=()=>{void cfUseLocal()}

F0446 | $('cfUseLocal').onclick

O05761 L1515:859 void cfUseLocal()

F0001 | anonymous callback

O05762 L1515:878 $('cloudModeBtn').onclick=switchCloudMode

O05763 L1515:920 $('localFallbackBtn').onclick=()=>{if(!authWorking&&!cloudRestoring&&!cloudRecovering){configureAccoun
tMode(false);setAuthMode('register')}}

F0447 | $('localFallbackBtn').onclick

O05764 L1515:958 !authWorking&&!cloudRestoring&&!cloudRecovering

F0001 | anonymous callback

O05769 L1515:1061 $('cloudBackupBtn').onclick=cloudExportBackup

O05770 L1515:1107 $('tabLogin').onclick=()=>setAuthMode('login')

O05771 L1515:1154 $('tabRegister').onclick=()=>setAuthMode('register')

O05772 L1515:1207 $('authForm').onsubmit=onAuth

O05773 L1515:1237 $('forgotPasswordBtn').onclick=cloudSendRecovery

O05774 L1515:1286 $('resetForm').onsubmit=cloudUpdatePassword

O05775 L1515:1330 $('guestBtn').onclick=()=>{game.user="LOCALIZED_8214f290";game.guest=true;initGame(null)}

F0450 | $('guestBtn').onclick

O05776 L1515:1357 game.user="LOCALIZED_8214f290"

O05777 L1515:1372 game.guest=true

F0451 | anonymous callback

O05778 L1515:1457 b.onclick=()=>showModal(b.dataset.panel)

F0001 | anonymous callback

O05779 L1515:1499 $('gateNews').onclick=()=>void globalThis.PetalUI.welcome()

F0453 | $('gateNews').onclick

O05780 L1515:1525 void globalThis.PetalUI.welcome()

F0001 | anonymous callback

O05781 L1515:1559 $('panelClose').onclick=closeModal

F0454 | anonymous callback

O05782 L1515:1648 e.target===$('panelLayer')

F0001 | anonymous callback

O05783 L1515:1690 $('respawnBtn').onclick=()=>{clearTransientState();game.p.x=cx;game.p.y=cy;game.p.hp=maxHP();game.p.lastHit=game.time;game.portalCool=3;game.bullets=[];game.friendlyShots=[];game.drops=[];resetEquip();game.dead=false;game.paused=false;$('death').classList.add('hidden');saveGame();toast("LOCALIZED_69321ed9")}

F0455 | $('respawnBtn').onclick

O05784 L1515:1741 game.p.x=cx

O05785 L1515:1753 game.p.y=cy

O05786 L1515:1765 game.p.hp=maxHP()

O05787 L1515:1783 game.p.lastHit=game.time

O05788 L1515:1808 game.portalCool=3

O05789 L1515:1826 game.bullets=[]

O05790 L1515:1842 game.friendlyShots=[]

O05791 L1515:1864 game.drops=[]

O05792 L1515:1891 game.dead=false

O05793 L1515:1907 game.paused=false

F0001 | anonymous callback

O05794 L1515:1988 $('mapBtn').onclick=()=>openPanel('map')

O05795 L1515:2029 $('bagBtn').onclick=()=>openPanel('bag',0)

O05796 L1515:2072 $('craftBtn').onclick=()=>openPanel('craft')

O05797 L1515:2117 $('probabilityBtn').onclick=()=>openPanel('probability')

O05798 L1515:2174 $('codexBtn').onclick=()=>openPanel('codex')

O05799 L1515:2219 $('questBtn').onclick=()=>openPanel('quests')

O05800 L1515:2265 $('questStrip').onclick=()=>openPanel('quests')

O05801 L1515:2313 $('collectionGoal').onclick=()=>openPanel('bag',0)

O05802 L1515:2364 $('deckBtn').onclick=swapLoadout

O05803 L1515:2397 $('orbitBtn').onclick=reverseOrbit

O05804 L1515:2432 $('touchDash').onclick=performDash

O05805 L1515:2467 $('statsBtn').onclick=()=>openPanel('stats')

O05806 L1515:2512 $('helpBtn').onclick=()=>openPanel('help')

O05807 L1515:2555 $('pauseBtn').onclick=()=>openPanel('pause')

O05808 L1515:2600 $('logoutBtn').onclick=logout

O05809 L1515:2630 mini.onclick=()=>openPanel('map')

F0468 | updateHostingStatus

O05810 L1518:59 !label||!notice

O05813 L1519:14 typeof navigator==='undefined'||navigator.onLine!==false

O05817 L1520:1 label.textContent="LOCALIZED_441ce3de"+(online?'':"LOCALIZED_2a792398")

O05820 L1520:70 notice.style.display='none'

F0469 | partyRoomName

O05821 L1527:35 /^[a-z0-9-]{3,18}$/i.test(raw)?raw.toLowerCase():null

F0470 | partySend

O05822 L1528:65 !ws||ws.readyState!==1

O05825 L1528:112 topic||'realtime:petal-realms:'+party.room

O05827 L1528:215 party.seq++

O05828 L1528:232 dest==='phoenix'?{}:{join_ref:party.joinRef}

F0471 | partySnapshot

O05830 L1529:28 !game.active||!game.p

O05833 L1529:73 cloudSession?.uid?.slice(0,8)||''

F0472 | partyPresence

O05834 L1530:52 !p

F0473 | leaveParty

O05835 L1531:98 party.heartbeat=null

O05836 L1531:119 party.socket=null

O05837 L1531:137 party.peers={}

O05838 L1531:152 party.state='offline'

O05839 L1531:174 party.room=null

O05840 L1531:190 party.lastSend=0

F0474 | joinParty

O05841 L1532:65 !name

O05842 L1532:112 !cloudSession?.uid||!game.active

O05845 L1532:185 typeof WebSocket==='undefined'

O05847 L1532:289 party.room=name

O05848 L1532:305 party.state='connecting'

O05849 L1532:330 party.joinRef=String(party.seq)

O05850 L1532:383 CLOUD_CONFIG.url.replace(/^http/,'ws')+'/realtime/v1/websocket?apikey='+encodeURIComponent(CLOUD_CONFIG.publishableKey)+'&vsn=1.0.0'

O05853 L1532:517 party.socket=ws

O05854 L1532:533 party.alias=cloudSession.uid.slice(0,8)

O05855 L1533:1 ws.onopen=()=>{if(party.socket!==ws)return;partySend('phx_join',{config:{broadcast:{ack:false,self:false},presence:{key:cloudSession.uid,enabled:true},private:false},access_token:token});party.heartbeat=setInterval(()=>{partySend('heartbeat',{},'phoenix');if(game.active&&!game.paused)partyPresence()},18000)}

F0475 | ws.onopen

O05856 L1533:19 party.socket!==ws

O05857 L1533:188 party.heartbeat=setInterval(()=>{partySend('heartbeat',{},'phoenix');if(game.active&&!game.paused)partyPresence()},18000)

F0476 | anonymous callback

O05858 L1533:260 game.active&&!game.paused

F0474 | joinParty

O05860 L1534:1 ws.onmessage=e=>{if(party.socket!==ws)return;let m;try{m=JSON.parse(e.data)}catch{return};if(!m||m.topic!=='realtime:petal-realms:'+party.room)return;if(m.event==='phx_reply'&&m.payload?.status==='error'){party.state='error';return}if(m.event==='phx_reply'&&m.payload?.status==='ok'){party.state='online';partyPresence()}
if(m.event==='presence_state'){party.peers={};for(const [key,v]of Object.entries(m.payload||{})){const x=v?.metas?.[0]?.user||v?.metas?.[0];if(key!==cloudSession?.uid&&x?.zone&&Number.isFinite(x.x)&&Number.isFinite(x.y))party.peers[key]={...x,x:clamp(x.x,0,W),y:clamp(x.y,0,H)}}} if(m.event==='presence_diff'){for(const [key]of Object.entries(m.payload?.leaves||{}))delete party.peers[key];for(const [key,v]of Object.entries(m.payload?.joins||{})){const x=v?.metas?.[0]?.user||v?.metas?.[0];if(key!==cloudSession?.uid&&x?.zone&&Number.isFinite(x.x)&&Number.isFinite(x.y))party.peers[key]={...x,x:clamp(x.x,0,W),y:clamp(x.y,0,H)}}}
if(m.event==='broadcast'&&m.payload?.event==='position'){const x=m.payload.payload;if(x?.uid&&x.uid!==cloudSession?.uid&&x.zone&&Number.isFinite(x.x)&&Number.isFinite(x.y)){party.peers[x.uid]={...x,x:clamp(x.x,0,W),y:clamp(x.y,0,H),receivedAt:Date.now()};if(Object.keys(party.peers).length>16)delete party.peers[Object.keys(party.peers)[0]]}}

F0477 | ws.onmessage

O05861 L1534:21 party.socket!==ws

O05862 L1534:56 m=JSON.parse(e.data)

O05863 L1534:94 !m||m.topic!=='realtime:petal-realms:'+party.room

O05867 L1534:154 m.event==='phx_reply'&&m.payload?.status==='error'

O05870 L1534:206 party.state='error'

O05871 L1534:236 m.event==='phx_reply'&&m.payload?.status==='ok'

O05874 L1534:285 party.state='online'

O05875 L1535:4 m.event==='presence_state'

O05876 L1535:32 party.peers={}

O05877 L1535:82 m.payload||{}

O05878 L1535:106 v?.metas?.[0]?.user||v?.metas?.[0]

O05879 L1535:144 key!==cloudSession?.uid&&x?.zone&&Number.isFinite(x.x)&&Number.isFinite(x.y)

O05883 L1535:221 party.peers[key]={...x,x:clamp(x.x,0,W),y:clamp(x.y,0,H)}

O05884 L1536:4 m.event==='presence_diff'

O05885 L1536:64 m.payload?.leaves||{}

O05886 L1536:87 delete party.peers[key]

O05887 L1536:146 m.payload?.joins||{}

O05888 L1536:177 v?.metas?.[0]?.user||v?.metas?.[0]

O05889 L1536:215 key!==cloudSession?.uid&&x?.zone&&Number.isFinite(x.x)&&Number.isFinite(x.y)

O05893 L1536:292 party.peers[key]={...x,x:clamp(x.x,0,W),y:clamp(x.y,0,H)}

O05894 L1537:4 m.event==='broadcast'&&m.payload?.event==='position'

O05897 L1537:87 x?.uid&&x.uid!==cloudSession?.uid&&x.zone&&Number.isFinite(x.x)&&Number.isFinite(x.y)

O05902 L1537:174 party.peers[x.uid]={...x,x:clamp(x.x,0,W),y:clamp(x.y,0,H),receivedAt:Date.now()}

O05903 L1537:259 Object.keys(party.peers).length>16

O05904 L1537:294 delete party.peers[Object.keys(party.peers)[0]]

F0474 | joinParty

O05905 L1538:1 ws.onerror=()=>{if(party.socket===ws)party.state='error'}

F0478 | ws.onerror

O05906 L1538:20 party.socket===ws

O05907 L1538:38 party.state='error'

F0474 | joinParty

O05908 L1538:59 ws.onclose=()=>{if(party.socket===ws){if(party.heartbeat)clearInterval(party.heartbeat);party.heartbeat=null;party.state='disconnected';party.peers={}}}

F0479 | ws.onclose

O05909 L1538:78 party.socket===ws

O05910 L1538:147 party.heartbeat=null

O05911 L1538:168 party.state='disconnected'

O05912 L1538:195 party.peers={}

F0480 | updateParty

O05913 L1540:73 p.receivedAt&&Date.now()-p.receivedAt>10000

O05916 L1540:117 delete party.peers[id]

O05917 L1540:143 party.state!=='online'||game.time-party.lastSend<.5

O05921 L1540:202 party.lastSend=game.time

O05922 L1540:351 $('partyTag').textContent="LOCALIZED_5a7cac5f"+party.room+"LOCALIZED_1c0b0850"+Object.keys(party.peers).length

F0481 | drawRemotePlayers

O05926 L1541:75 r.zone!==game.zone||!visible(r.x,r.y,30)

O05929 L1541:136 ctx.globalAlpha=.65

O05930 L1541:197 ctx.fillStyle='#203e46'

O05931 L1541:221 ctx.textAlign='center'

O05932 L1541:244 ctx.font='bold 14px system-ui'

O05933 L1541:296 r.y+5

O05934 L1541:303 ctx.fillStyle='#edfffb'

O05935 L1541:327 ctx.font='11px system-ui'

O05936 L1541:366 "LOCALIZED_6ca6d4a5"+Math.floor(Number(r.level)||1)

O05938 L1541:410 r.y-31

F0482 | renderPartyPanel

O05939 L1542:51 $('panelTitle').textContent="LOCALIZED_91b50ebb"

O05940 L1542:153 desc.className='notice'

O05941 L1542:177 desc.textContent="LOCALIZED_c091d886"

O05942 L1542:318 status.textContent="LOCALIZED_86344fd2"+party.state+(party.room?"LOCALIZED_e9cf0ba4"+party.room:'')+"LOCALIZED_fcb3250d"+Object.keys(party.peers).length+"LOCALIZED_417d8c1d"

O05950 L1543:45 label.className='field'

O05951 L1543:69 label.textContent="LOCALIZED_55e56b80"

O05952 L1543:153 input.id='partyRoomInput'

O05953 L1543:179 input.value=party.room||'garden-1'

O05955 L1543:214 input.maxLength=18

O05956 L1543:309 row.className='route-list'

F0483 | anonymous callback

O05957 L1543:465 err.message||"LOCALIZED_89c4766a"

F0482 | renderPartyPanel

O05958 L1543:581 btn.type='button'

O05959 L1543:599 btn.textContent=text

O05960 L1543:620 btn.onclick=fn

F0001 | anonymous callback

O05961 L1544:0 window.PetalRealmsInfo=Object.freeze({version:'5.16-botanical-workshop',ruleset:globalThis.PetalRules.version,randomState:()=>game.random?.snapshot()||null,snapshot:()=>game.active?{zone:game.zone,position:{x:game.p.x,y:game.p.y},mode:game.mode,movement:game.settings.movement,goal:game.goal?{...game.goal,owned:collectionOwned()}:null,level:game.p.level,hp:game.p.hp,slots:game.slots.length,deck:game.deck,backup:game.backupSlots.length,mobs:game.mapData[game.zone].living.length,rarities:rarity.length,bands:ZONE_BANDS.length,shots:game.friendlyShots.length,guardians:Object.keys(GUARDIANS).length,petals:Object.keys(pets).length,armor:game.p.armorLayers,orbitDirection:game.p.orbitDirection,petalState:game.equipped.map(e=>({hp:e.hp,cd:e.cd})),magnetRadius:magnetRadius(),bodyRadius:game.p.r,seen:Object.values(game.codex).filter(x=>x.seen).length,dashCooldown:game.p.dashCooldown,quality:game.settings.quality,talents:{...game.talents},talentPoints:talentPoints(),boon:game.boon?{type:game.boon.type,remaining:Math.max(0,game.boon.until-game.time)}:null,shrines:game.mapData[game.zone].shrines.length,skills:Object.fromEntries(Object.keys(SKILL_CONFIG).map(k=>[k,Math.round(skillRemaining(k)*10)/10])),synergies:{...game.synergies},event:game.worldEvent?.type||null,partyPeers:Object.keys(party.peers).length,partyState:party.state,quest:{...game.journal[game.zone]},combo:{count:game.combo.count,bonus:Math.min(.5,Math.max(0,game.combo.count-1)*.05)},bossVisible:!$('bossHud').classList.contains('hidden'),paused:game.paused}:null})

F0485 | randomState

O05962 L1544:126 game.random?.snapshot()||null

F0486 | snapshot

O05963 L1544:169 game.active?{zone:game.zone,position:{x:game.p.x,y:game.p.y},mode:game.mode,movement:game.settings.movement,goal:game.goal?{...game.goal,owned:collectionOwned()}:null,level:game.p.level,hp:game.p.hp,slots:game.slots.length,deck:game.deck,backup:game.backupSlots.length,mobs:game.mapData[game.zone].living.length,rarities:rarity.length,bands:ZONE_BANDS.length,shots:game.friendlyShots.length,guardians:Object.keys(GUARDIANS).length,petals:Object.keys(pets).length,armor:game.p.armorLayers,orbitDirection:game.p.orbitDirection,petalState:game.equipped.map(e=>({hp:e.hp,cd:e.cd})),magnetRadius:magnetRadius(),bodyRadius:game.p.r,seen:Object.values(game.codex).filter(x=>x.seen).length,dashCooldown:game.p.dashCooldown,quality:game.settings.quality,talents:{...game.talents},talentPoints:talentPoints(),boon:game.boon?{type:game.boon.type,remaining:Math.max(0,game.boon.until-game.time)}:null,shrines:game.mapData[game.zone].shrines.length,skills:Object.fromEntries(Object.keys(SKILL_CONFIG).map(k=>[k,Math.round(skillRemaining(k)*10)/10])),synergies:{...game.synergies},event:game.worldEvent?.type||null,partyPeers:Object.keys(party.peers).length,partyState:party.state,quest:{...game.journal[game.zone]},combo:{count:game.combo.count,bonus:Math.min(.5,Math.max(0,game.combo.count-1)*.05)},bossVisible:!$('bossHud').classList.contains('hidden'),paused:game.paused}:null

F0489 | anonymous callback

O05967 L1544:1167 Math.round(skillRemaining(k)*10)/10

F0490 | anonymous callback

O05973 L1544:1581 game.active&&!game.guest

F0491 | anonymous callback

O05975 L1544:1674 cloudPending!==null&&!cloudConflict

F0001 | anonymous callback

O05978 L1544:1820 cloudRestoring=true

O05979 L1544:1840 $('authSubmit').disabled=true

O05980 L1544:1870 $('localModeBtn').disabled=true

O05981 L1544:1902 $('cfModeBtn').disabled=true

O05982 L1544:1931 void globalThis.PetalUI.install({"schema": 1, "version": "5.16.0", "title": "LOCALIZED_fd6b4e3c", "developer": "LOCALIZED_df0e63cd", "introduction": "LOCALIZED_83337ab6", "updatedAt": "2026-10-21T00:00:00Z", "highlights": ["LOCALIZED_43697708", "LOCALIZED_e42e1283", "LOCALIZED_613a680c", "LOCALIZED_61467dc4", "LOCALIZED_0e370861"]})().then(async()=>{const handled=await cloudRecoveryFromURL();if(!handled)await cfResume()}).finally(()=>{cloudRestoring=false;$('authSubmit').disabled=false;$('localModeBtn').disabled=false;$('cfModeBtn').disabled=false})

F0492 | anonymous callback

O05983 L1544:2449 !handled

F0493 | anonymous callback

O05984 L1544:2490 cloudRestoring=false

O05985 L1544:2511 $('authSubmit').disabled=false

O05986 L1544:2542 $('localModeBtn').disabled=false

O05987 L1544:2575 $('cfModeBtn').disabled=false

F0494 | anonymous callback

O05988 L1544:2648 cfPending&&!cfConflict

O05990 L1544:2672 cfRetry=0

F0001 | anonymous callback

O05991 L1545:0 window.PetalAccountStatus=Object.freeze({accountId:()=>cfAccount?.id||null})

F0495 | accountId

O05992 L1545:55 cfAccount?.id||null

F0496 | anonymous callback

O05993 L1545:119 e.persisted&&cfAccount

F0498 | now

O05994 L13:29 Date.now()/1000

F0499 | anonymous callback

O05995 L15:58 this.status=status

O05996 L15:78 this.retry=retry

F0503 | username

O05997 L23:5 typeof value!=='string'

O05999 L25:5 emailOnly||name.includes('@')

O06000 L26:54 parts[1]||''

O06001 L28:7 name.length>254||parts.length!==2||!local||local.length>64||! /^[a-z0-9.!#$%&'*+/=?^_`{|}~-]+$/i.test(local)||local.startsWith('.')||local.endsWith('.')||local.includes('..')||labels.length<2||!labels.every(x=>/^[a-z0-9](?:[a-z0-9-]{0,61}[a-z0-9])?$/i.test(x))||!/^[a-z]{2,63}$/i.test(labels.at(-1))

O06020 L32:5 !/^[\p{L}\p{N}_-]{2,18}$/u.test(name)||['__proto__','constructor','prototype'].includes(name)||/^(cloud-|cf-)/i.test(name)

F0505 | password

O06023 L36:5 typeof value!=='string'||value.length<8||value.length>128

F0508 | anonymous callback

O06029 L39:98 err?reject(err):resolve(key)

F0509 | verifyPasswordRuntime

O06030 L42:9 passwordRuntimeReady??=(async()=>{ const key=await derive('fixed-public-runtime-probe','fixed-public-salt'); if(key.toString('hex')!=='964969a05a3b287285cf7576af1ffcdc6bc82f271132dcc5915c14dd7bea6767')throw Error('Password runtime self-test failed'); })()

F0510 | anonymous callback

O06031 L44:7 key.toString('hex')!=='964969a05a3b287285cf7576af1ffcdc6bc82f271132dcc5915c14dd7bea6767'

F0512 | checkPassword

O06032 L52:5 typeof pass!=='string'||pass.length>128

O06036 L53:10 record?JSON.parse(record):{algorithm:'scrypt',version:1,...SCRYPT,salt:'00000000000000000000000000000000',key:'0'.repeat(64)}

O06037 L55:5 p.algorithm==='scrypt'&&p.version===1&&p.N===SCRYPT.N&&p.r===SCRYPT.r&&p.p===SCRYPT.p

O006046 L55:91 actual=await derive(pass,p.salt)

O006047 L56:10 p.algorithm==='pbkdf2-sha256'&&p.iterations===600000

O006050 L58:8 actual=await new
Promise((resolve,reject)=>pbkdf2(pass,p.salt,p.iterations,32,'sha256',(err,key)=>err?reject(err):resolve(key)))

F0514 | anonymous callback

O006051 L58:106 err?reject(err):resolve(key)

F0512 | checkPassword

O006052 L59:16 e.name==='NotSupportedError'

O006053 L62:9 actual.length===expected.length&&timingSafeEqual(actual,expected)&&!!record

F0515 | bodyOf

O006058 L65:5 !/^application\/json(?:;|$)/i.test(request.headers.get('Content-Type')||'')

O006060 L66:5 Number(request.headers.get('Content-Length'))>MAX_BODY

O006061 L67:45 !reader

O006062 L69:63 total+=value.length

O006063 L69:86 total>MAX_BODY

O006064 L70:97 offset+=c.length

O006065 L70:172 !body||typeof body!=='object'||Array.isArray(body)

F0516 | limit

O006070 L77:112 t+seconds

O006071 L78:5 row.attempts>max

O006072 L78:74 row.reset_at-t

F0517 | cookieToken

O006073 L81:15 request.headers.get('Cookie')||''

O006074 L82:9 match?.[1]||null

F0518 | accountFor

O006075 L85:38 !token

F0520 | sessionCookie

O006076 L93:56 remember?30*86400:12*3600

O006079 L95:99 now()+duration

O006080 L96:5 r.meta.changes!==1

O006081 L97:73 remember?'; Max-Age='+duration:''

F0521 | validateSave

O006083 L100:5 !payload||typeof
payload!=='object'||Array.isArray(payload)||payload.version!==5||!ZONES.has(payload.zone)

O006092 L102:5 !p||typeof p!=='object'||!Number.isSafeInteger(p.level)||p.level<1||p.level>999||!['xp','kills','hp'].eve
ry(k=>Number.isFinite(p[k])&&p[k]>=0&&p[k]<=1e12)

F0522 | anonymous callback

O006104 L102:116 Number.isFinite(p[k])&&p[k]>=0&&p[k]<=1e12

F0523 | item

O006108 L103:16 x&&typeof x==='object'&&PETALS.has(x.t)&&Number.isInteger(x.r)&&x.r>=0&&x.r<10

F0521 | validateSave

O006117 L104:5 !Array.isArray(payload.slots)||payload.slots.length>10||!payload.slots.every(item)

O006122 L105:5 payload.backupSlots!==undefined&&(!Array.isArray(payload.backupSlots)||payload.backupSlots.length>10||!pa
yload.backupSlots.every(item))

O006129 L107:5 !bag||typeof bag!=='object'||Array.isArray(bag)||Object.keys(bag).length>PETALS.size*10

O006137 L108:86 !m||!PETALS.has(m[1])||!Number.isInteger(n)||n<0||n>99999

O006146 L109:5 payload.goal!=null&&(!PETALS.has(payload.goal.t)||!Number.isInteger(payload.goal.r)||payload.goal.r<0||pa
yload.goal.r>9)

O06155 L111:5 new TextEncoder().encode(text).length>MAX_BODY

F0524 | putSave

O06156 L115:5 expected!==null&&(!Number.isSafeInteger(expected)||expected<1||expected>=Number.MAX_SAFE_INTEGER)

O06163 L117:12 expected===null ?await statement(db,'INSERT INTO saves(player_id,schema_version,revision,payload,updated_at) VALUES(?,5,1,?,?) ON CONFLICT(player_id) DO NOTHING RETURNING revision',id,text,t).first() :await statement(db,'UPDATE saves SET revision=revision+1,payload=?,updated_at=? WHERE player_id=? AND revision=? RETURNING revision',text,t,id,expected).first()

O06165 L120:5 !row

F0525 | migrate

O06166 L125:5 typeof token!=='string'||token.length>8192||token.length<20

O06172 L126:56 'Bearer '+token

O06173 L127:33 LEGACY_URL+'/auth/v1/user'

O06174 L128:5 !userResponse.ok

O06175 L130:5 !/^[a-f0-9-]{36}$/i.test(user?.id||'')

O06177 L132:16 linked.player_id!==account.id

O06178 L134:29 LEGACY_URL+'/rest/v1/petal_realms_saves?user_id=eq.'+encodeURIComponent(user.id)+'&select=revision,payload&limit=1'

O06181 L135:5 !response.ok

O06182 L136:38 !Array.isArray(rows)||rows.length!==1

F0526 | onRequest

O06185 L148:7 !env.PETAL_DB

O06186 L150:7 !['GET','POST','PUT'].includes(method)

O06187 L151:7 method!=='GET'&&request.headers.get('Origin')!==url.origin

O06190 L152:7 method==='GET'&&path==='session'

O06193 L152:101 a?publicAccount(a):null

O06194 L153:7 method==='GET'&&path==='health'

O06197 L155:7 account&&['save','auth/password','auth/logout','migrate'].includes(path)&&request.headers.get('X-Petal-Account')!==account.id

O06200 L157:7 path==='save'&&method==='GET'

O06203 L158:9 !account

O06204 L160:19 row?{accountId:account.id,revision:row.revision,payload:JSON.parse(row.payload),updatedAt:row.updated_at}:{accountId:account.id,revision:null,payload:null}

O06205 L162:7 method==='GET'

O06206 L163:40 request.headers.get('CF-Connecting-IP')||'unknown'

O06207 L164:7 path==='save'&&method==='PUT'

O06210 L165:9 !account

O06211 L166:21 'save:'+account.id

O06212 L168:7 method!=='POST'

O06213 L169:7 path==='auth/logout'

O06214 L173:7 path==='migrate'

O06215 L174:9 !account

O06216 L175:21 'migrate:'+account.id

O06217 L177:7 !['auth/register','auth/login','auth/password','auth/recover'].includes(path)

O06218 L178:19 'auth-ip:'+hash(ip)

O06219 L179:7 path==='auth/password'

O06220 L180:9 !account

O06221 L181:21 'password:'+account.id

O06222 L182:9 !await checkPassword(body.currentPassword,account.password_record)

O06223 L185:9 !changed

O06224 L186:94 body.remember===true

O06225 L188:38 path==='auth/register'

O06226 L189:19 'auth-name:'+hash(name)

O06227 L190:7 path==='auth/register'

O06228 L191:21 'register:'+hash(ip)

O06229 L194:9 !row

O06230 L195:106 body.remember===true

O06231 L198:7 path==='auth/recover'

O06232 L200:21 typeof body.recoveryCode==='string'?body.recoveryCode.trim().toLowerCase():''

O06235 L201:9 !row||!/^[a-f0-9]{64}$/.test(recovery)||!timingSafeEqual(Buffer.from(hash(recovery),'hex'),Buffer.from(row.recovery_hash,'hex'))

O06240 L204:9 !changed

O06241 L207:7 !await checkPassword(body.password,row?.password_record)

O06242 L208:91 body.remember===true

O06243 L210:7 error instanceof APIError

O06244 L210:81 error.retry?{'Retry-After':String(error.retry)}:{}

O06245 L211:48 error?.name||'Error'

F0528 | plan

O06246 L5:4 !weights[policy]||!Array.isArray(slots)||slots.length<1||slots.length>10

O06253 L7:43 !Object.hasOwn(defs,k)||!Number.isSafeInteger(n)||n<0||n>99999

O06260 L7:139 pool[k]=n

O06261 L8:26 !Object.hasOwn(defs,k)

O06262 L8:86 pool[k]=(pool[k]||0)+1

F0529 | anonymous callback

O06265 L9:105 !Number.isFinite(x)||x<0

F0530 | anonymous callback

O06268 L9:185 maxima[i]=Math.max(maxima[i],x)

F0532 | anonymous callback

O06269 L10:81 s+weights[policy][i]*(maxima[i]?x/maxima[i]:0)

F0533 | anonymous callback

O06273 L11:33 pool[k]>0

F0534 | anonymous callback

O06274 L11:56 score(b)-score(a)||(a<b?-1:a>b?1:0)

F0528 | plan

O06281 L12:74 slots.length-result.length

O06282 L12:102 j<n

O06283 L12:106 j++

O06284 L12:126 pool[k]--

O06285 L13:4 result.length!==slots.length

F0535 | anonymous callback

O06286 L15:32 n>99999

F0536 | anonymous callback

O06287 L16:81 n>0

F0537 | anonymous callback

O06288 L16:114 s+score(k)

F0538 | anonymous callback

O06289 L16:155 s+score(k)

F0541 | text

O06290 L19:15 typeof s==='string'&&s.length>0&&s.length<=240&&!/[<>\x00-\x08]/.test(s)

F0540 | release

O06298 L20:4 !x||x.schema!==1||!/^\d+\.\d+\.\d+$/.test(x.version)||!text(x.title)||!text(x.developer)||!text(x.introduction)||!/^\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}Z$/.test(x.updatedAt)||!Number.isFinite(Date.parse(x.updatedAt))||!Array.isArray(x.highlights)||!x.highlights.length||x.highlights.length>20||!x.highlights.every(text)

F0527 | anonymous callback

O06321 L23:0 globalThis.PetalWorkshop=Object.freeze({plan,release,weights})

F0545 | paintNews

O06322 L7:66 news||fallback

O06323 L7:84 !x

O06324 L8:1 $('releaseVersion').textContent='V'+x.version

O06326 L8:47 $('releaseTitle').textContent=x.title

O06327 L9:1 $('developerName').textContent=x.developer

O06328 L9:44 $('developerIntro').textContent=x.introduction

O06329 L10:1 $('releaseDate').textContent="LOCALIZED_7ee196e2"+new Date(x.updatedAt).toLocaleString('zh-CN',{timeZone:'Asia/Shanghai',hour12:false})+"LOCALIZED_5eb76ad8"

O06332 L11:71 li.textContent=text

O06333 L12:1 $('releaseState').textContent=state

O06334 L12:37 $('refreshRelease').disabled=!!loading

F0546 | refresh

O06337 L17:1 loading=(async()=>{try{const r=await fetch('release.json',{cache:'no-store',signal:controller.signal});if(!r.ok)throw Error('Unavailable');const raw=await r.text();if(raw.length>16384)throw Error('Too large');const x=globalThis.PetalWorkshop.release(JSON.parse(raw));if(!x)throw Error('Invalid');news=x;checkedAt=new Date();paintNews("LOCALIZED_0cfa70fe"+checkedAt.toLocaleTimeString('zh-CN',{hour12:false}));}catch{paintNews(news?"LOCALIZED_1ddd2800":"LOCALIZED_4d14fefe");}finally{clearTimeout(timer);loading=null;$('refreshRelease').disabled=false}})()

F0548 | anonymous callback

O06338 L17:107 !r.ok

O06339 L17:168 raw.length>16384

O06340 L17:271 !x

O06341 L17:297 news=x

O06342 L17:304 checkedAt=new Date()

O06343 L17:335 "LOCALIZED_0cfa70fe"+checkedAt.toLocaleTimeString('zh-CN',{hour12:false})

O06344 L17:421 news?"LOCALIZED_1ddd2800":"LOCALIZED_4d14fefe"

O06345 L17:496 loading=null

O06346 L17:509 $('refreshRelease').disabled=false

F0546 | refresh

O06347 L18:1 $('refreshRelease').disabled=true

F0549 | welcome

O06348 L22:35 welcomeResolve?.promise||Promise.resolve()

F0550 | anonymous callback

O06349 L23:42 resolve=r

F0549 | welcome

O06350 L23:53 welcomeResolve={resolve,promise}

O06351 L23:86 lastFocus=document.activeElement

O06352 L24:11 news?"LOCALIZED_9d40c820":"LOCALIZED_b782f736"

O06353 L24:102 void refresh()

F0552 | anonymous callback

O06354 L29:30 asking={resolve,origin}

O06355 L29:54 $('confirmTitle').textContent=title

O06356 L29:90 $('confirmMessage').textContent=message

O06357 L29:130 $('confirmDialog').returnValue='cancel'

F0553 | install

O06358 L32:1 fallback=globalThis.PetalWorkshop.release(feed)

O06359 L32:52 !fallback

O06360 L33:1 $('welcomeContinue').onclick=closeWelcome
O06361 L33:43 $('welcomeClose').onclick=closeWelcome
O06362 L33:82 $('refreshRelease').onclick=refresh

F0554 | anonymous callback

O06363 L34:76 welcomeResolve=null
O06364 L34:99 lastFocus?.isConnected&&!lastFocus.closest('.hidden')

F0553 | install

O06366 L35:1 $('confirmAccept').onclick=()=>{$('confirmDialog').close('accept')}
O06367 L35:69 $('confirmCancel').onclick=()=>{$('confirmDialog').close('cancel')}

F0557 | anonymous callback

O06368 L36:65 asking=null
O06369 L36:80 !a
O06370 L36:100 $('confirmDialog').returnValue==='accept'

F0558 | anonymous callback

O06371 L37:89 void welcome()

F0559 | anonymous callback

O06372 L38:54 !document.hidden&&$('welcomeDialog').open
O06374 L38:96 void refresh()

F0561 | anonymous callback

O06375 L40:21 !document.hidden&&$('welcomeDialog').open
O06377 L40:63 void refresh()

F0542 | anonymous callback

O06378 L43:0 globalThis.PetalUI=Object.freeze({install,welcome,refresh,ask})

F0564 | anonymous callback

O06379 L4:47 v&&typeof v==='object'&&freeze(v)

F0565 | finite

O06383 L6:24 !Number.isFinite(x)

F0566 | probability

O06384 L7:39 p<0||p>1

F0567 | integer

O06387 L8:35 !Number.isSafeInteger(n)||n<0||n>max

F0568 | normalize

O06392 L9:32 !values.length||values.some(x=>!Number.isFinite(x)||x<0)

F0569 | anonymous callback

O06394 L9:63 !Number.isFinite(x)||x<0

F0568 | normalize

O06397 L9:160 m<=0

F0570 | anonymous callback

O06398 L9:234 x/m

F0571 | anonymous callback

O06399 L9:264 s+x

F0572 | anonymous callback

```
O06400 L9:292 x/sum
```

F0573 | choose

```
O06401 L10:82 u===1
O06402 L10:148 i<p.length
O06403 L10:159 i++
O06404 L10:164 c+=p[i]
O06405 L10:175 u<c
```

F0574 | anonymous callback

```
O06406 L10:214 x>0
```

F0575 | streams

```
O06407 L11:161 (seed^Math.imul(i+1,0x9e3779b9))>>>0
O06410 L11:198 counts[key]=0
O06411 L11:212 out[key]=()=>{counts[key]++;state=(state+0x6D2B79F5)>>>0;let
t=state;t=Math.imul(t^(t>>>15),t|1);t^=t+Math.imul(t^(t>>>7),t|61);return ((t^(t>>>14))>>>0)/4294967296}
```

F0576 | out[key]

```
O06412 L11:226 counts[key]++
O06413 L11:240 state=(state+0x6D2B79F5)>>>0
O06416 L11:281 t=Math.imul(t^(t>>>15),t|1)
O06420 L11:309 t^=t+Math.imul(t^(t>>>7),t|61)
O06425 L11:347 ((t^(t>>>14))>>>0)/4294967296
```

F0575 | streams

```
O06429 L11:378 out.snapshot=()=>({version:config.version,seed,counts:{...counts}})
```

F0579 | add

```
O06430 L12:107 (mass.get(r)||0)+p
```

F0578 | dropRanks

```
O06432 L12:130 rank===7
O06433 L12:248 rank-loss
O06434 L12:265 !treasure
O06435 L12:391 .72*p
O06436 L12:410 r+1
O06437 L12:415 .28*p
O06438 L12:438 (boosted.get(rr)||0)+pp
```

F0582 | rewardStats

```
O06440 L13:62 !types.length||guardian&&types.length<2
O06444 L13:245 finite(value(type,row.rank))+(guardian?finite(value(types[1],Math.max(1,row.rank-1))):0)
O06447 L13:351 row.p/types.length*.82
O06449 L13:389 row.p/types.length*.18
O06451 L13:418 base+finite(value('basic',Math.max(0,row.rank-2)))
```

F0583 | anonymous callback

```
O06453 L13:505 s+x.p*x.value
```

F0584 | anonymous callback

```
O06455 L13:554 s+x.p*(x.value-mean)**2
```

F0585 | craftCDF

```
O06459 L14:76 n+1
O06460 L14:93 k<=n
O06461 L14:98 k++
```

```
O06462 L14:102 f[k]=p+(1-p)*(f[k-1]+f[k-2]+f[k-3]+f[k-4])/4
```

F0586 | craftAnalysis

```
O06474 L15:108 n+',0'
O06475 L15:132 k<attempts
O06476 L15:143 k++
```

F0587 | add

```
O06477 L15:203 stock+','+wins
O06479 L15:231 (next.get(key)||0)+mass
```

F0586 | craftAnalysis

```
O06481 L15:337 stock<5
O06482 L15:380 stock-5
O06483 L15:388 wins+1
O06484 L15:395 mass*p
O06485 L15:418 loss<=4
O06486 L15:426 loss++
O06487 L15:437 stock-loss
O06488 L15:453 mass*(1-p)/4
O06491 L15:467 states=next
O06492 L15:502 attempts+1
O06493 L15:629 outputMass[w]+=pr
O06494 L15:647 mass+=pr
O06495 L15:656 stock+=s*pr
O06497 L15:668 outputs+=w*pr
O06499 L15:685 w>0
O06500 L15:689 success+=pr
O06501 L15:731 p===0?null:f.findIndex(x=>x>=.95)
```

F0588 | anonymous callback

```
O06503 L15:757 x>=.95
```

F0586 | craftAnalysis

```
O06504 L15:836 p?5+2.5*(1-p)/p:Infinity
O06509 L15:870 p?1.25*(1-p)/p+6.25*(1-p)/(p*p):Infinity
O06518 L15:915 q95<0?null:q95
```

F0589 | criticalChance

```
O06520 L16:90 1-(1-config.critical.perCopy)**copies
```

F0590 | criticalFactor

```
O06523 L17:72 p===0
O06524 L17:117 u===1
O06525 L17:170 u<p?config.critical.multiplier:1
```

F0591 | slowFactor

```
O06527 L18:72 1-Math.min(config.frost.cap,copies*config.frost.perCopy)*(guardian?config.frost.guardianFactor:1)
```

F0592 | expectedUtility

```
O06531 L19:58 eta<0
O06532 L19:109 Math.abs(outcomes.reduce((s,x)=>s+probability(x.p),0)-1)>1e-12
```

F0593 | anonymous callback

```
O06534 L19:141 s+probability(x.p)
```

F0594 | anonymous callback

```
O06535 L19:247 s+x.p*(eta===0?finite(x.value):-Math.expm1(-eta*finite(x.value))/eta)
```

F0595 | weightedValue

O06543 L20:46 features.length!==weights.length||features.some(x=>!Number.isFinite(x)||x<0||x>1)

F0596 | anonymous callback

O06545 L20:97 !Number.isFinite(x)||x<0||x>1

F0597 | anonymous callback

O06550 L20:223 s+w*features[i]

F0600 | damageMoments

O06552 L22:80 raw<0||armor<0||hp!==null&&(!Number.isFinite(hp)||hp<0)
O06561 L22:243 hp===null?Infinity:hp
O06563 L22:284 1-p
O06564 L22:320 raw-armor
O06565 L22:368 raw*config.critical.multiplier-armor

F0601 | anonymous callback

O06567 L22:443 s+x.p*x.value

F0602 | anonymous callback

O06569 L22:492 s+x.p*(x.value-mean)**2

F0603 | anonymous callback

O06573 L22:564 x.p>0

F0605 | anonymous callback

O06574 L22:638 x.p>0

F0600 | damageMoments

O06575 L22:693 hp===null?null:outcomes.reduce((s,x)=>s+(x.value>=hp?x.p:0),0)

F0607 | anonymous callback

O06577 L22:731 s+(x.value>=hp?x.p:0)

F0608 | targetDropChance

O06580 L23:75 !Array.isArray(types)||!types.length||guardian&&types.length<2||typeof target!=='string'
O06589 L23:241 types.filter(t=>t===target).length/types.length

F0609 | anonymous callback

O06590 L23:257 t===target

F0610 | anonymous callback

O06591 L23:349 row.rank===wanted?share:0
O06593 L23:382 guardian&&types[1]===target&&Math.max(1,row.rank-1)===wanted
O06598 L23:449 target==='basic'&&Math.max(0,row.rank-2)===wanted?.18:0
O06603 L23:512 s+row.p*(forced?1:main+(1-main)*extra)

F0611 | frameAlpha

O06609 L24:63 dt<0
O06610 L24:109 dt===0||gain===0
O06613 L24:138 gain===1
O06614 L24:163 -Math.expm1(60*dt*Math.log1p(-gain))

F0612 | specialProfile

O06618 L25:162 speed<0||!['charger','sniper','venom','nova','summon','shield'].includes(type)
O06621 L25:315 type==='sniper'||type==='venom'
O06624 L25:349 (type==='venom'?c.venomSpeed:c.sniperSpeed)+Math.min(c.levelSpeedCap,level*c.levelSpeed)

O06628 L25:447 type==='charger'?c.chargeDuration:c.projectileLife

O06630 L25:518 type==='charger'?c.chargeWindup:c.windup

O06632 L25:574 projectile?v:speed*c.chargeMultiplier*(enraged?c.enrageSpeed:1)*frostFactor

O06637 L25:656
projectile?v*duration:type==='charger'?speed*c.chargeMultiplier*(enraged?c.enrageSpeed:1)*frostFactor*duration:0

O06646 L25:776 type==='nova'?c.novaRadius:0

O06648 L25:813 projectile&&rank>=3?[0,-c.spread,c.spread]:[0]

F0613 | specialAim

O06652 L26:36 finite(p.x)-finite(m.x)

O06653 L26:63 finite(p.y)-finite(m.y)

O06654 L26:112 m.specialType==='sniper'?Math.min(d/370,.65):0

O06657 L26:161 dx+finite(p.vx??0)*lead

O06660 L26:187 dy+finite(p.vy??0)*lead

O06663 L26:215 Math.hypot(x,y)||1

O06664 L26:244 x/len

O06665 L26:252 y/len

F0562 | anonymous callback

O06666 L27:2 globalThis.PetalRules=Object.freeze({version:config.version,config,normalize,choose,streams,dropRanks,rewardStats,craftCDF,craftAnalysis,criticalChance,criticalFactor,slowFactor,expectedUtility,weightedValue,evaluatePolicy,damageMoments,targetDropChance,frameAlpha,specialProfile,specialAim})

F0614 | anonymous callback

O06667 L4:10 Math.PI*2

F0615 | ellipse

O06668 L5:91 c.fillStyle=color

F0616 | halo

O06669 L6:47 c.globalAlpha=alpha

O06670 L6:152 color+'00'

O06671 L6:164 c.fillStyle=g

O06672 L6:189 x-r

O06673 L6:193 y-r

O06674 L6:197 r*2

O06675 L6:201 r*2

F0617 | terrain

O06676 L7:38 palettes[zone]||palettes.garden

O06677 L7:101 seed>>>0

F0618 | random

O06678 L7:128 s=(Math.imul(s,1664525)+1013904223)>>>0

O06681 L7:175 s/4294967296

F0617 | terrain

O06682 L8:13 i<180

O06683 L8:19 i++

O06684 L8:32 random()*512

O06685 L8:47 random()*512

O06686 L8:62 3+random()*26

O06688 L8:76 c.globalAlpha=.035+random()*.04

O06691 L8:124 k*.4

O06692 L8:129 i%2?p[0]:p[1]

O06694 L8:143 random()*TAU

O06695 L8:157 c.globalAlpha=1

O06696 L9:13 i<105

O06697 L9:19 i++

```
O06698 L9:32 random()*512
O06699 L9:47 random()*512
O06700 L9:62 3+random()*13
O06702 L9:78 random()*TAU
O06703 L9:129 c.globalAlpha=.15
O06704 L9:147 c.lineWidth=.75
O06705 L9:163 c.strokeStyle=i%3?p[0]:p[1]
O06708 L9:191 c.fillStyle=p[0]
O06709 L10:4 zone===='factory'
O06710 L10:22 c.globalAlpha=.10
O06711 L11:9 zone===='ocean'||zone===='sewer'
O06714 L11:64 -k
O06715 L11:86 -k*.3
O06717 L11:92 -3
O06718 L11:95 k*.3
O06719 L11:121 i%7===0
O06721 L12:9 zone===='desert'||zone===='ant'
O06724 L12:63 -k
O06725 L12:90 -k*.3
O06727 L12:122 -k*.2
O06729 L13:9 zone===='hel'
O06730 L13:23 c.globalAlpha=.18
O06731 L13:41 c.strokeStyle=i%4?p[1]:p[2]
O06734 L13:92 -k
O06735 L13:109 k*.3
O06736 L13:126 -k*.2
O06738 L14:31 k*.4
O06739 L14:56 -k*.8
O06741 L14:62 -k*.5
O06743 L14:70 -k
O06744 L14:93 k*.8
O06745 L14:98 -k*.5
O06747 L14:106 k*.4
O06748 L14:157 k*.2
O06749 L14:174 -k*.8
```

F0619 | landmark

```
O06751 L16:49 palettes[zone]||palettes.garden
O06752 L16:159 c.globalAlpha=.20
O06753 L16:193 r*.70
O06754 L16:199 wet?p[1]:p[0]
O06755 L16:214 c.globalAlpha=.35
O06756 L17:4 zone===='factory'
O06757 L17:22 c.strokeStyle=p[1]
O06758 L17:41 c.lineWidth=3
O06759 L17:65 -2
O06760 L17:68 k<=2
O06761 L17:73 k++
O06762 L17:91 k*r*.29-r*.1
O06766 L17:104 -r*.48
O06768 L17:111 r*.20
O06769 L17:117 r*.96
O06770 L17:124 c.strokeStyle=p[2]
O06771 L17:143 c.lineWidth=1
O06772 L17:170 -r*.8
O06774 L17:176 -r*.6
O06776 L17:182 r*1.6
O06777 L17:188 r*1.2
O06778 L18:28 r*.8
```

```
O06779 L18:33 r*.5
O06780 L18:44 c.strokeStyle=p[2]
O06781 L18:63 c.lineWidth=1.4
O06782 L18:91 k<3
O06783 L18:95 k++
O06784 L18:128 r*(.45+k*.16)
O06787 L18:142 r*(.22+k*.13)
O06790 L18:158 .18+(!reduced?time*.04:0)
O06794 L18:184 Math.PI*1.5
O06795 L18:220 k<4
O06796 L18:224 k++
O06797 L18:238 Math.cos(k*2.1)*r*.6
O06800 L18:259 Math.sin(k*2.1)*r*.25
O06803 L18:281 r*.1
O06804 L18:286 r*.045
O06805 L19:9 zone==='desert'||zone==='ant'||zone==='hel'
O06810 L19:54 c.strokeStyle=p[1]
O06811 L19:73 c.lineWidth=4
O06812 L19:97 -1
O06813 L19:100 k<=1
O06814 L19:105 k++
O06815 L19:130 k*r*.16
O06817 L19:138 r*k*.22
O06819 L19:146 r*.6
O06820 L19:151 Math.PI*1.1
O06821 L19:163 TAU*.96
O06822 L19:195 k<4
O06823 L19:199 k++
O06824 L19:212 k*1.7+l.seed
O06826 L19:235 Math.cos(a)*r*.62
O06828 L19:253 Math.sin(a)*r*.45
O06830 L19:271 r*.12
O06831 L19:277 r*.08
O06832 L20:18 k<11
O06833 L20:23 k++
O06834 L20:36 k*2.399+l.seed
O06836 L20:53 r*(.3+(k%3)*.2)
O06840 L20:71 Math.cos(a)*d
O06841 L20:87 Math.sin(a)*d*.65
O06843 L20:117 j<5
O06844 L20:121 j++
O06845 L20:134 j*TAU/5
O06847 L20:152 x+Math.cos(b)*5
O06849 L20:168 y+Math.sin(b)*5
O06851 L20:190 k%3?p[2]:'#efd1cb'
```

F0620 | anatomy

```
O06853 L22:81 reduced?0:Math.sin(time*4+(m.id||0))*.08
O06858 L22:131 c.lineCap='round'
O06859 L22:149 c.lineJoin='round'
O06860 L23:4 t==='rat'
O06861 L23:15 c.strokeStyle='#926d69'
O06862 L23:39 c.lineWidth=Math.max(1.5,r*.09)
O06864 L23:96 r*.5
O06865 L23:118 r*.8
O06866 L23:123 r*1.1
O06867 L23:129 r*.6
O06868 L23:134 r*1.5
O06869 L23:140 -r*.2
```

```
O06871 L23:146 r*1.35
O06872 L23:184 -1
O06873 L23:201 side*r*.46
O06875 L23:212 -r*.5
O06877 L23:218 r*.22
O06878 L23:224 r*.20
O06879 L23:251 side*r*.46
O06881 L23:262 -r*.5
O06883 L23:268 r*.13
O06884 L23:274 r*.12
O06885 L23:291 c.lineWidth=Math.max(.6,r*.022)
O06887 L23:335 k<2
O06888 L23:339 k++
O06889 L23:367 side*r*.2
O06891 L23:377 -r*.25
O06893 L23:394 side*r*.78
O06895 L23:405 (-.4+k*.2)*r
O06899 L24:28 c.strokeStyle='#5b4549'
O06900 L24:52 c.lineWidth=Math.max(2,r*.15)
O06902 L24:107 r*.48
O06903 L24:130 r*.75
O06904 L24:136 r*1.5
O06905 L24:142 -r*.7
O06907 L24:148 r*1.6
O06908 L24:154 -r*.25
O06910 L24:161 r*.76
O06911 L24:189 -r*.25
O06913 L24:196 r*.76
O06914 L24:202 r*.13
O06915 L24:208 r*.22
O06916 L24:224 -.4
O06917 L24:248 -1
O06918 L24:278 side*r*.5
O06920 L24:288 -r*.2
O06922 L24:304 side*r*.95
O06924 L24:315 -r*.75
O06926 L24:344 side*r*.97
O06928 L24:355 -r*.80
O06930 L24:362 r*.23
O06931 L24:368 r*.16
O06932 L24:384 side*.5
O06933 L25:24
c.strokeStyle=palettes[/garden/i.test(t)?'garden':/ocean/i.test(t)?'ocean':/factory/i.test(t)?'factory':'desert'][2]
O06937 L25:141 c.lineWidth=Math.max(1.2,r*.035)
O06939 L25:186 k<6
O06940 L25:190 k++
O06941 L25:203 k*TAU/6
O06943 L25:257 -r*.72
O06945 L25:274 r*.11
O06946 L25:280 -r*.99
O06948 L25:297 -r*.08
O06950 L25:304 -r*1.18
O06952 L26:97 t.includes('garden')||t.includes('jungle')
O06953 L27:14 c.strokeStyle='#435d65'
O06954 L27:38 c.lineWidth=r*.18
O06956 L27:75 -1
O06957 L27:105 side*r*.65
O06959 L27:116 -r*.25
O06961 L27:133 side*r*1.07
O06963 L27:145 -r*.05
```

```
O06965 L27:162 side*r*1.02
O06967 L27:174 r*.55
O06968 L27:202 side*r*1.02
O06970 L27:214 r*.55
O06971 L27:220 r*.2
O06972 L27:225 r*.19
O06973 L27:254 r*.13
O06974 L27:260 r*.48
O06975 L27:266 r*.47
O06976 L27:295 r*.13
O06977 L27:301 r*.34
O06978 L27:307 r*.34
O06979 L27:336 r*.13
O06980 L27:342 r*.22
O06981 L27:348 r*.22
O06982 L27:354 m.enraged?'#efb18c':'#89cfc6'
O06983 L28:29 k<7
O06984 L28:33 k++
O06985 L28:46 (k-3)*.32
O06987 L28:56 c.strokeStyle='#addbe2'
O06988 L28:80 c.lineWidth=Math.max(2,r*.06)
O06990 L28:133 Math.sin(a)*r*.7
O06992 L28:150 r*.5
O06993 L28:175 Math.sin(a)*r*1.3+w*r
O06997 L28:197 r*.98
O06998 L28:203 Math.sin(a)*r*1.2
O07000 L28:221 r*1.4
O07001 L28:251 r*.2
O07002 L28:256 r*.44
O07003 L28:262 r*.22
O07004 L29:37 -1
O07005 L29:55 k<3
O07006 L29:59 k++
O07007 L29:73 side*r*(.65+k*.12)
O07011 L29:92 r*(k*.25-.35)
O07014 L29:106 r*.25
O07015 L29:112 r*.12
O07016 L29:118 k%2?'#8caf74':'#c4cf8c'
O07018 L29:142 side*(.5+k*.4)
O07021 L29:170 r*.13
O07022 L29:176 r*.35
O07023 L29:182 r*.28
O07024 L29:199 c.strokeStyle='#e8d4a1'
O07025 L29:223 c.lineWidth=Math.max(1,r*.025)
O07027 L29:279 -r*.1
O07029 L29:297 r*.4
O07030 L29:312 -r*.23
O07032 L29:319 r*.03
O07033 L29:337 r*.15
O07034 L29:353 r*.23
O07035 L29:359 r*.03
O07036 L30:7 c.strokeStyle=t.includes('hel')?'#edb180':'#dbc79f'
O07038 L30:59 c.lineWidth=Math.max(2,r*.06)
O07040 L30:108 -1
O07041 L30:138 side*r*.5
O07043 L30:148 -r*.5
O07045 L30:164 side*r*.8
O07047 L30:174 -r*.85
O07049 L30:191 side*r*.95
O07051 L30:202 -r*.52
```

O07053 L30:233 r*.18
O07054 L30:239 r*.28
O07055 L30:245 r*.3
O07056 L30:250 t.includes('hel')?'#c47d66':'#c9b38a'
O07057 L32:49 body.rx*.90
O07058 L32:61 body.ry*.90
O07059 L32:91 c.globalAlpha=.32
O07060 L33:46 k<5
O07061 L33:50 k++
O07062 L33:63 k*2.4
O07063 L33:79 Math.sin(a)*r*.45
O07065 L33:97 Math.cos(a)*r*.4
O07067 L33:114 r*(.06+k*.011)
O07070 L33:129 r*(.08+k*.01)
O07073 L33:166 -r*.28
O07075 L33:173 -r*.18
O07077 L33:180 r*.16
O07078 L33:186 r*.32
O07079 L34:27 c.strokeStyle='#efd8a7'
O07080 L34:51 c.lineWidth=Math.max(1,r*.03)
O07082 L34:104 -r*.28
O07084 L34:125 r*.14
O07085 L34:141 r*.28
O07086 L34:161 -r*.04
O07088 L34:180 r*.4
O07089 L34:216 -1
O07090 L34:232 side*r*.39
O07092 L34:243 -r*.38
O07094 L34:250 r*.045
O07095 L34:257 r*.05
O07096 L35:54 k<5
O07097 L35:58 k++
O07098 L35:74 (-.4+k*.20)*r
O07102 L35:88 body.rx*.82
O07103 L35:100 r*.05
O07104 L36:34 c.strokeStyle='#f0ddb3'
O07105 L36:58 c.lineWidth=Math.max(1,r*.025)
O07107 L36:108 -1
O07108 L36:126 k<3
O07109 L36:130 k++
O07110 L36:158 side*r*.15
O07112 L36:169 -r*.2+k*r*.26
O07117 L36:203 side*r*.58
O07119 L36:214 r*.03+k*r*.23
O07123 L36:228 side*r*.44
O07125 L36:239 r*.26+k*r*.18
O07129 L37:39 c.strokeStyle='#f5e2ac'
O07130 L37:63 c.lineWidth=Math.max(1,r*.035)
O07132 L37:104 -1
O07133 L37:107 k<=1
O07134 L37:112 k++
O07135 L37:140 k*r*.2
O07137 L37:147 -r*.34
O07139 L37:164 k*r*.2
O07141 L37:171 r*.3
O07142 L39:4 m.frostUntil>time
O07143 L39:23 c.strokeStyle='#b2f2ff'
O07144 L39:47 c.lineWidth=Math.max(1,r*.03)
O07146 L39:105 body.rx+3
O07147 L39:115 body.ry+3

```
O07148 L39:157 k<3
O07149 L39:161 k++
O07150 L39:175 Math.cos(k*2.2)*r*.7
O07153 L39:196 Math.sin(k*2.2)*r*.7
```

F0621 | lighting

```
O07156 L40:60 palettes[zone]||palettes.garden
O07157 L40:167 p[2]+'18'
O07158 L40:197 p[0]+'00'
O07159 L40:225 p[1]+'20'
O07160 L40:236 c.fillStyle=g
O07161 L40:313 width*.48
O07162 L40:323 height*.44
O07163 L40:334 Math.min(width,height)*.22
O07164 L40:361 width*.5
O07165 L40:370 height*.5
O07166 L40:380 Math.max(width,height)*.64
O07167 L40:461 p[1]+'28'
O07168 L40:472 c.fillStyle=edge
O07169 L40:521 zone==='ocean'||zone==='jungle'
O07172 L40:554 c.globalAlpha=.045
O07173 L40:573 c.fillStyle='#fff9d6'
O07174 L40:607 reduced?0:Math.sin(time*.12)*15
O07177 L40:651 i<3
O07178 L40:655 i++
O07179 L40:683 width*(i*.39)+phase-90
O07183 L40:718 width*(i*.39)+phase-15
O07187 L40:753 width*(i*.39)+phase+160
O07191 L40:794 width*(i*.39)+phase+60
```

F0622 | rock

```
O07195 L41:38 palettes[zone]||palettes.garden
O07196 L41:124 r*.96
O07197 L41:177 -r
O07198 L41:180 -r
O07199 L41:205 p[2]+'38'
O07200 L41:234 p[0]+'00'
O07201 L41:262 p[1]+'45'
O07202 L41:273 c.fillStyle=g
O07203 L41:298 -r
O07204 L41:301 -r
O07205 L41:304 r*2
O07206 L41:308 r*2
O07207 L41:313 c.globalAlpha=.38
O07208 L41:331 c.strokeStyle=p[2]
O07209 L41:350 c.lineWidth=.9
O07210 L41:408 k<5
O07211 L41:412 k++
O07212 L41:425 k*2.4+o.a
O07214 L41:458 Math.cos(a)*r*.15
O07216 L41:476 Math.sin(a)*r*.15
O07218 L41:514 Math.cos(a+.2)*r*.4
O07221 L41:534 Math.sin(a+.2)*r*.4
O07224 L41:554 Math.cos(a)*r*.74
O07226 L41:572 Math.sin(a)*r*.74
O07228 L41:611 zone==='factory'||zone==='sewer'
O07231 L41:655 -1
O07232 L41:658 k<=1
O07233 L41:663 k++
```

```
O07234 L41:691 -r*.6
O07236 L41:697 k*r*.36
O07238 L41:715 r*.6
O07239 L41:720 k*r*.36
O07241 L41:758 k<4
O07242 L41:762 k++
O07243 L41:775 k*1.8+o.a
O07245 L41:795 Math.cos(a)*r*.6
O07247 L41:812 Math.sin(a)*r*.6
O07249 L41:829 r*.06
O07250 L41:835 r*.04
```

F0623 | traits

```
O07251 L42:156 t==='bee'||/wasp|dragonfly|mosquito|drone/.test(t)
```

F0624 | surface

```
O07253 L43:42 !detail
O07254 L43:91 palettes[zone]||palettes.garden
O07255 L43:160 body.rx*.93
O07256 L43:172 body.ry*.93
O07257 L43:202 c.lineWidth=Math.max(.7,r*.016)
O07259 L43:234 c.strokeStyle=p[2]+'42'
O07261 L43:258 c.lineCap='round'
O07262 L44:61 -1
O07263 L44:91 side*r*.12
O07265 L44:102 -r*.55
O07267 L44:119 side*r*.48
O07269 L44:130 -r*.34
O07271 L44:147 side*r*.48
O07273 L44:158 r*.31
O07274 L44:174 side*r*.15
O07276 L44:185 r*.53
O07277 L44:215 k<3
O07278 L44:219 k++
O07279 L44:233 r*.36
O07280 L44:239 (-.15+k*.16)*r
O07284 L44:254 r*.025
O07285 L44:261 r*.025
O07286 L45:45 c.globalAlpha=.35
O07287 L45:75 k<5
O07288 L45:79 k++
O07289 L45:92 k*2.399
O07290 L45:110 Math.cos(a)*r*.52
O07292 L45:128 Math.sin(a)*r*.47
O07294 L45:146 r*(.025+k*.005)
O07297 L45:162 r*(.04+k*.006)
O07300 L46:25 -1
O07301 L46:43 k<4
O07302 L46:47 k++
O07303 L46:60 (-.47+k*.29)*r
O07307 L46:98 side*r*.10
O07309 L46:128 side*r*.29
O07311 L46:139 y-r*.03
O07313 L46:147 side*r*.56
O07315 L46:158 y+r*.03
O07317 L46:166 side*body.rx*.86
O07319 L46:183 y+r*.15
O07321 L47:34 -r
O07322 L47:37 -r
```

```
O07323 L47:142 c.fillStyle=rim
O07324 L47:169 -r
O07325 L47:172 -r
O07326 L47:175 2*r
O07327 L47:179 2*r
```

F0625 | status

```
O07328 L48:45 m.frostUntil>time
O07329 L48:93 m.poisonTime>0
O07330 L48:138 m.shieldUntil>time
O07331 L48:187 !states.length
O07332 L48:239 c.lineWidth=1.7
O07333 L48:285 c.strokeStyle=color
O07334 L48:313 m.r+6+k*3
O07337 L48:349 j<3
O07338 L48:353 j++
O07339 L48:367 Math.cos(j*TAU/3+.12)*r
O07343 L48:391 Math.sin(j*TAU/3+.12)*r
O07347 L48:428 j*TAU/3+.12
O07350 L48:440 j*TAU/3+1.35
O07353 L48:478 j<=k
O07354 L48:483 j++
O07355 L48:488 c.fillStyle=color
O07356 L48:517 -4+j*4
O07359 L48:524 -r-4
```

F0626 | telegraph

```
O07361 L49:47 !(m.windup>0)||!profile
O07365 L49:133 1-m.windup/profile.windup
O07367 L49:238 type==='venom'?'#bedc93':type==='shield'?'#c6d5ed':'#f0c28d'
O07371 L49:329 c.strokeStyle=color
O07372 L49:349 c.lineWidth=1.6
O07373 L49:365 c.fillStyle=color+'12'
O07375 L50:4 type==='nova'
O07376 L50:163 -Math.PI/2
O07378 L50:174 -Math.PI/2+TAU*progress
O07382 L50:199 c.lineWidth=3
O07383 L51:9 type==='charger'
O07384 L51:71 -m.r
O07385 L51:76 profile.reach+m.r
O07386 L51:94 m.r*2
O07387 L51:187 profile.reach-8
O07388 L51:203 -5
O07389 L51:242 profile.reach-8
O07390 L51:281 -angle
O07391 L52:9 type==='sniper'||type==='venom'
O07394 L52:154 profile.reach+m.r
O07395 L52:196 c.globalAlpha=.48+progress*.38
O07398 L52:259 -angle
O07399 L53:1 c.globalAlpha=1
O07400 L53:59 m.r+10
O07401 L53:66 -Math.PI/2
O07403 L53:77 -Math.PI/2+TAU*progress
O07407 L53:102 c.lineWidth=2.6
O07408 L53:129 c.fillStyle='#18352fe8'
O07409 L53:165 m.r+18
O07410 L53:190 c.fillStyle=color
O07411 L53:208 c.font='bold 10px system-ui'
O07412 L53:237 c.textAlign='center'
```

```
O07413 L53:269 type==='summon'?'+':type==='shield'?"LOCALIZED_f88c1b77":'!'
O07417 L53:315 m.r+21
```

F0614 | anonymous callback

```
O07418 L54:1 globalThis.PetalArt=Object.freeze({palettes,terrain,landmark,anatomy,lighting,halo,telegraph,rock,surface,
status,traits})
```

Appendix C. Numerical call ledger

All 569 identified numeric/probability helper call sites, including calls that do not appear within an operator envelope. Duplicates with Appendix B are intentional, so no standalone numerical function call is silently omitted. All ordinary call sites are in Appendix D. Platform method internals are external contracts.

```
C00007 F0003 L241 Math.max(a,Math.min(b,v))
C00008 F0003 L241 Math.min(b,v)
C00009 F0005 L241 Math.hypot(a-c,b-d)
C00011 F0008 L242 Math.imul(n^(n>>>15),1|n)
C00012 F0008 L242 Math.imul(t^(t>>>7),61|t)
C00014 F0009 L243 Math.pow(1.72,i)
C00015 F0009 L243 Math.pow(2,i)
C00041 F0010 L305 Math.max(0,Math.floor((level-1)/2))
C00042 F0010 L305 Math.floor((level-1)/2)
C00051 F0011 L306 Math.max(0,Math.min(TALENTS[k].max,Math.floor(n)))
C00052 F0011 L306 Math.min(TALENTS[k].max,Math.floor(n))
C00053 F0011 L306 Math.floor(n)
C00054 F0011 L306 Math.min(v,budget)
C00056 F0014 L308 Math.max(0,talentEarned(game.p?.level||1)-Object.values(game.talents||{}).reduce((sum,n)=>sum+n,0))
C00062 F0018 L311 Math.min(.35,talentRank('reload')*.05)
C00067 F0019 L312 Math.min(maxHP(),old+(maxHP()-oldHP))
C00076 F0020 L313 Math.min(game.p.hp,maxHP())
C00082 F0021 L315 Math.max(0,shrine.readyAt-game.time)
C00084 F0022 L316 Math.min(maxHP(),game.p.hp+maxHP()*.30)
C00091 F0023 L317 dist(game.p.x,game.p.y,shrine.x,shrine.y)
C00149 F0036 L379 Math.min(raw[k],Date.now()+86400000)
C00151 F0037 L380 Math.max(0,Math.ceil(((mailTimers[action]||0)-Date.now())/1000))
C00152 F0037 L380 Math.ceil(((mailTimers[action]||0)-Date.now())/1000)
C00154 F0038 L381 Math.max(mailTimers[action]||0,Date.now()+Math.min(86400,Math.max(1,seconds))*1000)
C00156 F0038 L381 Math.min(86400,Math.max(1,seconds))
C00157 F0038 L381 Math.max(1,seconds)
C00171 F0040 L393 Math.max(authMode==='register'?signup:0,reset)
C00198 F0042 L412 Math.min(86400,Math.max(0,Math.ceil(seconds)))
C00199 F0042 L412 Math.max(0,Math.ceil(seconds))
C00200 F0042 L412 Math.ceil(seconds)
C00203 F0043 L415 Math.max(30,a.expires_in||3600)
C00208 F0044 L418 Math.max(30,a.expires_in||3600)
C00240 F0049 L439 Math.min(30000,2000*2**(cloudRetryCount-1))
C00350 F0056 L540 Math.floor((cloudSession.expires-Date.now())/1000)
C00441 F0074 L619 Math.min(86400,Math.max(0,Number(response.headers.get('Retry-After'))||0))
C00442 F0074 L619 Math.max(0,Number(response.headers.get('Retry-After'))||0)
C00479 F0079 L646 Math.max(e.retryAfter*1000,Math.min(30000,2000*2**cfRetry))
C00480 F0079 L646 Math.min(30000,2000*2**cfRetry)
C00781 F0120 L839 clamp((game.bags[k]||0)+n,0,99999)
C00784 F0123 L846 Math.min(1.75,1+.05*Math.floor(lvl/10))
C00785 F0123 L846 Math.floor(lvl/10)
C00788 F0127 L846 clamp(viewScale(game.p.level)/antennaMultiplier(),.38,1)
C00791 F0128 L847 Math.hypot((x-cx)/cx,(y-cy)/cy)
C00793 F0131 L850 Math.round((100+Math.floor((game.p.level-1)*7)+game.slots.reduce((n,s)=>n+(s.t==='cactus'?24*rarity[s
.r].scale:s.t==='soil'?42*rarity[s.r].scale:0),0))*talentModifier('vitality'))
C00794 F0131 L850 Math.floor((game.p.level-1)*7)
C00799 F0136 L853 Math.max(.55,Math.pow(.89,soilCount()))
C00800 F0136 L853 Math.pow(.89,soilCount())
C00809 F0145 L866 clamp(e.hp+max*.06*dt,0,max)
C00810 F0146 L868 Math.sin(game.time*5+i*.7+Math.PI/6)
C00818 F0152 L888 dist(m.x,m.y,p.x,p.y)
C00826 F0152 L889 Math.cos(a)
```

```
C00827 F0152 L889 Math.sin(a)
C00828 F0152 L889 Math.cos(a)
C00829 F0152 L889 Math.sin(a)
C00833 F0152 L891 Math.round(maxHP()*(.13+.06*syner('nature')))
C00836 F0152 L891 Math.min(maxHP(),p.hp+value)
C00842 F0153 L895 Math.max(0,(game.skills[name]||0)-game.time)
C00848 F0156 L898 Math.round(tierBoost*2)
C00849 F0156 L898 Math.ceil(m.hp*(1+tierBoost*.15))
C00850 F0156 L898 Math.ceil(m.dmg*(1+tierBoost*.10))
C00852 F0157 L900 Math.min(7,130-game.mapData[game.zone].living.length)
C00856 F0157 L901 Math.floor((game.random?.world||Math.random)()*bag.length)
C00861 F0157 L901 clamp(game.p.x+Math.cos(a)*d,55,W-55)
C00862 F0157 L901 Math.cos(a)
C00863 F0157 L901 clamp(game.p.y+Math.sin(a)*d,55,H-55)
C00864 F0157 L901 Math.sin(a)
C00865 F0157 L901 Math.min(6,Math.max(1,zones[game.zone].drop+Math.floor((game.random?.world||Math.random)()*3)))
C00866 F0157 L901 Math.max(1,zones[game.zone].drop+Math.floor((game.random?.world||Math.random)()*3))
C00867 F0157 L901 Math.floor((game.random?.world||Math.random)()*3)
C00872 F0159 L904 Math.min(maxHP(),game.p.hp+dt*2.2)
C00878 F0159 L907 clamp(game.p.x+Math.cos(a)*d,60,W-60)
C00879 F0159 L907 Math.cos(a)
C00880 F0159 L907 clamp(game.p.y+Math.sin(a)*d,60,H-60)
C00881 F0159 L907 Math.sin(a)
C00883 F0159 L909 dist(impact.x,impact.y,game.p.x,game.p.y)
C00886 F0161 L911 globalThis.PetalRules.specialProfile({type:m.specialType||traitOf(m.t),rank:m.rank||0,level:m.level||
1,speed:m.speed||0,enraged:!!m.enraged,frostFactor:m.frostUntil>game.time?m.frostFactor:1})
C00888 F0162 L912 Math.max(mobs[m.t].guardian?4.2:5.2,10.3-(m.rank||0)*.45-(mobs[m.t].guardian?1.8:0))
C00889 F0163 L913 specialProfileFor({...m,specialType:trait})
C00890 F0163 L913 Math.atan2(dy,dx)
C00892 F0163 L913 Math.cos(a)
C00893 F0163 L913 Math.sin(a)
C00894 F0163 L913 Math.cos(a)
C00895 F0163 L913 Math.sin(a)
C00896 F0164 L915 specialProfileFor(m)
C00897 F0164 L915 Math.hypot(dx,dy)
C00898 F0164 L916 globalThis.PetalRules.specialAim(m,p)
C00900 F0164 L918 Math.min(3,125-map.living.length)
C00901 F0164 L918 clamp(m.x+Math.cos(a)*95,70,W-70)
C00902 F0164 L918 Math.cos(a)
C00903 F0164 L918 clamp(m.y+Math.sin(a)*95,70,H-70)
C00904 F0164 L918 Math.sin(a)
C00907 F0165 L918 dist(xx,yy,o.x,o.y)
C00908 F0164 L918 Math.ceil(def.hp*(1+.045*m.level))
C00910 F0164 L918 Math.min(6,m.rank)
C00911 F0164 L918 Math.max(1,m.level-3)
C00912 F0164 L918 Math.ceil(def.dmg*(1+.045*m.level))
C00920 F0166 L925 specialProfileFor(m)
C00948 F0173 L932 clamp(Math.floor(+a.p?.level||1),1,999)
C00949 F0173 L932 Math.floor(+a.p?.level||1)
C00950 F0173 L932 clamp(Math.floor(+a.p?.xp||0),0,1e9)
C00951 F0173 L932 Math.floor(+a.p?.xp||0)
C00952 F0173 L932 clamp(Math.floor(+a.p?.kills||0),0,1e9)
C00953 F0173 L932 Math.floor(+a.p?.kills||0)
C00959 F0173 L932 clamp(Math.floor(v),0,99999)
C00960 F0173 L932 Math.floor(v)
C00966 F0173 L936 clamp(Math.floor(Number(j?.kills)||0),0,1e8)
C00967 F0173 L936 Math.floor(Number(j?.kills)||0)
C00970 F0173 L936 clamp(Math.floor(Number(v)||0),0,1e6)
C00971 F0173 L936 Math.floor(Number(v)||0)
C00981 F0173 L937 clamp(Math.floor(Number(d.kills)||0),0,100000000)
```

```
C00982 F0173 L937 Math.floor(Number(d.kills)||0)
C00984 F0173 L937 clamp(Math.floor(Number(d.maxLevel)||0),0,999)
C00985 F0173 L937 Math.floor(Number(d.maxLevel)||0)
C00989 F0173 L938 clamp(+a.p?.hp||100,1,maxHP())
C01018 F0177 L940 Math.floor(r()*5)
C01023 F0177 L941 dist(x,y,cx,cy)
C01025 F0179 L941 dist(x,y,p.x,p.y)
C01027 F0180 L941 dist(x,y,o.x,o.y)
C01029 F0177 L941 Math.floor(r()*3)
C01032 F0177 L942 Math.floor(o.x/GRID_SIZE)
C01033 F0177 L942 Math.floor(o.y/GRID_SIZE)
C01042 F0181 L943 dist(sx,sy,o.x,o.y)
C01044 F0182 L943 dist(sx,sy,o.x,o.y)
C01045 F0177 L943 dist(sx,sy,cx,cy)
C01047 F0183 L943 dist(sx,sy,o.x,o.y)
C01056 F0185 L944 Math.floor(r()*z.mob.length)
C01059 F0185 L946 clamp(game.p.x+rnd(-980,980),100,W-100)
C01060 F0185 L946 rnd(-980,980)
C01062 F0185 L946 clamp(game.p.y+rnd(-760,760),100,H-100)
C01063 F0185 L946 rnd(-760,760)
C01065 F0185 L947 dist(x,y,game.p?.x??cx,game.p?.y??cy)
C01069 F0187 L948 dist(x,y,o.x,o.y)
C01071 F0188 L948 dist(x,y,o.x,o.y)
C01073 F0189 L948 dist(x,y,o.x,o.y)
C01077 F0185 L949 Math.floor(r()*4)
C01081 F0185 L950 Math.round(data.hp*mul)
C01082 F0185 L950 Math.round(data.hp*mul)
C01083 F0185 L950 Math.round(data.dmg*(.84+level*.095)*(elite?1.20:1)*(superSpawn?1.7:1))
C01091 F0190 L954 dist(x,y,cx,cy)
C01093 F0191 L954 dist(x,y,p.x,p.y)
C01100 F0193 L959 Math.floor(x/GRID_SIZE)
C01101 F0193 L959 Math.floor(y/GRID_SIZE)
C01104 F0194 L960 clamp(a.x+dx,r+27,W-r-27)
C01105 F0194 L960 clamp(a.y+dy,r+27,H-r-27)
C01107 F0194 L960 Math.hypot(vx,vy)
C01108 F0194 L960 clamp(nx,r+27,W-r-27)
C01109 F0194 L960 clamp(ny,r+27,H-r-27)
C01111 F0195 L961 Math.hypot(dx,dy)
C01112 F0195 L961 Math.cos(game.p.angle)
C01113 F0195 L961 Math.sin(game.p.angle)
C01114 F0195 L961 Math.hypot(dx,dy)
C01117 F0196 L962 Math.min(game.p.hp,maxHP())
C01120 F0196 L962 Math.max(2.5,petalReloadTime(e.t))
C01128 F0197 L964 Math.max(1,protectedDamage-rootedArmor())
C01132 F0197 L964 Math.ceil(protectedDamage)
C01134 F0198 L965 Math.max(s.maxLevel,m.level||1)
C01135 F0199 L966 Math.min(6,m.band||0)
C01137 F0199 L966 Math.max(0,rank-(r<.73?1:r<.94?2:3))
C01138 F0200 L966 Math.min(14,(m.rank||0)*1.25+(m.tier||0)*2)
C01139 F0201 L966 armorOf(m)
C01140 F0201 L966 Math.max(1,dmg-armor)
C01142 F0201 L966 Math.round(dmg)
C01147 F0201 L966 Math.max(1,Math.round(baseXP*(1+combo.bonus)*(game.worldEvent?.type==='bloodmoon'?1.28:1)))
C01148 F0201 L966 Math.round(baseXP*(1+combo.bonus)*(game.worldEvent?.type==='bloodmoon'?1.28:1))
C01150 F0201 L966 Math.round(combo.bonus*100)
C01155 F0201 L966 clamp(game.p.hp+22,0,maxHP())
C01159 F0201 L966 Math.round(1/game.zoom*100)
C01161 F0201 L966 Math.floor((game.random?.loot||Math.random)()*mob.drop.length)
C01165 F0201 L966 Math.min(7,rank+1)
C01169 F0201 L966 Math.max(1,rank-1)
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C01174 F0201 L966 Math.max(0,rank-2)
C01177 F0202 L967 Math.min(100,c.count+1)
C01178 F0202 L967 Math.min(.5,Math.max(0,c.count-1)*.05)
C01179 F0202 L967 Math.max(0,c.count-1)
C01180 F0203 L968 Math.floor(lvl*18+lvl*lvl*.9)
C01186 F0205 L970 Math.cos(a)
C01187 F0205 L970 Math.sin(a)
C01210 F0208 L973 Math.hypot(mx,my)
C01211 F0208 L973 clamp((d-53)/95,0,1)
C01212 F0208 L973 clamp((d-53)/95,0,1)
C01213 F0208 L973 Math.hypot(dx,dy)
C01222 F0211 L991 Math.hypot(dx,dy)
C01223 F0211 L992 Math.max(o.rx,o.ry)
C01224 F0211 L993 Math.cos(o.angle)
C01225 F0211 L993 Math.sin(o.angle)
C01226 F0211 L997 Math.hypot(o.rx*u,o.ry*v)
C01228 F0212 L1004 Math.max(a,b)
C01229 F0212 L1006 Math.cos(shape.angle)
C01230 F0212 L1006 Math.sin(shape.angle)
C01231 F0212 L1007 Math.abs(dx*cs+dy*sn)
C01232 F0212 L1007 Math.abs(-dx*sn+dy*cs)
C01233 F0212 L1013 Math.max(a*px,b*py)
C01235 F0213 L1024 Math.max(c.rx,c.ry)
C01236 F0214 L1027 Math.atan2(vy,vx)
C01238 F0214 L1029 Math.atan2(Math.sin(goal-current),Math.cos(goal-current))
C01239 F0214 L1029 Math.sin(goal-current)
C01240 F0214 L1029 Math.cos(goal-current)
C01241 F0214 L1030 Math.min(1,dt*9)
C01242 F0215 L1032 Math.cos(m.a)
C01243 F0215 L1032 Math.sin(m.a)
C01248 F0219 L1035 clamp(e.hp/(def.hp*scale),0,1)
C01251 F0220 L1036 globalThis.PetalRules.criticalFactor(activeCopies('prism'),game.random.combat)
C01252 F0220 L1036 activeCopies('prism')
C01253 F0222 L1038 Math.max(0,game.portalCool-dt)
C01258 F0222 L1038 Math.max(0,p.dashCooldown-dt)
C01259 F0222 L1038 Math.max(0,p.dashIFrames-dt)
C01260 F0222 L1038 Math.max(0,game.deckCooldown-dt)
C01270 F0222 L1039 Math.max(0,p.glow-dt)
C01271 F0222 L1039 Math.max(0,p.hp-dt*(2.6+zones[game.zone].drop))
C01275 F0222 L1040 Math.min(10,p.armorLayers+1)
C01276 F0222 L1042 dist(mob.x,mob.y,p.x,p.y)
C01277 F0222 L1042 Math.max(520,game.width/game.zoom*.6)
C01279 F0222 L1042 Math.max(0,mob.hit-dt)
C01281 F0222 L1042 Math.max(1,d)
C01282 F0222 L1042 Math.max(1,d)
C01284 F0222 L1042 Math.atan2(dy,dx)
C01286 F0222 L1042 Math.cos(a+off)
C01287 F0222 L1042 Math.sin(a+off)
C01288 F0222 L1042 rnd(1.65,2.85)
C01290 F0222 L1042 rnd(1.2,2.8)
C01291 F0222 L1042 rnd(0,TAU)
C01292 F0222 L1042 rnd(.7,2.4)
C01293 F0222 L1042 Math.cos(mob.a)
C01294 F0222 L1042 Math.sin(mob.a)
C01295 F0222 L1042 Math.max(0,mob.chargeFor-dt)
C01300 F0222 L1044 Math.hypot(pdx,pdy)
C01303 F0222 L1044 Math.hypot(dx,dy)
C01304 F0222 L1044 Math.max(.5,min-d)
C01308 F0222 L1044 Math.hypot(dx,dy)
C01313 F0222 L1046 Math.hypot(dx,dy)
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C01318 F0222 L1047 Math.cos(angle)
C01319 F0222 L1047 Math.sin(angle)
C01320 F0222 L1047 lerp(e.x,tx,globalThis.PetalRules.frameAlpha(.25,dt))
C01321 F0222 L1047 globalThis.PetalRules.frameAlpha(.25,dt)
C01322 F0222 L1047 lerp(e.y,ty,globalThis.PetalRules.frameAlpha(.25,dt))
C01323 F0222 L1047 globalThis.PetalRules.frameAlpha(.25,dt)
C01327 F0222 L1047 dist(e.x,e.y,o.x,o.y)
C01333 F0222 L1049 criticalMultiplier()
C01334 F0222 L1049 rawPetalDamage(e)
C01336 F0222 L1049 Math.max(.5,m.dmg)
C01340 F0222 L1049 Math.round(dealt)
C01341 F0222 L1049 clamp(p.hp+Math.max(1,dealt*.07),0,maxHP())
C01342 F0222 L1049 Math.max(1,dealt*.07)
C01344 F0222 L1049 dist(a.x,a.y,m.x,m.y)
C01346 F0222 L1049 globalThis.PetalRules.slowFactor(Math.max(1,activeCopies('frost')),!!mobs[m.t].guardian)
C01347 F0222 L1049 Math.max(1,activeCopies('frost'))
C01348 F0222 L1049 activeCopies('frost')
C01349 F0222 L1049 Math.hypot(m.x-p.x,m.y-p.y)
C01351 F0222 L1050 dist(e.x,e.y,m.x,m.y)
C01352 F0222 L1050 Math.atan2(aim.y-e.y,aim.x-e.x)
C01354 F0222 L1050 Math.cos(a)
C01355 F0222 L1050 Math.sin(a)
C01360 F0222 L1052 Math.min(maxHP(),p.hp+dt*.6*syner('nature'))
C01365 F0222 L1055 Math.min(maxHP(),p.hp+dt*1.9*rs)
C01368 F0222 L1056 Math.min(maxHP(),p.hp+dt*3.1*rs)
C01370 F0222 L1057 Math.max(0,e.healWait-dt)
C01372 F0222 L1057 Math.round((e.t==='rose'?19:35)*rs)
C01373 F0222 L1057 Math.min(maxHP()-p.hp,heal)
C01376 F0222 L1057 Math.round(gain)
C01380 F0223 L1058 dist(b.x,b.y,o.x,o.y)
C01381 F0222 L1058 dist(b.x,b.y,e.x,e.y)
C01384 F0222 L1058 dist(b.x,b.y,p.x,p.y)
C01389 F0222 L1059 Math.hypot(dx,dy)
C01392 F0225 L1059 dist(shot.x,shot.y,o.x,o.y)
C01394 F0222 L1059 dist(shot.x,shot.y,m.x,m.y)
C01398 F0222 L1059 dist(shot.x,shot.y,n.x,n.y)
C01399 F0222 L1059 Math.atan2(next.y-shot.y,next.x-shot.x)
C01400 F0222 L1059 Math.cos(a)
C01401 F0222 L1059 Math.sin(a)
C01403 F0222 L1060 dist(drop.x,drop.y,p.x,p.y)
C01405 F0222 L1060 clamp((reach-pd)/Math.max(reach,1),.14,.8)
C01406 F0222 L1060 Math.max(reach,1)
C01407 F0222 L1060 dist(drop.x,drop.y,p.x,p.y)
C01417 F0222 L1063 lerp(game.camera.x,clamp(p.x-game.width/game.zoom/2,0,Math.max(0,W-game.width/game.zoom)),globalThis.PetalRules.frameAlpha(8/60,dt))
C01418 F0222 L1063 clamp(p.x-game.width/game.zoom/2,0,Math.max(0,W-game.width/game.zoom))
C01419 F0222 L1063 Math.max(0,W-game.width/game.zoom)
C01420 F0222 L1063 globalThis.PetalRules.frameAlpha(8/60,dt)
C01421 F0222 L1063 lerp(game.camera.y,clamp(p.y-game.height/game.zoom/2,0,Math.max(0,H-game.height/game.zoom)),globalThis.PetalRules.frameAlpha(8/60,dt))
C01422 F0222 L1063 clamp(p.y-game.height/game.zoom/2,0,Math.max(0,H-game.height/game.zoom))
C01423 F0222 L1063 Math.max(0,H-game.height/game.zoom)
C01424 F0222 L1063 globalThis.PetalRules.frameAlpha(8/60,dt)
C01431 F0233 L1066 dist(portal.x,portal.y,game.p.x,game.p.y)
C01438 F0233 L1066 clamp(game.portalTimer/.9*100,0,100)
C01451 F0237 L1070 Math.sin(game.time*2+shrine.x)
C01456 F0237 L1070 Math.ceil(shrineCooldown(shrine))
C01466 F0238 L1071 Math.floor(game.camera.x/84)
C01467 F0238 L1071 Math.min(W,game.camera.x+game.width/game.zoom+84)
C01469 F0238 L1071 Math.max(0,game.camera.y)

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C01471 F0238 L1071 Math.min(H,game.camera.y+game.height/game.zoom)
C01472 F0238 L1071 Math.floor(game.camera.y/84)
C01473 F0238 L1071 Math.min(H,game.camera.y+game.height/game.zoom+84)
C01475 F0238 L1071 Math.max(0,game.camera.x)
C01477 F0238 L1071 Math.min(W,game.camera.x+game.width/game.zoom)
C01500 F0238 L1076 dist(m.x,m.y,game.p.x,game.p.y)
C01517 F0238 L1080 clamp(f.life/f.max,0,1)
C01519 F0238 L1080 clamp(f.life/.85,0,1)
C01531 F0239 L1081 clamp(cx+cx*rho,140,W-160)
C01573 F0241 L1083 Math.cos(a)
C01574 F0241 L1083 Math.sin(a)
C01578 F0241 L1083 Math.cos(a)
C01579 F0241 L1083 Math.sin(a)
C01601 F0242 L1084 Math.sin(game.time*3+p.x)
C01606 F0242 L1084 Math.cos(a)
C01607 F0242 L1084 Math.sin(a)
C01619 F0243 L1085 Math.sin(game.time*3+l.ang)
C01630 F0244 L1086 Math.min(p.armorLayers,5)
C01648 F0244 L1086 clamp(p.hp/maxHP(),0,1)
C01776 F0248 L1101 clamp(amount,-1,1)
C01777 F0248 L1101 Math.abs(a)
C01780 F0249 L1101 Math.round(c+(target-c)*f).toString(16).padStart(2,'0')
C01781 F0249 L1101 Math.round(c+(target-c)*f).toString(16)
C01782 F0249 L1101 Math.round(c+(target-c)*f)
C01813 F0252 L1113 Math.sin(game.time*(3+m.speed*.025)+(m.id||0)*.73)
C01814 F0252 L1113 clamp(m.r*.055,1.05,3.5)
C01861 F0253 L1122 Math.cos(a)
C01862 F0253 L1122 Math.sin(a)
C01864 F0253 L1122 Math.cos(a)
C01865 F0253 L1122 Math.sin(a)
C01938 F0253 L1137 clamp(r*.105,1.5,5.3)
C01961 F0254 L1144 specialProfileFor(m)
C01962 F0254 L1145 dist(m.x,m.y,game.p.x,game.p.y)
C01965 F0254 L1146 clamp(m.hp/m.max,0,1)
C01966 F0254 L1146 clamp(r*.12,3,6)
C01971 F0254 L1147 Math.max(0,(r*2-2)*pct)
C01972 F0254 L1147 Math.max(1,h-2)
C01975 F0254 L1150 clamp(r*.32,10,14)
C01978 F0254 L1150 Math.round(m.dmg*(m.enraged?1.20:1))
C01985 F0255 L1153 Math.atan2(dy,dx)
C01986 F0255 L1153 clamp(x,pad,game.width-pad)
C01987 F0255 L1153 clamp(y,pad,game.height-pad)
C02016 F0258 L1157 Math.sin(game.time*5)
C02036 F0259 L1158 dist(p.x,p.y,m.x,m.y)
C02041 F0259 L1158 clamp(boss.hp/boss.max*100,0,100)
C02043 F0259 L1158 Math.ceil(Math.max(0,boss.hp))
C02044 F0259 L1158 Math.max(0,boss.hp)
C02048 F0260 L1159 Math.ceil(r)
C02059 F0260 L1159 Math.ceil(e.until-game.time)
C02069 F0263 L1160 clamp(p.hp/maxHP()*100,0,100)
C02072 F0263 L1160 clamp(p.xp/xpNeed(p.level)*100,0,100)
C02075 F0263 L1160 Math.ceil(p.hp)
C02076 F0263 L1160 Math.ceil(maxHP())
C02079 F0263 L1160 Math.floor(p.xp)
C02084 F0263 L1160 Math.round(9+p.level*.4)
C02093 F0263 L1160 Math.round(100/game.zoom)
C02100 F0263 L1160 Math.ceil(Math.max(0,game.boon.until-game.time))
C02101 F0263 L1160 Math.max(0,game.boon.until-game.time)
C02106 F0263 L1160 Math.min(j.kills,q.goal)
C02107 F0263 L1160 Math.min(100,j.kills/q.goal*100)
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C02118 F0263 L1161 Math.ceil(e.hp/(pets[e.t].hp*rarity[e.r].scale)*100)
C02123 F0263 L1161 clamp(e.cd/(e.reloadTotal||petalReloadTime(e.t))*100,0,100)
C02126 F0263 L1161 clamp(e.hp/(pets[e.t].hp*rarity[e.r].scale)*100,0,100)
C02127 F0263 L1162 dist(m.x,m.y,p.x,p.y)
C02130 F0263 L1162 Math.round(near.dmg)
C02131 F0263 L1162 Math.ceil(near.hp)
C02138 F0265 L1163 Math.round(p.dmg*rarity[s.r].scale)
C02139 F0265 L1163 Math.round(p.hp*rarity[s.r].scale)
C02160 F0271 L1167 Math.min(devicePixelRatio||1,game.settings.quality===='low'?1:game.settings.quality===='high'?2:Math.m
in(2,game.width<850?1.5:2))
C02161 F0271 L1167 Math.min(2,game.width<850?1.5:2)
C02162 F0271 L1167 Math.round(game.width*game.dpr)
C02163 F0271 L1167 Math.round(game.height*game.dpr)
C02165 F0271 L1167 clamp(game.p.x-game.width/game.zoom/2,0,Math.max(0,W-game.width/game.zoom))
C02166 F0271 L1167 Math.max(0,W-game.width/game.zoom)
C02167 F0271 L1167 clamp(game.p.y-game.height/game.zoom/2,0,Math.max(0,H-game.height/game.zoom))
C02168 F0271 L1167 Math.max(0,H-game.height/game.zoom)
C02170 F0272 L1168 Math.min((now-last)/1000,.08)
C02196 F0276 L1172 clamp(game.p.hp,0,maxHP())
C02221 F0285 L1174 Math.hypot(event.clientX-origin.x,event.clientY-origin.y)
C02277 F0298 L1223 Math.min(5,Math.trunc(numeric))
C02278 F0298 L1223 Math.trunc(numeric)
C02281 F0298 L1231 Math.floor((game.random?.craft||Math.random)()*4)
C02326 F0316 L1273 Math.round(d.dmg*scale)
C02327 F0316 L1273 Math.round(d.hp*scale)
C02344 F0323 L1285 globalThis.PetalRules.craftAnalysis(rate,stock,5)
C02352 F0325 L1289 armorOf(m)
C02353 F0325 L1289 activeCopies('prism')
C02354 F0325 L1291 globalThis.PetalRules.damageMoments(rawPetalDamage(e),armor,copies,Math.max(0,m.hp))
C02355 F0325 L1291 rawPetalDamage(e)
C02356 F0325 L1291 Math.max(0,m.hp)
C02363 F0329 L1295 dist(a.x,a.y,game.p.x,game.p.y)
C02364 F0329 L1295 dist(b.x,b.y,game.p.x,game.p.y)
C02365 F0327 L1297
globalThis.PetalRules.rewardStats(spec.drop,m.rank??0,!!spec.guardian,treasure,(_,r)=>rarity[r].scale)
C02366 F0327 L1297 globalThis.PetalRules.targetDropChance(spec.drop,m.rank??0,!!spec.guardian,treasure,goal.t,goal.r)
C02367 F0327 L1298 Math.round(dist(m.x,m.y,game.p.x,game.p.y))
C02368 F0327 L1298 dist(m.x,m.y,game.p.x,game.p.y)
C02370 F0327 L1298 Math.sqrt(stats.variance).toFixed(2)
C02371 F0327 L1298 Math.sqrt(stats.variance)
C02377 F0327 L1303 globalThis.PetalRules.criticalChance(activeCopies('prism'))
C02378 F0327 L1303 activeCopies('prism')
C02380 F0327 L1303 globalThis.PetalRules.slowFactor(activeCopies('frost'))
C02381 F0327 L1303 activeCopies('frost')
C02387 F0333 L1312 Math.max(.1,d.cd)
C02389 F0333 L1313
globalThis.PetalWorkshop.plan(game.bags,game.slots.map(s=>inventoryKey(s.t,s.r)),defs,workshopPolicy)
C02395 F0337 L1318 buildWorkshopPlan()
C02401 F0337 L1324 Math.round(plan.weights[0]*100)
C02402 F0337 L1324 Math.round(plan.weights[1]*100)
C02403 F0337 L1324 Math.round(plan.weights[2]*100)
C02411 F0337 L1326 workshopIdentity()
C02414 F0342 L1328 workshopIdentity()
C02420 F0342 L1329 Math.min(game.p.hp,maxHP())
C02437 F0344 L1334 Math.round(maxHP()*(.13+.06*syner('nature')))
C02473 F0348 L1349 Math.round(rank*t.gain*100)
C02489 F0348 L1358 Math.round(d.dmg*rarity[r].scale)
C02490 F0348 L1358 Math.round(d.hp*rarity[r].scale)
C02517 F0361 L1366 Math.min(rarity.length-1,r+1)
C02608 F0348 L1420 Math.ceil(p.hp)
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C02609 F0348 L1420 Math.ceil(maxHP())
C02611 F0348 L1420 Math.round(9+p.level*.4)
C02612 F0348 L1420 Math.round(rootedArmor())
C02614 F0348 L1420 Math.round(bodySpeedFactor()*100)
C02616 F0348 L1420 Math.round(magnetRadius())
C02618 F0348 L1420 Math.floor(p.xp)
C02620 F0348 L1421 Math.round(game.width/game.zoom)
C02621 F0348 L1421 Math.round(game.height/game.zoom)
C02622 F0348 L1421 Math.round(100/game.zoom)
C02624 F0348 L1422 Math.round(p.x)
C02625 F0348 L1422 Math.round(p.y)
C02627 F0348 L1423 Math.round(42*scale)
C02630 F0348 L1423 Math.round(19*scale)
C02631 F0348 L1423 Math.round(35*scale)
C02632 F0348 L1423 Math.round(q.dmg*scale)
C02633 F0348 L1423 Math.ceil(e.hp)
C02634 F0348 L1423 Math.round(q.hp*scale)
C02637 F0348 L1424 Math.min(j.kills,q.goal)
C02638 F0348 L1424 Math.min(2,z.drop+1)
C02641 F0386 L1424 Math.min(2,zones[id].drop+1)
C02645 F0386 L1424 Math.min(maxHP(),game.p.hp+30)
C02858 F0419 L1479 Math.hypot(dx,dy)
C02859 F0419 L1479 Math.min(d,limit)
C03044 F0471 L1529 Math.round(game.p.x)
C03045 F0471 L1529 Math.round(game.p.y)
C03072 F0477 L1535 clamp(x.x,0,W)
C03073 F0477 L1535 clamp(x.y,0,H)
C03078 F0477 L1536 clamp(x.x,0,W)
C03079 F0477 L1536 clamp(x.y,0,H)
C03082 F0477 L1537 clamp(x.x,0,W)
C03083 F0477 L1537 clamp(x.y,0,H)
C03102 F0481 L1541 Math.floor(Number(r.level)||1)
C03137 F0486 L1544 Math.max(0,game.boon.until-game.time)
C03141 F0489 L1544 Math.round(skillRemaining(k)*10)
C03144 F0486 L1544 Math.min(.5,Math.max(0,game.combo.count-1)*.05)
C03145 F0486 L1544 Math.max(0,game.combo.count-1)
C03178 F0498 L13 Math.floor(Date.now()/1000)
C03187 F0503 L24 value.normalize('NFKC')
C03248 F0516 L78 Math.max(1,row.reset_at-t)
C03401 F0526 L193 body.username.normalize('NFKC')
C03458 F0530 L9 Math.max(maxima[i],x)
C03462 F0534 L11 score(b)
C03463 F0534 L11 score(a)
C03464 F0528 L12 Math.min(pool[k],slots.length-result.length)
C03474 F0537 L16 score(k)
C03476 F0538 L16 score(k)
C03479 F0539 L16 score(k)
C03587 F0566 L7 finite(p)
C03591 F0568 L9 Math.max(...values)
C03595 F0573 L10 normalize(weights)
C03596 F0573 L10 probability(rng())
C03599 F0575 L11 integer(seed,4294967295)
C03601 F0575 L11 Math.imul(i+1,0x9e3779b9)
C03602 F0576 L11 Math.imul(t^(t>>>15),t|1)
C03603 F0576 L11 Math.imul(t^(t>>>7),t|61)
C03604 F0578 L12 integer(rank,7)
C03611 F0578 L12 Math.max(0,rank-loss)
C03613 F0578 L12 Math.min(7,r+1)
C03617 F0582 L13 dropRanks(rank,treasure)
C03618 F0582 L13 finite(value(type,row.rank))
```

```
C03620 F0582 L13 finite(value(types[1],Math.max(1,row.rank-1)))
C03622 F0582 L13 Math.max(1,row.rank-1)
C03624 F0582 L13 finite(value('basic',Math.max(0,row.rank-2)))
C03626 F0582 L13 Math.max(0,row.rank-2)
C03629 F0585 L14 probability(p)
C03630 F0585 L14 integer(n)
C03631 F0586 L15 probability(p)
C03632 F0586 L15 integer(n)
C03633 F0586 L15 integer(attempts,5)
C03645 F0586 L15 craftCDF(p,99999)
C03647 F0589 L16 integer(copies,999)
C03648 F0589 L16 Math.min(config.critical.cap,1-(1-config.critical.perCopy)**copies)
C03649 F0590 L17 criticalChance(copies)
C03650 F0590 L17 probability(rng())
C03652 F0591 L18 integer(copies,999)
C03653 F0591 L18 Math.min(config.frost.cap,copies*config.frost.perCopy)
C03654 F0592 L19 finite(eta)
C03655 F0592 L19 Math.abs(outcomes.reduce((s,x)=>s+probability(x.p),0)-1)
C03657 F0593 L19 probability(x.p)
C03659 F0594 L19 finite(x.value)
C03660 F0594 L19 Math.expm1(-eta*finite(x.value))
C03661 F0594 L19 finite(x.value)
C03665 F0595 L20 normalize(weights)
C03666 F0598 L21 expectedUtility(outcomes.map(x=>({p:x.p,value:weightedValue(x.features,weights)})),eta)
C03668 F0599 L21 weightedValue(x.features,weights)
C03669 F0600 L22 finite(raw)
C03670 F0600 L22 finite(armor)
C03672 F0600 L22 criticalChance(copies)
C03673 F0600 L22 Math.min(limit,Math.max(1,raw-armor))
C03674 F0600 L22 Math.max(1,raw-armor)
C03675 F0600 L22 Math.min(limit,Math.max(1,raw*config.critical.multiplier-armor))
C03676 F0600 L22 Math.max(1,raw*config.critical.multiplier-armor)
C03681 F0604 L22 Math.min(s,x.value)
C03684 F0606 L22 Math.max(s,x.value)
C03687 F0608 L23 integer(wanted,9)
C03690 F0608 L23 dropRanks(rank,treasure)
C03691 F0610 L23 Math.max(1,row.rank-1)
C03692 F0610 L23 Math.max(0,row.rank-2)
C03693 F0611 L24 probability(gain)
C03694 F0611 L24 finite(dt)
C03695 F0611 L24 Math.expm1(60*dt*Math.log1p(-gain))
C03696 F0611 L24 Math.log1p(-gain)
C03697 F0612 L25 integer(rank,7)
C03698 F0612 L25 integer(level,999)
C03699 F0612 L25 finite(speed)
C03700 F0612 L25 probability(frostFactor)
C03702 F0612 L25 Math.min(c.levelSpeedCap,level*c.levelSpeed)
C03703 F0613 L26 finite(p.x)
C03704 F0613 L26 finite(m.x)
C03705 F0613 L26 finite(p.y)
C03706 F0613 L26 finite(m.y)
C03707 F0613 L26 Math.hypot(dx,dy)
C03708 F0613 L26 Math.min(d/370,.65)
C03709 F0613 L26 finite(p.vx??0)
C03710 F0613 L26 finite(p.vy??0)
C03711 F0613 L26 Math.hypot(x,y)
C03724 F0618 L7 Math.imul(s,1664525)
C03780 F0619 L18 Math.cos(k*2.1)
C03781 F0619 L18 Math.sin(k*2.1)
C03786 F0619 L19 Math.cos(a)
```

```
C03787 F0619 L19 Math.sin(a)
C03788 F0619 L20 Math.cos(a)
C03789 F0619 L20 Math.sin(a)
C03791 F0619 L20 Math.cos(b)
C03792 F0619 L20 Math.sin(b)
C03796 F0620 L22 Math.sin(time*4+(m.id||0))
C03798 F0620 L23 Math.max(1.5,r*.09)
C03805 F0620 L23 Math.max(.6,r*.022)
C03811 F0620 L24 Math.max(2,r*.15)
C03826 F0620 L25 Math.max(1.2,r*.035)
C03849 F0620 L28 Math.max(2,r*.06)
C03852 F0620 L28 Math.sin(a)
C03854 F0620 L28 Math.sin(a)
C03855 F0620 L28 Math.sin(a)
C03860 F0620 L29 Math.max(1,r*.025)
C03869 F0620 L30 Math.max(2,r*.06)
C03883 F0620 L33 Math.sin(a)
C03884 F0620 L33 Math.cos(a)
C03887 F0620 L34 Math.max(1,r*.03)
C03899 F0620 L36 Math.max(1,r*.025)
C03905 F0620 L37 Math.max(1,r*.035)
C03911 F0620 L39 Math.max(1,r*.03)
C03916 F0620 L39 Math.cos(k*2.2)
C03917 F0620 L39 Math.sin(k*2.2)
C03926 F0621 L40 Math.min(width,height)
C03927 F0621 L40 Math.max(width,height)
C03931 F0621 L40 Math.sin(time*.12)
C03953 F0622 L41 Math.cos(a)
C03954 F0622 L41 Math.sin(a)
C03956 F0622 L41 Math.cos(a+.2)
C03957 F0622 L41 Math.sin(a+.2)
C03958 F0622 L41 Math.cos(a)
C03959 F0622 L41 Math.sin(a)
C03966 F0622 L41 Math.cos(a)
C03967 F0622 L41 Math.sin(a)
C03979 F0624 L43 Math.max(.7,r*.016)
C03990 F0624 L45 Math.cos(a)
C03991 F0624 L45 Math.sin(a)
C04009 F0625 L48 Math.cos(j*TAU/3+.12)
C04010 F0625 L48 Math.sin(j*TAU/3+.12)
C04015 F0626 L49 Math.max(0,Math.min(1,1-m.windup/profile.windup))
C04016 F0626 L49 Math.min(1,1-m.windup/profile.windup)
C04017 F0626 L49 globalThis.PetalRules.specialAim(m,player)
C04018 F0626 L49 Math.atan2(a.y,a.x)
```

Appendix D. Complete call-site index

All 4,060 CallExpression nodes, identified by owner and callee. Inline function invocations are named rather than reproducing an entire enclosed script. This is a call-site inventory, not a runtime trace; dynamic dispatch and branch reachability require semantic review. Constructor NewExpression nodes and vendor internals are outside this call count and are not claimed as omitted game formulas.

```
C00001 client1:root L238 inline function / computed callee (body separately inventoried)
C00002 F0002 L239 document.getElementById
C00003 F0001 L239 $
C00004 F0001 L239 canvas.getContext
C00005 F0001 L239 $
C00006 F0001 L239 mini.getContext
C00007 F0003 L241 Math.max
C00008 F0003 L241 Math.min
C00009 F0005 L241 Math.hypot
C00010 F0006 L241 game.random?.world||Math.random
C00011 F0008 L242 Math.imul
C00012 F0008 L242 Math.imul
C00013 F0001 L243 inline function / computed callee (body separately inventoried)
C00014 F0009 L243 Math.pow
C00015 F0009 L243 Math.pow
C00016 F0001 L245 Object.assign
C00017 F0001 L247 Object.assign
C00018 F0001 L255 Object.assign
C00019 F0001 L277 Object.assign
C00020 F0001 L277 mobs.wasp.drop.push
C00021 F0001 L277 mobs.helWasp.drop.push
C00022 F0001 L277 mobs.mechaWasp.drop.push
C00023 F0001 L277 mobs.dragonfly.drop.push
C00024 F0001 L277 mobs.starfish.drop.push
C00025 F0001 L279 mobs.soldier.drop.push
C00026 F0001 L279 mobs.scarab.drop.push
C00027 F0001 L279 mobs.sporeBloom.drop.push
C00028 F0001 L279 mobs.mantis.drop.push
C00029 F0001 L279 mobs.centipede.drop.push
C00030 F0001 L279 mobs.queenAnt.drop.push
C00031 F0001 L279 mobs.mosquito.drop.push
C00032 F0001 L279 mobs.dragonfly.drop.push
C00033 F0001 L279 mobs.drone.drop.push
C00034 F0001 L279 mobs.mechaFlower.drop.push
C00035 F0001 L291 zones.garden.mob.push
C00036 F0001 L291 zones.jungle.mob.push
C00037 F0001 L291 zones.sewer.mob.push
C00038 F0001 L291 zones.factory.mob.push
C00039 F0001 L294 mobs[t].drop.push
C00040 F0001 L295 mobs[t].drop.push
C00041 F0010 L305 Math.max
C00042 F0010 L305 Math.floor
C00043 F0011 L306 Object.fromEntries
C00044 F0011 L306 Object.keys(TALENTS).map
C00045 F0011 L306 Object.keys
C00046 F0011 L306 talentEarned
C00047 F0011 L306 Object.keys
C00048 F0011 L306 Number
C00049 F0011 L306 Object.prototype.hasOwnProperty.call
C00050 F0011 L306 Number.isFinite
C00051 F0011 L306 Math.max
C00052 F0011 L306 Math.min
```

```
C00053 F0011 L306 Math.floor
C00054 F0011 L306 Math.min
C00055 F0013 L307 Number
C00056 F0014 L308 Math.max
C00057 F0014 L308 talentEarned
C00058 F0014 L308 Object.values(game.talents||{}).reduce
C00059 F0014 L308 Object.values
C00060 F0016 L309 talentRank
C00061 F0017 L310 talentModifier
C00062 F0018 L311 Math.min
C00063 F0018 L311 talentRank
C00064 F0019 L312 talentPoints
C00065 F0019 L312 talentRank
C00066 F0019 L312 maxHP
C00067 F0019 L312 Math.min
C00068 F0019 L312 maxHP
C00069 F0019 L312 maxHP
C00070 F0019 L312 saveGame
C00071 F0019 L312 syncHud
C00072 F0019 L312 renderModal
C00073 F0019 L312 toast
C00074 F0019 L312 talentRank
C00075 F0020 L313 sanitizeTalents
C00076 F0020 L313 Math.min
C00077 F0020 L313 maxHP
C00078 F0020 L313 saveGame
C00079 F0020 L313 syncHud
C00080 F0020 L313 renderModal
C00081 F0020 L313 toast
C00082 F0021 L315 Math.max
C00083 F0022 L316 shrineCooldown
C00084 F0022 L316 Math.min
C00085 F0022 L316 maxHP
C00086 F0022 L316 maxHP
C00087 F0022 L316 floating
C00088 F0022 L316 pulse
C00089 F0022 L316 toast
C00090 F0023 L317 shrineCooldown
C00091 F0023 L317 dist
C00092 F0023 L317 activateShrine
C00093 F0025 L333 localStorage.removeItem
C00094 F0026 L334 localStorage.setItem
C00095 F0026 L334 JSON.stringify
C00096 F0027 L335 localStorage.getItem
C00097 F0027 L335 JSON.parse
C00098 F0030 L339 Number
C00099 F0030 L339 Number.isSafeInteger
C00100 F0030 L339 Error
C00101 F0031 L340 localStorage.getItem
C00102 F0031 L340 cloudOutboxKey
C00103 F0031 L340 JSON.parse
C00104 F0031 L340 Array.isArray
C00105 F0031 L340 Error
C00106 F0031 L340 cloudRev
C00107 F0031 L340 Error
C00108 F0032 L341 cloudRev
C00109 F0032 L341 new Date().toISOString
C00110 F0032 L341 localStorage.setItem
C00111 F0032 L341 cloudOutboxKey
C00112 F0032 L341 JSON.stringify
```

```
C00113 F0032 L341 localStorage.setItem
C00114 F0032 L341 cloudLegacyFlag
C00115 F0033 L342 localStorage.removeItem
C00116 F0033 L342 cloudOutboxKey
C00117 F0033 L342 localStorage.removeItem
C00118 F0033 L342 cloudLegacyFlag
C00119 F0034 L343 /^https:\/\/[^/]+\.supabase\.co$/i.test
C00120 F0034 L343 /^sb_publishable_/.test
C00121 F0035 L347 String(err?.code||'').toLowerCase
C00122 F0035 L347 String
C00123 F0035 L348 String
C00124 F0035 L349 raw.toLowerCase
C00125 F0035 L350 /invalid.login.credentials|invalid.grant|invalid.password|email.not.found|user.not.found/.test
C00126 F0035 L352 ['email_address_not_authorized','email_not_authorized'].includes
C00127 F0035 L352 /email.address.not.authorized|email.recipient.not.authorized/.test
C00128 F0035 L354 /email|smtp|mail/.test
C00129 F0035 L354 /sending|send|failed|error|provider|smtp/.test
C00130 F0035 L354 ['email_provider_error','email_send_failed'].includes
C00131 F0035 L356 /email.not.confirmed|email.not.verified|confirm.your.email/.test
C00132 F0035 L358
['over_email_send_rate_limit','over_request_rate_limit','over_sms_send_rate_limit','rate_limit_exceeded'].includes
C00133 F0035 L358 /too.many.requests|rate.limit|email.rate.limit|email.send.rate/.test
C00134 F0035 L360 /already.registered|already.exists/.test
C00135 F0035 L362 ['signup_disabled','email_provider_disabled'].includes
C00136 F0035 L362 /signups?.(are.)?disabled/.test
C00137 F0035 L364 ['weak_password','password_too_short'].includes
C00138 F0035 L364 /password.should.be.at.least|weak.password/.test
C00139 F0035 L366 /invalid.email|email.address.is.invalid/.test
C00140 F0035 L368 /failed.to.fetch|networkerror|network.request.failed|load.failed/.test
C00141 F0035 L370 /captcha/.test
C00142 F0035 L373 /[\u3400-\u9fff]/.test
C00143 F0035 L373 raw.slice
C00144 F0001 L379 inline function / computed callee (body separately inventoried)
C00145 F0036 L379 JSON.parse
C00146 F0036 L379 localStorage.getItem
C00147 F0036 L379 Number.isFinite
C00148 F0036 L379 Date.now
C00149 F0036 L379 Math.min
C00150 F0036 L379 Date.now
C00151 F0037 L380 Math.max
C00152 F0037 L380 Math.ceil
C00153 F0037 L380 Date.now
C00154 F0038 L381 Math.max
C00155 F0038 L381 Date.now
C00156 F0038 L381 Math.min
C00157 F0038 L381 Math.max
C00158 F0038 L381 localStorage.setItem
C00159 F0038 L381 JSON.stringify
C00160 F0038 L381 refreshMailControls
C00161 F0039 L383 String
C00162 F0039 L384 ['email_not_confirmed','email_address_not_authorized','email_not_authorized','email_provider_error','
email_send_failed','over_email_send_rate_limit','mail_cooldown'].includes
C00163 F0039 L385 /SMTP|??????|Confirm Email|??/.test
C00164 F0039 L385 String
C00165 F0040 L388 $
C00166 F0040 L388 $
C00167 F0040 L388 $
C00168 F0040 L390 mailWait
C00169 F0040 L390 mailWait
C00170 F0040 L392 mailSending.has
```

```
C00171 F0040 L393 Math.max
C00172 F0041 L396 mailWait
C00173 F0041 L396 mailSending.has
C00174 F0041 L397 mailSending.add
C00175 F0041 L398 cloudRequest
C00176 F0041 L398 mailPause
C00177 F0041 L400 mailPause
C00178 F0041 L401 mailPause
C00179 F0041 L401 mailPause
C00180 F0041 L402 mailSending.delete
C00181 F0041 L402 refreshMailControls
C00182 F0042 L405 cloudReady
C00183 F0042 L405 Error
C00184 F0042 L409 fetch
C00185 F0042 L409 JSON.stringify
C00186 F0042 L410 resp.text
C00187 F0042 L411 JSON.parse
C00188 F0042 L411 Error
C00189 F0042 L412 String
C00190 F0042 L412 String
C00191 F0042 L412 resp.headers?.get
C00192 F0042 L412 Number.isFinite
C00193 F0042 L412 Number
C00194 F0042 L412 Number
C00195 F0042 L412 Date.parse
C00196 F0042 L412 Date.now
C00197 F0042 L412 Number.isFinite
C00198 F0042 L412 Math.min
C00199 F0042 L412 Math.max
C00200 F0042 L412 Math.ceil
C00201 F0043 L415 Error
C00202 F0043 L415 Date.now
C00203 F0043 L415 Math.max
C00204 F0044 L416 Error
C00205 F0044 L416 Date.now
C00206 F0044 L417 cloudRequest
C00207 F0044 L418 Date.now
C00208 F0044 L418 Math.max
C00209 F0044 L418 cloudStoreSession
C00210 F0045 L419 cloudToken
C00211 F0045 L419 cloudRequest
C00212 F0045 L419 encodeURIComponent
C00213 F0045 L419 Array.isArray
C00214 F0045 L419 Error
C00215 F0045 L419 cloudRev
C00216 F0046 L420 setTimeout
C00217 F0047 L420 cloudFlush
C00218 F0048 L422 cloudReadOutbox
C00219 F0048 L422 cloudWriteOutbox
C00220 F0048 L422 $
C00221 F0048 L422 console.warn
C00222 F0048 L423 cloudSchedule
C00223 F0049 L427 cloudToken
C00224 F0049 L428 cloudRequest
C00225 F0049 L428 new Date().toISOString
C00226 F0049 L429 cloudRequest
C00227 F0049 L429 encodeURIComponent
C00228 F0049 L429 encodeURIComponent
C00229 F0049 L429 new Date().toISOString
C00230 F0049 L430 Array.isArray
```

```
C00231 F0049 L430 Error
C00232 F0049 L431 cloudRev
C00233 F0049 L432 cloudClearOutbox
C00234 F0049 L432 $
C00235 F0049 L433 cloudWriteOutbox
C00236 F0049 L435 $
C00237 F0049 L436 console.warn
C00238 F0049 L438 cloudSchedule
C00239 F0049 L439 cloudSchedule
C00240 F0049 L439 Math.min
C00241 F0050 L442 setCloudSession
C00242 F0050 L445 cloudRequest
C00243 F0050 L445 cloudToken
C00244 F0050 L446 Error
C00245 F0050 L447 cloudLoad
C00246 F0050 L447 cloudReadOutbox
C00247 F0050 L448 localStorage.getItem
C00248 F0050 L448 cloudLegacyFlag
C00249 F0050 L448 Error
C00250 F0050 L449 Error
C00251 F0050 L450 initGame
C00252 F0050 L450 $
C00253 F0050 L450 $
C00254 F0050 L451 cloudStoreSession
C00255 F0050 L452 cloudSchedule
C00256 F0050 L452 toast
C00257 F0051 L458 cloudRequest
C00258 F0051 L459 Error
C00259 F0051 L460 Error
C00260 F0051 L462 Error
C00261 F0052 L466 cloudReady
C00262 F0052 L466 Error
C00263 F0052 L467 /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test
C00264 F0052 L467 Error
C00265 F0052 L468 $
C00266 F0052 L468 cloudForgetStoredSession
C00267 F0052 L470 Error
C00268 F0052 L471 $
C00269 F0052 L471 Error
C00270 F0052 L472 cloudVerifiedSignupReady
C00271 F0052 L473 mailRequest
C00272 F0052 L473 encodeURIComponent
C00273 F0052 L474 $('localFallbackBtn')?.classList?.remove
C00274 F0052 L474 $
C00275 F0052 L474 $
C00276 F0052 L474 $
C00277 F0052 L474 $
C00278 F0052 L475 cloudEnter
C00279 F0052 L477 cloudEnter
C00280 F0052 L477 cloudRequest
C00281 F0053 L482 cloudReadStoredSession
C00282 F0053 L483 $
C00283 F0053 L483 $
C00284 F0053 L485 cloudRequest
C00285 F0053 L486 $
C00286 F0053 L487 cloudEnter
C00287 F0053 L488 $
C00288 F0053 L490 /(?:invalid|expired|revoked|401|403|refresh.token|token.not.found)/i.test
C00289 F0053 L490 String
C00290 F0053 L490 cloudForgetStoredSession
```

```
C00291 F0053 L491 $
C00292 F0053 L491 cloudAuthMessage
C00293 F0053 L492 $
C00294 F0054 L496 $('authName').value.trim
C00295 F0054 L496 $
C00296 F0054 L497 /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test
C00297 F0054 L497 $
C00298 F0054 L498 $
C00299 F0054 L498 $
C00300 F0054 L500 mailRequest
C00301 F0054 L500 encodeURIComponent
C00302 F0054 L501 $
C00303 F0054 L502 $
C00304 F0054 L502 cloudAuthMessage
C00305 F0054 L502 $('localFallbackBtn')?.classList.toggle
C00306 F0054 L502 $
C00307 F0054 L502 isMailUnavailable
C00308 F0054 L503 refreshMailControls
C00309 F0055 L506 cloudReady
C00310 F0055 L507 location.hash.slice
C00311 F0055 L508 hash.get
C00312 F0055 L508 hash.get
C00313 F0055 L508 hash.get
C00314 F0055 L509 ['recovery','signup','email','magiclink'].includes
C00315 F0055 L510 configureAccountMode
C00316 F0055 L511 history.replaceState
C00317 F0055 L512 $
C00318 F0055 L514 cloudRequest
C00319 F0055 L515 Error
C00320 F0055 L517 $
C00321 F0055 L517 cloudForgetStoredSession
C00322 F0055 L518 setCloudSession
C00323 F0055 L518 Number
C00324 F0055 L518 hash.get
C00325 F0055 L519 $
C00326 F0055 L519 $
C00327 F0055 L519 $
C00328 F0055 L520 $
C00329 F0055 L520 $
C00330 F0055 L522 $
C00331 F0055 L522 cloudForgetStoredSession
C00332 F0055 L523 cloudEnter
C00333 F0055 L523 Number
C00334 F0055 L523 hash.get
C00335 F0055 L525 $
C00336 F0055 L526 $
C00337 F0056 L530 e.preventDefault
C00338 F0056 L532 $
C00339 F0056 L532 $
C00340 F0056 L533 $
C00341 F0056 L534 $
C00342 F0056 L536 cloudRequest
C00343 F0056 L536 cloudToken
C00344 F0056 L537 Error
C00345 F0056 L538 $
C00346 F0056 L538 $
C00347 F0056 L539 $
C00348 F0056 L539 $
C00349 F0056 L539 $
C00350 F0056 L540 Math.floor
```

```
C00351 F0056 L540 Date.now
C00352 F0056 L541 cloudEnter
C00353 F0056 L542 $
C00354 F0056 L542 $
C00355 F0056 L542 $
C00356 F0056 L542 $
C00357 F0056 L542 cloudForgetStoredSession
C00358 F0057 L547 event.preventDefault
C00359 F0057 L548 $
C00360 F0057 L548 $
C00361 F0057 L548 $
C00362 F0057 L550 form.querySelector
C00363 F0057 L551 form.querySelector
C00364 F0057 L552 form.querySelector
C00365 F0057 L557 cloudRequest
C00366 F0057 L557 cloudToken
C00367 F0057 L558 Error
C00368 F0057 L559 cloudRequest
C00369 F0057 L560 Error
C00370 F0057 L561 cloudRequest
C00371 F0057 L562 Error
C00372 F0057 L563 form.reset
C00373 F0057 L564 cloudAuthMessage
C00374 F0057 L565 form.querySelector
C00375 F0058 L568 Object.keys(localStorage).filter
C00376 F0058 L568 Object.keys
C00377 F0059 L568 k.startsWith
C00378 F0058 L568 all[0].slice
C00379 F0058 L569 Error
C00380 F0058 L570 cloudReadOutbox
C00381 F0058 L570 localStorage.getItem
C00382 F0058 L571 Error
C00383 F0058 L572 new Date().toISOString
C00384 F0058 L572 JSON.parse
C00385 F0058 L573 JSON.stringify
C00386 F0058 L573 URL.createObjectURL
C00387 F0058 L573 document.createElement
C00388 F0058 L574 new Date().toISOString().slice
C00389 F0058 L574 new Date().toISOString
C00390 F0058 L574 document.body.append
C00391 F0058 L574 a.click
C00392 F0058 L574 a.remove
C00393 F0058 L574 setTimeout
C00394 F0060 L574 URL.revokeObjectURL
C00395 F0058 L575 $
C00396 F0058 L576 $
C00397 F0061 L578 $('cfRecoverForm')?.remove
C00398 F0061 L578 $
C00399 F0061 L578 $
C00400 F0061 L580 $
C00401 F0061 L580 $
C00402 F0061 L580 $
C00403 F0061 L580 $
C00404 F0061 L581 $
C00405 F0061 L582 $
C00406 F0061 L582 $
C00407 F0061 L583 $
C00408 F0061 L583 $
C00409 F0061 L584 $
C00410 F0061 L585 $
```

```
C00411 F0061 L585 $
C00412 F0061 L586 $
C00413 F0061 L586 $
C00414 F0061 L587 $('localFallbackBtn').classList.add
C00415 F0061 L587 $
C00416 F0061 L588 $
C00417 F0061 L589 $
C00418 F0061 L590 setAuthMode
C00419 F0061 L590 refreshMailControls
C00420 F0062 L593 $
C00421 F0062 L594 configureAccountMode
C00422 F0062 L594 cfConfigure
C00423 F0062 L594 configureAccountMode
C00424 F0001 L596 Object.freeze
C00425 F0063 L596 cloudReady
C00426 F0069 L603 cfLockRelease
C00427 F0070 L605 cfReleaseLock
C00428 F0070 L606 Error
C00429 F0071 L608 inline function / computed callee (body separately inventoried)
C00430 F0071 L608 navigator.locks.request
C00431 F0072 L609 reject
C00432 F0072 L609 Error
C00433 F0072 L610 resolve
C00434 F0074 L616 fetch
C00435 F0074 L616 AbortSignal.timeout
C00436 F0074 L616 JSON.stringify
C00437 F0074 L617 Error
C00438 F0074 L618 response.json
C00439 F0074 L618 Error
C00440 F0074 L619 Error
C00441 F0074 L619 Math.min
C00442 F0074 L619 Math.max
C00443 F0074 L619 Number
C00444 F0074 L619 response.headers.get
C00445 F0075 L622 localStorage.getItem
C00446 F0075 L622 cfOutboxKey
C00447 F0075 L623 JSON.parse
C00448 F0075 L623 Number.isSafeInteger
C00449 F0075 L623 Error
C00450 F0075 L623 validateLocalProgress
C00451 F0075 L624 Error
C00452 F0076 L626 setTimeout
C00453 F0077 L626 cfFlush
C00454 F0078 L629 $
C00455 F0078 L630 cfReadOutbox
C00456 F0078 L630 $
C00457 F0078 L630 new Date().toISOString
C00458 F0078 L630 localStorage.setItem
C00459 F0078 L630 cfOutboxKey
C00460 F0078 L630 JSON.stringify
C00461 F0078 L630 $
C00462 F0078 L630 cfSchedule
C00463 F0078 L631 $
C00464 F0078 L631 console.warn
C00465 F0079 L637 cfAPI
C00466 F0079 L640 localStorage.getItem
C00467 F0079 L640 cfOutboxKey
C00468 F0079 L640 JSON.stringify
C00469 F0079 L640 $
C00470 F0079 L640 localStorage.removeItem
```

```
C00471 F0079 L640 cfOutboxKey
C00472 F0079 L640 $
C00473 F0079 L641 localStorage.setItem
C00474 F0079 L641 cfOutboxKey
C00475 F0079 L641 JSON.stringify
C00476 F0079 L644 $
C00477 F0079 L645 $
C00478 F0079 L646 cfSchedule
C00479 F0079 L646 Math.max
C00480 F0079 L646 Math.min
C00481 F0079 L647 cfSchedule
C00482 F0080 L650 configureAccountMode
C00483 F0080 L650 $
C00484 F0080 L650 $
C00485 F0080 L650 $
C00486 F0080 L650 $
C00487 F0080 L650 $
C00488 F0080 L651 $
C00489 F0080 L652 $
C00490 F0080 L653 $
C00491 F0080 L653 $
C00492 F0080 L653 setAuthMode
C00493 F0081 L657 $('cfRecoveryCard')?.remove
C00494 F0081 L657 $
C00495 F0081 L658 document.createElement
C00496 F0081 L658 card.setAttribute
C00497 F0081 L660 document.body.append
C00498 F0081 L660 $
C00499 F0081 L660 card.showModal
C00500 F0081 L660 $('cfCloseRecovery').focus
C00501 F0081 L660 $
C00502 F0081 L660 card.addEventListener
C00503 F0082 L660 e.preventDefault
C00504 F0081 L661 $
C00505 F0083 L661 cfDownload
C00506 F0081 L662 $
C00507 F0084 L662 card.close
C00508 F0084 L662 card.remove
C00509 F0085 L664 URL.createObjectURL
C00510 F0085 L664 JSON.stringify
C00511 F0085 L664 document.createElement
C00512 F0085 L664 document.body.append
C00513 F0085 L664 a.click
C00514 F0085 L664 a.remove
C00515 F0085 L664 setTimeout
C00516 F0086 L664 URL.revokeObjectURL
C00517 F0087 L666 Error
C00518 F0087 L667 cfTakeLock
C00519 F0087 L669 cfAPI
C00520 F0087 L669 Error
C00521 F0087 L670 localStorage.setItem
C00522 F0087 L671 cfReadOutbox
C00523 F0087 L672 Error
C00524 F0087 L673 validateLocalProgress
C00525 F0087 L675 clearTimeout
C00526 F0087 L677 initGame
C00527 F0087 L677 $
C00528 F0087 L677 $
C00529 F0087 L677 $
C00530 F0087 L678 saveGame
```

```
C00531 F0087 L678 cfSchedule
C00532 F0087 L678 $
C00533 F0087 L679 cfReleaseLock
C00534 F0088 L682 e.preventDefault
C00535 F0088 L682 $
C00536 F0088 L682 $
C00537 F0088 L684 $
C00538 F0088 L684 $
C00539 F0088 L684 Error
C00540 F0088 L685 cfAPI
C00541 F0088 L685 $('authName').value.trim
C00542 F0088 L685 $
C00543 F0088 L685 $
C00544 F0088 L685 $
C00545 F0088 L686 cfShowRecovery
C00546 F0088 L687 cfEnter
C00547 F0088 L688 $
C00548 F0088 L689 $
C00549 F0089 L692 $
C00550 F0089 L693 cfAPI
C00551 F0089 L693 cfEnter
C00552 F0089 L693 $
C00553 F0089 L694 $
C00554 F0090 L697 saveGame
C00555 F0091 L700 setTimeout
C00556 F0090 L701 cfFlush
C00557 F0090 L701 cfAPI
C00558 F0090 L702 cfReleaseLock
C00559 F0090 L702 clearTimeout
C00560 F0090 L703 logout
C00561 F0090 L703 cfConfigure
C00562 F0090 L704 toast
C00563 F0092 L707 localStorage.getItem
C00564 F0092 L707 localStorage.getItem
C00565 F0092 L707 cfOutboxKey
C00566 F0092 L707 Error
C00567 F0092 L707 cfDownload
C00568 F0092 L708 $
C00569 F0093 L711 askGame
C00570 F0093 L713 cfAPI
C00571 F0093 L713 localStorage.getItem
C00572 F0093 L713 Error
C00573 F0093 L713 cfTakeLock
C00574 F0093 L713 cfExportPending
C00575 F0093 L713 cfReleaseLock
C00576 F0093 L714 localStorage.setItem
C00577 F0093 L714 localStorage.getItem
C00578 F0093 L714 cfOutboxKey
C00579 F0093 L714 localStorage.removeItem
C00580 F0093 L714 cfOutboxKey
C00581 F0093 L714 cfEnter
C00582 F0093 L715 cfReleaseLock
C00583 F0093 L715 $
C00584 F0094 L718 askGame
C00585 F0094 L721 cfAPI
C00586 F0094 L721 localStorage.getItem
C00587 F0094 L722 Error
C00588 F0094 L723 cfReadOutbox
C00589 F0094 L723 Error
C00590 F0094 L724 cfTakeLock
```

```
C00591 F0094 L724 cfAPI
C00592 F0094 L724 cfDownload
C00593 F0094 L725 cfExportPending
C00594 F0094 L725 cfReleaseLock
C00595 F0094 L726 cfAPI
C00596 F0094 L727 localStorage.removeItem
C00597 F0094 L727 cfOutboxKey
C00598 F0094 L727 cfEnter
C00599 F0094 L728 cfReleaseLock
C00600 F0094 L728 $
C00601 F0095 L731 event.preventDefault
C00602 F0095 L731 $
C00603 F0095 L731 form.querySelector
C00604 F0095 L732 $
C00605 F0095 L732 $
C00606 F0095 L733 cfAPI
C00607 F0095 L733 $
C00608 F0095 L733 $
C00609 F0095 L733 form.reset
C00610 F0095 L733 $
C00611 F0096 L736 $
C00612 F0096 L737 ['pause','settings'].includes
C00613 F0096 L738 b.insertAdjacentHTML
C00614 F0096 L739 $
C00615 F0096 L740 $
C00616 F0097 L740 saveGame
C00617 F0097 L740 cfFlush
C00618 F0096 L740 $
C00619 F0098 L740 cfExportPending
C00620 F0098 L740 toast
C00621 F0096 L741 $
C00622 F0099 L741 saveGame
C00623 F0099 L741 cfDownload
C00624 F0099 L741 JSON.parse
C00625 F0099 L741 localStorage.getItem
C00626 F0096 L742 $
C00627 F0100 L742 cfImportProgress
C00628 F0096 L742 $
C00629 F0096 L742 $
C00630 F0096 L744 b.insertAdjacentHTML
C00631 F0096 L744 $
C00632 F0101 L749 Error
C00633 F0101 L749 JSON.parse
C00634 F0101 L749 file.text
C00635 F0101 L749 Error
C00636 F0101 L749 validateLocalProgress
C00637 F0101 L750 Error
C00638 F0101 L750 askGame
C00639 F0101 L751 saveGame
C00640 F0101 L751 localStorage.setItem
C00641 F0101 L751 localStorage.getItem
C00642 F0101 L751 initGame
C00643 F0101 L751 saveGame
C00644 F0101 L751 toast
C00645 F0101 L752 toast
C00646 F0102 L755 localStorage.getItem
C00647 F0102 L755 Error
C00648 F0102 L755 JSON.parse
C00649 F0102 L755 validateLocalProgress
C00650 F0102 L755 askGame
```

```
C00651 F0102 L755 initGame
C00652 F0102 L755 saveGame
C00653 F0102 L755 toast
C00654 F0102 L755 toast
C00655 F0103 L758 event.preventDefault
C00656 F0103 L758 $
C00657 F0103 L758 form.querySelector
C00658 F0103 L759 $('cfMigrateName').value.trim
C00659 F0103 L759 $
C00660 F0103 L759 $
C00661 F0103 L760 $
C00662 F0103 L761 askGame
C00663 F0103 L764 saveGame
C00664 F0103 L764 cloudFlush
C00665 F0103 L764 Error
C00666 F0103 L764 cloudToken
C00667 F0103 L765 JSON.parse
C00668 F0103 L765 localStorage.getItem
C00669 F0103 L765 validateLocalProgress
C00670 F0103 L766 cfAPI
C00671 F0103 L766 cfAPI
C00672 F0103 L767 cfShowRecovery
C00673 F0103 L768 Error
C00674 F0103 L769 cfTakeLock
C00675 F0103 L770 cfAPI
C00676 F0103 L770 cfAPI
C00677 F0103 L771 cfEnter
C00678 F0103 L771 toast
C00679 F0103 L772 cfReleaseLock
C00680 F0103 L773 $
C00681 F0103 L773 $
C00682 F0103 L773 $
C00683 F0103 L773 $
C00684 F0104 L777 $
C00685 F0104 L777 $
C00686 F0104 L777 form.remove
C00687 F0104 L778 document.createElement
C00688 F0104 L778 panel.after
C00689 F0104 L779 $
C00690 F0105 L779 form.remove
C00691 F0106 L780 e.preventDefault
C00692 F0106 L780 $
C00693 F0106 L780 $
C00694 F0106 L780 $
C00695 F0106 L781 cfAPI
C00696 F0106 L781 $
C00697 F0106 L781 $
C00698 F0106 L781 $
C00699 F0106 L781 cfShowRecovery
C00700 F0106 L781 form.remove
C00701 F0106 L781 cfEnter
C00702 F0106 L781 $
C00703 F0106 L781 $
C00704 F0107 L785 parseLocalBackup
C00705 F0107 L785 JSON.stringify
C00706 F0107 L786 Object.hasOwn
C00707 F0107 L786 payload.slots.every
C00708 F0108 L786 itemOf
C00709 F0107 L786 Object.keys(payload.bag).some
C00710 F0107 L786 Object.keys
```

```
C00711 F0109 L786 itemOf
C00712 F0107 L786 Error
C00713 F0110 L791 Array.isArray
C00714 F0110 L792 Object.entries
C00715 F0110 L792 itemOf
C00716 F0112 L797 itemOf
C00717 F0112 L799 isPetalProtected
C00718 F0112 L800 saveGame
C00719 F0112 L800 renderModal
C00720 F0112 L800 toast
C00721 F0112 L800 isPetalProtected
C00722 F0112 L800 itemOf
C00723 F0113 L804 Error
C00724 F0113 L805 JSON.parse
C00725 F0113 L807 Number.isInteger
C00726 F0113 L808 Array.isArray
C00727 F0113 L808 Array.isArray
C00728 F0113 L808 Error
C00729 F0113 L809 Number.isFinite
C00730 F0113 L809 Error
C00731 F0113 L810 Object.values
C00732 F0113 L810 Number.isInteger
C00733 F0113 L810 Error
C00734 F0114 L815 saveGame
C00735 F0114 L815 readSave
C00736 F0114 L815 toast
C00737 F0114 L816 new Date().toISOString
C00738 F0114 L817 URL.createObjectURL
C00739 F0114 L817 JSON.stringify
C00740 F0114 L817 document.createElement
C00741 F0114 L818 new Date().toISOString().slice
C00742 F0114 L818 new Date().toISOString
C00743 F0114 L818 document.body.append
C00744 F0114 L818 a.click
C00745 F0114 L818 a.remove
C00746 F0114 L818 setTimeout
C00747 F0115 L818 URL.revokeObjectURL
C00748 F0114 L818 toast
C00749 F0116 L823 Error
C00750 F0116 L824 parseLocalBackup
C00751 F0116 L824 file.text
C00752 F0116 L824 Error
C00753 F0116 L825 askGame
C00754 F0116 L826 localStorage.getItem
C00755 F0116 L827 localStorage.setItem
C00756 F0116 L828 localStorage.setItem
C00757 F0116 L828 JSON.stringify
C00758 F0116 L828 initGame
C00759 F0116 L828 saveGame
C00760 F0116 L828 toast
C00761 F0116 L829 toast
C00762 F0117 L833 localStorage.getItem
C00763 F0117 L833 toast
C00764 F0117 L834 parseLocalBackup
C00765 F0117 L834 JSON.stringify
C00766 F0117 L834 JSON.parse
C00767 F0117 L835 askGame
C00768 F0117 L836 localStorage.setItem
C00769 F0117 L836 JSON.stringify
C00770 F0117 L836 initGame
```

```
C00771 F0117 L836 saveGame
C00772 F0117 L836 toast
C00773 F0117 L837 toast
C00774 F0119 L839 /^([a-zA-Z]+)@(0|[1-9]\d*)$/.exec
C00775 F0119 L839 Number
C00776 F0119 L839 Object.hasOwn
C00777 F0119 L839 Number.isInteger
C00778 F0120 L839 itemOf
C00779 F0120 L839 inventoryKey
C00780 F0120 L839 inventoryKey
C00781 F0120 L839 clamp
C00782 F0121 L840 inventoryKey
C00783 F0121 L840 addBag
C00784 F0123 L846 Math.min
C00785 F0123 L846 Math.floor
C00786 F0124 L846 game.slots.filter(s=>s.t==='antenna').reduce
C00787 F0124 L846 game.slots.filter
C00788 F0127 L846 clamp
C00789 F0127 L846 viewScale
C00790 F0127 L846 antennaMultiplier
C00791 F0128 L847 Math.hypot
C00792 F0130 L849 minZoneLevel
C00793 F0131 L850 Math.round
C00794 F0131 L850 Math.floor
C00795 F0131 L850 game.slots.reduce
C00796 F0131 L850 talentModifier
C00797 F0133 L851 game.slots.filter
C00798 F0135 L852 soilCount
C00799 F0136 L853 Math.max
C00800 F0136 L853 Math.pow
C00801 F0136 L853 soilCount
C00802 F0137 L854 game.equipped.filter
C00803 F0137 L854 equipped.reduce
C00804 F0140 L855 game.equipped.some
C00805 F0142 L856 game.equipped.reduce
C00806 F0142 L856 talentModifier
C00807 F0144 L861 syncHud
C00808 F0144 L861 toast
C00809 F0145 L866 clamp
C00810 F0146 L868 Math.sin
C00811 F0146 L868 talentModifier
C00812 F0148 L880 Object.entries
C00813 F0148 L880 items.has
C00814 F0149 L883 Object.entries
C00815 F0149 L883 slots.filter
C00816 F0150 L883 def.types.includes
C00817 F0152 L888 syner
C00818 F0152 L888 dist
C00819 F0152 L888 damageMob
C00820 F0152 L888 petalDamageMultiplier
C00821 F0152 L888 syner
C00822 F0152 L888 pulse
C00823 F0152 L889 petalDamageMultiplier
C00824 F0152 L889 syner
C00825 F0152 L889 game.friendlyShots.push
C00826 F0152 L889 Math.cos
C00827 F0152 L889 Math.sin
C00828 F0152 L889 Math.cos
C00829 F0152 L889 Math.sin
C00830 F0152 L889 pulse
```

```
C00831 F0152 L890 syner
C00832 F0152 L890 pulse
C00833 F0152 L891 Math.round
C00834 F0152 L891 maxHP
C00835 F0152 L891 syner
C00836 F0152 L891 Math.min
C00837 F0152 L891 maxHP
C00838 F0152 L891 floating
C00839 F0152 L891 pulse
C00840 F0152 L892 pulse
C00841 F0152 L893 toast
C00842 F0153 L895 Math.max
C00843 F0154 L897 Object.values(WORLD_EVENTS).reduce
C00844 F0154 L897 Object.values
C00845 F0154 L897 rng
C00846 F0154 L897 Object.entries
C00847 F0156 L898 spawnMob
C00848 F0156 L898 Math.round
C00849 F0156 L898 Math.ceil
C00850 F0156 L898 Math.ceil
C00851 F0157 L899 game.random?.world||Math.random
C00852 F0157 L900 Math.min
C00853 F0157 L900 spawnNearPlayer
C00854 F0157 L901 zones[game.zone].mob.map(x=>mobs[x].drop[0]).filter
C00855 F0157 L901 zones[game.zone].mob.map
C00856 F0157 L901 Math.floor
C00857 F0157 L901 game.random?.world||Math.random
C00858 F0157 L901 game.random?.world||Math.random
C00859 F0157 L901 game.random?.world||Math.random
C00860 F0157 L901 game.drops.push
C00861 F0157 L901 clamp
C00862 F0157 L901 Math.cos
C00863 F0157 L901 clamp
C00864 F0157 L901 Math.sin
C00865 F0157 L901 Math.min
C00866 F0157 L901 Math.max
C00867 F0157 L901 Math.floor
C00868 F0157 L901 game.random?.world||Math.random
C00869 F0157 L902 toast
C00870 F0159 L903 beginWorldEvent
C00871 F0159 L903 weightedEvent
C00872 F0159 L904 Math.min
C00873 F0159 L904 maxHP
C00874 F0159 L906 spawnNearPlayer
C00875 F0159 L907 game.random?.world||Math.random
C00876 F0159 L907 game.random?.world||Math.random
C00877 F0159 L907 e.meteors.push
C00878 F0159 L907 clamp
C00879 F0159 L907 Math.cos
C00880 F0159 L907 clamp
C00881 F0159 L907 Math.sin
C00882 F0159 L909 screenFx
C00883 F0159 L909 dist
C00884 F0159 L909 damagePlayer
C00885 F0159 L909 e.meteors.filter
C00886 F0161 L911 globalThis.PetalRules.specialProfile
C00887 F0161 L911 traitOf
C00888 F0162 L912 Math.max
C00889 F0163 L913 specialProfileFor
C00890 F0163 L913 Math.atan2
```

```
C00891 F0163 L913 game.bullets.push
C00892 F0163 L913 Math.cos
C00893 F0163 L913 Math.sin
C00894 F0163 L913 Math.cos
C00895 F0163 L913 Math.sin
C00896 F0164 L915 specialProfileFor
C00897 F0164 L915 Math.hypot
C00898 F0164 L916 globalThis.PetalRules.specialAim
C00899 F0164 L916 spawnSpecialProjectile
C00900 F0164 L918 Math.min
C00901 F0164 L918 clamp
C00902 F0164 L918 Math.cos
C00903 F0164 L918 clamp
C00904 F0164 L918 Math.sin
C00905 F0164 L918 nearbyRocks(map.rocks,xx,yy).some
C00906 F0164 L918 nearbyRocks
C00907 F0165 L918 dist
C00908 F0164 L918 Math.ceil
C00909 F0164 L918 map.living.push
C00910 F0164 L918 Math.min
C00911 F0164 L918 Math.max
C00912 F0164 L918 Math.ceil
C00913 F0164 L919 damagePlayer
C00914 F0164 L919 pulse
C00915 F0164 L920 floating
C00916 F0166 L922 game.random?.world||Math.random
C00917 F0166 L923 traitOf
C00918 F0166 L924 performMobSpecial
C00919 F0166 L924 specialCooldown
C00920 F0166 L925 specialProfileFor
C00921 F0167 L927 computeSynergies
C00922 F0167 L927 game.slots.map
C00923 F0168 L927 game.random?.world||Math.random
C00924 F0169 L928 slotCapacity
C00925 F0169 L928 game.slots.push
C00926 F0169 L928 game.backupSlots.push
C00927 F0169 L928 toast
C00928 F0169 L928 resetEquip
C00929 F0169 L928 fillSlots
C00930 F0169 L928 saveGame
C00931 F0171 L930 JSON.parse
C00932 F0171 L930 localStorage.getItem
C00933 F0172 L931 localStorage.setItem
C00934 F0172 L931 JSON.stringify
C00935 F0172 L931 $
C00936 F0172 L931 cfQueue
C00937 F0172 L931 JSON.parse
C00938 F0172 L931 localStorage.getItem
C00939 F0172 L931 cloudQueue
C00940 F0172 L931 JSON.parse
C00941 F0172 L931 localStorage.getItem
C00942 F0172 L931 $
C00943 F0173 L932 leaveParty
C00944 F0173 L932 crypto.getRandomValues
C00945 F0173 L932 globalThis.PetalRules.streams
C00946 F0173 L932 baseSave
C00947 F0173 L932 newPlayer
C00948 F0173 L932 clamp
C00949 F0173 L932 Math.floor
C00950 F0173 L932 clamp
```

```
C00951 F0173 L932 Math.floor
C00952 F0173 L932 clamp
C00953 F0173 L932 Math.floor
C00954 F0173 L932 sanitizeTalents
C00955 F0173 L932 game.random?.world||Math.random
C00956 F0173 L932 Object.entries
C00957 F0173 L932 itemOf
C00958 F0173 L932 Number.isFinite
C00959 F0173 L932 clamp
C00960 F0173 L932 Math.floor
C00961 F0173 L933 sanitizeProtectedPetals
C00962 F0173 L934 slotCapacity
C00963 F0173 L934 Array.from
C00964 F0173 L935 Array.from
C00965 F0173 L936 Object.keys
C00966 F0173 L936 clamp
C00967 F0173 L936 Math.floor
C00968 F0173 L936 Number
C00969 F0173 L936 Object.entries
C00970 F0173 L936 clamp
C00971 F0173 L936 Math.floor
C00972 F0173 L936 Number
C00973 F0173 L936 Object.fromEntries
C00974 F0173 L936 Object.entries(a.settings||{}).filter
C00975 F0173 L936 Object.entries
C00976 F0173 L936 ['auto','low','high'].includes
C00977 F0173 L936 ['mouse','keyboard'].includes
C00978 F0173 L936 sanitizeCollectionGoal
C00979 F0173 L936 collectionOwned
C00980 F0173 L937 Object.entries
C00981 F0173 L937 clamp
C00982 F0173 L937 Math.floor
C00983 F0173 L937 Number
C00984 F0173 L937 clamp
C00985 F0173 L937 Math.floor
C00986 F0173 L937 Number
C00987 F0173 L938 $
C00988 F0173 L938 resetEquip
C00989 F0173 L938 clamp
C00990 F0173 L938 maxHP
C00991 F0173 L938 effectiveZoom
C00992 F0173 L938 Object.keys
C00993 F0173 L938 buildMap
C00994 F0173 L938 $('gate').classList.add
C00995 F0173 L938 $
C00996 F0173 L938 $('ui').classList.remove
C00997 F0173 L938 $
C00998 F0173 L938 $('death').classList.add
C00999 F0173 L938 $
C01000 F0173 L938 $('panelLayer').classList.add
C01001 F0173 L938 $
C01002 F0173 L938 $
C01003 F0173 L938 cfAccount.username.split('@')[0].slice
C01004 F0173 L938 cfAccount.username.split
C01005 F0173 L938 $
C01006 F0173 L938 fillSlots
C01007 F0173 L938 resize
C01008 F0173 L938 syncHud
C01009 F0173 L938 resetInputs
C01010 F0173 L938 toast
```

```
C01011 F0177 L939 SEED
C01012 F0177 L939 z.ports.map
C01013 F0177 L940 decor.push
C01014 F0177 L940 r
C01015 F0177 L940 r
C01016 F0177 L940 r
C01017 F0177 L940 r
C01018 F0177 L940 Math.floor
C01019 F0177 L940 r
C01020 F0177 L941 r
C01021 F0177 L941 r
C01022 F0177 L941 r
C01023 F0177 L941 dist
C01024 F0177 L941 portals.some
C01025 F0179 L941 dist
C01026 F0177 L941 rocks.some
C01027 F0180 L941 dist
C01028 F0177 L941 rocks.push
C01029 F0177 L941 Math.floor
C01030 F0177 L941 r
C01031 F0177 L941 r
C01032 F0177 L942 Math.floor
C01033 F0177 L942 Math.floor
C01034 F0177 L942 lookup.has
C01035 F0177 L942 lookup.set
C01036 F0177 L942 lookup.get(key).push
C01037 F0177 L942 lookup.get
C01038 F0177 L942 ROCK_GRID.set
C01039 F0177 L943 createLandmarks
C01040 F0177 L943 [0,4,8].entries
C01041 F0177 L943 rocks.every
C01042 F0181 L943 dist
C01043 F0177 L943 portals.every
C01044 F0182 L943 dist
C01045 F0177 L943 dist
C01046 F0177 L943 shrines.every
C01047 F0183 L943 dist
C01048 F0177 L943 r
C01049 F0177 L943 r
C01050 F0177 L943 shrines.push
C01051 F0177 L943 spawnMob
C01052 F0177 L943 living.some
C01053 F0177 L943 spawnMob
C01054 F0185 L944 r
C01055 F0185 L944 map.living.some
C01056 F0185 L944 Math.floor
C01057 F0185 L944 r
C01058 F0185 L946 r
C01059 F0185 L946 clamp
C01060 F0185 L946 rnd
C01061 F0185 L946 r
C01062 F0185 L946 clamp
C01063 F0185 L946 rnd
C01064 F0185 L946 r
C01065 F0185 L947 dist
C01066 F0185 L947 regionBand
C01067 F0185 L948 nearbyRocks(map.rocks,x,y).some
C01068 F0185 L948 nearbyRocks
C01069 F0187 L948 dist
C01070 F0185 L948 map.portals.some
```

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C01071 F0188 L948 dist
C01072 F0185 L948 map.living.some
C01073 F0189 L948 dist
C01074 F0185 L949 regionBand
C01075 F0185 L949 r
C01076 F0185 L949 minZoneLevel
C01077 F0185 L949 Math.floor
C01078 F0185 L949 r
C01079 F0185 L949 r
C01080 F0185 L950 map.living.push
C01081 F0185 L950 Math.round
C01082 F0185 L950 Math.round
C01083 F0185 L950 Math.round
C01084 F0185 L950 r
C01085 F0185 L950 r
C01086 F0185 L950 r
C01087 F0185 L950 r
C01088 F0185 L950 r
C01089 F0190 L954 r
C01090 F0190 L954 r
C01091 F0190 L954 dist
C01092 F0190 L954 portals.some
C01093 F0191 L954 dist
C01094 F0190 L954 arr.push
C01095 F0190 L954 r
C01096 F0190 L954 r
C01097 F0192 L957 visible
C01098 F0192 L957 globalThis.PetalArt.landmark
C01099 F0193 L959 ROCK_GRID.get
C01100 F0193 L959 Math.floor
C01101 F0193 L959 Math.floor
C01102 F0193 L959 lookup.get
C01103 F0193 L959 out.push
C01104 F0194 L960 clamp
C01105 F0194 L960 clamp
C01106 F0194 L960 nearbyRocks
C01107 F0194 L960 Math.hypot
C01108 F0194 L960 clamp
C01109 F0194 L960 clamp
C01110 F0195 L961 moveInput
C01111 F0195 L961 Math.hypot
C01112 F0195 L961 Math.cos
C01113 F0195 L961 Math.sin
C01114 F0195 L961 Math.hypot
C01115 F0195 L961 pulse
C01116 F0195 L961 toast
C01117 F0196 L962 Math.min
C01118 F0196 L962 maxHP
C01119 F0196 L962 resetEquip
C01120 F0196 L962 Math.max
C01121 F0196 L962 petalReloadTime
C01122 F0196 L962 fillSlots
C01123 F0196 L962 saveGame
C01124 F0196 L962 toast
C01125 F0197 L964 activeBoon
C01126 F0197 L964 syner
C01127 F0197 L964 syner
C01128 F0197 L964 Math.max
C01129 F0197 L964 rootedArmor
C01130 F0197 L964 floating
```

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C01131 F0197 L964 floating
C01132 F0197 L964 Math.ceil
C01133 F0197 L964 die
C01134 F0198 L965 Math.max
C01135 F0199 L966 Math.min
C01136 F0199 L966 game.random?.loot||Math.random
C01137 F0199 L966 Math.max
C01138 F0200 L966 Math.min
C01139 F0201 L966 armorOf
C01140 F0201 L966 Math.max
C01141 F0201 L966 floating
C01142 F0201 L966 Math.round
C01143 F0201 L966 recordMob
C01144 F0201 L966 registerComboKill
C01145 F0201 L966 toast
C01146 F0201 L966 toast
C01147 F0201 L966 Math.max
C01148 F0201 L966 Math.round
C01149 F0201 L966 toast
C01150 F0201 L966 Math.round
C01151 F0201 L966 floating
C01152 F0201 L966 xpNeed
C01153 F0201 L966 xpNeed
C01154 F0201 L966 effectiveZoom
C01155 F0201 L966 clamp
C01156 F0201 L966 maxHP
C01157 F0201 L966 ensureSlots
C01158 F0201 L966 toast
C01159 F0201 L966 Math.round
C01160 F0201 L966 pulse
C01161 F0201 L966 Math.floor
C01162 F0201 L966 game.random?.loot||Math.random
C01163 F0201 L966 chooseDropRank
C01164 F0201 L966 game.random?.loot||Math.random
C01165 F0201 L966 Math.min
C01166 F0201 L966 game.drops.push
C01167 F0201 L966 game.random?.visual||Math.random
C01168 F0201 L966 game.drops.push
C01169 F0201 L966 Math.max
C01170 F0201 L966 game.random?.loot||Math.random
C01171 F0201 L966 game.drops.push
C01172 F0201 L966 game.random?.visual||Math.random
C01173 F0201 L966 game.random?.visual||Math.random
C01174 F0201 L966 Math.max
C01175 F0201 L966 pulse
C01176 F0201 L966 saveGame
C01177 F0202 L967 Math.min
C01178 F0202 L967 Math.min
C01179 F0202 L967 Math.max
C01180 F0203 L968 Math.floor
C01181 F0204 L969 game.floats.shift
C01182 F0204 L969 game.floats.push
C01183 F0205 L970 game.random?.visual||Math.random
C01184 F0205 L970 game.random?.visual||Math.random
C01185 F0205 L970 game.particles.push
C01186 F0205 L970 Math.cos
C01187 F0205 L970 Math.sin
C01188 F0205 L970 game.random?.visual||Math.random
C01189 F0205 L970 game.random?.visual||Math.random
C01190 F0205 L970 game.particles.splice
```

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C01191 F0206 L971 saveGame
C01192 F0206 L971 clearTransientState
C01193 F0206 L971 game.random?.world||Math.random
C01194 F0206 L971 resetEquip
C01195 F0206 L971 toast
C01196 F0206 L971 syncHud
C01197 F0206 L971 saveGame
C01198 F0207 L972 $
C01199 F0207 L972 $('death').classList.remove
C01200 F0207 L972 $
C01201 F0207 L972 saveGame
C01202 F0208 L973 keys.has
C01203 F0208 L973 keys.has
C01204 F0208 L973 keys.has
C01205 F0208 L973 keys.has
C01206 F0208 L973 keys.has
C01207 F0208 L973 keys.has
C01208 F0208 L973 keys.has
C01209 F0208 L973 keys.has
C01210 F0208 L973 Math.hypot
C01211 F0208 L973 clamp
C01212 F0208 L973 clamp
C01213 F0208 L973 Math.hypot
C01214 F0210 L982 /bee|wasp|dragonfly|mosquito|drone|wing/i.test
C01215 F0210 L983 /ant|termite|mantis|centipede|scorpion|rat/i.test
C01216 F0210 L984 /beetle|scarab|ladybug|crab/i.test
C01217 F0210 L985 /spider/i.test
C01218 F0210 L986 /mecha|barrel/i.test
C01219 F0210 L987 /flower|bloom/i.test
C01220 F0210 L988 Number.isFinite
C01221 F0211 L991 mobCollider
C01222 F0211 L991 Math.hypot
C01223 F0211 L992 Math.max
C01224 F0211 L993 Math.cos
C01225 F0211 L993 Math.sin
C01226 F0211 L997 Math.hypot
C01227 F0212 L1002 mobCollider
C01228 F0212 L1004 Math.max
C01229 F0212 L1006 Math.cos
C01230 F0212 L1006 Math.sin
C01231 F0212 L1007 Math.abs
C01232 F0212 L1007 Math.abs
C01233 F0212 L1013 Math.max
C01234 F0213 L1024 mobCollider
C01235 F0213 L1024 Math.max
C01236 F0214 L1027 Math.atan2
C01237 F0214 L1028 Number.isFinite
C01238 F0214 L1029 Math.atan2
C01239 F0214 L1029 Math.sin
C01240 F0214 L1029 Math.cos
C01241 F0214 L1030 Math.min
C01242 F0215 L1032 Math.cos
C01243 F0215 L1032 Math.sin
C01244 F0216 L1033 keys.has
C01245 F0216 L1033 keys.has
C01246 F0216 L1033 keys.has
C01247 F0217 L1034 game.equipped.filter
C01248 F0219 L1035 clamp
C01249 F0219 L1035 petalDamageMultiplier
C01250 F0219 L1035 syner
```

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C01251 F0220 L1036 globalThis.PetalRules.criticalFactor
C01252 F0220 L1036 activeCopies
C01253 F0222 L1038 Math.max
C01254 F0222 L1038 actMode
C01255 F0222 L1038 updateWorldEvents
C01256 F0222 L1038 updateParty
C01257 F0222 L1038 bodyRadius
C01258 F0222 L1038 Math.max
C01259 F0222 L1038 Math.max
C01260 F0222 L1038 Math.max
C01261 F0222 L1038 updateShrines
C01262 F0222 L1038 moveInput
C01263 F0222 L1038 syner
C01264 F0222 L1038 syner
C01265 F0222 L1038 bodySpeedFactor
C01266 F0222 L1038 talentModifier
C01267 F0222 L1038 activeBoon
C01268 F0222 L1038 slideCircle
C01269 F0222 L1038 slideCircle
C01270 F0222 L1039 Math.max
C01271 F0222 L1039 Math.max
C01272 F0222 L1039 die
C01273 F0222 L1039 effectiveZoom
C01274 F0222 L1040 rootActive
C01275 F0222 L1040 Math.min
C01276 F0222 L1042 dist
C01277 F0222 L1042 Math.max
C01278 F0222 L1042 recordMob
C01279 F0222 L1042 Math.max
C01280 F0222 L1042 updateMobSpecial
C01281 F0222 L1042 Math.max
C01282 F0222 L1042 Math.max
C01283 F0222 L1042 shoot
C01284 F0222 L1042 Math.atan2
C01285 F0222 L1042 shoot
C01286 F0222 L1042 Math.cos
C01287 F0222 L1042 Math.sin
C01288 F0222 L1042 rnd
C01289 F0222 L1042 neutralMotion
C01290 F0222 L1042 rnd
C01291 F0222 L1042 rnd
C01292 F0222 L1042 rnd
C01293 F0222 L1042 Math.cos
C01294 F0222 L1042 Math.sin
C01295 F0222 L1042 Math.max
C01296 F0222 L1042 updateMobFacing
C01297 F0222 L1042 slideCircle
C01298 F0222 L1042 mobRockRadius
C01299 F0222 L1042 damageMob
C01300 F0222 L1044 Math.hypot
C01301 F0222 L1044 mobCircleOverlap
C01302 F0222 L1044 mobSupportRadius
C01303 F0222 L1044 Math.hypot
C01304 F0222 L1044 Math.max
C01305 F0222 L1044 slideCircle
C01306 F0222 L1044 damagePlayer
C01307 F0222 L1044 damageMob
C01308 F0222 L1044 Math.hypot
C01309 F0222 L1044 mobSupportRadius
C01310 F0222 L1044 mobSupportRadius
```

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C01311 F0222 L1046 slideCircle
C01312 F0222 L1046 mobRockRadius
C01313 F0222 L1046 Math.hypot
C01314 F0222 L1046 mobSupportRadius
C01315 F0222 L1046 mobSupportRadius
C01316 F0222 L1046 slideCircle
C01317 F0222 L1047 orbitRadius
C01318 F0222 L1047 Math.cos
C01319 F0222 L1047 Math.sin
C01320 F0222 L1047 lerp
C01321 F0222 L1047 globalThis.PetalRules.frameAlpha
C01322 F0222 L1047 lerp
C01323 F0222 L1047 globalThis.PetalRules.frameAlpha
C01324 F0222 L1047 e.lastHits.clear
C01325 F0222 L1047 pulse
C01326 F0222 L1047 nearbyRocks
C01327 F0222 L1047 dist
C01328 F0222 L1047 petalContactRadius
C01329 F0222 L1047 petalReloadTime
C01330 F0222 L1049 mobCircleOverlap
C01331 F0222 L1049 petalContactRadius
C01332 F0222 L1049 e.lastHits.get
C01333 F0222 L1049 criticalMultiplier
C01334 F0222 L1049 rawPetalDamage
C01335 F0222 L1049 damageMob
C01336 F0222 L1049 Math.max
C01337 F0222 L1049 e.lastHits.set
C01338 F0222 L1049 petalReloadTime
C01339 F0222 L1049 floating
C01340 F0222 L1049 Math.round
C01341 F0222 L1049 clamp
C01342 F0222 L1049 Math.max
C01343 F0222 L1049 maxHP
C01344 F0222 L1049 dist
C01345 F0222 L1049 damageMob
C01346 F0222 L1049 globalThis.PetalRules.slowFactor
C01347 F0222 L1049 Math.max
C01348 F0222 L1049 activeCopies
C01349 F0222 L1049 Math.hypot
C01350 F0222 L1049 pulse
C01351 F0222 L1050 dist
C01352 F0222 L1050 Math.atan2
C01353 F0222 L1050 game.friendlyShots.push
C01354 F0222 L1050 Math.cos
C01355 F0222 L1050 Math.sin
C01356 F0222 L1050 petalDamageMultiplier
C01357 F0222 L1050 syner
C01358 F0222 L1052 syner
C01359 F0222 L1052 maxHP
C01360 F0222 L1052 Math.min
C01361 F0222 L1052 maxHP
C01362 F0222 L1052 syner
C01363 F0222 L1054 recoverPetalDurability
C01364 F0222 L1055 maxHP
C01365 F0222 L1055 Math.min
C01366 F0222 L1055 maxHP
C01367 F0222 L1056 maxHP
C01368 F0222 L1056 Math.min
C01369 F0222 L1056 maxHP
C01370 F0222 L1057 Math.max
```

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C01371 F0222 L1057 maxHP
C01372 F0222 L1057 Math.round
C01373 F0222 L1057 Math.min
C01374 F0222 L1057 maxHP
C01375 F0222 L1057 floating
C01376 F0222 L1057 Math.round
C01377 F0222 L1057 pulse
C01378 F0222 L1058 nearbyRocks(map.rocks,b.x,b.y).some
C01379 F0222 L1058 nearbyRocks
C01380 F0223 L1058 dist
C01381 F0222 L1058 dist
C01382 F0222 L1058 screenFx
C01383 F0222 L1058 petalReloadTime
C01384 F0222 L1058 dist
C01385 F0222 L1058 damagePlayer
C01386 F0222 L1058 floating
C01387 F0222 L1058 screenFx
C01388 F0222 L1058 game.bullets.filter
C01389 F0222 L1059 Math.hypot
C01390 F0222 L1059 nearbyRocks(map.rocks,shot.x,shot.y).some
C01391 F0222 L1059 nearbyRocks
C01392 F0225 L1059 dist
C01393 F0222 L1059 shot.hitIds.includes
C01394 F0222 L1059 dist
C01395 F0222 L1059 damageMob
C01396 F0222 L1059 shot.hitIds.push
C01397 F0222 L1059 shot.hitIds.includes
C01398 F0222 L1059 dist
C01399 F0222 L1059 Math.atan2
C01400 F0222 L1059 Math.cos
C01401 F0222 L1059 Math.sin
C01402 F0222 L1059 game.friendlyShots.filter
C01403 F0222 L1060 dist
C01404 F0222 L1060 magnetRadius
C01405 F0222 L1060 clamp
C01406 F0222 L1060 Math.max
C01407 F0222 L1060 dist
C01408 F0222 L1060 addBag
C01409 F0222 L1060 floating
C01410 F0222 L1060 toast
C01411 F0222 L1060 game.drops.filter
C01412 F0222 L1061 game.particles.filter
C01413 F0222 L1061 game.floats.filter
C01414 F0222 L1062 map.living.filter
C01415 F0222 L1062 spawnMob
C01416 F0222 L1062 updatePortals
C01417 F0222 L1063 lerp
C01418 F0222 L1063 clamp
C01419 F0222 L1063 Math.max
C01420 F0222 L1063 globalThis.PetalRules.frameAlpha
C01421 F0222 L1063 lerp
C01422 F0222 L1063 clamp
C01423 F0222 L1063 Math.max
C01424 F0222 L1063 globalThis.PetalRules.frameAlpha
C01425 F0222 L1063 syncHud
C01426 F0222 L1063 saveGame
C01427 F0231 L1064 game.bullets.push
C01428 F0232 L1065 pulse
C01429 F0233 L1066 $('portalBanner').classList.add
C01430 F0233 L1066 $
```

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C01431 F0233 L1066 dist
C01432 F0233 L1066 $('portalBanner').classList.add
C01433 F0233 L1066 $
C01434 F0233 L1066 $('portalBanner').classList.remove
C01435 F0233 L1066 $
C01436 F0233 L1066 $
C01437 F0233 L1066 $
C01438 F0233 L1066 clamp
C01439 F0233 L1066 transition
C01440 F0233 L1066 $('portalBanner').classList.add
C01441 F0233 L1066 $
C01442 F0236 L1069 ctx.beginPath
C01443 F0236 L1069 ctx.arc
C01444 F0236 L1069 ctx.fill
C01445 F0236 L1069 ctx.stroke
C01446 F0237 L1070 visible
C01447 F0237 L1070 shrineCooldown
C01448 F0237 L1070 ctx.save
C01449 F0237 L1070 ctx.translate
C01450 F0237 L1070 globalThis.PetalArt.halo
C01451 F0237 L1070 Math.sin
C01452 F0237 L1070 circle
C01453 F0237 L1070 circle
C01454 F0237 L1070 ctx.fillText
C01455 F0237 L1070 ctx.fillText
C01456 F0237 L1070 Math.ceil
C01457 F0237 L1070 shrineCooldown
C01458 F0237 L1070 ctx.restore
C01459 F0238 L1071 ctx.fillRect
C01460 F0238 L1071 ctx.save
C01461 F0238 L1071 ctx.scale
C01462 F0238 L1071 ctx.translate
C01463 F0238 L1071 drawRegionBands
C01464 F0238 L1071 paintTerrain
C01465 F0238 L1071 ctx.beginPath
C01466 F0238 L1071 Math.floor
C01467 F0238 L1071 Math.min
C01468 F0238 L1071 ctx.moveTo
C01469 F0238 L1071 Math.max
C01470 F0238 L1071 ctx.lineTo
C01471 F0238 L1071 Math.min
C01472 F0238 L1071 Math.floor
C01473 F0238 L1071 Math.min
C01474 F0238 L1071 ctx.moveTo
C01475 F0238 L1071 Math.max
C01476 F0238 L1071 ctx.lineTo
C01477 F0238 L1071 Math.min
C01478 F0238 L1071 ctx.stroke
C01479 F0238 L1071 ctx.strokeRect
C01480 F0238 L1071 ctx.strokeRect
C01481 F0238 L1072 drawLandmark
C01482 F0238 L1072 drawShrine
C01483 F0238 L1072 visible
C01484 F0238 L1072 drawDecor
C01485 F0238 L1073 visible
C01486 F0238 L1073 drawRock
C01487 F0238 L1074 visible
C01488 F0238 L1074 drawPortal
C01489 F0238 L1075 visible
C01490 F0238 L1075 drawLoot
```

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C01491 F0238 L1075 ctx.save
C01492 F0238 L1075 ctx.setLineDash
C01493 F0238 L1075 ctx.beginPath
C01494 F0238 L1075 ctx.arc
C01495 F0238 L1075 ctx.stroke
C01496 F0238 L1075 ctx.restore
C01497 F0238 L1076 visible
C01498 F0238 L1076 drawMob
C01499 F0238 L1076 drawMobHitbox
C01500 F0238 L1076 dist
C01501 F0238 L1076 ctx.save
C01502 F0238 L1076 ctx.setLineDash
C01503 F0238 L1076 ctx.beginPath
C01504 F0238 L1076 ctx.moveTo
C01505 F0238 L1076 ctx.lineTo
C01506 F0238 L1076 ctx.stroke
C01507 F0238 L1076 ctx.restore
C01508 F0238 L1077 visible
C01509 F0238 L1077 circle
C01510 F0238 L1077 circle
C01511 F0238 L1078 visible
C01512 F0238 L1078 circle
C01513 F0238 L1078 circle
C01514 F0238 L1079 drawRemotePlayers
C01515 F0238 L1079 drawFlower
C01516 F0238 L1079 drawPetal
C01517 F0238 L1080 clamp
C01518 F0238 L1080 circle
C01519 F0238 L1080 clamp
C01520 F0238 L1080 ctx.fillText
C01521 F0238 L1080 ctx.fillText
C01522 F0238 L1080 ctx.restore
C01523 F0238 L1080 globalThis.PetalArt.lighting
C01524 F0238 L1080 drawNavArrow
C01525 F0239 L1081 ctx.save
C01526 F0239 L1081 ctx.beginPath
C01527 F0239 L1081 ctx.ellipse
C01528 F0239 L1081 ctx.setLineDash
C01529 F0239 L1081 ctx.stroke
C01530 F0239 L1081 ctx.setLineDash
C01531 F0239 L1081 clamp
C01532 F0239 L1081 ctx.fillRect
C01533 F0239 L1081 ctx.fillText
C01534 F0239 L1081 ctx.restore
C01535 F0240 L1082 ctx.save
C01536 F0240 L1082 ctx.translate
C01537 F0240 L1082 ctx.rotate
C01538 F0240 L1082 ctx.beginPath
C01539 F0240 L1082 ctx.moveTo
C01540 F0240 L1082 ctx.lineTo
C01541 F0240 L1082 ctx.moveTo
C01542 F0240 L1082 ctx.lineTo
C01543 F0240 L1082 ctx.stroke
C01544 F0240 L1082 circle
C01545 F0240 L1082 ctx.beginPath
C01546 F0240 L1082 ctx.arc
C01547 F0240 L1082 ctx.stroke
C01548 F0240 L1082 circle
C01549 F0240 L1082 ctx.beginPath
C01550 F0240 L1082 ctx.arc
```

```
C01551 F0240 L1082 ctx.stroke
C01552 F0240 L1082 ctx.beginPath
C01553 F0240 L1082 ctx.moveTo
C01554 F0240 L1082 ctx.lineTo
C01555 F0240 L1082 ctx.lineTo
C01556 F0240 L1082 ctx.stroke
C01557 F0240 L1082 ctx.fillRect
C01558 F0240 L1082 circle
C01559 F0240 L1082 ctx.beginPath
C01560 F0240 L1082 ctx.moveTo
C01561 F0240 L1082 ctx.lineTo
C01562 F0240 L1082 ctx.moveTo
C01563 F0240 L1082 ctx.lineTo
C01564 F0240 L1082 ctx.stroke
C01565 F0240 L1082 ctx.restore
C01566 F0241 L1083 ctx.save
C01567 F0241 L1083 ctx.translate
C01568 F0241 L1083 circle
C01569 F0241 L1083 circle
C01570 F0241 L1083 circle
C01571 F0241 L1083 circle
C01572 F0241 L1083 circle
C01573 F0241 L1083 Math.cos
C01574 F0241 L1083 Math.sin
C01575 F0241 L1083 circle
C01576 F0241 L1083 circle
C01577 F0241 L1083 circle
C01578 F0241 L1083 Math.cos
C01579 F0241 L1083 Math.sin
C01580 F0241 L1083 circle
C01581 F0241 L1083 circle
C01582 F0241 L1083 circle
C01583 F0241 L1083 ctx.beginPath
C01584 F0241 L1083 ctx.moveTo
C01585 F0241 L1083 ctx.lineTo
C01586 F0241 L1083 ctx.lineTo
C01587 F0241 L1083 ctx.stroke
C01588 F0241 L1083 ctx.fillRect
C01589 F0241 L1083 ctx.strokeRect
C01590 F0241 L1083 circle
C01591 F0241 L1083 ctx.fillRect
C01592 F0241 L1083 ctx.strokeRect
C01593 F0241 L1083 ctx.beginPath
C01594 F0241 L1083 ctx.moveTo
C01595 F0241 L1083 ctx.lineTo
C01596 F0241 L1083 ctx.stroke
C01597 F0241 L1083 circle
C01598 F0241 L1083 circle
C01599 F0241 L1083 ctx.restore
C01600 F0241 L1083 globalThis.PetalArt.rock
C01601 F0242 L1084 Math.sin
C01602 F0242 L1084 ctx.save
C01603 F0242 L1084 ctx.translate
C01604 F0242 L1084 circle
C01605 F0242 L1084 ctx.beginPath
C01606 F0242 L1084 Math.cos
C01607 F0242 L1084 Math.sin
C01608 F0242 L1084 ctx.moveTo
C01609 F0242 L1084 ctx.lineTo
C01610 F0242 L1084 ctx.closePath
```

```
C01611 F0242 L1084 ctx.stroke
C01612 F0242 L1084 circle
C01613 F0242 L1084 circle
C01614 F0242 L1084 ctx.fillText
C01615 F0242 L1084 ctx.fillText
C01616 F0242 L1084 ctx.restore
C01617 F0243 L1085 ctx.save
C01618 F0243 L1085 ctx.translate
C01619 F0243 L1085 Math.sin
C01620 F0243 L1085 ctx.rotate
C01621 F0243 L1085 circle
C01622 F0243 L1085 ctx.beginPath
C01623 F0243 L1085 ctx.roundRect
C01624 F0243 L1085 ctx.fill
C01625 F0243 L1085 ctx.stroke
C01626 F0243 L1085 ctx.fillText
C01627 F0243 L1085 ctx.restore
C01628 F0244 L1086 ctx.save
C01629 F0244 L1086 ctx.translate
C01630 F0244 L1086 Math.min
C01631 F0244 L1086 ctx.beginPath
C01632 F0244 L1086 ctx.arc
C01633 F0244 L1086 ctx.stroke
C01634 F0244 L1086 circle
C01635 F0244 L1086 ctx.save
C01636 F0244 L1086 ctx.rotate
C01637 F0244 L1086 ctx.beginPath
C01638 F0244 L1086 ctx.ellipse
C01639 F0244 L1086 ctx.fill
C01640 F0244 L1086 ctx.stroke
C01641 F0244 L1086 ctx.restore
C01642 F0244 L1086 circle
C01643 F0244 L1086 circle
C01644 F0244 L1086 circle
C01645 F0244 L1086 ctx.beginPath
C01646 F0244 L1086 ctx.arc
C01647 F0244 L1086 ctx.stroke
C01648 F0244 L1086 clamp
C01649 F0244 L1086 maxHP
C01650 F0244 L1086 ctx.fillRect
C01651 F0244 L1086 ctx.fillRect
C01652 F0244 L1086 ctx.restore
C01653 F0245 L1087 ctx.save
C01654 F0245 L1087 ctx.translate
C01655 F0245 L1087 ctx.rotate
C01656 F0245 L1087 circle
C01657 F0245 L1087 ctx.beginPath
C01658 F0245 L1087 ctx.moveTo
C01659 F0245 L1087 ctx.lineTo
C01660 F0245 L1087 ctx.lineTo
C01661 F0245 L1087 ctx.lineTo
C01662 F0245 L1087 ctx.closePath
C01663 F0245 L1087 ctx.fill
C01664 F0245 L1087 ctx.stroke
C01665 F0245 L1087 circle
C01666 F0245 L1087 circle
C01667 F0245 L1087 circle
C01668 F0245 L1087 ctx.beginPath
C01669 F0245 L1087 ctx.moveTo
C01670 F0245 L1087 ctx.lineTo
```

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C01671 F0245 L1087 ctx.lineTo
C01672 F0245 L1087 ctx.lineTo
C01673 F0245 L1087 ctx.closePath
C01674 F0245 L1087 materialFill
C01675 F0245 L1087 ctx.fill
C01676 F0245 L1087 ctx.stroke
C01677 F0245 L1087 ctx.beginPath
C01678 F0245 L1087 ctx.moveTo
C01679 F0245 L1087 ctx.lineTo
C01680 F0245 L1087 ctx.moveTo
C01681 F0245 L1087 ctx.lineTo
C01682 F0245 L1087 ctx.stroke
C01683 F0245 L1087 ctx.save
C01684 F0245 L1087 ctx.rotate
C01685 F0245 L1087 ctx.beginPath
C01686 F0245 L1087 ctx.moveTo
C01687 F0245 L1087 ctx.lineTo
C01688 F0245 L1087 ctx.moveTo
C01689 F0245 L1087 ctx.lineTo
C01690 F0245 L1087 ctx.moveTo
C01691 F0245 L1087 ctx.lineTo
C01692 F0245 L1087 ctx.stroke
C01693 F0245 L1087 ctx.restore
C01694 F0245 L1087 ctx.beginPath
C01695 F0245 L1087 ctx.moveTo
C01696 F0245 L1087 ctx.lineTo
C01697 F0245 L1087 ctx.lineTo
C01698 F0245 L1087 ctx.lineTo
C01699 F0245 L1087 ctx.closePath
C01700 F0245 L1087 ctx.fill
C01701 F0245 L1087 ctx.stroke
C01702 F0245 L1087 ctx.beginPath
C01703 F0245 L1087 ctx.moveTo
C01704 F0245 L1087 ctx.quadraticCurveTo
C01705 F0245 L1087 ctx.stroke
C01706 F0245 L1087 ctx.beginPath
C01707 F0245 L1087 ctx.ellipse
C01708 F0245 L1087 ctx.fill
C01709 F0245 L1087 ctx.stroke
C01710 F0245 L1087 circle
C01711 F0245 L1087 circle
C01712 F0245 L1087 circle
C01713 F0245 L1087 circle
C01714 F0245 L1087 circle
C01715 F0245 L1087 circle
C01716 F0245 L1087 ['leaf','iris','poison'].includes
C01717 F0245 L1087 ctx.beginPath
C01718 F0245 L1087 ctx.ellipse
C01719 F0245 L1087 ctx.fill
C01720 F0245 L1087 ctx.stroke
C01721 F0245 L1087 ctx.beginPath
C01722 F0245 L1087 ctx.moveTo
C01723 F0245 L1087 ctx.lineTo
C01724 F0245 L1087 ctx.lineTo
C01725 F0245 L1087 ctx.closePath
C01726 F0245 L1087 ctx.fill
C01727 F0245 L1087 ctx.stroke
C01728 F0245 L1087 circle
C01729 F0245 L1087 circle
C01730 F0245 L1087 ctx.save
```

```
C01731 F0245 L1087 ctx.rotate
C01732 F0245 L1087 ctx.fillRect
C01733 F0245 L1087 ctx.restore
C01734 F0245 L1087 circle
C01735 F0245 L1087 circle
C01736 F0245 L1087 ctx.save
C01737 F0245 L1087 ctx.rotate
C01738 F0245 L1087 ctx.beginPath
C01739 F0245 L1087 ctx.ellipse
C01740 F0245 L1087 ctx.fill
C01741 F0245 L1087 ctx.stroke
C01742 F0245 L1087 ctx.restore
C01743 F0245 L1087 circle
C01744 F0245 L1087 ctx.beginPath
C01745 F0245 L1087 ctx.arc
C01746 F0245 L1087 ctx.stroke
C01747 F0245 L1087 ctx.save
C01748 F0245 L1087 ctx.setLineDash
C01749 F0245 L1087 ctx.beginPath
C01750 F0245 L1087 ctx.arc
C01751 F0245 L1087 petalContactRadius
C01752 F0245 L1087 ctx.stroke
C01753 F0245 L1087 ctx.restore
C01754 F0245 L1087 ctx.restore
C01755 F0246 L1089 mobCollider
C01756 F0246 L1089 ctx.save
C01757 F0246 L1089 ctx.translate
C01758 F0246 L1089 ctx.rotate
C01759 F0246 L1090 ctx.beginPath
C01760 F0246 L1090 ctx.ellipse
C01761 F0246 L1090 ctx.setLineDash
C01762 F0246 L1090 ctx.stroke
C01763 F0246 L1091 ctx.beginPath
C01764 F0246 L1091 ctx.moveTo
C01765 F0246 L1091 ctx.lineTo
C01766 F0246 L1091 ctx.stroke
C01767 F0246 L1091 ctx.restore
C01768 F0247 L1094 mobCollider
C01769 F0247 L1095 ctx.beginPath
C01770 F0247 L1095 ctx.ellipse
C01771 F0247 L1096 ctx.fill
C01772 F0247 L1097 ctx.stroke
C01773 F0001 L1100 matchMedia
C01774 F0248 L1101 parseInt
C01775 F0248 L1101 hex.slice
C01776 F0248 L1101 clamp
C01777 F0248 L1101 Math.abs
C01778 F0248 L1101 [n>>16,(n>>8)&255,n&255].map(c=>Math.round(c+(target-c)*f).toString(16).padStart(2,'0')).join
C01779 F0248 L1101 [n>>16,(n>>8)&255,n&255].map
C01780 F0249 L1101 Math.round(c+(target-c)*f).toString(16).padStart
C01781 F0249 L1101 Math.round(c+(target-c)*f).toString
C01782 F0249 L1101 Math.round
C01783 F0250 L1102 ctx.createRadialGradient
C01784 F0250 L1102 g.addColorStop
C01785 F0250 L1102 tint
C01786 F0250 L1102 g.addColorStop
C01787 F0250 L1102 g.addColorStop
C01788 F0250 L1102 tint
C01789 F0251 L1105 terrainMaterials.get
C01790 F0251 L1106 document.createElement
```

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C01791 F0251 L1106 tile.getContext
C01792 F0251 L1106 SEED
C01793 F0251 L1106 globalThis.PetalArt.terrain
C01794 F0251 L1107 random
C01795 F0251 L1107 random
C01796 F0251 L1107 random
C01797 F0251 L1107 c.beginPath
C01798 F0251 L1107 c.ellipse
C01799 F0251 L1107 random
C01800 F0251 L1107 c.fill
C01801 F0251 L1108 random
C01802 F0251 L1108 random
C01803 F0251 L1108 c.beginPath
C01804 F0251 L1108 c.moveTo
C01805 F0251 L1108 c.quadraticCurveTo
C01806 F0251 L1108 c.stroke
C01807 F0251 L1109 ctx.createPattern
C01808 F0251 L1109 terrainMaterials.set
C01809 F0251 L1111 ctx.save
C01810 F0251 L1111 ctx.fillRect
C01811 F0251 L1111 ctx.restore
C01812 F0252 L1113 mobCollider
C01813 F0252 L1113 Math.sin
C01814 F0252 L1113 clamp
C01815 F0253 L1115 m.t.toLowerCase
C01816 F0253 L1115 creatureVisualProfile
C01817 F0253 L1116 globalThis.PetalArt.traits
C01818 F0253 L1117 tint
C01819 F0253 L1118 t.includes
C01820 F0253 L1118 ctx.beginPath
C01821 F0253 L1118 ctx.moveTo
C01822 F0253 L1118 ctx.lineTo
C01823 F0253 L1118 ctx.lineTo
C01824 F0253 L1118 ctx.stroke
C01825 F0253 L1118 circle
C01826 F0253 L1118 tint
C01827 F0253 L1119 ctx.save
C01828 F0253 L1119 ctx.translate
C01829 F0253 L1119 ctx.rotate
C01830 F0253 L1119 ctx.beginPath
C01831 F0253 L1119 ctx.ellipse
C01832 F0253 L1119 ctx.fill
C01833 F0253 L1119 ctx.stroke
C01834 F0253 L1119 ctx.beginPath
C01835 F0253 L1119 ctx.moveTo
C01836 F0253 L1119 ctx.lineTo
C01837 F0253 L1119 ctx.moveTo
C01838 F0253 L1119 ctx.lineTo
C01839 F0253 L1119 ctx.stroke
C01840 F0253 L1119 ctx.restore
C01841 F0253 L1120 tint
C01842 F0253 L1120 ctx.beginPath
C01843 F0253 L1120 ctx.moveTo
C01844 F0253 L1120 ctx.quadraticCurveTo
C01845 F0253 L1120 ctx.stroke
C01846 F0253 L1120 circle
C01847 F0253 L1120 materialFill
C01848 F0253 L1120 tint
C01849 F0253 L1120 ctx.beginPath
C01850 F0253 L1120 ctx.moveTo
```

```
C01851 F0253 L1120 ctx.lineTo
C01852 F0253 L1120 ctx.stroke
C01853 F0253 L1121 tint
C01854 F0253 L1121 ctx.beginPath
C01855 F0253 L1121 ctx.moveTo
C01856 F0253 L1121 ctx.bezierCurveTo
C01857 F0253 L1121 ctx.stroke
C01858 F0253 L1122 tint
C01859 F0253 L1122 ctx.beginPath
C01860 F0253 L1122 ctx.moveTo
C01861 F0253 L1122 Math.cos
C01862 F0253 L1122 Math.sin
C01863 F0253 L1122 ctx.lineTo
C01864 F0253 L1122 Math.cos
C01865 F0253 L1122 Math.sin
C01866 F0253 L1122 ctx.stroke
C01867 F0253 L1123 /flower|bloom/.test
C01868 F0253 L1123 ctx.save
C01869 F0253 L1123 ctx.rotate
C01870 F0253 L1123 materialFill
C01871 F0253 L1123 tint
C01872 F0253 L1123 ctx.beginPath
C01873 F0253 L1123 ctx.ellipse
C01874 F0253 L1123 ctx.fill
C01875 F0253 L1123 ctx.stroke
C01876 F0253 L1123 tint
C01877 F0253 L1123 ctx.beginPath
C01878 F0253 L1123 ctx.moveTo
C01879 F0253 L1123 ctx.lineTo
C01880 F0253 L1123 ctx.stroke
C01881 F0253 L1123 ctx.restore
C01882 F0253 L1124 drawCreatureBody
C01883 F0253 L1124 materialFill
C01884 F0253 L1124 tint
C01885 F0253 L1126 ctx.save
C01886 F0253 L1126 ctx.beginPath
C01887 F0253 L1126 ctx.ellipse
C01888 F0253 L1126 ctx.clip
C01889 F0253 L1127 tint
C01890 F0253 L1127 ctx.fillRect
C01891 F0253 L1127 tint
C01892 F0253 L1127 ctx.strokeRect
C01893 F0253 L1127 circle
C01894 F0253 L1127 circle
C01895 F0253 L1127 circle
C01896 F0253 L1128 ctx.beginPath
C01897 F0253 L1128 ctx.moveTo
C01898 F0253 L1128 ctx.lineTo
C01899 F0253 L1128 ctx.stroke
C01900 F0253 L1128 circle
C01901 F0253 L1128 circle
C01902 F0253 L1129 tint
C01903 F0253 L1129 ctx.beginPath
C01904 F0253 L1129 ctx.moveTo
C01905 F0253 L1129 ctx.lineTo
C01906 F0253 L1129 ctx.stroke
C01907 F0253 L1130 /ant|termite|centipede|scorpion/.test
C01908 F0253 L1130 tint
C01909 F0253 L1130 ctx.beginPath
C01910 F0253 L1130 ctx.ellipse
```

```
C01911 F0253 L1130 ctx.stroke
C01912 F0253 L1131 t.includes
C01913 F0253 L1131 circle
C01914 F0253 L1131 tint
C01915 F0253 L1131 tint
C01916 F0253 L1131 ctx.beginPath
C01917 F0253 L1131 ctx.moveTo
C01918 F0253 L1131 ctx.lineTo
C01919 F0253 L1131 ctx.lineTo
C01920 F0253 L1131 ctx.stroke
C01921 F0253 L1132 /beetle|scarab/.test
C01922 F0253 L1132 tint
C01923 F0253 L1132 ctx.beginPath
C01924 F0253 L1132 ctx.moveTo
C01925 F0253 L1132 ctx.lineTo
C01926 F0253 L1132 ctx.moveTo
C01927 F0253 L1132 ctx.quadraticCurveTo
C01928 F0253 L1132 ctx.stroke
C01929 F0253 L1133 ctx.beginPath
C01930 F0253 L1133 ctx.ellipse
C01931 F0253 L1133 ctx.fill
C01932 F0253 L1133 ctx.beginPath
C01933 F0253 L1133 ctx.ellipse
C01934 F0253 L1133 ctx.stroke
C01935 F0253 L1134 ctx.restore
C01936 F0253 L1135 globalThis.PetalArt.surface
C01937 F0253 L1136 globalThis.PetalArt.anatomy
C01938 F0253 L1137 clamp
C01939 F0253 L1137 circle
C01940 F0253 L1137 circle
C01941 F0253 L1137 circle
C01942 F0253 L1138 ctx.beginPath
C01943 F0253 L1138 ctx.moveTo
C01944 F0253 L1138 ctx.lineTo
C01945 F0253 L1138 ctx.lineTo
C01946 F0253 L1138 ctx.lineTo
C01947 F0253 L1138 ctx.lineTo
C01948 F0253 L1138 ctx.stroke
C01949 F0253 L1139 drawCreatureBody
C01950 F0254 L1142 ctx.save
C01951 F0254 L1142 ctx.translate
C01952 F0254 L1142 ctx.beginPath
C01953 F0254 L1142 ctx.ellipse
C01954 F0254 L1142 ctx.fill
C01955 F0254 L1142 ctx.rotate
C01956 F0254 L1143 circle
C01957 F0254 L1144 drawCreatureModel
C01958 F0254 L1144 ctx.restore
C01959 F0254 L1144 globalThis.PetalArt.status
C01960 F0254 L1144 globalThis.PetalArt.telegraph
C01961 F0254 L1144 specialProfileFor
C01962 F0254 L1145 dist
C01963 F0254 L1146 ctx.save
C01964 F0254 L1146 ctx.translate
C01965 F0254 L1146 clamp
C01966 F0254 L1146 clamp
C01967 F0254 L1147 ctx.beginPath
C01968 F0254 L1147 ctx.roundRect
C01969 F0254 L1147 ctx.fill
C01970 F0254 L1147 ctx.fillRect
```

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C01971 F0254 L1147 Math.max
C01972 F0254 L1147 Math.max
C01973 F0254 L1148 circle
C01974 F0254 L1149 circle
C01975 F0254 L1150 clamp
C01976 F0254 L1150 ctx.strokeText
C01977 F0254 L1150 ctx.fillText
C01978 F0254 L1150 Math.round
C01979 F0254 L1150 ctx.strokeText
C01980 F0254 L1150 ctx.fillText
C01981 F0254 L1150 ctx.restore
C01982 F0255 L1153 nextHop
C01983 F0255 L1153 game.mapData[game.zone].portals.find
C01984 F0255 L1153 proj
C01985 F0255 L1153 Math.atan2
C01986 F0255 L1153 clamp
C01987 F0255 L1153 clamp
C01988 F0255 L1153 ctx.save
C01989 F0255 L1153 ctx.translate
C01990 F0255 L1153 ctx.rotate
C01991 F0255 L1153 ctx.beginPath
C01992 F0255 L1153 ctx.moveTo
C01993 F0255 L1153 ctx.lineTo
C01994 F0255 L1153 ctx.lineTo
C01995 F0255 L1153 ctx.lineTo
C01996 F0255 L1153 ctx.closePath
C01997 F0255 L1153 ctx.fill
C01998 F0255 L1153 ctx.stroke
C01999 F0255 L1153 ctx.restore
C02000 F0257 L1154 q.shift
C02001 F0257 L1154 seen.has
C02002 F0257 L1154 seen.add
C02003 F0257 L1154 q.push
C02004 F0258 L1155 mctx.fillRect
C02005 F0258 L1155 mctx.fillRect
C02006 F0258 L1156 mctx.beginPath
C02007 F0258 L1156 mctx.ellipse
C02008 F0258 L1156 mctx.setLineDash
C02009 F0258 L1156 mctx.stroke
C02010 F0258 L1156 mctx.setLineDash
C02011 F0258 L1157 mctx.fillRect
C02012 F0258 L1157 mctx.beginPath
C02013 F0258 L1157 mctx.arc
C02014 F0258 L1157 mctx.fill
C02015 F0258 L1157 mctx.save
C02016 F0258 L1157 Math.sin
C02017 F0258 L1157 mctx.beginPath
C02018 F0258 L1157 mctx.arc
C02019 F0258 L1157 mctx.fill
C02020 F0258 L1157 mctx.restore
C02021 F0258 L1157 mctx.save
C02022 F0258 L1157 shrineCooldown
C02023 F0258 L1157 mctx.fillRect
C02024 F0258 L1157 mctx.restore
C02025 F0258 L1157 Object.values
C02026 F0258 L1157 mctx.fillRect
C02027 F0258 L1157 mctx.beginPath
C02028 F0258 L1157 mctx.arc
C02029 F0258 L1157 mctx.fill
C02030 F0258 L1157 mctx.beginPath
```

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C02031 F0258 L1157 mctx.arc
C02032 F0258 L1157 mctx.fill
C02033 F0258 L1157 mctx.stroke
C02034 F0258 L1157 mctx.strokeRect
C02035 F0259 L1158 $
C02036 F0259 L1158 dist
C02037 F0259 L1158 hud.classList.toggle
C02038 F0259 L1158 hud.classList.toggle
C02039 F0259 L1158 $
C02040 F0259 L1158 $
C02041 F0259 L1158 clamp
C02042 F0259 L1158 $
C02043 F0259 L1158 Math.ceil
C02044 F0259 L1158 Math.max
C02045 F0260 L1159 document.querySelectorAll
C02046 F0260 L1159 skillRemaining
C02047 F0260 L1159 label.key.toUpperCase
C02048 F0260 L1159 Math.ceil
C02049 F0260 L1159 Object.entries(SET_FAMILIES).filter(([id])=>syner(id)).map
C02050 F0260 L1159 Object.entries(SET_FAMILIES).filter
C02051 F0260 L1159 Object.entries
C02052 F0261 L1159 syner
C02053 F0262 L1159 syner
C02054 F0260 L1159 $
C02055 F0260 L1159 names.join
C02056 F0260 L1159 $('eventRibbon').classList.toggle
C02057 F0260 L1159 $
C02058 F0260 L1159 $
C02059 F0260 L1159 Math.ceil
C02060 F0263 L1160 regionBand
C02061 F0263 L1160 updateSkillDock
C02062 F0263 L1160 updateCollectionGoal
C02063 F0263 L1160 $('partyBtn').classList.toggle
C02064 F0263 L1160 $
C02065 F0263 L1160 $
C02066 F0263 L1160 $('orbitBtn').setAttribute
C02067 F0263 L1160 $
C02068 F0263 L1160 $
C02069 F0263 L1160 clamp
C02070 F0263 L1160 maxHP
C02071 F0263 L1160 $
C02072 F0263 L1160 clamp
C02073 F0263 L1160 xpNeed
C02074 F0263 L1160 $
C02075 F0263 L1160 Math.ceil
C02076 F0263 L1160 Math.ceil
C02077 F0263 L1160 maxHP
C02078 F0263 L1160 $
C02079 F0263 L1160 Math.floor
C02080 F0263 L1160 xpNeed
C02081 F0263 L1160 $
C02082 F0263 L1160 $
C02083 F0263 L1160 $
C02084 F0263 L1160 Math.round
C02085 F0263 L1160 $
C02086 F0263 L1160 $
C02087 F0263 L1160 $
C02088 F0263 L1160 $
C02089 F0263 L1160 bandLabel
C02090 F0263 L1160 $
```

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C02091 F0263 L1160 $
C02092 F0263 L1160 $
C02093 F0263 L1160 Math.round
C02094 F0263 L1160 SLOT_MILESTONES.find
C02095 F0263 L1160 $
C02096 F0263 L1160 talentPoints
C02097 F0263 L1160 talentPoints
C02098 F0263 L1160 $
C02099 F0263 L1160 tag.classList.toggle
C02100 F0263 L1160 Math.ceil
C02101 F0263 L1160 Math.max
C02102 F0263 L1160 $
C02103 F0263 L1160 game.deckCooldown.toFixed
C02104 F0263 L1160 $
C02105 F0263 L1160 qi.classList.remove
C02106 F0263 L1160 Math.min
C02107 F0263 L1160 Math.min
C02108 F0263 L1160 $('dashTag').classList.remove
C02109 F0263 L1160 $
C02110 F0263 L1160 $
C02111 F0263 L1160 p.dashCooldown.toFixed
C02112 F0263 L1160 $('navGuide').classList.toggle
C02113 F0263 L1160 $
C02114 F0263 L1160 $
C02115 F0263 L1160 nextHop
C02116 F0263 L1161 $
C02117 F0263 L1161 e.cd.toFixed
C02118 F0263 L1161 Math.ceil
C02119 F0263 L1161 dom.querySelector
C02120 F0263 L1161 dom.classList.toggle
C02121 F0263 L1161 dom.setAttribute
C02122 F0263 L1161 dom.querySelector
C02123 F0263 L1161 clamp
C02124 F0263 L1161 petalReloadTime
C02125 F0263 L1161 dom.querySelector
C02126 F0263 L1161 clamp
C02127 F0263 L1162 dist
C02128 F0263 L1162 $
C02129 F0263 L1162 node.classList.toggle
C02130 F0263 L1162 Math.round
C02131 F0263 L1162 Math.ceil
C02132 F0263 L1162 updateBossHud
C02133 F0263 L1162 minimap
C02134 F0265 L1163 $
C02135 F0265 L1163 c.style.setProperty
C02136 F0265 L1163 c.replaceChildren
C02137 F0265 L1163 document.createElement
C02138 F0265 L1163 Math.round
C02139 F0265 L1163 Math.round
C02140 F0265 L1163 b.addEventListener
C02141 F0266 L1163 openPanel
C02142 F0265 L1163 c.appendChild
C02143 F0267 L1164 $
C02144 F0267 L1164 d.classList.add
C02145 F0267 L1164 clearTimeout
C02146 F0267 L1164 setTimeout
C02147 F0268 L1164 d.classList.remove
C02148 F0269 L1165 drawGateBackground
C02149 F0269 L1165 drawWorld
C02150 F0270 L1166 ctx.fillRect
```

```
C02151 F0270 L1166 SEED
C02152 F0270 L1166 r
C02153 F0270 L1166 r
C02154 F0270 L1166 r
C02155 F0270 L1166 circle
C02156 F0270 L1166 ctx.beginPath
C02157 F0270 L1166 ctx.moveTo
C02158 F0270 L1166 ctx.lineTo
C02159 F0270 L1166 ctx.stroke
C02160 F0271 L1167 Math.min
C02161 F0271 L1167 Math.min
C02162 F0271 L1167 Math.round
C02163 F0271 L1167 Math.round
C02164 F0271 L1167 ctx.setTransform
C02165 F0271 L1167 clamp
C02166 F0271 L1167 Math.max
C02167 F0271 L1167 clamp
C02168 F0271 L1167 Math.max
C02169 F0001 L1168 performance.now
C02170 F0272 L1168 Math.min
C02171 F0272 L1168 update
C02172 F0272 L1168 render
C02173 F0272 L1168 requestAnimationFrame
C02174 F0273 L1169 resetInputs
C02175 F0273 L1169 $('panelLayer').classList.remove
C02176 F0273 L1169 $
C02177 F0273 L1169 $
C02178 F0273 L1169 renderModal
C02179 F0273 L1169 $('panelClose').focus
C02180 F0273 L1169 $
C02181 F0274 L1170 document.body.classList.remove
C02182 F0274 L1170 $('toolsMenuBtn').setAttribute
C02183 F0274 L1170 $
C02184 F0274 L1170 $('panelLayer').classList.add
C02185 F0274 L1170 $
C02186 F0274 L1170 $
C02187 F0274 L1170 resetInputs
C02188 F0274 L1170 canvas.focus
C02189 F0275 L1171 showModal
C02190 F0276 L1172 itemOf
C02191 F0276 L1172 Number.isInteger
C02192 F0276 L1172 takeBag
C02193 F0276 L1172 toast
C02194 F0276 L1172 addBag
C02195 F0276 L1172 resetEquip
C02196 F0276 L1172 clamp
C02197 F0276 L1172 maxHP
C02198 F0276 L1172 effectiveZoom
C02199 F0276 L1172 fillSlots
C02200 F0276 L1172 saveGame
C02201 F0276 L1172 renderModal
C02202 F0276 L1172 toast
C02203 F0277 L1174 root.querySelectorAll
C02204 F0277 L1174 root.querySelectorAll
C02205 F0278 L1174 slots.forEach
C02206 F0279 L1174 s.classList.remove
C02207 F0277 L1174 slot.addEventListener
C02208 F0280 L1174 e.preventDefault
C02209 F0280 L1174 slot.classList.add
C02210 F0277 L1174 slot.addEventListener
```

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C02211 F0281 L1174 slot.classList.remove
C02212 F0277 L1174 slot.addEventListener
C02213 F0282 L1174 e.preventDefault
C02214 F0282 L1174 resetHighlights
C02215 F0282 L1174 e.dataTransfer.getData
C02216 F0282 L1174 equipInto
C02217 F0277 L1174 item.addEventListener
C02218 F0283 L1174 e.dataTransfer.setData
C02219 F0277 L1174 item.addEventListener
C02220 F0277 L1174 item.addEventListener
C02221 F0285 L1174 Math.hypot
C02222 F0285 L1174 document.createElement
C02223 F0285 L1174 itemOf
C02224 F0285 L1174 document.body.appendChild
C02225 F0285 L1174 event.preventDefault
C02226 F0285 L1174 resetHighlights
C02227 F0285 L1174 document.elementFromPoint(event.clientX,event.clientY)?.closest('[data-slot]')?.classList.add
C02228 F0285 L1174 document.elementFromPoint(event.clientX,event.clientY)?.closest
C02229 F0285 L1174 document.elementFromPoint
C02230 F0286 L1174 window.removeEventListener
C02231 F0286 L1174 window.removeEventListener
C02232 F0286 L1174 window.removeEventListener
C02233 F0286 L1174 document.elementFromPoint(event.clientX,event.clientY)?.closest
C02234 F0286 L1174 document.elementFromPoint
C02235 F0286 L1174 ghost?.remove
C02236 F0286 L1174 resetHighlights
C02237 F0286 L1174 performance.now
C02238 F0286 L1174 equipInto
C02239 F0284 L1174 window.addEventListener
C02240 F0284 L1174 window.addEventListener
C02241 F0284 L1174 window.addEventListener
C02242 F0287 L1179 (isCraft?craftSearch:bagSearch).trim().toLocaleLowerCase
C02243 F0287 L1179 (isCraft?craftSearch:bagSearch).trim
C02244 F0287 L1181 root.querySelectorAll
C02245 F0287 L1182 cards.slice().sort
C02246 F0287 L1182 cards.slice
C02247 F0288 L1183 Number
C02248 F0288 L1183 Number
C02249 F0288 L1183 Number
C02250 F0288 L1183 Number
C02251 F0288 L1185 a.dataset.name.localeCompare
C02252 F0288 L1186 a.dataset.name.localeCompare
C02253 F0287 L1188 ordered.forEach
C02254 F0289 L1188 String
C02255 F0287 L1191 Number
C02256 F0287 L1191 Number
C02257 F0287 L1192 card.dataset.name.toLocaleLowerCase().includes
C02258 F0287 L1192 card.dataset.name.toLocaleLowerCase
C02259 F0287 L1193 Number
C02260 F0287 L1194 isPetalProtected
C02261 F0287 L1197 root.querySelector
C02262 F0290 L1201 root.querySelector
C02263 F0290 L1201 root.querySelector
C02264 F0291 L1203 inventoryFilterCards
C02265 F0292 L1204 update
C02266 F0293 L1205 update
C02267 F0290 L1206 root.querySelector
C02268 F0294 L1207 update
C02269 F0290 L1208 root.querySelector
C02270 F0295 L1208 update
```

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C02271 F0290 L1209 update
C02272 F0296 L1214 rarity.map((r,i)=>'<option value="'+i+'">'+r.en+"LOCALIZED_588da410"+r.name+'</option>').join
C02273 F0296 L1214 rarity.map
C02274 F0298 L1222 itemOf
C02275 F0298 L1222 Number
C02276 F0298 L1223 Number.isFinite
C02277 F0298 L1223 Math.min
C02278 F0298 L1223 Math.trunc
C02279 F0298 L1225 isPetalProtected
C02280 F0298 L1230 game.random?.craft||Math.random
C02281 F0298 L1231 Math.floor
C02282 F0298 L1231 game.random?.craft||Math.random
C02283 F0298 L1232 takeBag
C02284 F0298 L1234 addBag
C02285 F0299 L1240 Number.isInteger
C02286 F0299 L1240 itemOf
C02287 F0299 L1240 inventoryKey
C02288 F0300 L1245 inventoryKey
C02289 F0300 L1245 [...game.slots,...game.backupSlots].filter
C02290 F0302 L1247 Object.keys(zones).filter
C02291 F0302 L1247 Object.keys
C02292 F0303 L1247 zones[id].mob.includes
C02293 F0304 L1249 Object.keys(zones).filter
C02294 F0304 L1249 Object.keys
C02295 F0305 L1249 [...zones[id].mob,GUARDIANS[id]].some
C02296 F0306 L1249 mobs[t]?.drop.includes
C02297 F0307 L1252 sanitizeCollectionGoal
C02298 F0307 L1253 collectionOwned
C02299 F0307 L1253 saveGame
C02300 F0307 L1253 syncHud
C02301 F0307 L1253 renderModal
C02302 F0307 L1254 toast
C02303 F0308 L1257 $
C02304 F0308 L1258 node.classList.toggle
C02305 F0308 L1259 collectionOwned
C02306 F0308 L1259 collectionSources
C02307 F0308 L1260 sources.map(id=>zones[id].name).slice(0,2).join
C02308 F0308 L1260 sources.map(id=>zones[id].name).slice
C02309 F0308 L1260 sources.map
C02310 F0308 L1261 toast
C02311 F0310 L1265 collectionOwned
C02312 F0310 L1265 collectionSources(g.t).map(id=>'<button type="button"
data-goal-zone="'+id+'">'+zones[id].name+"LOCALIZED_5854f3d1").join
C02313 F0310 L1265 collectionSources(g.t).map
C02314 F0310 L1265 collectionSources
C02315 F0312 L1268 root.querySelectorAll('[data-goal-zone]').forEach
C02316 F0312 L1268 root.querySelectorAll
C02317 F0314 L1268 closeModal
C02318 F0314 L1268 syncHud
C02319 F0314 L1268 toast
C02320 F0312 L1269 $
C02321 F0312 L1269 $
C02322 F0315 L1269 saveGame
C02323 F0315 L1269 syncHud
C02324 F0315 L1269 renderModal
C02325 F0316 L1272 itemOf
C02326 F0316 L1273 Math.round
C02327 F0316 L1273 Math.round
C02328 F0317 L1276 itemOf
C02329 F0317 L1277 petalBaseStats
```

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C02330 F0317 L1278 petalBaseStats
C02331 F0317 L1278 inventoryKey
C02332 F0317 L1280
Object.entries(SET_FAMILIES).filter(([,family])=>family.types.includes(t)).map(([,family])=>family.name).join
C02333 F0317 L1280 Object.entries(SET_FAMILIES).filter(([,family])=>family.types.includes(t)).map
C02334 F0317 L1280 Object.entries(SET_FAMILIES).filter
C02335 F0317 L1280 Object.entries
C02336 F0319 L1280 family.types.includes
C02337 F0317 L1281 delta
C02338 F0317 L1281 delta
C02339 F0317 L1281 isPetalProtected
C02340 F0317 L1281 isPetalProtected
C02341 F0321 L1283 inline function / computed callee (body separately inventoried)
C02342 F0321 L1283 mass.map
C02343 F0322 L1283 (p*100).toFixed
C02344 F0323 L1285 globalThis.PetalRules.craftAnalysis
C02345 F0324 L1285 (n*100).toFixed
C02346 F0323 L1286 pct
C02347 F0323 L1286 a.outputs.toFixed
C02348 F0323 L1286 pct
C02349 F0323 L1286 a.meanCost.toFixed
C02350 F0323 L1286 renderCraftDistribution
C02351 F0325 L1289 game.equipped.filter
C02352 F0325 L1289 armorOf
C02353 F0325 L1289 activeCopies
C02354 F0325 L1291 globalThis.PetalRules.damageMoments
C02355 F0325 L1291 rawPetalDamage
C02356 F0325 L1291 Math.max
C02357 F0325 L1291 a.mean.toFixed
C02358 F0325 L1291 (a.killChance*100).toFixed
C02359 F0325 L1292 armor.toFixed
C02360 F0327 L1295
map.living.filter(m=>!m.dead).sort((a,b)=>dist(a.x,a.y,game.p.x,game.p.y)-dist(b.x,b.y,game.p.x,game.p.y)).slice
C02361 F0327 L1295 map.living.filter(m=>!m.dead).sort
C02362 F0327 L1295 map.living.filter
C02363 F0329 L1295 dist
C02364 F0329 L1295 dist
C02365 F0327 L1297 globalThis.PetalRules.rewardStats
C02366 F0327 L1297 globalThis.PetalRules.targetDropChance
C02367 F0327 L1298 Math.round
C02368 F0327 L1298 dist
C02369 F0327 L1298 stats.mean.toFixed
C02370 F0327 L1298 Math.sqrt(stats.variance).toFixed
C02371 F0327 L1298 Math.sqrt
C02372 F0327 L1299 (chance*100).toFixed
C02373 F0327 L1300 [...new Set(spec.drop)].map(t=>pets[t].name).join
C02374 F0327 L1300 [...new Set(spec.drop)].map
C02375 F0327 L1300 renderDamageForecast
C02376 F0327 L1303 (globalThis.PetalRules.criticalChance(activeCopies('prism'))*100).toFixed
C02377 F0327 L1303 globalThis.PetalRules.criticalChance
C02378 F0327 L1303 activeCopies
C02379 F0327 L1303 ((1-globalThis.PetalRules.slowFactor(activeCopies('frost')))*100).toFixed
C02380 F0327 L1303 globalThis.PetalRules.slowFactor
C02381 F0327 L1303 activeCopies
C02382 F0332 L1307 globalThis.PetalUI.ask
C02383 F0333 L1311 Object.keys
C02384 F0333 L1311 game.slots.map
C02385 F0334 L1311 inventoryKey
C02386 F0333 L1312 itemOf
C02387 F0333 L1312 Math.max
```

```
C02388 F0333 L1312 support.has
C02389 F0333 L1313 globalThis.PetalWorkshop.plan
C02390 F0333 L1313 game.slots.map
C02391 F0335 L1313 inventoryKey
C02392 F0336 L1315 JSON.stringify
C02393 F0337 L1317 $
C02394 F0337 L1317 $
C02395 F0337 L1318 buildWorkshopPlan
C02396 F0337 L1321 Object.entries
C02397 F0337 L1323 plan.slots.entries
C02398 F0337 L1323 itemOf
C02399 F0337 L1324 plan.before.toFixed
C02400 F0337 L1324 plan.after.toFixed
C02401 F0337 L1324 Math.round
C02402 F0337 L1324 Math.round
C02403 F0337 L1324 Math.round
C02404 F0337 L1325 b.querySelectorAll('[data-policy]').forEach
C02405 F0337 L1325 b.querySelectorAll
C02406 F0339 L1325 renderModal
C02407 F0337 L1325 $
C02408 F0340 L1325 showModal
C02409 F0337 L1325 $
C02410 F0341 L1325 showModal
C02411 F0337 L1326 workshopIdentity
C02412 F0337 L1326 $
C02413 F0342 L1327 askGame
C02414 F0342 L1328 workshopIdentity
C02415 F0342 L1328 toast
C02416 F0342 L1328 renderModal
C02417 F0342 L1329 plan.slots.map
C02418 F0343 L1329 itemOf
C02419 F0342 L1329 resetEquip
C02420 F0342 L1329 Math.min
C02421 F0342 L1329 maxHP
C02422 F0342 L1329 effectiveZoom
C02423 F0342 L1329 fillSlots
C02424 F0342 L1329 saveGame
C02425 F0342 L1329 syncHud
C02426 F0342 L1329 renderModal
C02427 F0342 L1329 toast
C02428 F0344 L1333 $
C02429 F0344 L1333 petalDamageMultiplier
C02430 F0344 L1334 syner
C02431 F0344 L1334 ((35+level*3)*power*(1+.18*syner('assault'))).toFixed
C02432 F0344 L1334 syner
C02433 F0344 L1334 ((17+level*1.7)*power*(1+.25*syner('artillery'))).toFixed
C02434 F0344 L1334 syner
C02435 F0344 L1334 (4.5+syner('fortress')).toFixed
C02436 F0344 L1334 syner
C02437 F0344 L1334 Math.round
C02438 F0344 L1334 maxHP
C02439 F0344 L1334 syner
C02440 F0344 L1336 Object.entries
C02441 F0344 L1336 cfg.key.toUpperCase
C02442 F0344 L1336 skillRemaining
C02443 F0344 L1336 skillRemaining(k).toFixed
C02444 F0344 L1336 skillRemaining
C02445 F0344 L1339 Object.entries
C02446 F0344 L1339 syner
C02447 F0344 L1339 syner
```

```
C02448 F0344 L1339 d.types.map(t=>pets[t].name).join
C02449 F0344 L1339 d.types.map
C02450 F0344 L1340 $
C02451 F0346 L1345 $
C02452 F0346 L1345 $
C02453 F0346 L1345 $
C02454 F0346 L1346 document.querySelectorAll('[data-panel]').forEach
C02455 F0346 L1346 document.querySelectorAll
C02456 F0347 L1346 b.setAttribute
C02457 F0347 L1346 b.removeAttribute
C02458 F0346 L1347 renderModalContents
C02459 F0346 L1347 $
C02460 F0346 L1347 $(identity).focus
C02461 F0346 L1347 $
C02462 F0348 L1349 renderBuildWorkshop
C02463 F0348 L1349 renderSkillAtlas
C02464 F0348 L1349 $
C02465 F0348 L1349 $
C02466 F0348 L1349 renderProbabilityLab
C02467 F0348 L1349 renderPartyPanel
C02468 F0348 L1349 $
C02469 F0348 L1349 talentEarned
C02470 F0348 L1349 talentPoints
C02471 F0348 L1349 Object.entries
C02472 F0348 L1349 talentRank
C02473 F0348 L1349 Math.round
C02474 F0348 L1349 talentPoints
C02475 F0348 L1349 b.querySelectorAll('[data-talent]').forEach
C02476 F0348 L1349 b.querySelectorAll
C02477 F0350 L1349 spendTalent
C02478 F0348 L1349 $
C02479 F0351 L1349 askGame
C02480 F0351 L1349 resetTalents
C02481 F0348 L1351 $
C02482 F0348 L1351 renderCollectionGuide
C02483 F0348 L1356 itemOf
C02484 F0348 L1356 sortedBag
C02485 F0348 L1356 renderPetalWorkbench
C02486 F0348 L1357 inventoryControls
C02487 F0348 L1358 sortedBag
C02488 F0348 L1358 itemOf
C02489 F0348 L1358 Math.round
C02490 F0348 L1358 Math.round
C02491 F0348 L1358 isPetalProtected
C02492 F0348 L1359 sortedBag
C02493 F0348 L1360 $
C02494 F0352 L1360 showModal
C02495 F0348 L1360 bindCollectionGuide
C02496 F0348 L1360 bindInventoryFilters
C02497 F0348 L1361 b.querySelectorAll('[data-deck]').forEach
C02498 F0348 L1361 b.querySelectorAll
C02499 F0354 L1361 renderModal
C02500 F0348 L1362 b.querySelectorAll('[data-slot]').forEach
C02501 F0348 L1362 b.querySelectorAll
C02502 F0356 L1362 renderModal
C02503 F0348 L1363 b.querySelectorAll('[data-equip]').forEach
C02504 F0348 L1363 b.querySelectorAll
C02505 F0358 L1363 performance.now
C02506 F0358 L1363 renderModal
C02507 F0348 L1364 $
```

```
C02508 F0348 L1364 $
C02509 F0359 L1364 equipInto
C02510 F0348 L1365 $
C02511 F0348 L1365 $
C02512 F0360 L1365 togglePetalProtection
C02513 F0348 L1366 $
C02514 F0348 L1366 $
C02515 F0361 L1366 itemOf
C02516 F0361 L1366 setCollectionGoal
C02517 F0361 L1366 Math.min
C02518 F0348 L1367 $
C02519 F0348 L1367 $
C02520 F0362 L1367 showModal
C02521 F0348 L1368 bindBagDrag
C02522 F0348 L1371 $
C02523 F0348 L1372 itemOf
C02524 F0348 L1372 itemOf
C02525 F0348 L1378 isPetalProtected
C02526 F0348 L1378 (rate*100).toFixed
C02527 F0348 L1378 isPetalProtected
C02528 F0348 L1378 isPetalProtected
C02529 F0348 L1378 isPetalProtected
C02530 F0348 L1379 renderCraftAnalysis
C02531 F0348 L1379 isPetalProtected
C02532 F0348 L1380 inventoryControls
C02533 F0348 L1382 sortedBag
C02534 F0348 L1382 itemOf
C02535 F0348 L1383 isPetalProtected
C02536 F0348 L1385 bindInventoryFilters
C02537 F0348 L1386 b.querySelectorAll('[data-craft]').forEach
C02538 F0348 L1386 b.querySelectorAll
C02539 F0364 L1386 renderModal
C02540 F0348 L1387 $
C02541 F0348 L1387 $
C02542 F0348 L1388 $
C02543 F0365 L1388 togglePetalProtection
C02544 F0366 L1391 itemOf
C02545 F0366 L1392 craftBatch
C02546 F0366 L1394 toast
C02547 F0366 L1395 saveGame
C02548 F0366 L1395 renderModal
C02549 F0367 L1398 itemOf
C02550 F0367 L1399 isPetalProtected
C02551 F0367 L1399 Number
C02552 F0367 L1402 askGame
C02553 F0367 L1403 craftBatch
C02554 F0367 L1405 toast
C02555 F0367 L1406 saveGame
C02556 F0367 L1406 renderModal
C02557 F0348 L1409 $
C02558 F0348 L1409 Object.entries
C02559 F0348 L1409 [id,to].sort().join
C02560 F0348 L1409 [id,to].sort
C02561 F0348 L1409 shown.has
C02562 F0348 L1409 shown.add
C02563 F0348 L1409 Object.entries
C02564 F0348 L1409 Object.entries
C02565 F0348 L1409 Object.entries
C02566 F0348 L1409 bandLabel
C02567 F0348 L1409 bandLabel
```

```
C02568 F0348 L1409 b.querySelectorAll('[data-nav]').forEach
C02569 F0348 L1409 b.querySelectorAll
C02570 F0369 L1409 closeModal
C02571 F0369 L1409 toast
C02572 F0348 L1410 $
C02573 F0348 L1411 Object.values(game.codex).filter
C02574 F0348 L1411 Object.values
C02575 F0348 L1411 Object.values(game.codex).reduce
C02576 F0348 L1411 Object.values
C02577 F0348 L1411 Object.keys
C02578 F0348 L1412 Object.keys
C02579 F0348 L1414 Object.entries
C02580 F0348 L1414 creatureRegions(t).map(id=>zones[id].name).join
C02581 F0348 L1414 creatureRegions(t).map
C02582 F0348 L1414 creatureRegions
C02583 F0348 L1415 creatureRegions(select).map(id=>zones[id].name).join
C02584 F0348 L1415 creatureRegions(select).map
C02585 F0348 L1415 creatureRegions
C02586 F0348 L1416 d.drop.map(t=>pets[t].name).join
C02587 F0348 L1416 d.drop.map
C02588 F0348 L1416 rarity.map((r,i)=>'<option value="'+i+'">'+r.name+'</option>').join
C02589 F0348 L1416 rarity.map
C02590 F0348 L1416 d.drop.map(t=>'<button type="button"
data-track-drop="'+t+"LOCALIZED_2f1529af"+pets[t].name+'</button>').join
C02591 F0348 L1416 d.drop.map
C02592 F0348 L1417 $
C02593 F0348 L1417 String
C02594 F0348 L1417 b.querySelectorAll('[data-track-drop]').forEach
C02595 F0348 L1417 b.querySelectorAll
C02596 F0378 L1417 setCollectionGoal
C02597 F0378 L1417 Number
C02598 F0378 L1417 $
C02599 F0348 L1417 b.querySelectorAll('[data-filter]').forEach
C02600 F0348 L1417 b.querySelectorAll
C02601 F0380 L1417 renderModal
C02602 F0348 L1417 b.querySelectorAll('[data-creature]').forEach
C02603 F0348 L1417 b.querySelectorAll
C02604 F0382 L1417 renderModal
C02605 F0348 L1418 $
C02606 F0348 L1418 regionBand
C02607 F0348 L1418 game.slots.filter
C02608 F0348 L1420 Math.ceil
C02609 F0348 L1420 Math.ceil
C02610 F0348 L1420 maxHP
C02611 F0348 L1420 Math.round
C02612 F0348 L1420 Math.round
C02613 F0348 L1420 rootedArmor
C02614 F0348 L1420 Math.round
C02615 F0348 L1420 bodySpeedFactor
C02616 F0348 L1420 Math.round
C02617 F0348 L1420 magnetRadius
C02618 F0348 L1420 Math.floor
C02619 F0348 L1420 xpNeed
C02620 F0348 L1421 Math.round
C02621 F0348 L1421 Math.round
C02622 F0348 L1421 Math.round
C02623 F0348 L1421 SLOT_MILESTONES.find
C02624 F0348 L1422 Math.round
C02625 F0348 L1422 Math.round
C02626 F0348 L1422 bandLabel
```

```
C02627 F0348 L1423 Math.round
C02628 F0348 L1423 (1.9*scale).toFixed
C02629 F0348 L1423 (3.1*scale).toFixed
C02630 F0348 L1423 Math.round
C02631 F0348 L1423 Math.round
C02632 F0348 L1423 Math.round
C02633 F0348 L1423 Math.ceil
C02634 F0348 L1423 Math.round
C02635 F0348 L1424 $
C02636 F0348 L1424 Object.entries
C02637 F0348 L1424 Math.min
C02638 F0348 L1424 Math.min
C02639 F0348 L1424 b.querySelectorAll('[data-claim]').forEach
C02640 F0348 L1424 b.querySelectorAll
C02641 F0386 L1424 Math.min
C02642 F0386 L1424 addBag
C02643 F0386 L1424 xpNeed
C02644 F0386 L1424 xpNeed
C02645 F0386 L1424 Math.min
C02646 F0386 L1424 maxHP
C02647 F0386 L1424 ensureSlots
C02648 F0386 L1424 viewScale
C02649 F0386 L1424 saveGame
C02650 F0386 L1424 renderModal
C02651 F0386 L1424 syncHud
C02652 F0386 L1424 toast
C02653 F0348 L1425 $
C02654 F0348 L1425 $
C02655 F0348 L1425 $
C02656 F0348 L1425 $
C02657 F0348 L1425 $
C02658 F0387 L1425 resetInputs
C02659 F0387 L1425 saveGame
C02660 F0348 L1425 $
C02661 F0348 L1425 $
C02662 F0388 L1425 resize
C02663 F0388 L1425 saveGame
C02664 F0348 L1425 $
C02665 F0389 L1425 saveGame
C02666 F0389 L1425 renderModal
C02667 F0348 L1425 $
C02668 F0390 L1425 saveGame
C02669 F0390 L1425 renderModal
C02670 F0348 L1425 $
C02671 F0391 L1425 saveGame
C02672 F0391 L1425 renderModal
C02673 F0348 L1425 $
C02674 F0392 L1425 saveGame
C02675 F0392 L1425 renderModal
C02676 F0348 L1425 $
C02677 F0393 L1425 closeModal
C02678 F0393 L1425 swapLoadout
C02679 F0348 L1426 $
C02680 F0348 L1427 $
C02681 F0348 L1427 $
C02682 F0394 L1427 showModal
C02683 F0348 L1427 $
C02684 F0395 L1427 showModal
C02685 F0348 L1427 $
C02686 F0396 L1427 showModal
```

```
C02687 F0348 L1427 $
C02688 F0348 L1427 $
C02689 F0397 L1427 saveGame
C02690 F0397 L1427 toast
C02691 F0348 L1427 $
C02692 F0348 L1428 b.insertAdjacentHTML
C02693 F0348 L1428 renderCollectionGuide
C02694 F0348 L1428 bindCollectionGuide
C02695 F0348 L1430 b.insertAdjacentHTML
C02696 F0348 L1431 $
C02697 F0348 L1431 $
C02698 F0348 L1431 $
C02699 F0398 L1431 importLocalProgress
C02700 F0348 L1433 cfPanelControls
C02701 F0399 L1435 Object.entries(game.bags).filter(([k,n])=>itemOf(k)&&n>0).sort
C02702 F0399 L1435 Object.entries(game.bags).filter
C02703 F0399 L1435 Object.entries
C02704 F0400 L1435 itemOf
C02705 F0401 L1435 itemOf
C02706 F0401 L1435 itemOf
C02707 F0401 L1435 pets[at].name.localeCompare
C02708 F0402 L1435 sortedBag
C02709 F0402 L1435 itemOf
C02710 F0402 L1435 grouped.has
C02711 F0402 L1435 grouped.set
C02712 F0402 L1435 grouped.get
C02713 F0402 L1435 Array.from
C02714 F0402 L1435 grouped.values
C02715 F0403 L1436 toast
C02716 F0403 L1436 cfLogout
C02717 F0403 L1436 leaveParty
C02718 F0403 L1436 saveGame
C02719 F0403 L1436 cloudForgetStoredSession
C02720 F0403 L1436 cloudFlush().finally
C02721 F0403 L1436 cloudFlush
C02722 F0403 L1436 $
C02723 F0403 L1436 $('panelLayer').classList.add
C02724 F0403 L1436 $
C02725 F0403 L1436 $('ui').classList.add
C02726 F0403 L1436 $
C02727 F0403 L1436 $('death').classList.add
C02728 F0403 L1436 $
C02729 F0403 L1436 $('gate').classList.remove
C02730 F0403 L1436 $
C02731 F0403 L1436 $
C02732 F0403 L1436 setAuthMode
C02733 F0403 L1436 $
C02734 F0403 L1436 keys.clear
C02735 F0405 L1437 JSON.parse
C02736 F0405 L1437 localStorage.getItem
C02737 F0405 L1437 Array.isArray
C02738 F0406 L1438 Array.from(arr,b=>b.toString(16).padStart(2,'0')).join
C02739 F0406 L1438 Array.from
C02740 F0407 L1438 b.toString(16).padStart
C02741 F0407 L1438 b.toString
C02742 F0408 L1439 Error
C02743 F0408 L1439 crypto.subtle.importKey
C02744 F0408 L1439 enc.encode
C02745 F0408 L1439 Uint8Array.from
C02746 F0408 L1439 salt.match(/.{2}/g).map
```

```
C02747 F0408 L1439 salt.match
C02748 F0409 L1439 parseInt
C02749 F0408 L1439 crypto.subtle.deriveBits
C02750 F0408 L1439 toHex
C02751 F0410 L1443 $
C02752 F0410 L1444 $
C02753 F0410 L1444 $
C02754 F0410 L1445 $
C02755 F0410 L1446 $('tabLogin').classList.toggle
C02756 F0410 L1446 $
C02757 F0410 L1446 $('tabRegister').classList.toggle
C02758 F0410 L1446 $
C02759 F0410 L1447 $
C02760 F0410 L1448 $
C02761 F0410 L1449 $
C02762 F0410 L1449 $
C02763 F0410 L1450 $
C02764 F0410 L1450 $
C02765 F0410 L1451 $
C02766 F0410 L1452 $
C02767 F0410 L1452 $
C02768 F0410 L1452 $
C02769 F0410 L1452 $
C02770 F0410 L1452 $
C02771 F0410 L1452 $
C02772 F0410 L1452 $
C02773 F0410 L1452 $
C02774 F0410 L1453 $
C02775 F0410 L1453 $
C02776 F0410 L1454 refreshMailControls
C02777 F0411 L1456 e.preventDefault
C02778 F0411 L1456 cfAuthenticate
C02779 F0411 L1456 $
C02780 F0411 L1456 $('authForm').setAttribute
C02781 F0411 L1456 $
C02782 F0411 L1456 $
C02783 F0411 L1456 cloudAuth
C02784 F0411 L1456 $('authName').value.trim
C02785 F0411 L1456 $
C02786 F0411 L1456 $
C02787 F0411 L1456 $
C02788 F0411 L1456 $
C02789 F0411 L1456 cloudAuthMessage
C02790 F0411 L1456 $('localFallbackBtn').classList.toggle
C02791 F0411 L1456 $
C02792 F0411 L1456 isMailUnavailable
C02793 F0411 L1456 $('authForm').removeAttribute
C02794 F0411 L1456 $
C02795 F0411 L1456 refreshMailControls
C02796 F0411 L1456 $('authName').value.trim
C02797 F0411 L1456 $
C02798 F0411 L1456 $
C02799 F0411 L1456 $
C02800 F0411 L1456 /^[\p{L}\p{N}_-]{2,18}$/u.test
C02801 F0411 L1456 ['__proto__','constructor','prototype'].includes
C02802 F0411 L1456 $
C02803 F0411 L1456 $
C02804 F0411 L1456 accounts
C02805 F0411 L1456 $
C02806 F0411 L1456 Error
```

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C02807 F0411 L1456 /^cloud-/i.test
C02808 F0411 L1456 Error
C02809 F0411 L1456 Object.prototype.hasOwnProperty.call
C02810 F0411 L1456 Error
C02811 F0411 L1456 toHex
C02812 F0411 L1456 crypto.getRandomValues
C02813 F0411 L1456 hashPass
C02814 F0411 L1456 localStorage.setItem
C02815 F0411 L1456 JSON.stringify
C02816 F0411 L1456 Object.prototype.hasOwnProperty.call
C02817 F0411 L1456 Error
C02818 F0411 L1456 hashPass
C02819 F0411 L1456 Error
C02820 F0411 L1456 $
C02821 F0411 L1456 $
C02822 F0411 L1456 $
C02823 F0411 L1456 initGame
C02824 F0411 L1456 readSave
C02825 F0411 L1456 saveGame
C02826 F0411 L1456 $
C02827 F0412 L1459 /^Key[A-Z]$/.test
C02828 F0412 L1459 e.code.slice(3).toLowerCase
C02829 F0412 L1459 e.code.slice
C02830 F0412 L1460 /^Digit[0-9]$/.test
C02831 F0412 L1460 e.code.slice
C02832 F0412 L1462 String(e.key||'').toLowerCase
C02833 F0412 L1462 String
C02834 F0413 L1464 node?.closest
C02835 F0414 L1466 keys.clear
C02836 F0414 L1467 heldTouch.values
C02837 F0414 L1467 refs.clear
C02838 F0414 L1468 $('touchAtk')?.classList.remove
C02839 F0414 L1468 $
C02840 F0414 L1468 $('touchDef')?.classList.remove
C02841 F0414 L1468 $
C02842 F0414 L1469 $
C02843 F0414 L1469 $
C02844 F0416 L1473 heldTouch.set
C02845 F0416 L1474 node.addEventListener
C02846 F0417 L1474 e.preventDefault
C02847 F0417 L1474 node.setPointerCapture
C02848 F0417 L1474 refs.add
C02849 F0417 L1474 node.classList.add
C02850 F0418 L1475 refs.delete
C02851 F0418 L1475 node.classList.remove
C02852 F0416 L1476 node.addEventListener
C02853 F0416 L1476 node.addEventListener
C02854 F0416 L1476 node.addEventListener
C02855 F0001 L1478 $
C02856 F0001 L1478 $
C02857 F0419 L1479 js.getBoundingClientRect
C02858 F0419 L1479 Math.hypot
C02859 F0419 L1479 Math.min
C02860 F0001 L1480 js.addEventListener
C02861 F0420 L1480 e.preventDefault
C02862 F0420 L1480 js.setPointerCapture
C02863 F0420 L1480 joyMove
C02864 F0001 L1481 js.addEventListener
C02865 F0421 L1481 e.preventDefault
C02866 F0421 L1481 joyMove
```

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C02867 F0001 L1483 js.addEventListener
C02868 F0001 L1483 js.addEventListener
C02869 F0001 L1483 js.addEventListener
C02870 F0001 L1484 bindTouchButton
C02871 F0001 L1484 $
C02872 F0001 L1484 bindTouchButton
C02873 F0001 L1484 $
C02874 F0001 L1485 canvas.addEventListener
C02875 F0001 L1486 canvas.addEventListener
C02876 F0423 L1486 e.preventDefault
C02877 F0423 L1486 canvas.focus
C02878 F0423 L1486 canvas.setPointerCapture
C02879 F0001 L1487 window.addEventListener
C02880 F0001 L1488 window.addEventListener
C02881 F0001 L1488 window.addEventListener
C02882 F0001 L1489 document.addEventListener
C02883 F0425 L1489 e.target.closest
C02884 F0425 L1489 canvas.focus
C02885 F0001 L1490 canvas.addEventListener
C02886 F0001 L1491 document.addEventListener
C02887 F0427 L1491 isEditing
C02888 F0427 L1491 e.preventDefault
C02889 F0001 L1492 document.addEventListener
C02890 F0428 L1492 isEditing
C02891 F0428 L1492 e.preventDefault
C02892 F0001 L1493 document.addEventListener
C02893 F0429 L1493 e.target.closest
C02894 F0429 L1493 isEditing
C02895 F0429 L1493 e.preventDefault
C02896 F0001 L1494 document.addEventListener
C02897 F0430 L1494 e.target.closest
C02898 F0430 L1494 isEditing
C02899 F0430 L1494 e.preventDefault
C02900 F0001 L1495 window.addEventListener
C02901 F0431 L1496
[...$('panelLayer').querySelectorAll('button:not(:disabled),input:not(:disabled),select,summary,a[href]')].filter
C02902 F0431 L1496 $('panelLayer').querySelectorAll
C02903 F0431 L1496 $
C02904 F0432 L1496 x.getClientRects
C02905 F0431 L1496 controls.at
C02906 F0431 L1496 $('panelLayer').contains
C02907 F0431 L1496 $
C02908 F0431 L1496 e.preventDefault
C02909 F0431 L1496 last?.focus
C02910 F0431 L1496 $('panelLayer').contains
C02911 F0431 L1496 $
C02912 F0431 L1496 e.preventDefault
C02913 F0431 L1496 first?.focus
C02914 F0431 L1497 keyForEvent
C02915 F0431 L1498 e.preventDefault
C02916 F0431 L1498 closeModal
C02917 F0431 L1498 openPanel
C02918 F0431 L1499 isEditing
C02919 F0431 L1501 [' ','arrowup','arrowdown','arrowleft','arrowright'].includes
C02920 F0431 L1501 e.preventDefault
C02921 F0431 L1503 ['w','a','s','d','arrowup','arrowdown','arrowleft','arrowright'].includes
C02922 F0431 L1503 keys.add
C02923 F0431 L1504 keys.add
C02924 F0431 L1506 Object.keys(SKILL_CONFIG).find
C02925 F0431 L1506 Object.keys
```

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C02926 F0431 L1506 activeSkill
C02927 F0431 L1508 openPanel
C02928 F0431 L1509 performDash
C02929 F0431 L1509 swapLoadout
C02930 F0431 L1509 reverseOrbit
C02931 F0431 L1510 openPanel
C02932 F0431 L1510 /^[0-9]$/.test
C02933 F0431 L1510 openPanel
C02934 F0431 L1510 Number
C02935 F0001 L1512 window.addEventListener
C02936 F0434 L1512 keys.delete
C02937 F0434 L1512 keyForEvent
C02938 F0001 L1512 window.addEventListener
C02939 F0001 L1513 window.visualViewport?.addEventListener
C02940 F0435 L1513 isEditing
C02941 F0435 L1513 resize
C02942 F0001 L1514 document.addEventListener
C02943 F0436 L1514 saveGame
C02944 F0436 L1514 resetInputs
C02945 F0001 L1515 $
C02946 F0437 L1515 document.body.classList.toggle
C02947 F0437 L1515 $('toolsMenuBtn').setAttribute
C02948 F0437 L1515 $
C02949 F0437 L1515 String
C02950 F0001 L1515 $
C02951 F0438 L1515 document.body.classList.toggle
C02952 F0438 L1515 $('skillMenuBtn').setAttribute
C02953 F0438 L1515 $
C02954 F0438 L1515 String
C02955 F0001 L1515 $
C02956 F0439 L1515 openPanel
C02957 F0001 L1515 $
C02958 F0440 L1515 openPanel
C02959 F0001 L1515 document.querySelectorAll('[data-active]').forEach
C02960 F0001 L1515 document.querySelectorAll
C02961 F0441 L1515 b.addEventListener
C02962 F0442 L1515 activeSkill
C02963 F0001 L1515 $
C02964 F0443 L1515 cfConfigure
C02965 F0443 L1515 cfResume
C02966 F0001 L1515 $
C02967 F0444 L1515 configureAccountMode
C02968 F0001 L1515 $
C02969 F0001 L1515 $
C02970 F0001 L1515 $
C02971 F0445 L1515 cfUseCloud
C02972 F0001 L1515 $
C02973 F0446 L1515 cfUseLocal
C02974 F0001 L1515 $
C02975 F0001 L1515 $
C02976 F0447 L1515 configureAccountMode
C02977 F0447 L1515 setAuthMode
C02978 F0001 L1515 $
C02979 F0001 L1515 $
C02980 F0448 L1515 setAuthMode
C02981 F0001 L1515 $
C02982 F0449 L1515 setAuthMode
C02983 F0001 L1515 $
C02984 F0001 L1515 $
C02985 F0001 L1515 $
```

```
C02986 F0001 L1515 $
C02987 F0450 L1515 initGame
C02988 F0001 L1515 document.querySelectorAll('[data-panel]').forEach
C02989 F0001 L1515 document.querySelectorAll
C02990 F0452 L1515 showModal
C02991 F0001 L1515 $
C02992 F0453 L1515 globalThis.PetalUI.welcome
C02993 F0001 L1515 $
C02994 F0001 L1515 $('panelLayer').addEventListener
C02995 F0001 L1515 $
C02996 F0454 L1515 $
C02997 F0454 L1515 closeModal
C02998 F0001 L1515 $
C02999 F0455 L1515 clearTransientState
C03000 F0455 L1515 maxHP
C03001 F0455 L1515 resetEquip
C03002 F0455 L1515 $('death').classList.add
C03003 F0455 L1515 $
C03004 F0455 L1515 saveGame
C03005 F0455 L1515 toast
C03006 F0001 L1515 $
C03007 F0456 L1515 openPanel
C03008 F0001 L1515 $
C03009 F0457 L1515 openPanel
C03010 F0001 L1515 $
C03011 F0458 L1515 openPanel
C03012 F0001 L1515 $
C03013 F0459 L1515 openPanel
C03014 F0001 L1515 $
C03015 F0460 L1515 openPanel
C03016 F0001 L1515 $
C03017 F0461 L1515 openPanel
C03018 F0001 L1515 $
C03019 F0462 L1515 openPanel
C03020 F0001 L1515 $
C03021 F0463 L1515 openPanel
C03022 F0001 L1515 $
C03023 F0001 L1515 $
C03024 F0001 L1515 $
C03025 F0001 L1515 $
C03026 F0464 L1515 openPanel
C03027 F0001 L1515 $
C03028 F0465 L1515 openPanel
C03029 F0001 L1515 $
C03030 F0466 L1515 openPanel
C03031 F0001 L1515 $
C03032 F0467 L1515 openPanel
C03033 F0468 L1518 $
C03034 F0468 L1518 $
C03035 F0001 L1522 window.addEventListener
C03036 F0001 L1523 window.addEventListener
C03037 F0001 L1524 updateHostingStatus
C03038 F0469 L1527 /^[a-z0-9-]{3,18}$/i.test
C03039 F0469 L1527 raw.toLowerCase
C03040 F0470 L1528 ws.send
C03041 F0470 L1528 JSON.stringify
C03042 F0470 L1528 String
C03043 F0471 L1529 cloudSession?.uid?.slice
C03044 F0471 L1529 Math.round
C03045 F0471 L1529 Math.round
```

```
C03046 F0471 L1529 Date.now
C03047 F0472 L1530 partySnapshot
C03048 F0472 L1530 partySend
C03049 F0473 L1531 clearInterval
C03050 F0473 L1531 socket.close
C03051 F0473 L1531 $('partyTag').classList.add
C03052 F0473 L1531 $
C03053 F0474 L1532 partyRoomName
C03054 F0474 L1532 Error
C03055 F0474 L1532 Error
C03056 F0474 L1532 Error
C03057 F0474 L1532 cloudToken
C03058 F0474 L1532 leaveParty
C03059 F0474 L1532 String
C03060 F0474 L1532 CLOUD_CONFIG.url.replace
C03061 F0474 L1532 encodeURIComponent
C03062 F0474 L1532 cloudSession.uid.slice
C03063 F0475 L1533 partySend
C03064 F0475 L1533 setInterval
C03065 F0476 L1533 partySend
C03066 F0476 L1533 partyPresence
C03067 F0477 L1534 JSON.parse
C03068 F0477 L1534 partyPresence
C03069 F0477 L1535 Object.entries
C03070 F0477 L1535 Number.isFinite
C03071 F0477 L1535 Number.isFinite
C03072 F0477 L1535 clamp
C03073 F0477 L1535 clamp
C03074 F0477 L1536 Object.entries
C03075 F0477 L1536 Object.entries
C03076 F0477 L1536 Number.isFinite
C03077 F0477 L1536 Number.isFinite
C03078 F0477 L1536 clamp
C03079 F0477 L1536 clamp
C03080 F0477 L1537 Number.isFinite
C03081 F0477 L1537 Number.isFinite
C03082 F0477 L1537 clamp
C03083 F0477 L1537 clamp
C03084 F0477 L1537 Date.now
C03085 F0477 L1537 Object.keys
C03086 F0477 L1537 Object.keys
C03087 F0479 L1538 clearInterval
C03088 F0480 L1540 Object.entries
C03089 F0480 L1540 Date.now
C03090 F0480 L1540 partySend
C03091 F0480 L1540 partySnapshot
C03092 F0480 L1540 $('partyTag').classList.remove
C03093 F0480 L1540 $
C03094 F0480 L1540 $
C03095 F0480 L1540 Object.keys
C03096 F0481 L1541 Object.values
C03097 F0481 L1541 visible
C03098 F0481 L1541 ctx.save
C03099 F0481 L1541 circle
C03100 F0481 L1541 ctx.fillText
C03101 F0481 L1541 ctx.fillText
C03102 F0481 L1541 Math.floor
C03103 F0481 L1541 Number
C03104 F0481 L1541 ctx.restore
C03105 F0482 L1542 $
```

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C03106 F0482 L1542 $
C03107 F0482 L1542 b.replaceChildren
C03108 F0482 L1542 document.createElement
C03109 F0482 L1542 b.append
C03110 F0482 L1542 document.createElement
C03111 F0482 L1542 Object.keys
C03112 F0482 L1542 b.append
C03113 F0482 L1543 document.createElement
C03114 F0482 L1543 document.createElement
C03115 F0482 L1543 label.append
C03116 F0482 L1543 b.append
C03117 F0482 L1543 document.createElement
C03118 F0483 L1543 joinParty
C03119 F0483 L1543 toast
C03120 F0483 L1543 renderPartyPanel
C03121 F0483 L1543 toast
C03122 F0484 L1543 leaveParty
C03123 F0484 L1543 renderPartyPanel
C03124 F0482 L1543 document.createElement
C03125 F0482 L1543 row.append
C03126 F0482 L1543 b.append
C03127 F0001 L1544 Object.freeze
C03128 F0485 L1544 game.random?.snapshot
C03129 F0486 L1544 collectionOwned
C03130 F0486 L1544 Object.keys
C03131 F0486 L1544 Object.keys
C03132 F0486 L1544 game.equipped.map
C03133 F0486 L1544 magnetRadius
C03134 F0486 L1544 Object.values(game.codex).filter
C03135 F0486 L1544 Object.values
C03136 F0486 L1544 talentPoints
C03137 F0486 L1544 Math.max
C03138 F0486 L1544 Object.fromEntries
C03139 F0486 L1544 Object.keys(SKILL_CONFIG).map
C03140 F0486 L1544 Object.keys
C03141 F0489 L1544 Math.round
C03142 F0489 L1544 skillRemaining
C03143 F0486 L1544 Object.keys
C03144 F0486 L1544 Math.min
C03145 F0486 L1544 Math.max
C03146 F0486 L1544 $('bossHud').classList.contains
C03147 F0486 L1544 $
C03148 F0001 L1544 window.addEventListener
C03149 F0490 L1544 saveGame
C03150 F0490 L1544 resetInputs
C03151 F0001 L1544 window.addEventListener
C03152 F0491 L1544 cloudSchedule
C03153 F0001 L1544 cfConfigure
C03154 F0001 L1544 setInterval
C03155 F0001 L1544 resize
C03156 F0001 L1544 requestAnimationFrame
C03157 F0001 L1544 $
C03158 F0001 L1544 $
C03159 F0001 L1544 $
C03160 F0001 L1544 inline function / computed callee (body separately inventoried)
C03161 F0001 L1544 inline function / computed callee (body separately inventoried)
C03162 F0001 L1544 inline function / computed callee (body separately inventoried)
C03163 F0001 L1544 globalThis.PetalUI.install
C03164 F0492 L1544 cloudRecoveryFromURL
C03165 F0492 L1544 cfResume
```

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C03166 F0493 L1544 $
C03167 F0493 L1544 $
C03168 F0493 L1544 $
C03169 F0001 L1544 window.addEventListener
C03170 F0494 L1544 cfSchedule
C03171 F0001 L1545 Object.freeze
C03172 F0001 L1545 window.addEventListener
C03173 F0496 L1545 location.reload
C03174 account-api:root L6 Object.freeze
C03175 F0497 L12 createHash('sha256').update(value).digest
C03176 F0497 L12 createHash('sha256').update
C03177 F0497 L12 createHash
C03178 F0498 L13 Math.floor
C03179 F0498 L13 Date.now
C03180 F0499 L15 super
C03181 F0501 L18 db.prepare(sql).bind
C03182 F0501 L18 db.prepare
C03183 F0502 L20 Response.json
C03184 F0503 L23 fail
C03185 F0503 L24 value.normalize('NFKC').trim().toLowerCase
C03186 F0503 L24 value.normalize('NFKC').trim
C03187 F0503 L24 value.normalize
C03188 F0503 L25 name.includes
C03189 F0503 L26 name.split
C03190 F0503 L27 domain.split
C03191 F0503 L28 /^[a-z0-9.!#$%&'*+/=?^_`{|}~-]+$/i.test
C03192 F0503 L28 local.startsWith
C03193 F0503 L28 local.endsWith
C03194 F0503 L28 local.includes
C03195 F0503 L28 labels.every
C03196 F0504 L28 /^[a-z0-9](?:[a-z0-9-]{0,61}[a-z0-9])?$/i.test
C03197 F0503 L28 /^[a-z]{2,63}$/i.test
C03198 F0503 L28 labels.at
C03199 F0503 L28 fail
C03200 F0503 L32 /^[\p{L}\p{N}_-]{2,18}$/u.test
C03201 F0503 L32 ['__proto__','constructor','prototype'].includes
C03202 F0503 L32 /^(cloud-|cf-)/i.test
C03203 F0503 L32 fail
C03204 F0505 L36 fail
C03205 F0507 L39 scrypt
C03206 F0508 L39 reject
C03207 F0508 L39 resolve
C03208 F0509 L42 inline function / computed callee (body separately inventoried)
C03209 F0510 L43 derive
C03210 F0510 L44 key.toString
C03211 F0510 L44 Error
C03212 F0511 L48 randomBytes(16).toString
C03213 F0511 L48 randomBytes
C03214 F0511 L49 JSON.stringify
C03215 F0511 L49 (await derive(pass,salt)).toString
C03216 F0511 L49 derive
C03217 F0512 L53 JSON.parse
C03218 F0512 L53 '0'.repeat
C03219 F0512 L55 derive
C03220 F0513 L58 pbkdf2
C03221 F0514 L58 reject
C03222 F0514 L58 resolve
C03223 F0512 L59 fail
C03224 F0512 L60 fail
C03225 F0512 L61 Buffer.from
```

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C03226 F0512 L62 timingSafeEqual
C03227 F0515 L65 /^application\/json(?:;|$)/i.test
C03228 F0515 L65 request.headers.get
C03229 F0515 L65 fail
C03230 F0515 L66 Number
C03231 F0515 L66 request.headers.get
C03232 F0515 L66 fail
C03233 F0515 L67 request.body?.getReader
C03234 F0515 L67 fail
C03235 F0515 L69 reader.read
C03236 F0515 L69 reader.cancel
C03237 F0515 L69 fail
C03238 F0515 L69 chunks.push
C03239 F0515 L70 bytes.set
C03240 F0515 L70 JSON.parse
C03241 F0515 L70 new TextDecoder().decode
C03242 F0515 L70 Array.isArray
C03243 F0515 L70 Error
C03244 F0515 L71 fail
C03245 F0516 L74 now
C03246 F0516 L75 inline function / computed callee (body separately inventoried)
C03247 F0516 L75 statement
C03248 F0516 L78 Math.max
C03249 F0517 L81 (request.headers.get('Cookie')||'').match
C03250 F0517 L81 request.headers.get
C03251 F0518 L85 cookieToken
C03252 F0518 L86 inline function / computed callee (body separately inventoried)
C03253 F0518 L86 statement
C03254 F0518 L87 hash
C03255 F0518 L87 now
C03256 F0520 L91 statement(db,'DELETE FROM sessions WHERE token_hash IN (SELECT token_hash FROM sessions WHERE
expires_at<=? LIMIT 100)',now()).run
C03257 F0520 L91 statement
C03258 F0520 L91 now
C03259 F0520 L92 statement(db,'DELETE FROM rate_limits WHERE bucket IN (SELECT bucket FROM rate_limits WHERE
reset_at<=? LIMIT 100)',now()).run
C03260 F0520 L92 statement
C03261 F0520 L92 now
C03262 F0520 L93 randomBytes(32).toString
C03263 F0520 L93 randomBytes
C03264 F0520 L94 inline function / computed callee (body separately inventoried)
C03265 F0520 L94 statement
C03266 F0520 L95 hash
C03267 F0520 L95 now
C03268 F0520 L96 fail
C03269 F0521 L100 Array.isArray
C03270 F0521 L100 ZONES.has
C03271 F0521 L100 fail
C03272 F0521 L102 Number.isSafeInteger
C03273 F0521 L102 ['xp','kills','hp'].every
C03274 F0522 L102 Number.isFinite
C03275 F0521 L102 fail
C03276 F0523 L103 PETALS.has
C03277 F0523 L103 Number.isInteger
C03278 F0521 L104 Array.isArray
C03279 F0521 L104 payload.slots.every
C03280 F0521 L104 fail
C03281 F0521 L105 Array.isArray
C03282 F0521 L105 payload.backupSlots.every
C03283 F0521 L105 fail
```

```
C03284 F0521 L107 Array.isArray
C03285 F0521 L107 Object.keys
C03286 F0521 L107 fail
C03287 F0521 L108 Object.entries
C03288 F0521 L108 /^([a-zA-Z]+)@(0|[1-9])$/.exec
C03289 F0521 L108 PETALS.has
C03290 F0521 L108 Number.isInteger
C03291 F0521 L108 fail
C03292 F0521 L109 PETALS.has
C03293 F0521 L109 Number.isInteger
C03294 F0521 L109 fail
C03295 F0521 L110 JSON.stringify
C03296 F0521 L111 new TextEncoder().encode
C03297 F0521 L111 fail
C03298 F0524 L115 Number.isSafeInteger
C03299 F0524 L115 fail
C03300 F0524 L116 validateSave
C03301 F0524 L116 now
C03302 F0524 L118 statement(db,'INSERT INTO saves(player_id,schema_version,revision,payload,updated_at)
VALUES(?,5,1,?,?) ON CONFLICT(player_id) DO NOTHING RETURNING revision',id,text,t).first
C03303 F0524 L118 statement
C03304 F0524 L119 statement(db,'UPDATE saves SET revision=revision+1,payload=?,updated_at=? WHERE player_id=? AND
revision=? RETURNING revision',text,t,id,expected).first
C03305 F0524 L119 statement
C03306 F0524 L120 fail
C03307 F0525 L125 fail
C03308 F0525 L127 fetch
C03309 F0525 L127 AbortSignal.timeout
C03310 F0525 L128 fail
C03311 F0525 L129 userResponse.json
C03312 F0525 L130 /^[a-f0-9-]{36}$/i.test
C03313 F0525 L130 fail
C03314 F0525 L131 statement(db,'SELECT player_id FROM legacy_links WHERE legacy_uid=?',user.id).first
C03315 F0525 L131 statement
C03316 F0525 L132 fail
C03317 F0525 L133 statement(db,'SELECT player_id FROM saves WHERE player_id=?',account.id).first
C03318 F0525 L133 statement
C03319 F0525 L133 fail
C03320 F0525 L134 fetch
C03321 F0525 L134 encodeURIComponent
C03322 F0525 L134 AbortSignal.timeout
C03323 F0525 L135 fail
C03324 F0525 L136 response.json
C03325 F0525 L136 Array.isArray
C03326 F0525 L136 fail
C03327 F0525 L137 validateSave
C03328 F0525 L139 db.batch
C03329 F0525 L140 statement
C03330 F0525 L140 now
C03331 F0525 L141 statement
C03332 F0525 L141 now
C03333 F0525 L142 fail
C03334 F0526 L148 fail
C03335 F0526 L149 url.pathname.slice
C03336 F0526 L150 ['GET','POST','PUT'].includes
C03337 F0526 L150 reply
C03338 F0526 L151 request.headers.get
C03339 F0526 L151 fail
C03340 F0526 L152 accountFor
C03341 F0526 L152 reply
```

```
C03342 F0526 L152 publicAccount
C03343 F0526 L153 db.prepare('SELECT id FROM players LIMIT 1').first
C03344 F0526 L153 db.prepare
C03345 F0526 L153 verifyPasswordRuntime
C03346 F0526 L153 reply
C03347 F0526 L154 accountFor
C03348 F0526 L155 ['save','auth/password','auth/logout','migrate'].includes
C03349 F0526 L155 request.headers.get
C03350 F0526 L156 fail
C03351 F0526 L158 fail
C03352 F0526 L159 statement(db,'SELECT revision,payload,updated_at FROM saves WHERE player_id=?',account.id).first
C03353 F0526 L159 statement
C03354 F0526 L160 reply
C03355 F0526 L160 JSON.parse
C03356 F0526 L162 fail
C03357 F0526 L163 bodyOf
C03358 F0526 L163 request.headers.get
C03359 F0526 L165 fail
C03360 F0526 L166 limit
C03361 F0526 L166 reply
C03362 F0526 L166 putSave
C03363 F0526 L168 fail
C03364 F0526 L170 cookieToken
C03365 F0526 L170 statement(db,'DELETE FROM sessions WHERE token_hash=?',hash(token)).run
C03366 F0526 L170 statement
C03367 F0526 L170 hash
C03368 F0526 L171 reply
C03369 F0526 L174 fail
C03370 F0526 L175 limit
C03371 F0526 L175 reply
C03372 F0526 L175 migrate
C03373 F0526 L177 ['auth/register','auth/login','auth/password','auth/recover'].includes
C03374 F0526 L177 fail
C03375 F0526 L178 limit
C03376 F0526 L178 hash
C03377 F0526 L180 fail
C03378 F0526 L181 limit
C03379 F0526 L182 checkPassword
C03380 F0526 L182 fail
C03381 F0526 L183 makePassword
C03382 F0526 L183 password
C03383 F0526 L184 inline function / computed callee (body separately inventoried)
C03384 F0526 L184 statement
C03385 F0526 L185 fail
C03386 F0526 L186 reply
C03387 F0526 L186 sessionCookie
C03388 F0526 L188 username
C03389 F0526 L189 limit
C03390 F0526 L189 hash
C03391 F0526 L191 limit
C03392 F0526 L191 hash
C03393 F0526 L192 makePassword
C03394 F0526 L192 password
C03395 F0526 L192 randomUUID
C03396 F0526 L192 randomBytes(32).toString
C03397 F0526 L192 randomBytes
C03398 F0526 L193 inline function / computed callee (body separately inventoried)
C03399 F0526 L193 statement
C03400 F0526 L193 body.username.normalize('NFKC').trim
C03401 F0526 L193 body.username.normalize
```

```
C03402 F0526 L193 hash
C03403 F0526 L193 now
C03404 F0526 L194 fail
C03405 F0526 L195 reply
C03406 F0526 L195 publicAccount
C03407 F0526 L195 sessionCookie
C03408 F0526 L197 statement(db,'SELECT * FROM players WHERE username=?',name).first
C03409 F0526 L197 statement
C03410 F0526 L199 password
C03411 F0526 L200 body.recoveryCode.trim().toLowerCase
C03412 F0526 L200 body.recoveryCode.trim
C03413 F0526 L201 /^[a-f0-9]{64}$/.test
C03414 F0526 L201 timingSafeEqual
C03415 F0526 L201 Buffer.from
C03416 F0526 L201 hash
C03417 F0526 L201 Buffer.from
C03418 F0526 L201 fail
C03419 F0526 L202 randomBytes(32).toString
C03420 F0526 L202 randomBytes
C03421 F0526 L202 makePassword
C03422 F0526 L203 inline function / computed callee (body separately inventoried)
C03423 F0526 L203 statement
C03424 F0526 L203 hash
C03425 F0526 L204 fail
C03426 F0526 L205 reply
C03427 F0526 L205 publicAccount
C03428 F0526 L205 sessionCookie
C03429 F0526 L207 checkPassword
C03430 F0526 L207 fail
C03431 F0526 L208 reply
C03432 F0526 L208 publicAccount
C03433 F0526 L208 sessionCookie
C03434 F0526 L210 reply
C03435 F0526 L210 String
C03436 F0526 L211 console.error
C03437 F0526 L212 reply
C03438 workshop:root L2 inline function / computed callee (body separately inventoried)
C03439 F0527 L3 Object.freeze
C03440 F0527 L3 Object.freeze
C03441 F0527 L3 Object.freeze
C03442 F0527 L3 Object.freeze
C03443 F0528 L5 Array.isArray
C03444 F0528 L5 Error
C03445 F0528 L6 Object.create
C03446 F0528 L6 Object.keys(defs).sort
C03447 F0528 L6 Object.keys
C03448 F0528 L7 Object.entries
C03449 F0528 L7 Object.hasOwn
C03450 F0528 L7 Number.isSafeInteger
C03451 F0528 L7 Error
C03452 F0528 L8 Object.hasOwn
C03453 F0528 L8 Error
C03454 F0528 L9 v.some
C03455 F0529 L9 Number.isFinite
C03456 F0528 L9 Error
C03457 F0528 L9 v.forEach
C03458 F0530 L9 Math.max
C03459 F0531 L10 [defs[k].attack,defs[k].renewal,defs[k].support].reduce
C03460 F0528 L11 keys.filter(k=>pool[k]>0).sort
C03461 F0528 L11 keys.filter
```

```
C03462 F0534 L11 score
C03463 F0534 L11 score
C03464 F0528 L12 Math.min
C03465 F0528 L12 result.push
C03466 F0528 L13 Error
C03467 F0528 L15 Object.values(pool).some
C03468 F0528 L15 Object.values
C03469 F0528 L15 Error
C03470 F0528 L16 Object.fromEntries
C03471 F0528 L16 Object.entries(pool).filter
C03472 F0528 L16 Object.entries
C03473 F0528 L16 slots.reduce
C03474 F0537 L16 score
C03475 F0528 L16 result.reduce
C03476 F0538 L16 score
C03477 F0528 L16 Object.fromEntries
C03478 F0528 L16 candidates.map
C03479 F0539 L16 score
C03480 F0541 L19 /[<>\x00-\x08]/.test
C03481 F0540 L20 /^\d+\.\d+\.\d+$/.test
C03482 F0540 L20 text
C03483 F0540 L20 text
C03484 F0540 L20 text
C03485 F0540 L20 /^\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}Z$/.test
C03486 F0540 L20 Number.isFinite
C03487 F0540 L20 Date.parse
C03488 F0540 L20 Array.isArray
C03489 F0540 L20 x.highlights.every
C03490 F0527 L23 Object.freeze
C03491 interface:root L2 inline function / computed callee (body separately inventoried)
C03492 F0543 L3 document.getElementById
C03493 F0544 L5 $
C03494 F0544 L5 $('welcomeDialog').close
C03495 F0544 L5 $
C03496 F0545 L7 $
C03497 F0545 L7 root.replaceChildren
C03498 F0545 L8 $
C03499 F0545 L8 $
C03500 F0545 L9 $
C03501 F0545 L9 $
C03502 F0545 L10 $
C03503 F0545 L10 new Date(x.updatedAt).toLocaleString
C03504 F0545 L11 document.createElement
C03505 F0545 L11 root.append
C03506 F0545 L12 $
C03507 F0545 L12 $
C03508 F0546 L16 setTimeout
C03509 F0547 L16 controller.abort
C03510 F0546 L17 inline function / computed callee (body separately inventoried)
C03511 F0548 L17 fetch
C03512 F0548 L17 Error
C03513 F0548 L17 r.text
C03514 F0548 L17 Error
C03515 F0548 L17 globalThis.PetalWorkshop.release
C03516 F0548 L17 JSON.parse
C03517 F0548 L17 Error
C03518 F0548 L17 paintNews
C03519 F0548 L17 checkedAt.toLocaleTimeString
C03520 F0548 L17 paintNews
C03521 F0548 L17 clearTimeout
```

```
C03522 F0548 L17 $
C03523 F0546 L18 $
C03524 F0549 L21 $('gate').classList.contains
C03525 F0549 L21 $
C03526 F0549 L21 Promise.resolve
C03527 F0549 L22 $
C03528 F0549 L22 Promise.resolve
C03529 F0549 L24 paintNews
C03530 F0549 L24 $('welcomeDialog').showModal
C03531 F0549 L24 $
C03532 F0549 L24 $('welcomeContinue').focus
C03533 F0549 L24 $
C03534 F0549 L24 refresh
C03535 F0551 L27 Promise.resolve
C03536 F0552 L29 $
C03537 F0552 L29 $
C03538 F0552 L29 $
C03539 F0552 L29 $('confirmDialog').showModal
C03540 F0552 L29 $
C03541 F0552 L29 $('confirmCancel').focus
C03542 F0552 L29 $
C03543 F0553 L32 globalThis.PetalWorkshop.release
C03544 F0553 L32 Error
C03545 F0553 L33 $
C03546 F0553 L33 $
C03547 F0553 L33 $
C03548 F0553 L34 $('welcomeDialog').addEventListener
C03549 F0553 L34 $
C03550 F0554 L34 welcomeResolve?.resolve
C03551 F0554 L34 lastFocus.closest
C03552 F0554 L34 lastFocus.focus
C03553 F0553 L35 $
C03554 F0555 L35 $('confirmDialog').close
C03555 F0555 L35 $
C03556 F0553 L35 $
C03557 F0556 L35 $('confirmDialog').close
C03558 F0556 L35 $
C03559 F0553 L36 $('confirmDialog').addEventListener
C03560 F0553 L36 $
C03561 F0557 L36 a.resolve
C03562 F0557 L36 $
C03563 F0557 L36 a.origin.focus
C03564 F0553 L37 new MutationObserver(()=>{if($('gate').classList.contains('hidden'))closeWelcome();else void
welcome()}).observe
C03565 F0558 L37 $('gate').classList.contains
C03566 F0558 L37 $
C03567 F0558 L37 closeWelcome
C03568 F0558 L37 welcome
C03569 F0553 L37 $
C03570 F0553 L38 document.addEventListener
C03571 F0559 L38 $
C03572 F0559 L38 refresh
C03573 F0553 L39 document.addEventListener
C03574 F0560 L39 document.querySelector
C03575 F0560 L39 e.stopImmediatePropagation
C03576 F0553 L40 setInterval
C03577 F0561 L40 $
C03578 F0561 L40 refresh
C03579 F0542 L43 Object.freeze
C03580 rules:root L2 inline function / computed callee (body separately inventoried)
```

```
C03581 F0563 L4 Object.values(o).forEach
C03582 F0563 L4 Object.values
C03583 F0564 L4 freeze
C03584 F0563 L4 Object.freeze
C03585 F0562 L5 freeze
C03586 F0565 L6 Number.isFinite
C03587 F0566 L7 finite
C03588 F0567 L8 Number.isSafeInteger
C03589 F0568 L9 values.some
C03590 F0569 L9 Number.isFinite
C03591 F0568 L9 Math.max
C03592 F0568 L9 values.map
C03593 F0568 L9 scaled.reduce
C03594 F0568 L9 scaled.map
C03595 F0573 L10 normalize
C03596 F0573 L10 probability
C03597 F0573 L10 rng
C03598 F0573 L10 p.findLastIndex
C03599 F0575 L11 integer
C03600 F0575 L11 ['world','loot','craft','combat','visual'].entries
C03601 F0575 L11 Math.imul
C03602 F0576 L11 Math.imul
C03603 F0576 L11 Math.imul
C03604 F0578 L12 integer
C03605 F0579 L12 mass.set
C03606 F0579 L12 mass.get
C03607 F0578 L12 add
C03608 F0578 L12 add
C03609 F0578 L12 add
C03610 F0578 L12 add
C03611 F0578 L12 Math.max
C03612 F0578 L12 [...mass].map
C03613 F0578 L12 Math.min
C03614 F0578 L12 boosted.set
C03615 F0578 L12 boosted.get
C03616 F0578 L12 [...boosted].map
C03617 F0582 L13 dropRanks
C03618 F0582 L13 finite
C03619 F0582 L13 value
C03620 F0582 L13 finite
C03621 F0582 L13 value
C03622 F0582 L13 Math.max
C03623 F0582 L13 outcomes.push
C03624 F0582 L13 finite
C03625 F0582 L13 value
C03626 F0582 L13 Math.max
C03627 F0582 L13 outcomes.reduce
C03628 F0582 L13 outcomes.reduce
C03629 F0585 L14 probability
C03630 F0585 L14 integer
C03631 F0586 L15 probability
C03632 F0586 L15 integer
C03633 F0586 L15 integer
C03634 F0587 L15 next.set
C03635 F0587 L15 next.get
C03636 F0586 L15 key.split(',').map
C03637 F0586 L15 key.split
C03638 F0586 L15 add
C03639 F0586 L15 add
C03640 F0586 L15 add
```

```
C03641 F0586 L15 Array(attempts+1).fill
C03642 F0586 L15 Array
C03643 F0586 L15 key.split(',').map
C03644 F0586 L15 key.split
C03645 F0586 L15 craftCDF
C03646 F0586 L15 f.findIndex
C03647 F0589 L16 integer
C03648 F0589 L16 Math.min
C03649 F0590 L17 criticalChance
C03650 F0590 L17 probability
C03651 F0590 L17 rng
C03652 F0591 L18 integer
C03653 F0591 L18 Math.min
C03654 F0592 L19 finite
C03655 F0592 L19 Math.abs
C03656 F0592 L19 outcomes.reduce
C03657 F0593 L19 probability
C03658 F0592 L19 outcomes.reduce
C03659 F0594 L19 finite
C03660 F0594 L19 Math.expm1
C03661 F0594 L19 finite
C03662 F0595 L20 features.some
C03663 F0596 L20 Number.isFinite
C03664 F0595 L20 normalize(weights).reduce
C03665 F0595 L20 normalize
C03666 F0598 L21 expectedUtility
C03667 F0598 L21 outcomes.map
C03668 F0599 L21 weightedValue
C03669 F0600 L22 finite
C03670 F0600 L22 finite
C03671 F0600 L22 Number.isFinite
C03672 F0600 L22 criticalChance
C03673 F0600 L22 Math.min
C03674 F0600 L22 Math.max
C03675 F0600 L22 Math.min
C03676 F0600 L22 Math.max
C03677 F0600 L22 outcomes.reduce
C03678 F0600 L22 outcomes.reduce
C03679 F0600 L22 outcomes.filter(x=>x.p>0).reduce
C03680 F0600 L22 outcomes.filter
C03681 F0604 L22 Math.min
C03682 F0600 L22 outcomes.filter(x=>x.p>0).reduce
C03683 F0600 L22 outcomes.filter
C03684 F0606 L22 Math.max
C03685 F0600 L22 outcomes.reduce
C03686 F0608 L23 Array.isArray
C03687 F0608 L23 integer
C03688 F0608 L23 types.filter
C03689 F0608 L23 dropRanks(rank,treasure).reduce
C03690 F0608 L23 dropRanks
C03691 F0610 L23 Math.max
C03692 F0610 L23 Math.max
C03693 F0611 L24 probability
C03694 F0611 L24 finite
C03695 F0611 L24 Math.expm1
C03696 F0611 L24 Math.log1p
C03697 F0612 L25 integer
C03698 F0612 L25 integer
C03699 F0612 L25 finite
C03700 F0612 L25 probability
```

```
C03701 F0612 L25 ['charger','sniper','venom','nova','summon','shield'].includes
C03702 F0612 L25 Math.min
C03703 F0613 L26 finite
C03704 F0613 L26 finite
C03705 F0613 L26 finite
C03706 F0613 L26 finite
C03707 F0613 L26 Math.hypot
C03708 F0613 L26 Math.min
C03709 F0613 L26 finite
C03710 F0613 L26 finite
C03711 F0613 L26 Math.hypot
C03712 F0562 L27 Object.freeze
C03713 art:root L2 inline function / computed callee (body separately inventoried)
C03714 F0615 L5 c.beginPath
C03715 F0615 L5 c.ellipse
C03716 F0615 L5 c.fill
C03717 F0616 L6 c.save
C03718 F0616 L6 c.createRadialGradient
C03719 F0616 L6 g.addColorStop
C03720 F0616 L6 g.addColorStop
C03721 F0616 L6 c.fillRect
C03722 F0616 L6 c.restore
C03723 F0617 L7 c.clearRect
C03724 F0618 L7 Math.imul
C03725 F0617 L8 random
C03726 F0617 L8 random
C03727 F0617 L8 random
C03728 F0617 L8 random
C03729 F0617 L8 ellipse
C03730 F0617 L8 random
C03731 F0617 L9 random
C03732 F0617 L9 random
C03733 F0617 L9 random
C03734 F0617 L9 random
C03735 F0617 L9 c.save
C03736 F0617 L9 c.translate
C03737 F0617 L9 c.rotate
C03738 F0617 L10 c.strokeRect
C03739 F0617 L10 c.beginPath
C03740 F0617 L10 c.moveTo
C03741 F0617 L10 c.lineTo
C03742 F0617 L10 c.stroke
C03743 F0617 L11 c.beginPath
C03744 F0617 L11 c.moveTo
C03745 F0617 L11 c.bezierCurveTo
C03746 F0617 L11 c.stroke
C03747 F0617 L11 ellipse
C03748 F0617 L12 c.beginPath
C03749 F0617 L12 c.moveTo
C03750 F0617 L12 c.quadraticCurveTo
C03751 F0617 L12 c.stroke
C03752 F0617 L12 ellipse
C03753 F0617 L13 c.beginPath
C03754 F0617 L13 c.moveTo
C03755 F0617 L13 c.lineTo
C03756 F0617 L13 c.lineTo
C03757 F0617 L13 c.stroke
C03758 F0617 L14 c.beginPath
C03759 F0617 L14 c.moveTo
C03760 F0617 L14 c.quadraticCurveTo
```

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C03761 F0617 L14 c.quadraticCurveTo
C03762 F0617 L14 c.fill
C03763 F0617 L14 c.stroke
C03764 F0617 L14 c.beginPath
C03765 F0617 L14 c.moveTo
C03766 F0617 L14 c.lineTo
C03767 F0617 L14 c.stroke
C03768 F0617 L14 c.restore
C03769 F0619 L16 c.save
C03770 F0619 L16 c.translate
C03771 F0619 L16 /ocean|sewer|jungle/.test
C03772 F0619 L16 ellipse
C03773 F0619 L17 c.strokeRect
C03774 F0619 L17 c.strokeRect
C03775 F0619 L18 ellipse
C03776 F0619 L18 c.beginPath
C03777 F0619 L18 c.ellipse
C03778 F0619 L18 c.stroke
C03779 F0619 L18 ellipse
C03780 F0619 L18 Math.cos
C03781 F0619 L18 Math.sin
C03782 F0619 L19 c.beginPath
C03783 F0619 L19 c.arc
C03784 F0619 L19 c.stroke
C03785 F0619 L19 ellipse
C03786 F0619 L19 Math.cos
C03787 F0619 L19 Math.sin
C03788 F0619 L20 Math.cos
C03789 F0619 L20 Math.sin
C03790 F0619 L20 ellipse
C03791 F0619 L20 Math.cos
C03792 F0619 L20 Math.sin
C03793 F0619 L20 ellipse
C03794 F0619 L21 c.restore
C03795 F0620 L22 m.t.toLowerCase
C03796 F0620 L22 Math.sin
C03797 F0620 L22 c.save
C03798 F0620 L23 Math.max
C03799 F0620 L23 c.beginPath
C03800 F0620 L23 c.moveTo
C03801 F0620 L23 c.bezierCurveTo
C03802 F0620 L23 c.stroke
C03803 F0620 L23 ellipse
C03804 F0620 L23 ellipse
C03805 F0620 L23 Math.max
C03806 F0620 L23 c.beginPath
C03807 F0620 L23 c.moveTo
C03808 F0620 L23 c.lineTo
C03809 F0620 L23 c.stroke
C03810 F0620 L24 t.includes
C03811 F0620 L24 Math.max
C03812 F0620 L24 c.beginPath
C03813 F0620 L24 c.moveTo
C03814 F0620 L24 c.bezierCurveTo
C03815 F0620 L24 c.stroke
C03816 F0620 L24 ellipse
C03817 F0620 L24 c.beginPath
C03818 F0620 L24 c.moveTo
C03819 F0620 L24 c.lineTo
C03820 F0620 L24 c.stroke
```

```
C03821 F0620 L24 ellipse
C03822 F0620 L25 /guardian/.test
C03823 F0620 L25 /garden/i.test
C03824 F0620 L25 /ocean/i.test
C03825 F0620 L25 /factory/i.test
C03826 F0620 L25 Math.max
C03827 F0620 L25 c.save
C03828 F0620 L25 c.rotate
C03829 F0620 L25 c.beginPath
C03830 F0620 L25 c.moveTo
C03831 F0620 L25 c.lineTo
C03832 F0620 L25 c.lineTo
C03833 F0620 L25 c.stroke
C03834 F0620 L25 c.restore
C03835 F0620 L26 t.includes
C03836 F0620 L26 t.includes
C03837 F0620 L26 t.includes
C03838 F0620 L26 t.includes
C03839 F0620 L26 t.includes
C03840 F0620 L27 c.beginPath
C03841 F0620 L27 c.moveTo
C03842 F0620 L27 c.lineTo
C03843 F0620 L27 c.lineTo
C03844 F0620 L27 c.stroke
C03845 F0620 L27 ellipse
C03846 F0620 L27 ellipse
C03847 F0620 L27 ellipse
C03848 F0620 L27 ellipse
C03849 F0620 L28 Math.max
C03850 F0620 L28 c.beginPath
C03851 F0620 L28 c.moveTo
C03852 F0620 L28 Math.sin
C03853 F0620 L28 c.quadraticCurveTo
C03854 F0620 L28 Math.sin
C03855 F0620 L28 Math.sin
C03856 F0620 L28 c.stroke
C03857 F0620 L28 ellipse
C03858 F0620 L29 ellipse
C03859 F0620 L29 ellipse
C03860 F0620 L29 Math.max
C03861 F0620 L29 c.beginPath
C03862 F0620 L29 c.moveTo
C03863 F0620 L29 c.lineTo
C03864 F0620 L29 c.moveTo
C03865 F0620 L29 c.lineTo
C03866 F0620 L29 c.lineTo
C03867 F0620 L29 c.stroke
C03868 F0620 L30 t.includes
C03869 F0620 L30 Math.max
C03870 F0620 L30 c.beginPath
C03871 F0620 L30 c.moveTo
C03872 F0620 L30 c.lineTo
C03873 F0620 L30 c.lineTo
C03874 F0620 L30 c.stroke
C03875 F0620 L30 ellipse
C03876 F0620 L30 t.includes
C03877 F0620 L32 c.save
C03878 F0620 L32 c.beginPath
C03879 F0620 L32 c.ellipse
C03880 F0620 L32 c.clip
```

```
C03881 F0620 L33 /slime|jelly|bubble/.test
C03882 F0620 L33 ellipse
C03883 F0620 L33 Math.sin
C03884 F0620 L33 Math.cos
C03885 F0620 L33 ellipse
C03886 F0620 L34 /spider/.test
C03887 F0620 L34 Math.max
C03888 F0620 L34 c.beginPath
C03889 F0620 L34 c.moveTo
C03890 F0620 L34 c.lineTo
C03891 F0620 L34 c.lineTo
C03892 F0620 L34 c.moveTo
C03893 F0620 L34 c.lineTo
C03894 F0620 L34 c.stroke
C03895 F0620 L34 ellipse
C03896 F0620 L35 /ant|termite|centipede/.test
C03897 F0620 L35 ellipse
C03898 F0620 L36 /scarab|beetle/.test
C03899 F0620 L36 Math.max
C03900 F0620 L36 c.beginPath
C03901 F0620 L36 c.moveTo
C03902 F0620 L36 c.quadraticCurveTo
C03903 F0620 L36 c.stroke
C03904 F0620 L37 /mecha|drone|barrel/.test
C03905 F0620 L37 Math.max
C03906 F0620 L37 c.beginPath
C03907 F0620 L37 c.moveTo
C03908 F0620 L37 c.lineTo
C03909 F0620 L37 c.stroke
C03910 F0620 L38 c.restore
C03911 F0620 L39 Math.max
C03912 F0620 L39 c.beginPath
C03913 F0620 L39 c.ellipse
C03914 F0620 L39 c.stroke
C03915 F0620 L39 ellipse
C03916 F0620 L39 Math.cos
C03917 F0620 L39 Math.sin
C03918 F0620 L39 c.restore
C03919 F0621 L40 c.save
C03920 F0621 L40 c.createLinearGradient
C03921 F0621 L40 g.addColorStop
C03922 F0621 L40 g.addColorStop
C03923 F0621 L40 g.addColorStop
C03924 F0621 L40 c.fillRect
C03925 F0621 L40 c.createRadialGradient
C03926 F0621 L40 Math.min
C03927 F0621 L40 Math.max
C03928 F0621 L40 edge.addColorStop
C03929 F0621 L40 edge.addColorStop
C03930 F0621 L40 c.fillRect
C03931 F0621 L40 Math.sin
C03932 F0621 L40 c.beginPath
C03933 F0621 L40 c.moveTo
C03934 F0621 L40 c.lineTo
C03935 F0621 L40 c.lineTo
C03936 F0621 L40 c.lineTo
C03937 F0621 L40 c.closePath
C03938 F0621 L40 c.fill
C03939 F0621 L40 c.restore
C03940 F0622 L41 c.save
```

```
C03941 F0622 L41 c.translate
C03942 F0622 L41 c.beginPath
C03943 F0622 L41 c.arc
C03944 F0622 L41 c.clip
C03945 F0622 L41 c.createLinearGradient
C03946 F0622 L41 g.addColorStop
C03947 F0622 L41 g.addColorStop
C03948 F0622 L41 g.addColorStop
C03949 F0622 L41 c.fillRect
C03950 F0622 L41 /garden|jungle/.test
C03951 F0622 L41 c.beginPath
C03952 F0622 L41 c.moveTo
C03953 F0622 L41 Math.cos
C03954 F0622 L41 Math.sin
C03955 F0622 L41 c.quadraticCurveTo
C03956 F0622 L41 Math.cos
C03957 F0622 L41 Math.sin
C03958 F0622 L41 Math.cos
C03959 F0622 L41 Math.sin
C03960 F0622 L41 c.stroke
C03961 F0622 L41 c.beginPath
C03962 F0622 L41 c.moveTo
C03963 F0622 L41 c.lineTo
C03964 F0622 L41 c.stroke
C03965 F0622 L41 ellipse
C03966 F0622 L41 Math.cos
C03967 F0622 L41 Math.sin
C03968 F0622 L41 c.restore
C03969 F0623 L42 String(type).toLowerCase
C03970 F0623 L42 String
C03971 F0623 L42 /ant|termite|mantis|centipede|scorpion|spider|beetle|scarab|ladybug/.test
C03972 F0623 L42 /wasp|dragonfly|mosquito|drone/.test
C03973 F0623 L42 /mecha|drone|barrel|factory/.test
C03974 F0624 L43 m.t.toLowerCase
C03975 F0624 L43 c.save
C03976 F0624 L43 c.beginPath
C03977 F0624 L43 c.ellipse
C03978 F0624 L43 c.clip
C03979 F0624 L43 Math.max
C03980 F0624 L44 /mecha|drone|factory|barrel/.test
C03981 F0624 L44 c.beginPath
C03982 F0624 L44 c.moveTo
C03983 F0624 L44 c.lineTo
C03984 F0624 L44 c.lineTo
C03985 F0624 L44 c.lineTo
C03986 F0624 L44 c.stroke
C03987 F0624 L44 ellipse
C03988 F0624 L45 /slime|jelly|bubble|ocean/.test
C03989 F0624 L45 ellipse
C03990 F0624 L45 Math.cos
C03991 F0624 L45 Math.sin
C03992 F0624 L46 c.beginPath
C03993 F0624 L46 c.moveTo
C03994 F0624 L46 c.bezierCurveTo
C03995 F0624 L46 c.stroke
C03996 F0624 L47 c.createLinearGradient
C03997 F0624 L47 rim.addColorStop
C03998 F0624 L47 rim.addColorStop
C03999 F0624 L47 rim.addColorStop
C04000 F0624 L47 c.fillRect
```

```
C04001 F0624 L47 c.restore
C04002 F0625 L48 states.push
C04003 F0625 L48 states.push
C04004 F0625 L48 states.push
C04005 F0625 L48 c.save
C04006 F0625 L48 c.translate
C04007 F0625 L48 c.beginPath
C04008 F0625 L48 c.moveTo
C04009 F0625 L48 Math.cos
C04010 F0625 L48 Math.sin
C04011 F0625 L48 c.arc
C04012 F0625 L48 c.stroke
C04013 F0625 L48 c.fillRect
C04014 F0625 L48 c.restore
C04015 F0626 L49 Math.max
C04016 F0626 L49 Math.min
C04017 F0626 L49 globalThis.PetalRules.specialAim
C04018 F0626 L49 Math.atan2
C04019 F0626 L49 c.save
C04020 F0626 L49 c.translate
C04021 F0626 L50 c.beginPath
C04022 F0626 L50 c.arc
C04023 F0626 L50 c.fill
C04024 F0626 L50 c.setLineDash
C04025 F0626 L50 c.stroke
C04026 F0626 L50 c.setLineDash
C04027 F0626 L50 c.beginPath
C04028 F0626 L50 c.arc
C04029 F0626 L50 c.stroke
C04030 F0626 L51 c.rotate
C04031 F0626 L51 c.beginPath
C04032 F0626 L51 c.roundRect
C04033 F0626 L51 c.fill
C04034 F0626 L51 c.setLineDash
C04035 F0626 L51 c.stroke
C04036 F0626 L51 c.setLineDash
C04037 F0626 L51 c.beginPath
C04038 F0626 L51 c.moveTo
C04039 F0626 L51 c.lineTo
C04040 F0626 L51 c.lineTo
C04041 F0626 L51 c.stroke
C04042 F0626 L51 c.rotate
C04043 F0626 L52 c.rotate
C04044 F0626 L52 c.save
C04045 F0626 L52 c.rotate
C04046 F0626 L52 c.beginPath
C04047 F0626 L52 c.moveTo
C04048 F0626 L52 c.lineTo
C04049 F0626 L52 c.setLineDash
C04050 F0626 L52 c.stroke
C04051 F0626 L52 c.restore
C04052 F0626 L52 c.rotate
C04053 F0626 L53 c.setLineDash
C04054 F0626 L53 c.beginPath
C04055 F0626 L53 c.arc
C04056 F0626 L53 c.stroke
C04057 F0626 L53 ellipse
C04058 F0626 L53 c.fillText
C04059 F0626 L53 c.restore
C04060 F0614 L54 Object.freeze
```

Appendix E. Complete numeric literal registry

All 3,824 numeric literal occurrences, grouped by function owner and value. Each group lists every N identifier and line, preserving occurrence count. A number is a source literal, not automatically a balancing coefficient: array indices, timeouts, colors and algorithm constants have different roles. Consult the expression envelope or exact source line for its operation and unit. Negative values may appear as unary minus applied to a positive literal; the operator ledger preserves that sign.

F0001

2 x12 | N00001@L240, N00004@L240, N00005@L240, N00336@L284, N00368@L288, N00458@L321, N00459@L321, N00703@L882, N00704@L882, N00705@L882, N00706@L882, N00707@L882

6800 x1 | N00002@L240

5000 x1 | N00003@L240

F0008

0x6d2b79f5 x1 | N00006@L242

0 x2 | N00007@L242, N00013@L242

15 x1 | N00008@L242

1 x1 | N00009@L242

7 x1 | N00010@L242

61 x1 | N00011@L242

14 x1 | N00012@L242

4294967296 x1 | N00014@L242

F0009

1.72 x1 | N00015@L243

7 x1 | N00016@L243

1 x1 | N00017@L243

.64 x1 | N00018@L243

2 x1 | N00019@L243

F0001

34 x4 | N00020@L244, N00052@L244, N00108@L248, N00273@L270

13 x6 | N00021@L244, N00035@L244, N00141@L260, N00153@L260, N00165@L261, N00257@L267

2.1 x1 | N00022@L244

16 x13 | N00023@L244, N00045@L244, N00047@L244, N00079@L244, N00091@L244, N00095@L245, N00105@L245, N00129@L256, N00147@L260, N00233@L266, N00243@L266, N00395@L293, N00398@L293

30 x5 | N00024@L244, N00128@L256, N00606@L842, N00612@L843, N00753@L896

6 x3 | N00025@L244, N00101@L245, N00599@L842

3.4 x1 | N00026@L244

17 x12 | N00027@L244, N00051@L244, N00055@L244, N00059@L244, N00067@L244, N00083@L244, N00111@L248, N00123@L251, N00131@L256, N00177@L262, N00185@L262, N00241@L266

40 x1 | N00028@L244

7 x4 | N00029@L244, N00113@L249, N00125@L252, N00601@L842

2.0 x1 | N00030@L244

18 x16 | N00031@L244, N00053@L244, N00063@L244, N00103@L245, N00115@L249, N00127@L252, N00135@L257, N00157@L261, N00173@L261, N00191@L262, N00221@L265, N00431@L314, N00433@L314, N00611@L843, N00752@L896, N00756@L896

11 x3 | N00032@L244, N00037@L244, N00229@L265

31 x3 | N00033@L244, N00258@L267, N00269@L269

4.3 x1 | N00034@L244

60 x1 | N00036@L244

3.3 x1 | N00038@L244

20 x6 | N00039@L244, N00071@L244, N00119@L250, N00203@L263, N00604@L842, N00700@L881

27 x4 | N00040@L244, N00205@L264, N00207@L264, N00303@L277

9 x3 | N00041@L244, N00085@L244, N00605@L842

1.9 x1 | N00042@L244

19 x13 | N00043@L244, N00049@L244, N00065@L244, N00075@L244, N00087@L244, N00099@L245, N00107@L245, N00121@L251, N00143@L260, N00199@L263, N00231@L265, N00245@L267, N00396@L293

38 x1 | N00044@L244

2.2 x2 | N00046@L244, N00102@L245

49 x2 | N00048@L244, N00298@L277

2.7 x3 | N00050@L244, N00054@L244, N00082@L244

26 x9 | N00056@L244, N00092@L245, N00163@L261, N00187@L262, N00195@L263, N00209@L264, N00239@L266, N00251@L267, N00261@L267

24 x10 | N00057@L244, N00088@L244, N00167@L261, N00175@L261, N00183@L262, N00215@L264, N00259@L267, N00299@L277, N00749@L896, N00755@L896

3.1 x1 | N00058@L244

80 x1 | N00060@L244

21 x10 | N00061@L244, N00077@L244, N00193@L263, N00201@L263, N00217@L264, N00219@L264, N00249@L267, N00400@L293, N00754@L896, N00757@L896

3.5 x1 | N00062@L244

35 x1 | N00064@L244

2.4 x2 | N00066@L244, N00106@L245

58 x2 | N00068@L244, N00166@L261

29 x1 | N00069@L244

3.7 x1 | N00070@L244

73 x2 | N00072@L244, N00160@L261

12 x7 | N00073@L244, N00161@L261, N00237@L266, N00297@L277, N00393@L293, N00602@L842, N00699@L881

3.0 x3 | N00074@L244, N00114@L249, N00122@L251

41 x1 | N00076@L244

2.5 x2 | N00078@L244, N00110@L248

32 x2 | N00080@L244, N00100@L245

14 x6 | N00081@L244, N00181@L262, N00189@L262, N00197@L263, N00394@L293, N00698@L881

44 x4 | N00084@L244, N00112@L249, N00124@L252, N00279@L271

3 x2 | N00086@L244, N00344@L285

4 x2 | N00089@L244, N00609@L843

2.3 x1 | N00090@L244

23 x7 | N00093@L245, N00151@L260, N00155@L260, N00235@L266, N00247@L267, N00253@L267, N00305@L277

2.9 x1 | N00094@L245

37 x2 | N00096@L245, N00285@L273

15 x5 | N00097@L245, N00149@L260, N00301@L277, N00399@L293, N00758@L896

2.6 x3 | N00098@L245, N00126@L252, N00134@L257

48 x3 | N00104@L245, N00140@L260, N00281@L272

22 x8 | N00109@L248, N00120@L251, N00139@L260, N00171@L261, N00223@L265, N00225@L265, N00255@L267, N00397@L293

54 x1 | N00116@L250

10 x4 | N00117@L250, N00133@L257, N00607@L842, N00610@L843

2.8 x2 | N00118@L250, N00130@L256

39 x2 | N00132@L257, N00277@L271

63 x1 | N00136@L260

8 x3 | N00137@L260, N00145@L260, N00603@L842

66 x2 | N00138@L260, N00614@L843

107 x1 | N00142@L260

45 x4 | N00144@L260, N00168@L261, N00267@L268, N00275@L270

103 x2 | N00146@L260, N00194@L263

82 x1 | N00148@L260

100 x5 | N00150@L260, N00218@L264, N00430@L314, N00432@L314, N00434@L314

96 x2 | N00152@L260, N00158@L261

86 x2 | N00154@L260, N00270@L269

115 x2 | N00156@L261, N00220@L265

28 x6 | N00159@L261, N00263@L267, N00265@L268, N00307@L277, N00702@L881, N00750@L896

67 x3 | N00162@L261, N00196@L263, N00286@L273

95 x2 | N00164@L261, N00176@L262

5 x8 | N00169@L261, N00402@L298, N00404@L299, N00406@L300, N00408@L301, N00410@L302, N00412@L303, N00600@L842

69 x2 | N00170@L261, N00188@L262

118 x1 | N00172@L261

51 x2 | N00174@L261, N00287@L273

98 x1 | N00178@L262

25 x5 | N00179@L262, N00211@L264, N00213@L264, N00701@L881, N00751@L896

105 x2 | N00180@L262, N00244@L267

85 x2 | N00182@L262, N00200@L263

140 x1 | N00184@L262

70 x2 | N00186@L262, N00262@L267

126 x1 | N00190@L262

106 x1 | N00192@L263

145 x1 | N00198@L263

122 x4 | N00202@L263, N00206@L264, N00214@L264, N00321@L281

180 x3 | N00204@L264, N00248@L267, N00256@L267

166 x1 | N00208@L264

113 x1 | N00210@L264

138 x1 | N00212@L264

110 x3 | N00216@L264, N00232@L266, N00296@L277

124 x1 | N00222@L265

280 x3 | N00224@L265, N00354@L286, N00362@L287

55 x1 | N00226@L265

33 x2 | N00227@L265, N00289@L274

75 x1 | N00228@L265

101 x1 | N00230@L265

119 x1 | N00234@L266

130 x1 | N00236@L266

56 x1 | N00238@L266

53 x2 | N00240@L266, N00295@L275

148 x1 | N00242@L266

117 x1 | N00246@L267

102 x1 | N00250@L267

120 x1 | N00252@L267

134 x1 | N00254@L267

225 x1 | N00260@L267

480 x1 | N00264@L268

84 x1 | N00266@L268

620 x1 | N00268@L269

47 x1 | N00271@L269

700 x2 | N00272@L270, N00310@L281

91 x1 | N00274@L270

730 x1 | N00276@L271

112 x1 | N00278@L271

1100 x1 | N00280@L272

83 x1 | N00282@L272

52 x1 | N00283@L272

830 x1 | N00284@L273

770 x1 | N00288@L274

88 x1 | N00290@L274

46 x2 | N00291@L274, N00613@L843

1080 x1 | N00292@L275

43 x1 | N00293@L275

77 x1 | N00294@L275

170 x1 | N00300@L277

72 x1 | N00302@L277

210 x1 | N00304@L277

68 x1 | N00306@L277

0 x28 | N00308@L281, N00429@L314, N00442@L320, N00443@L320, N00445@L320, N00446@L320, N00447@L320, N00448@L320, N00449@L320, N00451@L320, N00452@L320, N00453@L320, N00454@L320, N00455@L320, N00456@L320, N00460@L321, N00461@L321, N00465@L336, N00466@L336, N00535@L599, N00536@L599, N00608@L843, N01040@L951, N02536@L1168, N02552@L1171, N02553@L1171, N02559@L1173, N02792@L1526

2230 x1 | N00309@L281

610 x1 | N00311@L281

1480 x1 | N00312@L281

2090 x1 | N00313@L281

1430 x1 | N00314@L281

1260 x2 | N00315@L281, N00353@L286

1510 x1 | N00316@L281

1240 x2 | N00317@L281, N00341@L284

460 x1 | N00318@L281

2190 x1 | N00319@L281

350 x1 | N00320@L281

1 x9 | N00322@L282, N00328@L283, N00352@L286, N00360@L287, N00450@L320, N00457@L320, N00598@L842, N02793@L1526, N02833@L1544

340 x1 | N00323@L282

690 x1 | N00324@L282

2040 x1 | N00325@L282

1400 x2 | N00326@L282, N00334@L283

213 x1 | N00327@L282

1940 x1 | N00329@L283

380 x4 | N00330@L283, N00365@L287, N00369@L288, N00371@L288

580 x1 | N00331@L283

1420 x1 | N00332@L283

2130 x1 | N00333@L283

314 x1 | N00335@L283

310 x1 | N00337@L284

420 x2 | N00338@L284, N00340@L284

2100 x2 | N00339@L284, N00349@L285

1550 x1 | N00342@L284

415 x1 | N00343@L284

360 x2 | N00345@L285, N00357@L286

530 x1 | N00346@L285

1220 x1 | N00347@L285

260 x1 | N00348@L285

1280 x1 | N00350@L285

516 x1 | N00351@L285

2030 x1 | N00355@L286

1330 x1 | N00356@L286

1300 x1 | N00358@L286

617 x1 | N00359@L286

1250 x1 | N00361@L287

2120 x2 | N00363@L287, N00373@L288

900 x1 | N00364@L287

1350 x1 | N00366@L287

718 x1 | N00367@L287

500 x1 | N00370@L288

1290 x1 | N00372@L288

950 x1 | N00374@L288

819 x1 | N00375@L288

.50 x1 | N00376@L290

.40 x2 | N00377@L290, N00387@L290

.83 x1 | N00378@L290

.21 x1 | N00379@L290

.33 x1 | N00380@L290

.70 x1 | N00381@L290

.77 x1 | N00382@L290

.62 x1 | N00383@L290

.66 x1 | N00384@L290

.87 x1 | N00385@L290

.27 x2 | N00386@L290, N00616@L844

.10 x1 | N00388@L290

.23 x1 | N00389@L290

.53 x1 | N00390@L290

.12 x1 | N00391@L290

150 x1 | N00392@L292

.06 x2 | N00401@L298, N00405@L300
.05 x2 | N00403@L299, N00407@L301
.04 x1 | N00409@L302
.20 x1 | N00411@L303

F0010

0 x1 | N00413@L305
1 x1 | N00414@L305
2 x1 | N00415@L305

F0012

0 x1 | N00416@L306

F0011

0 x3 | N00417@L306, N00418@L306, N00419@L306

F0013

0 x1 | N00420@L307

F0014

0 x2 | N00421@L308, N00423@L308
1 x1 | N00422@L308

F0016

1 x1 | N00424@L309

F0018

1 x1 | N00425@L311
.35 x1 | N00426@L311
.05 x1 | N00427@L311

F0019

1 x1 | N00428@L312

F0021

0 x1 | N00435@L315

F0022

0 x1 | N00436@L316
.30 x1 | N00437@L316
55 x1 | N00438@L316
16 x1 | N00439@L316

F0023

0 x1 | N00440@L317
82 x1 | N00441@L317

F0001

65 x1 | N00444@L320

F0026

1 x1 | N00462@L334

F0027

1 x1 | N00463@L335
10 x1 | N00464@L335

F0030

1 x1 | N00467@L339

F0031

1 x1 | N00468@L340

F0032

1 x1 | N00469@L341

F0035

0 x1 | N00470@L373
240 x1 | N00471@L373

F0036

86400000 x1 | N00472@L379

F0037

0 x2 | N00473@L380, N00474@L380
1000 x1 | N00475@L380

F0038

0 x1 | N00476@L381
86400 x1 | N00477@L381
1 x1 | N00478@L381
1000 x1 | N00479@L381

F0039

429 x1 | N00480@L385

F0040

0 x3 | N00481@L391, N00482@L392, N00483@L393

F0041

60 x2 | N00484@L398, N00487@L400
429 x1 | N00485@L399
3600 x1 | N00486@L400

F0042

1000 x1 | N00488@L412
0 x3 | N00489@L412, N00491@L412, N00492@L412
86400 x1 | N00490@L412

F0043

30 x1 | N00493@L415
3600 x1 | N00494@L415
1000 x1 | N00495@L415
0 x1 | N00496@L415

F0044

60000 x1 | N00497@L416
30 x1 | N00498@L418
3600 x1 | N00499@L418
1000 x1 | N00500@L418

F0045

0 x2 | N00501@L419, N00502@L419

F0046

1200 x1 | N00503@L420

F0049

1 x4 | N00504@L428, N00506@L429, N00508@L430, N00516@L439
5 x2 | N00505@L428, N00507@L429
0 x2 | N00509@L431, N00510@L431
1200 x1 | N00511@L438
6 x1 | N00512@L439
30000 x1 | N00513@L439
2000 x1 | N00514@L439
2 x1 | N00515@L439

F0050

300 x1 | N00517@L452

F0052

254 x1 | N00518@L467
6 x1 | N00519@L467
8 x1 | N00520@L470

F0055

1 x1 | N00521@L507
3600 x2 | N00522@L518, N00523@L523

F0056

8 x1 | N00524@L533
1000 x1 | N00525@L540

F0057

8 x1 | N00526@L553

F0058

1 x1 | N00527@L568
0 x2 | N00528@L568, N00530@L574
2 x1 | N00529@L573
10 x1 | N00531@L574
3000 x1 | N00532@L574

F0061

254 x1 | N00533@L580
18 x1 | N00534@L580

F0074

15000 x1 | N00537@L616
86400 x1 | N00538@L619
0 x2 | N00539@L619, N00540@L619

F0075

1 x2 | N00541@L623, N00542@L623

F0076

1500 x1 | N00543@L626

F0078

1 x1 | N00544@L630

F0079

0 x2 | N00545@L639, N00554@L647
409 x1 | N00546@L644
401 x2 | N00547@L645, N00548@L646
6 x1 | N00549@L646
1000 x1 | N00550@L646
30000 x1 | N00551@L646
2000 x1 | N00552@L646
2 x1 | N00553@L646

F0080

254 x1 | N00555@L650

F0085

2 x1 | N00556@L664
3000 x1 | N00557@L664

F0087

0 x1 | N00558@L674
300 x1 | N00559@L678

F0091

25 x1 | N00560@L700

F0094

1 x1 | N00561@L724

F0099

1 x1 | N00562@L741

F0100

0 x1 | N00563@L742

F0101

1048576 x1 | N00564@L749
1 x1 | N00565@L749

F0103

409 x1 | N00566@L766

F0107

1 x1 | N00567@L785
10 x1 | N00568@L786

F0110

0 x1 | N00569@L792
400 x1 | N00570@L792

F0112

0 x2 | N00571@L797, N00573@L800
1 x1 | N00572@L797

F0113

1048576 x1 | N00574@L804
1 x2 | N00575@L806, N00577@L807
5 x1 | N00576@L806
999 x1 | N00578@L807

0 x2 | N00579@L809, N00580@L810
99999 x1 | N00581@L810

F0114

1 x1 | N00582@L816
2 x1 | N00583@L817
0 x1 | N00584@L818
10 x1 | N00585@L818
3000 x1 | N00586@L818

F0116

1048576 x1 | N00587@L823

F0117

1 x1 | N00588@L834

F0119

1 x1 | N00589@L839
2 x1 | N00590@L839
0 x1 | N00591@L839

F0120

1 x1 | N00592@L839
0 x2 | N00593@L839, N00594@L839
99999 x1 | N00595@L839

F0121

1 x1 | N00596@L840
0 x1 | N00597@L840

F0001

.15 x1 | N00615@L844
.39 x1 | N00617@L844
.52 x1 | N00618@L844
.65 x1 | N00619@L844
.78 x1 | N00620@L844

F0122

6 x1 | N00621@L845

F0123

1 x2 | N00622@L846, N00624@L846
1.75 x1 | N00623@L846
.05 x1 | N00625@L846
10 x1 | N00626@L846

F0124

1 x1 | N00627@L846

F0126

.12 x1 | N00628@L846
.018 x1 | N00629@L846

F0124

0 x1 | N00630@L846

F0127

.38 x1 | N00631@L846

1 x1 | N00632@L846

F0128

0 x1 | N00633@L847
6 x1 | N00634@L847

F0129

1 x1 | N00635@L848
4 x1 | N00636@L848

F0130

3 x1 | N00637@L849

F0131

100 x1 | N00638@L850
1 x1 | N00639@L850
7 x1 | N00640@L850

F0132

24 x1 | N00641@L850
42 x1 | N00642@L850
0 x1 | N00643@L850

F0131

0 x1 | N00644@L850

F0135

24 x1 | N00645@L852
7 x1 | N00646@L852

F0136

.55 x1 | N00647@L853
.89 x1 | N00648@L853

F0138

0 x2 | N00649@L854, N00650@L854

F0139

2.5 x1 | N00651@L854

F0137

0 x1 | N00652@L854

F0141

0 x2 | N00653@L855, N00654@L855

F0142

38 x1 | N00655@L856

F0143

0 x2 | N00656@L856, N00659@L856
95 x1 | N00657@L856
15 x1 | N00658@L856

F0142

0 x1 | N00660@L856

F0144

1 x4 | N00661@L860, N00662@L860, N00663@L860, N00664@L861

F0145

0 x4 | N00665@L864, N00666@L864, N00668@L864, N00671@L866

1.2 x2 | N00667@L864, N00669@L864

.06 x1 | N00670@L866

F0146

135 x1 | N00672@L868

51 x1 | N00673@L868

92 x1 | N00674@L868

42 x1 | N00675@L868

35 x1 | N00676@L868

5 x1 | N00677@L868

.7 x1 | N00678@L868

6 x1 | N00679@L868

F0147

24 x1 | N00680@L870

100 x1 | N00681@L870

1 x2 | N00682@L870, N00686@L870

0 x13 | N00683@L870, N00684@L870, N00685@L870, N00688@L870, N00689@L870, N00690@L870, N00691@L870, N00692@L870, N00693@L870, N00694@L870, N00695@L870, N00696@L870, N00697@L870

252 x1 | N00687@L870

F0149

4 x1 | N00708@L883

2 x1 | N00709@L883

1 x1 | N00710@L883

F0151

0 x1 | N00711@L884

F0152

0 x5 | N00712@L886, N00726@L889, N00734@L889, N00735@L889, N00737@L889

170 x1 | N00713@L888

30 x1 | N00714@L888

35 x1 | N00715@L888

3 x1 | N00716@L888

1 x2 | N00717@L888, N00724@L889

.18 x1 | N00718@L888

.25 x2 | N00719@L888, N00725@L889

.75 x1 | N00720@L888

20 x1 | N00721@L888

17 x1 | N00722@L889

1.7 x1 | N00723@L889

10 x2 | N00727@L889, N00728@L889

38 x2 | N00729@L889, N00730@L889

500 x2 | N00731@L889, N00732@L889

430 x1 | N00733@L889

9 x1 | N00736@L889

12 x2 | N00738@L889, N00740@L890

4.5 x1 | N00739@L890

.13 x1 | N00741@L891

.06 x1 | N00742@L891

60 x1 | N00743@L891

14 x1 | N00744@L891

7 x1 | N00745@L892
16 x1 | N00746@L892

F0153

0 x2 | N00747@L895, N00748@L895

F0154

0 x2 | N00759@L897, N00760@L897

F0156

0 x2 | N00761@L898, N00764@L898
130 x1 | N00762@L898
1 x3 | N00763@L898, N00766@L898, N00768@L898
2 x1 | N00765@L898
.15 x1 | N00767@L898
.10 x1 | N00769@L898

F0157

3 x2 | N00770@L899, N00788@L901
48 x1 | N00771@L899
42 x1 | N00772@L899
0 x2 | N00773@L900, N00778@L901
7 x2 | N00774@L900, N00779@L901
130 x1 | N00775@L900
1 x2 | N00776@L900, N00787@L901

F0158

0 x1 | N00777@L901

F0157

75 x1 | N00780@L901
210 x1 | N00781@L901
55 x4 | N00782@L901, N00783@L901, N00784@L901, N00785@L901
6 x1 | N00786@L901
47 x1 | N00789@L901

F0159

2.2 x1 | N00790@L904
3.1 x1 | N00791@L905
6.3 x1 | N00792@L905
0 x1 | N00793@L906
2 x1 | N00794@L906
1 x1 | N00795@L906
50 x1 | N00796@L907
150 x1 | N00797@L907
60 x4 | N00798@L907, N00799@L907, N00800@L907, N00801@L907
1.1 x1 | N00802@L907
95 x1 | N00803@L907
24 x1 | N00804@L909
6 x1 | N00805@L909

F0161

0 x2 | N00806@L911, N00808@L911
1 x2 | N00807@L911, N00809@L911

F0162

4.2 x1 | N00810@L912
5.2 x1 | N00811@L912

10.3 x1 | N00812@L912

0 x2 | N00813@L912, N00816@L912

.45 x1 | N00814@L912

1.8 x1 | N00815@L912

.75 x1 | N00817@L912

1 x1 | N00818@L912

F0163

0 x1 | N00819@L913

10 x1 | N00820@L913

8 x1 | N00821@L913

.48 x1 | N00822@L913

.6 x1 | N00823@L913

F0164

1 x6 | N00824@L915, N00836@L918, N00839@L918, N00841@L918, N00845@L918, N00846@L918

0 x10 | N00825@L918, N00843@L918, N00844@L918, N00847@L918, N00848@L918, N00849@L918, N00850@L918, N00851@L918, N00853@L918, N00854@L918

3 x3 | N00826@L918, N00828@L918, N00840@L918

125 x1 | N00827@L918

95 x2 | N00829@L918, N00832@L918

70 x4 | N00830@L918, N00831@L918, N00833@L918, N00834@L918

F0165

25 x1 | N00835@L918

F0164

.045 x2 | N00837@L918, N00842@L918

6 x2 | N00838@L918, N00852@L918

.65 x1 | N00855@L919

20 x1 | N00856@L919

4 x1 | N00857@L920

44 x1 | N00858@L920

F0166

5 x1 | N00859@L922

3 x1 | N00860@L922

0 x3 | N00861@L924, N00862@L924, N00863@L924

590 x1 | N00864@L925

F0167

0 x1 | N00865@L927

F0168

0 x2 | N00866@L927, N00867@L927

1.8 x1 | N00868@L927

.5 x1 | N00869@L927

F0169

0 x2 | N00870@L928, N00871@L928

F0170

1 x2 | N00872@L929, N00879@L929

0 x9 | N00873@L929, N00874@L929, N00880@L929, N00881@L929, N00882@L929, N00883@L929, N00884@L929, N00885@L929, N00886@L929

100 x1 | N00875@L929

8 x1 | N00876@L929

3 x1 | N00877@L929

2 x1 | N00878@L929

F0172

5 x1 | N00887@L931

F0173

1 x6 | N00888@L932, N00894@L932, N00895@L932, N00910@L936, N00911@L936, N00928@L938

0 x27 | N00889@L932, N00890@L932, N00892@L932, N00893@L932, N00897@L932, N00898@L932, N00900@L932, N00901@L932, N00905@L932, N00906@L932, N00912@L936, N00913@L936, N00914@L936, N00915@L936, N00917@L936, N00918@L936, N00920@L936, N00921@L937, N00922@L937, N00924@L937, N00925@L937, N00930@L938, N00931@L938, N00932@L938, N00933@L938, N00934@L938, N00935@L938

1.3 x1 | N00891@L932

999 x2 | N00896@L932, N00926@L937

1e9 x2 | N00899@L932, N00902@L932

65 x1 | N00903@L932

45 x1 | N00904@L932

99999 x1 | N00907@L932

F0174

0 x1 | N00908@L934

F0175

0 x1 | N00909@L935

F0173

1e8 x1 | N00916@L936

1e6 x1 | N00919@L936

100000000 x1 | N00923@L937

100 x1 | N00927@L938

2 x1 | N00929@L938

18 x1 | N00936@L938

F0178

2500 x1 | N00937@L939

1760 x1 | N00938@L939

51 x1 | N00939@L939

F0177

0 x6 | N00940@L940, N00949@L941, N00961@L943, N00964@L943, N00974@L943, N00975@L943

950 x1 | N00941@L940

75 x3 | N00942@L940, N00944@L940, N00965@L943

150 x2 | N00943@L940, N00945@L940

2 x1 | N00946@L940

10 x1 | N00947@L940

5 x1 | N00948@L940

285 x1 | N00950@L941

115 x2 | N00951@L941, N00953@L941

230 x2 | N00952@L941, N00954@L941

19 x1 | N00955@L941

33 x1 | N00956@L941

260 x1 | N00957@L941

F0179

135 x1 | N00958@L941

F0180

13 x1 | N00959@L941

F0177

3 x1 | N00960@L941

4 x1 | N00962@L943

8 x1 | N00963@L943

F0181

115 x1 | N00966@L943

F0182

145 x1 | N00967@L943

F0177

300 x1 | N00968@L943

F0183

240 x1 | N00969@L943

F0177

290 x2 | N00970@L943, N00972@L943

580 x2 | N00971@L943, N00973@L943

125 x1 | N00976@L943

F0185

145 x1 | N00977@L944

.025 x1 | N00978@L944

0 x13 | N00979@L945, N01005@L949, N01024@L950, N01025@L950, N01028@L950, N01029@L950, N01030@L950, N01031@L950, N01033@L950, N01036@L950, N01037@L950, N01038@L950, N01039@L950

60 x1 | N00980@L945

.66 x1 | N00981@L946

980 x2 | N00982@L946, N00983@L946

100 x6 | N00984@L946, N00985@L946, N00986@L946, N00990@L946, N00991@L946, N00992@L946

200 x2 | N00987@L946, N00993@L946

760 x2 | N00988@L946, N00989@L946

360 x1 | N00994@L947

310 x1 | N00995@L947

2 x2 | N00996@L947, N01027@L950

F0187

10 x1 | N00997@L948

F0188

42 x1 | N00998@L948

F0189

9 x1 | N00999@L948

F0185

6 x1 | N01000@L949

.003 x1 | N01001@L949

7 x2 | N01002@L949, N01035@L950

4 x1 | N01003@L949

18 x1 | N01004@L949

.05 x1 | N01006@L949

.017 x1 | N01007@L949

.82 x1 | N01008@L949

.15 x1 | N01009@L949

1.45 x1 | N01010@L949

1 x7 | N01011@L949, N01013@L949, N01015@L950, N01019@L950, N01021@L950, N01022@L950, N01032@L950

2.4 x1 | N01012@L949

1.16 x1 | N01014@L950

.84 x1 | N01016@L950
.095 x1 | N01017@L950
1.20 x1 | N01018@L950
1.7 x1 | N01020@L950
.04 x1 | N01023@L950
3 x2 | N01026@L950, N01034@L950

F0190

12 x1 | N01041@L953
0 x2 | N01042@L954, N01043@L954
430 x1 | N01044@L954
860 x1 | N01045@L954
400 x1 | N01046@L954
800 x1 | N01047@L954
18 x1 | N01048@L954
450 x1 | N01049@L954

F0191

160 x1 | N01050@L954

F0190

75 x1 | N01051@L954
90 x1 | N01052@L954
2 x1 | N01053@L954
99 x1 | N01054@L954

F0192

32 x1 | N01055@L957

F0193

1 x4 | N01056@L959, N01057@L959, N01058@L959, N01059@L959

F0194

27 x8 | N01060@L960, N01061@L960, N01062@L960, N01063@L960, N01071@L960, N01072@L960, N01073@L960, N01074@L960
0 x2 | N01064@L960, N01069@L960
2 x1 | N01065@L960
1 x3 | N01066@L960, N01068@L960, N01070@L960
.001 x1 | N01067@L960

F0195

0 x1 | N01075@L961
.12 x1 | N01076@L961
1 x1 | N01077@L961
.20 x1 | N01078@L961
.27 x1 | N01079@L961
5.2 x1 | N01080@L961
10 x1 | N01081@L961

F0196

0 x3 | N01082@L962, N01085@L962, N01087@L962
1 x1 | N01083@L962
2.6 x1 | N01084@L962
2.5 x1 | N01086@L962

F0197

0 x4 | N01088@L964, N01099@L964, N01104@L964, N01105@L964
.32 x1 | N01089@L964
.69 x1 | N01090@L964

1 x6 | N01091@L964, N01093@L964, N01095@L964, N01096@L964, N01098@L964, N01100@L964

.72 x1 | N01092@L964

.58 x1 | N01094@L964

.09 x1 | N01097@L964

58 x1 | N01101@L964

.5 x1 | N01102@L964

35 x1 | N01103@L964

F0198

0 x2 | N01106@L965, N01107@L965

1 x1 | N01108@L965

F0199

6 x2 | N01109@L966, N01113@L966

0 x2 | N01110@L966, N01118@L966

7 x2 | N01111@L966, N01116@L966

.68 x1 | N01112@L966

.96 x1 | N01114@L966

5 x1 | N01115@L966

.12 x1 | N01117@L966

.73 x1 | N01119@L966

1 x1 | N01120@L966

.94 x1 | N01121@L966

2 x1 | N01122@L966

3 x1 | N01123@L966

F0200

8 x1 | N01124@L966

0 x3 | N01125@L966, N01127@L966, N01129@L966

14 x1 | N01126@L966

1.25 x1 | N01128@L966

2 x1 | N01130@L966

F0201

1 x12 | N01131@L966, N01138@L966, N01140@L966, N01144@L966, N01145@L966, N01147@L966, N01152@L966, N01158@L966, N01163@L966, N01168@L966, N01169@L966, N01170@L966

.12 x1 | N01132@L966

0 x11 | N01133@L966, N01135@L966, N01136@L966, N01137@L966, N01143@L966, N01154@L966, N01157@L966, N01172@L966, N01178@L966, N01181@L966, N01184@L966

8 x2 | N01134@L966, N01139@L966

4 x1 | N01141@L966

10 x4 | N01142@L966, N01149@L966, N01167@L966, N01182@L966

1.28 x1 | N01146@L966

5 x2 | N01148@L966, N01183@L966

20 x1 | N01150@L966

100 x2 | N01151@L966, N01159@L966

3 x1 | N01153@L966

34 x3 | N01155@L966, N01174@L966, N01176@L966

22 x1 | N01156@L966

18 x1 | N01160@L966

.28 x1 | N01161@L966

7 x1 | N01162@L966

80 x2 | N01164@L966, N01171@L966

42 x1 | N01165@L966

32 x1 | N01166@L966

.18 x1 | N01173@L966

17 x2 | N01175@L966, N01177@L966

2 x1 | N01179@L966

28 x1 | N01180@L966

F0202

100 x1 | N01185@L967
1 x3 | N01186@L967, N01187@L967, N01191@L967
7 x1 | N01188@L967
.5 x1 | N01189@L967
0 x1 | N01190@L967
.05 x1 | N01192@L967

F0203

35 x1 | N01193@L968
18 x1 | N01194@L968
.9 x1 | N01195@L968

F0204

55 x1 | N01196@L969
.85 x1 | N01197@L969

F0205

8 x1 | N01198@L970
0 x2 | N01199@L970, N01208@L970
30 x1 | N01200@L970
130 x1 | N01201@L970
.3 x1 | N01202@L970
.5 x1 | N01203@L970
.8 x1 | N01204@L970
2 x1 | N01205@L970
3 x1 | N01206@L970
220 x2 | N01207@L970, N01209@L970

F0206

55 x1 | N01210@L971
38 x1 | N01211@L971
0 x3 | N01212@L971, N01213@L971, N01215@L971
2.7 x1 | N01214@L971
2 x2 | N01216@L971, N01217@L971

F0207

0 x2 | N01218@L972, N01219@L972

F0208

0 x4 | N01220@L973, N01221@L973, N01231@L973, N01235@L973
1 x7 | N01222@L973, N01223@L973, N01224@L973, N01225@L973, N01232@L973, N01236@L973, N01237@L973
2 x2 | N01226@L973, N01227@L973
53 x3 | N01228@L973, N01229@L973, N01233@L973
95 x2 | N01230@L973, N01234@L973

F0001

.80 x1 | N01238@L976

F0209

.74 x1 | N01239@L977

F0210

1 x1 | N01240@L979
.84 x2 | N01241@L981, N01242@L981
.80 x2 | N01243@L982, N01250@L985
.61 x1 | N01244@L982
.65 x1 | N01245@L983

.93 x1 | N01246@L983

.88 x1 | N01247@L984

.71 x1 | N01248@L984

.81 x1 | N01249@L985

.76 x1 | N01251@L986

.77 x1 | N01252@L986

.78 x2 | N01253@L987, N01254@L987

0 x1 | N01255@L988

F0211

1e-8 x1 | N01256@L992

F0212

1 x2 | N01257@L1008, N01264@L1017

0 x3 | N01258@L1009, N01259@L1013, N01261@L1014

2 x1 | N01260@L1013

18 x1 | N01262@L1014

.5 x2 | N01263@L1015, N01265@L1019

F0214

.009 x1 | N01266@L1026

2 x1 | N01267@L1027

1 x1 | N01268@L1030

9 x1 | N01269@L1030

F0215

180 x1 | N01270@L1032

.65 x2 | N01271@L1032, N01272@L1032

.5 x2 | N01273@L1032, N01274@L1032

F0218

0 x2 | N01275@L1034, N01276@L1034

F0219

1 x4 | N01277@L1035, N01279@L1035, N01280@L1035, N01283@L1035

0 x1 | N01278@L1035

.09 x1 | N01281@L1035

1.17 x1 | N01282@L1035

F0221

0 x7 | N01284@L1037, N01285@L1037, N01286@L1037, N01287@L1037, N01288@L1037, N01289@L1037, N01290@L1037

F0222

0 x75 | N01291@L1038, N01292@L1038, N01293@L1038, N01294@L1038, N01295@L1038, N01296@L1038, N01307@L1038, N01315@L1039, N01316@L1039, N01318@L1039, N01323@L1040, N01324@L1040, N01325@L1042, N01329@L1042, N01330@L1042, N01331@L1042, N01332@L1042, N01333@L1042, N01334@L1042, N01336@L1042, N01343@L1042, N01351@L1042, N01355@L1042, N01358@L1042, N01359@L1042, N01364@L1042, N01365@L1042, N01370@L1042, N01371@L1042, N01372@L1042, N01373@L1044, N01387@L1044, N01397@L1044, N01399@L1046, N01401@L1046, N01402@L1046, N01403@L1046, N01410@L1046, N01412@L1046, N01413@L1046, N01414@L1047, N01417@L1047, N01418@L1047, N01422@L1047, N01423@L1047, N01424@L1047, N01425@L1049, N01428@L1049, N01429@L1049, N01430@L1049, N01431@L1049, N01432@L1049, N01438@L1049, N01439@L1049, N01443@L1049, N01456@L1049, N01457@L1050, N01458@L1050, N01467@L1050, N01473@L1050, N01479@L1054, N01483@L1057, N01486@L1057, N01493@L1058, N01494@L1058, N01497@L1058, N01498@L1058, N01501@L1059, N01522@L1062, N01526@L1063, N01527@L1063, N01531@L1063, N01532@L1063, N01535@L1063, N01538@L1063

.79 x1 | N01297@L1038

1 x35 | N01298@L1038, N01299@L1038, N01301@L1038, N01303@L1038, N01306@L1038, N01312@L1039, N01313@L1039, N01314@L1039, N01322@L1040, N01337@L1042, N01338@L1042, N01350@L1042, N01366@L1042, N01367@L1042, N01368@L1042, N01369@L1042, N01376@L1044, N01384@L1044, N01386@L1044, N01391@L1044, N01393@L1044, N01395@L1044, N01404@L1046, N01408@L1046, N01427@L1049, N01434@L1049, N01436@L1049, N01440@L1049, N01441@L1049, N01448@L1049, N01450@L1049, N01453@L1049, N01477@L1050, N01506@L1060, N01514@L1060

.07 x2 | N01300@L1038, N01442@L1049

1.22 x1 | N01302@L1038

.87 x1 | N01304@L1038

.94 x1 | N01305@L1038

920 x2 | N01308@L1038, N01309@L1038

1.55 x1 | N01310@L1039

2.65 x1 | N01311@L1039

2.6 x2 | N01317@L1039, N01474@L1050

2 x10 | N01319@L1040, N01320@L1040, N01392@L1044, N01400@L1046, N01405@L1046, N01455@L1049, N01472@L1050, N01504@L1060, N01525@L1063, N01530@L1063

10 x4 | N01321@L1040, N01426@L1049, N01433@L1049, N01468@L1050

.35 x3 | N01326@L1042, N01380@L1044, N01381@L1044

520 x2 | N01327@L1042, N01335@L1042

.6 x2 | N01328@L1042, N01478@L1052

280 x1 | N01339@L1042

175 x1 | N01340@L1042

.42 x2 | N01341@L1042, N01342@L1042

440 x1 | N01344@L1042

.32 x2 | N01345@L1042, N01346@L1042

1.65 x1 | N01347@L1042

2.85 x1 | N01348@L1042

.60 x1 | N01349@L1042

1.2 x1 | N01352@L1042

2.8 x1 | N01353@L1042

.43 x1 | N01354@L1042

1.6 x2 | N01356@L1042, N01357@L1042

.7 x1 | N01360@L1042

2.4 x3 | N01361@L1042, N01445@L1049, N01470@L1050

.46 x2 | N01362@L1042, N01363@L1042

.5 x8 | N01374@L1044, N01375@L1044, N01377@L1044, N01398@L1044, N01406@L1046, N01411@L1046, N01419@L1047, N01437@L1049

.65 x3 | N01378@L1044, N01379@L1044, N01495@L1058

.39 x1 | N01382@L1044

1.20 x1 | N01383@L1044

1.18 x1 | N01385@L1044

.56 x1 | N01388@L1044

9 x1 | N01389@L1044

.4 x1 | N01390@L1044

.01 x2 | N01394@L1044, N01396@L1044

.001 x2 | N01407@L1046, N01409@L1046

.25 x2 | N01415@L1047, N01416@L1047

4 x1 | N01420@L1047

13 x1 | N01421@L1047

.24 x1 | N01435@L1049

5 x2 | N01444@L1049, N01454@L1049

125 x1 | N01446@L1049

.2 x1 | N01447@L1049

3.8 x1 | N01449@L1049

37 x2 | N01451@L1049, N01520@L1061

7 x2 | N01452@L1049, N01492@L1057

570 x4 | N01459@L1050, N01461@L1050, N01463@L1050, N01465@L1050

480 x2 | N01460@L1050, N01466@L1050

420 x3 | N01462@L1050, N01464@L1050, N01503@L1059

12 x2 | N01469@L1050, N01537@L1063

1.4 x1 | N01471@L1050

1.95 x1 | N01475@L1050

.84 x1 | N01476@L1050

1.9 x1 | N01480@L1055

.75 x1 | N01481@L1056

3.1 x1 | N01482@L1056

.97 x3 | N01484@L1057, N01517@L1061, N01518@L1061

.50 x1 | N01485@L1057

19 x1 | N01487@L1057

35 x1 | N01488@L1057

6 x2 | N01489@L1057, N01515@L1060
11 x1 | N01490@L1057
48 x1 | N01491@L1057
.9 x1 | N01496@L1058
3.5 x1 | N01499@L1058
45 x1 | N01500@L1058
300 x1 | N01502@L1059
36 x1 | N01505@L1060
.14 x1 | N01507@L1060
.8 x1 | N01508@L1060
170 x2 | N01509@L1060, N01511@L1060
500 x2 | N01510@L1060, N01512@L1060
38 x1 | N01513@L1060

F0227

0 x1 | N01516@L1060

F0228

0 x1 | N01519@L1061

F0229

0 x1 | N01521@L1061

F0222

112 x1 | N01523@L1062
1.3 x1 | N01524@L1062
8 x2 | N01528@L1063, N01533@L1063
60 x2 | N01529@L1063, N01534@L1063
.13 x1 | N01536@L1063

F0231

195 x1 | N01539@L1064
20 x1 | N01540@L1064
8 x1 | N01541@L1064
.70 x1 | N01542@L1064
1.20 x1 | N01543@L1064
1 x1 | N01544@L1064
2.5 x1 | N01545@L1064

F0232

4 x1 | N01546@L1065

F0233

0 x4 | N01547@L1066, N01549@L1066, N01550@L1066, N01553@L1066
.42 x1 | N01548@L1066
.9 x2 | N01551@L1066, N01555@L1066
100 x2 | N01552@L1066, N01554@L1066

F0235

60 x1 | N01556@L1068

F0236

1 x1 | N01557@L1069
0 x1 | N01558@L1069

F0237

65 x1 | N01559@L1070

0 x9 | N01560@L1070, N01561@L1070, N01562@L1070, N01567@L1070, N01570@L1070, N01571@L1070, N01574@L1070, N01577@L1070, N01579@L1070

94 x1 | N01563@L1070

.24 x1 | N01564@L1070

1 x1 | N01565@L1070

.35 x1 | N01566@L1070

2 x1 | N01568@L1070

3 x2 | N01569@L1070, N01576@L1070

39 x1 | N01572@L1070

2.5 x1 | N01573@L1070

24 x1 | N01575@L1070

9 x1 | N01578@L1070

57 x1 | N01580@L1070

F0238

0 x11 | N01581@L1071, N01582@L1071, N01588@L1071, N01593@L1071, N01604@L1072, N01606@L1072, N01614@L1075, N01616@L1076, N01628@L1079, N01629@L1080, N01633@L1080

1 x9 | N01583@L1071, N01624@L1077, N01627@L1078, N01630@L1080, N01631@L1080, N01634@L1080, N01635@L1080, N01636@L1080, N01637@L1080

84 x8 | N01584@L1071, N01585@L1071, N01586@L1071, N01587@L1071, N01589@L1071, N01590@L1071, N01591@L1071, N01592@L1071

36 x1 | N01594@L1071

8 x3 | N01595@L1071, N01596@L1071, N01620@L1076

16 x2 | N01597@L1071, N01598@L1071

3 x3 | N01599@L1071, N01605@L1072, N01611@L1075

35 x3 | N01600@L1071, N01601@L1071, N01615@L1076

70 x2 | N01602@L1071, N01603@L1071

15 x1 | N01607@L1072

10 x1 | N01608@L1073

105 x1 | N01609@L1074

40 x1 | N01610@L1075

7 x1 | N01612@L1075

6 x2 | N01613@L1075, N01619@L1076

.32 x1 | N01617@L1076

470 x1 | N01618@L1076

2 x1 | N01621@L1076

20 x1 | N01622@L1077

4 x2 | N01623@L1077, N01626@L1078

24 x1 | N01625@L1078

.85 x1 | N01632@L1080

F0239

6 x1 | N01638@L1081

1 x3 | N01639@L1081, N01640@L1081, N01650@L1081

0 x4 | N01641@L1081, N01642@L1081, N01646@L1081, N01648@L1081

2.2 x1 | N01643@L1081

10 x1 | N01644@L1081

20 x1 | N01645@L1081

7 x1 | N01647@L1081

.07 x1 | N01649@L1081

.92 x1 | N01651@L1081

2 x1 | N01652@L1081

140 x1 | N01653@L1081

160 x1 | N01654@L1081

102 x1 | N01655@L1081

17 x1 | N01656@L1081

204 x1 | N01657@L1081

30 x1 | N01658@L1081

5 x1 | N01659@L1081

F0240

1.8 x1 | N01660@L1082

0 x19 | N01661@L1082, N01665@L1082, N01669@L1082, N01670@L1082, N01673@L1082, N01674@L1082, N01676@L1082, N01680@L1082, N01682@L1082, N01683@L1082, N01686@L1082, N01687@L1082, N01692@L1082, N01693@L1082, N01694@L1082, N01696@L1082, N01697@L1082, N01698@L1082, N01700@L1082

.7 x2 | N01662@L1082, N01666@L1082

.45 x2 | N01663@L1082, N01664@L1082

.4 x3 | N01667@L1082, N01699@L1082, N01701@L1082

.67 x1 | N01668@L1082

.5 x3 | N01671@L1082, N01690@L1082, N01691@L1082

2.1 x1 | N01672@L1082

.62 x1 | N01675@L1082

.8 x2 | N01677@L1082, N01689@L1082

2 x3 | N01678@L1082, N01679@L1082, N01681@L1082

.15 x1 | N01684@L1082

.85 x1 | N01685@L1082

.2 x1 | N01688@L1082

.3 x1 | N01695@L1082

F0241

2 x7 | N01702@L1083, N01708@L1083, N01721@L1083, N01727@L1083, N01734@L1083, N01743@L1083, N01782@L1083

5 x3 | N01703@L1083, N01716@L1083, N01717@L1083

4 x1 | N01704@L1083

1 x3 | N01705@L1083, N01774@L1083, N01775@L1083

0 x16 | N01706@L1083, N01707@L1083, N01715@L1083, N01722@L1083, N01723@L1083, N01725@L1083, N01726@L1083, N01728@L1083, N01735@L1083, N01736@L1083, N01741@L1083, N01742@L1083, N01747@L1083, N01748@L1083, N01780@L1083, N01781@L1083

.23 x1 | N01709@L1083

.25 x2 | N01710@L1083, N01738@L1083

.63 x1 | N01711@L1083

.26 x1 | N01712@L1083

.15 x1 | N01713@L1083

.47 x5 | N01714@L1083, N01760@L1083, N01761@L1083, N01762@L1083, N01763@L1083

.45 x4 | N01718@L1083, N01719@L1083, N01776@L1083, N01778@L1083

.61 x1 | N01720@L1083

.4 x2 | N01724@L1083, N01745@L1083

6 x2 | N01729@L1083, N01730@L1083

.57 x1 | N01731@L1083

.55 x1 | N01732@L1083

.24 x1 | N01733@L1083

3 x5 | N01737@L1083, N01744@L1083, N01751@L1083, N01764@L1083, N01769@L1083

.22 x1 | N01739@L1083

.52 x1 | N01740@L1083

.5 x2 | N01746@L1083, N01749@L1083

.65 x1 | N01750@L1083

.76 x4 | N01752@L1083, N01753@L1083, N01756@L1083, N01757@L1083

1.52 x4 | N01754@L1083, N01755@L1083, N01758@L1083, N01759@L1083

.85 x2 | N01765@L1083, N01770@L1083

.7 x2 | N01766@L1083, N01771@L1083

1.7 x2 | N01767@L1083, N01772@L1083

1.4 x2 | N01768@L1083, N01773@L1083

.6 x2 | N01777@L1083, N01779@L1083

.2 x2 | N01783@L1083, N01784@L1083

.48 x1 | N01785@L1083

F0242

3 x2 | N01786@L1084, N01792@L1084

2 x3 | N01787@L1084, N01796@L1084, N01803@L1084

0 x11 | N01788@L1084, N01789@L1084, N01791@L1084, N01793@L1084, N01802@L1084, N01809@L1084, N01810@L1084, N01812@L1084, N01813@L1084, N01815@L1084, N01817@L1084

13 x1 | N01790@L1084

6 x2 | N01794@L1084, N01795@L1084

.4 x2 | N01797@L1084, N01799@L1084

.5 x2 | N01798@L1084, N01807@L1084

.64 x1 | N01800@L1084

.17 x1 | N01801@L1084

.45 x1 | N01804@L1084

.14 x1 | N01805@L1084

3.4 x1 | N01806@L1084

1 x1 | N01808@L1084

.58 x1 | N01811@L1084

.35 x1 | N01814@L1084

5 x1 | N01816@L1084

30 x1 | N01818@L1084

F0243

3 x2 | N01819@L1085, N01824@L1085

4 x1 | N01820@L1085

0 x3 | N01821@L1085, N01822@L1085, N01830@L1085

20 x1 | N01823@L1085

14 x2 | N01825@L1085, N01826@L1085

28 x2 | N01827@L1085, N01828@L1085

7 x1 | N01829@L1085

1 x1 | N01831@L1085

F0244

.25 x1 | N01832@L1086

.68 x1 | N01833@L1086

0 x14 | N01834@L1086, N01838@L1086, N01839@L1086, N01841@L1086, N01842@L1086, N01843@L1086, N01845@L1086, N01849@L1086, N01853@L1086, N01854@L1086, N01855@L1086, N01856@L1086, N01864@L1086, N01871@L1086

2 x2 | N01835@L1086, N01848@L1086

5 x3 | N01836@L1086, N01859@L1086, N01862@L1086

.5 x1 | N01837@L1086

18 x1 | N01840@L1086

12 x1 | N01844@L1086

6 x4 | N01846@L1086, N01847@L1086, N01876@L1086, N01881@L1086

.82 x1 | N01850@L1086

.48 x1 | N01851@L1086

.67 x1 | N01852@L1086

4 x2 | N01857@L1086, N01865@L1086

8 x2 | N01858@L1086, N01861@L1086

3.4 x2 | N01860@L1086, N01863@L1086

7 x1 | N01866@L1086

.15 x2 | N01867@L1086, N01868@L1086

2.5 x1 | N01869@L1086

1 x2 | N01870@L1086, N01872@L1086

28 x2 | N01873@L1086, N01878@L1086

22 x2 | N01874@L1086, N01879@L1086

56 x2 | N01875@L1086, N01880@L1086

.38 x1 | N01877@L1086

F0245

.25 x2 | N01882@L1087, N01916@L1087

1 x5 | N01883@L1087, N01931@L1087, N01960@L1087, N01963@L1087, N02038@L1087

3 x9 | N01884@L1087, N01932@L1087, N01957@L1087, N01975@L1087, N01982@L1087, N01990@L1087, N02009@L1087, N02044@L1087, N02045@L1087

.84 x1 | N01885@L1087

2.6 x1 | N01886@L1087

2 x4 | N01887@L1087, N01986@L1087, N01994@L1087, N02022@L1087

.7 x8 | N01888@L1087, N01892@L1087, N01920@L1087, N01923@L1087, N01927@L1087, N01929@L1087, N01964@L1087, N01965@L1087

.6 x3 | N01889@L1087, N01893@L1087, N01898@L1087

.45 x2 | N01890@L1087, N01917@L1087

0 x61 | N01891@L1087, N01895@L1087, N01897@L1087, N01915@L1087, N01919@L1087, N01921@L1087, N01922@L1087, N01924@L1087, N01925@L1087, N01926@L1087, N01928@L1087, N01930@L1087, N01933@L1087, N01936@L1087, N01937@L1087, N01938@L1087, N01939@L1087, N01943@L1087, N01947@L1087, N01950@L1087, N01951@L1087, N01954@L1087, N01956@L1087, N01958@L1087, N01962@L1087, N01966@L1087, N01967@L1087, N01971@L1087, N01972@L1087, N01973@L1087, N01979@L1087, N01980@L1087, N01983@L1087, N01984@L1087, N01987@L1087, N01988@L1087, N01991@L1087, N01992@L1087, N01995@L1087, N01996@L1087, N02000@L1087, N02001@L1087, N02006@L1087, N02007@L1087, N02013@L1087, N02019@L1087, N02020@L1087, N02023@L1087, N02024@L1087, N02026@L1087, N02029@L1087, N02033@L1087, N02034@L1087, N02035@L1087, N02036@L1087, N02040@L1087, N02041@L1087, N02043@L1087, N02047@L1087, N02048@L1087, N02049@L1087

.35 x3 | N01894@L1087, N01899@L1087, N01901@L1087

.16 x1 | N01896@L1087

.15 x2 | N01900@L1087, N01902@L1087

.22 x5 | N01903@L1087, N01905@L1087, N01912@L1087, N01941@L1087, N01945@L1087

.38 x3 | N01904@L1087, N01906@L1087, N02037@L1087

.3 x1 | N01907@L1087

1.5 x1 | N01908@L1087

.37 x2 | N01909@L1087, N01911@L1087

.12 x2 | N01910@L1087, N01913@L1087

.48 x1 | N01914@L1087

1.7 x2 | N01918@L1087, N02039@L1087

6 x2 | N01934@L1087, N01935@L1087

.65 x2 | N01940@L1087, N01944@L1087

.83 x2 | N01942@L1087, N01946@L1087

.9 x3 | N01948@L1087, N01952@L1087, N01997@L1087

.86 x3 | N01949@L1087, N01953@L1087, N01981@L1087

4 x5 | N01955@L1087, N01999@L1087, N02012@L1087, N02016@L1087, N02042@L1087

.5 x3 | N01959@L1087, N01969@L1087, N01970@L1087

.62 x1 | N01961@L1087

.98 x1 | N01968@L1087

.92 x1 | N01974@L1087

.28 x1 | N01976@L1087

.26 x1 | N01977@L1087

.23 x1 | N01978@L1087

.43 x1 | N01985@L1087

.78 x1 | N01989@L1087

.4 x1 | N01993@L1087

.63 x1 | N01998@L1087

.72 x3 | N02002@L1087, N02004@L1087, N02021@L1087

.75 x2 | N02003@L1087, N02005@L1087

.74 x1 | N02008@L1087

5 x5 | N02010@L1087, N02011@L1087, N02025@L1087, N02027@L1087, N02028@L1087

8 x3 | N02014@L1087, N02015@L1087, N02017@L1087

9 x1 | N02018@L1087

.47 x1 | N02030@L1087

.33 x1 | N02031@L1087

.53 x1 | N02032@L1087

1.4 x1 | N02046@L1087

F0246

0 x7 | N02050@L1090, N02051@L1090, N02052@L1090, N02053@L1090, N02057@L1091, N02058@L1091, N02059@L1091

1.6 x1 | N02054@L1090

6 x1 | N02055@L1090

4 x1 | N02056@L1090

2 x1 | N02060@L1091

F0247

2 x1 | N02061@L1093

0 x4 | N02062@L1095, N02063@L1095, N02064@L1095, N02065@L1095

F0248

1 x3 | N02066@L1101, N02069@L1101, N02070@L1101

7 x1 | N02067@L1101

16 x2 | N02068@L1101, N02074@L1101

0 x2 | N02071@L1101, N02072@L1101

255 x3 | N02073@L1101, N02076@L1101, N02077@L1101

8 x1 | N02075@L1101

F0249

16 x1 | N02078@L1101

2 x1 | N02079@L1101

F0250

.33 x1 | N02080@L1102

.4 x1 | N02081@L1102

.06 x1 | N02082@L1102

0 x3 | N02083@L1102, N02084@L1102, N02086@L1102

1.1 x1 | N02085@L1102

.40 x1 | N02087@L1102

.48 x1 | N02088@L1102

1 x1 | N02089@L1102

.38 x1 | N02090@L1102

F0251

512 x1 | N02091@L1106

813 x2 | N02092@L1106, N02093@L1106

0 x3 | N02094@L1107, N02102@L1107, N02104@L1108

95 x1 | N02095@L1107

256 x4 | N02096@L1107, N02097@L1107, N02106@L1108, N02107@L1108

1 x1 | N02098@L1107

12 x1 | N02099@L1107

3 x1 | N02100@L1107

.43 x1 | N02101@L1107

.65 x1 | N02103@L1108

18 x1 | N02105@L1108

8 x1 | N02108@L1108

6 x1 | N02109@L1108

14 x1 | N02110@L1108

2 x1 | N02111@L1108

F0252

17 x1 | N02112@L1113

0 x2 | N02113@L1113, N02116@L1113

3 x1 | N02114@L1113

.025 x1 | N02115@L1113

.73 x1 | N02117@L1113

.055 x1 | N02118@L1113

1.05 x1 | N02119@L1113

3.5 x1 | N02120@L1113

F0253

.50 x3 | N02121@L1117, N02206@L1122, N02207@L1122

4 x2 | N02122@L1118, N02191@L1121

2 x6 | N02123@L1118, N02129@L1118, N02131@L1118, N02184@L1121, N02185@L1121, N02203@L1122

3 x2 | N02124@L1118, N02194@L1121

0 x33 | N02125@L1118, N02156@L1119, N02157@L1119, N02158@L1119, N02198@L1122, N02210@L1123, N02215@L1123, N02219@L1123, N02220@L1123, N02222@L1123, N02224@L1123, N02227@L1126, N02228@L1126, N02231@L1126, N02232@L1126, N02244@L1127, N02248@L1127, N02258@L1128, N02259@L1128, N02284@L1130, N02287@L1130, N02288@L1130, N02289@L1131, N02297@L1131, N02302@L1132, N02303@L1132, N02305@L1132, N02313@L1133, N02315@L1133, N02316@L1133, N02319@L1133, N02341@L1138, N02347@L1139

1 x15 | N02126@L1118, N02127@L1118, N02128@L1118, N02132@L1118, N02133@L1118, N02144@L1119, N02145@L1119, N02165@L1120, N02166@L1120, N02251@L1127, N02252@L1127, N02253@L1127, N02254@L1127, N02326@L1137, N02327@L1137

.37 x1 | N02130@L1118
.07 x1 | N02134@L1118
.58 x3 | N02135@L1118, N02163@L1119, N02213@L1123
.96 x5 | N02136@L1118, N02140@L1118, N02229@L1126, N02230@L1126, N02236@L1127
.17 x3 | N02137@L1118, N02141@L1118, N02334@L1137
1.16 x2 | N02138@L1118, N02197@L1121
.12 x2 | N02139@L1118, N02155@L1119
.7 x6 | N02142@L1118, N02174@L1120, N02176@L1120, N02247@L1127, N02277@L1129, N02278@L1129
.10 x1 | N02143@L1118
.40 x1 | N02146@L1119
.18 x3 | N02147@L1119, N02193@L1121, N02204@L1122
.56 x3 | N02148@L1119, N02276@L1129, N02283@L1130
.13 x3 | N02149@L1119, N02268@L1128, N02286@L1130
.6 x6 | N02150@L1119, N02169@L1120, N02178@L1120, N02238@L1127, N02312@L1133, N02314@L1133
.32 x1 | N02151@L1119
.30 x6 | N02152@L1119, N02256@L1127, N02261@L1128, N02263@L1128, N02281@L1130, N02310@L1133
.44 x2 | N02153@L1119, N02240@L1127
.77 x3 | N02154@L1119, N02192@L1121, N02342@L1138
.51 x2 | N02159@L1119, N02218@L1123
.70 x1 | N02160@L1119
.15 x5 | N02161@L1119, N02186@L1121, N02221@L1123, N02274@L1129, N02311@L1133
.25 x7 | N02162@L1119, N02177@L1120, N02188@L1121, N02233@L1127, N02328@L1137, N02330@L1137, N02333@L1137
.29 x3 | N02164@L1119, N02265@L1128, N02267@L1128
.2 x2 | N02167@L1120, N02299@L1131
.16 x2 | N02168@L1120, N02273@L1129
.1 x1 | N02170@L1120
1.1 x3 | N02171@L1120, N02180@L1120, N02187@L1121
.35 x2 | N02172@L1120, N02322@L1137
1.07 x2 | N02173@L1120, N02175@L1120
.4 x2 | N02179@L1120, N02298@L1131
.88 x2 | N02181@L1120, N02242@L1127
.98 x2 | N02182@L1120, N02195@L1121
.66 x2 | N02183@L1120, N02332@L1137
.45 x1 | N02189@L1121
.36 x2 | N02190@L1121, N02309@L1133
.27 x1 | N02196@L1121
12 x2 | N02199@L1122, N02201@L1122
5 x2 | N02200@L1122, N02202@L1122
.22 x3 | N02205@L1122, N02275@L1129, N02282@L1130
1.12 x2 | N02208@L1122, N02209@L1122
6 x2 | N02211@L1123, N02212@L1123
.65 x3 | N02214@L1123, N02239@L1127, N02294@L1131
.53 x1 | N02216@L1123
.31 x1 | N02217@L1123
.28 x3 | N02223@L1123, N02290@L1131, N02308@L1133
.83 x1 | N02225@L1123
.48 x2 | N02226@L1124, N02234@L1127
.42 x1 | N02235@L1127
.87 x3 | N02237@L1127, N02317@L1133, N02318@L1133
.38 x4 | N02241@L1127, N02260@L1128, N02262@L1128, N02293@L1131
.79 x1 | N02243@L1127
.06 x2 | N02245@L1127, N02249@L1127
.24 x3 | N02246@L1127, N02339@L1138, N02343@L1138
.11 x1 | N02250@L1127
.34 x1 | N02255@L1127
.045 x1 | N02257@L1127
.39 x2 | N02264@L1128, N02266@L1128
.025 x1 | N02269@L1128
.03 x1 | N02270@L1128
.035 x1 | N02271@L1128

.55 x1 | N02272@L1129

.43 x1 | N02279@L1130

.85 x1 | N02280@L1130

.89 x1 | N02285@L1130

.33 x3 | N02291@L1131, N02337@L1138, N02345@L1138

.20 x2 | N02292@L1131, N02295@L1131

.19 x2 | N02296@L1131, N02300@L1131

.47 x1 | N02301@L1132

.3 x3 | N02304@L1132, N02307@L1132, N02335@L1137

.05 x1 | N02306@L1132

.9 x1 | N02320@L1133

1.6 x1 | N02321@L1133

.105 x1 | N02323@L1137

1.5 x1 | N02324@L1137

5.3 x1 | N02325@L1137

1.15 x1 | N02329@L1137

.08 x1 | N02331@L1137

.21 x1 | N02336@L1137

.67 x2 | N02338@L1138, N02346@L1138

.91 x2 | N02340@L1138, N02344@L1138

F0254

2 x5 | N02348@L1142, N02367@L1147, N02373@L1147, N02374@L1147, N02376@L1147

.24 x1 | N02349@L1142

1.05 x1 | N02350@L1142

.74 x1 | N02351@L1142

0 x16 | N02352@L1142, N02353@L1142, N02354@L1142, N02355@L1143, N02356@L1143, N02359@L1144, N02361@L1146, N02372@L1147, N02381@L1149, N02382@L1149, N02390@L1150, N02391@L1150, N02392@L1150, N02394@L1150, N02398@L1150, N02400@L1150

6 x2 | N02357@L1143, N02365@L1146

1.5 x1 | N02358@L1143

290 x1 | N02360@L1145

1 x6 | N02362@L1146, N02370@L1147, N02375@L1147, N02380@L1148, N02385@L1149, N02397@L1150

.12 x1 | N02363@L1146

3 x4 | N02364@L1146, N02368@L1147, N02379@L1148, N02389@L1150

13 x1 | N02366@L1147

.3 x1 | N02369@L1147

12 x1 | N02371@L1147

.81 x1 | N02377@L1148

.77 x1 | N02378@L1148

.85 x1 | N02383@L1149

4 x1 | N02384@L1149

.32 x1 | N02386@L1150

10 x1 | N02387@L1150

14 x1 | N02388@L1150

22 x2 | N02393@L1150, N02395@L1150

1.20 x1 | N02396@L1150

18 x2 | N02399@L1150, N02401@L1150

F0255

65 x1 | N02402@L1153

2 x3 | N02403@L1153, N02404@L1153, N02405@L1153

17 x1 | N02406@L1153

0 x2 | N02407@L1153, N02411@L1153

12 x2 | N02408@L1153, N02412@L1153

11 x2 | N02409@L1153, N02413@L1153

7 x1 | N02410@L1153

F0258

0 x10 | N02414@L1155, N02415@L1155, N02421@L1156, N02422@L1156, N02423@L1156, N02434@L1157, N02442@L1157, N02443@L1157, N02455@L1157, N02458@L1157

4 x4 | N02416@L1155, N02417@L1155, N02440@L1157, N02454@L1157

8 x2 | N02418@L1155, N02419@L1155

5 x5 | N02420@L1156, N02426@L1156, N02427@L1156, N02438@L1157, N02457@L1157

1 x2 | N02424@L1156, N02444@L1157

2 x2 | N02425@L1156, N02433@L1157

1.5 x2 | N02428@L1157, N02429@L1157

3 x8 | N02430@L1157, N02431@L1157, N02435@L1157, N02446@L1157, N02447@L1157, N02450@L1157, N02451@L1157, N02456@L1157

20 x1 | N02432@L1157

.75 x1 | N02436@L1157

.25 x1 | N02437@L1157

6 x5 | N02439@L1157, N02448@L1157, N02449@L1157, N02452@L1157, N02453@L1157

2.7 x1 | N02441@L1157

.35 x1 | N02445@L1157

F0259

1150 x1 | N02459@L1158

100 x2 | N02460@L1158, N02462@L1158

0 x2 | N02461@L1158, N02463@L1158

F0260

.05 x1 | N02464@L1159

0 x1 | N02465@L1159

F0263

1 x6 | N02466@L1160, N02467@L1160, N02481@L1160, N02485@L1160, N02488@L1161, N02491@L1161

100 x12 | N02468@L1160, N02470@L1160, N02471@L1160, N02473@L1160, N02477@L1160, N02482@L1160, N02483@L1160, N02489@L1161, N02494@L1161, N02496@L1161, N02498@L1161, N02500@L1161

0 x14 | N02469@L1160, N02472@L1160, N02479@L1160, N02480@L1160, N02484@L1160, N02486@L1161, N02487@L1161, N02490@L1161, N02492@L1161, N02493@L1161, N02495@L1161, N02497@L1161, N02499@L1161, N02502@L1162

2 x1 | N02474@L1160

9 x1 | N02475@L1160

.4 x1 | N02476@L1160

10 x1 | N02478@L1160

620 x1 | N02501@L1162

F0265

0 x1 | N02503@L1163

1 x2 | N02504@L1163, N02505@L1163

F0267

2600 x1 | N02506@L1164

F0270

0 x5 | N02507@L1166, N02508@L1166, N02510@L1166, N02515@L1166, N02517@L1166

777 x1 | N02509@L1166

140 x1 | N02511@L1166

14 x1 | N02512@L1166

2 x1 | N02513@L1166

.25 x1 | N02514@L1166

85 x1 | N02516@L1166

1 x1 | N02518@L1166

F0271

1 x2 | N02519@L1167, N02520@L1167

2 x5 | N02521@L1167, N02522@L1167, N02525@L1167, N02530@L1167, N02533@L1167

850 x1 | N02523@L1167

1.5 x1 | N02524@L1167

0 x8 | N02526@L1167, N02527@L1167, N02528@L1167, N02529@L1167, N02531@L1167, N02532@L1167, N02534@L1167, N02535@L1167

F0272

1000 x1 | N02537@L1168
.08 x1 | N02538@L1168
0 x3 | N02539@L1168, N02548@L1168, N02549@L1168
1 x3 | N02540@L1168, N02543@L1168, N02545@L1168
60 x3 | N02541@L1168, N02544@L1168, N02546@L1168
4 x2 | N02542@L1168, N02547@L1168

F0275

0 x2 | N02550@L1171, N02551@L1171

F0276

0 x4 | N02554@L1172, N02555@L1172, N02556@L1172, N02558@L1172
2.5 x1 | N02557@L1172

F0285

12 x1 | N02560@L1174
0 x1 | N02561@L1174

F0286

500 x1 | N02562@L1174

F0287

0 x1 | N02563@L1189
5 x1 | N02564@L1194
1 x1 | N02565@L1194

F0298

1 x4 | N02566@L1221, N02574@L1227, N02580@L1231, N02582@L1234
0 x8 | N02567@L1223, N02569@L1223, N02570@L1224, N02571@L1224, N02572@L1224, N02573@L1224, N02575@L1228, N02576@L1229
5 x3 | N02568@L1223, N02577@L1229, N02579@L1231
7 x1 | N02578@L1230
4 x1 | N02581@L1231

F0300

0 x2 | N02583@L1244, N02584@L1245

F0307

0 x1 | N02585@L1253

F0308

0 x1 | N02586@L1260
2 x1 | N02587@L1260

F0317

0 x3 | N02588@L1277, N02591@L1281, N02593@L1281

F0318

0 x2 | N02589@L1279, N02590@L1279

F0317

1 x1 | N02592@L1281

F0322

100 x2 | N02594@L1283, N02595@L1283
2 x1 | N02596@L1283

F0323

5 x1 | N02597@L1285

F0324

100 x1 | N02598@L1285
2 x1 | N02599@L1285

F0323

3 x1 | N02600@L1286
2 x1 | N02601@L1286

F0326

0 x2 | N02602@L1289, N02603@L1289

F0325

0 x1 | N02604@L1291
2 x2 | N02605@L1291, N02608@L1292
100 x1 | N02606@L1291
1 x1 | N02607@L1291

F0327

0 x5 | N02609@L1295, N02611@L1297, N02612@L1297, N02613@L1298, N02623@L1303
6 x1 | N02610@L1295
2 x4 | N02614@L1298, N02615@L1298, N02617@L1299, N02620@L1303
100 x4 | N02616@L1299, N02618@L1299, N02619@L1303, N02622@L1303
1 x1 | N02621@L1303

F0333

.1 x1 | N02624@L1312
0 x1 | N02625@L1312

F0337

0 x2 | N02626@L1322, N02630@L1324
1 x2 | N02627@L1323, N02632@L1324
3 x2 | N02628@L1324, N02629@L1324
100 x3 | N02631@L1324, N02633@L1324, N02635@L1324
2 x1 | N02634@L1324

F0342

2.5 x1 | N02636@L1329

F0344

170 x1 | N02637@L1334
30 x1 | N02638@L1334
35 x1 | N02639@L1334
3 x1 | N02640@L1334
1 x6 | N02641@L1334, N02643@L1334, N02646@L1334, N02648@L1334, N02650@L1334, N02654@L1336
.18 x1 | N02642@L1334
17 x1 | N02644@L1334
1.7 x1 | N02645@L1334
.25 x1 | N02647@L1334
4.5 x1 | N02649@L1334
.13 x1 | N02651@L1334
.06 x1 | N02652@L1334
0 x1 | N02653@L1336

F0346

0 x1 | N02655@L1345

1 x1 | N02656@L1345

F0348

100 x4 | N02657@L1349, N02684@L1378, N02728@L1420, N02729@L1421

5 x3 | N02658@L1349, N02689@L1378, N02692@L1383

0 x26 | N02659@L1352, N02660@L1352, N02662@L1352, N02663@L1353, N02664@L1355, N02666@L1356, N02668@L1356, N02669@L1356, N02672@L1373, N02673@L1373, N02677@L1374, N02681@L1378, N02688@L1378, N02691@L1379, N02701@L1409, N02705@L1409, N02709@L1409, N02718@L1409, N02720@L1411, N02721@L1411, N02722@L1414, N02724@L1416, N02725@L1417, N02730@L1423, N02739@L1424, N02750@L1425

1 x19 | N02661@L1352, N02665@L1355, N02667@L1356, N02674@L1374, N02675@L1374, N02676@L1374, N02678@L1375, N02680@L1378, N02682@L1378, N02683@L1378, N02693@L1383, N02703@L1409, N02707@L1409, N02711@L1409, N02723@L1416, N02733@L1423, N02735@L1423, N02738@L1423, N02741@L1424

F0361

1 x2 | N02670@L1366, N02671@L1366

F0348

7 x1 | N02679@L1375

.01 x1 | N02685@L1378

3 x1 | N02686@L1378

2 x3 | N02687@L1378, N02716@L1409, N02740@L1424

10 x1 | N02690@L1378

F0366

1 x2 | N02694@L1392, N02696@L1394

7 x1 | N02695@L1394

F0367

0 x1 | N02697@L1399

10 x1 | N02698@L1399

5 x1 | N02699@L1403

7 x1 | N02700@L1405

F0348

700 x3 | N02702@L1409, N02706@L1409, N02710@L1409

360 x3 | N02704@L1409, N02708@L1409, N02712@L1409

26 x1 | N02713@L1409

21 x1 | N02714@L1409

4 x2 | N02715@L1409, N02717@L1409

6 x1 | N02719@L1409

9 x1 | N02726@L1420

.4 x1 | N02727@L1420

42 x1 | N02731@L1423

1.9 x1 | N02732@L1423

3.1 x1 | N02734@L1423

19 x1 | N02736@L1423

35 x2 | N02737@L1423, N02743@L1424

60 x1 | N02742@L1424

F0386

2 x2 | N02744@L1424, N02746@L1424

1 x1 | N02745@L1424

60 x1 | N02747@L1424

35 x1 | N02748@L1424

30 x1 | N02749@L1424

F0398

0 x1 | N02751@L1431

F0400

0 x1 | N02752@L1435

F0401

0 x2 | N02753@L1435, N02754@L1435

F0402

0 x1 | N02755@L1435

F0407

16 x1 | N02756@L1438
2 x1 | N02757@L1438

F0409

16 x1 | N02758@L1439

F0408

120000 x1 | N02759@L1439
256 x1 | N02760@L1439

F0410

8 x2 | N02761@L1450, N02765@L1452
6 x1 | N02762@L1450
4 x1 | N02763@L1450
254 x1 | N02764@L1452

F0411

4 x1 | N02766@L1456
8 x1 | N02767@L1456
16 x1 | N02768@L1456

F0412

3 x1 | N02769@L1459
5 x1 | N02770@L1460

F0414

0 x1 | N02771@L1466

F0418

0 x1 | N02772@L1475

F0419

2 x2 | N02773@L1479, N02774@L1479
.31 x1 | N02775@L1479
1 x1 | N02776@L1479
0 x1 | N02777@L1479

F0422

0 x1 | N02778@L1482

F0423

0 x1 | N02779@L1486
2 x1 | N02780@L1486

F0424

0 x1 | N02781@L1487

2 x1 | N02782@L1487

F0425

0 x1 | N02783@L1489

F0431

0 x2 | N02784@L1496, N02786@L1508

1 x2 | N02785@L1496, N02788@L1510

9 x1 | N02787@L1510

F0455

3 x1 | N02789@L1515

F0457

0 x1 | N02790@L1515

F0463

0 x1 | N02791@L1515

F0470

1 x1 | N02794@L1528

F0471

0 x1 | N02795@L1529

8 x1 | N02796@L1529

F0473

0 x1 | N02797@L1531

1000 x1 | N02798@L1531

F0474

0 x1 | N02799@L1532

8 x1 | N02800@L1532

F0475

18000 x1 | N02801@L1533

F0477

0 x11 | N02802@L1535, N02803@L1535, N02804@L1535, N02805@L1535, N02806@L1536, N02807@L1536, N02808@L1536, N02809@L1536, N02810@L1537, N02811@L1537, N02813@L1537

16 x1 | N02812@L1537

F0480

10000 x1 | N02814@L1540

.5 x1 | N02815@L1540

F0481

30 x1 | N02816@L1541

.65 x1 | N02817@L1541

21 x1 | N02818@L1541

2 x1 | N02819@L1541

5 x1 | N02820@L1541

1 x1 | N02821@L1541

31 x1 | N02822@L1541

F0482

18 x1 | N02823@L1543

F0486

0 x2 | N02824@L1544, N02828@L1544

F0489

10 x2 | N02825@L1544, N02826@L1544

F0486

.5 x1 | N02827@L1544
1 x1 | N02829@L1544
.05 x1 | N02830@L1544

F0491

300 x1 | N02831@L1544

F0001

1000 x1 | N02832@L1544

F0494

0 x1 | N02834@L1544
300 x1 | N02835@L1544

account-api:root

16384 x1 | N02836@L6
8 x1 | N02837@L6
5 x1 | N02838@L6
33554432 x1 | N02839@L6
1048576 x1 | N02840@L7

F0498

1000 x1 | N02841@L13

F0499

0 x1 | N02842@L15

F0502

200 x1 | N02843@L19

F0503

400 x3 | N02844@L23, N02852@L28, N02853@L32
0 x1 | N02845@L26
1 x2 | N02846@L26, N02851@L28
254 x1 | N02847@L28
2 x2 | N02848@L28, N02850@L28
64 x1 | N02849@L28

F0505

8 x1 | N02854@L36
128 x1 | N02855@L36
400 x1 | N02856@L36

F0507

32 x1 | N02857@L39

F0511

16 x1 | N02858@L48
1 x1 | N02859@L49

F0512

128 x1 | N02860@L52
1 x2 | N02861@L53, N02863@L55
64 x1 | N02862@L53
600000 x1 | N02864@L56

F0513

32 x1 | N02865@L58

F0512

503 x2 | N02866@L59, N02867@L60

F0515

415 x1 | N02868@L65
413 x2 | N02869@L66, N02872@L69
400 x2 | N02870@L67, N02874@L71
0 x2 | N02871@L68, N02873@L70

F0516

429 x1 | N02875@L78
1 x1 | N02876@L78

F0517

1 x1 | N02877@L82

F0520

32 x1 | N02878@L93
30 x1 | N02879@L93
86400 x1 | N02880@L93
12 x1 | N02881@L93
3600 x1 | N02882@L93
1 x1 | N02883@L96
409 x1 | N02884@L96

F0521

5 x1 | N02885@L100
400 x7 | N02886@L100, N02891@L102, N02895@L104, N02897@L105, N02899@L107, N02903@L108, N02906@L109
1 x2 | N02887@L102, N02900@L108
999 x1 | N02888@L102

F0522

0 x1 | N02889@L102
1e12 x1 | N02890@L102

F0523

0 x1 | N02892@L103
10 x1 | N02893@L103

F0521

10 x3 | N02894@L104, N02896@L105, N02898@L107
0 x2 | N02901@L108, N02904@L109
99999 x1 | N02902@L108
9 x1 | N02905@L109

413 x1 | N02907@L111

F0524

1 x1 | N02908@L115
400 x1 | N02909@L115
409 x1 | N02910@L120

F0525

8192 x1 | N02911@L125
20 x1 | N02912@L125
400 x1 | N02913@L125
10000 x2 | N02914@L127, N02919@L134
401 x2 | N02915@L128, N02916@L130
409 x3 | N02917@L132, N02918@L133, N02925@L142
502 x1 | N02920@L135
1 x1 | N02921@L136
404 x1 | N02922@L136
0 x2 | N02923@L137, N02924@L140

F0526

503 x2 | N02926@L148, N02963@L212
5 x3 | N02927@L149, N02939@L175, N02952@L191
405 x2 | N02928@L150, N02936@L168
403 x1 | N02929@L151
409 x4 | N02930@L156, N02948@L185, N02955@L194, N02959@L204
401 x7 | N02931@L158, N02933@L165, N02938@L174, N02944@L180, N02947@L182, N02957@L201, N02961@L207
404 x2 | N02932@L162, N02941@L177
120 x1 | N02934@L166
60 x1 | N02935@L166
200 x4 | N02937@L171, N02949@L186, N02960@L205, N02962@L208
3600 x2 | N02940@L175, N02953@L191
30 x1 | N02942@L178
300 x3 | N02943@L178, N02946@L181, N02951@L189
10 x1 | N02945@L181
15 x1 | N02950@L189
32 x2 | N02954@L192, N02958@L202
201 x1 | N02956@L195

F0527

.8 x1 | N02964@L3
.15 x2 | N02965@L3, N02967@L3
.05 x1 | N02966@L3
.35 x1 | N02968@L3
.5 x1 | N02969@L3
.4 x1 | N02970@L3
.3 x2 | N02971@L3, N02972@L3

F0528

1 x2 | N02973@L5, N02978@L8
10 x1 | N02974@L5
0 x8 | N02975@L7, N02977@L8, N02979@L9, N02980@L9, N02981@L9, N02989@L12, N02992@L16, N02993@L16
99999 x1 | N02976@L7

F0529

0 x1 | N02982@L9

F0532

0 x1 | N02983@L10

F0531

0 x1 | N02984@L10

F0533

0 x1 | N02985@L11

F0534

1 x2 | N02986@L11, N02987@L11
0 x1 | N02988@L11

F0535

99999 x1 | N02990@L15

F0536

0 x1 | N02991@L16

F0541

0 x1 | N02994@L19
240 x1 | N02995@L19

F0540

1 x2 | N02996@L20, N02998@L21
20 x1 | N02997@L20

F0546

5000 x1 | N02999@L16

F0548

16384 x1 | N03000@L17

F0553

60000 x1 | N03001@L40

F0562

0.15 x1 | N03002@L5
0.6 x1 | N03003@L5
1.6 x1 | N03004@L5
0.12 x1 | N03005@L5
0.35 x1 | N03006@L5
1.2 x1 | N03007@L5
0.5 x1 | N03008@L5
28 x1 | N03009@L5
18 x2 | N03010@L5, N03012@L5
21 x1 | N03011@L5
15 x1 | N03013@L5
0.4 x1 | N03014@L5
0.3 x1 | N03015@L5
0.2 x1 | N03016@L5
0.1 x1 | N03017@L5
1.1 x1 | N03018@L5
0.95 x1 | N03019@L5
0.52 x1 | N03020@L5
3 x1 | N03021@L5
1.33 x1 | N03022@L5
340 x1 | N03023@L5
270 x1 | N03024@L5
1.7 x1 | N03025@L5

120 x1 | N03026@L5
2.3 x1 | N03027@L5
0.18 x1 | N03028@L5
210 x1 | N03029@L5

F0566

0 x1 | N03030@L7
1 x1 | N03031@L7

F0567

99999 x1 | N03032@L8
0 x1 | N03033@L8

F0569

0 x1 | N03034@L9

F0568

0 x2 | N03035@L9, N03036@L9

F0573

1 x1 | N03037@L10
0 x2 | N03038@L10, N03039@L10

F0574

0 x1 | N03040@L10

F0575

4294967295 x1 | N03041@L11
1 x1 | N03042@L11
0x9e3779b9 x1 | N03043@L11
0 x2 | N03044@L11, N03045@L11

F0576

0x6D2B79F5 x1 | N03046@L11
0 x2 | N03047@L11, N03053@L11
15 x1 | N03048@L11
1 x1 | N03049@L11
7 x1 | N03050@L11
61 x1 | N03051@L11
14 x1 | N03052@L11
4294967296 x1 | N03054@L11

F0578

7 x4 | N03055@L12, N03057@L12, N03062@L12, N03074@L12

F0579

0 x1 | N03056@L12

F0578

6 x1 | N03058@L12
.68 x1 | N03059@L12
5 x1 | N03060@L12
.28 x2 | N03061@L12, N03076@L12
.04 x1 | N03063@L12
0 x3 | N03064@L12, N03072@L12, N03077@L12
.12 x1 | N03065@L12
1 x2 | N03066@L12, N03075@L12
.61 x1 | N03067@L12

2 x1 | N03068@L12
.21 x1 | N03069@L12
3 x1 | N03070@L12
.06 x1 | N03071@L12
.72 x1 | N03073@L12

F0582

2 x2 | N03078@L13, N03086@L13
1 x3 | N03079@L13, N03080@L13, N03081@L13
0 x4 | N03082@L13, N03085@L13, N03087@L13, N03089@L13
.82 x1 | N03083@L13
.18 x1 | N03084@L13

F0584

2 x1 | N03088@L13

F0585

1 x3 | N03090@L14, N03092@L14, N03093@L14
5 x1 | N03091@L14
2 x1 | N03094@L14
3 x1 | N03095@L14
4 x2 | N03096@L14, N03097@L14

F0586

5 x5 | N03098@L15, N03099@L15, N03103@L15, N03104@L15, N03120@L15
1 x8 | N03100@L15, N03105@L15, N03106@L15, N03108@L15, N03110@L15, N03122@L15, N03124@L15, N03126@L15
0 x9 | N03101@L15, N03111@L15, N03112@L15, N03113@L15, N03114@L15, N03115@L15, N03116@L15, N03118@L15, N03127@L15

F0587

0 x1 | N03102@L15

F0586

4 x2 | N03107@L15, N03109@L15
99999 x1 | N03117@L15

F0588

.95 x1 | N03119@L15

F0586

2.5 x1 | N03121@L15
1.25 x1 | N03123@L15
6.25 x1 | N03125@L15

F0589

999 x1 | N03128@L16
1 x2 | N03129@L16, N03130@L16

F0590

0 x1 | N03131@L17
1 x3 | N03132@L17, N03133@L17, N03134@L17

F0591

999 x1 | N03135@L18
1 x2 | N03136@L18, N03137@L18

F0592

0 x4 | N03138@L19, N03139@L19, N03140@L19, N03144@L19
1 x1 | N03141@L19

1e-12 x1 | N03142@L19

F0594

0 x1 | N03143@L19

F0596

0 x1 | N03145@L20
1 x1 | N03146@L20

F0595

0 x1 | N03147@L20

F0598

0 x1 | N03148@L21

F0600

0 x7 | N03149@L22, N03150@L22, N03151@L22, N03155@L22, N03157@L22, N03160@L22, N03162@L22
1 x3 | N03152@L22, N03153@L22, N03154@L22

F0602

2 x1 | N03156@L22

F0603

0 x1 | N03158@L22

F0605

0 x1 | N03159@L22

F0607

0 x1 | N03161@L22

F0608

2 x1 | N03163@L23
9 x1 | N03164@L23

F0610

0 x3 | N03165@L23, N03169@L23, N03172@L23
1 x5 | N03166@L23, N03167@L23, N03168@L23, N03173@L23, N03174@L23
2 x1 | N03170@L23
.18 x1 | N03171@L23

F0608

0 x1 | N03175@L23

F0611

0 x4 | N03176@L24, N03177@L24, N03178@L24, N03179@L24
1 x2 | N03180@L24, N03181@L24
60 x1 | N03182@L24

F0612

0 x7 | N03183@L25, N03185@L25, N03189@L25, N03192@L25, N03193@L25, N03195@L25, N03196@L25
1 x4 | N03184@L25, N03186@L25, N03190@L25, N03191@L25
7 x1 | N03187@L25
999 x1 | N03188@L25
3 x1 | N03194@L25

F0613

370 x1 | N03197@L26

.65 x1 | N03198@L26

0 x3 | N03199@L26, N03200@L26, N03201@L26

1 x1 | N03202@L26

F0614

2 x1 | N03203@L4

F0615

0 x2 | N03204@L5, N03205@L5

F0616

.2 x1 | N03206@L6

0 x2 | N03207@L6, N03208@L6

1 x1 | N03209@L6

2 x2 | N03210@L6, N03211@L6

F0617

0 x25 | N03212@L7, N03213@L7, N03216@L7, N03221@L8, N03231@L8, N03234@L9, N03243@L9, N03245@L9, N03247@L10, N03248@L10, N03255@L11, N03260@L11, N03262@L11, N03263@L11, N03264@L11, N03268@L12, N03269@L12, N03271@L12, N03281@L13, N03282@L13, N03285@L14, N03289@L14, N03292@L14, N03294@L14, N03296@L14

512 x6 | N03214@L7, N03215@L7, N03223@L8, N03224@L8, N03236@L9, N03237@L9

F0618

1664525 x1 | N03217@L7

1013904223 x1 | N03218@L7

0 x1 | N03219@L7

4294967296 x1 | N03220@L7

F0617

180 x1 | N03222@L8

3 x6 | N03225@L8, N03238@L9, N03242@L9, N03257@L11, N03259@L11, N03265@L11

26 x1 | N03226@L8

.035 x1 | N03227@L8

.04 x1 | N03228@L8

.4 x3 | N03229@L8, N03286@L14, N03293@L14

2 x4 | N03230@L8, N03266@L11, N03267@L11, N03280@L13

1 x5 | N03232@L8, N03233@L8, N03244@L9, N03276@L12, N03279@L13

105 x1 | N03235@L9

13 x2 | N03239@L9, N03250@L10

.15 x1 | N03240@L9

.75 x1 | N03241@L9

.10 x1 | N03246@L10

28 x1 | N03249@L10

4 x5 | N03251@L10, N03252@L10, N03254@L10, N03273@L12, N03278@L13

22 x1 | N03253@L10

.3 x4 | N03256@L11, N03258@L11, N03270@L12, N03283@L13

7 x1 | N03261@L11

.2 x3 | N03272@L12, N03284@L13, N03295@L14

1.1 x1 | N03274@L12

.7 x1 | N03275@L12

.18 x1 | N03277@L13

.8 x3 | N03287@L14, N03290@L14, N03297@L14

.5 x2 | N03288@L14, N03291@L14

F0619

.20 x2 | N03298@L16, N03312@L17

0 x15 | N03299@L16, N03300@L16, N03303@L16, N03320@L18, N03321@L18, N03324@L18, N03327@L18, N03329@L18, N03330@L18, N03335@L18, N03338@L18, N03340@L18, N03358@L19, N03366@L20, N03373@L20

.70 x1 | N03301@L16

1 x8 | N03302@L16, N03305@L17, N03315@L17, N03349@L19, N03351@L19, N03352@L19, N03365@L19, N03384@L20

.35 x1 | N03304@L16

3 x4 | N03306@L17, N03328@L18, N03370@L20, N03380@L20

2 x6 | N03307@L17, N03308@L17, N03314@L17, N03325@L18, N03348@L18, N03381@L20

.29 x1 | N03309@L17

.1 x2 | N03310@L17, N03346@L18

.48 x1 | N03311@L17

.96 x2 | N03313@L17, N03357@L19

.8 x2 | N03316@L17, N03322@L18

.6 x3 | N03317@L17, N03343@L18, N03355@L19

1.6 x1 | N03318@L17

1.2 x1 | N03319@L17

.5 x1 | N03323@L18

1.4 x1 | N03326@L18

.45 x2 | N03331@L18, N03362@L19

.16 x2 | N03332@L18, N03353@L19

.22 x2 | N03333@L18, N03354@L19

.13 x1 | N03334@L18

.18 x1 | N03336@L18

.04 x1 | N03337@L18

1.5 x1 | N03339@L18

4 x4 | N03341@L18, N03350@L19, N03359@L19, N03378@L20

2.1 x2 | N03342@L18, N03344@L18

.25 x1 | N03345@L18

.045 x1 | N03347@L18

1.1 x1 | N03356@L19

1.7 x1 | N03360@L19

.62 x1 | N03361@L19

.12 x1 | N03363@L19

.08 x1 | N03364@L19

11 x1 | N03367@L20

2.399 x1 | N03368@L20

.3 x1 | N03369@L20

.2 x1 | N03371@L20

.65 x1 | N03372@L20

5 x4 | N03374@L20, N03375@L20, N03376@L20, N03377@L20

2.6 x1 | N03379@L20

2.4 x2 | N03382@L20, N03383@L20

F0620

0 x36 | N03385@L22, N03387@L22, N03391@L23, N03411@L23, N03420@L24, N03447@L25, N03450@L25, N03469@L27, N03473@L27, N03477@L27, N03481@L28, N03493@L28, N03499@L29, N03510@L29, N03516@L29, N03518@L29, N03522@L29, N03536@L30, N03540@L32, N03541@L32, N03544@L32, N03545@L32, N03547@L33, N03564@L34, N03565@L34, N03568@L34, N03569@L34, N03571@L34, N03579@L35, N03581@L35, N03590@L36, N03611@L39, N03612@L39, N03615@L39, N03616@L39, N03617@L39

4 x2 | N03386@L22, N03624@L39

.08 x3 | N03388@L22, N03454@L25, N03554@L33

1.5 x3 | N03389@L23, N03396@L23, N03423@L24

.09 x1 | N03390@L23

.5 x10 | N03392@L23, N03402@L23, N03406@L23, N03435@L24, N03443@L24, N03488@L28, N03508@L29, N03530@L30, N03531@L30, N03560@L33

.8 x2 | N03393@L23, N03532@L30

1.1 x1 | N03394@L23

.6 x2 | N03395@L23, N03409@L23

.2 x9 | N03397@L23, N03413@L23, N03417@L23, N03436@L24, N03467@L27, N03494@L28, N03593@L36, N03605@L37, N03607@L37

1.35 x1 | N03398@L23

1 x21 | N03399@L23, N03400@L23, N03433@L24, N03434@L24, N03457@L27, N03458@L27, N03497@L29, N03498@L29, N03514@L29, N03528@L30, N03529@L30, N03561@L34, N03573@L34, N03574@L34, N03586@L36, N03588@L36, N03589@L36, N03601@L37, N03603@L37, N03604@L37, N03609@L39

.46 x2 | N03401@L23, N03405@L23
.22 x5 | N03403@L23, N03431@L24, N03479@L27, N03480@L27, N03496@L28
.20 x2 | N03404@L23, N03583@L35
.13 x6 | N03407@L23, N03430@L24, N03470@L27, N03474@L27, N03478@L27, N03511@L29
.12 x3 | N03408@L23, N03502@L29, N03506@L29
.022 x1 | N03410@L23
2 x7 | N03412@L23, N03418@L24, N03444@L25, N03485@L28, N03507@L29, N03526@L30, N03623@L39
.25 x6 | N03414@L23, N03426@L24, N03428@L24, N03460@L27, N03503@L29, N03505@L29
.78 x1 | N03415@L23
.4 x7 | N03416@L23, N03432@L24, N03509@L29, N03519@L29, N03551@L33, N03572@L34, N03582@L35
.15 x3 | N03419@L24, N03523@L29, N03592@L36
.48 x2 | N03421@L24, N03471@L27
.75 x2 | N03422@L24, N03438@L24
.7 x4 | N03424@L24, N03487@L28, N03620@L39, N03622@L39
1.6 x1 | N03425@L24
.76 x2 | N03427@L24, N03429@L24
.95 x2 | N03437@L24, N03534@L30
.97 x1 | N03439@L24
.80 x1 | N03440@L24
.23 x4 | N03441@L24, N03520@L29, N03524@L29, N03597@L36
.16 x2 | N03442@L24, N03558@L33
1.2 x2 | N03445@L25, N03491@L28
.035 x2 | N03446@L25, N03602@L37
6 x2 | N03448@L25, N03449@L25
.72 x1 | N03451@L25
.11 x1 | N03452@L25
.99 x1 | N03453@L25
1.18 x1 | N03455@L25
.18 x4 | N03456@L27, N03537@L30, N03557@L33, N03600@L36
.65 x2 | N03459@L27, N03501@L29
1.07 x1 | N03461@L27
.05 x3 | N03462@L27, N03578@L34, N03585@L35
1.02 x2 | N03463@L27, N03465@L27
.55 x2 | N03464@L27, N03466@L27
.19 x1 | N03468@L27
.47 x1 | N03472@L27
.34 x3 | N03475@L27, N03476@L27, N03606@L37
7 x1 | N03482@L28
3 x6 | N03483@L28, N03500@L29, N03591@L36, N03613@L39, N03614@L39, N03618@L39
.32 x3 | N03484@L28, N03546@L32, N03559@L33
.06 x3 | N03486@L28, N03527@L30, N03552@L33
1.3 x1 | N03489@L28
.98 x1 | N03490@L28
1.4 x1 | N03492@L28
.44 x2 | N03495@L28, N03598@L36
.35 x2 | N03504@L29, N03512@L29
.28 x5 | N03513@L29, N03538@L30, N03556@L33, N03563@L34, N03567@L34
.025 x2 | N03515@L29, N03587@L36
.1 x1 | N03517@L29
.03 x5 | N03521@L29, N03525@L29, N03562@L34, N03596@L36, N03610@L39
.85 x1 | N03533@L30
.52 x1 | N03535@L30
.3 x2 | N03539@L30, N03608@L37
.90 x2 | N03542@L32, N03543@L32
5 x2 | N03548@L33, N03580@L35
2.4 x1 | N03549@L33
.45 x1 | N03550@L33
.011 x1 | N03553@L33
.01 x1 | N03555@L33
.14 x1 | N03566@L34

.04 x1 | N03570@L34

.39 x1 | N03575@L34

.38 x1 | N03576@L34

.045 x1 | N03577@L34

.82 x1 | N03584@L35

.26 x2 | N03594@L36, N03599@L36

.58 x1 | N03595@L36

2.2 x2 | N03619@L39, N03621@L39

F0621

0 x13 | N03625@L40, N03626@L40, N03627@L40, N03630@L40, N03633@L40, N03634@L40, N03641@L40, N03644@L40, N03645@L40, N03647@L40, N03650@L40, N03654@L40, N03657@L40

2 x1 | N03628@L40

.55 x1 | N03629@L40

1 x4 | N03631@L40, N03632@L40, N03642@L40, N03643@L40

.48 x1 | N03635@L40

.44 x1 | N03636@L40

.22 x1 | N03637@L40

.5 x2 | N03638@L40, N03639@L40

.64 x1 | N03640@L40

.045 x1 | N03646@L40

.12 x1 | N03648@L40

15 x2 | N03649@L40, N03656@L40

3 x1 | N03651@L40

.39 x4 | N03652@L40, N03655@L40, N03658@L40, N03660@L40

90 x1 | N03653@L40

160 x1 | N03659@L40

60 x1 | N03661@L40

F0622

0 x7 | N03662@L41, N03663@L41, N03665@L41, N03666@L41, N03669@L41, N03677@L41, N03694@L41

.96 x1 | N03664@L41

2 x5 | N03667@L41, N03672@L41, N03673@L41, N03675@L41, N03701@L41

.4 x3 | N03668@L41, N03683@L41, N03685@L41

1 x4 | N03670@L41, N03671@L41, N03688@L41, N03689@L41

.38 x1 | N03674@L41

.9 x1 | N03676@L41

5 x1 | N03678@L41

2.4 x1 | N03679@L41

.15 x2 | N03680@L41, N03681@L41

.2 x2 | N03682@L41, N03684@L41

.74 x2 | N03686@L41, N03687@L41

.6 x4 | N03690@L41, N03692@L41, N03697@L41, N03698@L41

.36 x2 | N03691@L41, N03693@L41

4 x1 | N03695@L41

1.8 x1 | N03696@L41

.06 x1 | N03699@L41

.04 x1 | N03700@L41

F0624

0 x8 | N03702@L43, N03703@L43, N03706@L43, N03707@L43, N03721@L44, N03730@L45, N03741@L46, N03752@L47

.93 x2 | N03704@L43, N03705@L43

.7 x1 | N03708@L43

.016 x1 | N03709@L43

2 x4 | N03710@L43, N03728@L44, N03755@L47, N03756@L47

1 x5 | N03711@L44, N03712@L44, N03739@L46, N03740@L46, N03754@L47

.12 x1 | N03713@L44

.55 x1 | N03714@L44

.48 x2 | N03715@L44, N03717@L44

.34 x1 | N03716@L44

.31 x1 | N03718@L44

.15 x3 | N03719@L44, N03724@L44, N03751@L46

.53 x1 | N03720@L44

3 x1 | N03722@L44

.36 x1 | N03723@L44

.16 x1 | N03725@L44

.025 x3 | N03726@L44, N03727@L44, N03735@L45

.35 x1 | N03729@L45

5 x1 | N03731@L45

2.399 x1 | N03732@L45

.52 x1 | N03733@L45

.47 x2 | N03734@L45, N03743@L46

.005 x1 | N03736@L45

.04 x1 | N03737@L45

.006 x1 | N03738@L45

4 x1 | N03742@L46

.29 x2 | N03744@L46, N03746@L46

.10 x1 | N03745@L46

.03 x2 | N03747@L46, N03749@L46

.56 x1 | N03748@L46

.86 x1 | N03750@L46

.4 x1 | N03753@L47

F0625

0 x6 | N03757@L48, N03758@L48, N03764@L48, N03770@L48, N03771@L48, N03776@L48

1 x1 | N03759@L48

2 x3 | N03760@L48, N03780@L48, N03781@L48

1.7 x1 | N03761@L48

6 x1 | N03762@L48

3 x6 | N03763@L48, N03765@L48, N03766@L48, N03768@L48, N03772@L48, N03774@L48

.12 x3 | N03767@L48, N03769@L48, N03773@L48

1.35 x1 | N03775@L48

4 x3 | N03777@L48, N03778@L48, N03779@L48

F0626

0 x15 | N03782@L49, N03783@L49, N03787@L50, N03788@L50, N03789@L50, N03792@L50, N03793@L50, N03797@L51, N03803@L51, N03806@L52, N03807@L52, N03813@L53, N03814@L53, N03819@L53, N03823@L53

1 x3 | N03784@L49, N03785@L49, N03812@L53

1.6 x1 | N03786@L49

8 x6 | N03790@L50, N03801@L51, N03804@L51, N03808@L52, N03809@L52, N03822@L53

5 x4 | N03791@L50, N03800@L51, N03802@L51, N03805@L51

2 x5 | N03794@L50, N03795@L50, N03798@L51, N03816@L53, N03817@L53

3 x1 | N03796@L50

6 x1 | N03799@L51

.48 x1 | N03810@L52

.38 x1 | N03811@L52

10 x1 | N03815@L53

2.6 x1 | N03818@L53

18 x1 | N03820@L53

11 x1 | N03821@L53

21 x1 | N03824@L53

Appendix F. Randomness and entropy register

All 93 identified candidate call sites are classified through lexical declarations and parameter defaults: seeded gameplay/presentation draws, default unseeded draws/helpers, cryptographic entropy or seed factories. A SEED construction is not a random draw. The register is syntactic coverage of the frozen code, not a claim that Math.random is cryptographically secure or reproducible. UI and legacy paths may be dormant in production; inventory presence alone does not establish reachability.

Kind	Call sites
domain-separated seeded draw (Math.random fallback before session)	46
injected pseudorandom draw	3
cryptographic entropy	7
seed factory, not a draw	3
seeded pseudorandom draw	25
seeded cosmetic draw	9

```
RNG001 client1:root L238 | domain-separated seeded draw (Math.random fallback before session) | inline function invocation; body separately inventoried | binding: None
RNG002 F0006 L241 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG003 F0154 L897 | injected pseudorandom draw | rng() | binding: None
RNG004 F0157 L899 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG005 F0157 L901 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG006 F0157 L901 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG007 F0157 L901 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG008 F0157 L901 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG009 F0159 L907 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG010 F0159 L907 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG011 F0166 L922 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG012 F0168 L927 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG013 F0173 L932 | cryptographic entropy | crypto.getRandomValues(new Uint32Array(1)) | binding: None
RNG014 F0173 L932 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None
RNG015 F0177 L939 | seed factory, not a draw | SEED(z.seed) | binding: (n)=>()=>{n=(n+0x6d2b79f5)|0;let t=Math.imul(n^(n>>>15),1|n);t^=t+Math.imul(t^(t>>>7),61|t);return((t^(t>>>14))>>>0)/4294967296}
RNG016 F0177 L940 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG017 F0177 L940 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG018 F0177 L940 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG019 F0177 L940 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG020 F0177 L940 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG021 F0177 L941 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG022 F0177 L941 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG023 F0177 L941 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG024 F0177 L941 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG025 F0177 L941 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG026 F0177 L943 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
RNG027 F0177 L943 | seeded pseudorandom draw | r() | binding: SEED(z.seed)
```

RNG028 F0185 L944 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG029 F0185 L944 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG030 F0185 L946 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG031 F0185 L946 | domain-separated seeded draw (Math.random fallback before session) | rnd(-980,980) | binding: (a,b)=>a+(game.random?.world||Math.random)()*(b-a)

RNG032 F0185 L946 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG033 F0185 L946 | domain-separated seeded draw (Math.random fallback before session) | rnd(-760,760) | binding: (a,b)=>a+(game.random?.world||Math.random)()*(b-a)

RNG034 F0185 L946 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG035 F0185 L949 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG036 F0185 L949 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG037 F0185 L949 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG038 F0185 L950 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG039 F0185 L950 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG040 F0185 L950 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG041 F0185 L950 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG042 F0185 L950 | domain-separated seeded draw (Math.random fallback before session) | r() | binding: game.random?.world||Math.random

RNG043 F0190 L954 | seeded pseudorandom draw | r() | binding: Injected from buildMap SEED(z.seed)

RNG044 F0190 L954 | seeded pseudorandom draw | r() | binding: Injected from buildMap SEED(z.seed)

RNG045 F0190 L954 | seeded pseudorandom draw | r() | binding: Injected from buildMap SEED(z.seed)

RNG046 F0190 L954 | seeded pseudorandom draw | r() | binding: Injected from buildMap SEED(z.seed)

RNG047 F0199 L966 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.loot||Math.random)() | binding: None

RNG048 F0201 L966 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.loot||Math.random)() | binding: None

RNG049 F0201 L966 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.loot||Math.random)() | binding: None

RNG050 F0201 L966 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.visual||Math.random)() | binding: None

RNG051 F0201 L966 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.loot||Math.random)() | binding: None

RNG052 F0201 L966 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.visual||Math.random)() | binding: None

RNG053 F0201 L966 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.visual||Math.random)() | binding: None

RNG054 F0205 L970 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.visual||Math.random)() | binding: None

RNG055 F0205 L970 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.visual||Math.random)() | binding: None

RNG056 F0205 L970 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.visual||Math.random)() | binding: None

RNG057 F0205 L970 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.visual||Math.random)() | binding: None

RNG058 F0206 L971 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.world||Math.random)() | binding: None

RNG059 F0222 L1042 | domain-separated seeded draw (Math.random fallback before session) | rnd(1.65,2.85) | binding: (a,b)=>a+(game.random?.world||Math.random)()*(b-a)

RNG060 F0222 L1042 | domain-separated seeded draw (Math.random fallback before session) | rnd(1.2,2.8) | binding: (a,b)=>a+(game.random?.world||Math.random)()*(b-a)

RNG061 F0222 L1042 | domain-separated seeded draw (Math.random fallback before session) | rnd(0,TAU) | binding: (a,b)=>a+(game.random?.world||Math.random)()*(b-a)

RNG062 F0222 L1042 | domain-separated seeded draw (Math.random fallback before session) | rnd(.7,2.4) | binding: (a,b)=>a+(game.random?.world||Math.random)()*(b-a)

RNG063 F0251 L1106 | seed factory, not a draw | SEED(zones[game.zone].seed+813) | binding: (n)=>()=>{n=(n+0x6d2b79f5)|0;let

t=Math.imul(n^(n>>>15),1|n);t^=t+Math.imul(t^(t>>>7),61|t);return((t^(t>>>14))>>>0)/4294967296}

RNG064 F0251 L1107 | seeded pseudorandom draw | random() | binding: SEED(zones[game.zone].seed+813)

RNG065 F0251 L1107 | seeded pseudorandom draw | random() | binding: SEED(zones[game.zone].seed+813)

RNG066 F0251 L1107 | seeded pseudorandom draw | random() | binding: SEED(zones[game.zone].seed+813)

RNG067 F0251 L1107 | seeded pseudorandom draw | random() | binding: SEED(zones[game.zone].seed+813)

RNG068 F0251 L1108 | seeded pseudorandom draw | random() | binding: SEED(zones[game.zone].seed+813)

RNG069 F0251 L1108 | seeded pseudorandom draw | random() | binding: SEED(zones[game.zone].seed+813)

RNG070 F0270 L1166 | seed factory, not a draw | SEED(777) | binding: (n)=>()=>{n=(n+0x6d2b79f5)|0;let t=Math.imul(n^(n>>>15),1|n);t^=t+Math.imul(t^(t>>>7),61|t);return((t^(t>>>14))>>>0)/4294967296}

RNG071 F0270 L1166 | seeded pseudorandom draw | r() | binding: SEED(777)

RNG072 F0270 L1166 | seeded pseudorandom draw | r() | binding: SEED(777)

RNG073 F0270 L1166 | seeded pseudorandom draw | r() | binding: SEED(777)

RNG074 F0298 L1230 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.craft||Math.random)() | binding: None

RNG075 F0298 L1231 | domain-separated seeded draw (Math.random fallback before session) | (game.random?.craft||Math.random)() | binding: None

RNG076 F0411 L1456 | cryptographic entropy | crypto.getRandomValues(new Uint8Array(16)) | binding: None

RNG077 F0485 L1544 | domain-separated seeded draw (Math.random fallback before session) | game.random?.snapshot() | binding: None

RNG078 F0511 L48 | cryptographic entropy | randomBytes(16) | binding: None

RNG079 F0520 L93 | cryptographic entropy | randomBytes(32) | binding: None

RNG080 F0526 L192 | cryptographic entropy | randomUUID() | binding: None

RNG081 F0526 L192 | cryptographic entropy | randomBytes(32) | binding: None

RNG082 F0526 L202 | cryptographic entropy | randomBytes(32) | binding: None

RNG083 F0573 L10 | injected pseudorandom draw | rng() | binding: None

RNG084 F0590 L17 | injected pseudorandom draw | rng() | binding: None

RNG085 F0617 L8 | seeded cosmetic draw | random() | binding: ()=>{s=(Math.imul(s,1664525)+1013904223)>>>0;return s/4294967296}

RNG086 F0617 L8 | seeded cosmetic draw | random() | binding: ()=>{s=(Math.imul(s,1664525)+1013904223)>>>0;return s/4294967296}

RNG087 F0617 L8 | seeded cosmetic draw | random() | binding: ()=>{s=(Math.imul(s,1664525)+1013904223)>>>0;return s/4294967296}

RNG088 F0617 L8 | seeded cosmetic draw | random() | binding: ()=>{s=(Math.imul(s,1664525)+1013904223)>>>0;return s/4294967296}

RNG089 F0617 L8 | seeded cosmetic draw | random() | binding: ()=>{s=(Math.imul(s,1664525)+1013904223)>>>0;return s/4294967296}

RNG090 F0617 L9 | seeded cosmetic draw | random() | binding: ()=>{s=(Math.imul(s,1664525)+1013904223)>>>0;return s/4294967296}

RNG091 F0617 L9 | seeded cosmetic draw | random() | binding: ()=>{s=(Math.imul(s,1664525)+1013904223)>>>0;return s/4294967296}

RNG092 F0617 L9 | seeded cosmetic draw | random() | binding: ()=>{s=(Math.imul(s,1664525)+1013904223)>>>0;return s/4294967296}

RNG093 F0617 L9 | seeded cosmetic draw | random() | binding: ()=>{s=(Math.imul(s,1664525)+1013904223)>>>0;return s/4294967296}

Appendix G. CSS and database computation register

CSS calculations use CSS pixel/percentage/environment units, resolved by the browser layout engine; they are not world-unit movement or reward formulas. The 97 calc/min/max/clamp occurrences include nested occurrences and are listed separately. Their style_line positions refer to the extracted inline style text. SQL entries retain parameter placeholders; bind order and validation remain in the service function registry.

```
CSS001 style L3 calc(16px + env(safe-area-inset-top))
CSS002 style L3 min(320px,34vw)
CSS003 style L9 calc(115px + env(safe-area-inset-top))
CSS004 style L9 min(250px,48vw)
CSS005 style L10 calc(156px + env(safe-area-inset-top))
CSS006 style L11 calc(64px + env(safe-area-inset-top))
CSS007 style L18 calc(16px + env(safe-area-inset-top))
CSS008 style L22 calc(16px + env(safe-area-inset-top))
CSS009 style L23 calc(22px + env(safe-area-inset-bottom))
CSS010 style L24 calc(17px + env(safe-area-inset-top))
CSS011 style L25 calc(101px + env(safe-area-inset-top))
CSS012 style L25 min(480px,85vw)
CSS013 style L27 calc(25px + env(safe-area-inset-left))
CSS014 style L27 calc(44px + env(safe-area-inset-bottom))
CSS015 style L27 calc(22px + env(safe-area-inset-right))
CSS016 style L27 calc(51px + env(safe-area-inset-bottom))
CSS017 style L28 min(890px,100%)
CSS018 style L29 clamp(35px,5vw,54px)
CSS019 style L31 min(780px,100%)
CSS020 style L31 min(760px,90dvh)
CSS021 style L31 min(370px,100%)
CSS022 style L32 calc(12px + env(safe-area-inset-bottom))
CSS023 style L32 calc(48px + env(safe-area-inset-bottom))
CSS024 style L33 calc(8px + env(safe-area-inset-top))
CSS025 style L33 calc(9px + env(safe-area-inset-top))
CSS026 style L33 min(740px,100%)
CSS027 style L34 calc(11px + env(safe-area-inset-bottom))
CSS028 style L34 calc(11px + env(safe-area-inset-left))
CSS029 style L34 calc(75px + env(safe-area-inset-bottom))
CSS030 style L34 calc(10px + env(safe-area-inset-right))
CSS031 style L34 calc(85px + env(safe-area-inset-bottom))
CSS032 style L34 min(13.3vw,48px)
CSS033 style L34 calc(169px + env(safe-area-inset-bottom))
CSS034 style L35 calc(25px + env(safe-area-inset-bottom))
CSS035 style L35 calc(28px + env(safe-area-inset-bottom))
CSS036 style L37 calc(65px + env(safe-area-inset-top))
CSS037 style L37 min(310px,40vw)
CSS038 style L41 min(850px,97vw)
CSS039 style L57 calc(83px + env(safe-area-inset-top))
CSS040 style L58 calc(117px + env(safe-area-inset-top))
CSS041 style L58 calc(9px + env(safe-area-inset-bottom))
CSS042 style L58 calc(100vw - 10px)
CSS043 style L58 min(17vw,44px)
CSS044 style L58 calc(130px + env(safe-area-inset-bottom))
CSS045 style L58 calc(139px + env(safe-area-inset-bottom))
CSS046 style L58 calc(203px + env(safe-area-inset-bottom))
CSS047 style L59 min(8vw,39px)
CSS048 style L61 calc(170px + env(safe-area-inset-top))
CSS049 style L63 calc(102px + env(safe-area-inset-bottom))
```

CSS050 style L69 calc(135px + env(safe-area-inset-top))
CSS051 style L70 calc(113px + env(safe-area-inset-top))
CSS052 style L72 calc(133px + env(safe-area-inset-top))
CSS053 style L76 calc(177px + env(safe-area-inset-top))
CSS054 style L82 calc(165px + env(safe-area-inset-bottom))
CSS055 style L84 calc(12px + env(safe-area-inset-top))
CSS056 style L85 calc(225px + env(safe-area-inset-bottom))
CSS057 style L86 calc(248px + env(safe-area-inset-bottom))
CSS058 style L87 calc(199px + env(safe-area-inset-bottom))
CSS059 style L87 calc(245px + env(safe-area-inset-bottom))
CSS060 style L87 calc(72px + env(safe-area-inset-top))
CSS061 style L101 calc(100vw - 12px)
CSS062 style L114 min(520px,94vw)
CSS063 style L122 calc(100vw - 12px)
CSS064 style L122 calc(7px + env(safe-area-inset-bottom))
CSS065 style L123 calc(224px + env(safe-area-inset-bottom))
CSS066 style L123 calc(216px + env(safe-area-inset-bottom))
CSS067 style L124 calc(342px + env(safe-area-inset-bottom))
CSS068 style L124 calc(318px + env(safe-area-inset-bottom))
CSS069 style L125 calc(325px + env(safe-area-inset-bottom))
CSS070 style L130 min(62vw,650px)
CSS071 style L130 calc(100vw - 380px)
CSS072 style L132 calc(16px + env(safe-area-inset-bottom))
CSS073 style L132 calc(8px + env(safe-area-inset-right))
CSS074 style L132 calc(12px + env(safe-area-inset-bottom))
CSS075 style L132 calc(8px + env(safe-area-inset-left))
CSS076 style L135 calc(110px + env(safe-area-inset-left))
CSS077 style L135 calc(100vw - 220px - env(safe-area-inset-left) - env(safe-area-inset-right))
CSS078 style L135 calc(3px + env(safe-area-inset-bottom))
CSS079 style L137 calc(110px + env(safe-area-inset-bottom))
CSS080 style L142 calc(204px + env(safe-area-inset-bottom))
CSS081 style L142 calc(8px + env(safe-area-inset-left))
CSS082 style L142 calc(212px + env(safe-area-inset-bottom))
CSS083 style L142 calc(8px + env(safe-area-inset-right))
CSS084 style L143 calc(300px + env(safe-area-inset-bottom))
CSS085 style L145 calc(8px + env(safe-area-inset-right))
CSS086 style L145 calc(8px + env(safe-area-inset-left))
CSS087 style L147 calc(100vw - 40px)
CSS088 style L176 min(100%,270px)
CSS089 style L186 min(960px,100%)
CSS090 style L193 clamp(24px,5vw,64px)
CSS091 style L194 min(680px,calc(100vw - 28px))
CSS092 style L194 calc(100vw - 28px)
CSS093 style L194 calc(100dvh - 28px)
CSS094 style L194 calc(100dvh - 140px)
CSS095 style L196 calc(100dvh - 18px)
CSS096 style L197 calc(100dvh - 12px)
CSS097 style L197 calc(100dvh - 92px)

Runtime SQL operations

SQL001 F0516 L75 `INSERT INTO rate_limits(bucket,attempts,reset_at) VALUES(?,1,?) ON CONFLICT(bucket) DO UPDATE SET attempts=CASE WHEN reset_at<=? THEN 1 ELSE attempts+1 END, reset_at=CASE WHEN reset_at<=? THEN excluded.reset_at ELSE reset_at END RETURNING attempts,reset_at`

SQL002 F0518 L86 `SELECT p.id,p.username,p.display_name,p.password_record,p.recovery_hash,p.credential_version FROM sessions s JOIN players p ON p.id=s.player_id WHERE s.token_hash=? AND s.expires_at>? AND s.credential_version=p.credential_version`

SQL003 F0520 L91 DELETE FROM sessions WHERE token_hash IN (SELECT token_hash FROM sessions WHERE expires_at<=? LIMIT 100)

SQL004 F0520 L92 DELETE FROM rate_limits WHERE bucket IN (SELECT bucket FROM rate_limits WHERE reset_at<=? LIMIT 100)

SQL005 F0520 L94 `INSERT INTO sessions(token_hash,player_id,credential_version,expires_at) SELECT ?,id,credential_version,? FROM players WHERE id=? AND credential_version=?`

```
SQL006 F0524 L118 INSERT INTO saves(player_id,schema_version,revision,payload,updated_at) VALUES(?,5,1,?,?) ON CONFLICT(player_id) DO NOTHING RETURNING revision
SQL007 F0524 L119 UPDATE saves SET revision=revision+1,payload=?,updated_at=? WHERE player_id=? AND revision=? RETURNING revision
SQL008 F0525 L131 SELECT player_id FROM legacy_links WHERE legacy_uid=?
SQL009 F0525 L133 SELECT player_id FROM saves WHERE player_id=?
SQL010 F0525 L140 INSERT INTO legacy_links(legacy_uid,player_id,source_revision,migrated_at) VALUES(?,?,?,?)
SQL011 F0525 L141 INSERT INTO saves(player_id,schema_version,revision,payload,updated_at) VALUES(?,5,1,?,?)
SQL012 F0526 L153 SELECT id FROM players LIMIT 1
SQL013 F0526 L159 SELECT revision,payload,updated_at FROM saves WHERE player_id=?
SQL014 F0526 L170 DELETE FROM sessions WHERE token_hash=?
SQL015 F0526 L184 UPDATE players SET password_record=?,credential_version=credential_version+1 WHERE id=? AND credential_version=? RETURNING credential_version
SQL016 F0526 L193 INSERT INTO players(id,username,display_name,password_record,recovery_hash,created_at) VALUES(?,?,?,?,?,?) ON CONFLICT(username) DO NOTHING RETURNING id,username,display_name,credential_version
SQL017 F0526 L197 SELECT * FROM players WHERE username=?
SQL018 F0526 L203 UPDATE players SET password_record=?,recovery_hash=?,credential_version=credential_version+1 WHERE id=? AND credential_version=? AND recovery_hash=? RETURNING credential_version
```

Current database schema contracts

Table	Fields and constraints
players	id primary key; username unique/non-null; display_name, password_record, recovery_hash and created_at non-null; credential_version integer defaults to1
sessions	token_hash primary key; player_id foreign key with cascade; credential_version and expires_at required; expiry index
saves	player_id primary key/foreign key with cascade; schema_version=5; revision>=1; valid JSON payload <=1,048,576 bytes; updated_at required
legacy_links	legacy_uid primary key; player_id unique foreign key/cascade; optional source_revision; migrated_at required
rate_limits	bucket primary key; attempts and reset_at required; expiry index

Foreign-key enforcement is enabled by PRAGMA. Legacy Supabase saves use one auth.users-linked UUID row, schema range1..100, revision>=1, object JSONB payload <=1 MiB and timestamp default now(). RLS allows authenticated users SELECT/INSERT/UPDATE only on their own user_id, with no client DELETE grant. These legacy guards are provenance contracts, not the current Cloudflare login mechanism.

Schema and deployment boundaries

The declared source scope includes both current Cloudflare account/save schema and legacy Supabase migration. The legacy SQL is retained for audit compatibility, not a current hosting recommendation. Foreign keys, uniqueness, revision guards and atomic update eligibility remain mandatory even though they are not stochastic formulas. netlify.toml is a legacy deployment configuration artifact; its presence does not prove a hosting service was retired. The production basis for this handbook is Cloudflare.

Appendix H. Coverage, limitations and change control

Inventory assertion	Result
function_nodes	626
operator_nodes	7418
expression_envelopes	3795
calls	4060
numeric_calls	569
numeric_literals	3824
random_candidates	93
css_computations	97
all_operators_have_envelope	True
all_calls_have_known_owner	True

The source implements exact probability analysis, seeded domains, bounded critical/frost mechanics and frozen profiles. It does not automatically optimize every player action by expected value. The source remains client-authoritative for combat and inventory. Publishing a specification does not make gameplay deterministic, calibrated, fair against cheating or server-authoritative. C2 and C3 require implementation and separate evidence.

AST coverage is bounded to the files listed in the source pin. It is not whole-program semantic proof. The lexical state-reference index does not resolve every alias or dynamic property; function inputs, source envelopes and hand-reviewed contracts are jointly necessary. CSS literal colors and browser layout internals are not counted as JavaScript numerical literals. Future runtime files, constructors with gameplay behavior, imported dependencies and generated formulas require inventory expansion.

Change procedure: pin a new commit; compare function/operator/call/literal and RNG inventories; assign a version to changed probability and utility profiles; evaluate complete outcome distributions; review reset/clock/order changes; execute source-oracle and numerical checks; inspect every rendered page; then publish the revised handbook. No silent rebalance, source substitution or retroactive change to a recorded probability is permitted.

Normative completion criterion

A mechanism is standardized only when its input domain, units, transition order, distribution, output, composition, exceptional cases, verification and trace are documented. A live release may claim C2 only after implementing versioned probability/utility profiles and demonstrating behavior against these contracts. C3 additionally requires authoritative reward and inventory state transitions. This edition supplies the foundation and source audit; remaining implementation work is explicitly visible.

Normative interface supplement

Scope and authority

This supplement specifies the V5.17 interface transition corrections. The 333-page Edition 3.2 remains a frozen V5.16 source inventory, not a census of V5.17 syntax. This supplement supersedes its overlapping presentation and input descriptions only. M001-M138 keep their identifiers and classifications. No new combat, loot, crafting, reward or preference-score distribution is introduced.

The normative implementation is site/index.html: resetInterfaceState, showModal, closeModal, renderModal, renderPanelFooter, bindBagDrag, resetInputs, equipInto, initGame and logout. The PDF supplement records exact source hashes; publication is a separate verification step.

S1-01: Session-scoped presentation state

Let U = (menus, filters, selection, scroll positions, build preference). On an actual game initialization or completed logout, reset U to the collapsed menus, empty queries, all rarities, rarity ordering, first slot, current deck, no item preview, no craft selection, balanced preference and empty scroll-position map. Set both menu aria-expanded values to false. U is transient and is not a player-save schema extension.

S1-02: Panel position ownership

Let q be the current panel and P[q] its stored vertical offset in CSS pixels. Before opening a panel or closing it, record its current offset. Opening q restores max(0, P[q]) subject to the browser's normal scroll-range clamp. An unvisited panel starts at zero. Rerendering a panel preserves its offset. A new session clears P. Offsets never control simulation time, inventory or RNG.

Normative interface supplement

S1-03: Semantic focus identity

Before rerendering, capture a focused control's DOM id or one of its stable data keys: equip, craft, slot, deck, policy or talent. After rendering, focus the surviving corresponding control with preventScroll. If a removed in-panel control has no successor, focus the panel close button. This is a presentation identity, not an account authorization token. Native modal dialogs retain their existing focus isolation.

S1-04: Persistent equipment action

The bag footer is outside the scrolling panel body. It displays the selected item's rarity and family, the target current/backup deck and one-based slot number. Its action invokes the same equipInto(key, slot) path as the inspection workbench. It does not mint items or bypass insufficient-stock checks, backup-deck selection, the existing reload cost or the HP clamp. With no available selected item, disable the action. Other panels hide the footer.

For valid item k and replaced item j, the bag transition is b'[k] = b[k] - 1 and b'[j] = b[j] + 1, with the selected slot replaced by k. When j = k, the net bag count is unchanged. The sum of bag, current-deck and backup-deck counts is conserved per key, within the existing inventory domain. If k differs from j and the return stack b[j] is already 99999, reject the entire swap before taking k. A same-key swap at the cap is permitted because its net change is zero. This prevents silent loss of the replaced item.

Normative interface supplement

S1-05: Scroll and drag partition

Card bodies use vertical browser panning. A touch/pen drag begins only from the explicitly marked grip; the grip uses touch-action: none. Mouse native drag-and-drop remains available on the card. After grip displacement exceeds 12 CSS pixels, show a noninteractive drag ghost. A valid pointer-up over a slot in the same live bag root invokes equipment. Cancellation never equips. After a completed drag, suppress the generated card click for 500 milliseconds.

S1-06: Drag lifetime and isolation

At most one touch drag owns global move/up/cancel listeners. Beginning a replacement drag, resetting game input, switching or closing panels, rerendering, window blur or page hide clears those listeners, ghost and highlights. A late pointer-up from a discarded drag cannot equip into a newly created panel. Unknown item keys and invalid slot indices remain rejected by equipInto.

Normative interface supplement

Probability and persistence boundary

For fixed valid state x and interface action a, the interface-only transition has kernel K(x'|x,a) = 1 for its determined successor. UI render, filtering, scrolling and focus restoration do not consume loot, craft or combat RNG. Equipment application retains its existing simulation-side reload behavior and must not be described as RNG-free. The only ruleset.json change is the version label. Account API behavior, database schema and save schema 5 are unchanged, apart from the health endpoint version.

Verification and limits

tests/browser_v517.mjs exercises five Chromium viewports: 1280x800, 390x844, 844x390, 320x568 and 568x320. The suite verifies semantic keyboard focus, per-panel scroll ownership, footer visibility, equipment stock use and conservation on its bounded fixture, mobile browser touch panning, grip-only dragging, stale-drag cleanup, logout state reset and lack of runtime errors. The V5.16 regression suite continues to verify all 14 panels and the build workshop. CI additionally runs the existing account, recovery, CAS, controls, collision and probability tests. Emulation does not certify hardware Safari behavior or all possible save states.

Source identity and change boundary

Runtime source commit: 363a1faa631b10f88b9dfc5cd4539df4e5257fa5

site/index.html

SHA-256 3ffe190579e2047c6bb81f6723612ccb90d6cbaff670fcf2f4b9128f6930f6f8

site/rules.js

SHA-256 9b4c517f96711fff9348b86f245aadbf5c8b4f053d03342e51ef4df6efdbae73

site/release.json

SHA-256 83101f9f0272733e2ccd015dd6e1e7cb3d2366871a48030a8dd0b5460301b9ef

functions/api/[[path]].js

SHA-256 dc7d3a61a14f915016a58d6df767e509e6aeda5870d68d88afb50f16f0184a90

docs/standard/ruleset.json

SHA-256 fd97dc99617778d079acaa6e1b95f2b75161e545db1ec600467e0c4ba32e1d30

docs/standard/mechanisms.json

SHA-256 bc932a72100b50df396d67fc7cc0f8bfe7545b9b2fcbda662de20243f5b62fbf

docs/standard/V5.17_INTERFACE_CONTINUITY.md

SHA-256 47c9c88e67566ed95aff1bc02980e0c4d01bba7984d1c9c59e114da91bb947b6

tests/browser_v517.mjs

SHA-256 f1833a093efff499f61d0e33a4de60734dd852e97f9e6b38b3fdcf1eb00c35bd

Conformance and limits

The current browser regression contains 128 assertions across five viewport sizes. The preceding V5.16 suite contains 245 assertions. Source, test result and deployment verification are separate evidence dimensions; this document does not turn an unexecuted check into a pass.

Equipment safety

For distinct item keys, reject a swap if the replaced item cannot return to its existing bag stack. The rejection occurs before consuming the requested item. Same-key swaps at the 99999 cap have zero net stock change. Neither operation bypasses existing reload, HP or save behavior.

Compatibility

The native account database and save schema 5 remain compatible. Only the API health version label changes. The probability profile coefficients are unchanged; its version label follows the release. No production player account is created by the isolated browser suite.

Reproduction

Run scripts/build-interface-supplement.py with the frozen Edition 3.2 baseline PDF, an output path and the exact V5.17 source commit. The source script checks the baseline page count and verifies that every original page retains its text in the consolidated output. ReportLab invariant mode removes timestamp variability from newly generated pages.

Baseline PDF SHA-256: 392b465ca4b2259eabdc332ca36a8ec2caf9cb9c4906b88b7b2db56bbc01eb2b

Hardware boundary

Chromium mobile emulation and injected browser touch events do not certify physical iPhone, iPad or Safari behavior. Successful public byte and health checks establish publication of matching files, not a complete live-account playthrough.

Normative presentation supplement

Scope and reference evidence

V5.18 introduces an original outline-art presentation profile. It does not claim to reproduce Florr's complete interface, assets, server logic or numerical rules. On 22 October 2026 the official https://florr.io/ page was opened in the browser. The guest entry, animated floating-petal background, green ground and top-left utility icons were directly visible. After the guest action the page stayed on Connecting and logged "Ran out of transport methods to try". One reload and guest retry did not establish a playable session. Combat, map layout, enemy behavior, crafting odds and settings beyond that screen were not verified. A generic transport failure is not evidence of bot detection.

The original 333-page standard and Supplement S1 remain source-frozen descriptions of earlier versions. S2 governs the V5.18 presentation changes below; their old syntax ledgers must not be called a census of V5.18. All 138 existing mechanism identifiers and status classifications remain registered. No combat or reward probability coefficient changes.

S2-01: Bounded presentation profile

The settings projection accepts visualStyle in {field, botanical}, hud in {compact, full}, motion in {auto, reduced}, and strict Boolean showGrid and orbitGuide. Missing or invalid values map respectively to field, compact, auto, false and false. This projection runs when a save enters the game and after presentation selection. It extends the existing settings object inside save schema 5, not the account identity, database schema or progression model. Effective reduced motion is the logical OR of the operating-system media preference and motion == reduced.

Normative presentation supplement

S2-02: Rendering-state separation

Let G be the gameplay state, U the presentation profile, t the presentation clock and I the output image. Rendering is I = R(G,U,t), with G unchanged. Rendering, thumbnails and settings projection must not draw from the world, loot, craft, combat or visual gameplay streams. Deterministic terrain decoration uses a private local LCG, s[n+1] = (1664525*s[n] + 1013904223) mod 2^32. Its seed is the existing biome seed plus 813. This private decoration sequence is not a drop distribution or probability-balance source. Presentation caches are keyed by both style and biome to prevent cross-style material reuse.

S2-03: Shared original artwork and scale

All 29 registered petal families use one original procedural outline renderer for world art and the inventory, forging, workbench and equipment thumbnails. World size remains the existing petal size times PETAL_WORLD_SCALE. UI thumbnails are rasterized from the same vector paths onto their own high-resolution canvas; icon dimensions never change a collider. The eight field biome palettes are distinct. Both world and thumbnail creature rendering use the existing species body ellipses, anatomy and size. The optional target canvas is passed explicitly so a codex portrait cannot write onto the world canvas. Decorative legs, wings, crowns and tails remain outside collision semantics.

Normative presentation supplement

S2-04: Clock and motion policy

Avatar gaze uses existing velocity, blink uses a deterministic sine phase, and hurt expression uses the existing last-hit clock. None modifies movement or HP. Creature locomotion phase remains derived from simulation time, species speed and identity. Reduced motion freezes optional blink, creature gait, portal spin and bob, loot bob, gate drift and UI transitions. It does not freeze combat, authoritative projectiles, orbital collision motion, cooldowns or danger telegraphs. The gate drift clock comes from performance.now without advancing game.time. Motion preferences must not become an advantage in crafting or loot odds.

S2-05: HUD and backup-deck access

Compact HUD retains HP, XP, save status, biome information where space permits, map access, inventory access, skill access and pause. Extra numeric panels and tools are disclosed through existing menus. Colored equipment cards retain the rarity label through accessible names and show current durability/reload state. On fine-pointer desktop layouts the backup strip opens manual equipment for the selected backup slot; it does not perform a combat deck swap, consume an item or bypass the existing reload cost. Coarse-pointer layouts retain backup access through the bag and hide this extra strip to protect movement controls.

Normative presentation supplement

S2-06: Optional guides and gameplay invariants

The grid is visual only. The orbit guide uses the actual distance between the first live petal and player, not a replacement collision or attack-radius formula. Field biome rings retain positional rarity-band information without the large world-text plaques; the HUD and map remain the information sources. No viewport resize, style switch, glyph replacement or portrait rendering may alter mobCollider, petalContactRadius, damage, hit ordering, crafting, drops, save revisions or inventory conservation. Existing constraints and failure boundaries continue to apply.

Verification and limitations

The browser suite uses 1280x800, 390x844, 844x390, 320x568 and 568x320 Chromium viewports. It checks settings controls, sanitization defaults, compact/full disclosure, all 29 petal thumbnails, 43 codex portraits, world-pixel isolation of thumbnails, gameplay-state and RNG preservation, backup-slot routing, eight biome views and 129 species-size collider comparisons. Existing V5.16 and V5.17 regressions continue to cover panels, native dialogs, scrolling, touch grip behavior and safe equipment return stacks. Emulated Chromium does not certify physical Safari behavior. The unavailable Florr gameplay session prevents any claim of verified one-to-one gameplay parity.

Source identity and change boundary

Runtime source commit: 08771687513e480a2cacd84c52ef1a6327b154fa

site/index.html

SHA-256 c4aabf812d7d3e0acb44d8e09e81e10c6f213afc67d3ed493cd04262ed55da1c

site/art.js

SHA-256 284f0008c2b7612c2db2018169f47dd7306e25c6754db793ddf3fe13ac1dff85

site/rules.js

SHA-256 f48c1ecb485da6da35ac6d2e71cdaa11268a4f8e4a6aed095dc0b8a1380f4790

site/release.json

SHA-256 04096b15e42672ff7277a8cd34e5a5bfcf8e25fba3abb9c20ad3fdf0019c6ebd

functions/api/[[path]].js

SHA-256 2080fc49108d45440aacd91cc5418f9b4870bc9c7f9c1dd3903d15566a47bd2c

docs/standard/ruleset.json

SHA-256 9f6b248f6d90e55e4fb5c6ebe4028eebbda7ad59ecc17f33fc05a38dbbc6d8a5

docs/standard/mechanisms.json

SHA-256 b042a4b96f0053b002a2cee2e1190d0204038903fd8af64841758faf061c05a5

docs/standard/V5.18_ORBITAL_FIELD.md

SHA-256 e4a8c67f8446a477947105dc2ccf8a7b255c3fc8c48dee75266c777efd15dc42

tests/browser_v518.mjs

SHA-256 1cfeabeae276f0289cc6188668ed7271cc67990c1c059822bb5680a92a35c763

Conformance and limits

The V5.18 browser suite exercises five viewport sizes, all 29 petal thumbnails, 43 creature portraits and 129 species-size collider cases. It verifies settings, backup-slot routing, world-pixel isolation and gameplay-state/RNG invariance. The earlier V5.16 and V5.17 suites remain mandatory release regressions.

Unchanged probability coefficients

The canonical ruleset version label follows V5.18. All numerical probability-profile coefficients are unchanged. Collision radii, hit resolution, crafting, loot and native save schema 5 are retained. Reduced decorative motion is separate from combat simulation and danger timing.

Reproduction

Run scripts/build-interface-supplement.py BASE.pdf OUTPUT.pdf SOURCE_COMMIT --field using the preserved 340-page S1 baseline. The builder verifies every preserved page text. ReportLab invariant mode and a fixed source commit allow a fresh duplicate build to be compared byte for byte.

Baseline PDF SHA-256: 76554a990faf5ad852af100aa45d2c5601e536a716507a7cc1d3c6ab3df8b3e8

Evidence boundary

Local and CI results, Cloudflare publication and public health are separately verified. An uploaded PDF is not evidence that the game is deployed. Emulated Chromium is not physical Safari testing. Florr combat, maps, enemy algorithms and crafting odds remain unverified because the guest world connection did not succeed.

Normative combat supplement

Scope and authority

S3 governs the V5.19 runtime changes and supersedes conflicting descriptions in earlier editions only for the behaviors explicitly listed here. The 138 registered mechanism identifiers retain their implemented, derived or proposed classifications. Save schema 5 and the account database remain compatible. No crafting, drop, critical, frost, skill-cooldown or event-weight coefficient is changed. Geometry and terminal-state corrections can change realized combat outcomes; they must not be described as purely decorative changes.

The official Florr entry page was directly observed. Its guest connection did not reach a playable world. Search verified the official game's combat-and-collection description, but community guides and third-party databases are not official formula specifications. This release uses original art and independent rules; it does not certify complete Florr parity. The original 347 manual pages remain frozen historical evidence, with their original source pins and limitations.

S3-01: Terminal combat state

Let D be the death flag and H the player's health. Once a fatal event is committed, D=1, H=0 and pause=1. Within the current fixed tick, the implementation returns at the next explicit boundary after fatal world-event damage, poison, a monster special, body contact or a hostile projectile. Subsequent healing, retaliation, experience awards, friendly-shot resolution, collection and portal transitions must not run in that tick. The fatal source itself may already have changed position, visual feedback or its own cooldown before this boundary. This is ordered execution, not rollback of the entire tick.

For D=1, the ordinary update map is absorbing: U_dt(G)=G. Player and monster damage entry points respect terminal state. Only the explicit respawn transition returns the player to an active life. Respawn restores maximum health, clears transient effects and held input, resets equipment, clears transient shots and loot, snaps the camera to the safe position and restores canvas focus. Existing skill cooldowns remain on their previous simulation clock. Death does not introduce an inventory-loss rule.

S3-02: Input ownership and probability-one transitions

Each captured touch pointer belongs to its joystick or action control. Canceling pointer p removes p from that owner's held set. Other held touch pointers remain active until their own release, cancellation or capture loss. Defense retains priority over attack. Blur, background transitions, menu entry, death and respawn neutralize all input. Non-touch pointer cancellation retains the existing global reset.

These are deterministic state transitions with probability one, not new random trials. A death/respawn transition must leave movement=(0,0) and action=normal unless a new eligible input arrives. A rendered pressed state must match the underlying held set. This ownership rule extends M003; it does not change joystick normalization, movement speed, attack radius or reload cost.

Normative combat supplement

S3-03: Damage acceptance and effective feedback

damagePlayer returns an acceptance Boolean A. A=0 for death, an active dash immunity interval or the existing 0.32-second body-hit grace; in these cases the projectile contact does not attach venom. For an accepted surviving venom hit, A=1 and venomUntil=t+3.5. The hostile projectile is consumed on contact regardless of A. Poison already present before a dash continues ticking; immunity is not a poison cleanse. Fatal contact enters S3-01 before any further status write.

For monster HP H>=0, requested damage X and armor B, mitigated damage is Y=max(1,X-B). The effective damage receipt is L=min(H,Y). Floating damage feedback reports rounded L; a batched exchange reports round(sum_i L_i). Crit count and exchange count remain separate indicators. The existing fang heal and lightning chain continue to use their declared requested-damage basis, so a feedback correction must not silently retune those effects. Negative HP from a prior poison tick contributes zero additional effective HP loss while the ordinary death pipeline records that kill.

S3-04: Shared body geometry

Friendly projectile contact now calls mobCircleOverlap(m,x,y,r), the same oriented ellipse-circle predicate used by orbital melee petals. For a target ellipse with axes a,b and facing theta, rotate the projectile center into the body frame; an interior point hits. For an exterior point, contact occurs when the closest ellipse-boundary distance is less than r. The existing solver uses eighteen bisection iterations. A conservative outer circle remains a rejection shortcut inside that solver, not the final contact decision.

Wing, leg, tail, horn and other decorative extents remain excluded. A projectile beside the short body axis must not hit merely because it lies within m.r+r. A genuine torso contact still applies the ordinary damage pipeline and grape hit-ID/bounce budget. This remains a sampled fixed-step contact test, not continuous collision detection: very fast or unusually small future projectiles require a separate swept-collision contract.

Normative combat supplement

S3-05: Bounded progression

The authoritative saved-level range remains 1<=level<=999. The runtime progression loop consumes a next-level cost only while level<999. At the cap, experience is retained, bounded by the existing loader ceiling of 10^9. For earned experience E, xp'=min(10^9,xp+E) precedes the bounded progression loop. For levels below the cap, the existing requirement 35+floor(18*l+0.9*l^2) remains unchanged.

The last legal transition 998->999 remains reachable. A subsequent reward must not create level 1000 and thereby make the next cloud save fail validation. This ceiling is an operational compatibility constraint, not a claim that 999 is the maximum level of the reference game. All rarity, reward and slot-capacity laws retain their prior definitions.

S3-06: Cloud-save retry state

The pending outbox remains owned by the same account and base revision until a matching successful upload clears or rebases it. HTTP 400,401,403,409 and413 do not automatically schedule another attempt inside the failed flush. Missing status,408,429 and5xx are retryable. For retry index k=1..6, d_k=max(1000*R,min(30000,2000*2^k)) milliseconds, where a missing Retry-After value gives R=0. Thus an ordinary transient failure starts at four seconds; a server's longer Retry-After is honored.

At most six retries are scheduled by that consecutive-failure sequence. A successful upload resets the retry counter. Only a confirmed successful upload may trigger the finalizer's scheduling path for a newer generation created during that upload. A fresh autosave or explicit online event is a separate trigger; this is not a permanent account lockout or a global lifetime request quota. Invalid or expired-session failures preserve pending bytes and show a recovery/export instruction. No authentication, account-identity or compare-and-swap boundary is bypassed.

Normative combat supplement

S3-07: Readable encounter presentation

All 29 loot families use the same original procedural petal outlines as equipment and inventory. Rarity remains a surrounding card border. Reduced decorative motion freezes optional loot bob and rotation. Rendering does not change item identity, collection radius, lifetime, rarity, count or gameplay RNG.

The guardian bar, event notice, player status and side tools occupy separate screen lanes. Portrait compact mode temporarily suppresses the biome card during a guardian encounter, leaving the map and guardian identity available. Short landscape mode narrows the central guardian strip. Desktop event notices appear below the guardian bar. Native DOM text remains separate from world canvas art. These layouts are tested at 1280x800,390x844,844x390,320x568 and568x320; safe-area insets remain part of CSS positioning.

S3-08: Verification and remaining boundaries

The new combat browser suite reproduces terminal poison, meteor, body, nova and projectile deaths; held-input cleanup; accepted versus blocked venom; rotated-body projectile misses and hits; level and experience bounds; effective damage feedback; and encounter overlap cases. The retry suite exercises permanent failures, transient backoff, Retry-After, exhaustion and successful upload generations. Earlier interface, rendering, crafting and account suites remain required regressions.

These checks use isolated fixtures and Chromium mobile emulation. They do not certify physical iOS Safari, every touch gesture, all adversarial imported values, or production database availability. Native cloud accounts store client-authored single-player progress; server-authoritative combat, a shared multiplayer economy, full ordered replay, pity and general elemental matrices remain outside this release. Publication, runtime health, source identity and local verification are distinct claims. A source-pinned PDF is not proof of deployment.

Source identity and change boundary

Local source commit: e0ece6621fd394858a4d5b8ce37b6da09c58db67

site/index.html

SHA-256 b31472af43b059d825737362b80065adbfbe02155478284ebf7bf6032960d4ae

site/rules.js

SHA-256 64cb9e4d8b8b8a80a47cbaf38c41e3824916519de07de4a7910d97f62b373930

site/release.json

SHA-256 2f2ee3cd8fae717e630e1bc356cb6fd57ef9a8f605fcdea5fc7e3970f29a4892

functions/api/[[path]].js

SHA-256 149e926ed9eafdc10d35841413f592f93f3d2153951781fb5b05170cb6eea0a1

docs/standard/ruleset.json

SHA-256 43116dd8db0d19900723e9e59675125b55fe56037b23086adfc87c7b61e0f5cc

docs/standard/mechanisms.json

SHA-256 9339b0a36a15441dd5bfa6e2c4c60918a1aedc2ae94230fa1288be009dc92a1f

docs/standard/V5.19_COMBAT_CONSISTENCY.md

SHA-256 a106f4f1abff4b0a52ec888901d13bfee5ec31d80f07373b3648577f7eb94945

tests/browser_v519.mjs

SHA-256 c860138ee2d664e1dfd5c71aec43d9a050d67973b1378e8e6a396d7b18aef3fb

tests/v519_sync_retry.cjs

SHA-256 774442635818aab7179b4f854aa94da5c79ac58c1379d3f096f3382074640ccb

Conformance, evidence and reproduction

The V5.19 browser suite contains 89 checks across combat fixtures and five encounter-layout viewports. The touch/keyboard continuity suite contains 46 checks. Prior V5.16, V5.17 and V5.18 browser suites contain 245, 128 and 130 checks. The retry suite contains 11 cases and the native account API suite contains 12 isolated cases. These counts describe specific executed checks, not exhaustive proof of all game states.

Probability and safety boundary

The probability-profile coefficients are unchanged except for the version label. Terminal-state and geometry corrections can change realized survival, contact and reward sequences. Effective damage feedback is distinct from the existing requested-damage basis of fang healing and lightning chaining. No account or database schema migration is required.

Reproduction

Run scripts/build-combat-supplement.py BASE.pdf OUTPUT.pdf SOURCE_COMMIT with the frozen 347-page S2 baseline. The builder checks page count, preserved text, source hashes and added-page bounds. Invariant ReportLab output and a fixed commit allow a fresh duplicate build to be compared byte for byte.

Baseline PDF SHA-256: 013e7d9842fcfcbcf48c2a27fc4366f7ec2597c4ead37308c942f6baacd24132

Reference evidence

Official reference: https://florr.io/ (entry observed; guest transport failed). The official search description confirms enemies and rare-petal collection; it supplies no numerical server specification. Community wikis, fan databases and hypothetical mechanics are not promoted to official authority.

Local checks do not establish production publication, server-authoritative multiplayer or physical Safari behavior. The QA report tracks those boundaries independently.

花瓣之境・玩家手册

V5.21 / 2026-10-27。本文说明当前实现，不把同类游戏的社区机制当作本游戏规则。数值均来自当前源代码；历史手册中的旧封面、候选状态与源码计数属于相应历史版本。

01 上手与操作

先选择角色类型，再开始探索。建议第一次进入先使用键盘移动或手机摇杆，在花园中心熟悉攻击、防御和闪避。世界地图的连线代表传送门关系，不代表点击后直接传送。

设备/ 操作	方法
电脑移动	WASD / 方向键；设置中可切换鼠标跟随或仅键盘
攻击/ 防御	左键或空格攻击；右键或Shift 防御
手机	左侧摇杆移动；按住攻击/ 防御，点击闪避与技能
主要面板	B 背包，C 合成，M 地图，J 图鉴，T 任务，X 天赋
战斗快捷键	Q 闪避；H 反转环绕；R 切换构筑；Z / V / F / E / G 主动技能
暂停与返回	Esc 暂停/ 返回；手机使用暂停按钮；弹窗内先阅读确认内容

精简HUD 下部分按钮放在“更多”和“技能”菜单中。屏幕较矮时可滚动面板，横向导航也可滑动。操作按钮被菜单隐藏，不代表对应功能不可用。

02 攻防、技能与状态

花瓣同时承担输出与耐久消耗。攻击时轨道外扩，防御时收拢；破损花瓣等待自己的装填倒计时后恢复。防御姿态下，本体和该花瓣连续1.2 秒未受伤后，完整花瓣每秒修复6% 耐久；已经破损的花瓣不会因此跳过装填。

技能/ 按键	冷却	当前基础行为
花瓣风暴Z	14 秒	半径170；伤害(35 + 3×等级)；普通目标眩晕0.75 秒，守卫0.25 秒
螺旋散射V	12 秒	10 枚弹体；单枚基础伤害17 + 1.7×等级；射程430
守护屏障F	20 秒	基础持续4.5 秒；受到的伤害乘以0.58
生长治愈E	25 秒	基础回复至多最大生命的13%；满血不消耗冷却
花瓣超载G	28 秒	持续7 秒；接触花瓣伤害乘以1.17

天赋与套装可以改变上述伤害、范围、治疗或持续时间，技能面板显示当前构筑数值。闪避持续0.20 秒，基础无敌时间0.27 秒，冷却5.2 秒。

本次修正：眩晕会停止正在执行的冲锋，取消特殊攻击前摇；恢复行动后，特殊攻击需要重新预警。已经离手的弹体和实体碰撞仍然存在，不能把眩晕理解为敌人消失或玩家完全无敌。冰霜花瓣提供减速，不等同于风暴的眩晕。

03 花瓣目录1 / 2

下表为普通品质的基础单次伤害、耐久与破损装填秒数，不是实际DPS。属性随品质缩放，实战还受耐久比例、护甲、暴击、天赋、套装和命中次数影响。

花瓣/ 标识	伤害	耐久	装填秒	基础定位
基础/ basic	13	34	2.1	均衡的入门花瓣。
玫瑰/ rose	6	30	3.4	受伤时定时释放一次治疗，冷却后可以再次恢复。
绿叶/ leaf	7	40	2	装备时每秒持续恢复生命，不限战斗姿态。
蜂刺/ stinger	31	11	4.3	伤害高，但容易损坏。
仙人掌/ cactus	11	60	3.3	装备时增加生命值上限。
气泡/ bubble	9	27	1.9	命中时将敌人推远。
砂砾/ sand	16	38	2.2	沙漠地区的高效率花瓣。
蟹钳/ claw	19	49	2.7	耐久与伤害兼顾。
鸢尾/ iris	18	34	2.7	对目标造成持续伤害。
闪电/ lightning	24	26	3.1	命中后对附近敌人连锁放电。
齿轮/ gear	21	80	3.5	重型防御花瓣。
尖牙/ fang	19	35	2.4	命中时少量吸取生命。
熔岩/ magma	29	58	3.7	高伤害熔岩花瓣。
贝壳/ shell	12	73	3	坚固的防御型花瓣。
触角/ antenna	21	41	2.5	装备后增强视野，品质越高，加成越强。

03 花瓣目录2 / 2

下表为普通品质的基础单次伤害、耐久与破损装填秒数，不是实际DPS。属性随品质缩放，实战还受耐久比例、护甲、暴击、天赋、套装和命中次数影响。

花瓣/ 标识	伤害	耐久	装填秒	基础定位
毒液/ poison	14	32	2.7	造成持续的中毒伤害。
海星瓣/ starpetal	9	44	3	生命低于75%时持续恢复生命。
花蜜/ nectar	4	24	2.3	生命低于一半时释放较强治疗，冷却较长。
导弹/ missile	23	26	2.9	黄蜂掉落，自动发射有限射程的导弹。
葡萄/ grape	15	37	2.6	射出弹射果实，命中后弹向其他怪物（最多再弹两次）。
绒球/ puff	6	32	2.2	释放安抚气息，削弱普通怪物对玩家的仇恨。
花冠/ bloom	16	48	2.4	野生花朵掉落，兼顾攻防。
玻璃/ glass	22	34	2.5	不会因碰撞破损；同一个目标每秒最多命中一次，不能进行十连击。
根须/ root	7	44	3	每两秒累积一层护甲（最多十层）；玩家受击消耗一层。
泥土/ soil	10	54	2.8	显著提高本体生命与体积，但每片降低移动速度。
飞翼/ wing	19	22	3	花瓣在轨道内外往复飞行，短时攻击距离很远。
磁石/ magnet	7	44	2.6	吸引更远的掉落花瓣，使拾取路线更安全。
棱晶/ prism	16	30	2.8	完整棱晶为接触攻击提供暴击：每片独立15%，合并上限60%，暴击伤害×1.6。
霜叶/ frost	10	39	2.6	命中后减速1.2秒：每片完整霜叶12%，上限35%；守卫效果减半。

04 品质与合成

阶	品质	升级单次成功率	属性缩放
0	普通/ common	64.000%	1.000
1	优秀/ unusual	32.000%	1.720
2	稀有/ real	16.000%	2.958
3	史诗/ epic	8.000%	5.088
4	传说/ legend	4.000%	8.752
5	神话/ mythic	2.000%	15.054
6	究极/ ultra	1.000%	25.892

阶	品质	升级单次成功率	属性缩放
7	超凡/ super	100.000%	44.535
8	独一/ unique	0.250%	76.600
9	永恒/ eternal	终点，不可再合	131.752

每次至少需要5 张同类同品质材料。成功消耗5 张并产生1 张下一品质；失败等概率消耗1、2、3 或4 张，不产生新花瓣。每次独立，没有隐藏保底。超凡到独一必成；独一到永恒仍是低概率步骤。

方式	停止条件	确认范围
单合	最多1 次尝试	当前材料组；成功与失败成本在按钮旁显示
批量合	最多5 次；不足5 张或产物满仓立即停止	先确认当前种类、品质、库存与损失规则
全合	当前组剩余不足5 张或产物达到99999 张	只消耗选中组，不处理其他组，不把新产物连续升级

例如库存12 张：第一次成功后余7 张，第二次失败损失3 张后余4 张，操作停止；合计尝试2 次、产出1 张、消耗8 张。全合的实际尝试次数取决于每次损耗，不是简单把初始库存除以五。

左侧搜索或按品质筛选，右侧查看选中材料、配方、概率与按钮。已经装备的花瓣不在可消耗背包中；材料保护同时阻止单合、批量和全合。产物满仓时不会继续扣料。概率详情里的图表示“最多五次”的分布，不是全合的总体分布。

05 构筑与成长

背包先选当前/ 备用构筑和槽位，再选择花瓣比较属性。替换后旧花瓣返还背包；拖拽需使用卡片上的拖动柄。切换A/B 构筑有2.6 秒冷却，当前花瓣重新装填至少2.5 秒。材料锁保护合成材料，不等于锁定装备槽。

达到等级	槽位数量
1	6
5	7
12	8
20	9
30	10

天赋	每级收益	最高等级
生命之根	每级增加6% 最大生命值	5
锋芒	每级增加5% 花瓣攻击伤害	5
延展	每级扩大6% 花瓣轨道半径	5

天赋	每级收益	最高等级
复苏	每级减少5% 破损花瓣装填时间	5
轻盈	每级增加4% 移动速度	5
采集	每级增加20% 掉落吸附距离	5

升级需要经验XP(L)=35+floor(18L+0.9L^2)，等级上限999。每两个等级增加一个天赋点，准确式为floor((L-1)/2)，所以首点在LV3。当前六项天赋合计最多投入30 点；LV61 后继续升级可能产生暂时无法投入的点，这是后期成长内容缺口，不应解释成已经完成的突破系统。

同系列2 / 4 片会激活一阶/ 二阶套装。伤害、耐久、回复和移动各有代价，图鉴与技能面板用于核对当前构筑，工作台评分不代表绝对最优装备。

06 区域与探索

区域	可连接区域	守卫
花园	海洋、沙漠、丛林、蚁穴、下水道、工厂	花园守卫·甲虫王
海洋	花园、丛林	深海巨蟹
沙漠	花园、蚁穴、地狱	赤砂蝎王
丛林	花园、海洋、地狱	密林螳螂王
地狱	沙漠、丛林、工厂	熔核守卫
蚁穴	花园、沙漠、下水道	古代蚁后
下水道	花园、工厂、蚁穴	深沟吞噬者
工厂	花园、下水道、地狱	机械巨像

地图从中心到边缘划分七档危险带。中心相对安全，边缘的等级、品质和精英压力更大；实际敌人等级由区域、危险带及生成随机量共同决定。图鉴记录已发现物种、击败数量和最高遭遇等级。当前共有43 种生物定义，其中包含区域守卫。

花园、海洋、沙漠、丛林、地狱、蚁穴、下水道与工厂构成探索网络。选择目标后应到游戏中的对应传送门；任务、掉落、收集目标和构筑升级共同形成循环。世界事件包含群落入侵、稀有花瓣雨、危险狂潮、复苏季风和落星预警。事件持续时间不同，选择概率不等于游玩时间占比。

区域守卫低生命时会狂暴。先观察攻击预警，再决定输出、防御或撤退。死亡后从当前区域安全点重新生长，装备与背包保留；仍需以存档状态确认最近进度是否已保存。

07 存档、备份与恢复

角色类型	存储与恢复	边界
Cloudflare 云端角色	邮箱作为登录标识；密码登录；恢复码找回；本机副本和云端版本同步	邮箱不用于发验证或找回邮件；云端写入需要有效会话和正确版本
本机角色	当前浏览器、当前网站来源的本地存储；可导出/ 导入进度JSON	不自动跨设备同步；清理浏览器数据前先备份

角色类型	存储与恢复	边界
旧邮箱云角色	保留旧账号入口及显式迁移流程	不能把新云角色、本机角色和旧邮箱角色当作同一份存档

本机写入成功不等于云端同步成功。以“云端已同步”等明确状态为准；离线时保留待同步副本。云端版本冲突会停止自动覆盖，摘要分别显示等级、库存、区域、版本与时间，不能只看等级或片数决定哪份更好。

正常本机保存会保留上一份有效副本；相同的自动保存不会挤掉恢复点。当前存档损坏时，先读取有效副本；两份都不可用时停止载入，不能静默创建一级角色覆盖原进度。通过本机账号密码校验后，可直接选择之前导出的角色进度文件恢复，损坏原记录保留一份。

合成先结算到独立库存副本，再写入本机。写入失败会撤销本次材料和产物变化；已经成功写入后，即使云端排队失败，也不会错误撤销结果。导出直接读取当前页面进度，避免存储失败时导出旧磁盘副本。

处理冲突前先下载备份并确认文件已保存。选择云端会保留本机回退；选择本机会先备份云端，再按当前版本尝试写入。不会自动把两份库存相加。恢复码文件、待同步文件和角色进度文件用途不同，不能互相冒充导入。不要分享密码、Cookie、恢复码或私有进度。

08 战术训练与常见问题

模式	目标与资源	边界
四面楚歌	每波识别并穿过金色突破区，再清理敌群；方向随三波轮换	侧翼冲锋、前排牵制和远程压制迫使选择突破方向
背水一战	以35% 生命进入，三波共用3 次主动技能；花瓣治疗仍可用	需要在恢复、防守和爆发之间分配有限次数

训练有1.5 秒方向预警、180 秒总时限和最多24 个训练敌人的上界。训练复制当前构筑到独立场景，不写主线，不发永久奖励；退出会恢复主线现场与随机状态。刷新会结束训练，并从已保存的主线进度继续。

现象	处理
合成按钮不可用	检查是否选中材料、至少五张、材料未保护、非最高品质、产物未满仓
提示本机保存失败	保留当前页面，立即导出当前进度；检查浏览器存储权限和容量
云端版本冲突	先导出备份，再比较；不要重复强行覆盖或清理待同步记录
账号入口不同	确认使用的是新云端、本机还是旧邮箱角色；不同来源的网站也有不同本机存储
手机掉帧或耗电	使用省电画质、减少动画或关闭装饰粒子；这不应隐藏攻击预警与实际冷却
找不到某按钮	精简HUD 中展开“更多”或“技能”；也可用对应快捷键

当前房间功能属于实验性位置预览，不是已经完成的服务器权威多人战斗。物理iPhone Safari、Android 和低性能设备的长时间体验，需要真机验收；模拟触控通过不等于这些平台已经验收。

S4 / S5 现行机制与工程补编

S4 对应V5.20 的结算演出、技能反馈和独立训练；S5 对应V5.21 的存档恢复、单/ 批量/ 全合及本次控制状态修复。本补编仅覆盖明确更新的条款，其余历史数学说明按原有适用条件阅读。

S-01 概率与数值契约

合成成功概率为p；成功消耗5，失败损耗C_f 均匀分布于{1,2,3,4}。在持续补充同阶材料、独立固定概率且目标有产物空位的条件下，首次成功的期望材料成本为E[C]=5+2.5(1-p)/p。它不是有限库存的成功保证。

来源阶	品质	持续补料的平均首次成功成本
0	普通	6.406
1	优秀	10.312
2	稀有	18.125
3	史诗	33.750
4	传说	65.000
5	神话	127.500
6	究极	252.500
7	超凡	5.000
8	独一	1002.500

固定最多五次的有限库存分布必须逐步检查剩余材料；独立Bernoulli 的1-(1-p)^5 仅在五次尝试都能进行时才适用。全合终止条件不同，不使用五次分布冒充全合统计。

伤害预测区分名义伤害、扣护甲后的实际伤害与剩余生命截断后的有效伤害。耐久、暴击、套装、护盾和无敌窗口须按真实调用顺序计算。单次期望伤害不能直接当作稳定DPS；工作台效用权重也不是概率。

等级上限、槽位里程碑和天赋预算保持现有规则；本轮没有调整掉落、经验、合成成功率或装备强度。后期天赋点闲置、部分构筑在护甲下落入最低伤害等问题属于已识别的后续平衡议题。

S-02 状态、提交与动画

路径	不变量
合成	单个界面操作只接受一次；材料及产物在同一库存副本结算；本机提交成功后才呈现结果
写入失败	恢复原库存；不展示已获得产物；已消耗的会话随机抽样不会回退
云端排队失败	不能把已成功的本机提交改判失败；队列持有脱离游戏可变对象的快照
演出取消	跳过、关闭、降低动画或切到后台都不能再次结算、抽样或撤销已保存结果
眩晕	中断当前冲锋并取消特殊前摇；眩晕期间不能派发特殊攻击；恢复后需要新的特殊预警

路径	不变量
版本冲突	以账号、界面、角色对象与版本共同绑定确认；不自动合并库存，不借用另一份待同步版本

Cloudflare API 使用expectedRevision 条件写入。版本不符返回冲突，客户端保留待同步记录。服务器的updatedAt 为Unix 秒，本机记录为ISO 时间；摘要分别按对应格式解析，避免有效时间显示为未知。健康接口返回完整版本5.21.0，供部署验收与页面版本比较。

服务端负责身份、会话、输入范围、版本和写入约束；当前客户端战斗没有被服务端重演，不把这套单机存档结构称作防作弊竞技经济。

S-03 代码与验证边界

模块	职责
site/index.html	主循环、实体、输入、角色、各面板、账号与同步编排
site/rules.js / docs/standard	概率模型、条件预测、固定步长换算与可追溯契约
site/save-store.js	本机有效副本、损坏检测及原记录保留
site/interface.js / workshop.js	确认对话框、公告、构筑规划与库存守恒
site/art.js / effects.js / feedback.*	共享矢量美术、技能图形与已结算回执的演出
site/trials.js	无永久奖励的训练模式和确定性波次布局
functions/api/[[path]].js	Cloudflare 账号、Cookie 会话、限流、条件存档及显式迁移
database / wrangler.jsonc	既有存储结构与环境绑定；数据库名不等于环境身份

验证层	能证明什么	不能代替什么
Node + 内存SQLite 适配器	API 状态、校验、并发条件与身份逻辑	不是workerd，也不证明线上CPU 和D1 行为
Wrangler 编译/ 本地 runtime	可构建与对应运行时可执行性	不是线上真实部署和绑定
浏览器fixture 回归	实际渲染、输入、布局、状态转换	不是跨设备服务或物理手机长稳测试
隔离预览站	真实Workers/D1、会话、跨上下文保存与冲突	不能代替真实旧用户数据迁移验收
生产只读检查	提交、版本、函数、绑定、匿名健康、公开资源字节	不应为了测试而写入真实玩家账号

完整审核按文件角色记录：现行程序和高风险路径深查；测试与配置执行对应门禁；历史HTML、ZIP、旧QA和旧手册保留归档身份。第三方依赖使用锁文件和漏洞扫描，不声称逐行审完全部依赖源码。

S-04 发布、回滚与复现

当前正式入口https://petal-realms.pages.dev/ 。发布前先核对PR 的当前head、CI 和预览部署；完成隔离账号验收后才合并。部署成功还需检查新提交、uses_functions、数据库绑定、/api/health、匿名会话与页面资源的实际字节。

环境	PETAL_DB ID	用途
生产	92c62437-5604-46c2-821c-8999cf739d90	既有玩家库，历史名称带preview；不可因名称交换库
预览	9ec11aae-8c72-4e78-a32e-79b9525e7d1d	隔离验收账号与测试存档

本轮保持角色version 5 和既有数据库结构，不重建、清空或批量迁移生产存档。页面代码回退不会回退数据库；出现问题优先保留数据与导出，再修复或回退页面。旧页面和新数据之间的兼容性必须单独判断。

开发验证入口：npm ci；npm test；node scripts/build-rules.cjs --check；npm run build:worker；GitHub 完整浏览器工作流。真实站smoke 使用tests/browser_live.mjs，并比较手册、保存模块和页面资源；不创建生产云账号。

文档复现：python3 scripts/build-release-handbook.py。需要PyMuPDF、XeLaTeX、Node、fc-match及经哈希校验的Noto CJK 衬线/ 无衬线字体。构建不缩小正文掩盖溢出；新增内容连续分页，历史354 页的文本和绘制内容流必须保持。

发布回执写明实际提交、部署和检查结果。手册中的版本声明描述软件目标，不预先替代真实发布回执。

S-05 来源与一致性

本次运行时文件集指纹SHA-256：

8312ef62908bd0edc01f62541c739e7aedfe3faf80a2812923087f5bb22e09a8

指纹由以下文件的SHA-256 映射按路径排序后计算，不包含PDF 自身或会变化的发布回执，避免自引用。文件和完整摘要存于docs/manual/V5.21_MANUAL_MANIFEST.json。

来源文件	SHA-256（前24 位）
docs/standard/mechanisms.json	5d708fd203e629d0d331f8ab
docs/standard/ruleset.json	546d8d1968a9a9ce8b10a52c
functions/api/[[path]].js	8001fa8ab551867cbcc0b05f
package-lock.json	52de1f89b8aceb4e613644e9
package.json	3a6a7563255b427a677ca974
site/art.js	284f0008c2b7612c2db20181

来源文件	SHA-256（前24 位）
site/effects.js	671ce93f57e13277936481a6
site/feedback.css	f2a0f5f55d3ebb91580ab14e
site/feedback.js	477d35d8ba2a05aaf1a88b86
site/index.html	6f4b31b97948e7534749c2cf
site/interface.js	72f47d051d2645be738e7f65
site/release.json	7f6e3b1df6fdb952678bcaaa
site/rules.js	b64c92ccf6ee56afed1d1fb8
site/save-store.js	aef1b1f055c5588410932f7f
site/trials.js	23f97914af9e2f583978f31e
site/workshop.js	ecb9bf79ef5590e02a52c3a2
wrangler.jsonc	8c1ecd2fb0e760a66abdb01c

历史正文来源：354 页V5.19 手册页副本。原Library 文档SHA-256 为d2a34c075c4c213c950953a2e1af067691d6afaf53ff1a1f2c2ef25a21ae3732。仓库页副本只用于可复现补编，不携带原整体文档的 Content Credentials；新组合文件不冒用旧文件的签署或生成凭据。

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